

SANLORENZO

SD118



OWNER'S MANUAL

Sab BATTICAL

VALLETTA



Sanlorenzo Spa

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SANLORENZO

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UNI EN ISO 9001:2015
UNI EN 9100:2016
EN ISO 17100:2017



WELCOME ON BOARD



Section **A**

INTRODUCTION

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Foreword to the use of the Manual

Congratulations,

on having chosen **SANLORENZO**, one of the world's most prestigious names in yachting. The entire staff at SANLORENZO thanks you and considers your purchase the best reward to our constant commitment to producing the most advanced vessels from a technical and aesthetic perspective.

Every design decision guiding the construction of your boat was made with a view to achieving harmony, style, high performance, comfort, safety and reliability. In addition, at SANLORENZO, customer satisfaction has always been one of our central objectives and we are certain that the care taken in putting together this manual is tangible proof.

Prior to operating on the vessel and the equipment, read the Manual carefully, in order to acquire an adequate familiarity with the systems and their operation, in order to avoid hazard to personnel and risks of costly damages.

Passion for cruising and the ship prestige as Owner, guest, crew are incentives to regularly scheduled maintenance which will guarantee long navigation periods, long operational life and safety improvement.

The maintenance operations described in the Manual are simple, but should be performed by authorised and qualified technical personnel only, according to standard procedures of each supplier equipment and in compliance with national and international regulations.

For specific activities, it is suggested to refer to specialised technicians, or to call our Service Department.



Please, keep this Manual carefully in a safe, dry and easily accessible place for an easy consultation. When you decide to change boat, deliver this Manual to the new Owner in its integrity.



*The boat in your possession could differ in some way from the boat presented in this Manual as a result of special installations requested, equipment installed after compilation of this Manual, or modifications made after the boat left the yard.
If you have any doubts or if you find substantial inconsistencies with the descriptions given here, contact a builder's assistance centre immediately to receive the necessary instructions.*

Manual Introduction

This Manual is written in compliance with Directive **2003/44/CE**, UNI EN **ISO 10240** and in accordance with Directive **RINA**.

The purpose of this Manual is to give information and to help You to use and drive Your Yacht in full safety and with complete satisfaction.

The Manual contains detailed explanations about the ship, its systems and equipment, as well as information about the Yacht practical use and maintenance.

We recommend you to read this Manual carefully and to get knowledge and familiarity with the ship before going at sea for the first time.

**IN THE EVENT OF LOSS OF OR DAMAGE TO THE MANUAL,
THE SANLORENZO SHIPYARD WILL ALWAYS BE ABLE TO SUPPLY A COPY**



SANLORENZO S.p.A. shipyard declines all responsibilities for any damages to third parties and for eventual discrepancies between the Manual and reality.

Guarantees and assistance request procedure

SANLORENZO assistance is at your complete disposal. For any question, please contact directly:

SANLORENZO SERVICE DEPARTMENT

**SANLORENZO S.p.A.
Via Luigi Salvatori, 58
55049 - Viareggio (LU) - ITALY
TEL.: +39 0584 38071
FAX: +39 0584 380784**

SANLORENZO has selected all main equipment and accessories among the most reliable ones and those that ensure also an easy availability of spare parts.

Such components are covered by their own warranty issued by relevant manufacturers; any repair should be carried out by specialised technicians, authorised by the shipyard, otherwise the warranty will be void and the equipment may get damaged.



SANLORENZO declines all responsibilities for the installation and operation of electrical, electronic and mechanical equipment improperly installed by third parties in a fashion not authorised by the boat Builder.

SANLORENZO declines all responsibilities if third parties tampered with equipment installed by the boat builder. Such tampering or unauthorised installations will make the guarantee to become void, and even cause damages to the boat and the people on board.

SANLORENZO declines all responsibilities for damages due to improper preservation and poor maintenance.



For all legal aspects related to the boat warranty, refer only to the sale contract which specify all warranty clauses applicable.

Yacht's identification data

Builder:

SANLORENZO Shipyard

Via Luigi Salvatori, 58

55049 Viareggio (LU) - ITALY

Phone: +39 0584 38071

Fax: +39 0584 380784

E-mail: info@sanlorenzoyacht.com

Model:

SD 118

Contruction:

SD 118_102

CIN:

Class notation:

RINA PLEASURE YACHT UNRESTRICTED NAVIGATION
C ⚡ HULL • MACH, Y_{CH}

Classification Society:

RINA

Name:

SabBaTiCal

Flag:

MALTA

Port of Registry:

VALLETTA

Certificates present on board

The following certificates must be kept on board:

Classification Certificate issued by the Classification Society for the hull and the machinery.

Builder's Certificate and **Commercial Invoice** issued by the Shipyard.

International Tonnage Certificate issued by the Classification Society.

Notices

Major relevance sentences and some important requirements are highlighted by symbols defined as follows:



WARNING

Indicating that special procedures are required.

Failure to comply with such notice may cause abnormal operations.



CAUTION

Indicating a reminder to safety procedures or not safe operations.

Failure to comply with such notices may cause injuries to personnel or damages to the boat and its equipment.



PERSONNEL SAFETY

DANGER *Indicating a serious hazard.*

Failure to comply with such notice may cause serious injuries or death.



ENVIRONMENT PROTECTION

Indicating behaviours assumed to avoid risks of environmental pollution.

Important notes



Do not modify the yacht and its systems as it may compromise the safety.

To customize and modify existing machinery, please refer to the Shipyard for receiving necessary instructions.



The yacht is delivered by the Shipyard with all safety equipment required by International regulations; the Operator should equip the yacht with equipment required by the Country where the yacht is registered, as well as arranging inspections to verify efficiency and validity.

Everyone on board should know where life jackets storing compartments are and how to wear them; before starting a cruise, Crew should suitably inform the Guests.

Life rafts are located on the upper deck. For use and maintenance instructions please refer to specific manual supplied by the manufacturer, which is delivered separately.

We recommend you to read this manual carefully and make sure that all periodical inspections are carried out.



The manual provides general information regarding the behaviour that crew and guest have to hold on board.

In particular, as far as use and operating procedures of yacht's systems and equipment are concerned, please do refer to the single manufacturer's manuals.

It is assumed that the crew is fully aware of the contents of the above mentioned manuals and the site has no liability for damages to property and/or people caused by negligence, neglect and/or incompetence in the use/maintenance of the yacht's systems or by the inobservance of the specific requirements of the single equipment's manual.



We recommend to guard the yacht throughout the day and especially in the night in order to intervene promptly in case of serious failures that might occur.

Use of technical documentation



SANLORENZO suggests to read carefully all documents supplied by manufacturers of each system; any question concerning use and maintenance should be forwarded to Service Departments listed in the suppliers' documents. However, if necessary some intervention may be carried out by yacht's personnel, after having read the operation manual.



*This manual should not be considered as a guide for repair or extraordinary or in depth maintenance interventions.
You can find further detailed information in the specific manuals. However, for maintenance operations and interventions on the various components, we suggest you contact SANLORENZO or an authorized service centre of the equipment installed on Your boat.*



*Always contact only especially trained and competent personnel for maintenance and repair of your boat.
The modifications that can have an impact on the safety characteristics of the boat must be assessed, executed and documented by personnel trained in the material.
SANLORENZO cannot be held responsible for any unauthorized modifications made to its boats.*



*All the plan listed or included in this Manual are intended for reference since not bearing RINA approval stamp.
Official plan stamped by RINA should be considered.*

Dimensions and characteristics DATA

The Motor Yacht **SD118-102** has the following main characteristics:

(L _{OA}) Length overall	m ft.	35.75 117.29
(L _{WL}) Length at water line (half load)	m ft.	33.80 110.89
Length between perpendiculars	m ft.	31.65 103.84
(B _{MAX}) Beam max	m ft.	7.90 25.92
(D) Draught (full load)	m ft.	1.96 6.43
Gross Tonnage	t	298
Deadrise	°	23.5
Maximum speed (at half load)	kN	19±1
Autonomy	Nautical Miles	3000



Never exceed the maximum loading capacity and the maximum number of people on board.

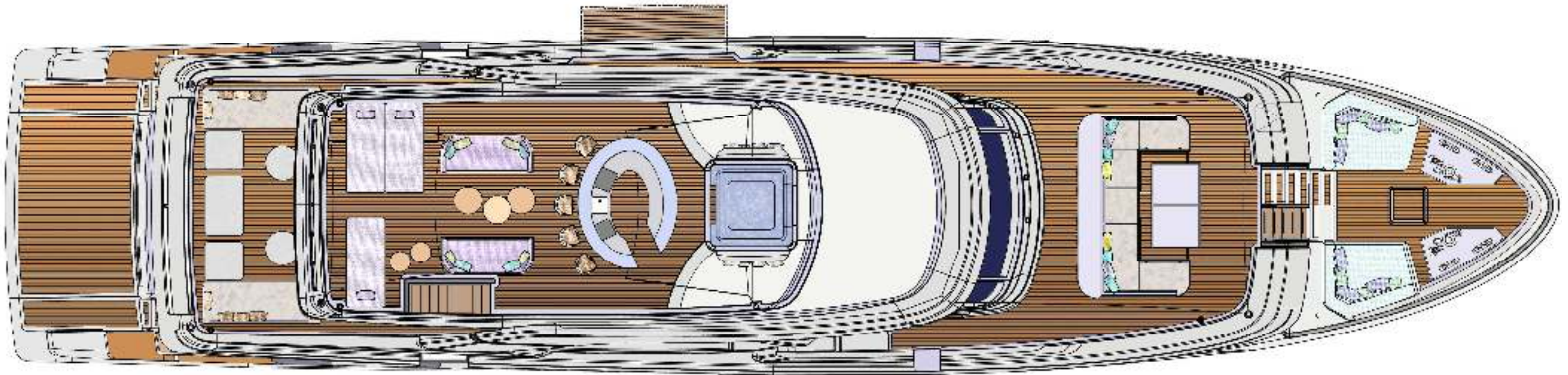
Make sure that materials are distributed on board in a suitable manner.

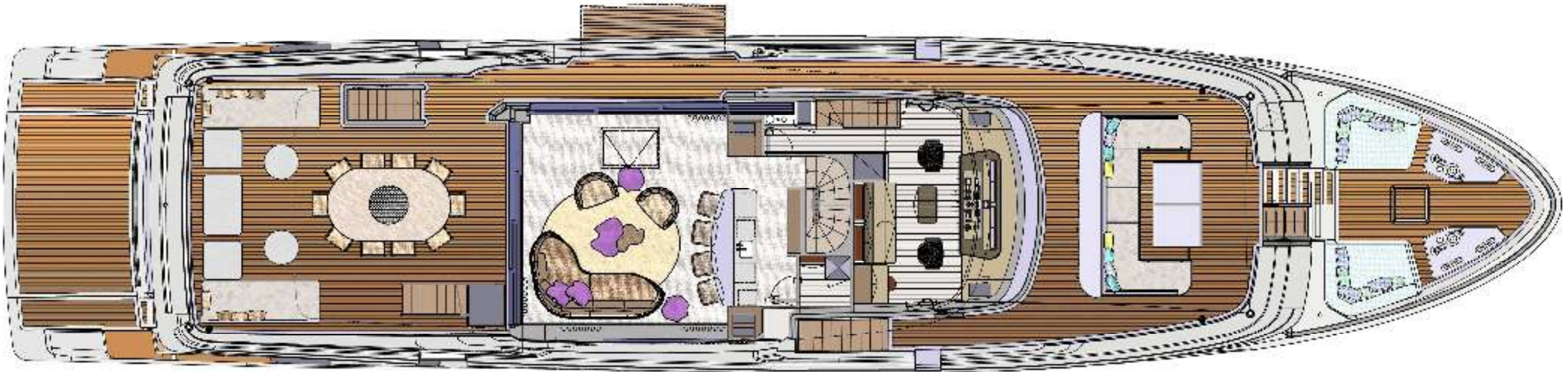
Main technical data

Hull type:	semi displacement
Construction material:	composite
Accommodation for Owner / Guests	8 (+2) persons in 4 cabins (Lower Deck) 2 persons in 1 cabin (Main Deck)
Accommodation for crew	7 persons in 4 cabins (Lower Deck)
• Main engines (n°2)	kW 1,193 (1,600 Hp) rpm 2,300 type CAT C32 ACERT
• Gear box (n°2)	type ZF 2360 (R = 4:1)
• Range	2,500 n.m. approx. (at 11 knots + 1 with n°1 D.G. working at 50% of load)
• Fuel tanks	lt. 37,000 total
• Fresh water tank	lt. 4,000
• Black & Grey water tank	lt. 3,000
• Power supply	n°2 KOHLER Mod. 70EFOZDJ (70kW, 400-230 VAC, 50 Hz, three-phase)
• Power supply	n°1 400 VAC 3P+T dock connection
• Chiller Unit	n°2 144.000 BTU/h (CONDARIA)
• Water maker	n°2 IDROMAR MOD. MC5S (250 l/h)
• Bow Thruster	n°1 45 KW – Thrust 7.1-8.1 kN – 230 V/400V 3ph (CMC MARINE – TP60)
• Stabilizer	n°2 Loads HS80-A (5.0 kW) with Fin 190LR (1.90 m ²) (CMC MARINE – HS80+190LR)
• Propeller	n°2 5 fix blades

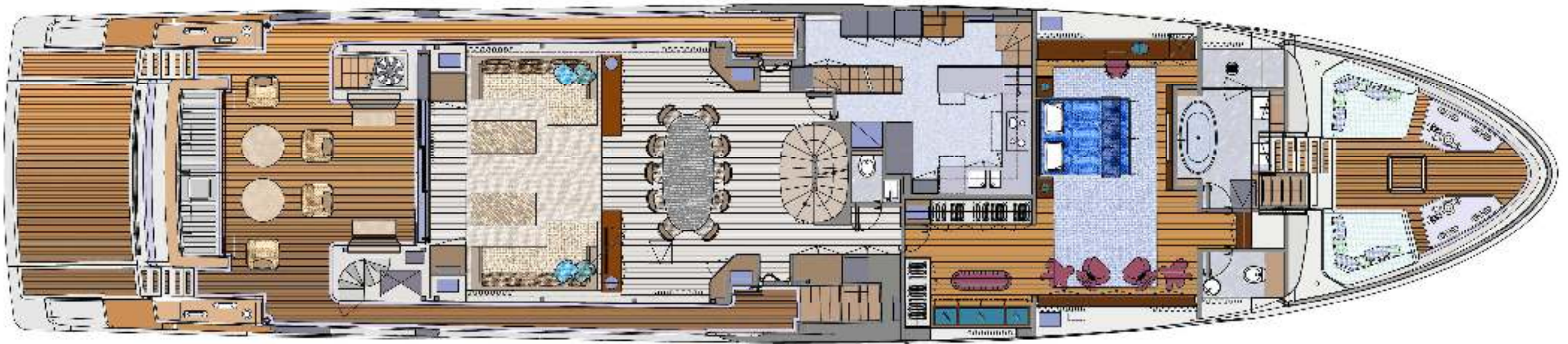
GENERAL PLANS – Profile

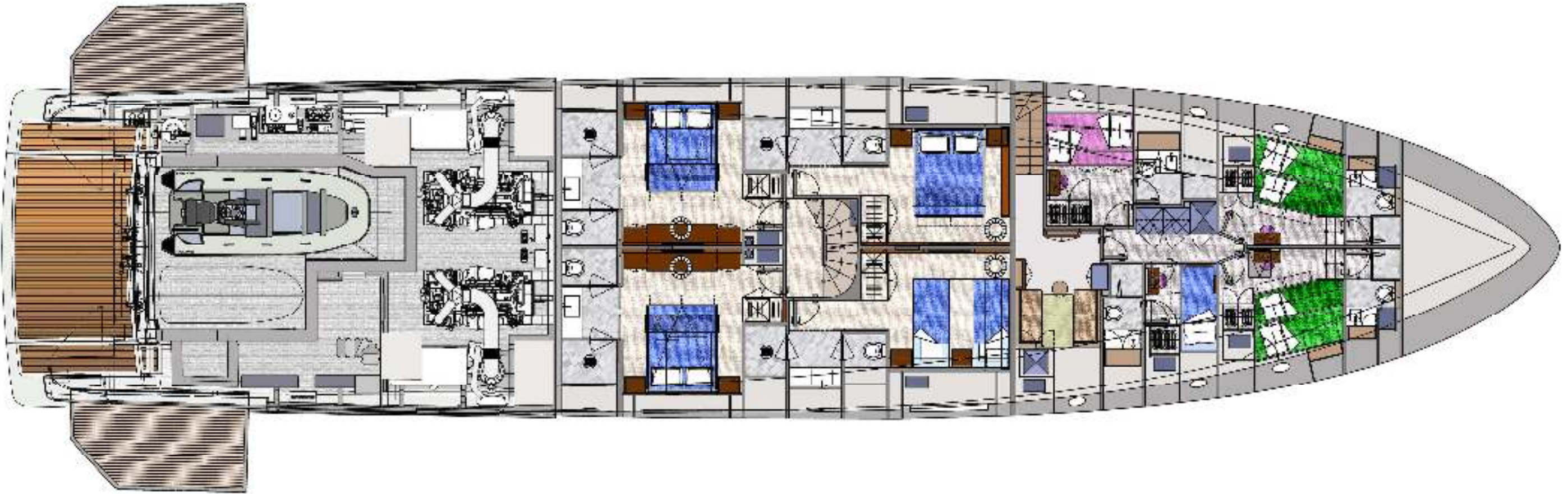


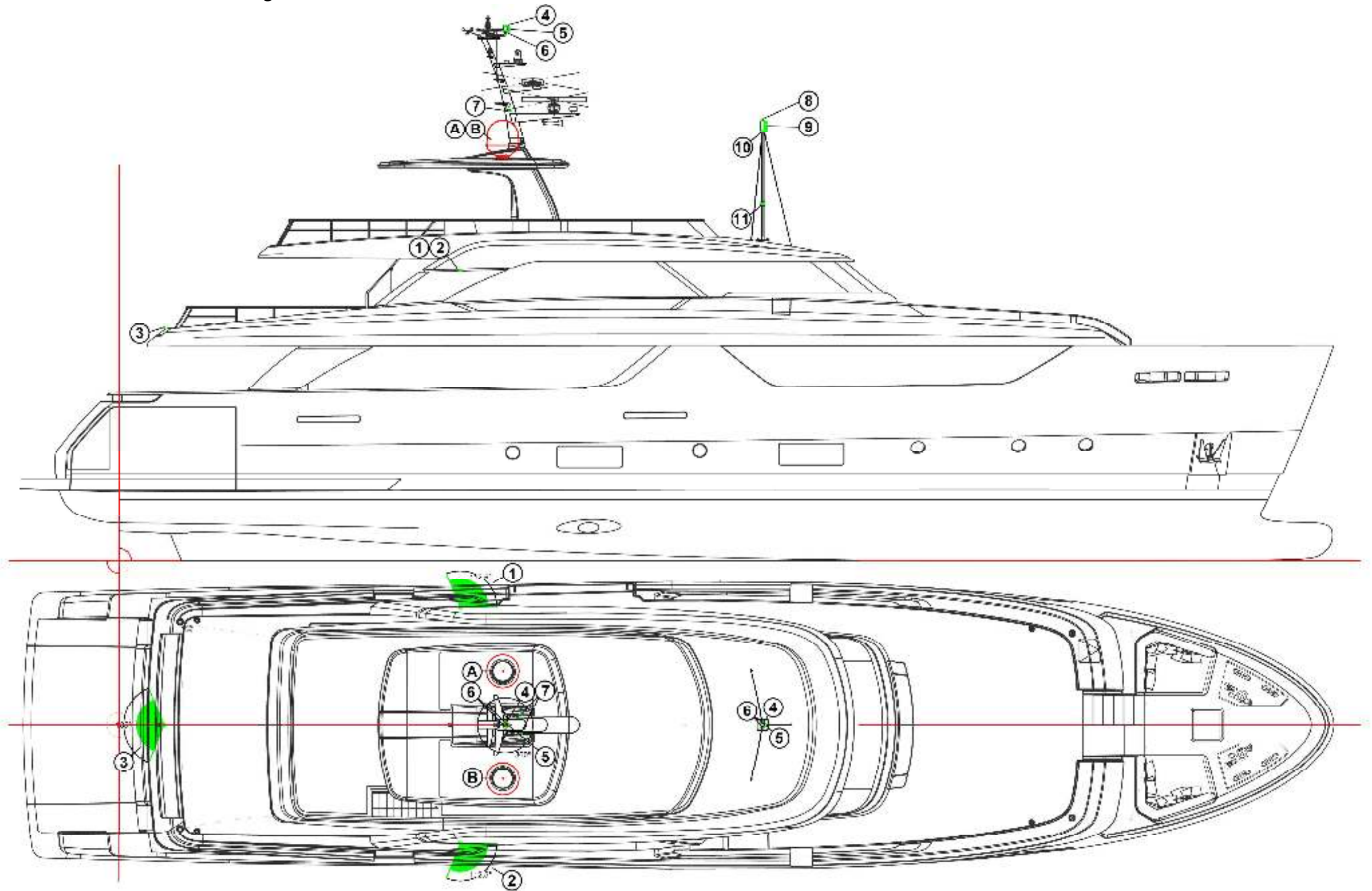
GENERAL PLANS – Flying Bridge

GENERAL PLANS – Upper Deck

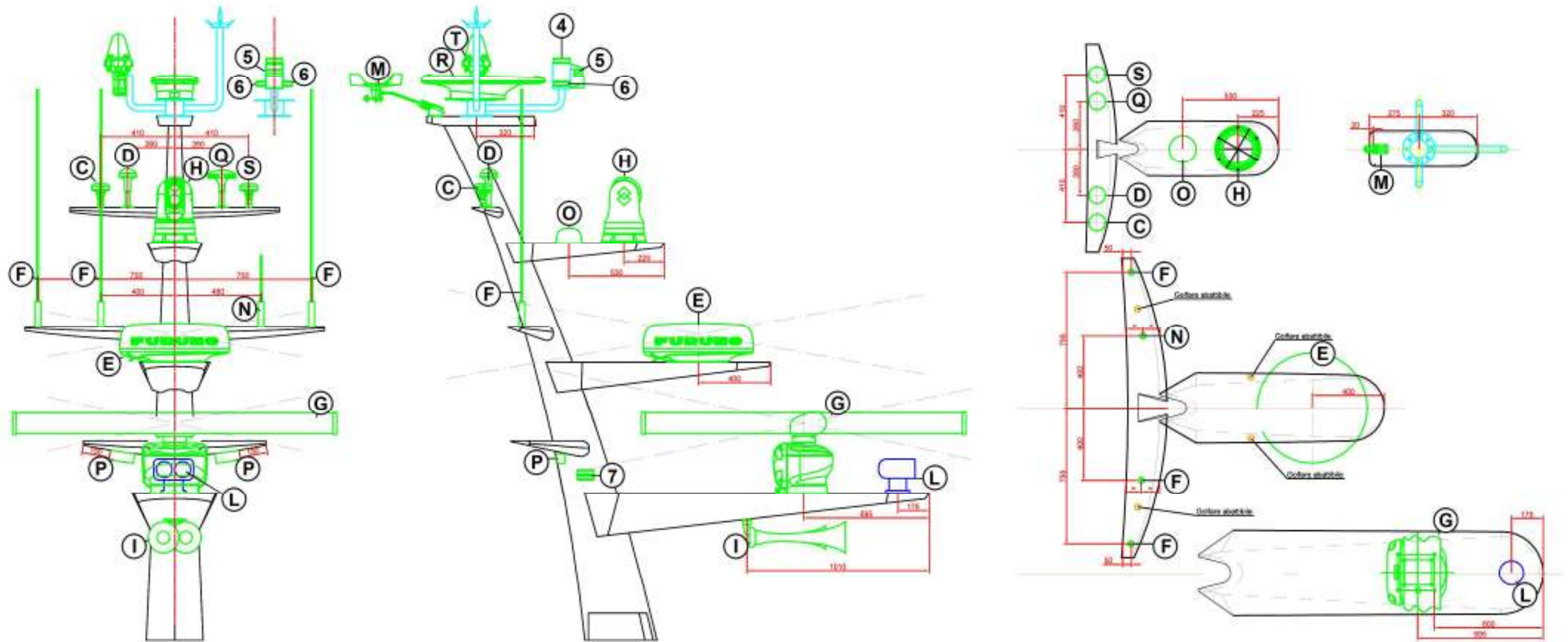
GENERAL PLANS – Main Deck



GENERAL PLANS – Lower Deck

GENERAL PLANS – Mast Arrangement

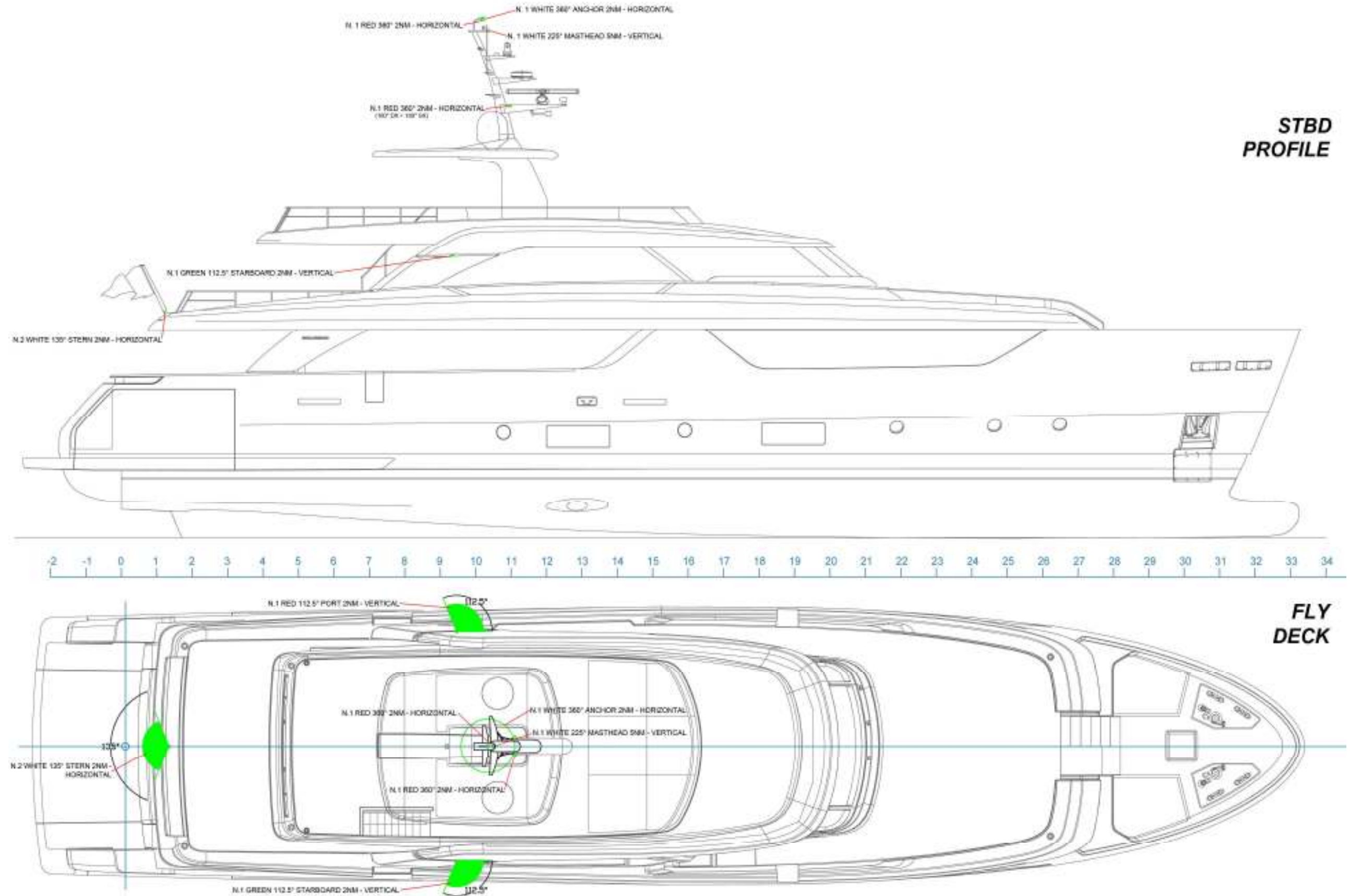
GENERAL PLANS – Mast Arrangement



POS. Item	DESCRIPTION Description	FACTORY Brand	QUANTITY Quantity	MODELLO Model	FORNITORE Supplier	CODICE V. V. Code	CODICE SUPPORTO Support. Code
A	RADOME KUH TV8		1			0300022	0300034
B	RADOME KUH TV8 V.DTC		1			0300038	0300034
C	ANTENNA WPA-1000 - 10W/100V/1000W/1000W		1		FURUKO	0300017	0300017
D	GPS NAVIGATOR GP170/GFADLTS		1		FURUKO	0300129	0300129
E	GPS40 HXT FURLNO RADAR SENSOR RADOME		1		FURUKO	0300129	
F	VHF 21111111111111111111		2		FURUKO	0300129	0300129
G	ANTENNA RADAR ON ASSUMBLA XN200 - 0.02		1		FURUKO	0300129	
H	LITTOCAMERA FURUKO FURUKO		1		FURUKO	0300129	
I	TRANSIMBOLA SUILO SUILO SUILO		1		INDIMATI S.P.A.	0300129	0300129
L	PROIETTORE JARSCO 7 24V		1		INDIMATI S.P.A.	0300129	
M	ANTENNA RADAR 1-2001		1		FURUKO	0300129	
N	ANTENNA RADAR 1-2001		1		FURUKO	0300129	
O	TADE 15 100		1		INDIMATI S.P.A.	0300129	
P	SERENA CON LAMPADINE 1000A M		2		VARINER SRL	0300129	0300129
Q	NAVITE FURUKO HX-2000 1000A M		1		FURUKO	0300129	0300129
R	SATELLITE COMPASS 50 70		1		FURUKO	0300129	0300129
S	TRANSIMBOLA SUILO SUILO SUILO		1		FURUKO	0300129	0300129
T	INDIMATI SUILO SUILO SUILO		1		FURUKO	0300129	0300129

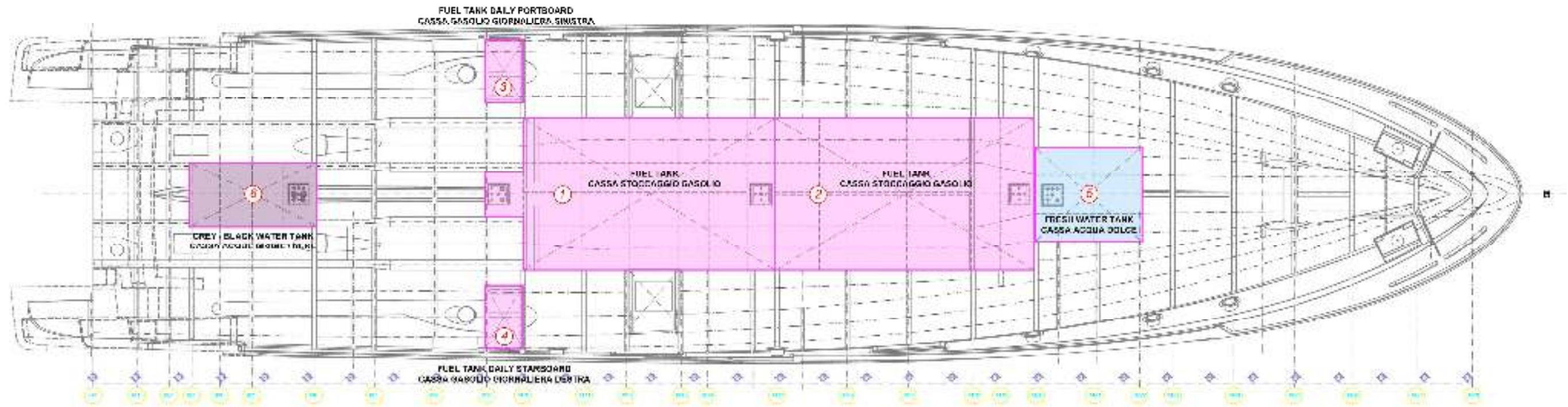
POS. Item	DESCRIPTION Description	FACTORY Brand	QUANTITY Quantity	MODELLO Model	FORNITORE Supplier	CODICE V. V. Code	CODICE SUPPORTO Support. Code
1	RADOME KUH TV8		1		FURUKO	0300022	0300034
2	RADOME KUH TV8 V.DTC		1		FURUKO	0300038	0300034
3	ANTENNA WPA-1000 - 10W/100V/1000W/1000W		1		FURUKO	0300017	0300017
4	GPS NAVIGATOR GP170/GFADLTS		1		FURUKO	0300129	0300129
5	GPS40 HXT FURLNO RADAR SENSOR RADOME		1		FURUKO	0300129	
6	VHF 21111111111111111111		2		FURUKO	0300129	0300129
7	ANTENNA RADAR ON ASSUMBLA XN200 - 0.02		1		FURUKO	0300129	
8	LITTOCAMERA FURUKO FURUKO		1		FURUKO	0300129	
9	TRANSIMBOLA SUILO SUILO SUILO		1		INDIMATI S.P.A.	0300129	0300129
10	PROIETTORE JARSCO 7 24V		1		INDIMATI S.P.A.	0300129	
11	ANTENNA RADAR 1-2001		1		FURUKO	0300129	
12	ANTENNA RADAR 1-2001		1		FURUKO	0300129	
13	TADE 15 100		1		INDIMATI S.P.A.	0300129	
14	SERENA CON LAMPADINE 1000A M		2		VARINER SRL	0300129	0300129
15	NAVITE FURUKO HX-2000 1000A M		1		FURUKO	0300129	0300129
16	SATELLITE COMPASS 50 70		1		FURUKO	0300129	0300129
17	TRANSIMBOLA SUILO SUILO SUILO		1		FURUKO	0300129	0300129
18	INDIMATI SUILO SUILO SUILO		1		FURUKO	0300129	0300129

GENERAL PLANS – Navigation Lights

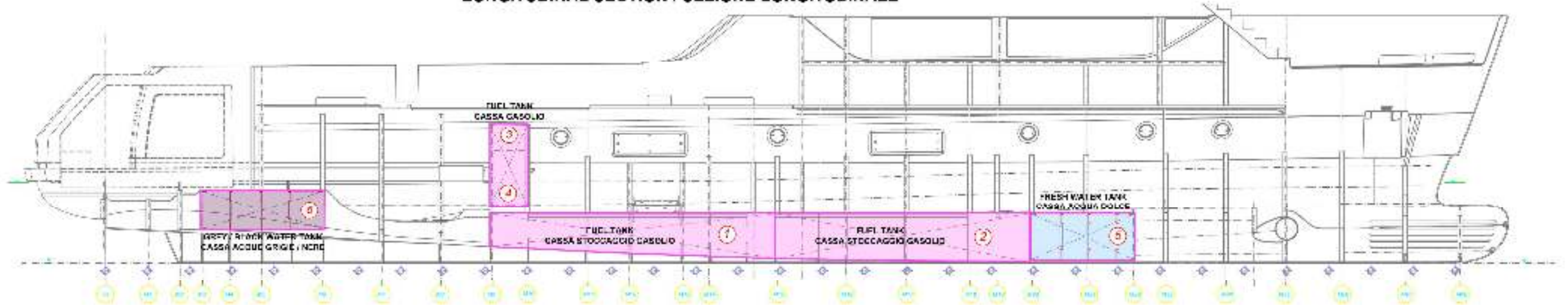


GENERAL PLANS – Capacity Plan

PLAN VIEW / VISTA IN PIANTA

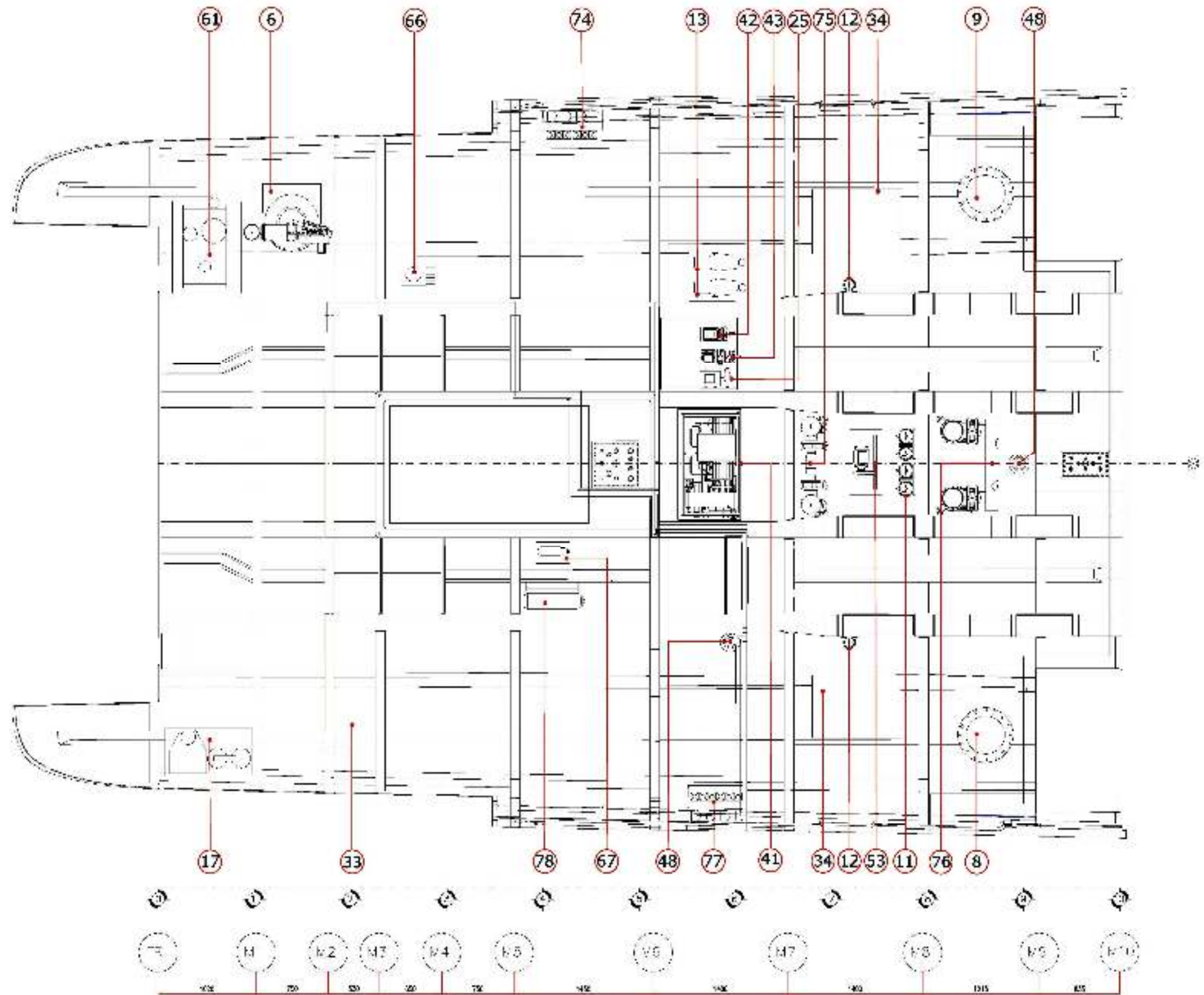


LONGITUDINAL SECTION / SEZIONE LONGITUDINALE

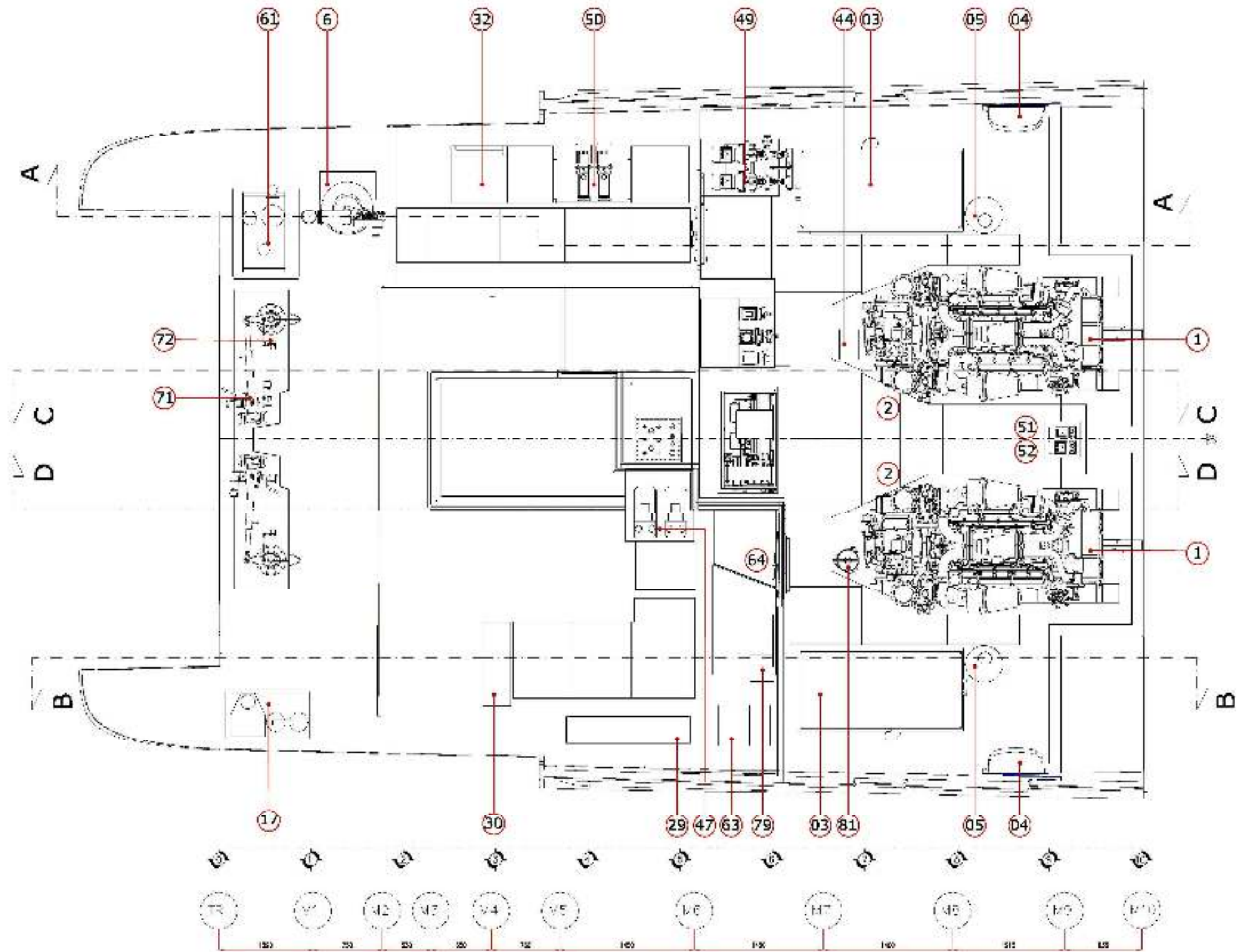


1	FUEL TANK AFT / CASSA GASOLIO POPPA – 14.8 m ³	3	DAILY TANK PS / CASSA GASOLIO GIORNALIERA – 1.9 m ³	5	FRESH WATER TANK / CASSA ACQUA DOLCE– 4.0 m ³
2	FUEL TANK FORE / CASSA GASOLIO PRUA – 15.0 m ³	4	DAILY TANK STBD / CASSA GASOLIO GIORNALIERA – 1.9 m ³	6	GREY&BLACK WATER TANK / CASSA ACQUE GRIGIE-NERE– 3.0 m ³

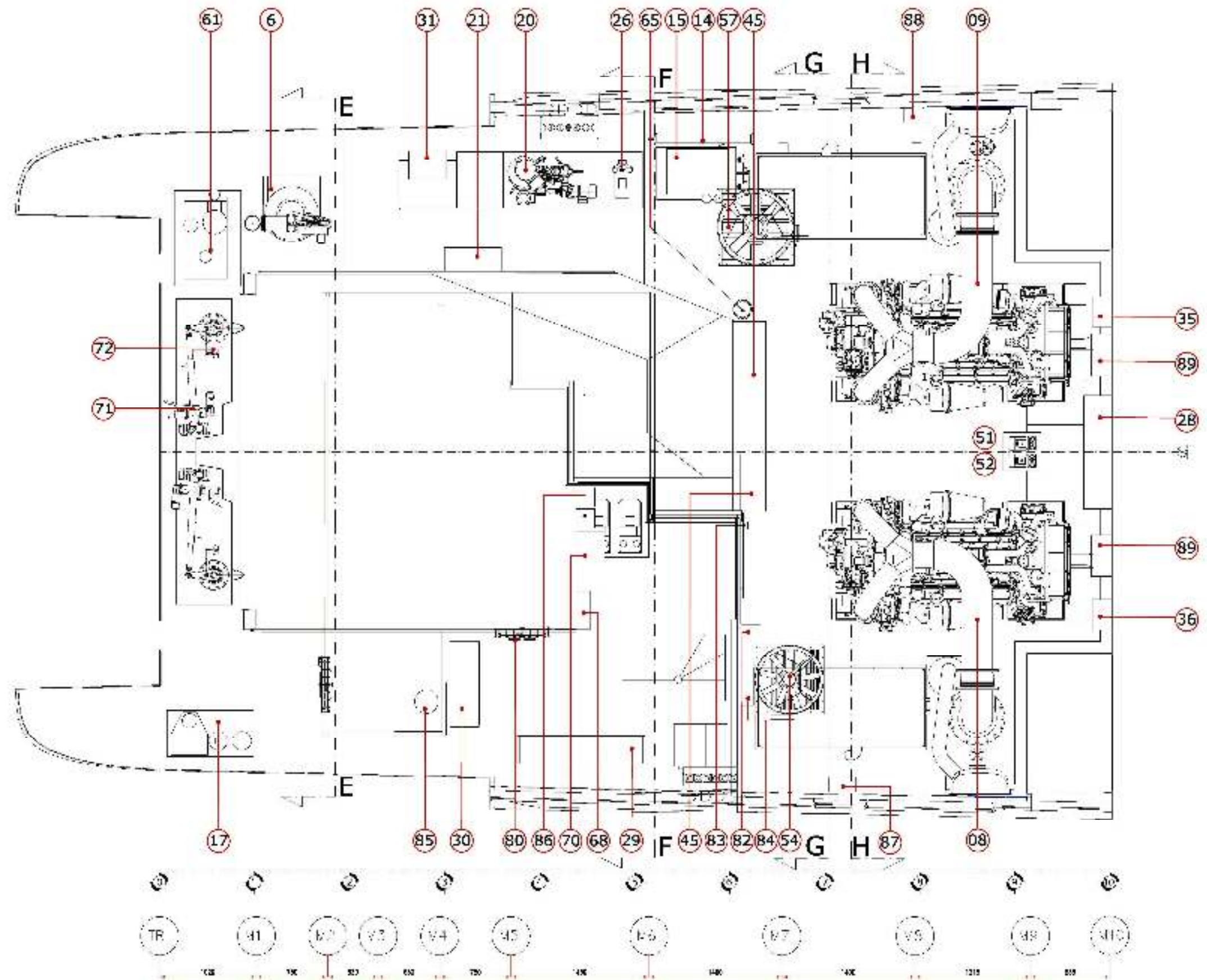
GENERAL PLANS - Engine Room - Bottom view



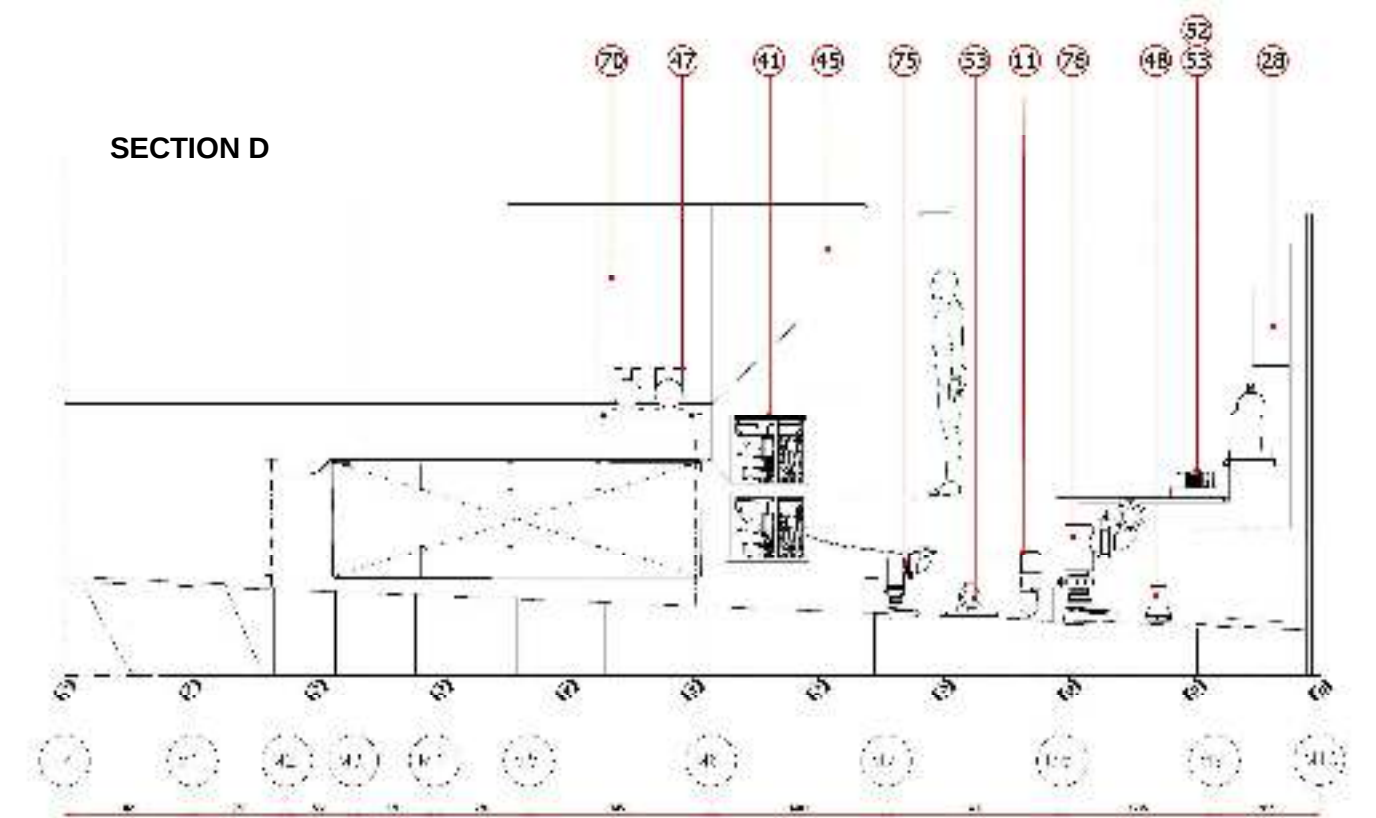
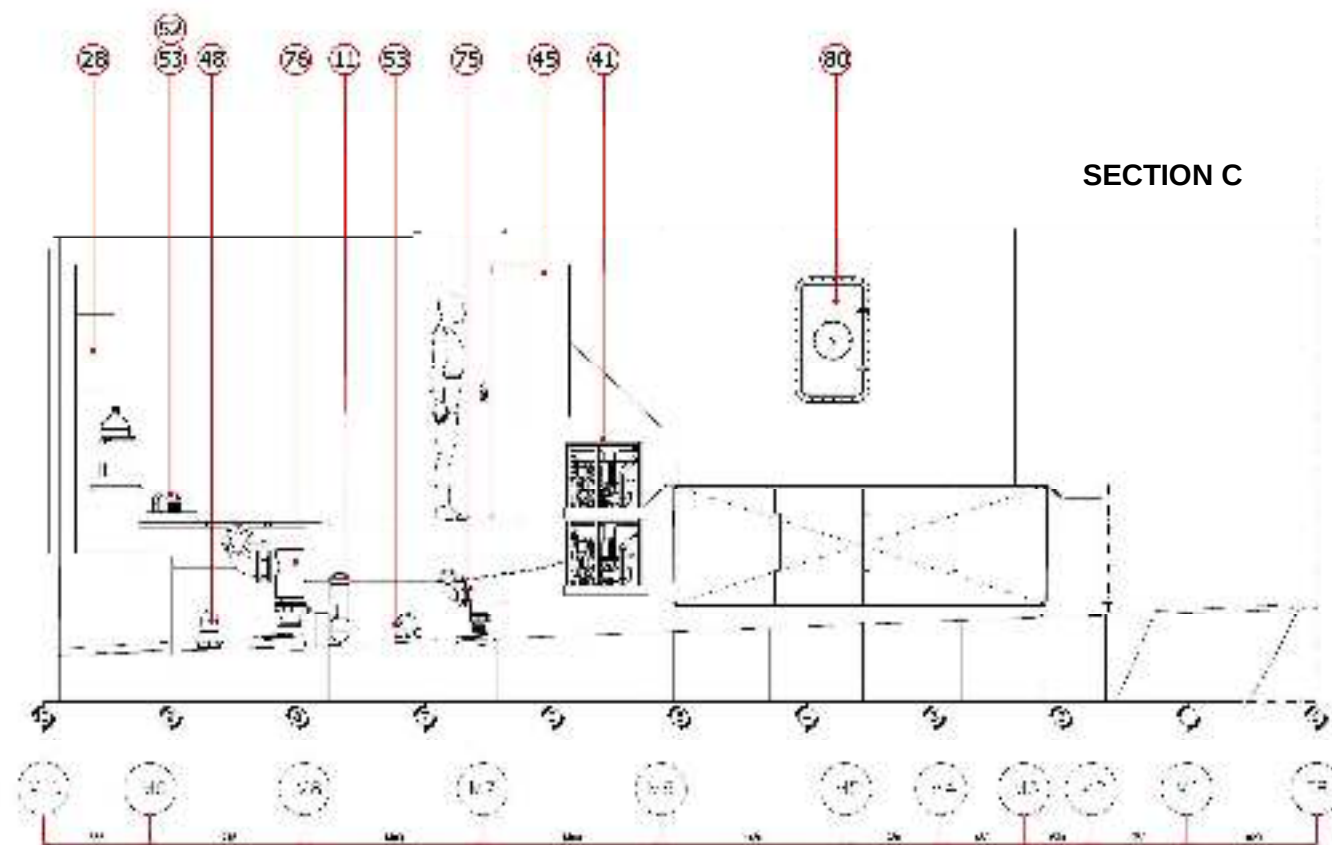
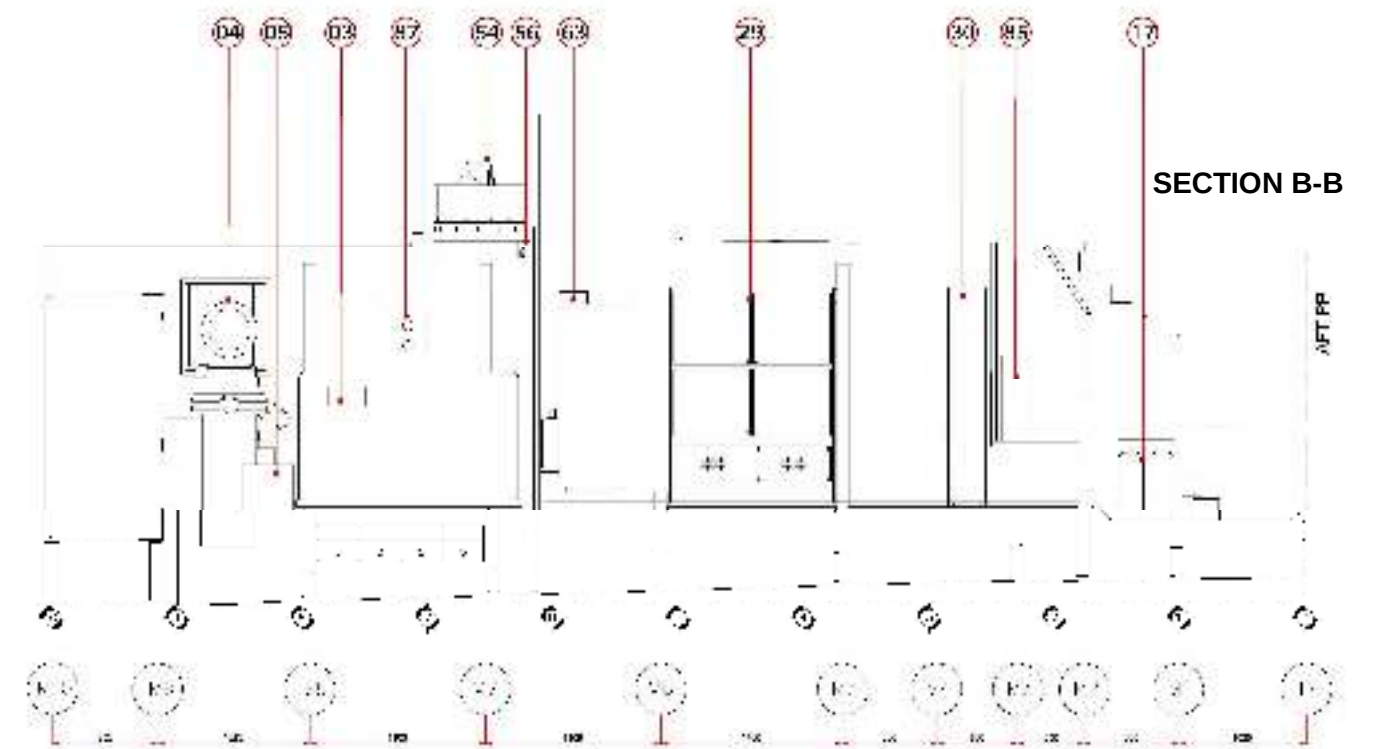
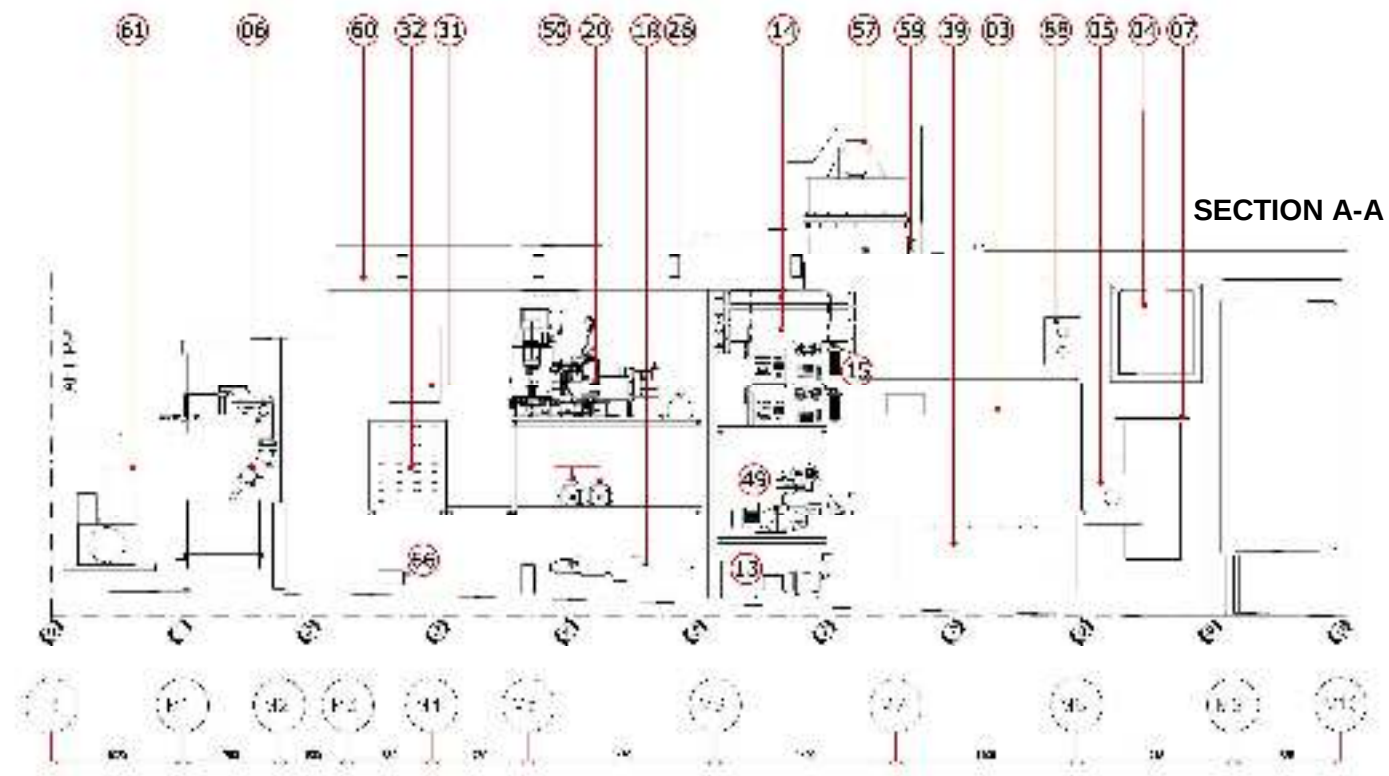
GENERAL PLANS – Engine Room – Floor view



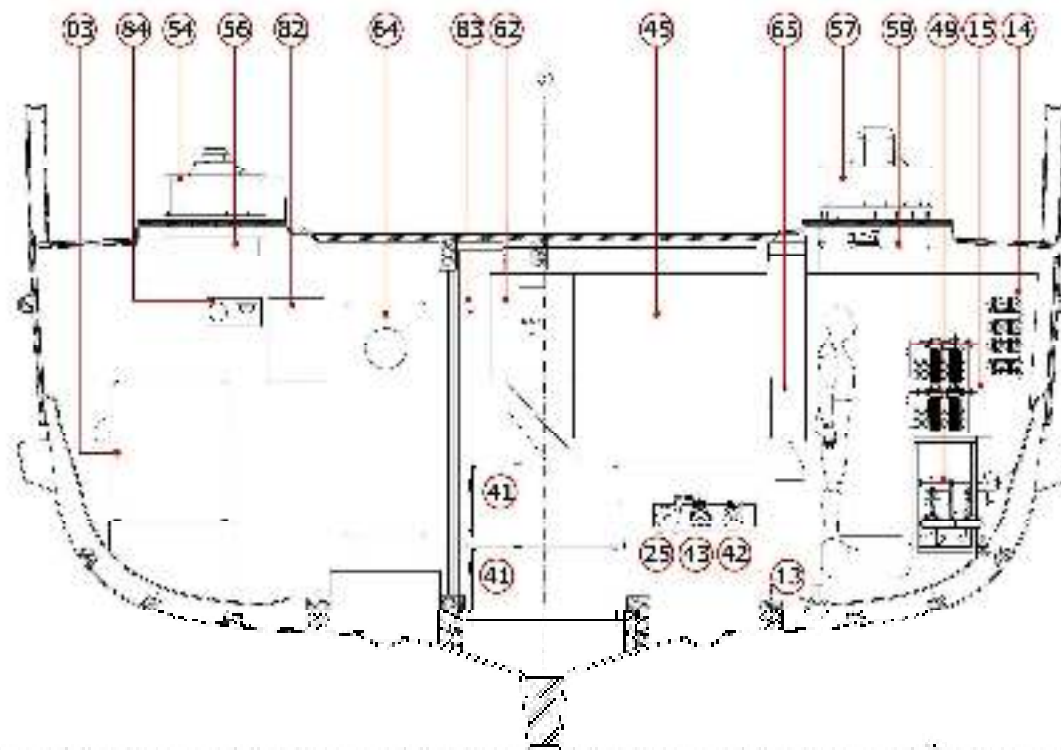
GENERAL PLANS - Engine Room - Ceiling view



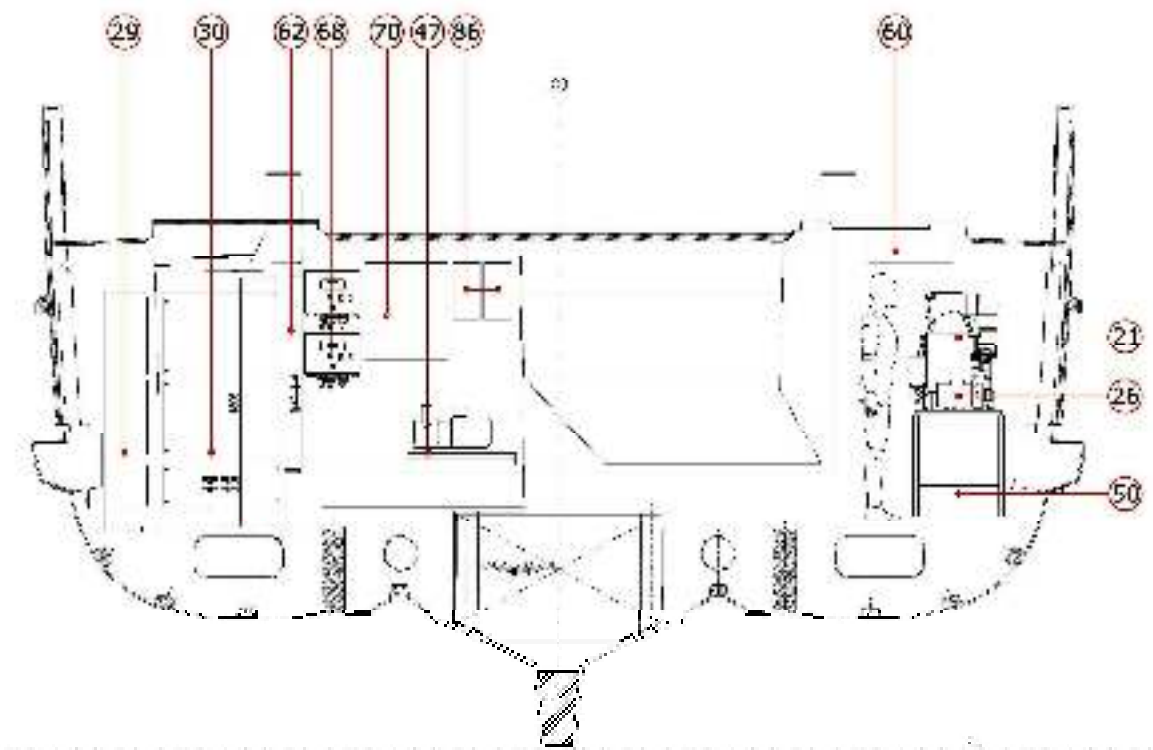
GENERAL PLANS – Engine Room – Longitudinal Section



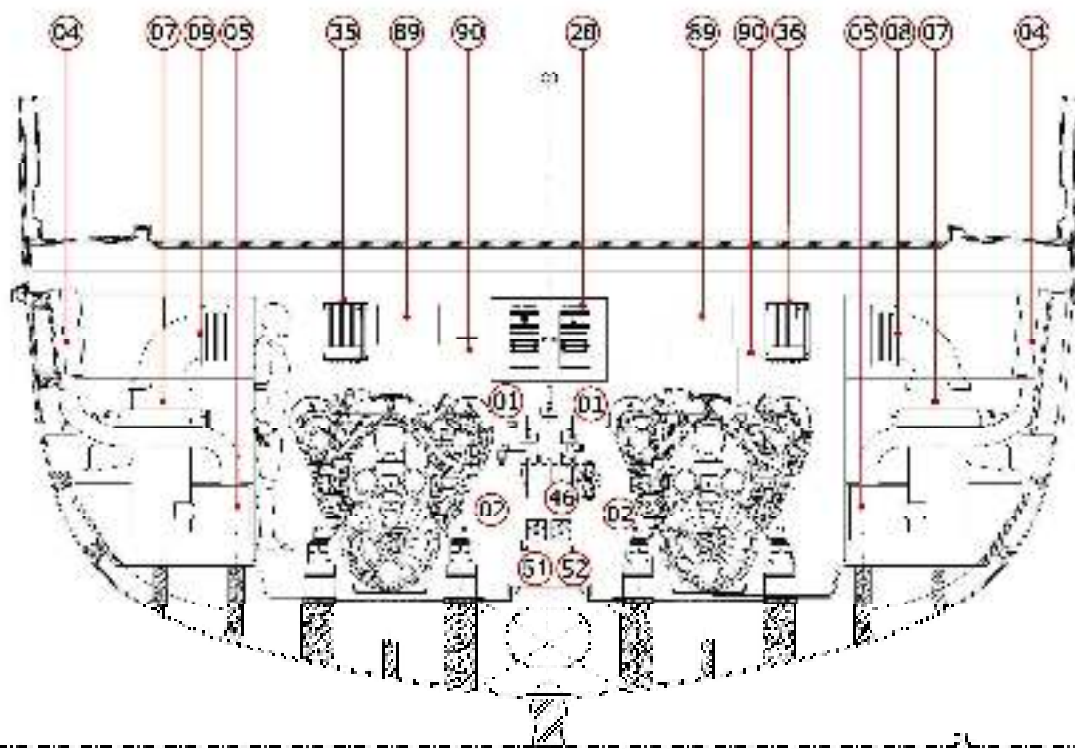
GENERAL PLANS - Engine Room - Sections



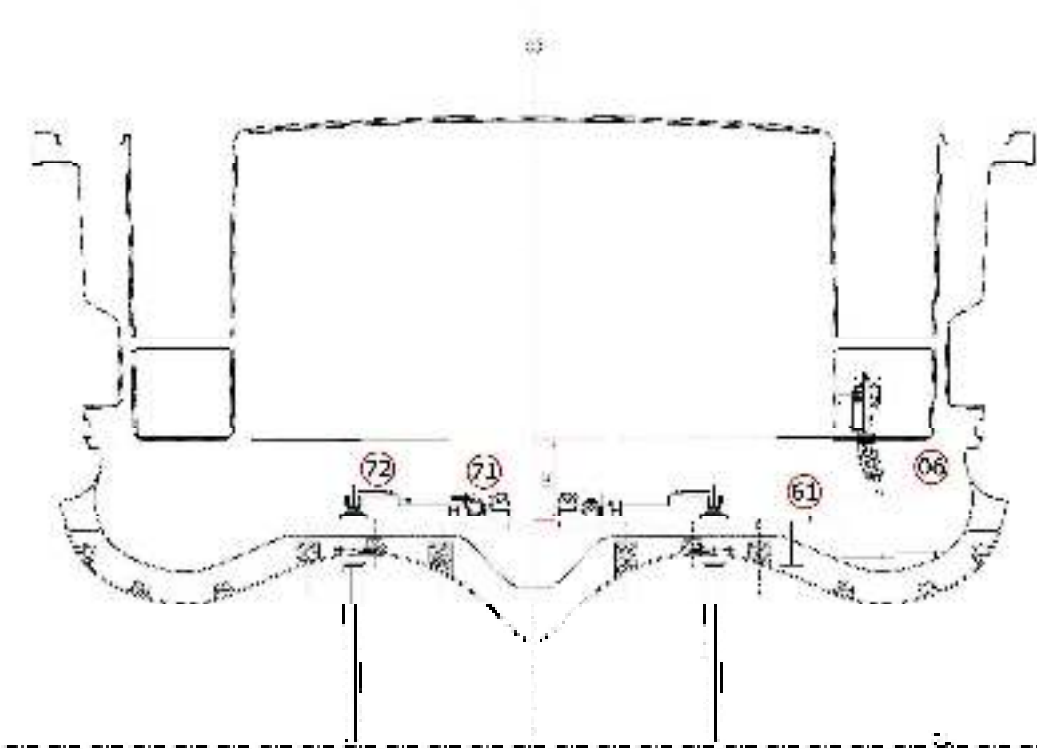
M6+910
LOOKING AFT
SEC. G-G



M6
LOOKING AFT
SEC. F-F



M10
LOOKING FWD
SEC. H-H



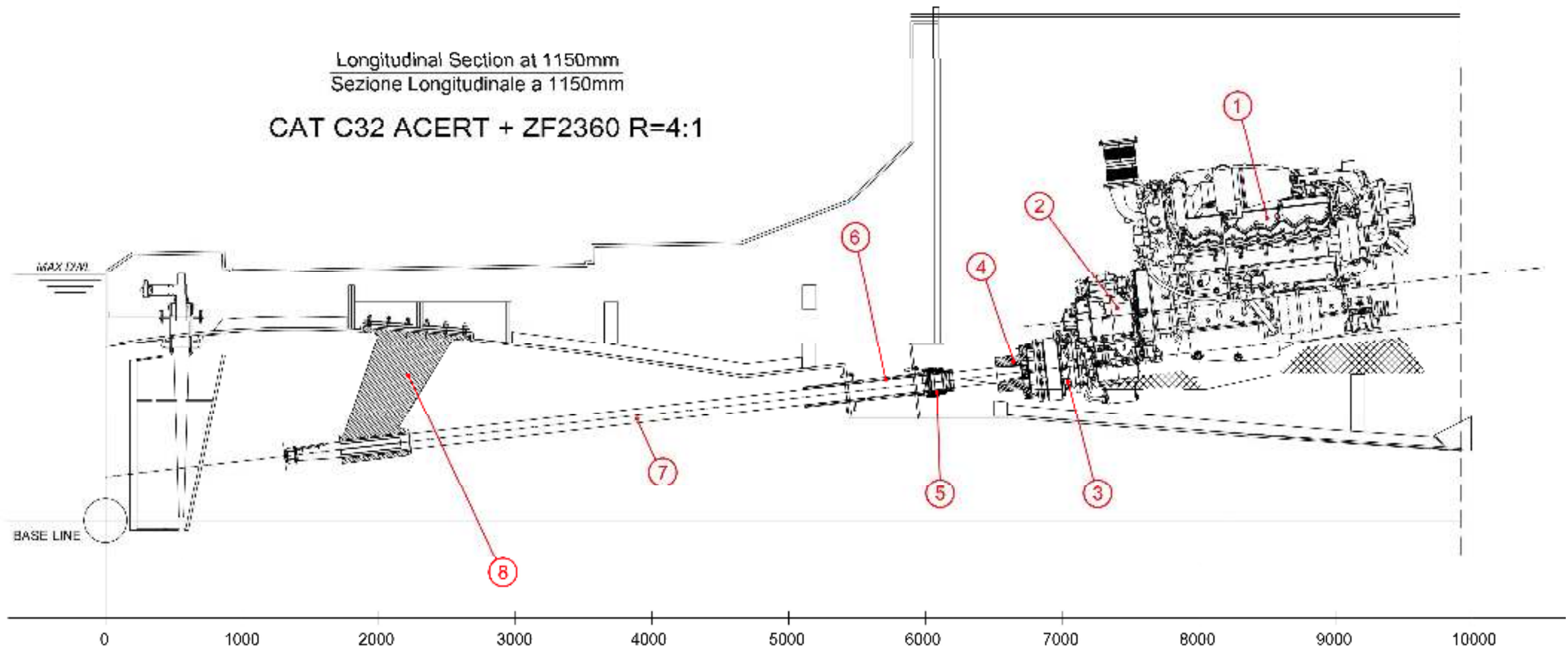
M1
LOOKING AFT
SEC. E-E

GENERAL PLANS – Engine Room – Legend

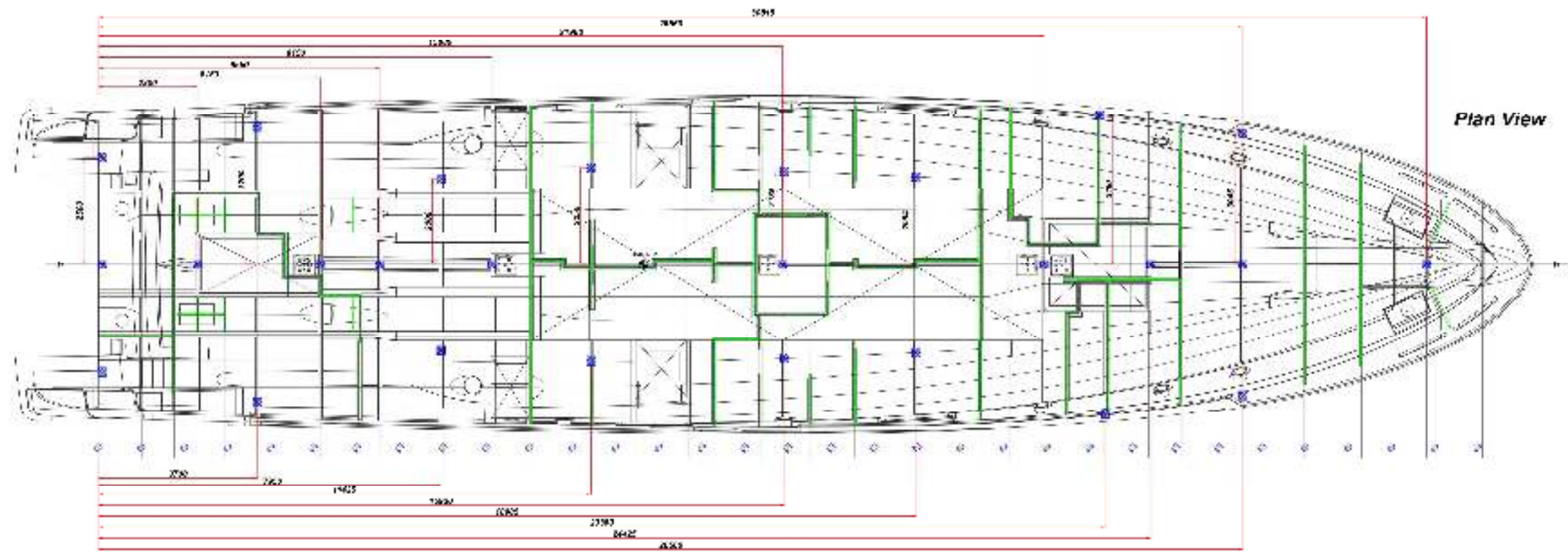
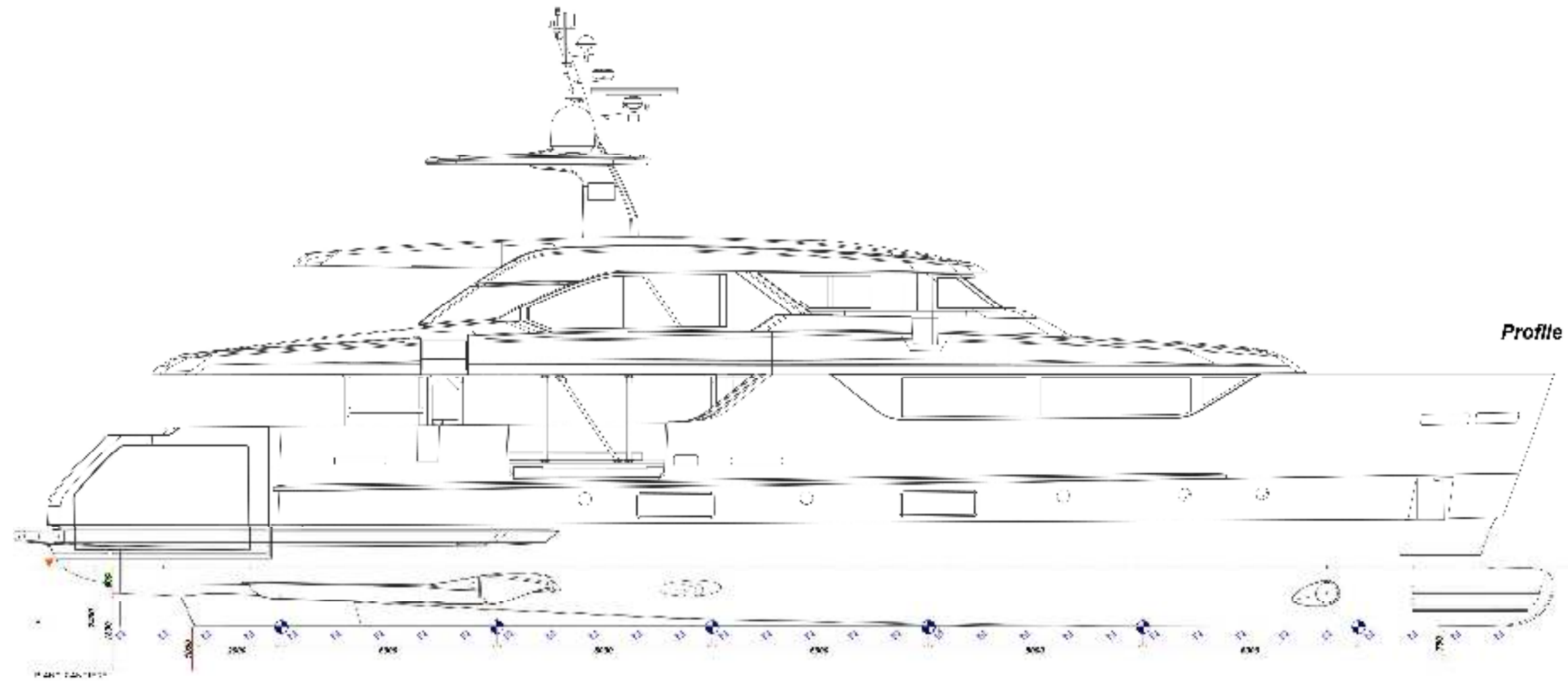
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90	TAVOLA IN LEGNO DI NATTRESSAMENTO CON 2000 LITRI DI TACCA	2	CAT
89	OPETROLIO IN VASO IN PIASTRA IN STALLER CONTROL UNIT	2	AVENTICS
08	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER B4 100% STUCCO IN STALLER CONTROL UNIT	1	B&B
87	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER B4 100% STUCCO IN STALLER CONTROL UNIT	1	B&B
86	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	2	CMC MARINE
85	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	CATF
84	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	B&B
83	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	B&B
82	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	ALFALAVAL
81	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	ALFALAVAL
80	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	GALISTRI
79	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	CMC
78	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	GIANNESCHI
77	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	SANLORENZO
76	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	2	F.LLI COCCA
75	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	2	F.LLI COCCA
74	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	SANLORENZO
72	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	2	CMC
71	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	2	CMC
70	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	CMC
69	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	2	CMC
68	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	2	CMC
67	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	KAHLENBERG
66	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	OSCOLATI
65	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	ATS
64	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	OFF D. RFI D
63	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	CATF
02	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	GALISTRI
61	STUCCO ANTIRIFLESSO ANTIRIFLESSO PER STUCCO ANTIRIFLESSO ANTIRIFLESSO PER	1	GALISTRI

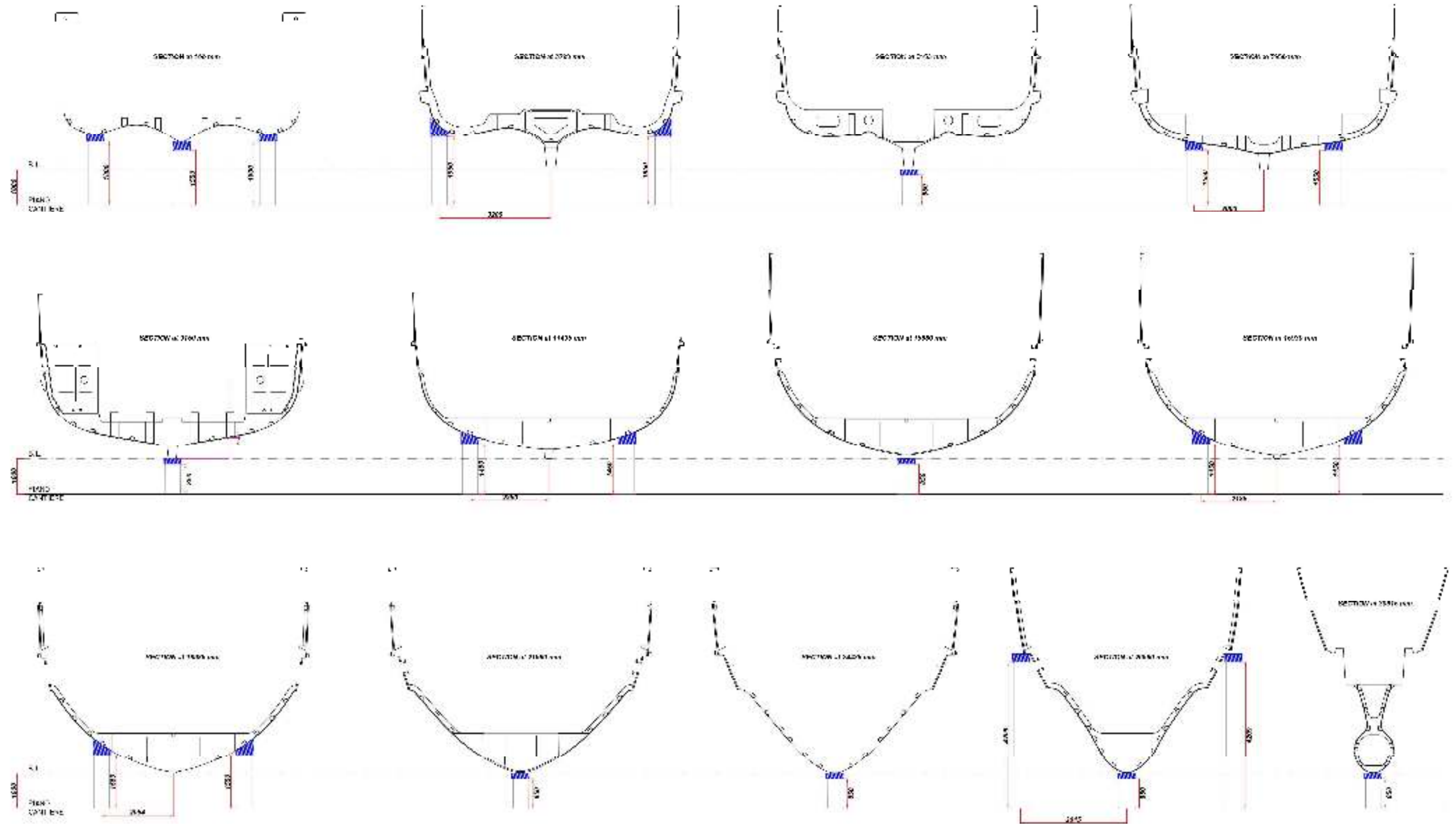
Shaft Line



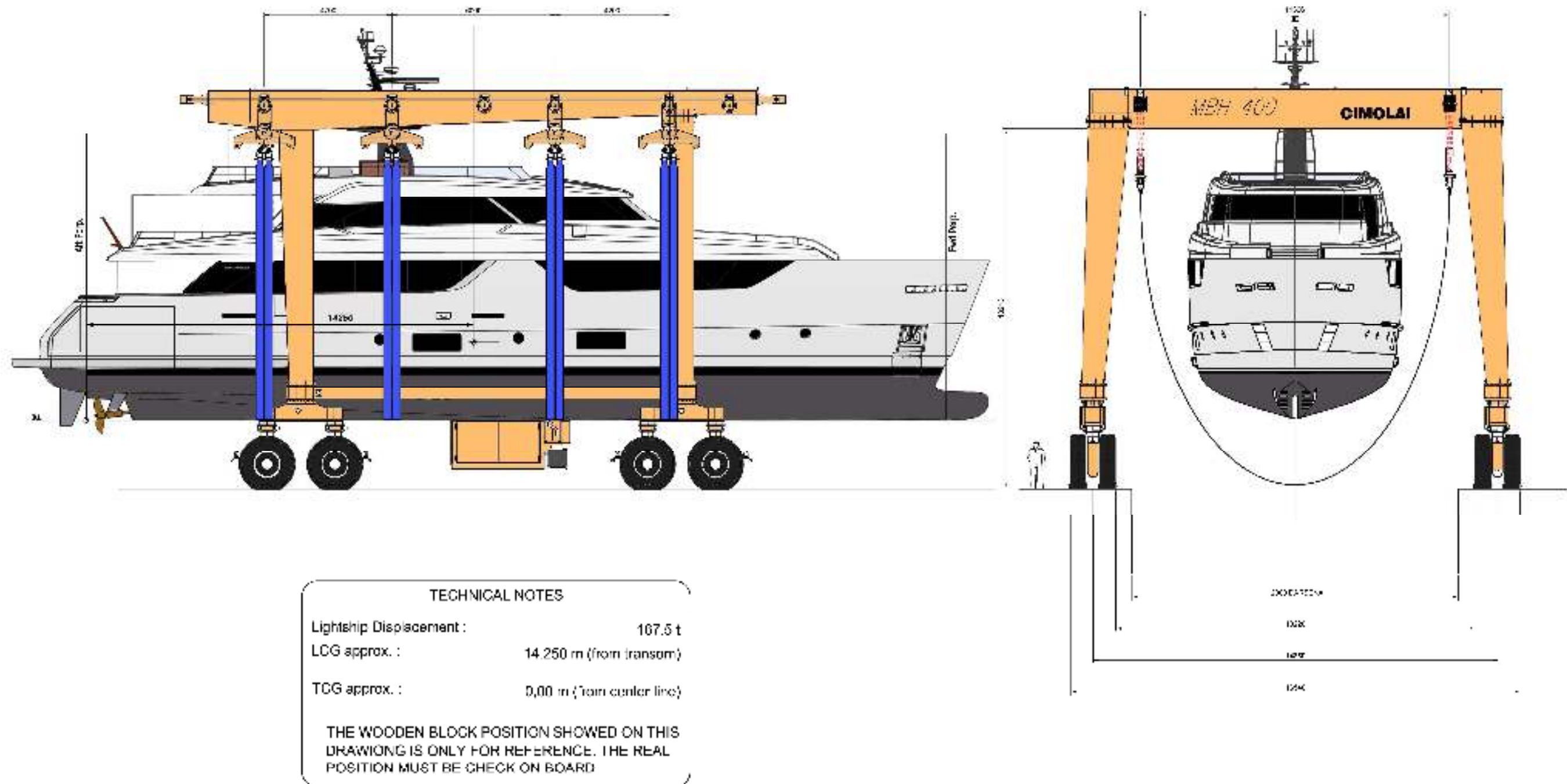
#	Q.	DE SCRIZIONE	DESCRIPTION	Fornitore & Supplier
①	2	Motore CAT C32 ACERT 1153 Kw@2300 rpm	Main Propulsion CAT C32 ACERT 1153 Kw@2300 rpm	CGT
②	2	Riduttore ZF 2360 R=4:1	Gearbox ZF 2360 R=4:1	CGT
③	2	Giunto Propflex	Joint Propflex	CGI
④	2	Mancione asse D.110 MAN016A.00	Gearbox Flange D.110 MAN016A.00	WORLDYACHTS
⑤	2	Tenuta meccanica MTM 908 D.110	Packing Gland MTM 908 D.110	WORLDYACHTS
⑥	2	Astuccio D.110 AST047.00	Shaft Case D.110 AST047.00	WORLDYACHTS
⑦	2	Asse elica D.110 L5475 mm AP052.00	Propeller Shaft D.110 L5475 mm AP052.00	WORLDYACHTS
⑧	2	Supporto asse D. 110	Shaft Bracket D. 110	WORLDYACHTS

Docking Plan

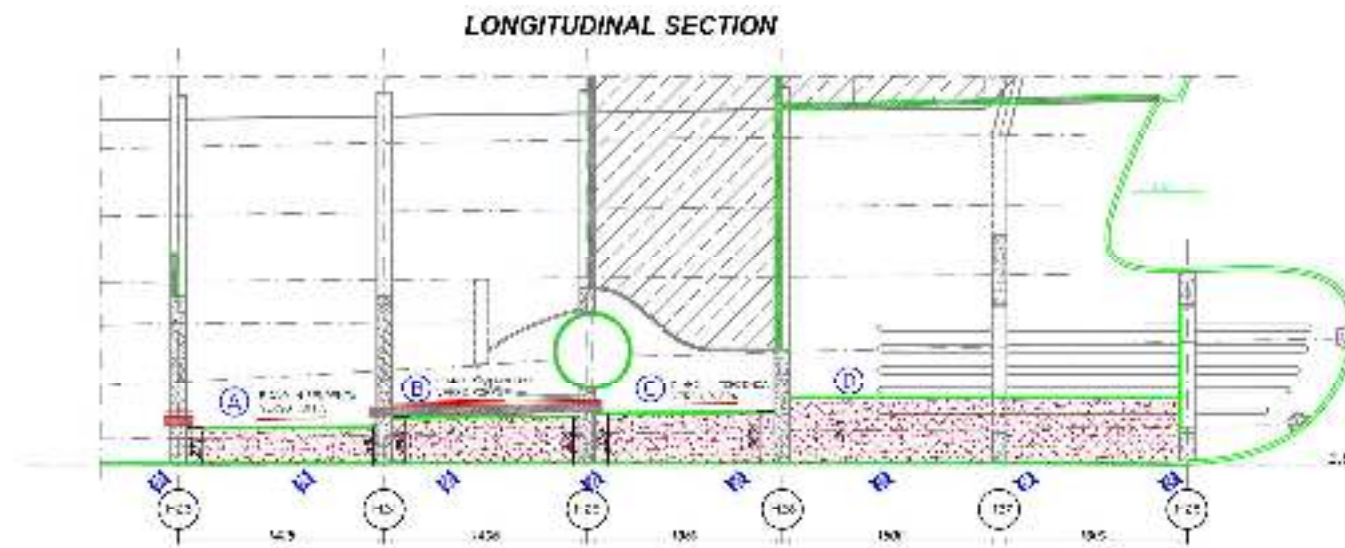
Docking Plan - Sections



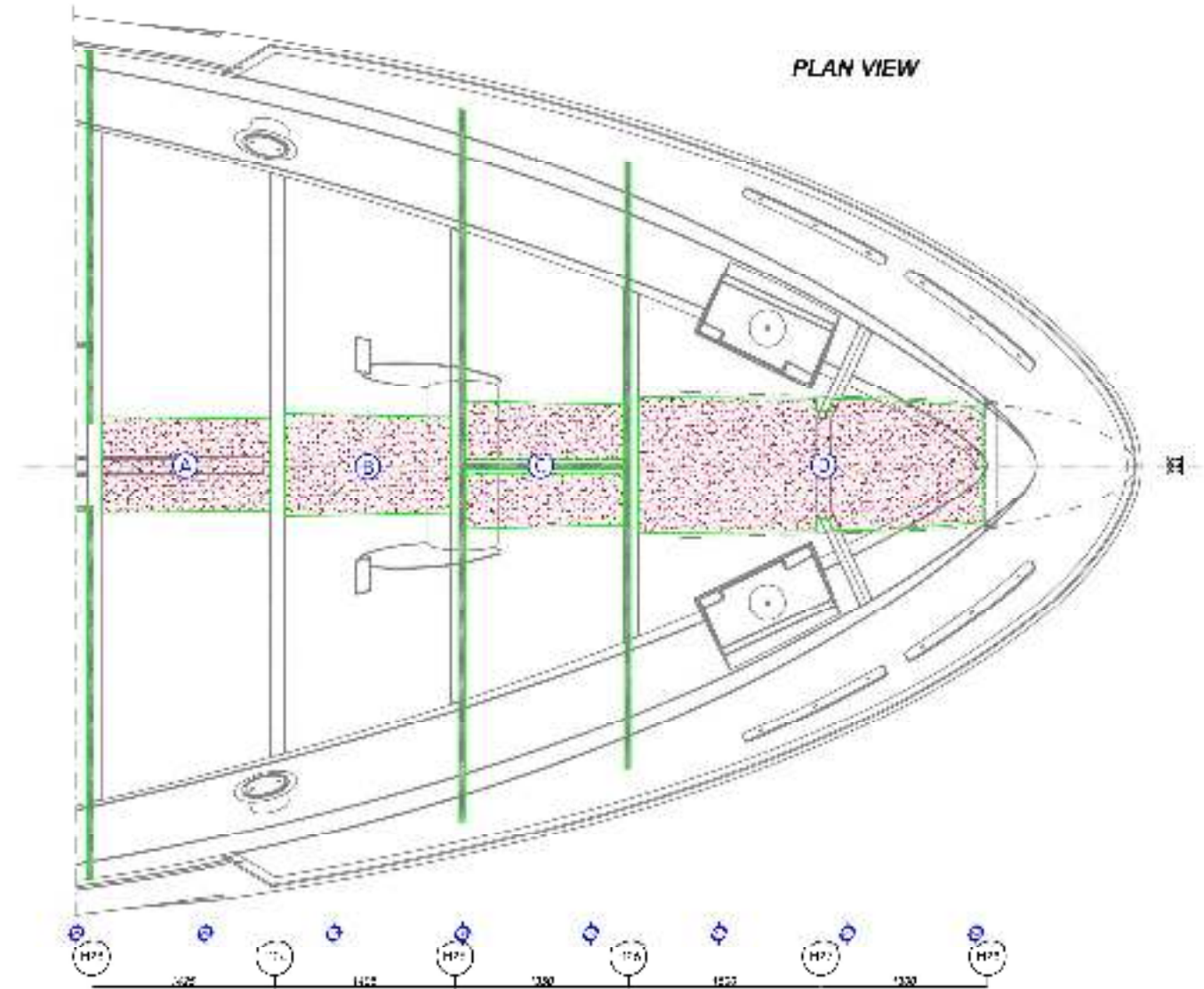
Lifting Plan



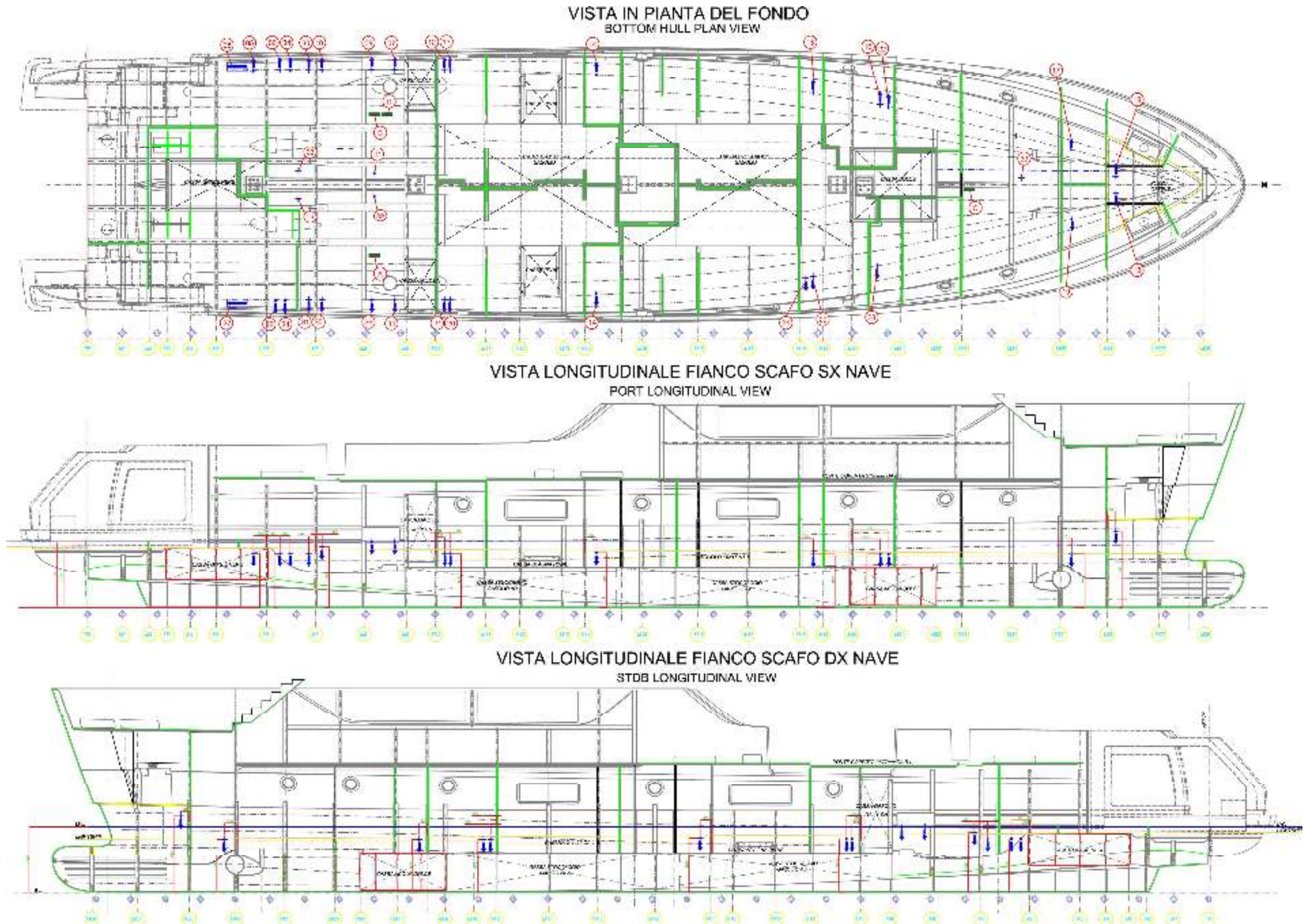
Ballast Plan



PIANO ZAVORRA SD118							
	Peso (kg)	Xg (m)	Yg (m) pos/ds	Zg (m)	Mom X (kg/m)	Mom Y (kg/m)	Mom Z (kg/m)
A	1920,00	26,84	0,00	0,17	36760,00	0,00	265,00
B	1920,00	27,25	0,00	0,17	40975,00	0,00	265,00
C	1920,00	28,63	0,00	0,20	42945,00	0,00	300,00
D	6520,00	30,67	0,00	0,30	129355,00	0,00	1362,00
TOTALE PESO	11000,00	28,27	0,00	0,25	321935,00	0,00	2092,00



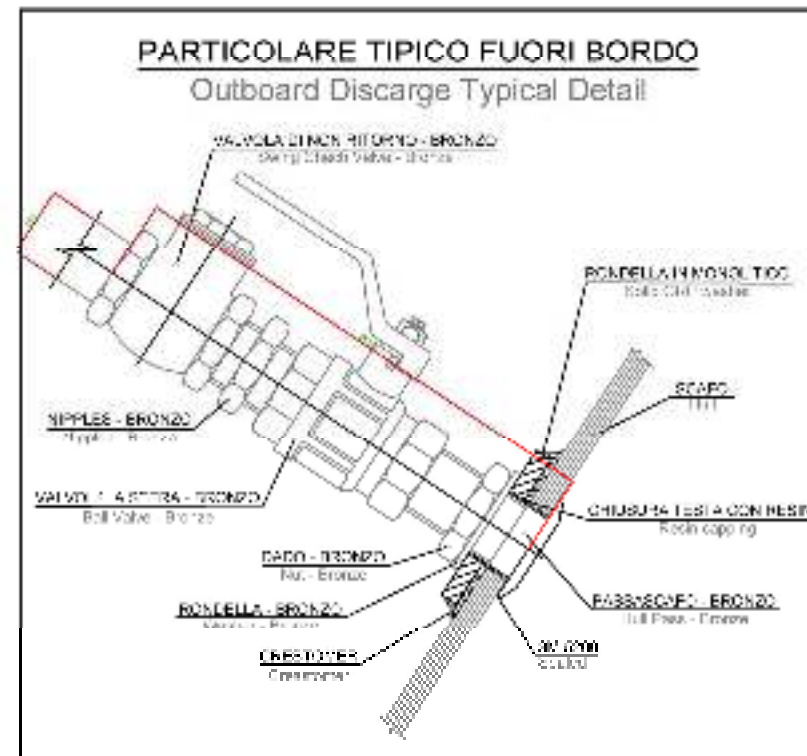
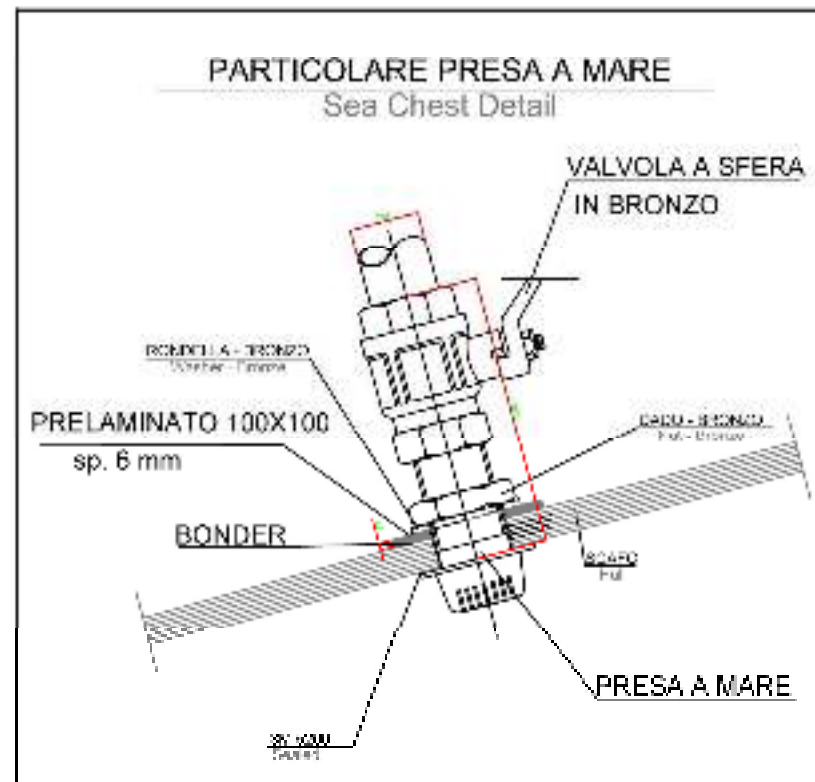
Hull Exhausts



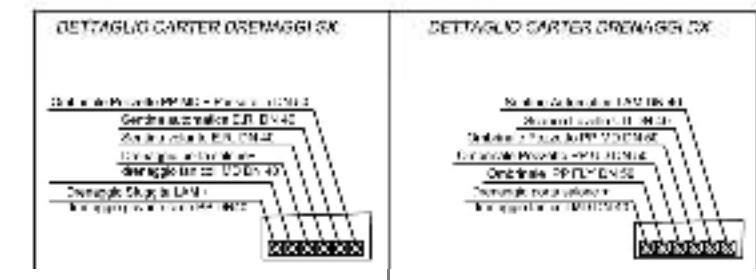
Hull Exhausts – Details

DESCRIZIONE - Description				DESCRIZIONE - Description				DESCRIZIONE - Description			
ITEM	IMPIANTO - System	PISCAPPO - Hull Connection	Q.TA'-Q.Ty	ITEM	IMPIANTO - System	PISCAPPO - Hull Connection	Q.TA'-Q.Ty	ITEM	IMPIANTO - System	PISCAPPO - Hull Connection	Q.TA'-Q.Ty
01-02	PRESSA MARE WATER PRESSURE GAGE	DN50 - (BRONZO - BRONZE)	2	17	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	34	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1
03-04	PRESSA MARE WATER PRESSURE GAGE	DN50 - (BRONZO - BRONZE)	2	18	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	2				
05	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	19	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
06	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	20	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
07-11	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	2	21	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
08-27	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	2	22	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
09-26	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	2	23	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
10-25	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	2	24	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
12	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	28	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
13	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	29	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				
14	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	30-31	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	2				
16	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	32	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	2				
18	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1	33	SCARICO F.E. OVERBOARD	DN50 - (BRONZO - BRONZE)	1				

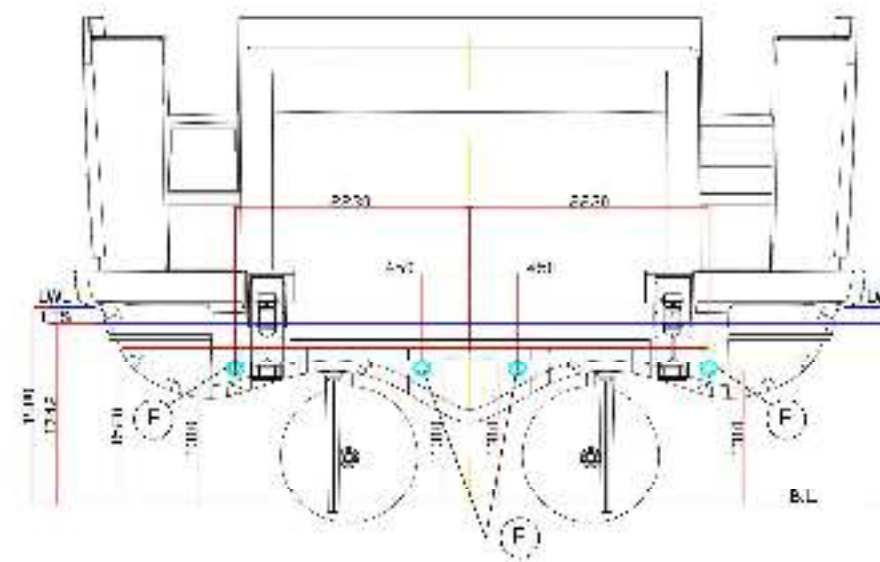
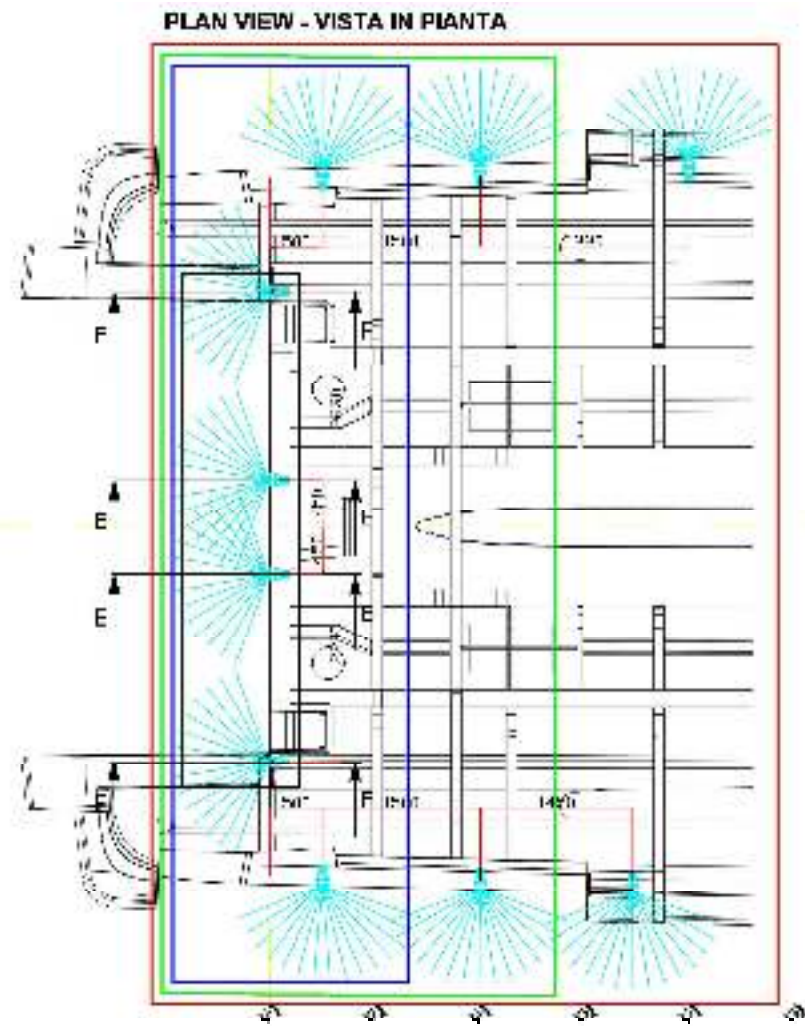
DESCRIZIONE - Description			
ITEM	IMPIANTO - System	PISCAPPO - Hull Connection	Q.TA'-Q.Ty
A	PASTIGIA POROSA-100V	DN50 - (BRONZO - BRONZE)	1
B	PASTIGIA POROSA-ELECTRONIC	DN50 - (BRONZO - BRONZE)	1
C	PASTIGIA POROSA-ELECTRONIC DI BORE	DN50 - (BRONZO - BRONZE)	1
D	PASTIGIA POROSA-PARAFUMINE	DN50 - (BRONZO - BRONZE)	1



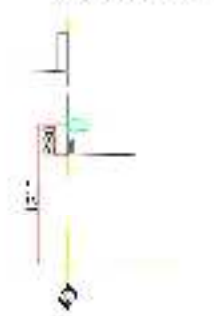
- | NOTE GENERALI / General notes | |
|--------------------------------------|---|
| 1) | DO NOT DISPLACE A SEAL OR BARRIER LAYER DURING CONSTRUCTION OF THE THRU-HULL WILL BE COVERED WITH LIQUID RESIN. |
| 2) | SIGILLARE IL PASSAGGIO A SCARICO IN CANTO
THE THRU-HULL WILL BE SEALED WITH AN EPOXY |
| 3) | L'INTERFACCIA SOTTO IL SCUDO IN FIBRA DI VETRO DEVE ESSERE REFORZATA CON UN COMPOSITO IN CARBONIO O Kevlar. |
| 4) | LE QUOTE INDICANO CONTAVERGHE A BORDO
PLEASE LISTEN UPON WITH A RAIL TO BE USED FOR OVERLAP |



Anodes and Underwater Lights Arrangement



SECTION E-E



SECTION F-F



SECTION A-A



SECTION B-B



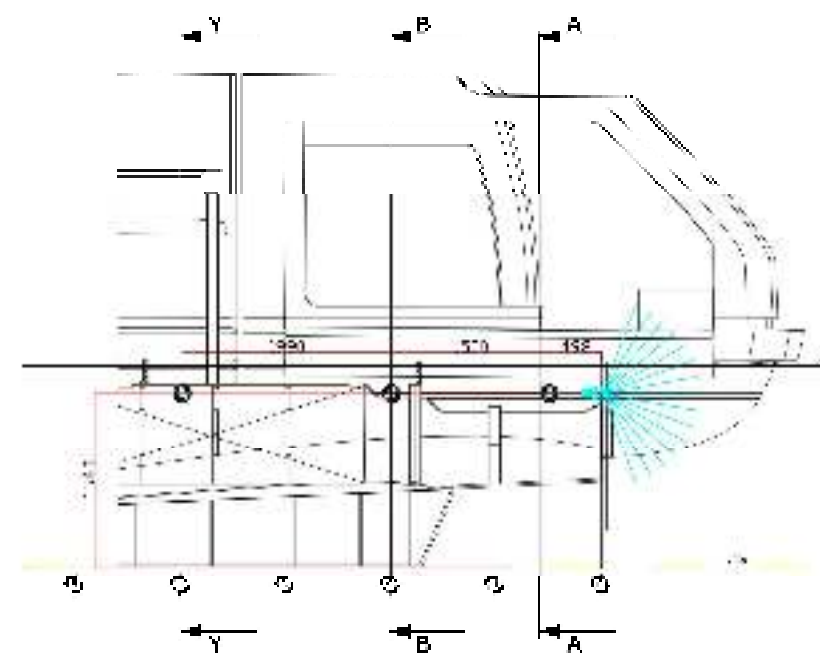
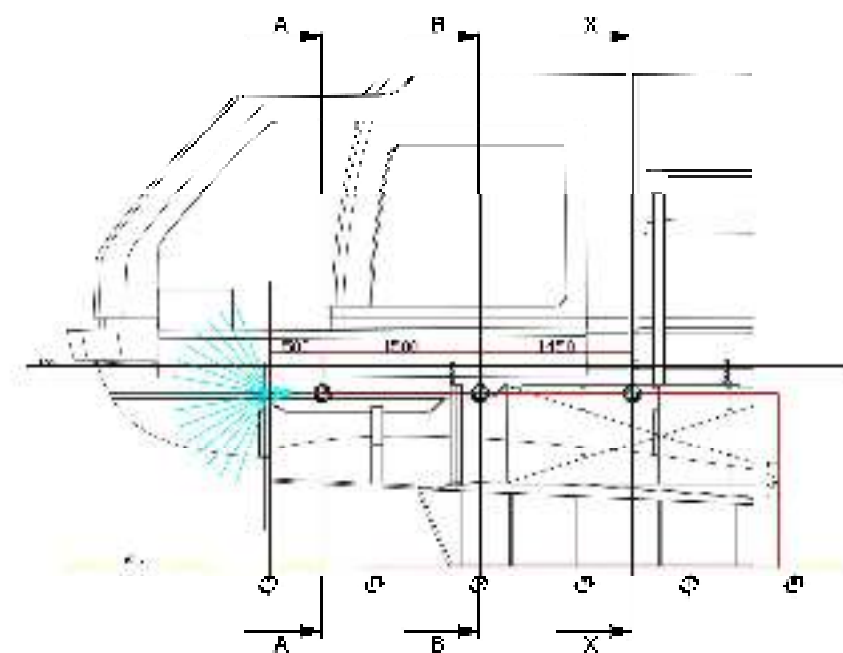
SECTION X-X



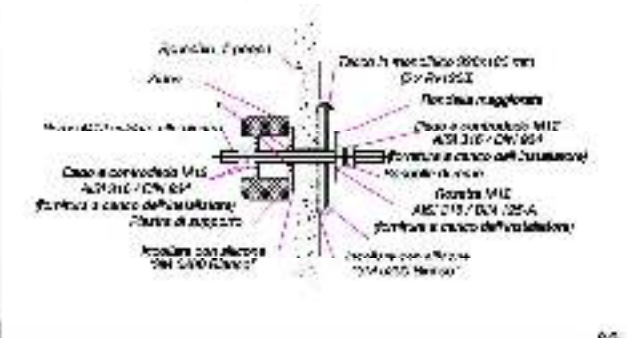
SECTION Y-Y



VEDI PIANTA PER ARRE COLORATE	Q. LUCI DA INSTALLARE
— LUCE SUB OCEANICO	4
— LUCE SUB OCEANICO	6
— LUCE SUB OCEANICO	8
— LUCE SUB OCEANICO	10



DETAGLIO FISSAGGIO ZINCHI
mod.FC-6 completo cod. art. 03070007



Section **B**

OPERATION

Main navigation rules
Vessel steering rule
Navigation with reduced visibility
Navigation in bad weather conditions
Instructions and checks
General information
Navigation set up procedures
Check to be carried out during and after navigation
Main engines start up
Main engines stop
Steering system
Generators start up
Electric power supply
Radio communications – General information
Emergency radio calls (VHF – FM-8900S Furuno)
Mooring and unmooring
Anchoring
Unattended mooring
Leaving the mooring
Safety equipments
Fire prevention rules
Collision
"Man overboard" recovery
Running aground
Exhaust gas emissions
Towing in case of failure
Environmental pollution prevention rules
Risk areas
Transom equipment handling

Main navigation rules

This paragraph is only a brief description of the main navigation rules.

As this manual does not contain all rules, it is advisable to acquire an adequate knowledge before taking control of the yacht.

The two groups of rules concerning navigation are:

National Rules: these rules regulate navigation in inland waters; each country has its own regulations.

International Rules: these rules are applied to all yachts cruising in open sea and in waters navigable by open sea yachts.

Only international rules are mentioned in this manual, therefore it is advisable that you will contact the dealer for reading eventual applicable local laws and regulations. As not all situations are provided for always use common sense.

Main definitions

It is extremely important to understand the following definitions concerning navigation.

Vessel:

the term “vessel” is referred to any floating mean, included seaplanes, used or able to be used as a mean of transport on the water.

Vessel underway:

a vessel is underway when is neither anchored, moored nor aground.

Mechanical propulsion vessel:

any engine-powered vessel.

Cruising vessel:

the term is referred to all vessels provided with sails, on condition that the engine, if present, is not used (not operating).

Fishing vessel:

the term is referred to all vessels which are undergoing fishing operations with nets, ropes, lines or other equipment that may reduce manoeuvrability, but is not referred to vessels involved in fishing with lines or other equipment that do not limit manoeuvrability.

Vessel not under control:

the term is referred to all vessels that, in exceptional conditions, are not able to manoeuvre according to regulations, and therefore they are not able to maintain the correct distance from another vessel.

Vessel with reduced manoeuvrability:

the term is referred to all vessels which, due to their activity, cannot manoeuvre freely, according to regulations stated in this Manual, and therefore they are not able to maintain the correct distance from another vessel.

Vessel steering rule

Safety speed:

all vessels must always move at a safety speed in order to allow rapid intervention for avoiding an eventual collision and for stopping at a certain distance, according to circumstances and conditions.

Risk of collision:

the vessels must use all their means available, adequate to the circumstances and conditions, for defining a risk of collision. In case of doubt, the risk must be considered existent.

Vessel in sight

There are three modes of sighting another vessel at sea:

- *approach*;
- *crossing*;
- *overtaking*.

In general, the vessel with reduced manoeuvrability has right of way.
Leave her course free and pass a stern.

The vessel that has right of way is referred as ***privileged vessel***.
This vessel can keep her speed and course.

The ***penalized vessel*** is the vessel that must control her speed and/or course in order to maintain a correct distance from the privileged vessel.

Approach

When approaching another vessel that is cruising in parallel direction, both vessels must control their speed and maintain the course.

When two mechanical propulsion vessels are approaching on intersecting or nearly-intersecting courses, so as to create a risk of collision, each vessel must change course towards starboard, in order to pass on the port side of the other vessel.

Crossing

When two mechanical propulsion vessels are crossing, so as to create a risk of collision, the vessel that sees the other vessel on her starboard side must manoeuvre and, if possible, avoiding to pass at the bow of the other vessel.

Overtaking

A vessel is overtaking another one when coming from a direction greater than 225 degrees abaft the other, so she can see only the stern light of the other vessel, and no side lights.

When you are overtaking a slower vessel that is cruising along your course, you are the penalized vessel.

Carry out any operation in order to avoid collision and overtake on port or starboard side. Transmit your intentions activating the horn twice if you will overtake on port side, or once if you will overtake on starboard side.

Responsibility between vessels

It is necessary to respect the following regulations, if not in conflict with other rules:

Mechanical propulsion vessel, when underway, must give way to:

- ✓ vessels that are not able to steer;
- ✓ vessels with reduced manoeuvrability;
- ✓ fishing vessels;
- ✓ cruising vessels.

Cruising vessels, when underway, must give way to:

- ✓ vessels that are not able to steer;
- ✓ vessels with reduced manoeuvrability;
- ✓ fishing vessels.

Fishing vessels when underway, must give way as much as possible to:

- ✓ vessels that are not able to steer;
- ✓ vessels with reduced manoeuvrability.



|| Even if you have right of way, you are not exempt for avoiding a collision.

Navigation lights

Rules concerning navigation lights must be respected between sunset and sunrise, and during this period no other lights must be visible, except those that cannot be exchanged for the lights specified in this manual. Although lighting is provided by the manufacturer, the Owner/Captain are responsible for respecting the local rules.

Navigation lights

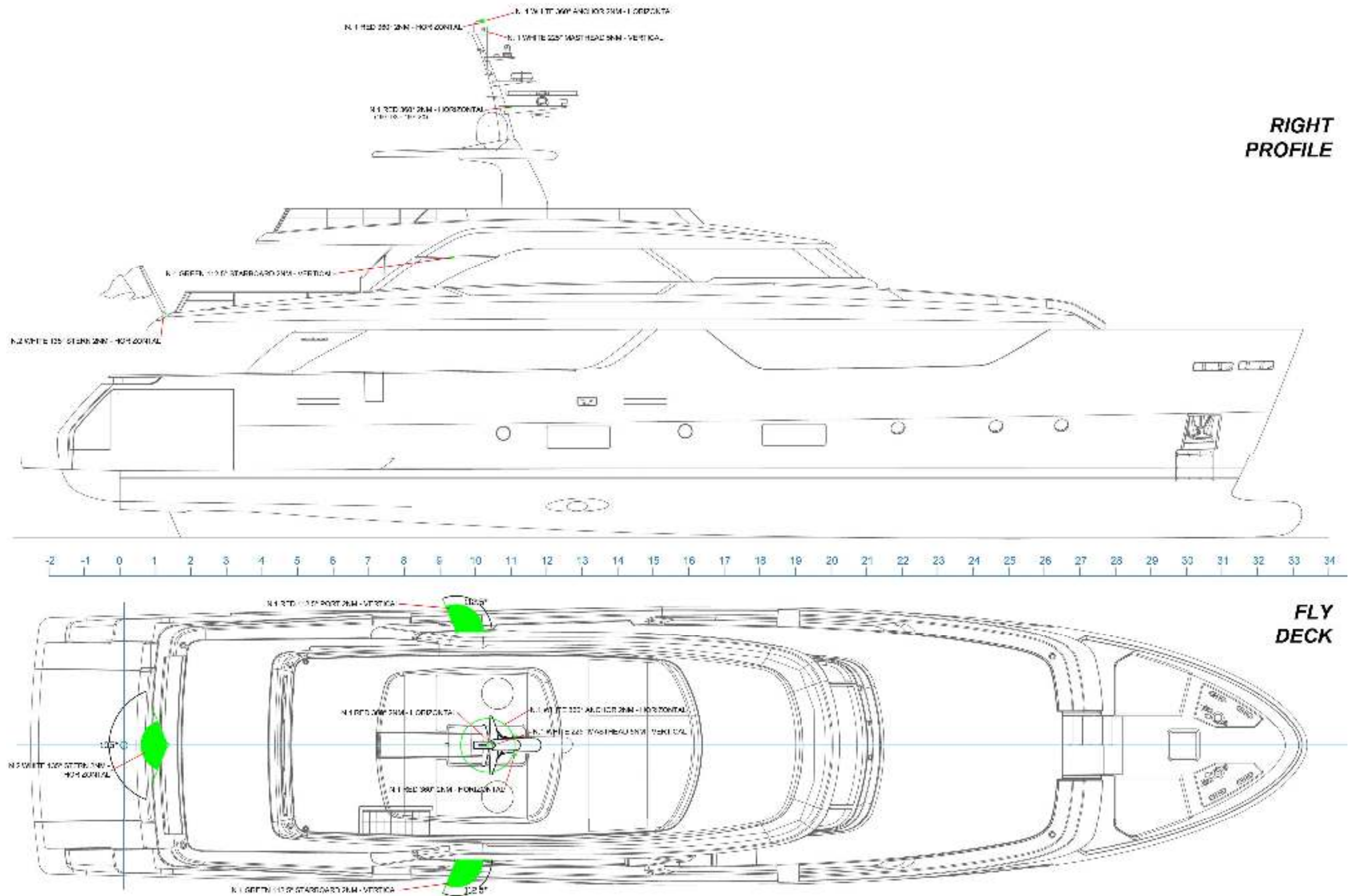
- ***Masthead light:***
white light located on the longitudinal axis of the vessel, visible from bow, to port and starboard sides (fix light with a luminous sector of 225°).
- ***Light visible around the whole horizon:***
white light located on the mast, visible from any direction (fix light with a luminous sector of 360°).
- ***Side lights:***
a red light on port side and a green light on starboard side (fix lights with a luminous sector of 112.5°). Vessels shorter than 20 metres can have the side lights grouped in a single light located on the centerline, from bow to stern.
- ***Stern light:***
white light, located on the stern, visible from abaft (fix light with a luminous sector of 135°).
- ***Anchor light:***
white light visible from any direction.

Remember that national rules concerning navigation lights can vary slightly from international rules. Therefore you should gather information concerning local regulations.

Night navigation requires more care.

All rules are applicable, but independently from who has right of way, it is advisable to reduce the speed and maintain a safe distance from other vessels. It is advisable to remember that intense lights reduce visual capacity at night.

Navigation Lights



Navigation with reduced visibility

Avoid going underway in case of fog; if the visibility is reduced due to fog, mist or rain, a vessel underway must reduce her speed, stay on the look-out and send a warning signal by activating the horn or the whistle, emitting a prolonged sound (approximately 5 seconds) every two minutes.

Maintain your course and speed.

Wear floating safety equipment.

Incite the passengers to stay on the look-out.

A vessel moored in a reduced visibility area must activate the horn for five seconds, at intervals of one minute

If you hear a fog warning signal at the bow or stern, slow down or stop the vessel, then proceed with great care until the risk of collision has ended.

International regulations for preventing collisions at sea (COLREG 1972)

The pneumatic whistle (horn) installed onboard the yacht, responds adequately to requirements of International Regulations for Preventing Collisions at Sea (COLREG 72).

An extract of the International Regulations for Preventing Collisions at Sea is shown herein after.

Application

(Rule no.1) – These Rules shall apply to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels.

Responsibility

(Rule no.2) – Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules.

Definitions

(Rule no.32)



the term "short blast" means a blast of about one second's duration



the term "prolonged blast" means a blast of from four to six seconds' duration

Manoeuvring and warning signals

(Rule no. 34)

■	one short blast to mean "I am altering my course to starboard"
■ ■	two short blasts to mean "I am altering my course to port"
■ ■ ■	three short blasts to mean "I am operating astern propulsion"
■ ■ ■ ■	two prolonged blasts followed by one short blast to mean "I intend to overtake you on your starboard side"
■ ■ ■ ■ ■	two prolonged blasts followed by two short blasts to mean "I intend to overtake you on your port side"
■ ■ ■ ■ ■ ■	the vessel about to be overtaken shall indicate her agreement with one prolonged, one short, one prolonged and one short blast
■ ■ ■ ■ ■ ■ ■	five short and rapid blasts "I have doubts concerning the manoeuvre"
■ ■ ■ ■ ■ ■ ■ ■	one prolonged blast "vessel approaching a channel bend"
■ ■ ■ ■ ■ ■ ■ ■ ■	one prolonged blast "vessel responding to the previous signal"

Signals in restricted visibility

(Rule no.35 and no.37)

■ ■ ■ ■ ■	one prolonged blast at intervals of two minutes "power-driven vessel making way through the water"
■ ■ ■ ■ ■ ■ ■ ■	two prolonged blasts in succession with an interval of two seconds between them, repeated every two minutes "power-driven vessel underway but stopped and making no way through the water"
■ ■ ■ ■ ■ ■ ■ ■ ■	one prolonged blast and two short blasts in succession, at intervals of two minutes "vessel not under command, a vessel restricted in her ability to manoeuvre or a vessel engaged in towing"
■ ■ ■ ■ ■ ■ ■ ■ ■ ■	one prolonged blast and three short blasts in succession, at intervals of two minutes "last vessel of the tow, responding to the signal made by the towing vessel"
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■	one short, one prolonged and one short blast "vessel at anchor giving warning of her position and of the possibility of collision to an approaching vessel"
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■	five seconds of continuous blast, at intervals of one minute "vessel at anchor, signaling her position"
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■	three brief blasts in rapid succession "vessel aground"
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■	four brief blasts "pilot vessel engaged on pilotage duty"
■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■	one continuous blast "danger and request of aid"

Navigation in bad weather conditions

Your **SD 118** yacht is designed to be safe and comfortable, both with calm sea and in bad weather. In any case, the navigation safety depends mainly on the Captain, who should either not set off or reduce the speed, sometimes considerably, run the boat with appropriate attitude and choosing the most favourable route.

During a navigation in bad weather, it is important to make sure that all fittings, doors, drawers and any other part which may move after yacht's sudden movements are carefully stored and fixed, to avoid damages and injuries.

Machinery reliability, ensured by perfect maintenance, before-cruise inspections, skilled Captain's presence, is highly important during bad weather.

Above mentioned precautions should also be applied for night navigation, which must be considered as particular navigation.

SANLORENZO S.p.A. shipyard has always paid maximum attention to components which ensure navigation safety, both during design and during hull construction, by means of innovative technical methodologies and long experience in the nautical field, such as:

- hull appropriate shape;
- propulsion engine with suitable free power;
- rudders and stabilising fins efficiency;
- high stability of the yacht;
- correctly designed and carefully manufactured structures;
- particular attention must be paid for all loading conditions;
- easily reachable safety equipment;
- adequate navigation and reconnaissance instruments (radar, autopilot, GPS, depth-sounder, etc.).



Before setting off, it is necessary to be aware of the sea and weather conditions you will find along the transfer route and in the area you want to reach.



While at sea, all doors and hatches on deck giving access to below-deck accommodation must be kept closed.



In case of flooding, try to stop the incoming water by using a wooden plug and/or cloths, if possible from the exterior.



When underway, keep the rear aft door normally closed, in order to prevent that exhaust gas and water splashes get into the interior compartments in particular wind direction and sea conditions. These precautions will improve comfort and silence inside the yacht.



This Yacht has been designed and built in compliance with the Relevant Classification Society Rules and is in all respect seaworthy and capable of navigation without restrictions.

The Builder however warns the Buyer about the dangers and difficulties of navigation in adverse sea and weather conditions.

Such navigation must be carefully afforded by the Yacht's Captain who will be solely and exclusively responsible for leaving any port or shelter notwithstanding adverse sea and weather conditions forecast or adverse sea and weather conditions in act and for adopting the appropriate speed, engine revolutions and course.

The Builder declines any responsibility for improper use and navigation of the Yacht in adverse sea and weather conditions.



SANLORENZO S.p.A. shipyard declines any responsibility for the improper use of the boat, in relation to the weather conditions.

Instructions and checks

General information

This section of the Manual describes how to behave and operate in various situation which may occur during navigation.

Some rules should be always kept in mind, to ensure a safe operation of the Yacht.

- Check that the individual and collective rescue equipment are available and in good conditions.
- Keep a safe distance when anchoring.
- Check that all the boat safety devices are in good conditions and no maintenance activity is overdue.
- In case of the fire fighting system operation do not ventilate the engine room, until the fire has been completely extinguished and specific checks have been carried out (smoke reduction, temperature of surrounding surfaces is not rising, etc.). Carefully carry out a visual inspection through the watertight engine room access door. If the inspection gives a positive result, ventilate the room by means of electric fans and electric extractors, before entering.
- Ventilate the below deck compartment before entering, if portable extinguishers have been used.
- When operating the gangway, be sure that nobody is in the way, and do not allow access until the handrails have been fixed.
- Oils, used filters, emulsions, coolants, electrolytes are all harmful products: avoid contacts with the skin and do not dispose of them in the environment.
- In engine room, be cautious with hot and moving parts.
- When entering the engine room, wear individual protection means such as earmuffs, gloves, etc..
- Do not use open flames and do not smoke, when handling fuel or lubricants.
- Do not dispose of fuel oil into the environment.
- Replace the fresh water stored in the tanks frequently and apply disinfectants as needed.
- Do not exceed speed limits in harbour and confined waters.

- Reduce speed in proximity of other vessels or swimmers. If necessary, set the engines to neutral.
- Adjust speed according to sea conditions.
- Reduce speed before leaving the control station, if you are the only helmsman available.
- Before connecting the shore power supply, switch the switchboard main breaker off.
- Handle hot oils carefully, in order to prevent serious burns.
- Secure engines, shafts and generators prior to performing any maintenance activity.
- Open the heat exchangers very carefully in order to prevent serious burns.
- Do not perform any work inside energised switchboard or generator: serious risk of electric shock.
- Do not inhale exhaust fumes.
- When disconnecting a battery make sure that battery charger is off. If the battery charger is operating, disconnect it, remove the negative cable from the battery and then remove the positive cable. To reconnect the battery charger operate in the reverse order (positive first, then the negative).
- Never disconnect batteries when the generators and/or main engines are running.
- Operate immediately and in a suitable manner on parts which show corrosion traces.

Navigation set up procedures

Preliminary checks

- Gather information on weather forecast and warnings along the route.
- Consult the pilot book;
- Consult the navigation charts, in particular the distances, routes, dangerous sea bottoms and shallow waters.
- Calculate the quantity of fuel required, included a suitable reserve.
- Consider the cruise duration.
- Make sure that both personal and yacht's documents and certifications are onboard.
- Make sure that safety equipment is efficient; in good conditions and complying to local regulations.
- Check that all items required are on board (provisions, navigation charts, documents, flares, first aid kit, etc.).
- Check the safety equipment, as indicated herein after:
 - check that life jackets are in good conditions, that the inflating device is efficient and that the jackets are in the right location and easily accessible (do not obstruct hatches with anything);
 - check that the collective life rafts are easily extractable and that its mooring and anti-capsize line is in good conditions;
 - ensure that the life belt is in its correct location and fitted with relevant safety rope and luminous buoy;
 - check extinguishers charge level: the extinguisher is charged when the pressure gauge indicator is in the green sector.



The designated Captain must ensure that all passengers are perfectly aware of safety equipment location and use (fire extinguishers, life rafts, life belts, etc.).

Check the safety equipment prior to leaving port, and be sure that all items are good conditions. Memorise their location and become familiar as how to use them.

The short time spent for such checks may become very useful in case of emergency.

- Check navigation lights and horn operation.
- Check anchor windlass and chain stopper operation.
- Check efficiency of instrumentation such as Plotter - VHF - depth-sounder - compass, etc..

- Check that the loads are distributed evenly, and the boat has a correct trim; the store-room and other material loading may alter the boat trim, and particularly the list. Distribute evenly and secure all items, in order to avoid sudden load conditions.
- Check that all loose items are properly secured, both on the main deck and below deck.
- Check proper closing of portholes, doors and hatches.
- Make sure that the tender is secured.
- Make sure that no protection covers are preventing correct operation of dashboards, control levers and steering wheels on the flying bridge control station.
- Make sure that external walk-arounds are free from objects, which may obstacle the movement during manoeuvres.
- Check engines, gear boxes and generators oil levels: top up if necessary.
- Check the engines and generators cooling water level: top up if necessary.
- Check gangway, rudders, Bow-Thruster hydraulic oil levels: top up if necessary.



For all maintenance and refilling activities, refer to the specific Manufacturer Manual.

- Make sure that the fuel separator filters are clean: drain any water in the filters by operating the drain valve.
- Check the liquid level (fuel, fresh water) in the tanks.
- Check correct operation of bilge pumps (automatic and manual) which should be used to discharge water in case of flooding; operate the switches located on the bridge switchboard which allow pumps automatic start.



Before starting each cruise, check eventual presence of oily water in the bilge below the main engines, as that space is watertight and is not sucked by the bilge system.



Therefore, in case of need, suck and collect the oily water in special containers. It is strictly forbidden to discharge oily water into the sea. Each marina is provided with special areas for waste disposal.

- Check fire fighting pump operativeness by verifying the water discharge through the fire fighting branch of chain washing circuit, from the two hawspipes.
- Check that all sea chests filters (used for engines and generators cooling, as well as for air conditioning and service systems supply) are clean. If the filters are dirty, close the hull valves (valve handle perpendicular to pipe line), remove and clean the filter baskets, reassemble the filters and open the hull valves.
- Check stabilising fins operation.
- Operate the Bow-Thruster in both directions, one at a time, for a few seconds (remember that the yacht is still moored).
- Check rudder operation (put the rudder hard-over to port and starboard, and then back to zero).
- Make sure that the engine room extraction system is operating.
- Check that no flammable or improper materials are stowed in the engine room.
- Check that the engines and generators fuel systems are switch on (valves open).
- Make sure that sea water intake valves for engines/generators cooling and air conditioning system are open.
- Start the generators, pre-heat the engine for a few minutes, and then give load to the generators from the main switchboard.
- Check the battery charge level on the main switchboard. If necessary, recharge the batteries.
- Insert start up battery breakers and engines electronic control breakers.
- Energise all 24V circuits necessary for navigation and services from the bridge switchboard (and on the main switchboard, if necessary);
- Start the engines with two gear box in "neutral".
- Disconnect the shore connections.
- Ease off the mooring lines and check that everything is ready for the unmooring operations (all lines are clear, anchor chains are not entangled with other boats chains, etc.).

Main engines start up

Regular checks should be carried out on a regular basis, before starting the propeller engines, to ensure that:

1. the sea cock valves of the cooling system are completely open;
2. the outlet fuel valve is open;
3. there are no oil leaks;
4. there are no leaks in the cooling system;
5. the belts are not dangerously worn or broken (periodic check);
6. there are no loose wiring or connections and the cables are not worn or frayed (periodic check);
7. move the start switch on the ON position;
8. move the ignition lever switch on the START position;
9. press the starter push-button.

Don't keep the starter motor running for more thirty ten seconds; wait for at least 2 minutes before trying again.

Start the first engine. Once this is running, check that it is working correctly, and that no warnings are being shown on the display. Once it has been assessed that everything is working correctly, the second engine can be started.

Check the oil pressure after the start of the engines and ensure that it increases to its normal operating value within 15 seconds. Should the oil pressure remain low, the engine should be stopped immediately.

After each engine has been started, check discharge outlet of the cooling system, and ensure that the cooling water is circulating correctly.

Allow the engine to run without any load for at least three to five minutes, or at least until the cooling system temperatures start to increase.

Check that the engine oil pressure value has reached the normal operating level before applying any load to the engine (shifting into gear and increasing the number of revolutions).



Before starting up the engines, disconnect the battery charger from the engine ignition batteries.



The main propeller engines shall be started with the gear box in neutral and the gas levers at their minimum level.

Running the engines at their minimum level for extended periods could cause a lowering of the engines' temperatures, which could give rise to light blue or white smoke. It is therefore recommended that long running periods at minimum levels be avoided.

Running at minimum levels will also cause a higher level of wear to the mechanical components of the engines, and can therefore be more detrimental as far as pollutant emissions are concerned.



- | | |
|--|--|
| <p>1 Port propulsion control lever</p> | <ul style="list-style-type: none"> • Propulsion control lever to engage/disengage and set the engine speed for the port engine. • Position indicates set gear state of the port gearbox. |
| <p>2 Starboard propulsion control lever</p> | <ul style="list-style-type: none"> • Propulsion control lever to engage/disengage and set the engine speed for the starboard engine. • Position indicates set gear state of the starboard gearbox. |
| <p>3 CMD LED</p> | <ul style="list-style-type: none"> • LED lights up when the CMD pushbutton is pressed. |
| <p>4 CMD</p> | <ul style="list-style-type: none"> • Button for selecting the commanding control stand. • Button for acknowledging alarms of the remote control system (LED (16, 18) and horn of the command unit). • Pushbutton to activate the warm-up function. |
| <p>5 COMMAND LED port</p> | <ul style="list-style-type: none"> • LED flashes when command of the port power train can be assumed. • LED lit when propulsion control lever is in command of the port power train. • LED flashes quickly when the warm-up function is activated. |
| <p>6 SCRS</p> | <ul style="list-style-type: none"> • Pushbutton to activate SMART CRUISING mode. |
| <p>7 COMMAND LED starboard</p> | <ul style="list-style-type: none"> • LED flashes slowly when command of the starboard power train can be assumed. • LED lit when propulsion control lever is in command of the starboard power train. • LED flashes quickly when the warm-up function is activated. |

- | | | |
|-----------|------------------------------|--|
| 8 | CRS | <ul style="list-style-type: none"> • Pushbutton to activate CRUISING mode. |
| 9 | CRS LED | <ul style="list-style-type: none"> • LED flashes when the engine speed has been limited due to mode change TRL ! CRS. • LED flashes when CRUISING mode is active. |
| 10 | Ahead mark | <ul style="list-style-type: none"> • The gearbox is engaged ahead when the pointer (23, 24) is level with or past this mark. |
| 11 | Neutral LED starboard | <ul style="list-style-type: none"> • LED lights up when the pointer (23) is at the level of this marking.
The starboard gear is disengaged (or will be disengaged). |
| 12 | SCRS LED | <ul style="list-style-type: none"> • LED flashes when SMART CRUISE mode is active. |
| 13 | Astern mark | <ul style="list-style-type: none"> • The gearbox is engaged astern when the pointer (23, 24) is level with or past this mark. |
| 14 | TRL LED | <ul style="list-style-type: none"> • LED flashes when TROLLING mode is active. |
| 15 | TRL | <ul style="list-style-type: none"> • Pushbutton to activate TROLLING mode. |
| 16 | Alarm LED starboard | <ul style="list-style-type: none"> • LED flashes when an unacknowledged summary alarm is received from the starboard RCS. • LED lights up when this alarm has been acknowledged and is active. |
| 17 | SYNC | <ul style="list-style-type: none"> • Pushbutton to activate SYNC or SMART SYNC function. |
| 18 | Alarm LED port | <ul style="list-style-type: none"> • LED flashes when an unacknowledged summary alarm is received from the port RCS. • LED lights up when this alarm has been acknowledged and is active. |
| 19 | AUX | <ul style="list-style-type: none"> • Pushbutton to activate an additional function (Installation and Commissioning Instructions). |
| 20 | AUX LED | <ul style="list-style-type: none"> • LED lights up when additional function is active. |
| 21 | SYNC LED | <ul style="list-style-type: none"> • LED lights up when the SYNC or SMART SYNC function is active. |
| 22 | Neutral LED port | <ul style="list-style-type: none"> • LED lights up when the pointer (24) is at the level of this marking. • The port gear is disengaged (or will be disengaged). |
| 23 | Starboard pointer | <ul style="list-style-type: none"> • The position of the pointer indicates the starboard engagement setting. |
| 24 | Port pointer | <ul style="list-style-type: none"> • The position of the pointer indicates the port engagement setting. |



Make sure that no one is standing by the exhausts and the mooring lines.

Main engines stop

Do not stop the engines immediately after heavy-duty operation, but leave them running for about five minutes at idle to balance the temperatures.

- The engine throttles should be in neutral position.
- Move the starter switch on the off position.
- Disconnect the electric power to the inspection and control system of the propellers.
- Disconnect the engine starter battery.
- Check eventual leakage through the shaft line stuffing box.
- Let the engine room ventilation system run for at least 30 minutes after engines stop, to cool down the room.



Remove the ignition keys to avoid that unauthorised personnel start up the engines accidentally.

Steering system

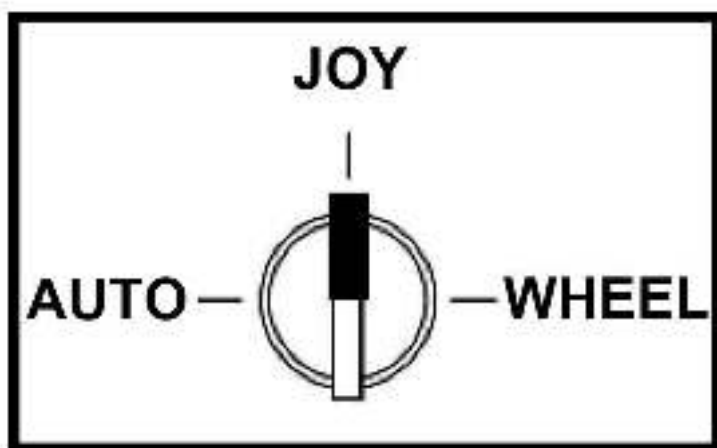
CMC **DIRECTA** system is an electric power steering system designed and tested with the most modern technologies and realized with proper materials in order to supply the best steering solution in any condition and maximum safety.

DIRECTA 100 R3 is an electric power steering system that fits to yachts with two separate rudders: each rudder, mechanically independent, is controlled by a single electric cylinder. CANopen protocol is implemented on DIRECTA System among the main system equipments. A transducer on each rudder axis is installed to reading the actual angle.

This system is provided with the necessary equipment for the wheelhouse manoeuvring (Pilothouse) and Flybridge control station. The Wings control station (Port & Stbd) are considered as optional.

The system consists of three different **CONTROL MODES**: AUTO Mode, WHEEL Mode and JOY Mode.

The control modes can be selected only from the selector in the wheelhouse.



NOTE: should the selector fail, the system automatically switch to JOY mode.

➤ AUTO MODE

This is the standard mode, in which the system is controlled by the PLC (via CANopen protocol).

The system can be commanded by the helmsman or by the Autopilot (autopilot is not supplied by CMC).

The MANUAL function allows the helmsman to command the steering system using the wheel (or the joysticks) of the station in command: the helmsman set (through the wheel or joystick) the required angle of the rudders (the set angle is shown in the displays of the command stations).

The AUTOPILOT function gives the control of the steering system to the Autopilot.

Both functions can be activated only from the control station which is in command.

The rudder angle range and the related rounds of the wheel can be set up during the boat start-up.

When the Autopilot is active, a quick rotation of the active wheel automatically switch the system in MANUAL and steering is controlled by the helmsman using the wheel.

This is a safety feature to allow the quick control of the helmsman to avoid an obstacle.



It is very important that the wheel of the station in command is safeguarded from unintended movement (to avoid unintended exit from AUTOPILOT function).

There are three possible CONTROL STATIONS for AUTO mode:

- Wheelhouse Control Station
- Flybridge Control Station
- Wing Control Station (optional).

NOTE: Only the touch screen of the Wheelhouse allows the system settings.

NOTE: Wheelhouse and Flybridge control stations have different hardware but are equivalent for the use of the system.

➤ WHEEL MODE



*The two rudders can only be controlled in **synchronous way**.*

The WHEEL mode is intended to be a back-up system for possible loss of CANopen protocol.

The WHEEL mode is activated when the selector position is on "Wheel".

In the WHEEL mode the PLC is excluded and the wheel directly communicate with the inverters to manage the system.

Should the CANopen protocol faults:

- the "CANopen Fault" alarm is shown,
- the PLC is excluded,
- the system keeps working, using the hardwired logic,
- a general alarm (acoustic and visible) is shown also in the other control stations.

When in WHEEL mode, the system can be controlled only by the wheel in the Wheelhouse station (nearby to the Auxiliary control station). The wheel on the Flybridge is not active.

➤ JOY MODE



*The two rudders can only be controlled in **synchronous way**.*

The JOY mode is a back-up system in case of loss of CANopen protocol and wheel failure.

The JOY mode is activated when the selector position is on “JOY” and the “Manual Override” indicator is lit on.

In JOY mode the system can be controlled only by the joysticks in the wheelhouse control station.

In JOY mode the joysticks directly communicate with Inverter and the wheels are not allowed to control the system.

In normal circumstances, the system - in JOY mode - is controlled by the PLC (via CANopen protocol).

Should the CANopen protocol faults:

- the “CANopen Fault” alarm is shown,
- the PLC is excluded,
- the system keeps working, using the hardwired logic,
- the rudders angle range is limited by the limit switches on each actuator,
- a general alarm (acoustic and visible) is shown also in the other control stations.

When in JOY mode, the system can be controlled only from the “Auxiliary control station”, made up of:

- 1 selector with 3-positions (AUTO-JOY-WHEEL);
- 2 ON/OFF joysticks for independent control of the two rudders;
- 1 light panel with 6 leds (3 led for each side) for specific alarms, as follows:
 - Power Unit in case of loss of power supply,
 - Control System when a fault in the control algorithm is displayed,
 - Motor Overload when the thermal protection of the motor activates,
 - 1 led indicator: is automatically lit when in WHEEL or JOY mode.

SAFETY FEATURES

In the steering system the two separate rudders are independent, each one controlled by an independent system.

Should one rudder or one system fails, the other rudder is used as a backup to steer the vessel.

In AUTO mode (conceived to be the “normal” operation mode) the rudders are operated in a synchronous way to reduce the helmsman’s workload.

For each system, 230VAC power supply is needed.

For each system, 24VDC power supply is needed.

In normal circumstances the system is commanded by the Wheelhouse control station or Flybridge control station or from the Wings control station (optional) or by the (external) Autopilot. Should a failure occur, there are several backup options.

The switch of command among control station is protected with a strict procedure to avoid unwilled changes.

Another safety feature is that the command can be transferred only with throttle in idle.

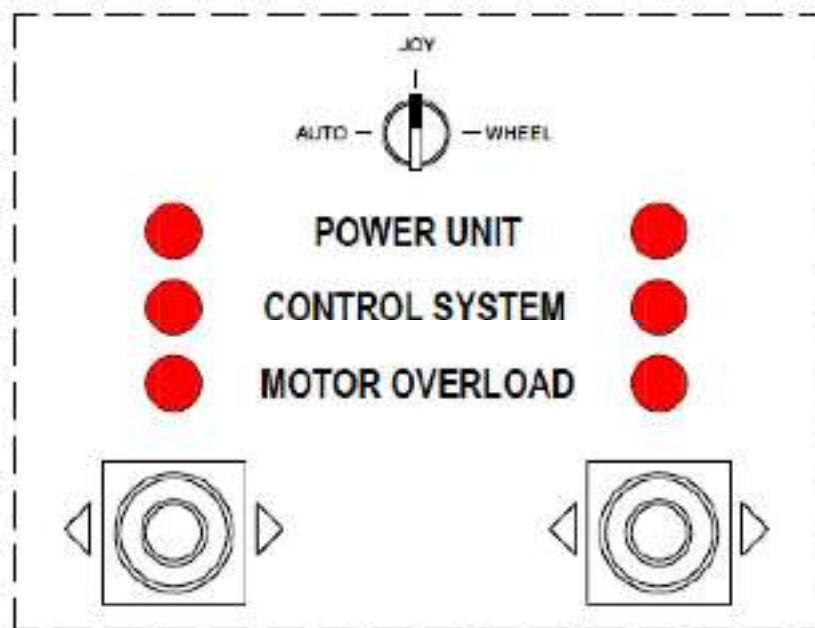
A safety feature allows to override the Flying Bridge control station or the optional control stations (Wings) and take back the command to the Wheelhouse control station.

In case of sudden need of human override over Autopilot, a safety feature allows to quickly regain the control of the vessel through the wheel.

At least one rudder command is guaranteed even with PLC or electrical failures.

Should a rudder cannot be moved by AC power supply, it can be always moved to zero position (to not hamper the control of the vessel) by manual means.

AUXILIARY CONTROL STATION



The AUXILIARY CONTROL STATION is located in the wheelhouse. It is part of the Wheelhouse control station and aimed at the emergency control of the system.

The AUXILIARY CONTROL STATION is composed of:

- one selector to switch between “AUTO” (command from PLC), “JOY” (command from Joysticks), or “WHEEL” (command from Wheel),
- two ON/OFF Joystick, each one commanding a rudder (only when in JOY mode),
- sets of 3 alarm leds (one set for each rudder).

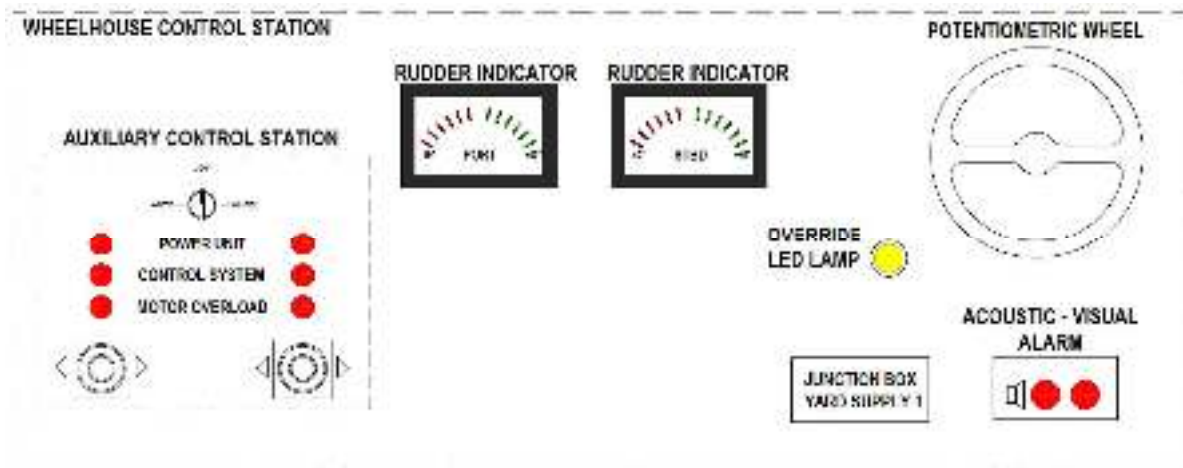
In case of PLC failure or CANopen protocol loss, the AUXILIARY CONTROL STATION allows to switch to the JOY (joystick) mode or WHEEL mode.

In case of operation under JOY mode, only the joysticks on the AUXILIARY CONTROL STATION are active.

CONTROL STATIONS

➤ WHEELHOUSE CONTROL STATION

The wheelhouse control station is the main control station.



The Wheelhouse control Station is the main control station and it is composed of:

- Auxiliary Control Station;
- Rudder Indicators (one for each rudder), hardwired: independent from CANopen network;
- Override Led Lamp (lit when in JOY or WHEEL mode);
- Steering wheel (named "Potentiometric Wheel" after its hardware), hardwired: independent from CANopen network;
- Acoustic – Visual Alarm (buzzer and red light), hardwired: independent from CANopen network;
- Touchscreen panel, independent from CANopen network.

NOTE:

- only from the wheelhouse's touch screen you can override the other control station and take back the command: keep the "TAKE COMMAND" icon tapped for 5 seconds;
- the Wheelhouse components keep working should CANopen network fail;
- the Wheelhouse components need at least the 24Vdc supply to keep working.

Touch Screen Panel – Home Page

The touch screen could be shared with a CMC Marine stabilizers system: in this case there are dedicated pages for each system; the main menu (**Home Page**) allows the access to the desired system page.

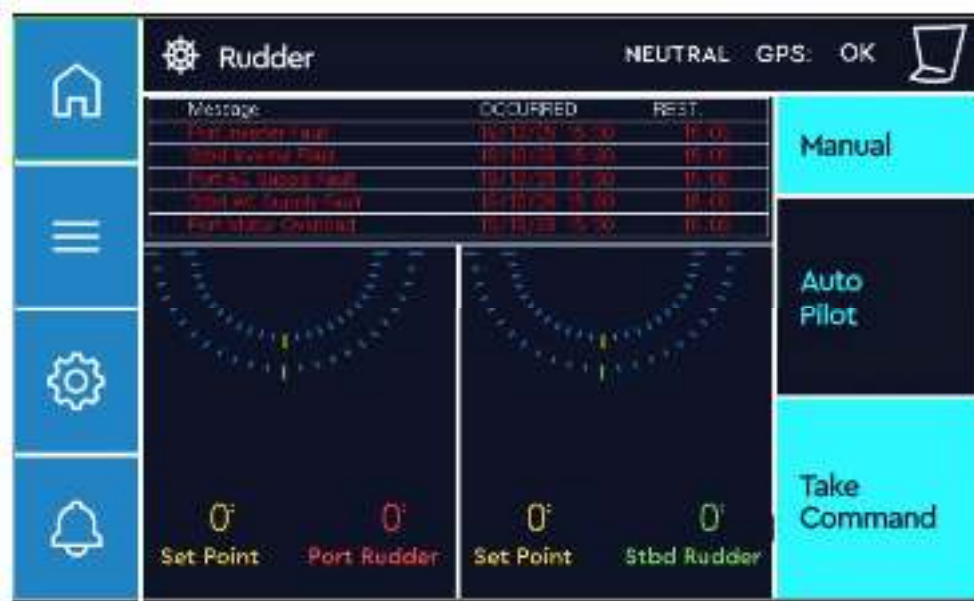
The left navigation bar of the Home Page is always visible and accessible.



Home Page



Touch Screen Panel – Steering Page



Steering Page

Buttons state:

- Blue letters on black background = OFF
- Black letters on blue background = ON

The rudder page is split in two to give dedicated indication for the PORT and for the STARBOARD rudder.

Rudders angle: indication of the actual rudder angle.

Set point angle: indication of the angle set by the helmsman.

Manual tap to switch on/off the MANUAL function (wheel in command).

Autopilot to switch on/off the AUTOPILOT function.

NOTE: The wheel keeps active and the light blue blinks until the Autopilot actually takes the command (the icon is lit only when the Autopilot takes command).

Alarm Page tap to go to Alarm Page. It works only on the active control station. In case of alarm, the icon blinks red until the alarm is not acknowledged.

Take Command: for the *request to change/confirmation between control stations*.

When a command station is in command, the “take command” icon is lit (blue background and black letters).

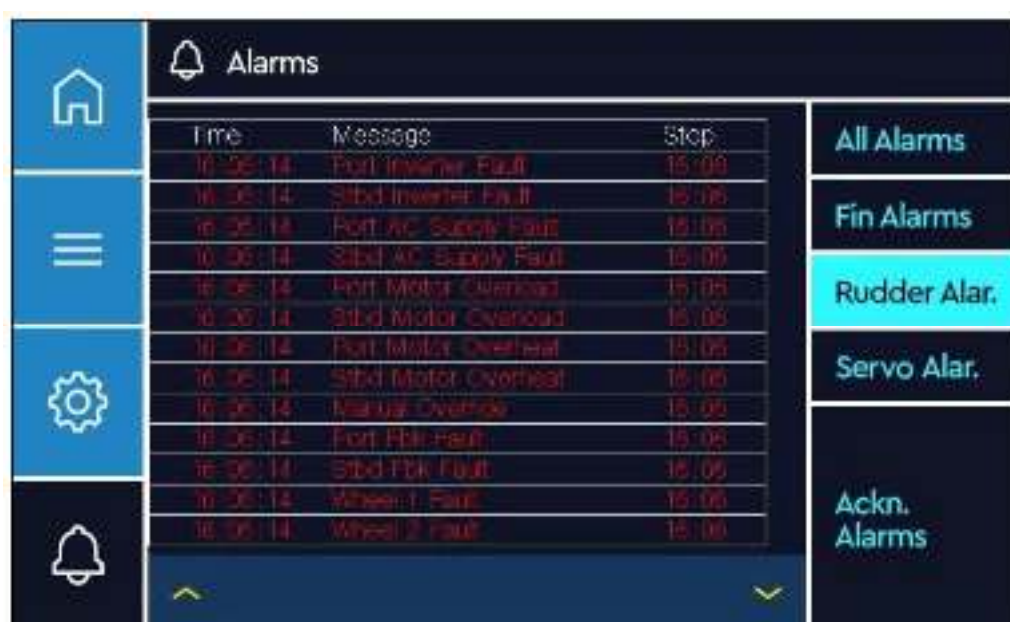
When the “take command” icon is blinking, the station is entitled to take command.

Any change of control station is allowed only when throttle is in idle.

Procedure to switch command to another station (either Flybridge station or Wings stations):

- A. Throttle to idle position.
- B. Tap on the “take command” icon of the station in command (on which the icon is lit).
- C. The “take command” icon starts blinking (on all the stations).
- D. Within 30 seconds tap the blinking “take command” icon on the required control station.
- E. When the “take command” icon stays lit (without blinking): the station is in command.

Touch Screen Panel – Alarm Page



Alarm Page

In this page is possible filter the alarms in four types:

- **All Alarms** → all alarms of the steering and stabilizer systems
- **Fin Alarms** → only alarms of the stabilizer systems (if required)
- **Rudder Alarms** → only alarms of the rudders systems
- **Servo Alarms** → only alarm of the servo-drivers

Acknowledge Alarms: tap to stop the buzzer. It works only on the active control station.

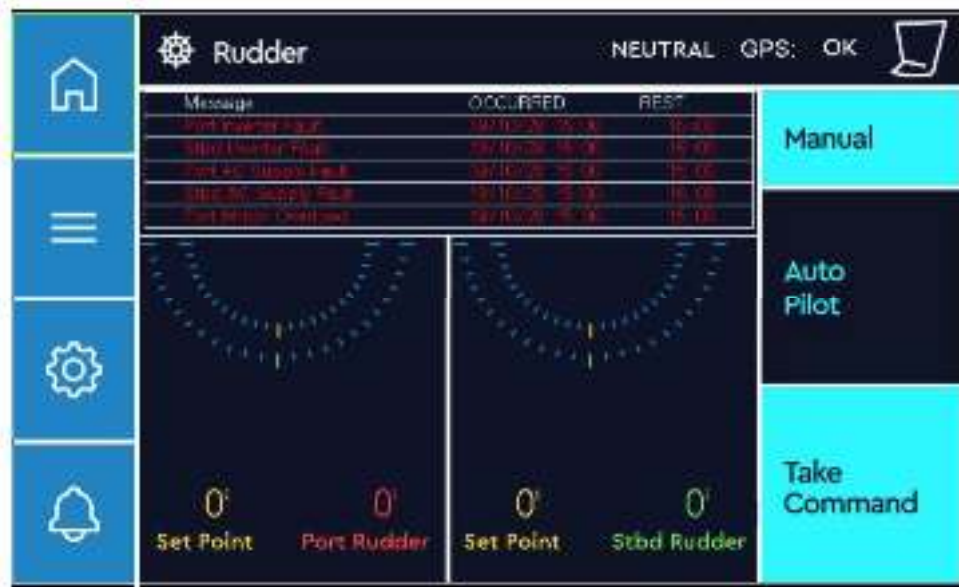
Alarm Index Legend

↑ ↓	Browse alarm index
TIME	Alarm time start
MESSAGE	Description of the alarm
STOP	Alarm time stop

Red → Alarm active and not fixed

Green → Alarm fixed

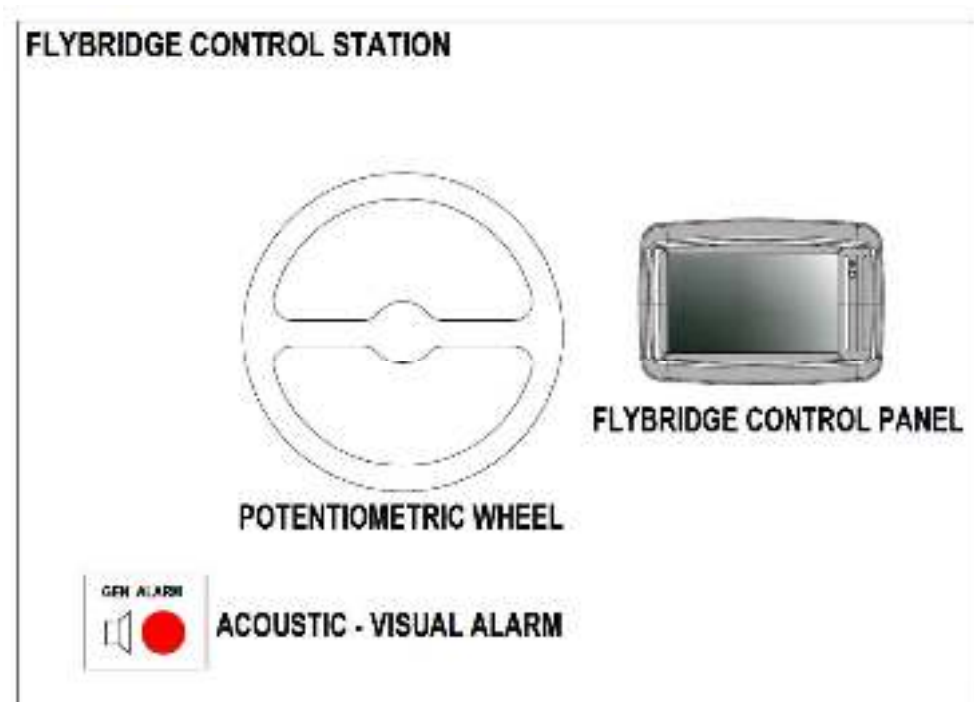
Touch Screen Panel – Steering Page



Steering Page

Should CANopen network fail, the touch screen keeps working but on the Steering page a “COMUNICATION FAULT” message is repeated as shown above.

➤ **FLYBRIDGE CONTROL STATION**



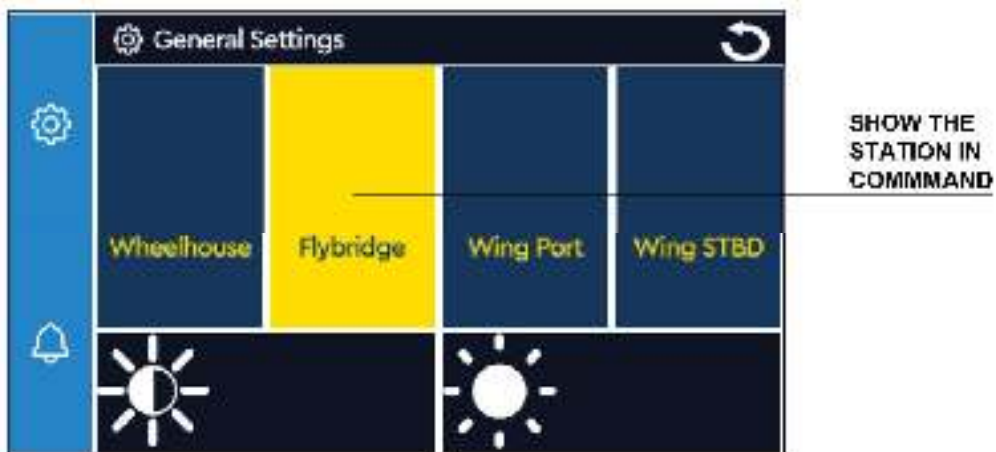
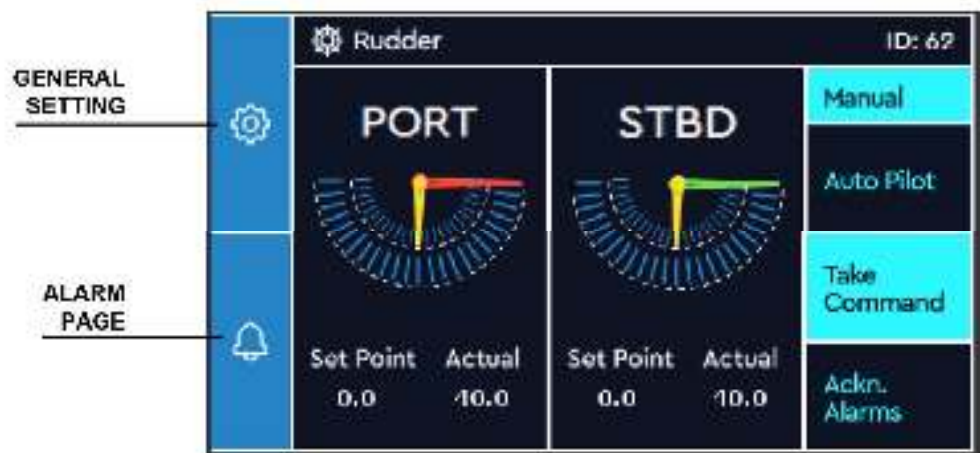
The FLYBRIDGE CONTROL STATION is composed of:

- External touchscreen Control Panel, *dependent from CANopen network*
- Steering wheel (named "Potentiometric Wheel" after its hardware), *dependent from CANopen network*
- Acoustic – Visual Alarm (buzzer and red light), *hardwired: independent from CANopen network*

NOTE:

- should the CANopen network fail, the Flybridge control stations are no longer operative,
- in the WHEEL mode, the Flybridge wheel is not active,
- the command cannot be shifted from the Wings station to the Flybridge station and vice-versa but must always pass through Wheelhouse control station.

Below, some screenshots taken from the touch screen control panel mounted in the flybridge and in the wing(s) control stations are shown.



Buttons state:

- Blue letters on black background = OFF
- Black letters on blue background = ON

The "Home page" of the Flybridge touch screen is similar to the "Rudder page" of the wing's screen.

The home page is split in two to give dedicated indication for the PORT and for the STARBOARD rudder.

Rudders angle: indication of the actual rudder angle.

Set point angle: indication of the angle set by the helmsman.

Manual tap to switch on/off the MANUAL function (wheel in command).

Autopilot to switch on/off the AUTOPILOT function. NOTE: The wheel keeps active and the light blue blinks until the Autopilot actually takes the command (the icon is lit only when the Autopilot takes command).

Alarm Page tap to go to Alarm Page. It works only on the active control station. In case of alarm, the icon blinks red until the alarm is not acknowledged.

General Settings: can be accessed also from a station which is not in command and shows which station is actually in command.

Take Command: for the *request to change/confirmation between control stations*.

When a command station is in command, the "take command" icon is lit (blue background and black letters).

When the "take command" icon is blinking, the station is entitled to take command.

Any change of control station, is allowed only when throttle is in idle.

Procedure to switch command to another station (either Flybridge station or Wings stations):

F. Throttle to idle position.

G. Tap on the "take command" icon of the station in command (on which the icon is lit).

H. The "take command" icon starts blinking (on all the stations).

I. Within 30 seconds tap the blinking "take command" icon on the required control station.

When the "take command" icon stays lit (without blinking): the station is in command.

Alarm List Page:

In this page the alarm index is shown.

Each alarm can be of two colors:

Red → Alarm active and not fixed

Green → Alarm fixed

EMERGENCY MODE



EMERGENCY MODE must be used with maximum attention, only if necessary and only by trained personnel.

In the Directa steering system the 2 separate rudders are independent, each one controlled by an independent system.

Should one rudder or its actuator or control system fails, the other rudder is used as a back-up to steer the vessel. The statutory emergency mode is – essentially - the use of the system with one single rudder active.

To ease the control of the vessel with one single rudder, it can be helpful to manually move the failed rudder as described below by 24Vdc auxiliary motor (**AUXILIARY OPERATION: 24V MODE**).

Depending on the vessel class approval and on the shipyard installation, some systems are further strengthened with an emergency 230Vac and an emergency 24Vdc supply.

To give a general non-exhaustive overview:

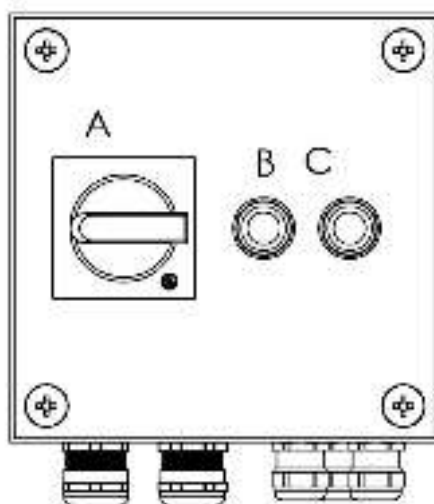
- the CANbus network is needed to control of the system from the optional control stations;
- the 230Vac supply is needed to normally move the actuators (electric cylinders), apart from their auxiliary operations in 24Vdc;
- the 24Vdc supply is needed to allow the control of the system in JOY mode or in WHEEL mode, to keep working the rudder indicators and the wheelhouse's touch screen and to operate the actuators in 24V mode.

Please refers to the troubleshooting paragraph for detailed informations.

Here a non-exhaustive list of hints in case of some fails:

- should one inverter or one actuator fail, it can be manually moved (with 24Vdc motors on other sizes), while the other one can still be automatically controlled by the system to steer the vessel;
- should the CANopen network fail, the system can be still controlled from the wheelhouse (in AUTO, WHEEL or JOY mode);
- should the CANopen network fail and then also the Wheelhouse's Wheel fail, the system can still be controlled from the wheelhouse in JOY mode;
- should the CANopen network fail and then also the Wheelhouse's Joysticks fail, the system can still be controlled from the wheelhouse in WHEEL mode;
- should the Wheelhouse's Wheel or Wheelhouse Joysticks fail when the CANopen network is still available, the system can still be controlled from the Flybridge control stations;
- should any 230Vac supply fail, the rudders can still be manually moved (with 24Vdc motors);
- should the main 230Vac supply fail with a 230Vac emergency supply available, one of the rudder (usually the port one) can still be automatically controlled by the system, while the other one can be manually moved (with 24Vdc motors).

AUXILIARY OPERATION: 24V MODE



The 2 Junction boxes, one for each side, contain the disconnect switch (A) “Emergency Mode” and 2 pushbutton (B and C).

In normal operation, the selector must be switch ON.

When the selector is switched to OFF: the actuator is disconnected from the inverter and from the control system (PLC), the clutch on the motor is released and the soft starter is ready to command the 24Vdc motor.

In case of emergency, it is switched to OFF to allow the manual positioning of the rudder.

The rudder angle is visible on graduate crown or on Local Control Panel on the inverter.



EMERGENCY MODE must be used with maximum attention, only if necessary and only by trained personnel.

In case of fault of 230V AC power supply, or of the inverter or of the brushless motor, the actuator cannot be automatically controlled by the system, but must be manually moved (and locked) to centre position.

For Directa 100 an alternative DC motor is connected to the actuator through a clutch. The motor can be operated through the pushbuttons on the Junction box.

Procedure for emergency mode for Directa 100

1. Stop the yacht or move it at minimum speed forward.
2. Turn off 230Vac main breaker switch present on main switchboard of the yacht.
3. Switch to OFF the disconnecter of the junction box of the rudder: the power supply from the inverter is off, permissive signal to start 24Vdc motor and the clutch of the secondary motor 24V is activated. This rudder is no longer controlled by operator control stations.
4. Use the pushbuttons present on the junction boxes to move the rudder to required position. The rudder angle is visible on graduate crown or on Local Control Panel in front of the inverter.
5. After this procedure, when the rudder is in the required position, turn ON the disconnecter switch on the JB and the 230Vac main breaker switch.

Steering System - Diagram

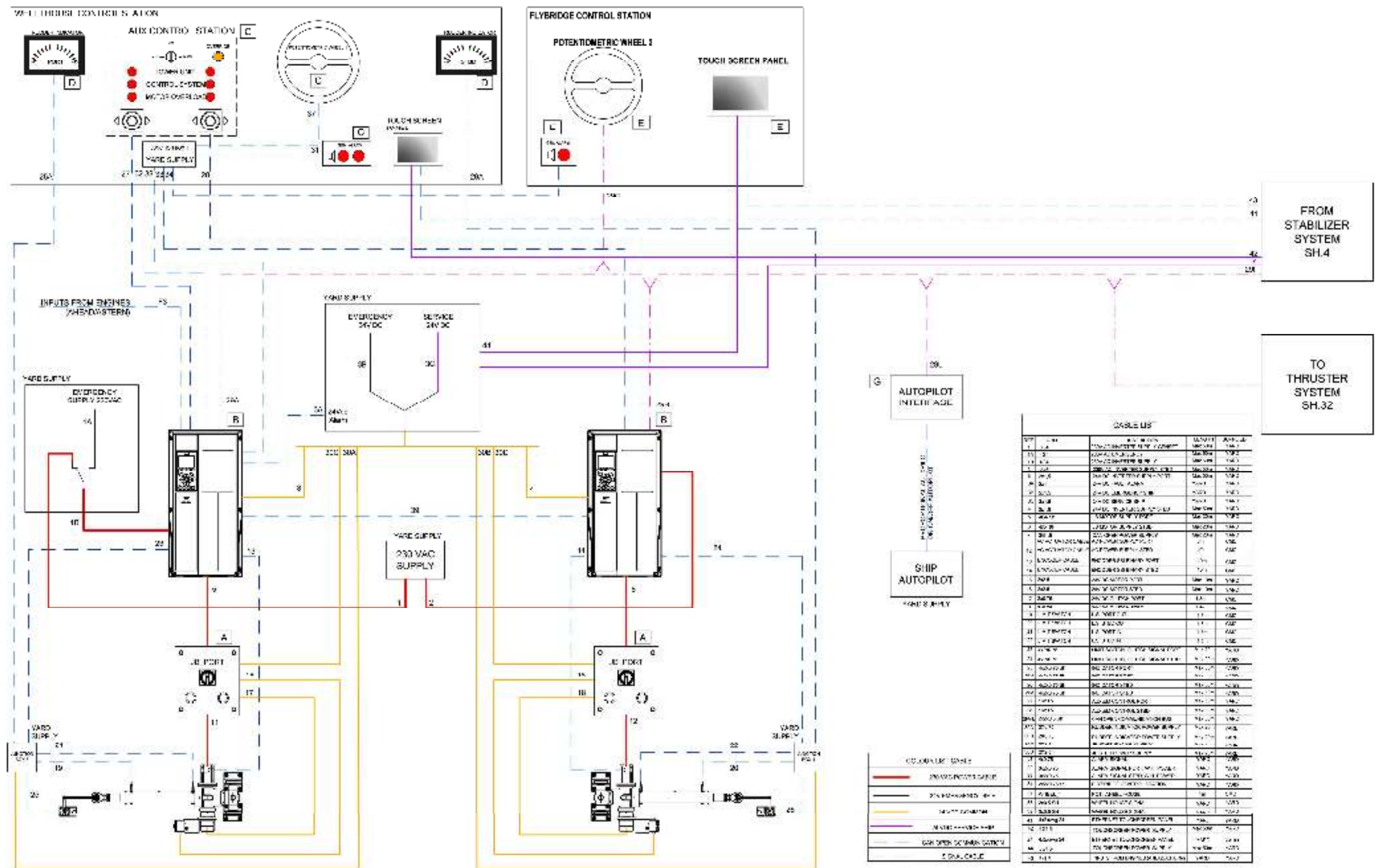


Table – Alarms

This is the list of the alarms that could be displayed on the touch screen monitor of the Wheelhouse and their possible solution.

The Acknowledge Alarm button stops the buzzer but works only on the active control station. The led is red until the alarm is active.

CONDITION (failure or alarm description)	System Status	Causes	Solution
The Touch Screen Monitor is off	No power supply to the Touch Screen.	The system can be controlled by the "Auxiliary control station" or from the Flybridge control station.	Check fuses on the ECO Switchboard.
	No power supply to the system.	The alarm "24Vdc Fault" is ON.	Check fuses on the ECO Switchboard and general switch for steering system.
Buzzer + Led activated on Flybridge or Wheelhouse	Fault/Lack of power supply to PLC.	The system is an alarm, the buzzer is active and red led alarm is ON.	Switch ON the Touch Screen and check the alarm list in alarm index. Push Ack Alarm button on the touchscreen panel for silence buzzer alarm.
			Shift to Joy Mode or Wheel Mode from the dedicated control station.
Manual Override	The system is in Joy or Wheel Mode. The manual override alarm is activated and the override led lamp is on.	The selector of the Auxiliary control station switched on Joy-stick or Wheel mode.	Use the joystick or the wheel to move the rudders. Switch to AUTO to remove the alarm.
Port/Stbd Inverter Fault	The system Port/Stbd does not move in AUTO, JOY or WHEEL mode.	Inverter or motor fault.	Check on the inverter panel the alarm details.
			If necessary, move the faulty rudder at center in MANUAL Mode and switch off the power supply to the inverter.

CONDITION (failure or alarm description)	System Status	Causes	Solution
Port/Stbd AC Supply Fault	The system can work only in MAN-UAL mode.	Lack of power supply.	Check the integrity of power supply system.
			Check the integrity of the connections on the inverter side.
			If necessary, use the MAN-UAL mode to move the rudders for emergency manoeuvres.
Port/Stbd Motor Overload	The system works normally.	The actuator exceeds the max power but the system keeps working.	Check the actuator and rudder integrity.
			A protracted use of the system in Overload condition may drive to a Motor Overheat alarm.
Port/Stbd Motor Overheat	The motor is in a trip condition. Its thermal protection is on.	The system exceeded the maximum of temperature on the motor. The thermal protection knocked in to avoid of the damage at the motor.	Check the sensors connection.
			Switch to JOY mode and steer with the rudder which is not in failure: if necessary, use the MANUAL mode to move to the center the rudder in failure.
			Call an authorized CMC marine service center for further investigation on the actuator.
FC OFF	The system does not work from any station.	The inverter display is set on HAND-ON/OFF mode.	Select the AUTO-ON Mode on the inverter display.
Limit Switch IN/OUT	The rudder cannot move further to Port/Stbd side.	The rudder has reached the maximum angle on Port/Stbd side.	Move the rudder in the opposite direction.
		The Limit switches is faulty.	Check the integrity of Limit switches connections.
Port/Stbd Encoder Error	In AUTO mode the ram does not move under request	In AUTO mode the ram does not move under request.	Check the integrity of the actuator encoder connections.
			Check the integrity of the actuator encoder connections.

CONDITION (failure or alarm description)	System Status	Causes	Solution
Wheel 1 Fault	The wheel of the main station (wheelhouse) is not working.	The rotary encoder of the wheel has no power supply or it is faulty.	Check the integrity of the connections and of the sensor.
			Change control station to control the system.
Wheel 2 Fault	The wheel of the Flybridge station is not working.	Canbus fault.	
		Sensor not properly connected.	Check the integrity of the cable connections.
		Sensor damaged.	Replace the sensor.
CANopen Fault	The system does not work in AUTO mode.	Fault in the CANbus network or in any of its component.	Check the cable connections.
			Restart the system.
Communication Fault (Port/Stbd side)	The rudder does not work in AUTO mode.	The PLC-MCO communication is interrupted.	Check the integrity of the connections.
			Restart the system.
24V Fault	The system works normally.	The 24V is not provided to one of the inverters.	Check the integrity of the electrical connections and of the 24V power source.
Emergency 24V Fault	The system works normally.	The Emergency 24V is not provided to one of the inverters.	Check the integrity of the electrical connections and of the Emergency 24V power source.

MAINTENANCE

MAINTENANCE PERIOD	ACTION	COMPONENT / AREA	METHOD
After 50 working hours	Check of screws/nuts tightening.	All mechanical components.	
	Temperatures check.	Linear actuators and inverters.	After 30 minutes of use in rough sea, check the temperatures of linear actuator (max temperature 85°C).
After 2 months/ 300 working hours	Visual inspection.	Linear Actuators.	Check for eventual lubricant leakage close to the actuators.
After 6 months/ 1000 working hours	Visual inspection.	Linear Actuators.	Check for eventual lubricant leakage close to the actuators.
Every year/ 2000 working hours	General check.	All parts.	Check & work by CMC Technician.

Generators start up

Before starting up the generators make sure that cooling system sea chest valves are completely open and check the switchboard settings.

Start the generators and make sure that cooling water circulation is correct; wait a few minutes before applying the load.

If the yacht is powered by shore system, de-energise the shore socket and transfer the load to the generators. Recover the shore connection cable and store it.

The yacht is provided with two generators, driven by Diesel engines, contained into sound-proof hoods.

The **KOHLER** generators (**70EFOZDJ**, 50kW, 400-230VAC, 50Hz, 3-phase) are installed in the engine room, on port and starboard sides, next to the aft bulkhead.

The generators are started up by means of their own batteries, stored into protection cases located close to the generators.

The generators suck combustion air directly from the engine room; the exhaust gas is cooled by means of the cooling water, then is separated in gas and water by a separator and discharged respectively in the side outlets and outboard.

The generators' Diesel engines are supplied by fuel sucked from the service tanks and filtered by special individual **RACOR** filters equipped with alarm sensors for clogged filter, with signalling on the bridge; the fuel in excess returns into the service tanks.

The engines are cooled down, through a heat exchanger, by sea water sucked through an independent sea chest, located below the generator and provided with check valve and filter.

The sea water intake filter should be cleaned with a periodicity according to the system use and to the pollution conditions of the sucked water (seaweed, mucilage, etc.).

Before cleaning the filter, remember to close the sea chest check valve. Reopen the valve, after having cleaned the filter.

Electric power supply

The power supply distribution is managed through the main switchboard, which is located in Engine Control Room the switchboard distributes power supply to various equipment, with specific circuits protected by magneto-thermal breakers.

The switchboard is equipped with power sources internal selection circuits, which should be known when operating the yacht.

- The yacht is equipped with special "seamless transfer" system which, by means of a proper command, can be switched from generator supply to earthline. Therefore when connecting to shore power supply the generator load is transferred to shore supply; the generators are then unloaded and can be stopped.
- Shore power can supply the whole yacht except for the air conditioning system.
- The generators are provided with a semi-automatic parallel system.

Radio communications – General information

Two types of radios are generally used onboard yachts:

1. Amplitude modulation (AM) radio, at medium frequency MF (between 1.6 and 3.8 MHz). The acronym SSB (Single Side Band) specifies that only half of the modulated wave is used.

This system presents the following characteristics:

- long-haul communications;
- duplex connection (different frequency used for transmission and reception);
- the quality of communication is compressed by a low immunity to interference.

2. Frequency modulation at very high frequency (VHF-FM) (between 156 and 174 MHz). At present, this is the most widely used method for low-range communications.

This system presents the following characteristics:

- “optic” port connection;
- simplex connection (same frequency used for transmission and reception);
- communication is less sensitive to interference, compared to MF system.

A limited number of channels have been established for VHF transmission, each one corresponding to a channel.

Radio communications protocol

VHF

Emergency frequency = channel 16

Call frequency = channel 9

MF

Emergency and call frequency = 2,182 MHz

Ordinary calls

- Repeat three times the name of the yacht or radio station that you are calling;
- identify yourself with the name of the radio station and with the name of your yacht, repeat three times;
- after receiving a reply, set the channel required by the ship or radio station that you have called.

Special calls

- **Safety call** – generally issued by the harbour radio station, with information concerning weather bulletin. The message starts with the word SECURITY or SECURITE repeated three times.

SECURITE – SECURITE – SECURITE

- **Urgent call** – generally issued by the yacht that require assistance, but that is not in imminent danger situation. The message starts with the word PAN repeated three times.

PAN – PAN – PAN

- **Emergency call** – starts with the word MAYDAY repeated three times; it means that the yacht is in danger and requires immediate assistance.

MAYDAY – MAYDAY – MAYDAY

General suggestions

In case of emergency or calamities, the communication lines are often overloaded and therefore not much effective. Radio equipment users often forget to use the appropriate protocol when using such lines. Some suggestions and rules to be followed when using bi-directional communications are shown herein after.

- Do not attempt to talk while another person is talking. First, listen. If someone is using the radio, wait until the transmission is finished.
Communications are carried out in “simplex”, that is to say “unidirectional” (it is not possible to receive and transmit simultaneously). When two people try to transmit simultaneously, it will result in a confused and indistinct sound. Therefore the transmission must be repeated, causing loss of precious time.
- In case of emergency situations, interrupt any radio communication.
- Radio contacts must be possibly short.
- Talk slowly and clearly. Keep the radio or the microphone approximately 2 centimetres from your mouth. It is not necessary to shout, talk normally. If you shout, you will distort the transmission.
- Make sure you know which channel you can use. Some of the 60 channels of your radio are reserved for commercial or government purposes; other channels are reserved for harbour operations, such as navigation, mooring and weather communications.
- When checking the radio at sea, it is not a good idea to ask another boat to call you. Instead, call a specific boat or station. If you don't know one, seek a channel to verify traffic and after the two stations have finished transmitting, call one and ask for a radio check. Most boaters respond promptly to such requests.

Terms to be used during transmission

Affirmative / Negative: YES or NO can be misunderstood when radio transmission is disturbed or weak.

Stop: word often stated by the Coast Guard for separating various messages during a single transmission.

Number: informs the listener that the radio station operator is transmitting a number.

Spelling: informs the listener that you intend to spell phonetically an important, unusual or difficult word. Use the Alpha Bravo code, described in the following table.

Over: you have ended your message and you are waiting a response.

Out: your transmission is ended and no response is expected. The phrase "Over and Out" is not correct and may be misunderstood.

Roger: you have received and understood the message.

Letter	Code
A	ALFA
B	BRAVO
C	CHARLIE
D	DELTA
E	ECHO
F	FOXTROT
G	GOLF
H	HOTEL
I	INDIA
J	JULIET
K	KILO
L	LIMA
M	MIKE
N	NOVEMBER
O	OSCAR
P	PAPA
Q	QUEBEC
R	ROMEO
S	SIERRA
T	TANGO
U	UNIFORM
V	VICTOR
W	WHISKEY
X	X-RAY
Y	YANKEE
Z	ZULU

Number	Code
0	NAADZERO
1	UNAONE
2	BISSOTWO
3	TERRATHREE
4	KARTEFOUR
5	PANTAFIVE
6	SOXISIX
7	SETTESEVEN
8	OKTOEIGHT
9	NOVENINE

Emergency radio calls (VHF – FM-8900S Furuno)

The FM-8900S is a cost-effective all-in-one marine VHF radio system consisting of a 25W VHF radiotelephone, a DSC modem, and a CH70 watchkeeping receiver. It complies with GMDSS carriage requirements for safety and general communications.

The FM-8900S offers semi-duplex voice communication on ITU channels in the marine mobile VHF band. The features include Scanning Dual Channels (DW) which allows a continuous watch on CH16 and another selected frequency.

Data is displayed on a large, easy-to-read color LCD. Operation is simplified by the use of few keys and easy-to-follow menus.

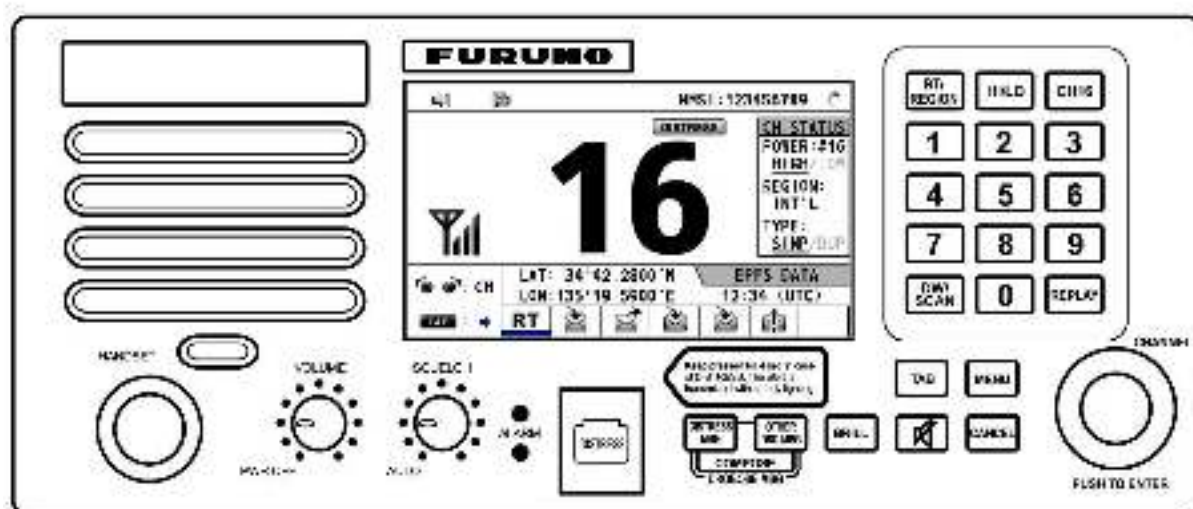
The built-in DSC function produces and receives digital selective callings for quick and efficient establishment of distress, urgency, safety and routine communications with other ships and coast stations that install any VHF DSC facilities.

Full Class-A DSC functions are provided for distress alert transmission and reception, as well as the general call formats (Individual telephone, All Ships, and Group call). Distress alert can be readily transmitted but an arrangement is provided to prevent accidental activation. The FM-8900S maintains a continuous watch on CH70 even while another VHF channel is in use. Aural and visual alarms are given to incoming DSC messages.



The SOS function can only be used in case of emergency and with the correct DSC number coded in the device.

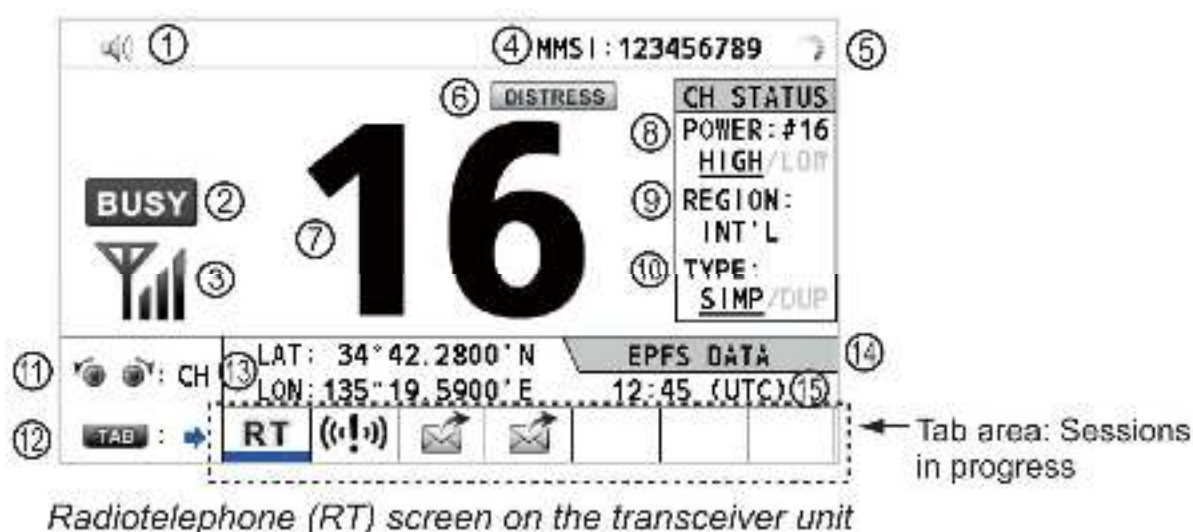
Sending a distress call in the absence of an imminent danger to the boat and/or its crew is a punishable offense.



Description of controls for transceiver unit

Control	Function
VOLUME/PWR knob	• Turns the power on or off. • Adjusts the volume.
CHANNEL/ENTER knob	• Rotate to select channel. • Rotate to select menu items or change the page in multi-page screens (e.g., log data); push to confirm a selection.
SQUELCH knob	Rotate to adjust the squelch. The squelch mutes the audio output in the absence of an incoming signal. AUTO position automatically reduces white noise.
DISTRESS key	Press and hold down the key four seconds to transmit the distress alert.
DISTRESS MSG key	Composes DSC TX message for DISTRESS ALERT.
OTHER DSC MSG key	Composes DSC TX message except DISTRESS ALERT and DROBOSE (Distress Relay On Behalf Of Someone Else).
DROBOSE MSG key	Composes DSC TX message for DROBOSE (Distress Relay On Behalf Of Someone Else). Press the DISTRESS MSG key and the OTHER DSC MSG key simultaneously.
BRILL key	Adjusts the brilliance.
TAB key	• Switches control to the tab area. • Switches the session.
 key	Turns the main speaker on or off.
MENU key	Opens/closes the menu.
CANCEL key	• Cancels the creation of the DSC message currently being created. • Silences the audio alarm. • Erases error message and pop-up message. • Returns previous layer in multi-layer menu. • Erases character input.
RT/REGION key	• Switches to the RT (radiotelephone) screen. • Opens/closes the option window for channel region.
HI/LO key	Changes the output power to high (25 W) or low (1 W).
CH16 key	Switches to the RT (radiotelephone) screen and sets CH16.
0 to 9 keys	• Enter alphabet, numeric or symbol. • Direct selection of corresponding function on menu and applicable screens.
DW/SCAN key	• Opens the option window for DW or SCAN. • Cancels DW or SCAN in process.
REPLAY key ALARM lamps	Opens the replay screen. Top: Flashes in red when receiving distress alert, distress and urgency messages. Bottom: Flashes in green when receiving safety and routine messages, and when daily test is completed. The flashing of a lamp for receiving a DSC message is in synch with the audio alarm. The flashing cycle for both top and bottom lamps is 200 msec (lighting) → 200 msec (off) > 200 msec (lighting) > 200 msec (off) > ...

Turn the power on, or press the **RT/REGION** key to show the radiotelephone (RT) screen. This is where you set up the transceiver unit, and communicate by voice



No.	Meaning
1	Loudspeaker on () or off ()
2	This icon appears when the channel is busy.
3	Intensity of reception (This icon does not appear while transmitting.)
4	Own ship's MMSI (nine digits)
5	Spinner rotates when the equipment is functioning normally.
6	This icon appears when the frequency is for distress.
7	Channel
8	Output power ([HIGH], [LOW])
9	Channel region ([INT'L], [USA], [CANADA], [INLAND-W], [PRIVATE])
10	Channel type ([SIMP]: Simplex, [DUP]: Duplex)
11	Guidance: Rotate the CHANNEL/ENTER knob to select channel.
12	Guidance: Press the TAB key to switch the session.
13	Own ship's position (LAT: Latitude, LON: Longitude) Method of data input [EPFS DATA]: The position and time data from EPFS. [EPFS (OFFLINE)]: Indicates no position data from EPFS for ten minutes. [EPFS (OLD)]: Indicates no position data from EPFS for four hours. [MANUAL INPUT]: Set the position and time data manually. [NO INFO]: No position and time data.
14	Time (UTC: universal time coordinated) of the position fix
15	



SAFETY INSTRUCTIONS

The user must read the appropriate safety instructions before attempting to install or operate the equipment.



WARNING

Indicates a condition that can cause death or serious injury if not avoided.



CAUTION

Indicates a condition that can cause minor or moderate injury if not avoided.



Warning, Caution



Prohibitive Action



Mandatory Action



WARNING



Do not open the equipment.

Hazardous voltage which can cause electrical shock, burn or serious injury exists inside the equipment. Only qualified personnel should work inside the equipment.



Do not approach the antenna closer than listed below when it is transmitting.

The antenna emits radio waves that can be harmful to the human body.

RF power density on antenna aperture	Distance	Description required by
100 W/m ²	0.12 m	IEC 60945
10 W/m ²	0.39 m	IEC 60945
2 W/m ²	0.80 m	MPE by FCC

(MPE: Minimum Permissible Exposure)



Immediately turn off the power at the switchboard if water leaks into the equipment or the equipment is emitting smoke or fire.

Continued use of the equipment can cause fire or electrical shock. Contact a FURUNO agent for service.



Do not disassemble or modify the equipment.

Fire, electrical shock or serious injury can result.



Do not place liquid-filled containers on the top of the equipment.

Fire or electrical shock can result if a liquid spills into the equipment.



WARNING



Do not operate the equipment with wet hands.

Electrical shock can result.



Turn off the power immediately if you feel the equipment is behaving abnormally.

Turn off the power at the switchboard if the equipment becomes abnormally warm or is emitting odd noises. Contact a FURUNO dealer or agent for advice.



Make sure no rain or water splash leaks into the equipment.

Fire or electrical shock can result if water leaks in the equipment.



Use the proper fuse.

Use of the wrong fuse can cause fire or electrical shock.



Any repair work must be done by a licensed radio technician.

Improper repair work can cause fire or electrical shock.



Do not operate the [DISTRESS] key except in case of a life-endangering situation on your vessel.

Operating the [DISTRESS] key transmits the distress alert. Accidental transmission may prevent search and rescue operations for actual emergency. If the distress alert is accidentally transmitted, contact the nearest station to cancel the alert.

⚠ CAUTION



If the distress alert is accidentally transmitted, contact the nearest coast station and inform them of the accidental transmission, providing the following data:

- a) Ship's name
- b) Ship's call sign and DSC number
- c) Position at time of transmission
- d) Time of transmission

⚠ CAUTION



Do not touch any part of the antenna when the equipment is transmitting.

Electrical shock can result.



Do not apply strong pressure to the LCD, which is made of glass.

Injury can result if the LCD breaks.

WARNING LABELS

Warning labels are attached to the equipment. Do not remove any label. If a label is missing or damaged, contact a FURUNO agent or dealer about replacement.



Name: Warning Label 1
Type: 86-003-1011-3
Code No.: 100-236-233-10



Name: Warning Label
Type: 14-055-1202-1
Code No.: 100-245-221-10



Name: High Temp Warning Label
Type: 05-009-2142-0
Code No.: 100-301-620-00

About the TFT LCD

The TFT LCD is constructed using the latest LCD techniques, and displays 99.99% of its pixels. The remaining 0.01% of the pixels may drop out or blink, however this is not an indication of malfunction.

DISTRESS ALERT

How to send a distress alert

Below is the procedure for transmitting a distress alert via radiotelephone. Transmit the distress alert when a life-endangering situation occurs on your vessel.

1. Open the **DISTRESS** key cover then press and hold the **DISTRESS** key for four seconds. The following screen appears.

Y141 [DIST] [HIGH] [RAMP] CH: 16

COUNTDOWN BEFORE SENDING

MSG TYPE : **DISTRESS ALERT**

DISTRESS ID: **123456789**

NATURE : **UNDESIGNATED**

LAT : **34°42.2800'N**

LOX/UTC : **135°19.5900'E / 12:34**

COMM MODE : **TELEPHONE**

DISTRESS BUTTON PRESSED!
KEEP PRESSED FOR 2 s

Countdown message

2. When the message "Sending DISTRESS ALERT." appears on the screen, release the **DISTRESS** key. The audio alarm sounds for two seconds.

DSC SEND

Sending
DISTRESS ALERT.

After the distress alert has been sent, the following screens appear in order.

Y141 [DIST] [HIGH] [RAMP] CH: 16

WAIT FOR ACK ELAPSED TIME: 00:00:00.55

DISTRESS ALERT RESEND: 10:00:00.55

DISTRESS ID: 123456789

NATURE: UNDESIGNATED

COMM: CH 16/TELEPHONE

LAT/LOX/UTC: 34°42.2800'N/135°19.5900'E/12:34

OWN DISTRESS

When distress acknowledge call is received by coast station

Y141 [DIST] [HIGH] [RAMP] CH: 16

ACKNOWLEDG-01 ELAPSED TIME: 00:00:00.55

DISTRESS ACK

PRIORITY: DISTRESS

DISTRESS ID: 123456789

FROM: DISTRESS ACK

NATURE: DISTRESS ACK

LAT/LOX/UTC: received! 12:34

COMM: CH 16/TELEPHONE

EOS: Stop alarm

OWN DISTRESS

3. The audio alarm sounds. Press the **CANCEL** key to silence the audio alarm.
4. Communicate with the coast station via radiotelephone (CH16) as below.
 - a) Say 'MAYDAY' three times.
 - b) Say 'This is ...' name of your ship and call sign three times.
 - c) Give nature of distress and assistance needed.
 - d) Give description of your ship (type, color, number of persons onboard, etc.).

Note: If you do not receive the distress alert acknowledge call, the equipment automatically re-transmits the distress alert after 3 min 30 seconds to 4 min 30 seconds. Then the equipment awaits the distress alert acknowledge call. This is repeated until the distress alert is acknowledged.

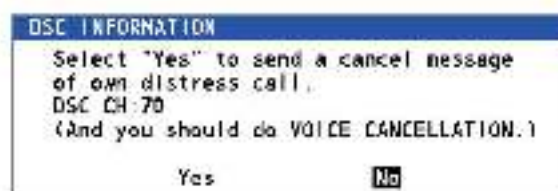
How to cancel the distress alert

You can cancel the distress alert while it is being sent or while waiting for its acknowledgement as follows.

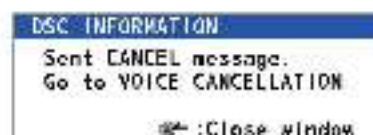
1. Rotate the **CHANNEL/ENTER** knob to select [CANCEL] in the user options area then push the knob.



The following message appears on the screen.



2. Rotate the **CHANNEL/ENTER** knob to select [Yes] then push the knob to send the distress cancel call on CH70. After transmitting the distress cancel call, the following message appears on the screen.



3. Push the **CHANNEL/ENTER** knob to erase the message. The following message appears on the screen.



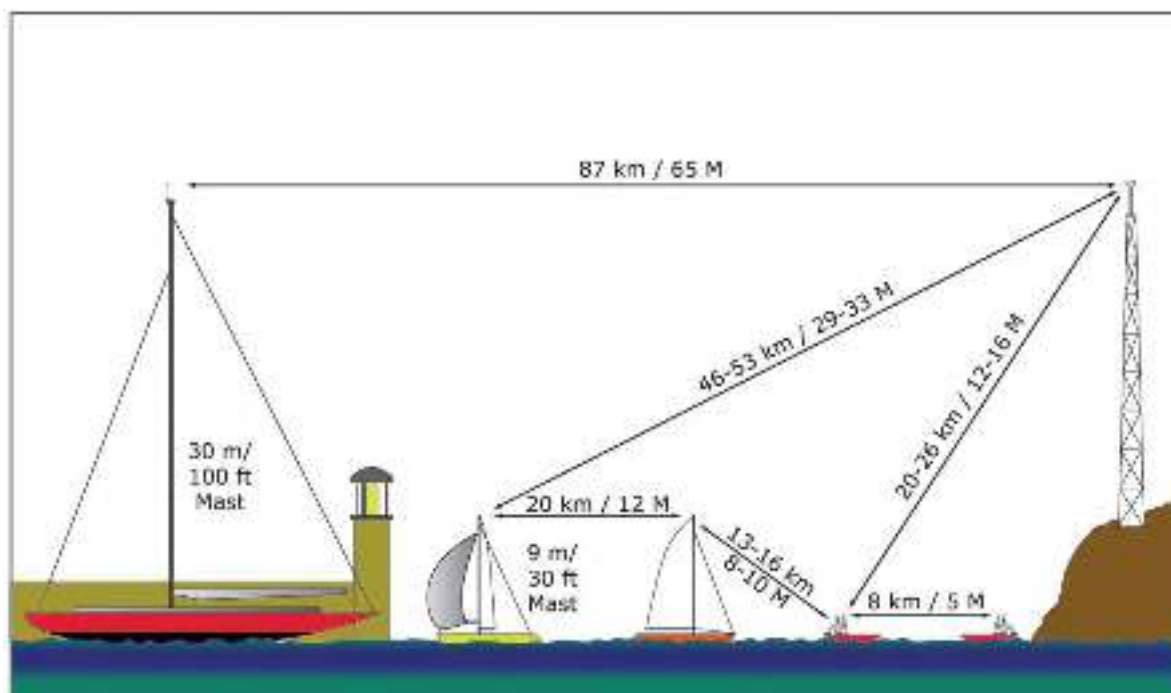
4. Communicate with all ships via radiotelephone referring to the message at step 3.
5. Push the **CHANNEL/ENTER** knob. The following message appears on the screen.



6. With [Finish] selected, push the **CHANNEL/ENTER** knob.

Transmission range

Because VHF signals travel in a straight line and are not reflected back off the ionosphere as lower frequency signals are, the range of VHF signals is limited to 'line of sight', beyond which the other vessel passes behind the curve of the Earth. Therefore, the range will increase greatly the higher above sea level the antenna is, as shown in figure (assuming maximum transmission power is used):



The typical ship to ship range of a fixed VHF radio such as the RS82 with a masthead antenna will be approximately 20 Km (12 miles). This will increase as height above sea level increases, or if the other radio user's antenna is at a greater height - note that the range between the yacht with the antenna mounted on a 9 m (30 Ft) mast and the shore station increases to 46-53 Km (29-33 Miles).

MAINTENANCE & TROUBLESHOOTING

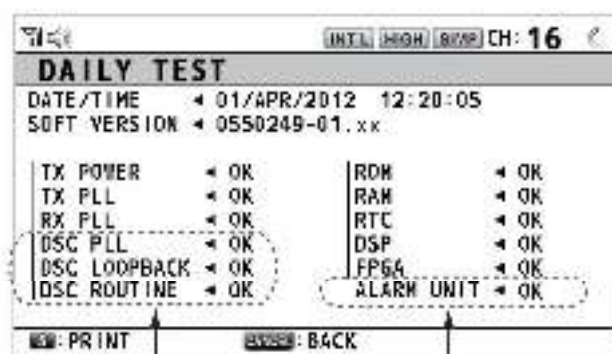
 WARNING	NOTICE
 ELECTRICAL SHOCK HAZARD Do not open the equipment. Only qualified personnel should work inside the equipment.	Do not apply paint, anti-corrosive sealant or contact spray to plastic parts or equipment coating. Those items contain products that can damage plastic parts and equipment coating.

Daily Test

Do the daily test to check the radiotelephone for proper operation.

Rotate the **CHANNEL/ENTER** knob to select [TEST] on the [MENU] screen then push the knob. The daily test starts. After the test is completed, the audio alarm sounds and the screen shown below appears. This screen shows:

- Test date
- Program version number
- Test results for TX power, TX PLL, RX PLL, DSC PLL, DSC loopback, DSC routine, ROM, RAM, RTC, DSP, FPGA and alarm unit (only when connecting the alarm unit), shown as [OK] or [NG] (No Good). For NG, contact your dealer for advice. The DSC test checks, using a DSC signal, the encode and decode functions of the signal processor.



INTL HIGH BWR CH: 16

DAILY TEST

DATE/TIME ◀ 01/APR/2012 12:20:05

SOFT VERSION ◀ 0550249-01.xx

TX POWER ◀ OK	ROM ◀ OK
TX PLL ◀ OK	RAM ◀ OK
RX PLL ◀ OK	RTC ◀ OK
DSC PLL ◀ OK	DSP ◀ OK
DSC LOOPBACK ◀ OK	FPGA ◀ OK
DSC ROUTINE ◀ OK	ALARM UNIT ◀ OK

PRINT BACK

These items are not available when not entering own ship's MMSI.

This item is not available when not connecting the alarm unit.

To print out the test result manually, press the **5** key. Automatic printing of the daily test is available.

Maintenance

Regular maintenance helps to keep your equipment in good condition and prevents future problems. Check the items shown in the table below.

Item	Check point	Remedy/Remarks
Antenna	Check for physical damage and corrosion.	Replace damaged parts.
Wire antenna	Check that the antenna is properly spanned and separated sufficiently from metallic structures.	If necessary, re-span antenna.
Insulators for antenna	Check for salt water deposits on insulators. Check that connection at the lead-in insulator is tight and rust-free.	Replace damaged insulator(s). Remove salt water deposits. Clean with fresh water, then dry. Remove rust, then tighten bolts and lock nuts. Cover metallic surface with sealing compound.
Transceiver unit	• Check ground connection. • Check connection at signal cable, coaxial cable, control cable, power cable and external equipment (including navigator). • Confirm that there are no objects on the top of the transceiver unit. • Remove dust from transceiver unit with soft cloth. Note: Do not use chemical cleaners to clean the transceiver unit; they can remove paint or markings and deform the equipment.	• Tighten the loosened connections; remove foreign materials from connectors. • Remove any objects. • Wipe the LCD carefully to prevent scratching, using tissue paper and an LCD cleaner. To remove dirt or salt deposits, use an LCD cleaner, wiping slowly with tissue paper so as to dissolve the dirt or salt. Change paper frequently so the salt or dirt does not scratch the LCD.
Power supply	Check that the supply voltage at transmission is within the rated range (21.6 to 31.2 VDC at the power connector).	If not within the range, check power source. Low voltage may cause erratic operation.
Power fuse	Check if a power fuse (7A) has blown.	If the fuse has blown, find out the cause before replacing it (Type: FGBO 125V 7A PBF). If it blows after replacement, contact your dealer.

Simple Troubleshooting

The table below provides possible problems and the means with which to restore normal operation. If normal operation cannot be restored, do not attempt to check inside the equipment. Any servicing should be referred to a qualified technician.

Problem	Probable cause	Remedy
Power cannot be turned on.	• Mains switchboard is off. • (DC) voltage is too high or too low. • Battery has discharged, or poor contact at terminals.	• Turn on the mains switchboard. • Check supply voltage. • Recharge the battery and tighten the battery terminals.
Display indications do not appear.	Display brilliance is too low.	Press the BRILL key to adjust the display brilliance.
Power is on but no sound from the main speaker.	Main speaker is off.	Press the  key to turn on the main speaker.
Output power reduced to LOW	Power is automatically reduced to protect against overheating due to continuous transmission.	Wait until the unit cools.

Warning and Caution Messages

The table below shows error messages, their meanings, and remedies. To delete the messages, press the **CANCEL** key.

Message	Meaning	Remedy
WARNING TX PLL UNLOCK! CH:XX Unable to transmit. [CANCEL]: Stop alarm	TX PLL unlock. Transmission is disabled.	Contact your dealer.
WARNING Unable to transmit! Hardware error occurs. Check alarm status. [CANCEL]: Close window	TX PLL unlock error. Transmission is disabled.	Contact your dealer.
WARNING RX PLL UNLOCK! CH:XX Unable to receive. [CANCEL]: Stop alarm	RX PLL unlock. Reception is disabled.	Contact your dealer.
WARNING DSC PLL UNLOCK! CH:70 Unable to receive DSC message. [CANCEL]: Stop alarm	DSC (CH70) PLL unlock.	Contact your dealer.
CAUTION EPFS error! [CANCEL]: Stop alarm	Position data is not input for ten minutes. Note: This message does not appear when [INPUT TYPE] in the [POSITION] menu is set to [MANUAL].	Enter the position.
CAUTION Position data is not updated. Position was older than 4H. Update it. [CANCEL]: Stop alarm	Position data has not been updated for 4H.	Enter the position.
CAUTION Lost position. Position was older than 23.5H. You must update position! [CANCEL]: Stop alarm	Position data has not been updated for 23.5H.	Enter the position.
CAUTION The unit will not transmit any DSC call until own ship's MMSI is entered. [CANCEL]: Stop alarm	You tried to send a DSC message but your MMSI has not been registered in the equipment.	Enter MMSI no. of your ship.
CAUTION Cannot print. Check printer. [CANCEL]: Stop alarm	Printer trouble. Cannot print.	Check the printer (connection, power, paper).
CAUTION Communication error! [CANCEL]: Stop alarm	Communication between the transceiver unit and a remote handset is lost for three seconds.	Check the connection with the remote handset. If unsolved, contact your dealer.
CAUTION TX power reduced. RF AMP heated. [CANCEL]: Stop alarm	RF amplifier is too hot. Transmission power is reduced.	Allow the transceiver unit to cool. If the message appears again, contact your dealer.
CAUTION System has rebooted.	Internal error. System is rebooted.	System automatically restarts. If the problem occurs again, contact your dealer.

Mooring and unmooring

Operations



Before each operation, check that engines, rudders, gear boxes, Bow-Thruster work properly.

The control stations steering wheels are not interlocked. However when manoeuvring and cruising, be sure that nobody is in proximity of the station not used and interferes with the commands.

Before starting the operation, make sure that passengers are not in way of the operations and remain in safe locations.



Do not use the swim platform cleats for mooring the yacht, as they should be used only for mooring the tender.

A basic rule to keep in mind is manoeuvring always the boat at low speed, in order to have time to react and evaluate the situation; and, in case of accidental hit with other vessels, no serious damage will occur.

Before unmooring check that:

- no other boat is manoeuvring;
- the mooring lines are not fouled;
- the fenders are correctly positioned and secured (in case of windy or swelling conditions, have a person ready with a fender to avoid damages);
- no floating objects or loose ropes are in your way and damage the propellers;
- the shore power supply cable and socket are stored on board;
- the gangway has been recovered.

If the boat is stern moored to the pier:

- undo the stern lines, haul in the chain, until distant from the pier, and head to the exit.

If the boat is side moored:

- undo the stern lines, haul in the bow line, pushing the stern away from the pier and head to the exit.

Unmooring

The boat is controlled by using the steering wheel that operates the rudders (the steering gear operation is independent from the engine operation) and by using the Bow-Thruster.

If needed and/or in restricted waters, steer the boat with the engines (varying rpm and/or reversing the gear box).

Do not leave the steering wheel unattended, particularly at high speed or in restricted waters.

Do not exceed the speed limits when operating in confined waters, harbours and wherever mandated.

Keep in mind that the rudder effect is proportional to the propeller rpm and the boat way, especially when running ahead.



Unplug the shore power supply cable.

Mooring

Before entering the harbour:

- check that the mooring lines and fenders are clear and ready to be used;
- check that the mooring berth and the berthing route are free from incoming, leaving or moored vessels;
- switch all necessary equipment on from the switchboard (anchor windlass, Bow-Thruster, etc.). Switch all equipment not required off;
- before reaching the harbour, stop in free waters and check the operation of reverse gear, Bow-Thruster;
- have the boat hook handy without obstructing any passage;
- check the operation of horn and swinging projector (in case of night mooring);
- in case of night mooring, have working torch light handy;
- make sure that passengers are of no obstacle to the operations and, if participating, they know whom to listen to, what to do and where to stay;
- check that black and grey water tanks have been emptied.

Reduce the speed when approaching the berth.

If the boat is to be stern moored to the pier:

- haul on the stern lines and the chain, until the boat is close to the pier. Use the engines for brief periods, avoiding high speeds and disturbing yachts moored nearby. Use the Bow Thruster if necessary.

If the boat is to be side moored:

- haul on bow and stern lines, until boat is close and parallel to the pier. Use the Bow-Thruster if necessary.

Once moored:

- shut the engines down, proceeding as described in the previous description;
- check that all indicator lights on the panel are off and remove the ignition keys;
- de-energise all unnecessary electrical equipment and check the switchboard status, as well as the voltmeters and amperometers readings;
- check the bilge pumps switch status and set them to automatic operation;
- connect the shore power supply;
- shut the generators down, once cooled down;
- wash the boat by using fresh water.
- Before leaving the yacht for a long time:
- check that the below deck lights are off;
- check that the navigation lights, projectors and external lights are off;
- check that anchor lights are on;
- check that all unnecessary equipment switches (plotter, radio, anchor windlass, etc.) are off;
- check that all equipment necessary for boarding and leaving the yacht are on and fixed;
- check that shore power cable is properly connected and has enough slack;
- check that all deck equipments (life jackets, boat hook, torches, etc.) are stowed correctly;
- check that no flammable liquid bottles and containers are open or loose;
- check that there are no food residues, which might go bad or clog the scuppers;
- check that the gangway is in the right position and properly secured;
- check that the boat is adequately moored (in case of bad weather, reinforce the mooring lines and make sure that the distance from other vessels is safe; and the fenders are properly fastened, etc.);
- disconnect the battery isolator switches of the starter motors, of the starters generators and electronic inspection/control systems, by moving them on the OFF position;
- make sure that all portholes, internal doors and below deck rooms are closed.

Anchoring

Anchoring and mooring operations are extremely important for the safety of your yacht, therefore experience and particular attention is required.

When anchoring and mooring, remember that the yacht may suddenly change direction due to gusts, sea conditions or waves caused by other yachts.

The anchor windlass can be controlled from the flying bridge, by means of switches. A windlass pendant pushbutton located near the equipment allows local control during anchor weighing/dropping.



Before using the anchor, remove the chain stopper, make sure that wildcat clutch is properly tighten and test anchor windlass operation in both directions from the control station that you intend to use.

The anchor windlass system is not equipped with safety limit switches, therefore it is advisable to operate the last metres of the chain from the local pushbutton panel, both when weighing the anchor and when almost the whole chain is released.

To avoid anchor windlass overheating, it is advisable to ease the weighing by moving the yacht slowly towards the chain, by using the engines, without overtaking the anchor to avoid hull damages.



It is advisable to operate the anchor windlass while the engines are running, in order to ensure availability of the great electric power required.



During windlass operation, be extremely careful of rotating components; keep them at a safe distance from your body and other objects.

Make sure that the electric motor is de-energised when operating the anchor windlass manually and when using the clutch releasing handle, as other people may accidentally activate the windlass from another control station.

Weighing the anchor

- 1.** Start the yacht's main engines.
- 2.** Make sure that the anchor windlass clutch is closed. The wildcat is coupled to the clutch main shaft. The clutch is engaged/released by operating the special removable handle (to be removed after operation) in the special slot.
- 3.** Operate the anchor windlass control from Your control station and start weighing the anchor.
- 4.** If the anchor windlass is excessively stressed and tends to stop, the anchor may be stranded. In this case the safety system should intervene, protecting the anchor windlass from excessive stress.
- 5.** If the anchor windlass continues to stop after having tried in various directions (moving with the engines), it is advisable to operate the yacht to set free the anchor.
- 6.** Check locally the recovery of the last few metres of chain, to avoid damages caused by anchor movement and swinging.

Dropping the anchor

1. The anchor can be dropped both by using the anchor windlass (with neutral wildcat) or by gravity.
2. To drop the anchor by gravity, hold the anchor with the chain stopper, open the clutch so the wildcat can rotate freely and let the anchor fall into the water. To slow down the anchor movement use the brake which is controlled by the handwheel.
3. To drop the anchor in a controlled manner operate the control pushbuttons. In this way the dropping can be controlled and the chain release is regular.
4. Once the anchor is settled, close the clutch and lock the chain by using the chain stopper.



The chain and the anchor may damage the yacht's bow, if the anchor windlass is operated without particular attention. It is advisable to control operation from the pendant push-button strip, therefore it should be possible to check the anchor speed and the anchor shank entrance/exit into/from the hawspipe. During this operation, excessive anchor skid or incorrect position of the anchor shank may damage the yacht's bow.



The chain is fixed to the yacht by means of the head chain stopper. If anchor weighing is impossible, it is possible to resume navigation by releasing the head chain stopper and letting the chain fall into the sea.

The entire anchorage area must be free of obstacles, in case of sudden variations of wind and/or current direction, above all during night anchorage.

At night, before dropping the anchor, ensure that proper operation of the white riding light.

Before dropping the anchor, check on the navigation chart: anchorage is forbidden in certain areas; in algae seabed, anchorage is unsafe and harmful for the environment; on rocky seabed, the anchor might run aground or get lost.

Anchorage must be carried out while engines are running.

Distance from obstacles or other vessels must be longer than the chain dropped (on 360°).

During anchorage, the windlass should be connected.

Windlass rotation direction should never be suddenly reversed.



During windlass operation, be extremely careful of rotating components; keep them at a safe distance to avoid damages and injuries.

Unattended mooring

If the boat is left unattended, operate as follows:

- close the sea water intakes;
- check the main switchboards and switch all unnecessary equipment off;
- make sure that all portholes, doors, windows and hatch locks are closed;
- make sure that bilges are clean and dry;
- make sure that black and grey water tanks are empty;
- make sure that fuel tanks valves are closed;
- check the shaft line stuffing box; in case of excessive dripping tighten the packing. Empty the bilge, if necessary;
- check that the boat is securely moored;



Disconnect the shore power cable, in particular if the boat is left unattended for a long period.

Re-charge the batteries periodically.

Check sea chests and overboard discharge valves thoroughly, to prevent flooding.

Check the electrical system status thoroughly, to prevent risks of fires.

Check periodically the mooring condition, mooring lines wear conditions and fenders internal pressure.

- after checking the charge level of the batteries, or after these have been charged, disconnect the battery isolator switches of the starter engines, of the starter generators, services and emergency, by moving them on the OFF position.

Leaving the mooring



Disconnect the shore power supply cable.

The boat is steered by using the steering wheel that operates the rudders. The steering gear operation is independent from the engine operation.

If needed and / or in restricted waters, steer the boat with the engines (varying and/or reversing the engine speed).

Do not leave the steering wheel unattended, particularly at high speed or in restricted waters.

Do not exceed the speed limits when operating in confined waters, harbours and wherever mandated.

Keep in mind that the rudder effect is proportional to the propeller speed and the boat way, especially when running ahead. As a consequence, rudder efficiency is high at high speed. On the contrary, when the engines are idling and the headway is low the rudder angle reaction is almost negligible.

Safety equipments

Your Yacht has been equipped by SANLORENZO S.p.A. shipyard with specific equipment to ensure the highest safety protection also in case of major danger.



Please refer to the local Harbour Authority for national prescriptions and Safety Rules variations.



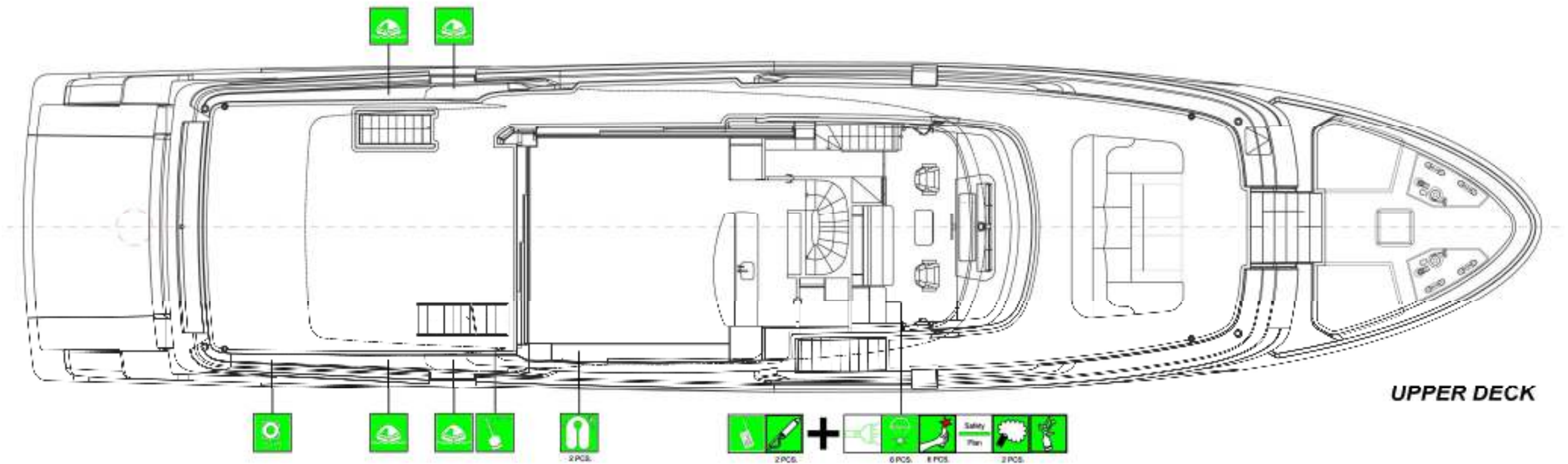
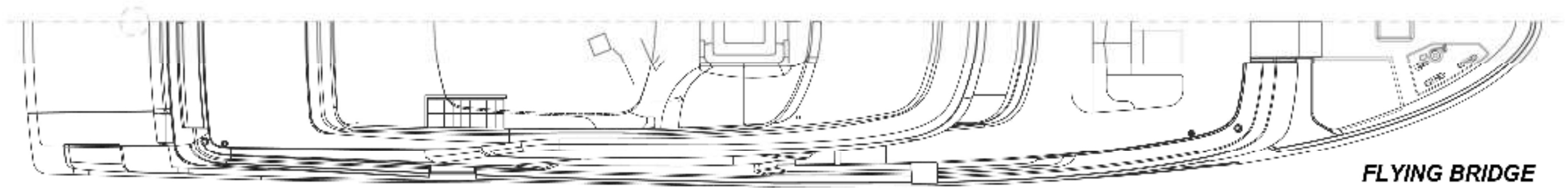
Above mentioned safety equipment should comply with local and international navigation regulations, and must be periodically checked by specialised firms and qualified personnel before the expiry date shown on each device.



*Keeping **escape routes** clear and always accessible may save lives in case of immediate yacht's abandonment.*

Yacht's personnel must make sure that such routes are always kept free and accessible.

Safety Equipment - Flying Bridge / Upper Deck





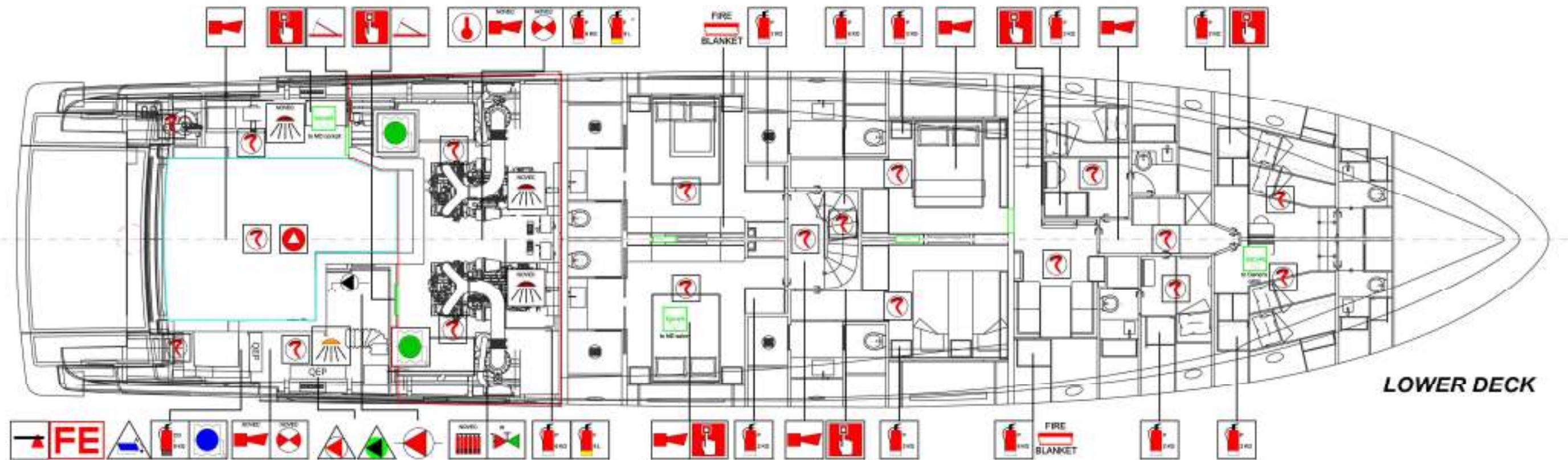
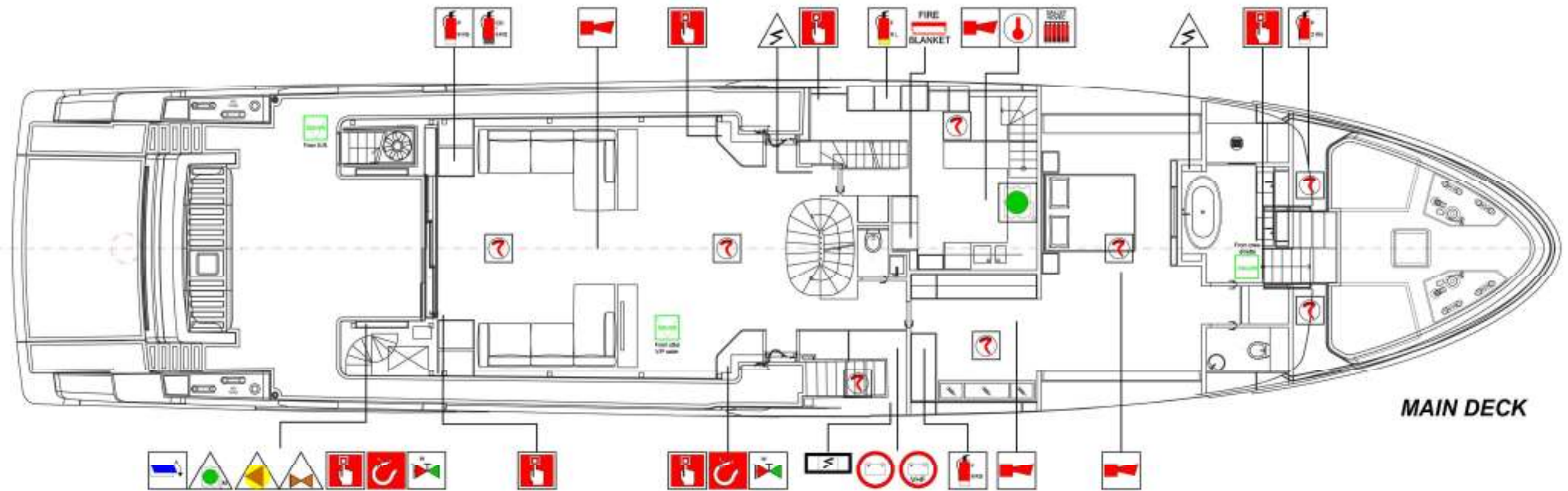
Safety Equipment - Legend

UNDER DECK	LOWER DECK	MAIN DECK	UPPER DECK	FLYING DECK		LIST OF SYMBOLS DESCRIPTION
			1			TWO-WAY VHF RADIO
		1				POSTER: showing survival craft and equipment operating instruction yes training manual and instructions for onboard maintenance life-saving signals and rescue poster
			1			SEARCH AND RESCUE RADAR TRANSPONDER (SART)
		2				LIFEBUOY WITH LIGHT AND SMOKE (M.O.B. SIGNALS)
			2			LINE-THROWING APPLIANCE
			20			IMMERSION SUITS
			2			LIFEBUOY WITH LIGHT
			1			LIFEBUOY WITH 30m BUOYANT LIFELINE
			1			EMBARKATION LADDER
			1			PLAN FOR LIFE SAVING APPLIANCES MEANS OF ESCAPE
			1			MUSTERSTATION
			1			BREATHING APPARATUS
			1			TORCHLIGHT WITH BATTERIES

UNDER DECK	LOWER DECK	MAIN DECK	UPPER DECK	FLYING DECK		LIST OF SYMBOLS
			1			FIRST AID (Kil Tab. A - DM n°279 28/05/1988)
			1			EPIRB
	14	4	2			N° LIFEJACKET WITH LIGHT AND WHISTLE
		2	2			CHILD'S LIFEJACKET
			6			ROCKET PARACHUTE FLARE
			6			RED HANDHELD FLARES
			2			SMOKE SIGNAL
			1			SEARCH AND RESCUE TRANSPONDER (S.A.R.T.)
			4			LIFERAFT (SOLAS "A" pack - 10 PERSONS)
			1			RESCUE BOAT



Safety Equipment - Main Deck / Lower Deck



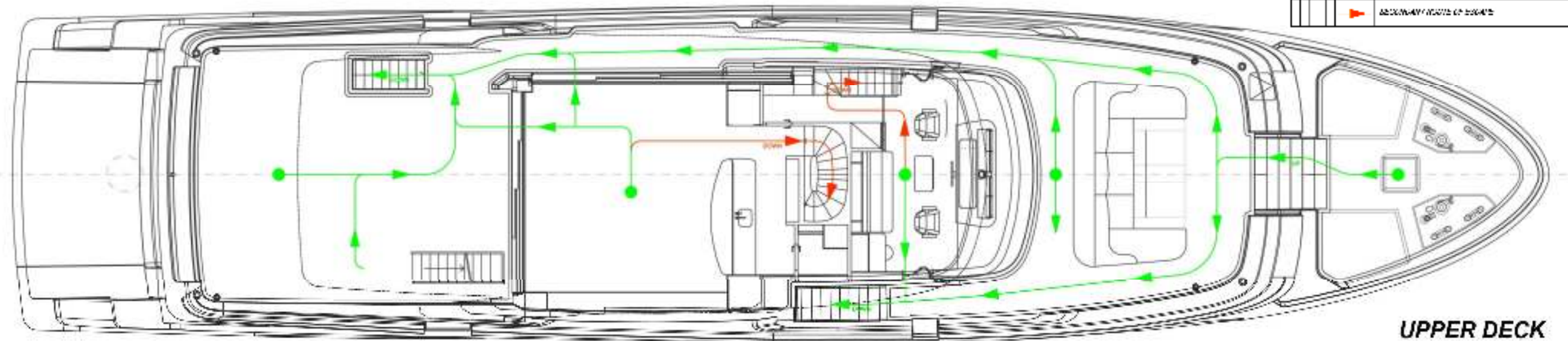
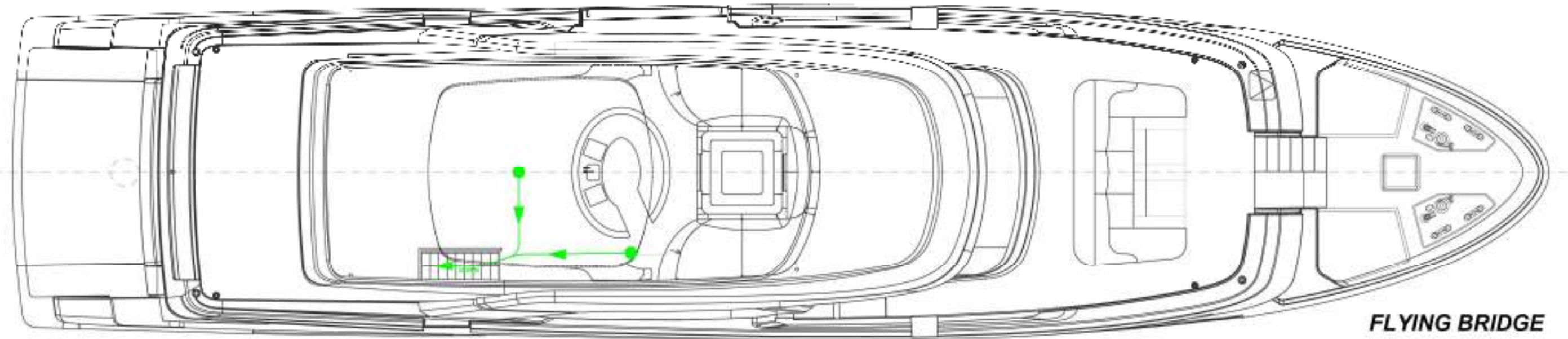
Safety Equipment - Legend

LOWER DECK	LOWER DECK	LOWER DECK	LOWER DECK	LOWER DECK	LIST OF SYMBOLS
					DESCRIPTION
4					B15-CLASS DIVISION
7					B15-CLASS HINGED FIRE DOOR
2	1				NOVEC FIXED FIRE EXTINGUISHING BATTERY
		1			PUSH BUTTON/SWITCH FOR GENERAL ALARM BUTTON
6	6	3			MANUALLY OPERATED CALL POINT
1					GAS DETECTOR
1	1				HEAT DETECTOR
2	6	8	4		SMOKE DETECTOR
7	4	3	1		FIRE HORN
	1	1			EMERGENCY SWITCHBOARD
		1			FIRE ALARM PANEL
2	1				SPACE PROTECTED BY NOVEC EXTINGUISHING SYSTEM
2	1				E.R. FIRE DAMPER
1					FIRE DAMPER (GARAGE)

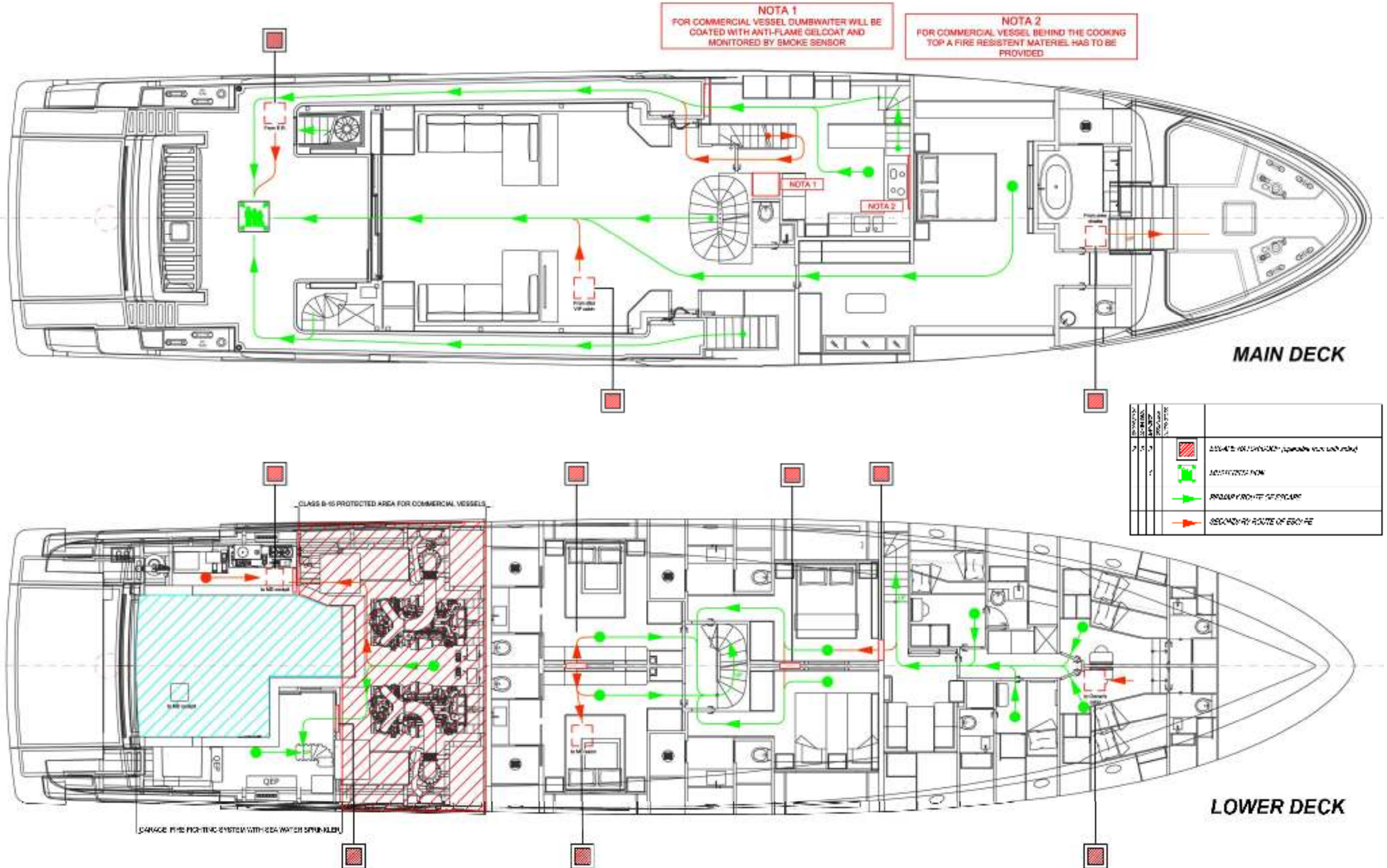
LOWER DECK	LOWER DECK	LOWER DECK	LOWER DECK	LOWER DECK	LIST OF SYMBOLS
					DESCRIPTION
	1				EMERGENCY RADIO VHF BATTERIES
	1				EMERGENCY BATTERIES
1					FIRE PUMP (min 18 m3/hr - max 30 m3/hr)
1					BILGE PUMP
	1				EMERGENCY FIRE/BILGE PUMP (min 15 m3/hr - max 30 m3/hr)
1	3	1			FIRE HYDRANT
	1				CLOSING DEVICE FOR VENTILATION (GARAGE AND GALLEY)
	1				E.R. FIRE DAMPER SHUT-OFF
	2				APPLIANCES POWER SHUT-OFF SWITCH
1					REMOTE CONTROL FOR FIRE PUMP
	1				FUEL PUMPS REMOTE SHUT OFF
1					REMOTE CONTROL FOR BILGE PUMP
	1				FUEL OIL QUICK CLOSING VALVES SHUT-OFF
1					MANUAL CLOSING DEVICE FOR VENTILATION IN-OUT

LOWER DECK	LOWER DECK	LOWER DECK	LOWER DECK	LOWER DECK	LIST OF SYMBOLS
					DESCRIPTION
3	1	2			POWDER EXTINGUISHER - 2 KG.
4	2	1	1		POWDER EXTINGUISHER - 8 KG.
2	1				FOAM EXTINGUISHER - 9 L.
	1				CO ₂ EXTINGUISHER - 5 KG.
1					CO ₂ EXTINGUISHER - 9 KG.
1					SPACE PROTECTED BY FIRE-EXTINGUISHING SYSTEM SPRINKLER OR HIGH PRESSURE WATER SYSTEM
1	3	1			HOSE BOX WITH SPRAY/JET FIRE NOZZLE, 1" M.
1		1			FLASHING LIGHT
		1			PLAN FOR FIRE PROTECTION
2	1				FIRE BLANKET
1					FIREMAN'S OUTFITS : the suit with gloves electrically non-conducting boots and gloves, head hat, fire cape, steel helmet, fireman's axe (insulated handle), approved extinguisher lamp, min parcel 3 pieces (weight 20kg)
1					FIREMAN'S AXE

Escape Routes - Flying Bridge / Upper Deck



Escape Routes - Main Deck / Lower Deck



Portable extinguishers

- Check at least every six months the extinguisher charge status and before use.
- Be cautious when using the extinguisher to fight electrical panels or equipment fires.
- Make sure that the discharge nozzle is not directed towards yourself or anybody else in the vicinity.
- Direct the discharge nozzle towards the fire bottom and open the discharge valve.
- Try to extinguish the flames and cool the equipment.
- After the extinguisher is used to suffocate the flames in closed spaces, carefully ventilate the room before going in to clean up the deposited powder.



The person in charge of the boat must make sure that all passengers know the locations and how to operate the extinguishers.



The signal rockets have an expiry date written on the casing. When reaching the expiring date refer to the rockets' suppliers for disposal.

Do not ignite them unnecessarily, you may alert emergency services without any reason.



Fire prevention rules

A regular and correct maintenance of the systems and a prudent behaviour of all passengers are indispensable measures for preventing any direct or indirect risks of fire.

Nearly all fires, if early detected, can be extinguished easily.

For these reasons, It is necessary to carry out preventive surveys on a regular basis and identify all fire possible sources, and in particular:

- Check the proper operation of all main equipment/systems;
- Visit all compartments and in particular the engine room frequently;
- If a system does not work correctly, identify the failure and take the appropriate corrective actions;
- Operate all systems and equipment as specified.

If a fire is detected, determine and remove the cause, if possible, (i.e. in case of a short circuit, de-energize the electrical system), extinguish the fire promptly and be vigilant to make sure that the fire does not break out again.



Do not use water for extinguishing electric or electronic equipment fires.



The ability to operate the fire extinguishers properly plays a key role when fighting a fire.

It is vital that the fire fighting operations are performed by people competent and specifically trained.

The **EUROPEAN REGULATION** classify fires as follows:

Class A: fires of SOLIDS (e.g. wood, plastic, fabric, stuffing, etc.), can be extinguished by using water, portable extinguishers and the dry-chemical fire fighting system (engine room).

Class B: fires of LIQUIDS (e.g. fuel oil, oil, gasoline, etc.), can be extinguished only by using portable extinguishers and the dry-chemical fire fighting system (engine room).

Class C: fires of ELECTRIC equipment (e.g. breakers, connectors, electrical cables, etc.), can be extinguished only by using portable extinguishers and the dry-chemical fire fighting system (engine room).

At all times the yacht must be equipped with portable extinguishers, located as shown on figure "Safety equipment arrangement".

The yacht Owner and the Captain are directly responsible for:

- having the fire extinguishers and fire fighting systems checked for maintenance as scheduled on their label, and replaced with similar or with equivalent or higher capacity equipment, when provided for by the existing rules;
- informing the crew about the location and use of fire extinguishers, fire fighting systems and emergency exits;
- ensuring that fire extinguishers are available also in passenger cabins.

Do not:

- obstruct passageways and emergency exits;
- obstruct the access to safety devices, such as fuel valves, electrical switches, etc.;
- obstruct access to fire extinguishers stowed inside cabinets;
- leave the boat unguarded when burners or heat generating equipment are on;
- use open flames;
- modify electrical or fuel supply systems, without consulting SANLORENZO S.p.A. shipyard beforehand;
- smoke in proximity or when handling flammable materials;
- stow highly flammable materials (such as fuel, diluent, etc.) in proximity of heat sources, such as engines, galley, halogen lamps, etc.;
- stow any flammable material in the engine room. Non-combustible materials may be stowed only if properly rigged, so they do not come accidentally into contact with engine rotating parts, or obstruct access to the engine room;
- keep dirty and do not inspect the bilge frequently for any oil or fuel leakage.

In addition to these requirements, SANLORENZO S.p.A. shipyard recommends what follows:

- do not smoke below deck and in engine room;
- do not drop liquids into bilge and keep it clean, especially in the engine room. In case of fuel leakage from piping of engines or the generators, operate as follows:
 - stop all engines Immediately;
 - locate the leakage origin and, if possible, repair it after having stopped all the fuel flow and closed all supply valves;
 - dry and clean the bilge before restarting the engines, paying attention not to spill fuel into the sea.
- do not stow flammable items in proximity of heat sources.

In case of fire, follow the following procedure:

1. Keep calm and do not spread panic among the passengers.
2. Stop the boat and close the sea intakes and drains.
3. Close the air intake in the engine room.
4. Stop any flammable liquid transfer operation.
5. Locate the fire and its origin.
6. Avoid breathing smoke.
7. Extinguish the fire, by following standard fire extinguishing techniques.



When fighting a fire, the ability to use the extinguishers effectively can make the difference between success and failure.

It is advisable to have the task performed by people qualified to handle unforeseen emergencies of this type.



Do not stow cans of gasoline or inflammable materials in closed spaces. There is a risk of explosion!



The Captain of a leisure craft must be familiar with the basic methods of putting out fires and using the extinguishers.



Never use a water jet to put out electrical or electronic apparatus fires.



Fire is one of the major on-board hazards on all boats. All the precautions needed to prevent fires from breaking out must thus be taken.



In case fire breaks out in proximity of electrical equipment, do not use water, but use the manual dry-powder fire extinguishers only.

After using the extinguishers, leave and ventilate the area immediately before re-entering, in order to prevent asphyxia and physical damages.



In case of a fire, disconnect electrically the area involved immediately the switches and all AC input magneto-thermal switches, to avoid feeding fires and short circuits.



In case of yacht abandonment, close all fuel, oil and black waters valves to reduce environmental pollution.

Collision

Observe carefully the presence of other vessels, in particular at night and when entering the harbour. Remember that stress caused by reflection, movement and vibrations, reduces significantly the reaction time. Study and comply with the navigation code, and be polite.

In case of collision:

- check the people on board and make sure that they are not injured;
- start immediately the bilge pump and, if necessary, start the manual bilge pump, in order to reduce the quantity of water entering the yacht;
- check eventual damages to the yacht's structure and estimate the loss;
- slow down or stop the yacht in order to reduce entrance of water, or maintain your speed, if you can maintain the leak above the waterline.

If the yacht does not sink, remain near it and require assistance, if the damage is serious.

• ALARM SIGNALS FOR THE CREW •	
■ EMERGENCY GENERAL ALARM	→ seven or more short hoats followed by a long one, produced by ship's hoater of horn and other sounds signals
■ MAN OVERBOARD	→ one long hoat
■ SERIOUS FIRE	→ two long hoats followed by quick and continuous alarm sounds
■ SHIP ABANDONMENT	→ prolonged alarm sounds, Captain's order transmitted through interphone

"Man overboard" recovery

A person fallen overboard must be saved before an eventual hypothermia or drowning. The floating time depends on critical conditions of the water, swimming capacity, bulky clothing, poisoning, eventual injuries and debilitation.

The "man overboard" recovery depends on the capacity of controlling the yacht (knowledge of speed and steering capacity of the yacht) and experience of rescue procedures.

Rescue is a combination of actions: reach the victim, establish a contact and recover the person on board.

Reach the victim

- Activate immediately the acoustic alarm and maintain sight contact with the victim.
- Slow down and direct towards the person in the sea. At night, direct the best light source available towards the victim.
- Throw a life belt into the sea. It will be useful to rescuers as a reference.

Contact

- Stop or slow down the yacht and sail around the victim.
- Try to approach going towards the wind or the waves.
- Keep the victim to the wheelhouse side, in order to maintain contact between pilot and victim.
- When you are next to the victim, stop the engines with engaged gear, in order to avoid propellers rotation.

Recovering the victim onboard

Help the victim boarding the yacht. Generally, the victim must be recovered onboard from the stern. If he is injured, an expert rescuer may wear a floating rescue equipment, with a safety line mounted, and dive into the sea in order to assist the victim.

Drowning

- Try to reach the drowning person as soon as possible, or throw a life belt. Swim towards a drowning person only as the last resource.
- Move the injured person with great care, as the backbone may be fractured.
- Try to resuscitate the victim. At least two people onboard should have the specific certification for cardio-pulmonary resuscitation.
- Keep the victim warm and require immediate assistance.

Running aground

It is possible to avoid running aground paying attention to signalling buoys and waves that generates above a sandbank.

If your yacht runs aground, the seriousness of the situation depends on the violence of the impact, and if the yacht remains grounded. In case of a single shock, check the hull and the propulsion equipment. If the yacht remains grounded, estimate the situation before reacting. In some cases, moving the yacht in reverse gear may cause further damages.

If you run aground:

- check eventual damages to the hull, to propulsion equipment and to steering systems;
- check the absence of leaks. If you are swamping water, it is more important to try to stop water entrance than trying to set the yacht free;
- estimate the water depth around the yacht and the type of seabed (sand, mud, rocks, etc.), in order to decide how to move the yacht;
- consider whether tide or stream may help to set the yacht free or if they contribute to further grounding.

Exhaust gas emissions

The main engines exhaust duct system, before outboard discharge, provides mixing the gasses with cooling water, by means of a rain chamber system (a ring, concentric to the gas duct, with small holes for water delivery and consequent mixing with exhaust gasses), in order to reduce:

- gas temperature near the hull opening;
- gas specific volume;
- gas smoke;
- exhaust noise.

Propulsion engines exhaust ducts are made in stainless steel and do not create counter-pressures on the engines exhausts, in all operative conditions and speeds.

In particular, each engine is equipped with two exhausts: a main submersed exhaust for normal speed and a secondary bulwarks exhaust located in the lateral basket for reverse and low speeds.

In particular, main engines submersed exhaust has been located in the most suitable position for minimising both turbulence induced by the exhaust and propellers efficiency.

The bulwarks exhaust is used in low speed conditions, when exhaust gasses counter-pressure is low.

All outboard discharges, both above and below the water line, are manufactured to ensure high efficiency, low visibility and cleanliness (they do not causes dark shades on the yacht's bulwarks).

All gas exhausts are equipped with: structures and components resistant to high temperature; flexible ducts supports to avoid weight and vibration stress. The ducts are externally covered with ceramic fibre insulating material, lined with stainless steel.

Generator exhausts are equipped with devices which separate water from residual gasses, before being discharged outboard.



Due to the particular shape of the submersed exhaust, do not increase the speed when cruising in reverse direction, preventing sea water from entering the discharge.

*When starting up the engines, make sure that water flows through the minimum exhaust: this is a symptom of engines cooling and exhaust correct operation, as the same water cools down also the exhausts. If water does not flow through the exhausts, increase the speed.
If the failure persists, refer to the Service Assistance.*

Strong smell and smoke traces coming from the exhaust insulation during the first start up should be considered normal.

Towing in case of failure

In case of towing another yacht, it is advisable to pass the towing rope through the cockpit fairleads and fixing both ends to the stern bollards.

The towing rope should be fixed in the middle of the above mentioned bight rope by means of a bowline knot.

The towing rope length should be set according to the sea condition and to the towed vessel weight, and in any case it should not be less than three times the towed vessel length.

If your yacht is unable to sail with its own means and needs to be towed, pass a rope through fore fairleads fixing both ends to the bow bollards.

The towing rope should be fixed in the middle of the above mentioned bight rope by means of a bowline knot.

The towing rope length should be set according to the sea condition and in any case it should not be less than three times the length of your yacht.



If it is necessary to tow another vessel, such operation is possible only with calm sea and calm wind, and only for boats which displacement is less than Your yacht. In case of emergency, if towing is not possible, bring help by taking on board the people from the other vessel, as much as provided and possible, and go to the nearest port.

In any case, inform immediately the Coast Guard.

Do not stand near the ropes during drawing or towing operations, since a potential breaking might be extremely dangerous.

During towing, check continuously the exhaust temperature, cooling water temperature and oil pressure of propulsion engines.

The towing vessel should maintain suitable speed and course according to the type of operation.

Environmental pollution prevention rules

Environmental pollution includes three categories:

- water pollution (caused by: waste disposal and liquids discharge into the sea);
- air pollution (caused by; noise, exhausts gasses);
- soil pollution (caused by: waste discharge to land).

Non-oily black waters (i.e. those waters containing human organic waste only) can be discharged in the open sea. In port, such waters should be collected in tanks and then discharged in the open sea, directly or into shore or trucked collecting systems.

Soil pollution is due to waste discharge to land.

91/689 EEC directive concerning disposal of hazardous waste, referring to pleasure boats, states in essence what follows:

- during navigation, do not discharge, in the open sea, any commercial or non-biodegradable substance;
- in port, normal waste is considered as urban waste; and must be closed hermetically in plastic bags and thrown into waste dumpsters.

Special waste include:

- oily waters and mixture (i.e. bilge water);
- oils (fuel, additives, and lubricants);
- toxic-noxious substances (such as battery acid, paints, anti-fouling, diluent, and their containers);
- spray cans containing C.F.C gases;
- batteries and exhausted batteries;
- expired signal flares;
- expired medical products;
- products containing lead, asbestos, etc..

Be aware that, as mandated by EEC regulations, you will be considered the holder of such waste until delivered to a Disposal Company authorised and persecuted in case of illegal dumping. If, in the harbour area, there are no proper dumpsters, the responsible authority for waste disposal is the Port Authority "Waste management" department.

In particular, some typical form of pollution caused by the yacht should be considered:

- noise delivered in the air and in the water;
- waves caused by hull and propellers;
- exhaust gasses emission;
- use of paints, detergents and other products for yacht's preservation, maintenance and efficiency.



Special waste should be placed into dedicated containers, or delivered to local authorised waste Disposal Company, in accordance with the requirements of the local Port Authority.

Risk areas

On the yacht there are some areas where must be very careful and in case wear proper individual protective means, in order to safeguard health of persons on board.

These areas are:

Engine room: high noise area, presence of moving parts, burning and falling risk.

Swimming platform: external area not protected against falling into the sea. While underway, this is a **"RESTRICTED AREA"**.

Flybridge/Sun pad area: area having falling protections short in height and extension.

In case of man overboard, the yacht is equipped with the following recovery/rescue means:

- Nr. 2 Life buoy with light and 30 mt. buoyant line.



During the navigation with bad weather conditions, do close all doors and portholes with the proper systems.



The access to the **"restricted areas"** of the superstructure roof is allowed only to crew members that wear proper individual protective means.



IMPORTANT: In case of rain do open the Jacuzzi (if present) swimming pool discharge.



IMPORTANT: Use of the Jacuzzi (if present) is only permitted when the boat is stationary in the port or at anchor and only if, under responsibility of the Captain, the weather conditions are considered appropriate. It is strictly forbidden to use the Jacuzzi when under way in any weather conditions.

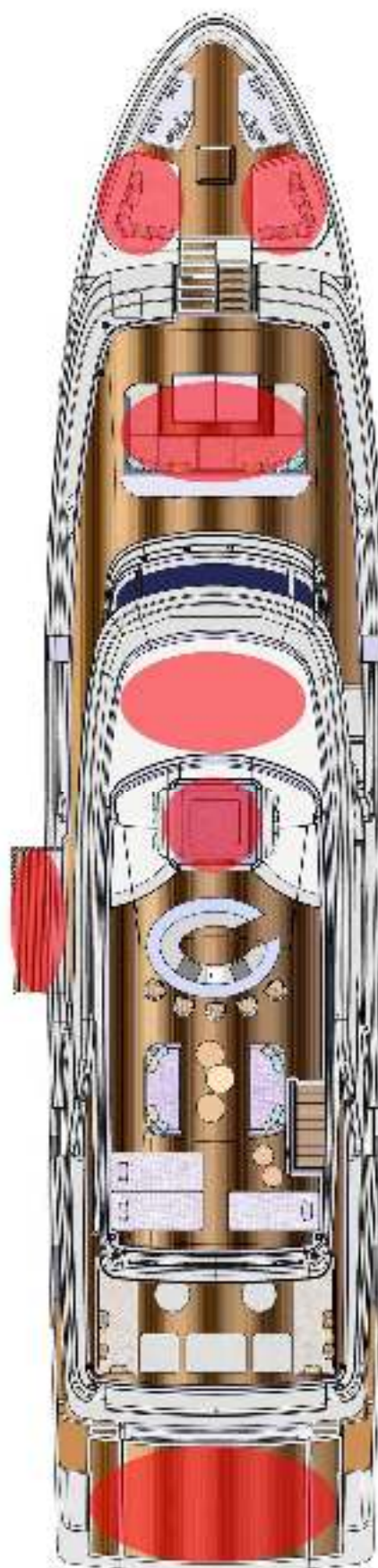


The mobile platform can be used only when it is correctly positioned and yacht has been stopped. It is absolutely forbidden to jump on the platform or load it with a large amount of weight.

Max 5 people (max Kg.400) only with mobile platform open.



IMPORTANT: during the navigation with any weather condition do not stay on the platform.



Transom Equipment Handling

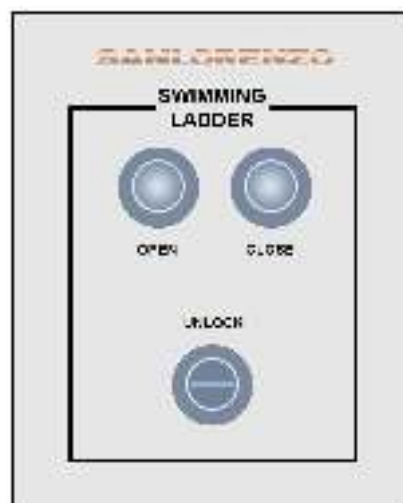
In proximity of the port descent ladders to the stern platform it is located a panel for the command/control of:

- Gangway (Rise - Lower - Out and In);
- Starboard Platform Beach (Open - Close);
- Port Platform Beach (Open - Close);
- Swimming Platform (Rise - Lower - Unlock);
- Garage (Open - Close);
- Lights (Cockpit - Stairs - Beach - Sub).

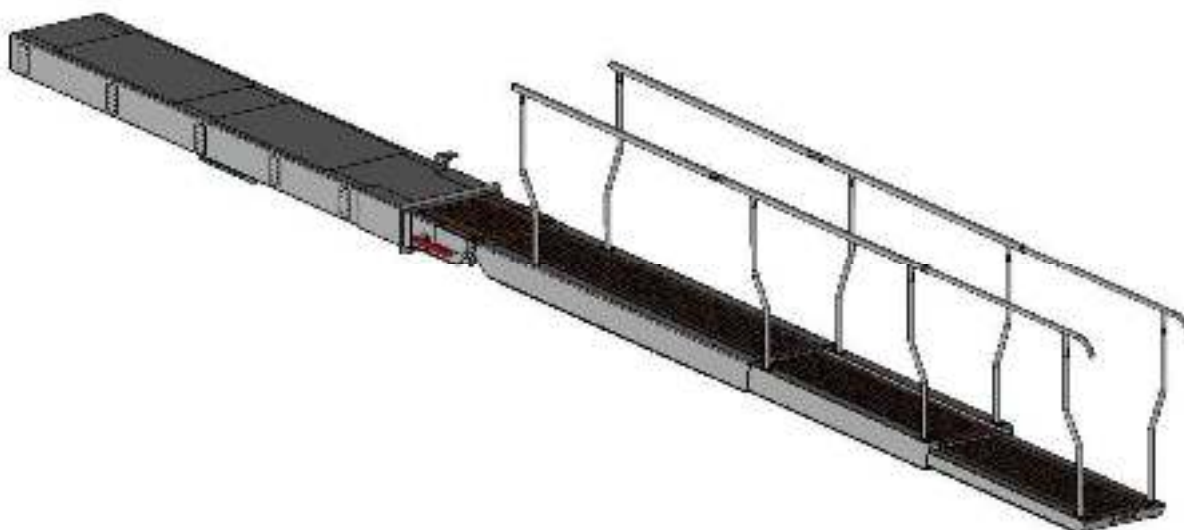


In proximity of the starboard descent ladders to the stern platform it is located a panel for the command/control of:

- Swimming Ladder (Open - Close - Unlock).



Gangway



Characteristic Data

Dimensions overboard:	Length 6.17 m
	Width 0.57 m
Maximum inclination:	-10° ÷ +15°
Maximum capacity:	150 kg

After each navigation it is advisable to wash the gangway with fresh water to eliminate the saline deposit and occasionally with neutral soap to eliminate greasy deposits.



Before carrying out any cleaning operation, disconnect the power supply.

Ensure that the work areas are well lit and free from subjects that could be a source of hindrance or danger.



Do not use solvent substances to clean the gangway (danger of pollution and possible damage to the external coating of the materials of the gangway and the boat).

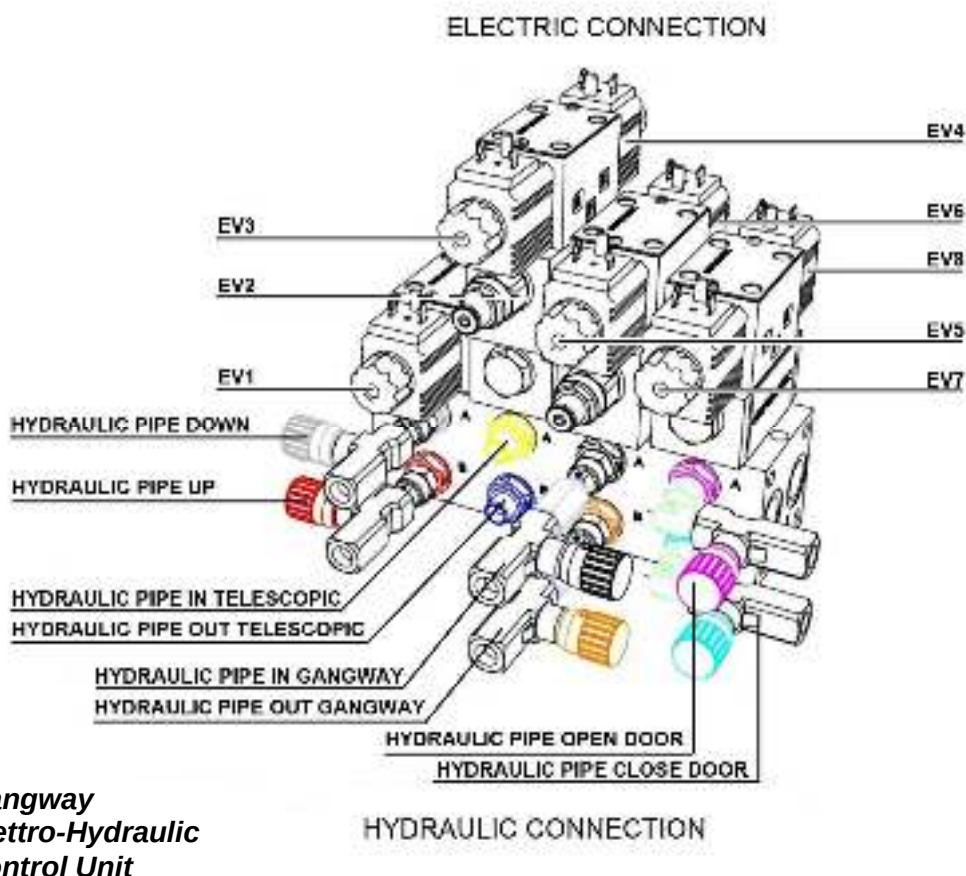
On stainless steel surfaces, avoid the use of abrasive products or objects that would damage the external glossy finish, exposing it to possible surface oxidation.

Gangway - Emergency

If the gangway does not work correctly by acting on the appropriate control panel or by using the supplied remote control, it is necessary to manually act on the solenoid valve block.

WARNING: acting manually on the solenoid valve block, the electrical limit switch are deactivated, therefore proceed with the maximum attention and/or with the help of a second person to view the moviment of the gangway.

- Solenoid Valve N°1 : OPEN Door Gangway
- Solenoid Valve N°2 : CLOSE Door Gangway
- Solenoid Valve N°3 : OUT Gangway
- Solenoid Valve N°4 : IN Gangway
- Solenoid Valve N°5 : OUT Telescopic Gangway
- Solenoid Valve N°6 : IN Telescopic Gangway
- Solenoid Valve N°7 : UP Gangway
- Solenoid Valve N°8 : DOWN Gangway

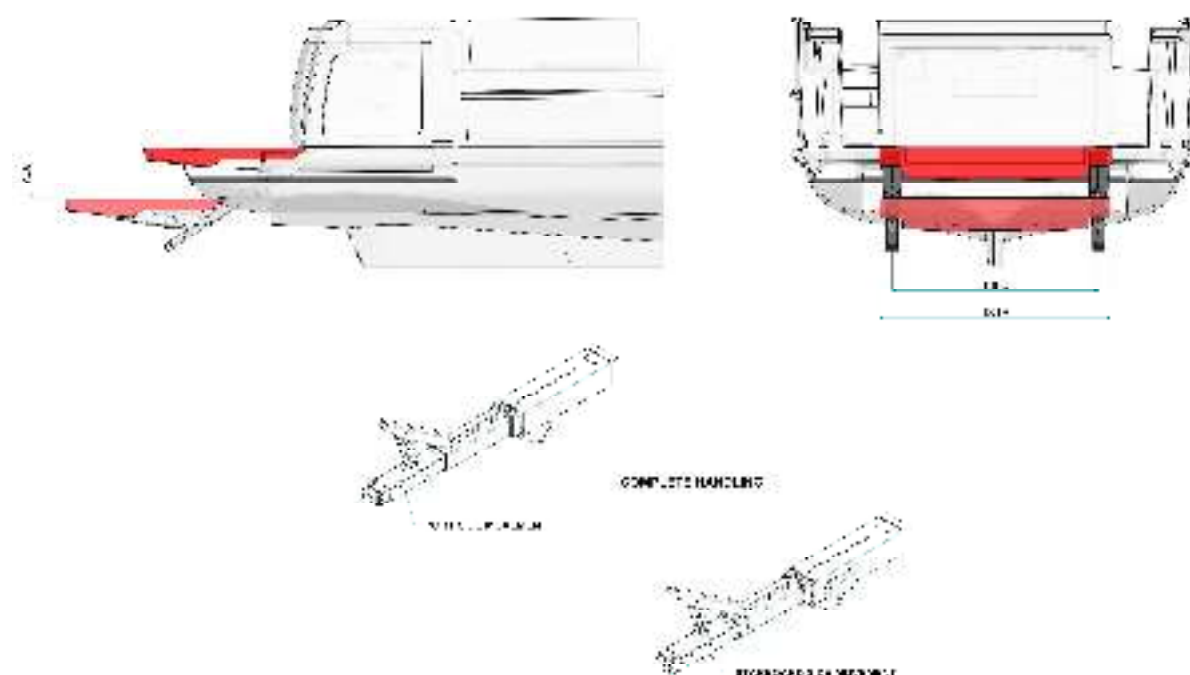
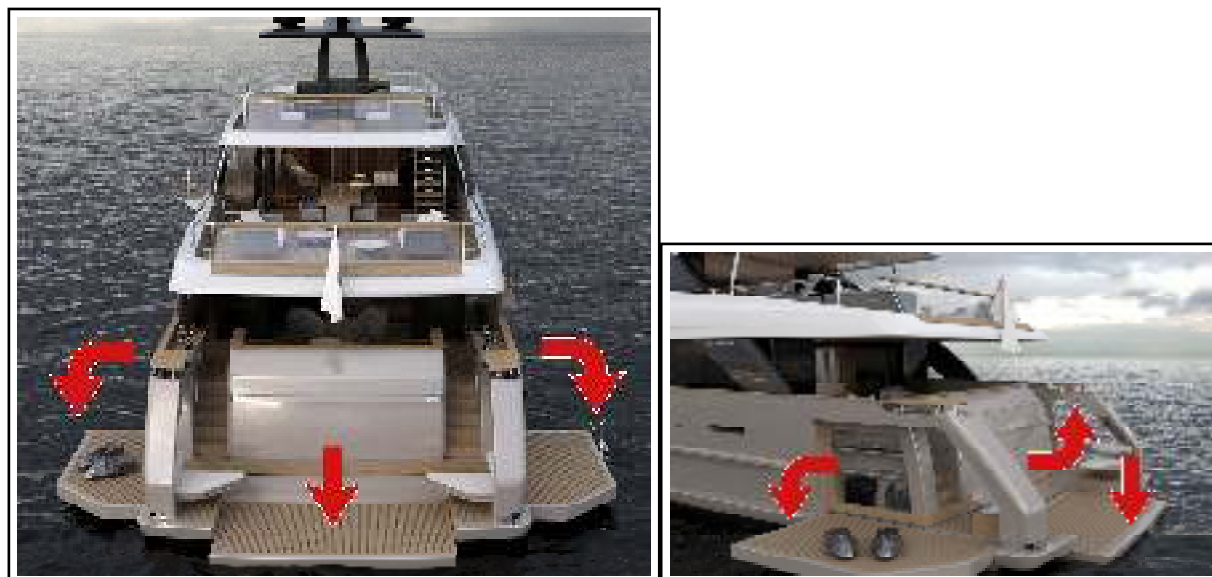


CAUTION: If you want to get the gangway OUT, by acting on the Solenoid Valve N°3, proceed with particular attention, the UP and DOWN pistons could be discharged (not under pressure) and therefore the gangway could fall down.

Starboard-Port Platform Beach & Swimming Platform

In addition to the swimming platform, this Yacht is equipped with two side beaches that can be controlled from a dedicated panel.

Always from the same panel it is possible to control the movement of the swimming platform. It is possible to lower the swimming platform by about 1 meter to facilitate the exit of the tender from the garage.

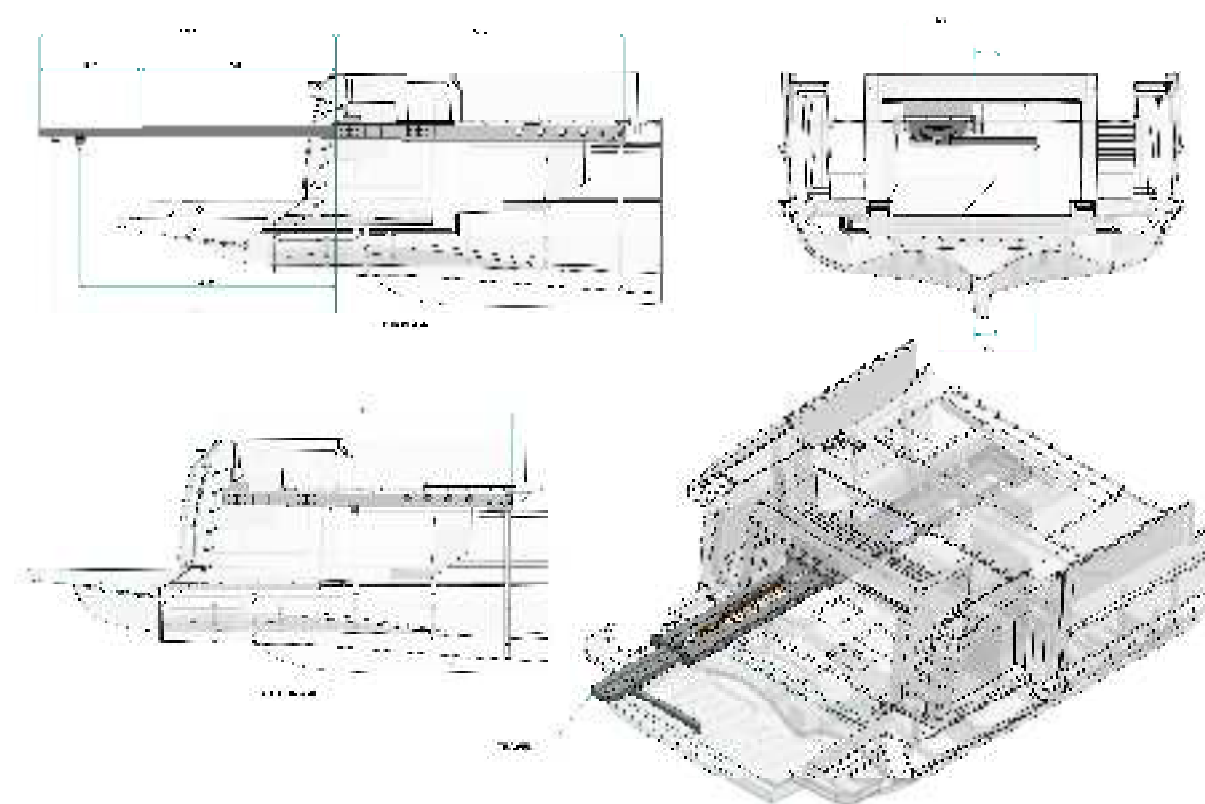


Garage

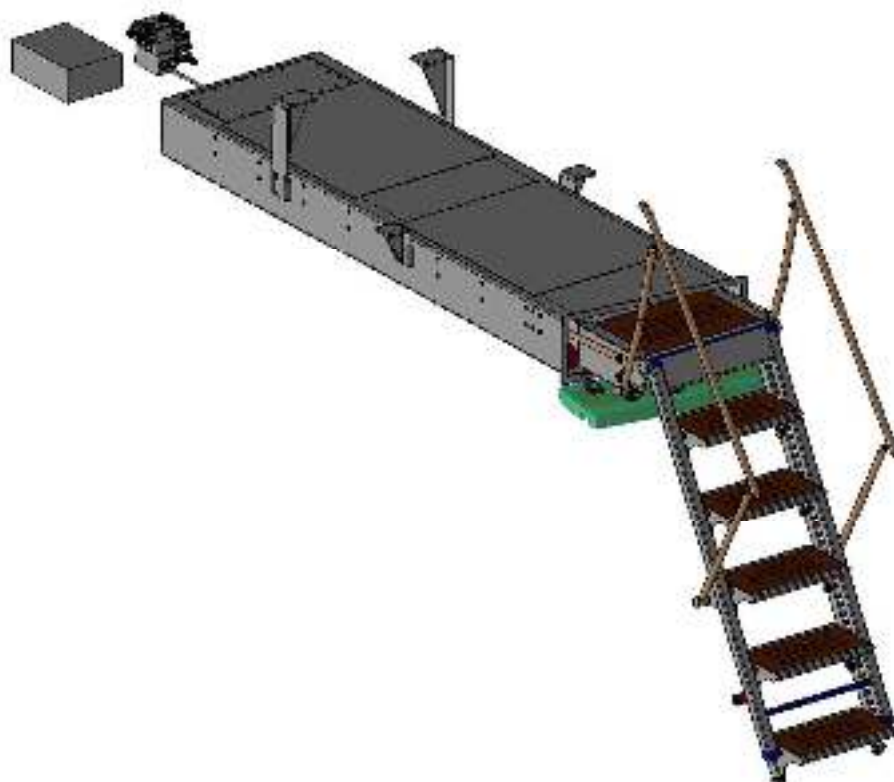


Before launching, recovering or moving the tender, seabobs and any water toys make sure that nobody is on board or in the area interested by the operations.



TRAVEL

Swimming Ladder



Characteristic Data

Dimensions overboard:	Length	1.8 m
	Width	0.63 m
Maximum inclination:		60°
Maximum capacity:		150 kg

After each navigation it is advisable to wash the swimming ladder with fresh water to eliminate the saline deposit and occasionally with neutral soap to eliminate greasy deposits.



Before carrying out any cleaning operation, disconnect the power supply.

Ensure that the work areas are well lit and free from subjects that could be a source of hindrance or danger.



Do not use solvent substances to clean the swimming ladder (danger of pollution and possible damage to the external coating of the materials of the swimming ladder and the boat).

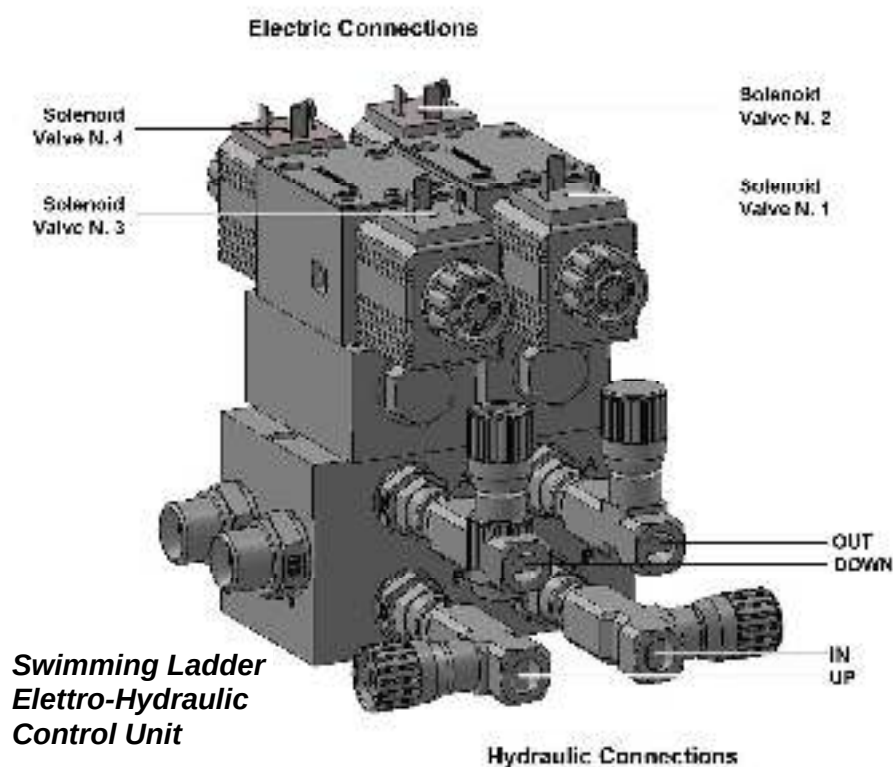
On stainless steel surfaces, avoid the use of abrasive products or objects that would damage the external glossy finish, exposing it to possible surface oxidation.

Swimming Ladder - Emergency

If the swimming ladder does not work correctly by acting on the appropriate control panel or by using the supplied remote control, it is necessary to manually act on the solenoid valve block.

WARNING: acting manually on the solenoid valve block, the electrical limit switch are deactivated, therefore proceed with the maximum attention and/or with the help of a second person to view the moviment of the swimming ladder.

- Solenoid Valve N°1 : OUT Swimming Ladder
- Solenoid Valve N°2 : IN Swimming Ladder
- Solenoid Valve N°3 : UP Swimming Ladder
- Solenoid Valve N°4 : DOWN Swimming Ladder



CAUTION: If you want to get the swimming ladder OUT, by acting on the Solenoid Valve N°1, proceed with particular attention, the UP and DOWN pistons could be discharged (not under pressure) and therefore the swimming ladder could fall down.

Section C

YACHT'S CONTROL STATIONS

Yacht's control stations

Wheelhouse

Main switchboard

Circuit Booklet

Shore Power Supply

Switchboard

Yacht's Control Stations

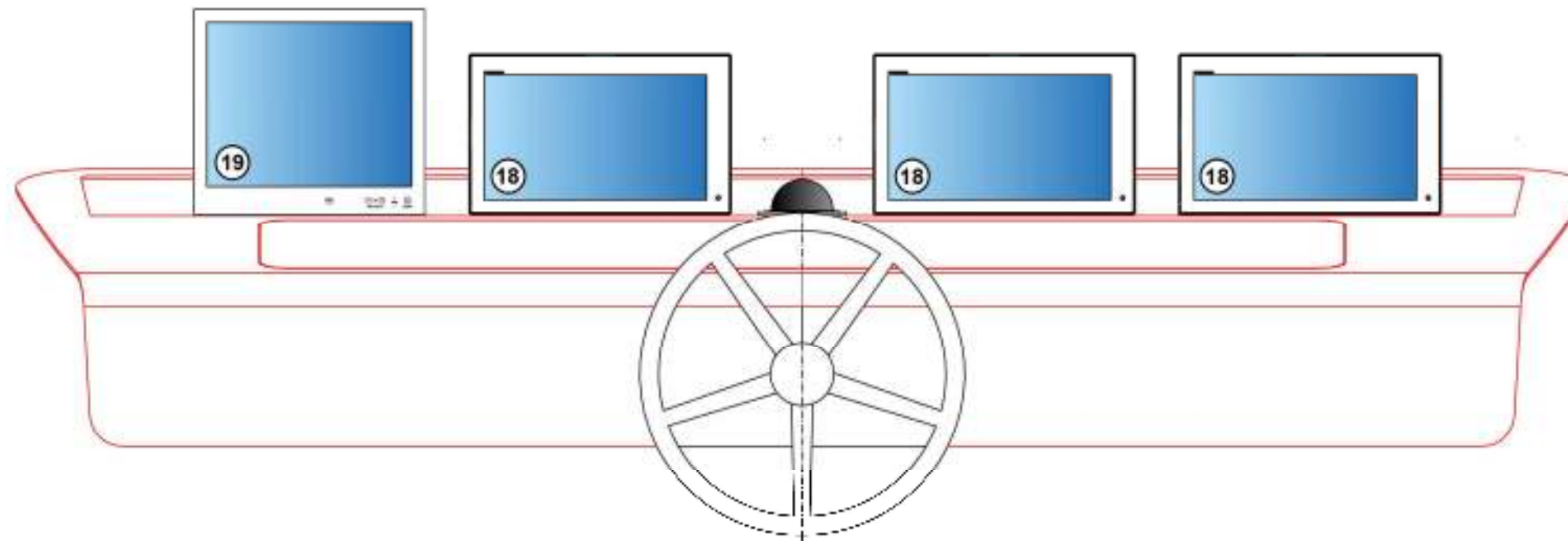
Wheelhouse

Wide and luminous internal room, equipped with wide windows which allows helmsman visibility on almost the whole horizon (except for aft sectors) where he can have the pleasure of operating this fabulous and prestigious yacht in free navigation, with consciousness of having the whole yacht under his control, staying comfortably in its seat.

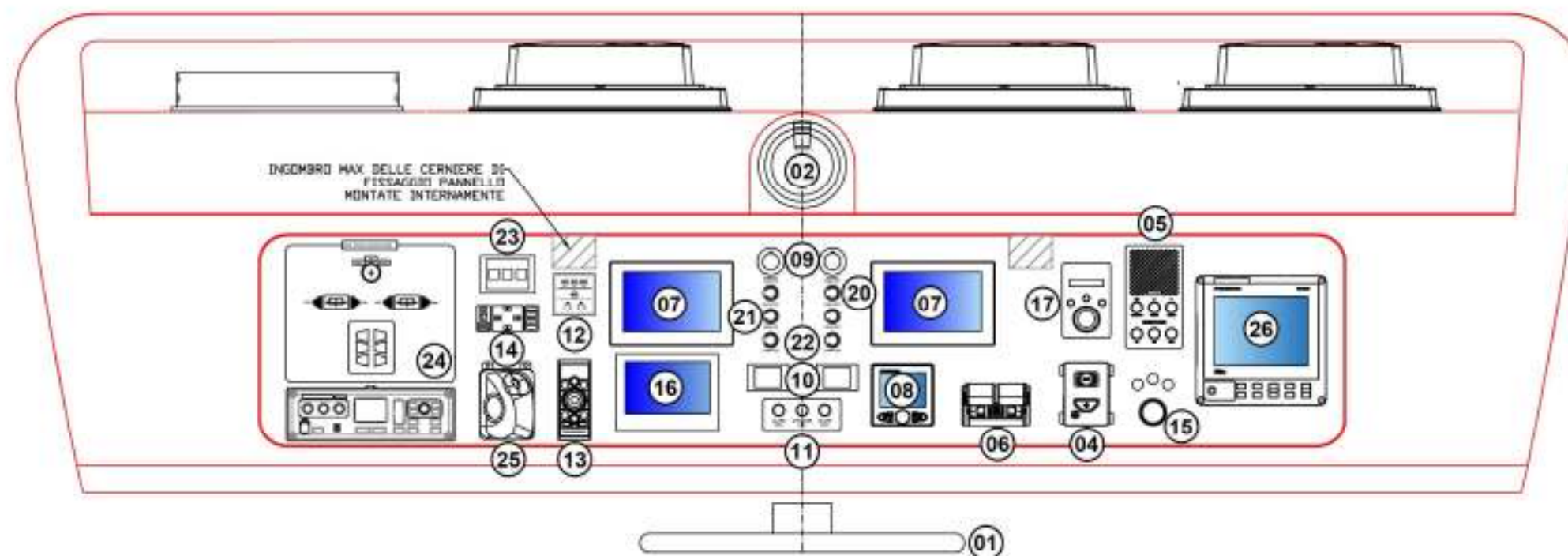
With this in mind, the room includes not only the control equipments for the propulsion machineries, but also the electronic equipment, for the monitoring and identification of the surroundings (such as Log, echo-sounder, radar, helm angle indicator); the internal and external communication equipments (VHF transceiver, intercom, etc.); the navigation support equipments (chart plotter, autopilot, magnetic compass, etc.); the passive safety control equipment of the yacht (CCTV monitor), smoke detector control unit, alarm and emergency control panel for the various vessel systems, such as navigation lights, bilge solenoid and pumps, etc..

A touch screen monitor has been installed on the bridge to handle the on-board systems requiring constant monitoring and not requiring local manual operations, such as: navigation light control, fuel level monitoring, grey/black water level monitoring, monitoring of bilge water alarms in the various internal compartments etc..

Wheelhouse - Control Station



VISTA FRONTALE PANNELLO STRUMENTAZIONE
Instrument panel front view



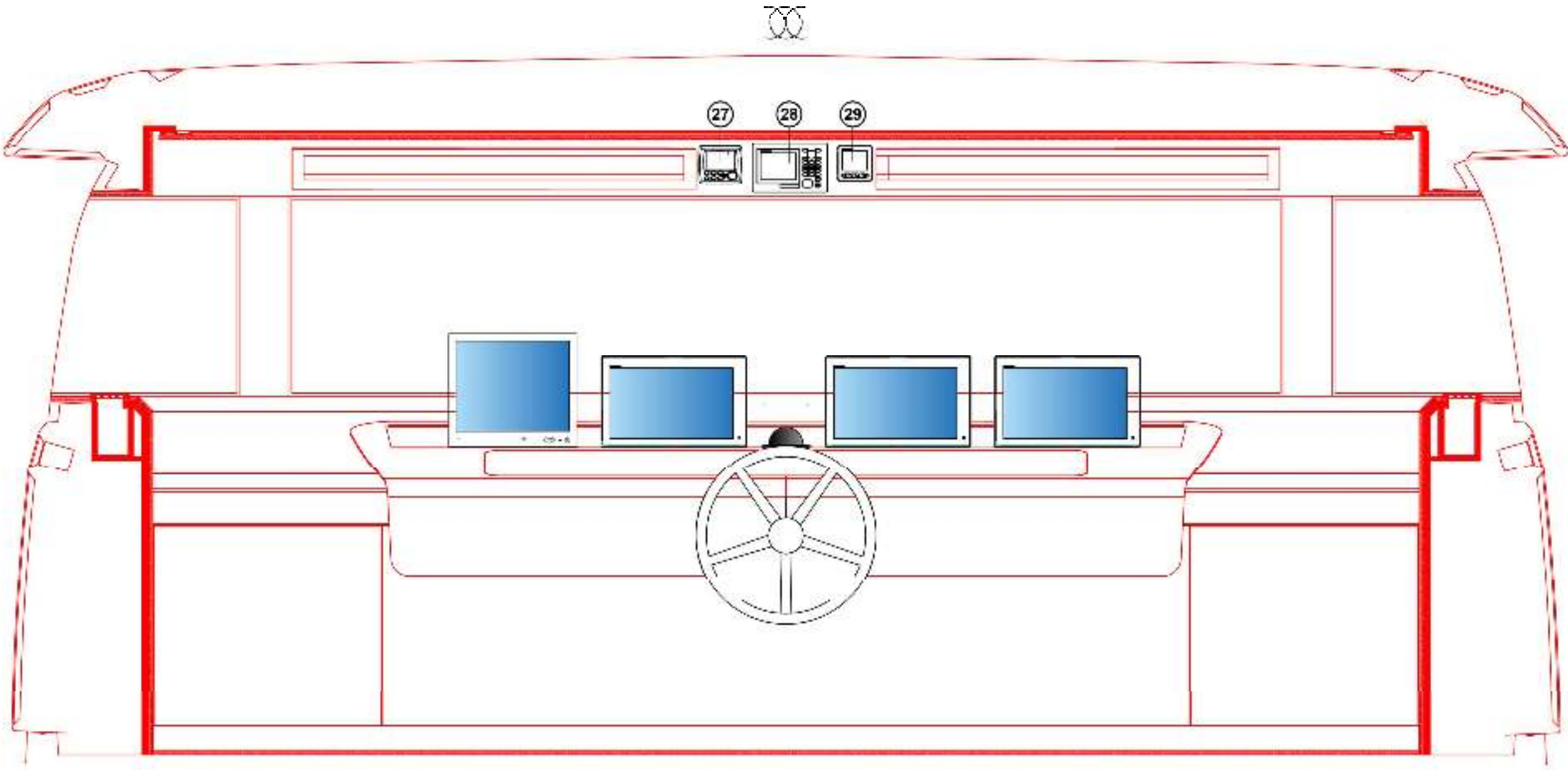
VISTA IN PIANTA PANNELLO STRUMENTAZIONE
Instrument panel plan view

POS	Q.TA Qty	DESCRIZIONE Description	FORNITORE Supplier
01	1	Helm wheel	METALSTYLE
02	1	Magnetic compass BU1	RIVIERA
03	1	VHF handset	FURUNO
04	1	Joystick bow + stern thruster	CMC MARINE
05	1	Service buttons	B&B SYSTEM
06	1	Engine bottle	CAT
07	2	Display for control stand CMD7	CAT
08	1	RAP 70110 A... Joystick controller	FLRJO
09	4	EMERGENCY Engine buttons	CAT
10	2	Rudder angle indicator	CMC MARINE
11	1	Service buttons	B&B SYSTEM
12	1	Wipers controls panel	SPEICH
13	1	MCU-004 Wired remote controller	FURUNO
14	1	Sanshin searchlight panel	SANSHIN
15	1	Trackball control CCTV	PANASONIC
16	1	Pannello touch screen GT2507-WTSD	CMC MARINE
17	1	JCU remote control FLIR	FLIR
18	4	Monitor T2T19F-MULTIFUNCTION DISPLAY	FURUNO
19	1	TOUCH MONITOR UNIT HD 19T22FLD	FURUNO
20	2	START Engine buttons	B&B SYSTEM
21	2	STOP Engine buttons	B&B SYSTEM
22	2	Engine dimmer light buttons	B&B SYSTEM
23	1	Horn switch panel	KAHLENBERG
24	1	Emergency steering panel	CMC MARINE
25	1	Control Unit + Trackball -Touch Monitor Unit	FURUNO
26	1	Echosounder FE-800	FURUNO

Wheelhouse - Control Station

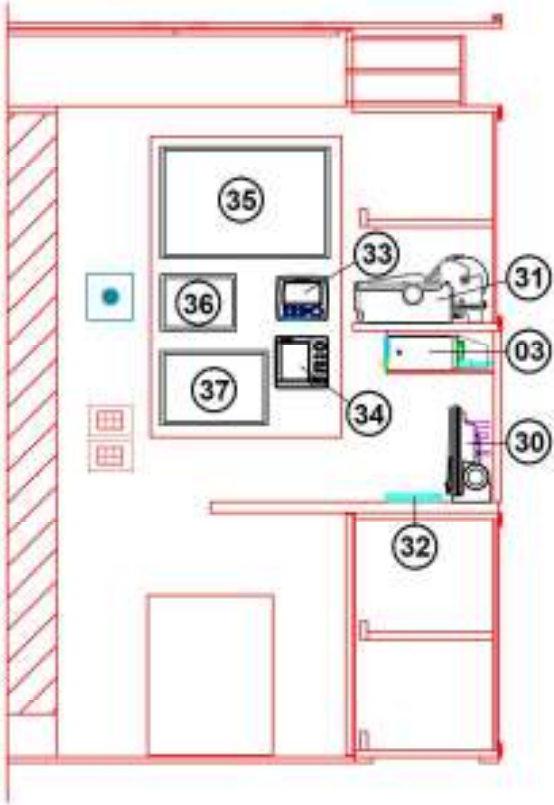
27	1	Satellite Compass SC-70	FURUNO
28	1	GPS-GP-170	FURUNO
29	1	FI-70-Remote Display	FURUNO

VISTA FRONTALE PANNELLO STRUMENTAZIONE ALTA
Instrument panel high side - front view

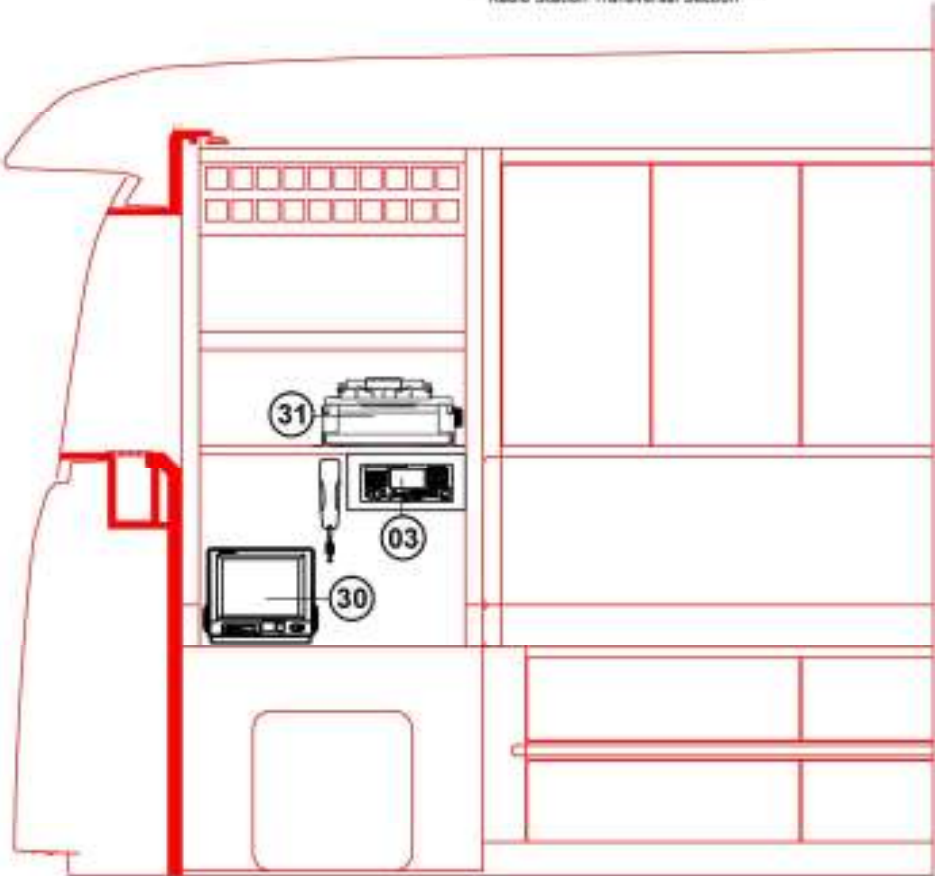


Wheelhouse - Control Station

VISTA LATERALE STAZIONE RADIO
Radio Station side view

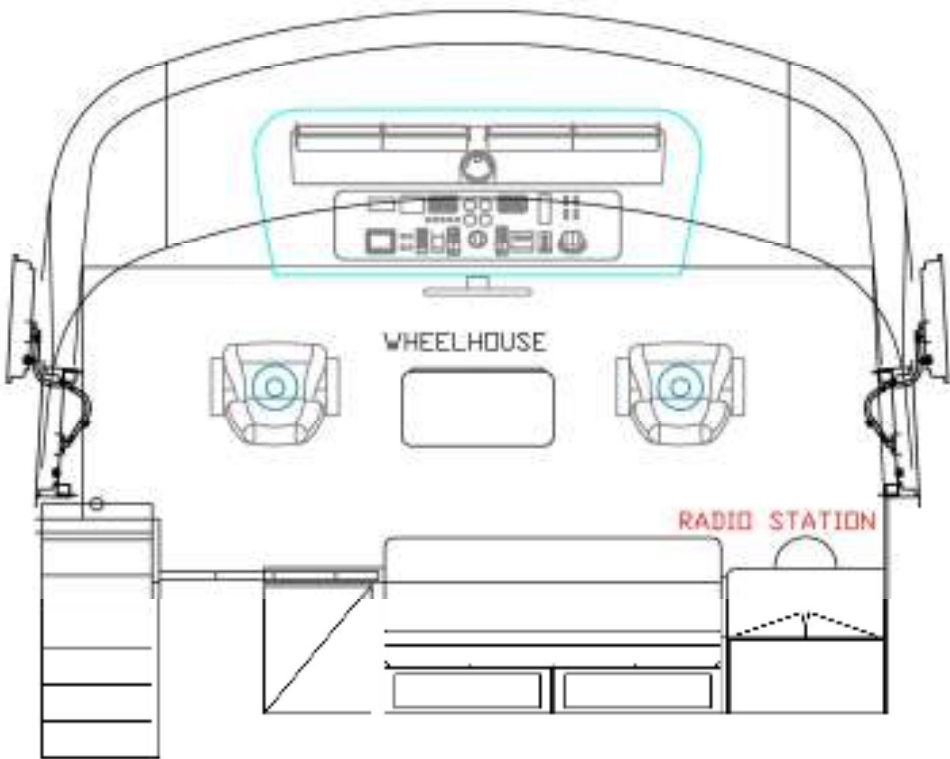
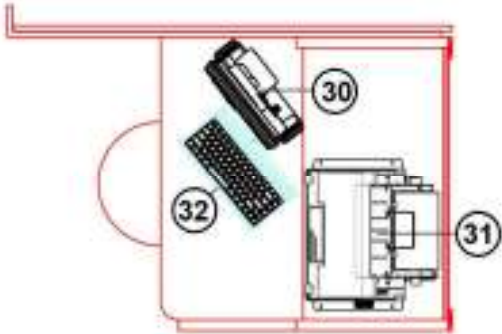


VISTA TRASVERSALE STAZIONE RADIO
Radio Station Transversal Section



03	1	VHF-DSC FM8900 S Handset+ Transceiver Unit	FURUNO
30	1	INMARSAT -C FLECOM 18 Terminal Unit	FURUNO
31	1	INMARSAT -C FLECOM 18 Printer Unit	FURUNO
32	1	INMARSAT -C FLECOM 18 Keyboard	FURUNO
33	1	BR-500 Bridge Navigation Alarm System	FURUNO
34	1	NAVTEX NX-700B	FURUNO
35	1	Monitoring System Screen	B&B
36	1	Garage Smoke & Fire Detection System	B&B
37	1	Fire Detection System Autronica	B&B

VISTA IN PIANTA STAZIONE RADIO
Radio Station plan view



Main characteristics

All electric cables are insulated with halogen free and fire resistant material, self-extinguishing and not propagating the fire, in accordance with IEC 30992; they are labelled to facilitate as much as possible the maintenance and are contained in self-extinguishing PVC sheaths and pipes, for further protection. In case of fire some essential circuits must continue to supply energy to the safety devices (emergency lights, alarm systems, water pumps for the firemen, etc.) The fire resistant cables made of tape mica + isolation g10 are in accordance with IEC 60331/IEC 20-36 (temperature of the flame = 750°C-test duration = 90 minutes in nominal voltage).



The electric system, as any other system onboard, is subject to stresses and vibrations of the hull and is in an environment with strong saline humidity and therefore requires periodical checks in order to ensure its good conditions.



The electric system, if not used or maintained correctly and neglected, statistically represents one of the most frequent causes of fire onboard.



MAIN LINES:

A.C. MAIN LINES:

- GENERATOR 1 KOHLER 70EFOZDJ - 50kW - 400-230Vac/50Hz/3-phase+Neutral
- GENERATOR 2 KOHLER 70EFOZDJ - 50kW - 400-230Vac/50Hz/3-phase+Neutral
- Aft Shore line 400V, 50Hz, 3-phase

D.C. MAIN LINES:

- Nº - EXIDE EP2100, 12V-240Ah batteries (Nº 2 for each engine)
- Nº - EXIDE ES1200, 12V-110Ah batteries (Nº 2 for each D.G.)
- Nº1 - EXIDE ES650, 12V-56Ah battery (radio emergency)
- Nº8 - EXIDE ES1100-6, 6V-200Ah batteries for emergency
- Nº24 - EXIDE 6EPZV330, 330Ah batteries for services

Gel batteries

The standard batteries installed on your yacht have been replaced with gel technology batteries (optional).

These batteries do not require any maintenance and, due to their safety and basic design, can supply the power required for various activities and to operate all yacht's systems and equipment.

The advantages of gel batteries can be summarised as follows:

- Maintenance-free;
- Possibility of being fully discharged without permanent damage;
- Maximum duration;
- No emission of acid vapours during recharge;
- Very short recharging time;
- Very low self-discharge (maintenance of 65% of the original charge after one year without being used);
- Installable in any position;
- No possibility of internal short-circuit.

Suggestions and warnings, for using correctly these batteries are:

- After a complete discharge of the battery, recharge them within a few days;
- In case of long inactivity, disconnect all equipment, in order to avoid that excessive discharges can damage the battery;
- If long inactivity periods are expected (more than two months) carry out a charge/discharge cycle;
- Never open the vent valves.

Cable type used for electrical installation

Shipboard cables according following rules:

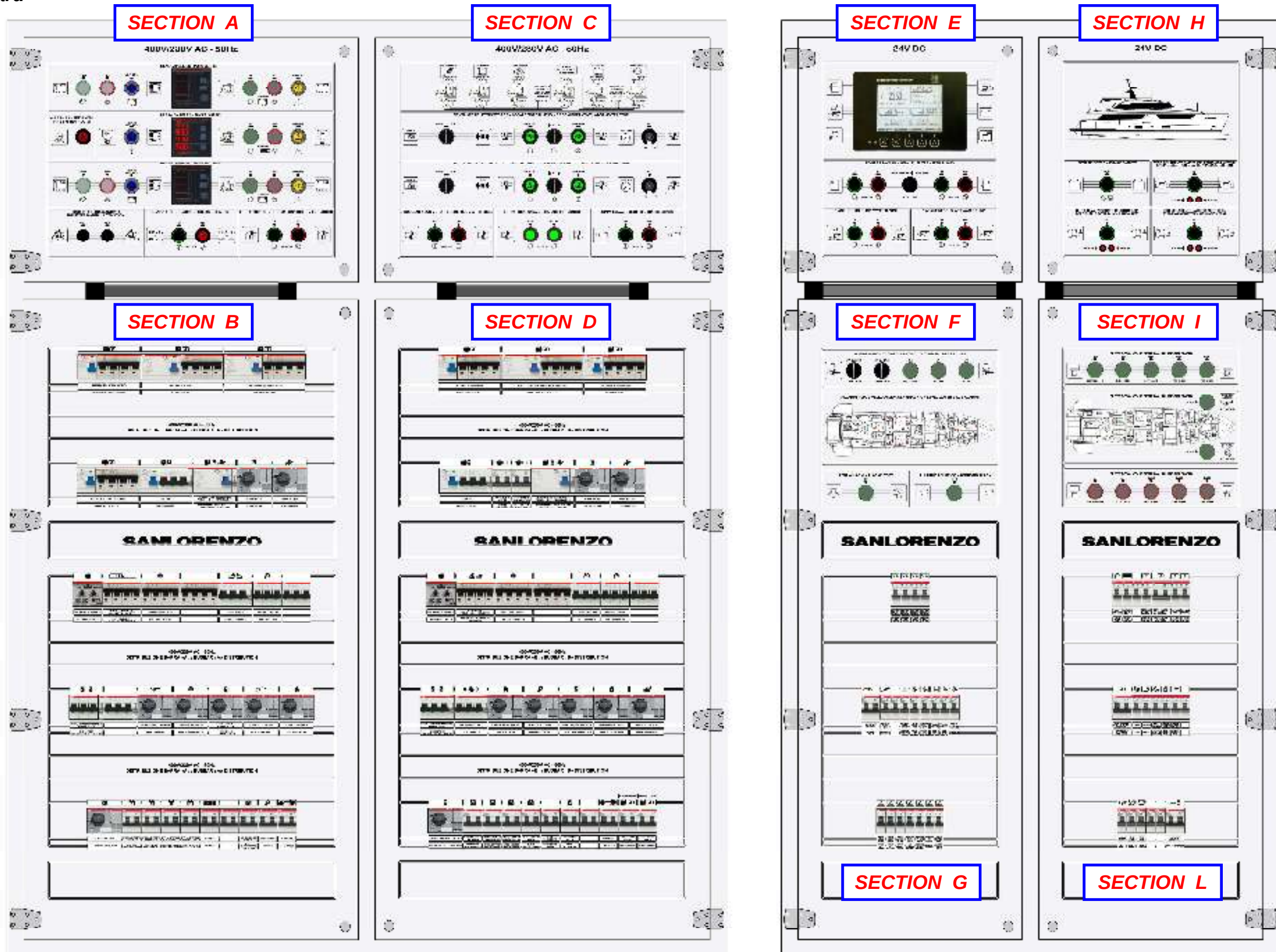
- IEC 60092-350 Edition 3.0 2008-02
- IEC 60092-507 Ed. 2.0 2008-01
- IEC 60332-1
- IEC 60332-3-22
- IEC 60331 (only for emergency consumers)

Main switchboard

Realised in compact shape, it carries externally on the front side a set of control panels and a set of power supply magneto-thermal switches divided into rows, which manage:

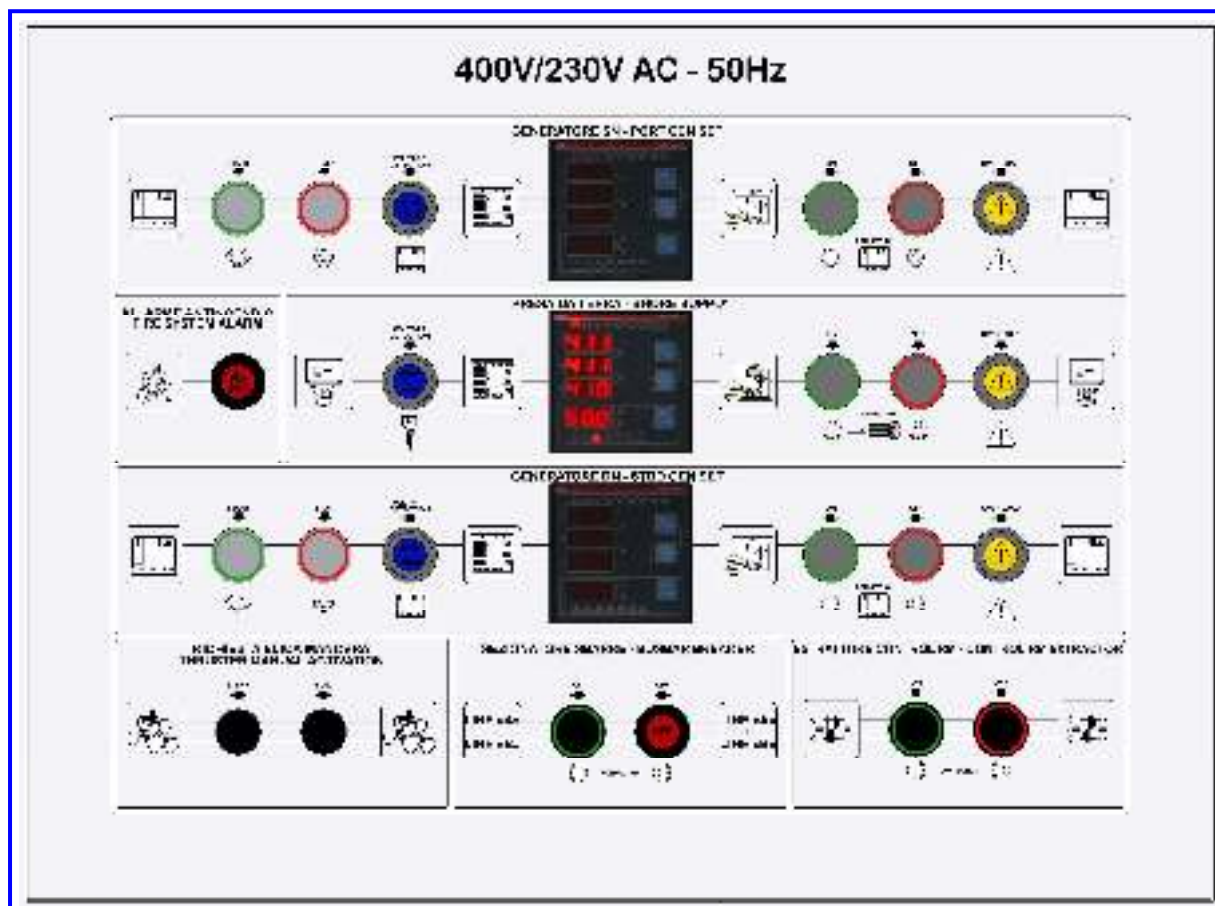
- **SECTION A** 400V/230VAC - 50Hz (Port Generator; Shore Supply; Starboard Generator; Busbar; Fire System Alarm; Thruster Manual Activation; Busbar Breaker; Control Room Extractor)
- **SECTION B** 400V/230VAC - 50Hz (Distribution Busbar A)
- **SECTION C** 400V/230VAC - 50Hz (Engine Room Port Ventilator/Extractor; Engine Room Starboard Ventilator/Extractor; Engine Room Air Circulating; Garage Extractor; Oil Transfer Pump)
- **SECTION D** 400V/230VAC - 50Hz (Starboard B)
- **SECTION E** 24VDC (Touch Screen; Fuel Transfer Pumps; Grey Water Pump; Black Water Pump)
- **SECTION F** 24VDC (Automatic Bilge Pumps; Alarms Test; Acoustic Alarm)
- **SECTION G** 24VDC (Distribution Services)
- **SECTION H** 24VDC (Service Battery; Engine Electronic Battery Insulation Test; Port Engine Battery Insulation Test; Starboard Engine Battery Insulation Test)
- **SECTION I** 24VDC (Bilge Solenoid)
- **SECTION L** 24VDC (Distribution Services)

Main Switchboard

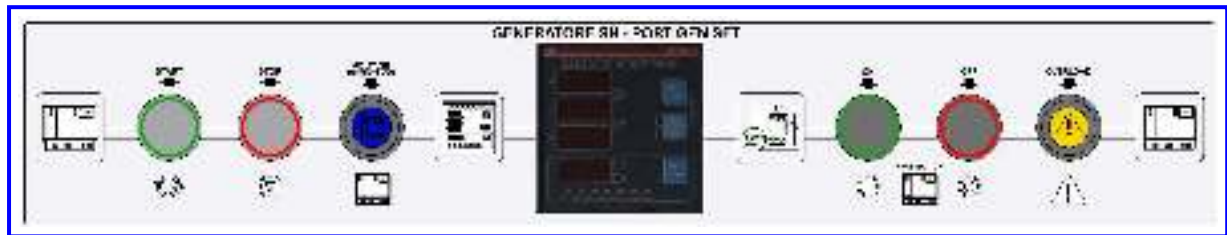


Main switchboard **SECTION A****400V/230VAC - 50Hz**

In order to make the description of switchboard easier, this section is divided into four parts:



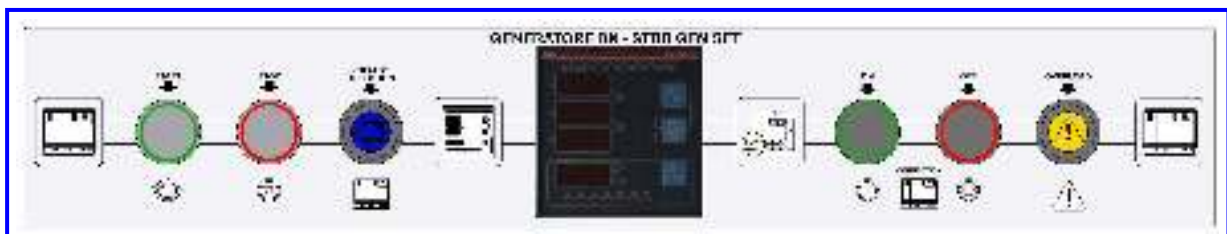
SECTION A1 – Port Generator Set.



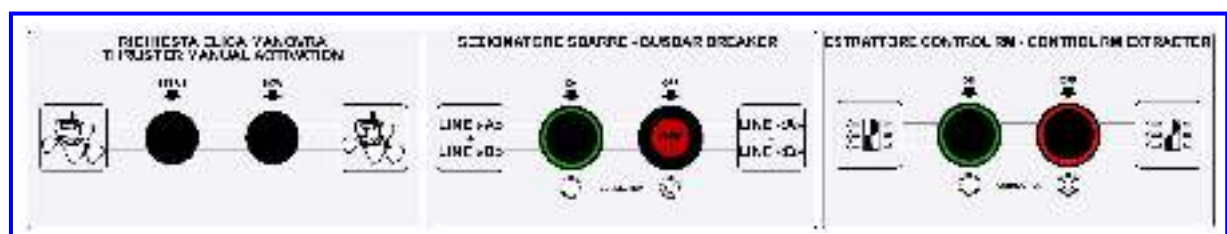
SECTION A2 – Fire System Alarm – Shore Supply.



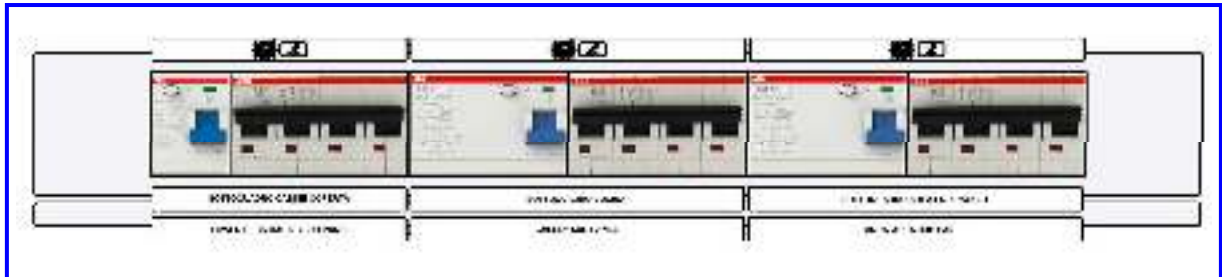
SECTION A3 – Starboard Generator Set.



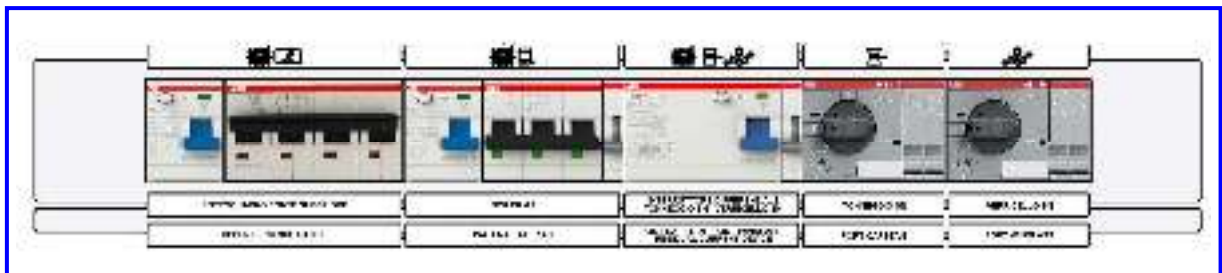
SECTION A4 – Thruster Manual Activation – Busbar Breaker – Control Room Extractor.



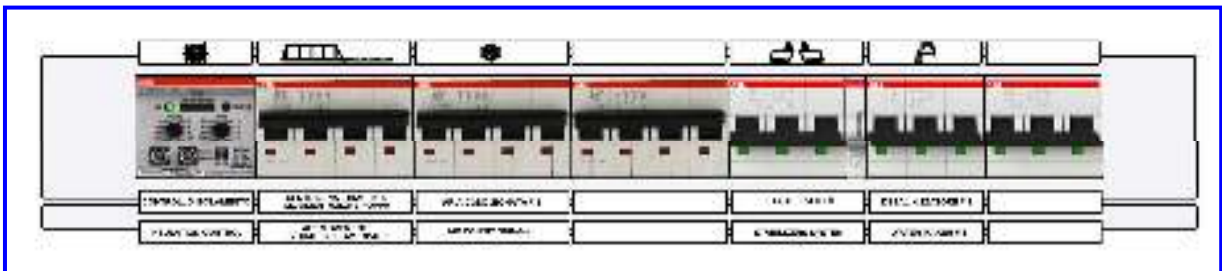
SECTION B1 - N3 Magneto-Differential Switch: Lower Deck Cabins Sub-Panel; Galley Sub-Panel; Crew area Sub-Panel.



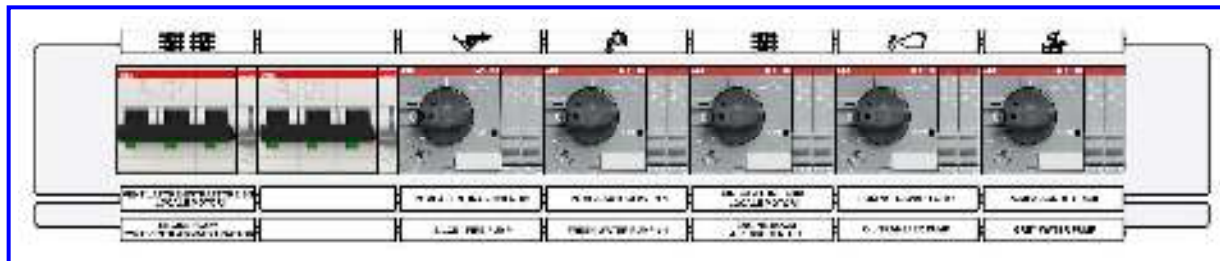
SECTION B2 - N3 Magneto-Differential Switch and N2 Protection Switch: Upper Deck Sub-Panel; Water Heater # 1; Port Capstain - Port Windlass (Residual Current Device); Port Capstain; Port Windlass.



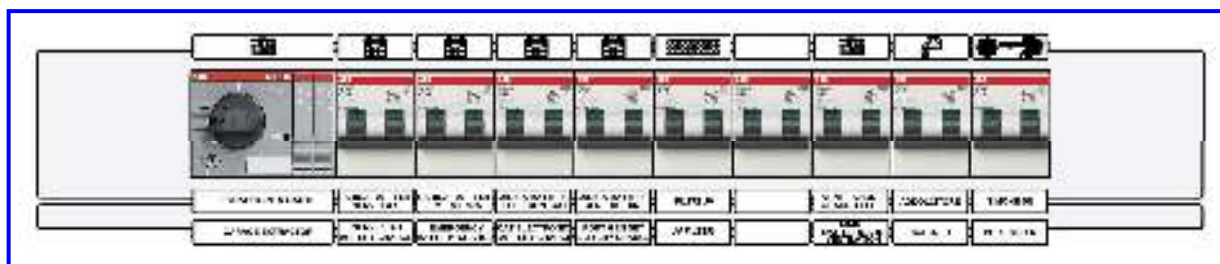
SECTION B3 - Panel Control and N6 Magneto Switch: Insulation Control; AFT Movement Hydraulic Powerpack; Air Conditioning # 1; Spare; Stabilizers System; Water Maker # 1; Spare.



SECTION B4 - N2 Magneto Switch and N° 5 Protection Switch: Engine Room Port Ventilator/Extractor; Spare; Bilge – Fire Pump; Fresh Water Pump # 1; Engine Room Air Circulator; Oil Transfer Pump; Grey Water Pump.



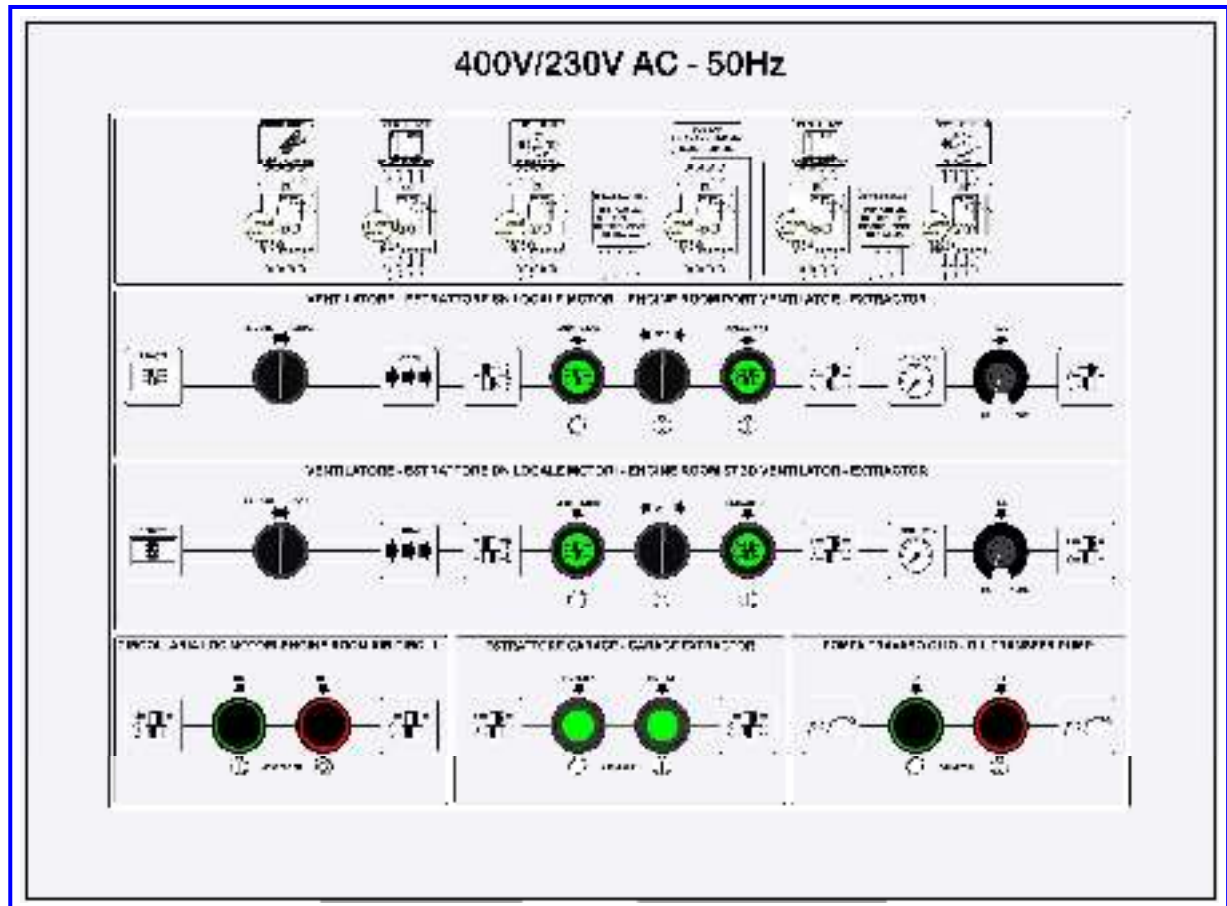
SECTION B5 - Protection Switch and N°9 Magneto Switch: Garage Extractor; Service # 1 Battery Charge; Emergency Battery Charge; CAT Electronic Battery Charge; Port Gen Set Battery Charge; UV Filter; Spare; Main Switch-board Ventilation; Softner; Port Steer.



Main switchboard **SECTION C**

400V/230VAC - 50Hz

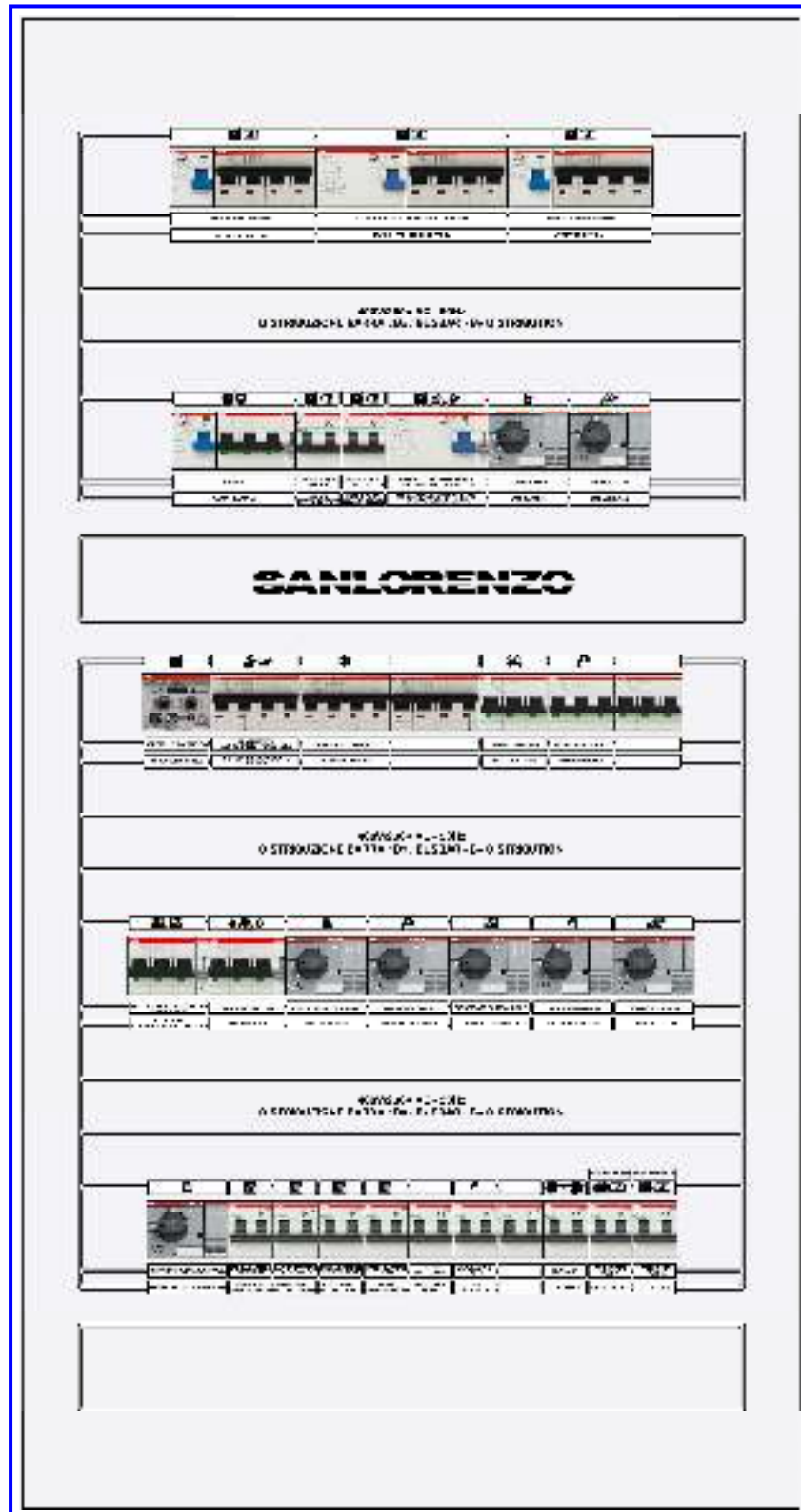
In order to make the description of switchboard easier, this section is divided into three parts:



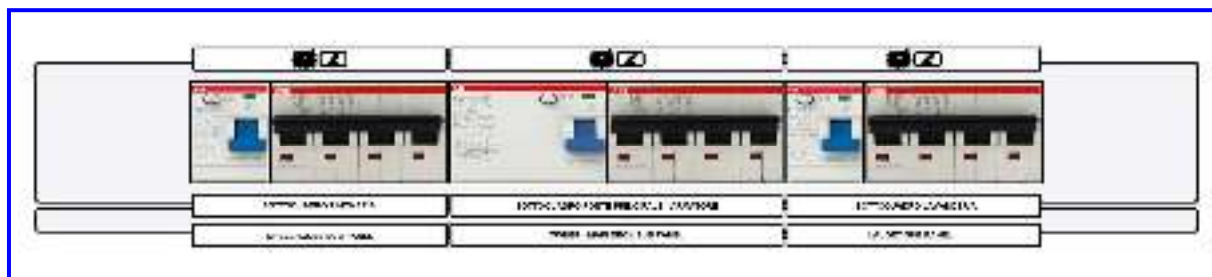
Main switchboard **SECTION D**

400V/230VAC - 50Hz Busbar "B" Distribution

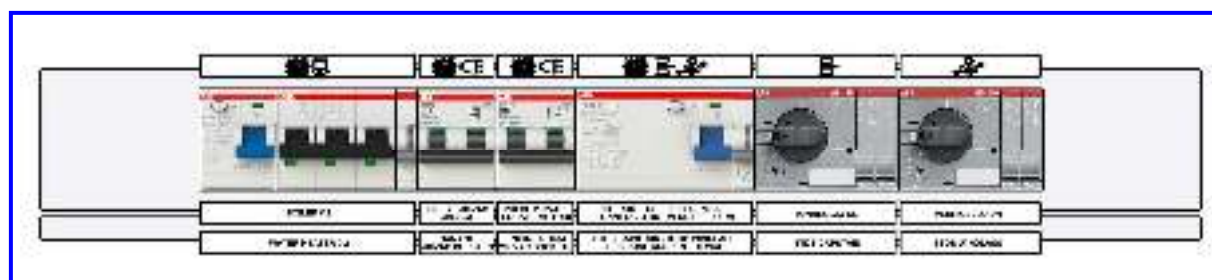
In order to make the description of switchboard easier, this section is divided into five parts:



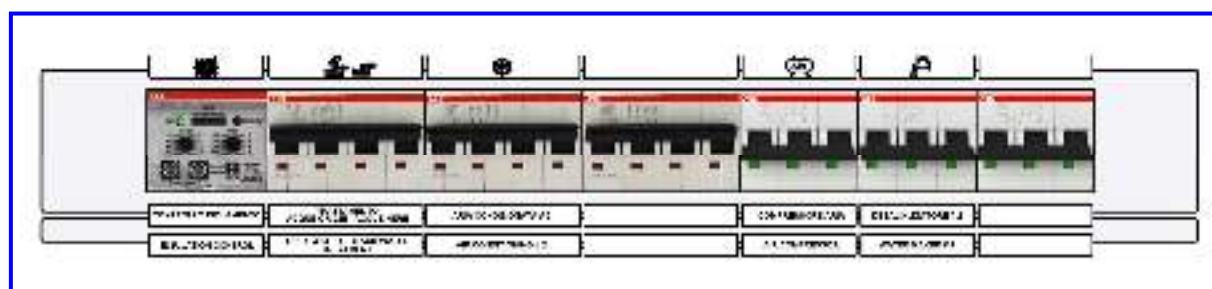
SECTION D1 – N3 Magneto-Differential Switch: Wheelhouse Sub-Panel; Owner / Main Deck Sub-Panel; Laundry Sub-Panel.



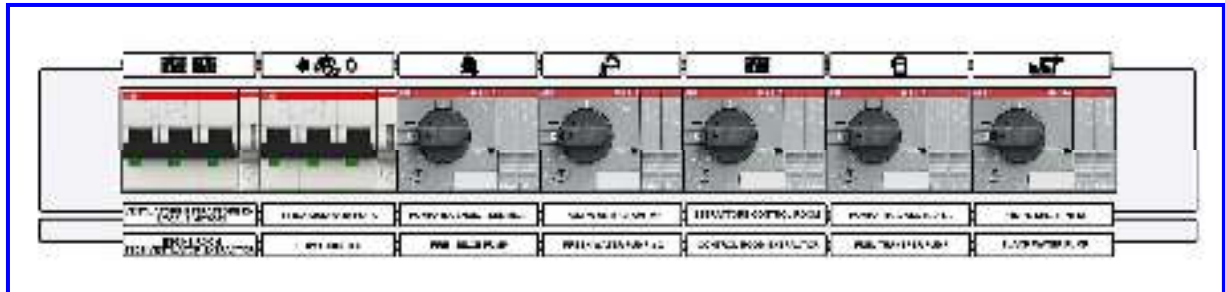
SECTION D2 – N4 Magneto-Differential Switch and N2 Protection Switch: Water Heater # 2; Garage 230VAC Sockets; Engine Room 230VAC Sockets; Starboard Capstain - Starboard Windlass (Residual Current Device); Starboard Capstain; Starboard Windlass.



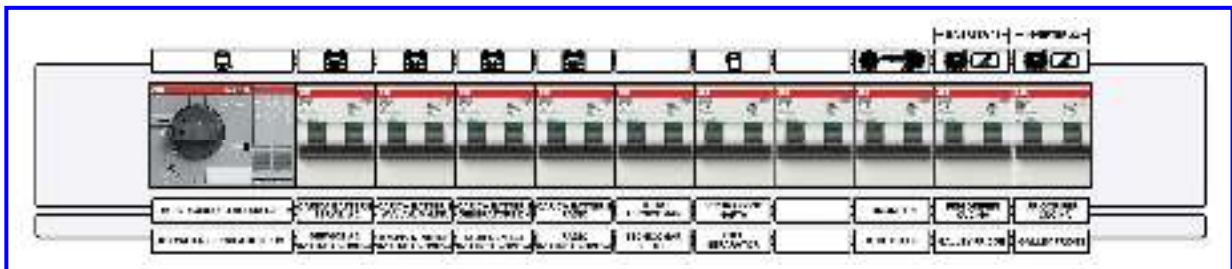
SECTION D3 – Panel Control and N6 Magneto Switch: Insulation Control; Grey Water – Black Water Treatment; Air Conditioning # 2; Spare; Air Compressor; Water Maker # 2; Spare.



SECTION D4 – N°2 Magneto Switch and N° 5 Protection Switch: Eng ine Room Star-board Ventilator/Extractor; Bow Thruster; Fire – Bilge Pump; Fresh Wa-ter # 2; Control Room Extractor; Fuel Transfer Pump; Black Water Pump.

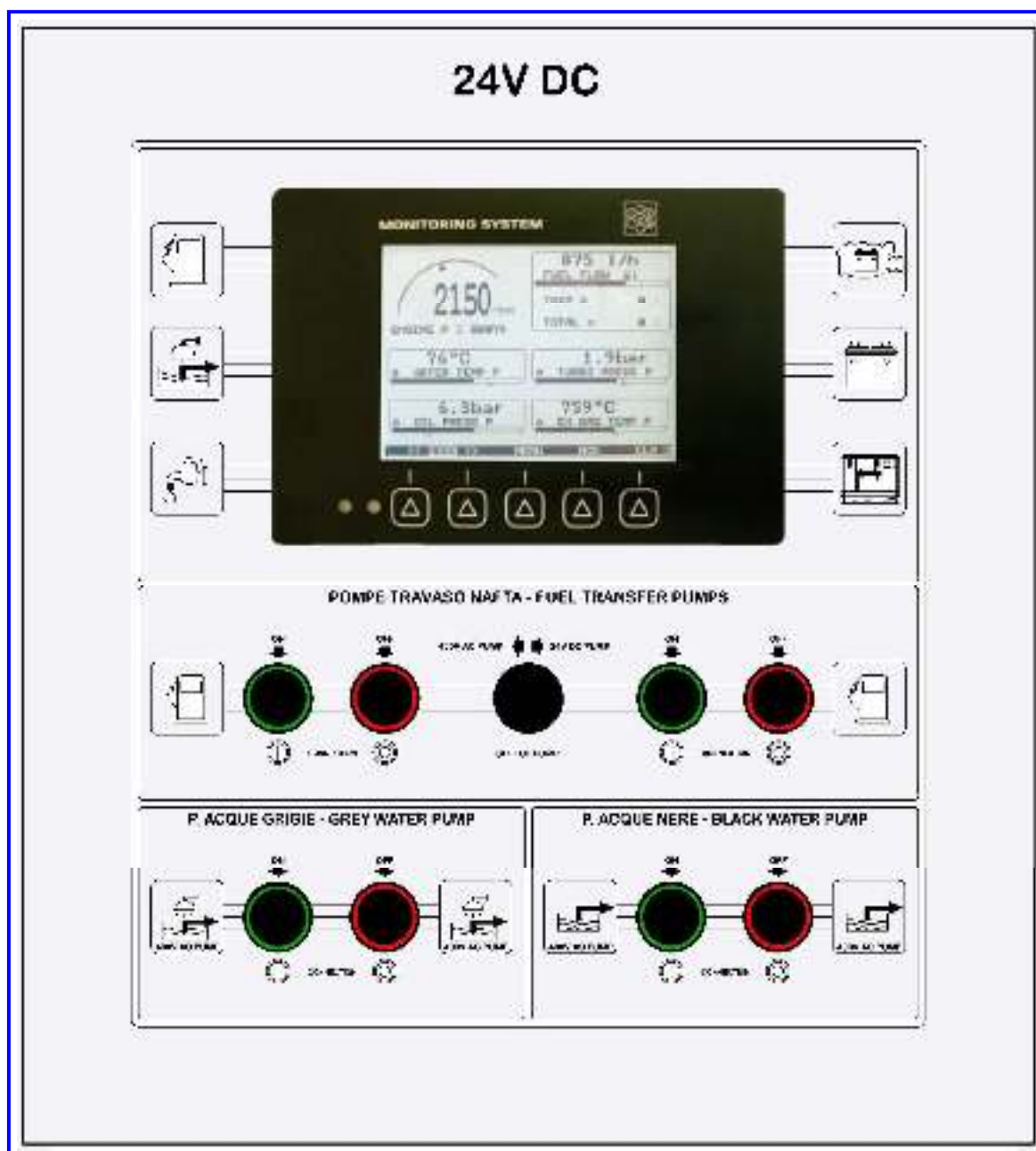


SECTION D5 – Protection Switch and N°10 Magneto Switch: Hot Wat er Circulation Pump; Service # 2 Battery Charge; MM.PP. Starter Battery Charge; Starboard Gen Set Battery Charge; Radio Battery Charge; Tecnicomar Filter; Fuel Separator; Spare; Starboard Steer; Galley Fridge Inverter #1; Galley Fridge Inverter #2.



Main switchboard **SECTION E****24V DC**

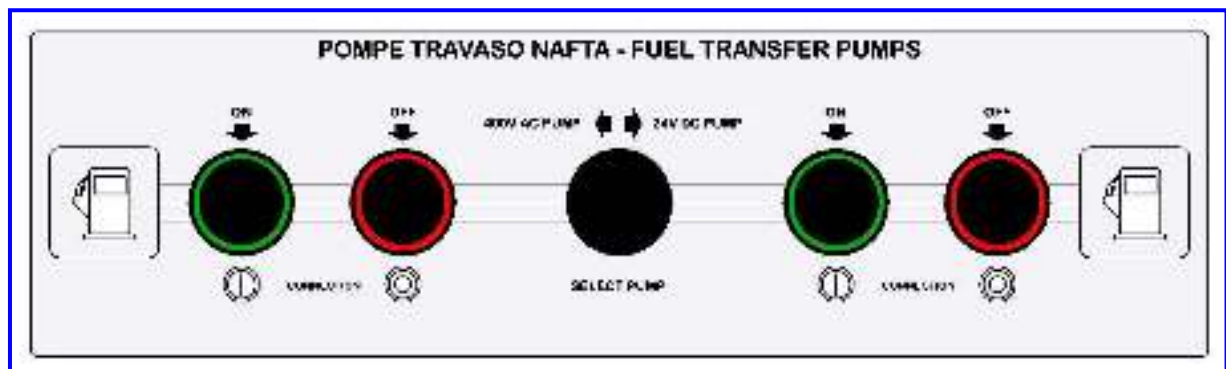
In order to make the description of switchboard easier, this section is divided into three parts:



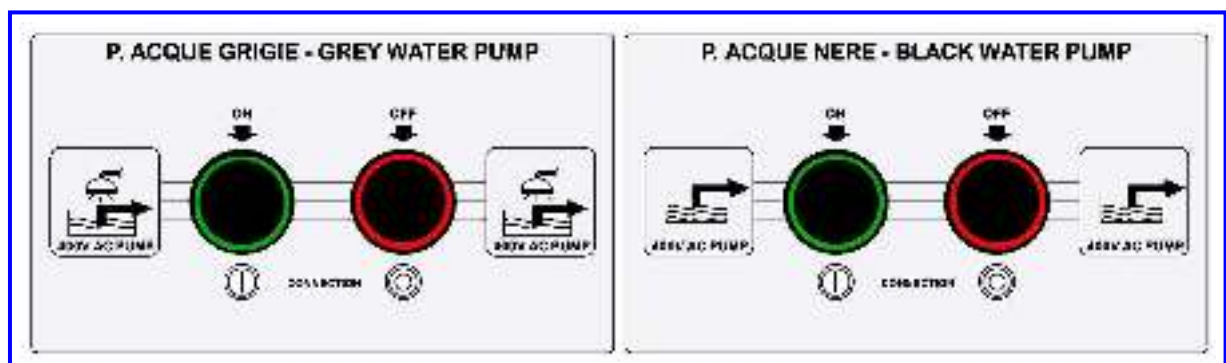
SECTION E1 – Touch Screen.



SECTION E2 – Fuel Transfer Pumps.

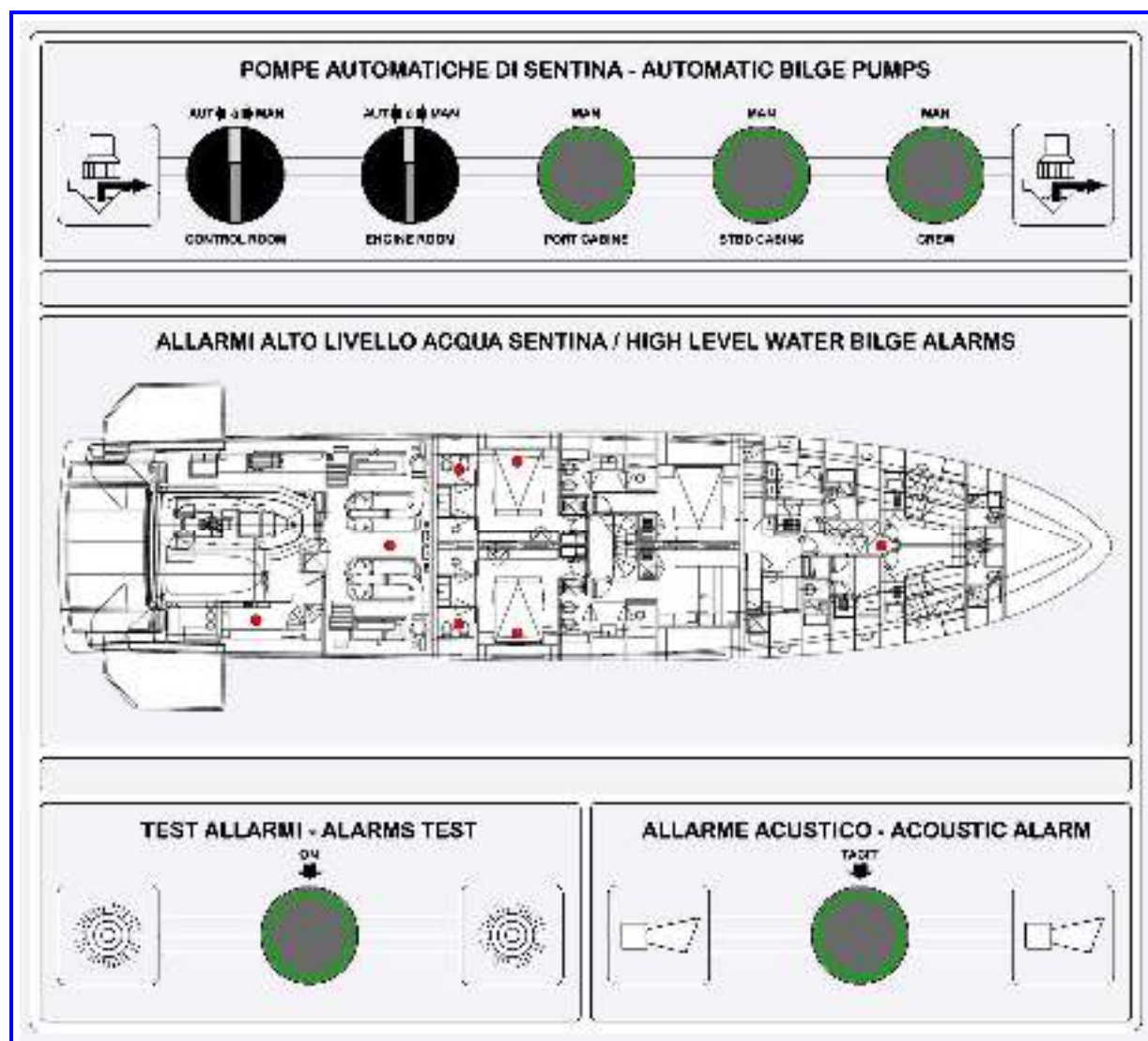


SECTION E3 – Grey Water Pump; Black Water Pump.

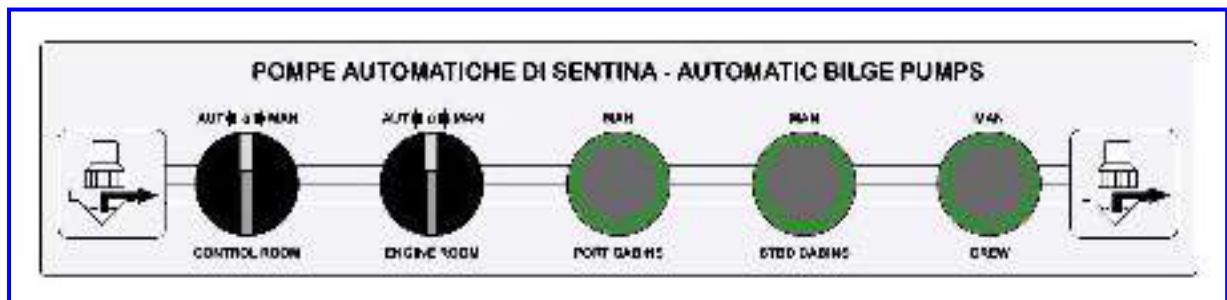


Main switchboard **SECTION F****24V DC**

In order to make the description of switchboard easier, this section is divided into three parts:



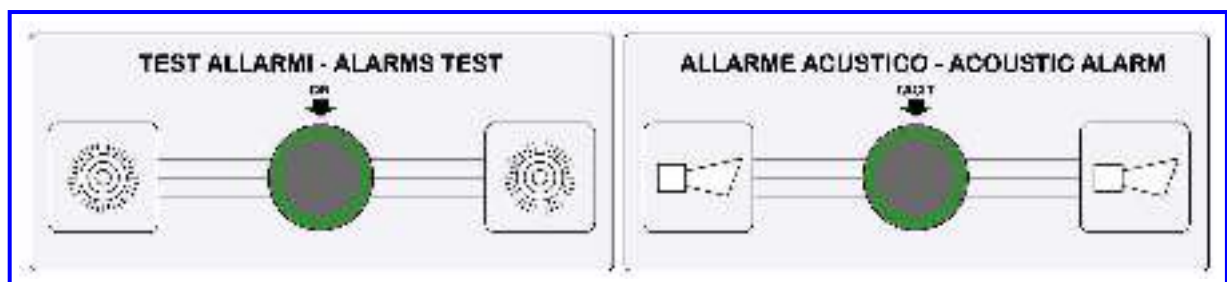
SECTION F1 – Automatic Bilge Pumps.



SECTION F2 – High Level Water Bilge Alarms.

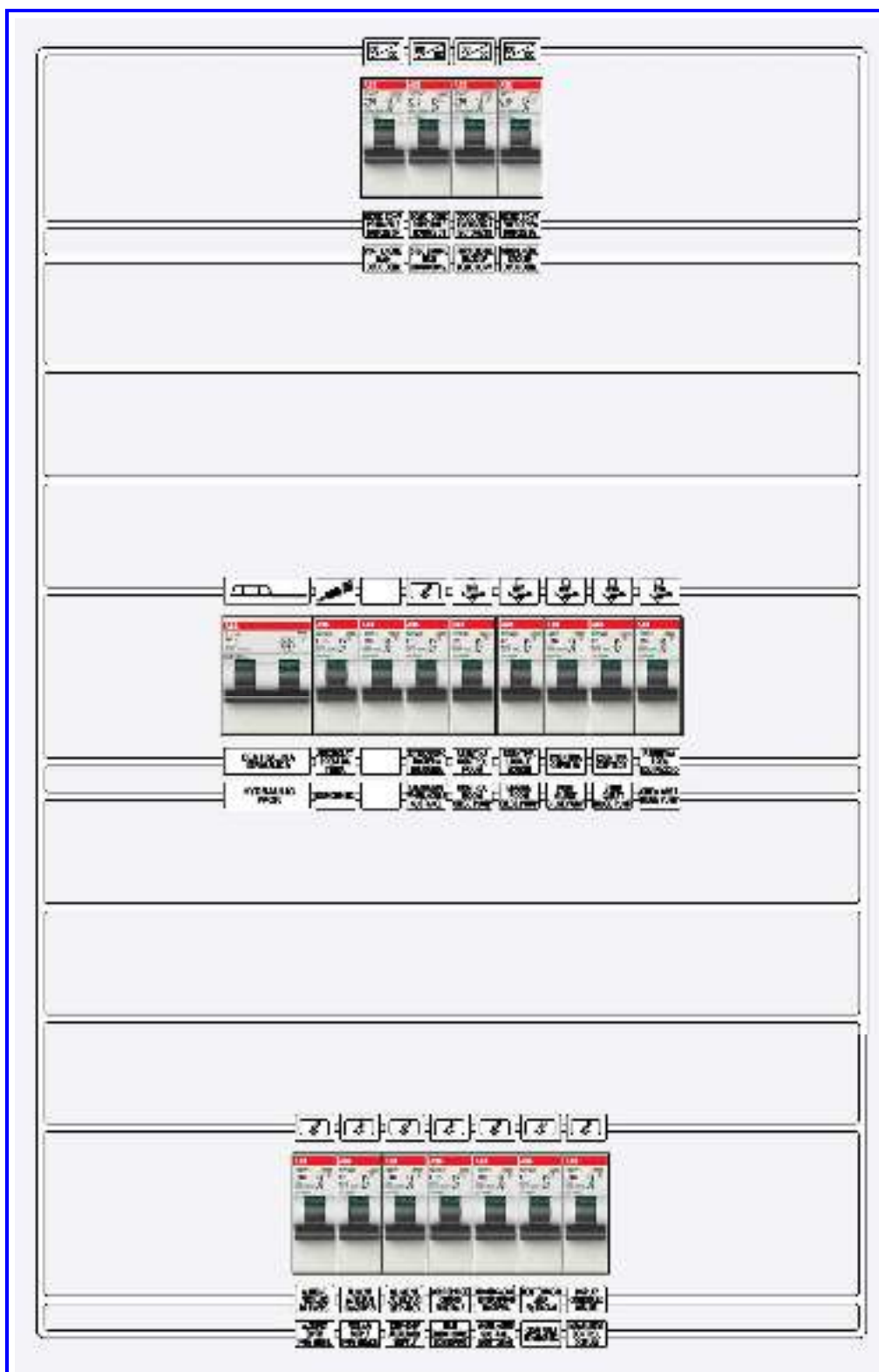


SECTION F3 – Alarms Test; Acoustic Alarm.



Main switchboard **SECTION G****24V DC distribution**

In order to make the description of switchboard easier, this section is divided into three parts:



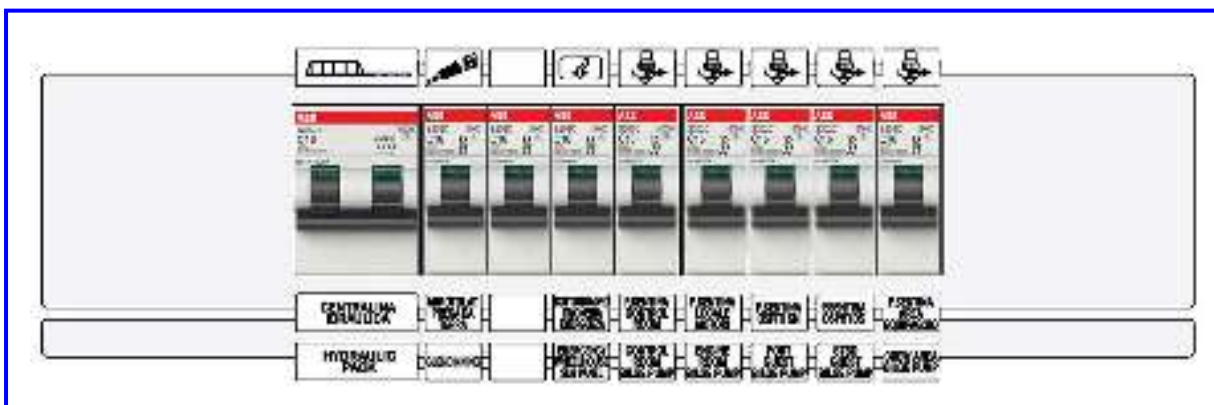
SECTION G1 – Magneto-thermal for Services 24VDC.

Port Engine Main DC/DC Converter; Starboard Engine Main DC/DC Converter; Port Engine Back-Up DC/DC Converter; Starboard Engine Back-Up DC/DC Converter.



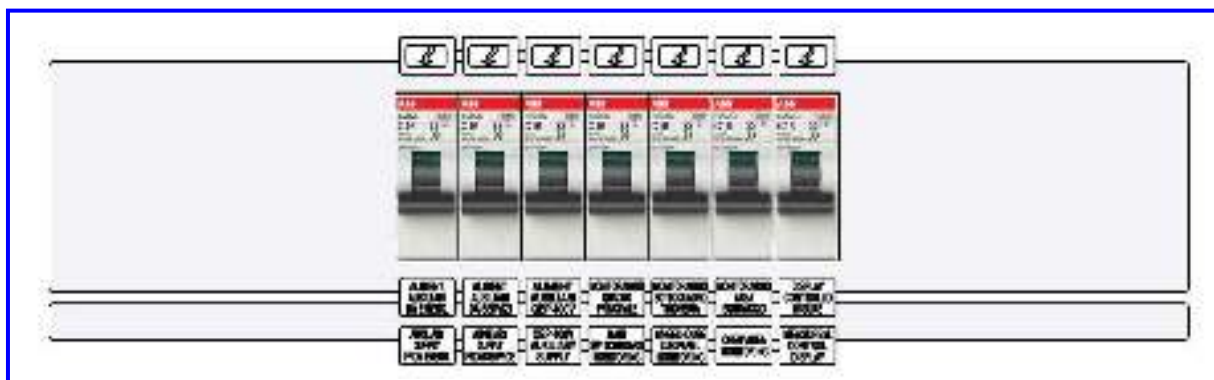
SECTION G2 – Magneto-thermal for Services 24VDC.

Hydraulic Pack; Glendinning; Spare; Emergency Wheelhouse Sub-Panel; Control Room Bilge Pump; Engine Room Bilge Pump; Port Guest Bilge Pump; Starboard Guest Bilge Pump; Crew Area Bilge Pump.



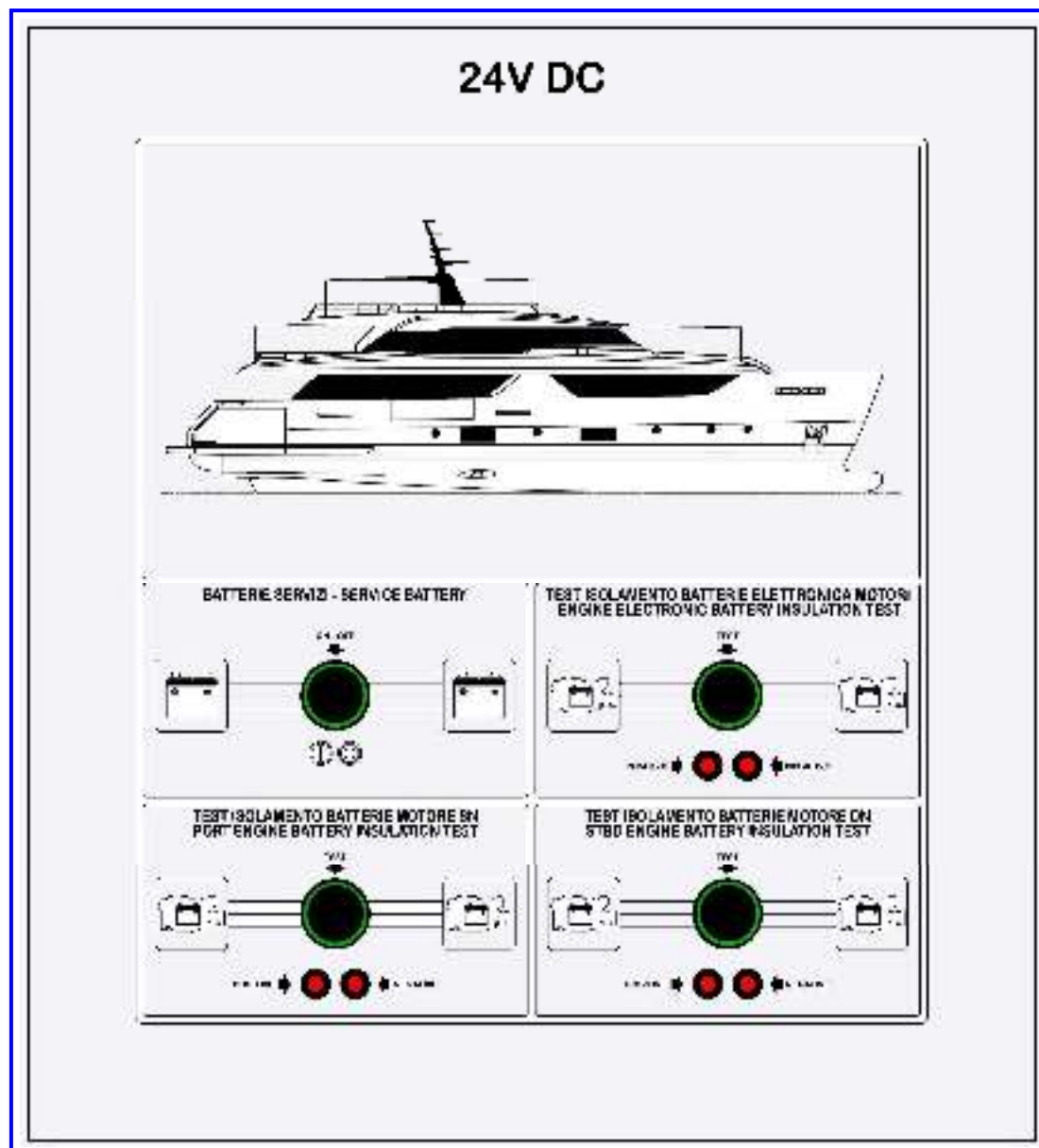
SECTION G3 – Magneto-thermal for Services 24VDC.

Auxiliary Supply From Emergency; Auxiliary Supply From Service; QEP 400V Auxiliary Supply; Main Switchboard Monitoring; Wheelhouse Sub-Panel Monitoring; Crew Area Monitoring; Measurement Control Display.

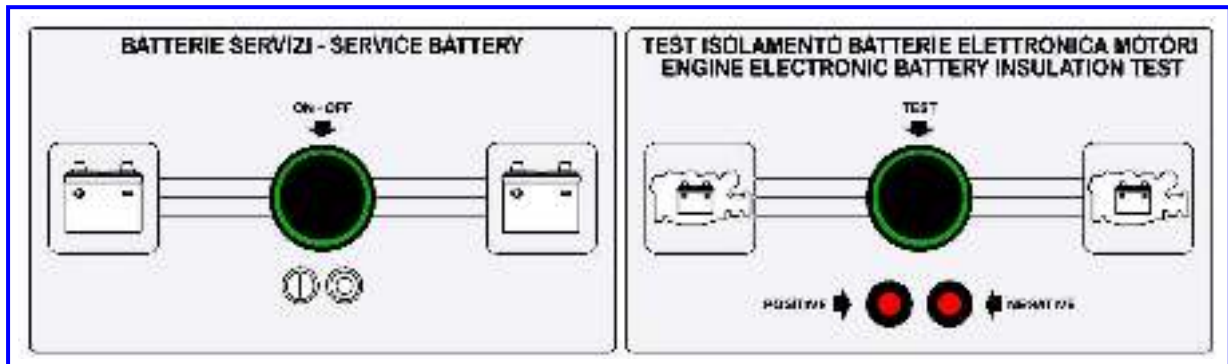


Main switchboard **SECTION H****24V DC**

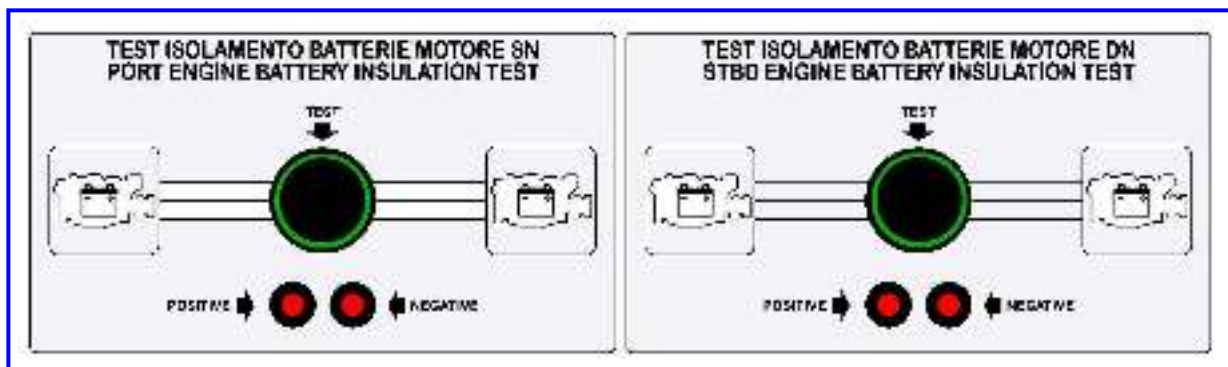
In order to make the description of switchboard easier, this section is divided into two parts:

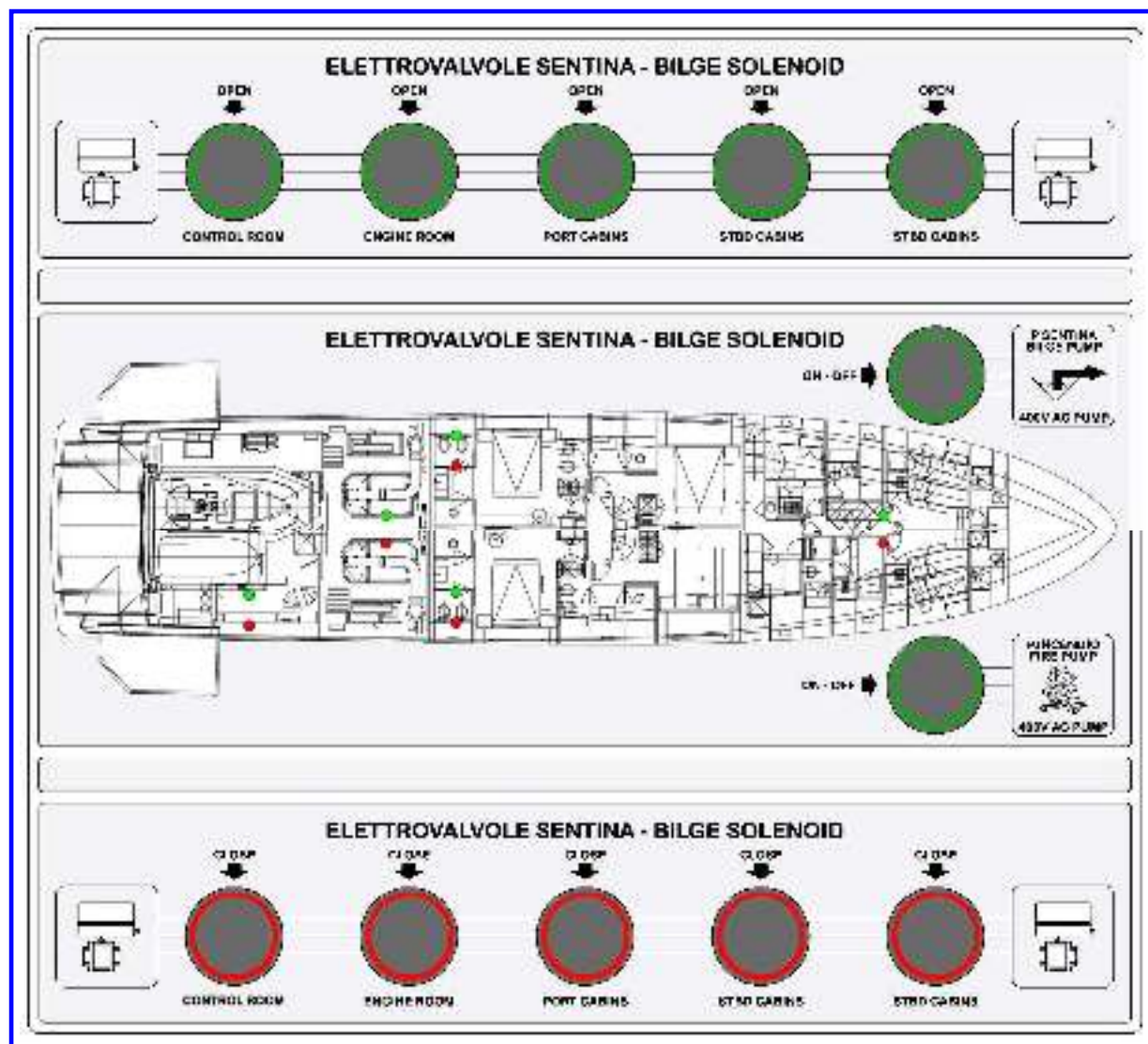


SECTION H1 – Service Battery; Engine Electronic Battery Insulation Test.

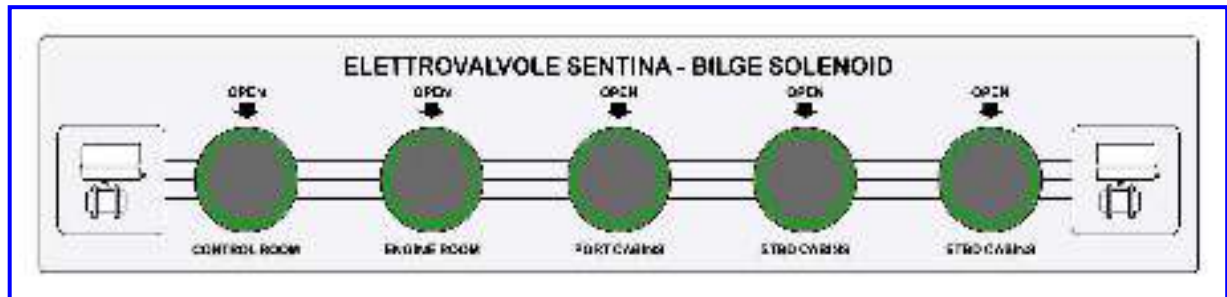


SECTION H2 – Port Engine Battery Insulation Test; Starboard Engine Battery Insulation Test.

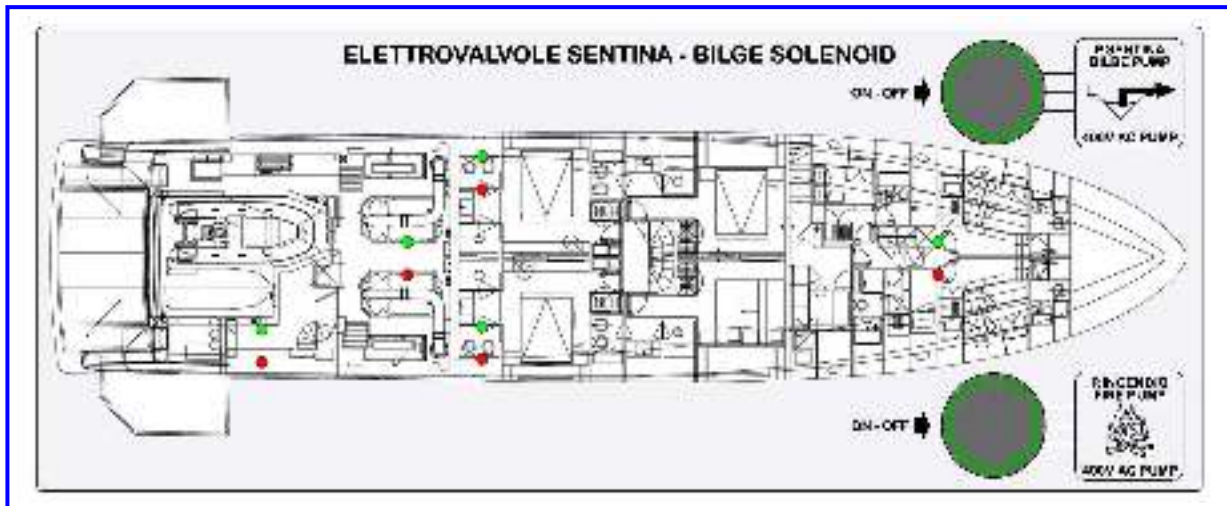


Main switchboard **SECTION I****Bilge Solenoid**

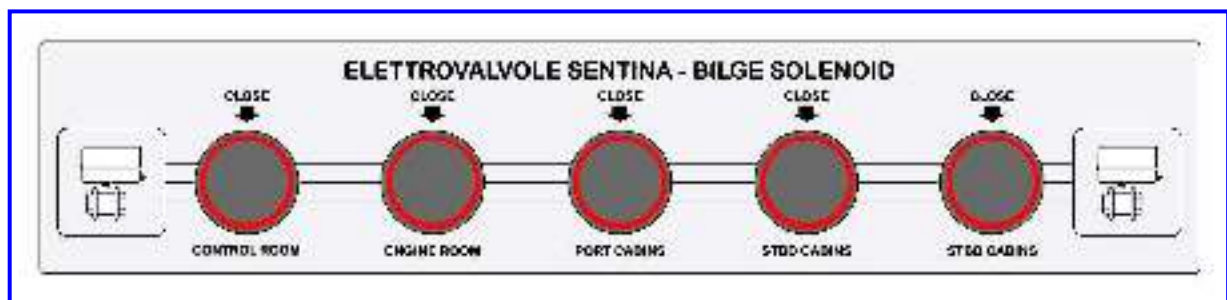
SECTION I1 – Bilge Solenoid - Activation.



SECTION I2 – Bilge Pump; Fire Pump.

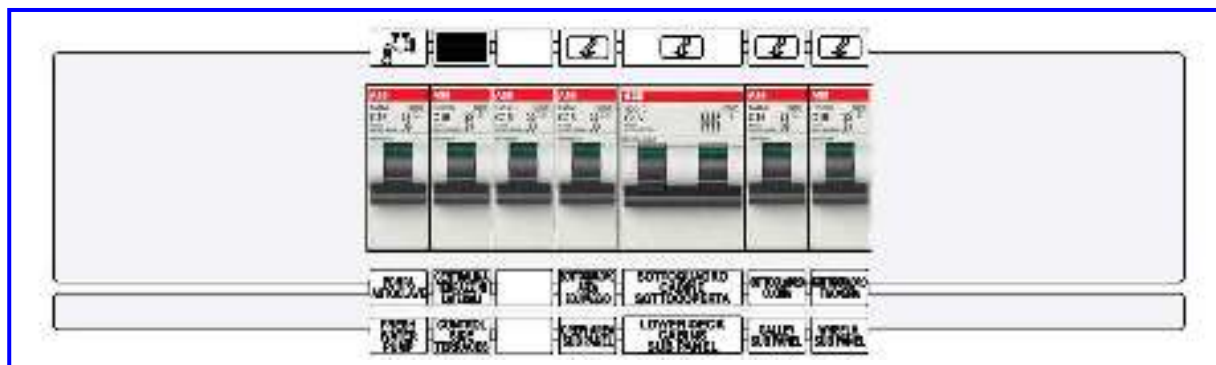


SECTION I3 – Bilge Solenoid – De-Activation.



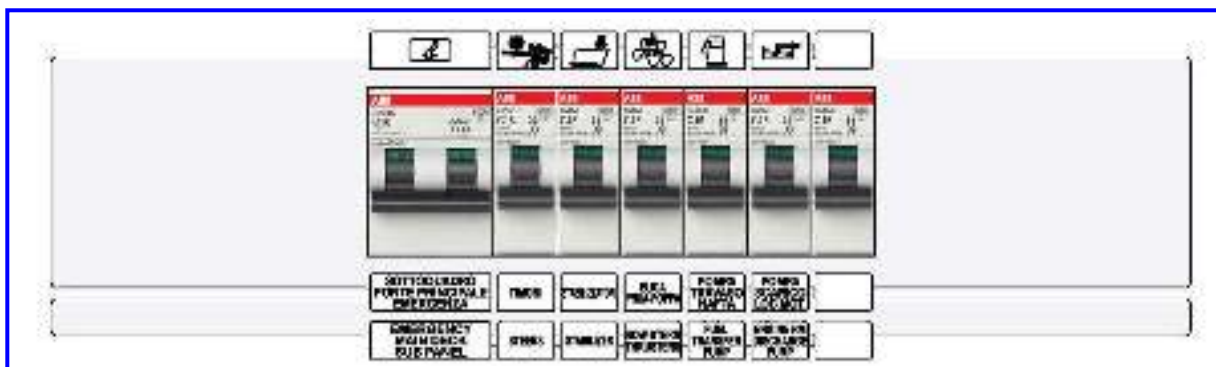
SECTION L1 – Magneto-thermal 24V DC distribution.

Fresh Water Pump; Control Side Terraces; Spare; Crew Area Sub-Panel; Lower Deck Cabins Sub-Panel; Galley Sub-Panel; Wheelhouse Sub-Panel.



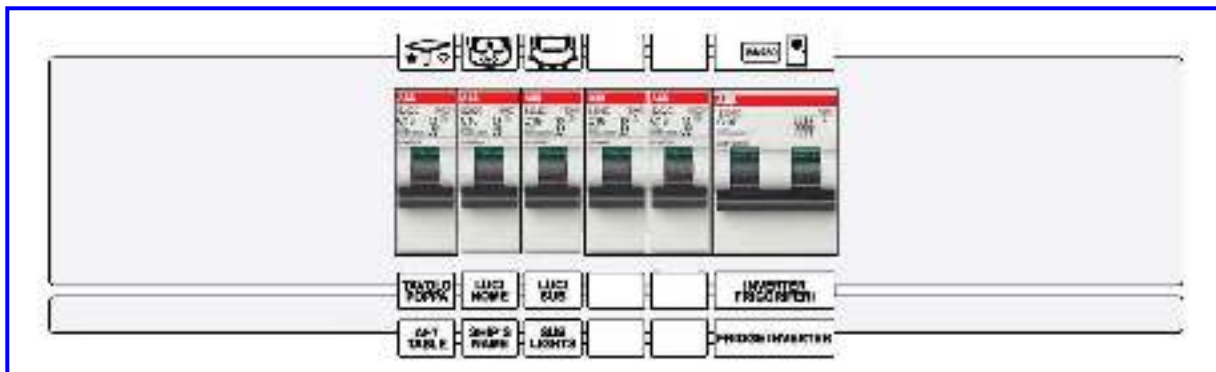
SECTION L2 – Magneto-thermal 24V DC distribution.

Emergency Main Deck Sub-Panel; Steers; Stabilizer; Bow-Stern Thruster; Fuel Transfer Pump; Engine Room Discharge Pump; Spare.



SECTION L3 – Magneto-thermal 24V DC distribution.

AFT Table; Ship's Name; Sub Lights; Spare; Spare; Fridge Inverter.



Electric System - Circuit Booklet (1 of 4)

B&B SYSTEM SRL																	YARD :		SAN LORENZO																																																																													
CIRCUIT BOOKLET																	HULL:		SD118-102																																																																													
VAC ENERGY SOURCES												VDC ENERGY SOURCES							<table><tr><th colspan="4">PARAMETERS</th></tr><tr><th colspan="2">AC (Ph):</th><th>400</th><th>AC 1Ph:</th><th>230</th></tr><tr><td colspan="2">MAX. DROP VAC 3Ph(%):</td><td>6</td><td colspan="2">24</td></tr><tr><td colspan="2">MAX. DROP VAC 1Ph(%):</td><td>6</td><td colspan="2">14</td></tr><tr><td colspan="4">FREQUENCY</td><td colspan="2"></td><td colspan="2">50</td><td colspan="4"></td></tr><tr><td colspan="4">VDC :</td><td colspan="2">24</td><td colspan="4"></td><td colspan="4"></td></tr><tr><td colspan="4">MAX. DROP VDC (%):</td><td colspan="2">10</td><td colspan="4">2</td><td colspan="4"></td></tr><tr><td colspan="4">TEMPERATUR: 45°</td><td colspan="2"></td><td colspan="4">Icc simmetric < 6KA</td><td colspan="4"></td></tr></table>					PARAMETERS				AC (Ph):		400	AC 1Ph:	230	MAX. DROP VAC 3Ph(%):		6	24		MAX. DROP VAC 1Ph(%):		6	14		FREQUENCY						50						VDC :				24										MAX. DROP VDC (%):				10		2								TEMPERATUR: 45°						Icc simmetric < 6KA							
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TEMPERATUR: 45°						Icc simmetric < 6KA																																																																																										
GENERATOR N° 1 (PORT)					400V - 3PH - 70kW - 50Hz					SERVICE BATTERIES					24VDC / 800Ah																																																																																	
GENERATOR N° 2 (STBD)					400V - 3PH - 70kW - 50Hz					EMERGENCY BATTERY					24VDC / 400Ah																																																																																	
SHORE POWER N° 1					400V - 3PH - 35kW - 50Hz																																																																																											
NOTA K1										NOTA K2																																																																																						
Correzione temperatura ambiente >45°C temperatura scilata 45°C										Correzione cavi in fascio																																																																																						
ENERGY SOURCES MAIN LINES																																																																																																
CIRCUIT		MACHINERY						CABLE									DROP			AUTOMATIC CIRCUIT BREAKER																																																																												
CIRC.	DENOMINATION	Ph	VOLT	POWER	Bars	cos FI	CURR	MARK/TYPE	L	N° WIRE	COND. SIZE	RES. 20°C IN DC	REACT	CARR.C (Iz)	K1	K2	CARR.C (Iz)	RES. 45°C	VOLTS	ΔV	ΔV TOT.	TYPE	ICC	MAKING CAP.	PROTECTION																																																																							
N°	SYMBOL		V	W			A	PRYSMIAN	MT.	N°	(n x mm2)	Ohm/ Km	Ohm/ Km	A (45°C)	45°C		A (Effect.)	Ohm/ Km	V	%	%		(KA)	(KA)	(A)																																																																							
	PORT GENERATOR	3	400	70.000		0,8	126,30	HFX-U 0,6/1 KV	10	4	1 x 50	0,387	0,101	196,00	1,00	0,70	137,2	0,43	0,13	0,03	0,03	Magnetot.	2,42	25	250																																																																							
	STDB GENERATOR	3	400	70.000		0,8	126,30	HFX-U 0,6/1 KV	15	4	1 x 50	0,492	0,101	196,00	1,00	0,70	137,2	0,54	0,23	0,05	0,06	Magnetot.	2,42	25	250																																																																							
	SHORE POWER 1	3	400	35.000		0,8	63,15	HFX-U 0,6/1 KV	13	4	1 x 35	0,664	0,086	157,00	1,00	0,70	109,9	0,73	0,13	0,03	0,03	Magnetot.	2,20	25	100																																																																							
	SERVICE BATTERIES		24	5.500			229	HFX-U 0,6/1 KV	7	1	1 x 120	0,664	0,09	339,00	1,00	1,00	339	0,73	0,14	0,60	0,60	Fuse	4,70	-	300																																																																							
AC 3Ph LOAD LINES FROM MAIN PANEL																																																																																																
CIRCUIT		MACHINERY						CABLE									DROP			AUTOMATIC CIRCUIT BREAKER																																																																												
CIRC.	DENOMINATION	Ph	VOLT	POWER	Bars	cos FI	CURR	MARK/TYPE	L	N° WIRE	COND. SIZE	RES. 20°C IN DC	REACT	CARR.C (Iz)	K1	K2	CARR.C (Iz)	RES. 45°C	VOLTS	ΔV	ΔV TOT.	TYPE	ICC	MAKING CAP.	PROTECTION																																																																							
N°	SYMBOL		V	W			A	PRYSMIAN	MT.	N°	(n x mm2)	Ohm/ Km	Ohm/ Km	A (45°C)	45°C		A (Effect.)	Ohm/ Km	V	%	%		(KA)	(KA)	(A)																																																																							
	MAIN BILGE PUMP	3	400	5.000	A	0,8	9,02	HFX-U 0,6/1 KV	5	1	3 x 6	3,08	0,084	36,00	1,00	1,00	36	3,38	0,12	0,03	0,09	Salvatore	2,01	6	10																																																																							
505	FIRE /BILGE PUMP	3	400	5.000	B	0,8	9,02	HFX-U 0,6/1 KV	7	1	3 x 6	3,08	0,084	36,00	1,00	1,00	36	3,38	0,17	0,04	0,10	Salvatore	2,01	6	10																																																																							
600	GREY WATER PUMP	3	400	1.100	A	0,8	1,98	HFX-U 0,6/1 KV	17	1	3 x 2,5	7,41	0,094	21,00	1,00	1,00	21	8,14	0,22	0,06	0,11	Salvatore	1,99	6	2,5																																																																							
605	BLACK WATER PUMP	3	400	1.100	B	0,8	1,98	HFX-U 0,6/1 KV	17	1	3 x 2,5	7,41	0,094	21,00	1,00	1,00	21	8,14	0,22	0,06	0,11	Salvatore	1,99	6	1,6																																																																							
800	VAC FUEL PUMP	3	400	750	B	0,8	1,35	HFX-U 0,6/1 KV	15	1	3 x 2,5	7,41	0,094	21,00	1,00	1,00	21	8,14	0,13	0,03	0,09	Salvatore	1,98	6	4																																																																							
805	FRESH WATER PUMP N1	3	400	1.100	A	0,8	1,98	HFX-U 0,6/1 KV	20	1	3 x 2,5	7,41	0,094	21,00	1,00	1,00	21	8,14	0,26	0,07	0,12	Salvatore	1,99	6	4																																																																							
810	FRESH WATER PUMP N2	3	400	1.100	B	0,8	1,98	HFX-U 0,6/1 KV	20	1	3 x 2,5	7,41	0,094	21,00	1,00	1,00	21	8,14	0,26	0,07	0,12	Salvatore	1,99	6	4																																																																							

Electric System - Circuit Booklet (2 of 4)

AC 3Ph LOAD LINES FROM MAIN PANEL																									
CIRCUIT		MACHINERY						CABLE										DROP			AUTOMATIC CIRCUIT BREAKER				
CIRC.	DENOMINATION	Ph	VOLT	POWER	Bars	cos FI	CURR	MARK/TYPE	L	N° WIRE	COND. SIZE	RES. 20°C IN DC	REACT	CARR.C (Iz)	K1	K2	CARR.C (Iz)	RES. 45°C	VOLTS	ΔV	ΔV TOT.	TYPE	ICC	MAKING CAP.	PROTECTION
N°	SYMBOL		V	W			A	PRYSMEAN	MT.	N°	(n x mm2)	Ohm/ Km	Ohm/ Km	A (45°C)	45°C		A (effect.)	Ohm/ Km	V	%	%		(KA)	(KA)	(A)
500	FRESH WATER MAKER N1	3	400	4.000	A	0,8	7,22	HFX-U 0,6/1 KV	18	1	3 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,53	0,13	0,19	Magnetot.	2,01	6	16
501	FRESH WATER MAKER N2	3	400	4.000	B	0,8	7,22	HFX-U 0,6/1 KV	18	1	3 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,53	0,13	0,19	Magnetot.	2,01	6	16
502	PORT REVERSIBLE FAN	3	400	4.000	A	0,8	7,22	HFX-U 0,6/1 KV	20	1	3 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,59	0,15	0,21	Magnetot.	2,01	6	16
503	STBD REVERSIBLE FAN	3	400	5.500	B	0,8	9,92	HFX-U 0,6/1 KV	9	1	3 x 6	3,08	0,084	36,00	1,00	1,00	36	3,38	0,25	0,06	0,12	Magnetot.	2,01	6	20
504	ENGINE ROOM CIRCULATION FAN	3	400	250	A	0,8	0,45	HFX-U 0,6/1 KV	3	1	3 x 2,5	7,41	0,094	21,00	1,00	1,00	21	8,14	0,01	0,00	0,06	Salvatore	1,98	6	1,6
505	CONTROL RM EXTRACTOR	3	400	750	B	0,8	1,35	HFX-U 0,6/1 KV	15	1	3 x 1,5	12,1	0,1	16,00	1,00	1,00	16	13,30	0,22	0,05	0,11	Salvatore	1,98	6	1,6
506	OIL TRANSFER PUMP	3	400	750	A	0,8	1,35	HFX-U 0,6/1 KV	15	1	3 x 1,5	12,1	0,1	16,00	1,00	1,00	16	13,30	0,22	0,05	0,11	Salvatore	1,98	6	1,6
507	STERN THRUSTER	3	400	20.000	A	0,8	36,09	HFX-U 0,6/1 KV	15	1	3 x 35	0,524	0,074	110,00	1,00	1,00	110	0,58	0,27	0,07	0,13	Magnetot.	2,11	25	63
508	BOW THRUSTER	3	400	40.000	B	0,8	72,17	HFX-U 0,6/1 KV	37	1	3 x 35	0,524	0,074	110,00	1,00	1,00	110	0,58	1,35	0,34	0,40	Magnetot.	2,23	25	80
509	PORT WINDLASS	3	400	5.500	A	0,8	9,92	HFX-U 0,6/1 KV	40	1	3 x 6	3,08	0,084	36,00	1,00	1,00	36	3,38	1,09	0,27	0,33	Salvatore	2,01	6	12,5
510	PORT CAPSTAN	3	400	4.000	A	0,8	7,22	HFX-U 0,6/1 KV	18	1	3 x 6	3,08	0,084	36,00	1,00	1,00	36	3,38	0,36	0,09	0,15	Salvatore	2,01	6	9
511	STBD WINDLASS	3	400	5.500	B	0,8	9,92	HFX-U 0,6/1 KV	35	1	3 x 6	3,08	0,084	36,00	1,00	1,00	36	3,38	0,96	0,24	0,30	Salvatore	2,01	6	12,5
512	STBD CAPSTAN	3	400	4.000	B	0,8	7,22	HFX-U 0,6/1 KV	5	1	3 x 6	3,08	0,084	36,00	1,00	1,00	36	3,38	0,10	0,02	0,08	Salvatore	2,01	6	9
513	AIR COMPRESSOR	3	400	3.500	B	0,8	6,31	HFX-U 0,6/1 KV	12	1	3 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,31	0,08	0,14	Magnetot.	2,00	6	16
514	AFT MOVEMENT HYDR. PACK	3	400	2.200	A	0,8	3,97	HFX-U 0,6/1 KV	7	1	4 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,11	0,03	0,09	Magnetot.	1,99	6	16
515	BLACK WT TREATMENT	3	400	3.500	B	0,8	6,31	HFX-U 0,6/1 KV	16	1	4 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,41	0,10	0,16	Magnetot.	2,00	6	20
516	LOWER DECK SUBPANEL	3	400	17.000	A	0,8	30,67	HFX-U 0,6/1 KV	20	1	4 x 10	1,83	0,079	50,00	1,00	1,00	50	2,01	1,02	0,25	0,31	Magnetot. Diff 30mA	2,09	6	40
517	GALLEY SUBPANEL	3	400	22.000	A	0,8	39,69	HFX-U 0,6/1 KV	30	1	4 x 25	0,727	0,075	89,00	0,87	1,00	77,43	0,80	0,81	0,20	0,26	Magnetot. Diff 30mA	2,12	6	63
518	WHEELHOUSE SUBPANEL	3	400	7.000	B	0,8	12,63	HFX-U 0,6/1 KV	35	1	4 x 16	1,15	0,075	67,00	0,87	1,00	58,29	1,26	0,47	0,12	0,18	Magnetot. Diff 30mA	2,02	6	40
519	LAUNDRY SUBPANEL	3	400	16.000	B	0,8	28,87	HFX-U 0,6/1 KV	35	1	4 x 16	1,15	0,075	67,00	1,00	1,00	67	1,26	1,07	0,27	0,33	Magnetot.	2,08	6	40
520	CREW SUBPANEL	3	400	10.000	A	0,8	18,04	HFX-U 0,6/1 KV	35	1	4 x 16	1,15	0,075	67,00	1,00	1,00	67	1,26	0,67	0,17	0,23	Magnetot. Diff 30mA	2,04	6	50
521	MAIN DECK-OWNER SUBPANEL	3	400	12.000	B	0,8	21,65	HFX-U 0,6/1 KV	27	1	4 x 25	0,727	0,075	89,00	1,00	1,00	89	0,80	0,40	0,10	0,16	Magnetot. Diff 30mA	2,06	6	63
522	UPPER DECK SUBPANEL	3	400	7.000	A	0,8	12,63	HFX-U 0,6/1 KV	35	1	4 x 16	1,15	0,075	67,00	1,00	1,00	67	1,26	0,47	0,12	0,18	Magnetot. Diff 30mA	2,02	6	32
523	BOILER 1	3	400	3.000	A	1	4,33	HFX-U 0,6/1 KV	15	1	3 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,33	0,08	0,14	Magnetot. Diff 30mA	2,00	6	16
524	BOILER 2	3	400	3.000	B	1	4,33	HFX-U 0,6/1 KV	35	1	3 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,77	0,19	0,25	Magnetot. Diff 30mA	2,00	6	16
525	AIR CONDITIONING N1	3	400	20.000	A	0,8	36,09	HFX-U 0,6/1 KV	20	1	4 x 16	1,15	0,075	67,00	1,00	1,00	67	1,26	0,76	0,19	0,25	Magnetot.	2,11	6	50
526	AIR CONDITIONING N2	3	400	10.000	B	0,8	18,04	HFX-U 0,6/1 KV	20	1	4 x 16	1,15	0,075	67,00	1,00	1,00	67	1,26	0,38	0,10	0,15	Magnetot.	2,04	6	50
527	STABILIZER	3	400	5.000	A	0,8	9,02	HFX-U 0,6/1 KV	15	1	3 x 10	1,83	0,079	50,00	1,00	1,00	50	2,01	0,22	0,06	0,11	Magnetot.	2,01	6	400
	HOT WT CIRCULATION PUMP	3	400	1.100	B	0,8	1,98	HFX-U 0,6/1 KV	15	1	3 x 4	4,61	0,088	28,00	1,00	1,00	28	5,07	0,12	0,03	0,09	Salvatore	1,22	6	3

Electric System - Circuit Booklet (3 of 4)

AC 1Ph LOAD LINES FROM MAIN PANEL																									
CIRCUIT		MACHINERY						CABLE										DROP			AUTOMATIC CIRCUIT BREAKER				
CIRC.	DENOMINATION	Ph	VOLT	POWER	BARS	COS FI	CURR	MARK/TYPE	L	Nº WIRE	COND. SIZE	RES. 20°C IN DC	REACT	CARR.C (I _a)	K1	K2	CARR.C (I _a)	RES. 45°C	VOLTS	ΔV	ΔV TOT.	TYPE	ICC	MAKING CAP.	PROTECTION
Nº	SYMBOL		V	W			A	DRYEMIAN	MT.	Nº	(n x mm2)	Ohm/ Km	Ohm/ Km	A (45°C)	45°C		A (Effect.)	Ohm/ Km	V	%	%		(KA)	(KA)	(A)
519	UV FILTER	1	230	500	A	0,8	2,72	HFX-U 0,6/1 KV	15	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,27	0,12	0,17	Magnetot.	1,99	6	10
518	GARAGE SOCKETS	1	230	1.500	B	0,8	8,15	HFX-U 0,6/1 KV	8	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,43	0,19	0,24	Magnetot. Diff. 30mA	2,01	6	16
516	ENGINE ROOM SOCKETS 230VAC	1	230	1.500	B	0,8	8,15	HFX-U 0,6/1 KV	10	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,54	0,23	0,29	Magnetot. Diff. 30mA	2,01	6	16
526	FUEL SEPARATOR	1	230	1.500	B	0,8	8,15	HFX-U 0,6/1 KV	18	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,60	0,26	0,32	Magnetot.	2,01	6	16
528	SERVICE BATT. CHARGER N1	1	230	1.800	A	0,8	9,78	HFX-U 0,6/1 KV	4	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,16	0,07	0,13	Magnetot.	2,01	6	16
530	SERVICE BATT. CHARGER N2	1	230	1.800	B	0,8	9,78	HFX-U 0,6/1 KV	4	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,16	0,07	0,13	Magnetot.	2,01	6	16
532	MTU BATT. CHARGER	1	230	1.000	A	0,8	5,43	HFX-U 0,6/1 KV	15	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,54	0,23	0,29	Magnetot.	2,00	6	16
534	STARTING BATT CHARGER	1	230	1.000	B	0,8	5,43	HFX-U 0,6/1 KV	15	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,54	0,23	0,29	Magnetot.	2,00	6	16
536	GENERATOR BATT CHARGER	1	230	1.000	B	0,8	5,43	HFX-U 0,6/1 KV	18	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,64	0,28	0,34	Magnetot.	2,00	6	16
540																							1,98		
542	EMERGENCY BATT.CHARGER	1	230	1.000	A	0,8	5,43	HFX-U 0,6/1 KV	25	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,56	0,24	0,30	Magnetot.	2,00	6	16
544	RADIO BATT. CHARGER	1	230	250	B	0,8	1,36	HFX-U 0,6/1 KV	25	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,22	0,10	0,16	Magnetot.	1,98	6	6
546	PORT STEER	1	230	2.250	A	0,8	12,23	HFX-U 0,6/1 KV	25	1	2 x 6	3,08	0,084	44,00	1,00	1,00	44	3,38	0,84	0,37	0,43	Magnet.	2,02	6	25
548	STBD STEER	1	230	2.250	B	0,8	12,23	HFX-U 0,6/1 KV	13	1	2 x 6	3,08	0,084	44,00	1,00	1,00	44	3,38	0,44	0,19	0,25	Magnet.	2,02	6	25
550	GALLEY REFRIGERATOR	1	230	2.500	B	0,8	13,59	HFX-U 0,6/1 KV	25	1	2 x 6	3,08	0,084	44,00	1,00	1,00	44	3,38	0,94	0,41	0,47	Magnet.	2,03	6	16
	SOFTENER	1	230	250	A	0,8	1,36	HFX-U 0,6/1 KV	12	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,11	0,05	0,10	Magnet.	1,98	6	16
	TECNICOMAR FILTER	1	230	250	B	0,8	1,36	HFX-U 0,6/1 KV	18	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,10	0,04	0,10	Magnet.	1,98	6	6

Electric System - Circuit Booklet (4 of 4)

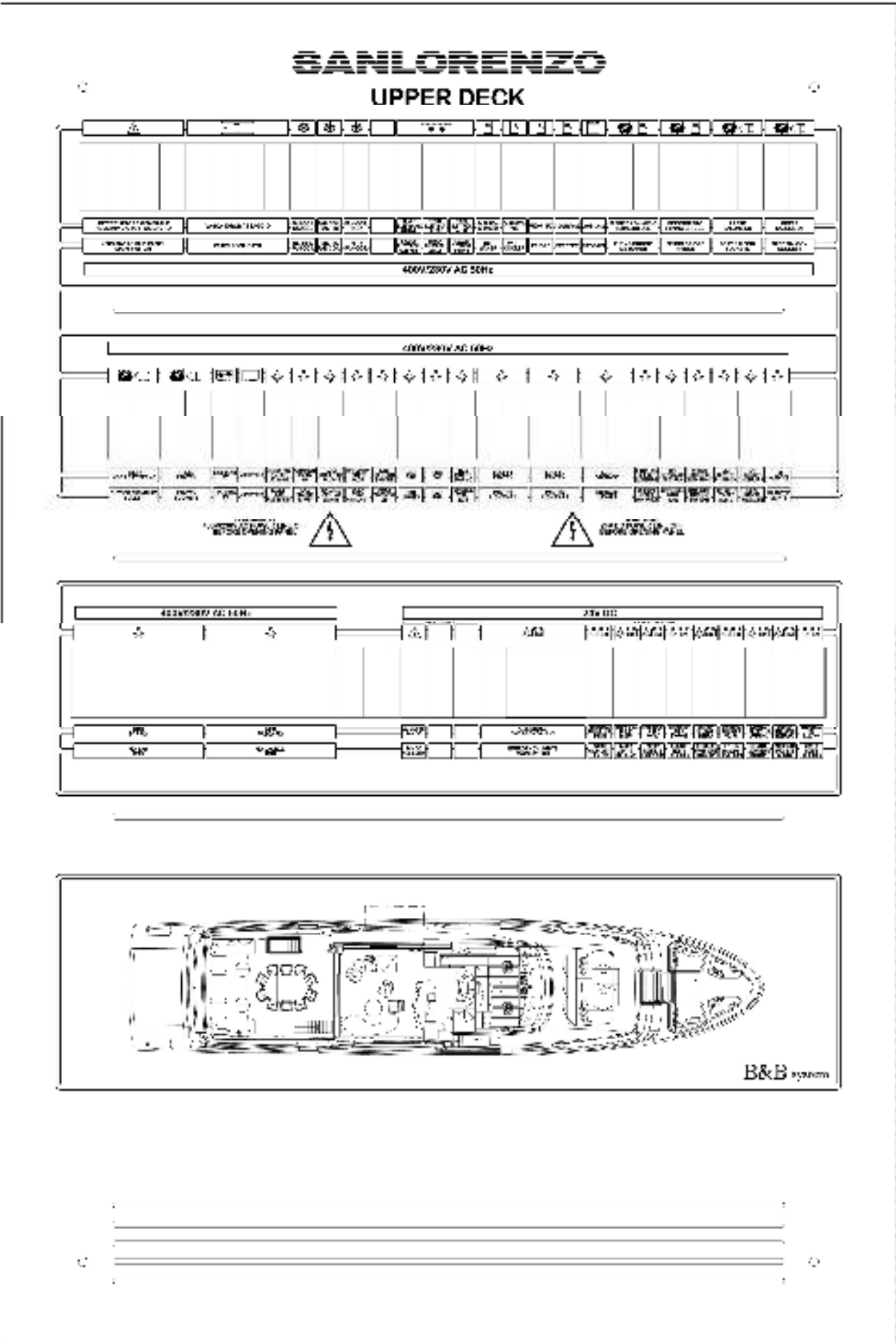
DC LOAD LINES FROM MAIN PANEL																								
CIRCUIT		MACHINERY					CABLE										DROP			AUTOMATIC CIRCUIT BREAKER				
CIRC.	DENOMINATION	Ph	VOLT	POWER	BATTERY	CURR	MARK/TYPE	L	N° WIRE	COND. SIZE	RES. 20°C IN DC	REACT	CARR.C (Ix)	K1	K2	CARR.C (Ix)	RES. 45°C	VOLTS	ΔV	ΔV TOT.	TYPE	ICC	MAKING CAP.	PROTECTION
N°	SYMBOL		V	W	SERV / EMERG	A	PRYSMIAN	MT.	N°	(n x mm2)	Ohm/ Km	Ohm/ Km	A (45°C)	45°C		A (Effect.)	Ohm/ Km	V	%	%		(KA)	(KA)	(A)
	HYDRAULIC PACK		24	1.500	SERV	62,50	HFX-U 0,6/1 KV	11	1	1 x 50	0,387	0,101	196,00	1,00	1,00	196	0,43	0,58	2,44	3,04	Magnetot.	4,88	6	2X80
	GANGWAY AUXILIARY		24	100	SERV	4,17	HFX-U 0,6/1 KV	18	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	1,22	5,09	5,69	Fuse	0,16	-	10
	SHORE S. REEL		24	100	SERV	4,17	HFX-U 0,6/1 KV	18	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	1,22	5,09	5,69	Fuse	0,16	-	10
	WH. SUBPANEL SERVICE DIRECT		24	300	SERV	12,50	HFX-U 0,6/1 KV	16	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	2,03	8,44	9,05	Fuse	0,30	-	16
	GARAGE CTRL RM BILGE PUMP		24	100	SERV	4,17	HFX-U 0,6/1 KV	18	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	1,22	5,09	5,69	Fuse	0,16	-	16
	ENGINE ROOM BILGE PUMP		24	100	SERV	4,17	HFX-U 0,6/1 KV	4	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	0,27	1,13	1,73	Fuse	0,74	-	16
	PORT GUEST BILGE PUMP		24	100	SERV	4,17	HFX-U 0,6/1 KV	25	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	1,70	7,07	7,67	Fuse	0,12	-	16
	STBD GUEST BILGE PUMP		24	100	SERV	4,17	HFX-U 0,6/1 KV	20	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	1,36	5,66	6,26	Fuse	0,15	-	16
	CREW AREA BILGE PUMP		24	100	SERV	4,17	HFX-U 0,6/1 KV	25	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	1,70	7,07	7,67	Fuse	0,12	-	16
	VDC FRESH WATER PUMP		24	1.400	SERV	58,33	HFX-U 0,6/1 KV	15	1	2 x 16	1,15	0,075	82,00	1,00	1,00	82	1,26	2,21	9,22	9,82	Magnetot.	1,27	6	80
	SIDE TERRACES CONTROL PANEL		24	1.000	SERV	41,67	HFX-U 0,6/1 KV	11	1	2 x 16	1,15	0,075	82,00	1,00	1,00	82	1,26	1,16	4,83	5,43	Magnetot.	1,73	6	2X25
	CREW SUBPANEL		24	1.800	SERV	75,00	HFX-U 0,6/1 KV	22	1	1 x 35	0,524	0,086	157,00	1,00	1,00	157	0,58	1,90	7,92	8,52	Magnetot.	1,89	6	2x50
	CABINS SUBPANEL		24	2.300	SERV	95,83	HFX-U 0,6/1 KV	22	1	1 x 50	0,387	0,101	196,00	1,00	1,00	196	0,43	1,79	7,47	8,07	Magnetot.	2,56	6	2x50
	GALLEY SUBPANEL		24	300	SERV	12,50	HFX-U 0,6/1 KV	16	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	2,03	8,44	9,05	Magnetot.	0,30	6	16
	MAIN DECK-EMERG. SUBPANEL		24	1.000	SERV	41,67	HFX-U 0,6/1 KV	20	1	2 x 16	1,15	0,075	82,00	1,00	1,00	82	1,26	2,11	8,78	9,38	Magnetot.	0,95	6	50
	WH. SUBPANEL		24	650	SERV	27,08	HFX-U 0,6/1 KV	20	1	2 x 10	1,83	0,079	61,00	1,00	1,00	61	2,01	2,18	9,08	9,68	Magnetot.	0,60	6	32
	VDC FUEL PUMP		24	750	SERV	31,25	HFX-U 0,6/1 KV	15	2	1 x 6	3,08	0,107	52,00	1,00	1,00	104	3,38	1,59	6,61	7,21	Magnetot.	0,47	6	32
	AUXILIARY STEERING		24	200	SERV	8,33	HFX-U 0,6/1 KV	20	1	2 x 16	1,15	0,075	82,00	1,00	1,00	82	1,26	0,42	1,76	2,36	Magnetot.	0,95	6	10
	ER DISCHARGE PUMP		24	100	SERV	4,17	HFX-U 0,6/1 KV	15	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,63	2,64	3,24	Magnetot.	0,32	6	16
	SUB LIGHTS		24	300	SERV	12,50	HFX-U 0,6/1 KV	8	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	1,01	4,22	4,82	Magnetot.	0,59	6	32
	SHIP'S NAME LIGHT		24	20	SERV	0,83	HFX-U 0,6/1 KV	10	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,08	0,35	0,95	Magnetot.	0,47	6	16
	STABILIZER CONTROL PANEL		24	200	SERV	8,33	HFX-U 0,6/1 KV	6	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,51	2,11	2,71	Magnetot.	0,79	6	16
	THRUSTERS CONTROL PANEL		24	250	EMERG	10,42	HFX-U 0,6/1 KV	6	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,63	2,64	3,24	Magnetot.	0,79	6	16
	AFT CONTROL PANEL LIGHT		24	20	SERV	0,83	HFX-U 0,6/1 KV	12	1	2 x 4	4,61	0,088	34,00	1,00	1,00	34	5,07	0,10	0,42	1,02	Magnetot.	0,39	6	16
	PORT ENGINE DC/DC CONV		24	200	SERV	8,33	HFX-U 0,6/1 KV	15	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	2,04	8,48	9,08	Magnetot.	0,20	6	20
	STBD ENGINE DC/DC CONV		24	200	SERV	8,33	HFX-U 0,6/1 KV	15	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	2,04	8,48	9,08	Magnetot.	0,20	6	20
	PORT ENGINE DC/DC BACK-UP CONV		24	200	SERV	8,33	HFX-U 0,6/1 KV	15	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	2,04	8,48	9,08	Magnetot.	0,20	6	20
	STBD ENGINE DC/DC BACK-UP CONV		24	200	SERV	8,33	HFX-U 0,6/1 KV	15	1	2 x 2,5	7,41	0,094	26,00	1,00	1,00	26	8,14	2,04	8,48	9,08	Magnetot.	0,20	6	20
	REFRIGERATOR INVERTER		24	2.000	SERV	83,33	HFX-U 0,6/1 KV	10	2	1 x 50	0,387	0,101	196,00	1,00	0,80	313,6	0,43	0,35	1,48	2,08	Magnet.	5,64	6	2X63

Shore Power Supply

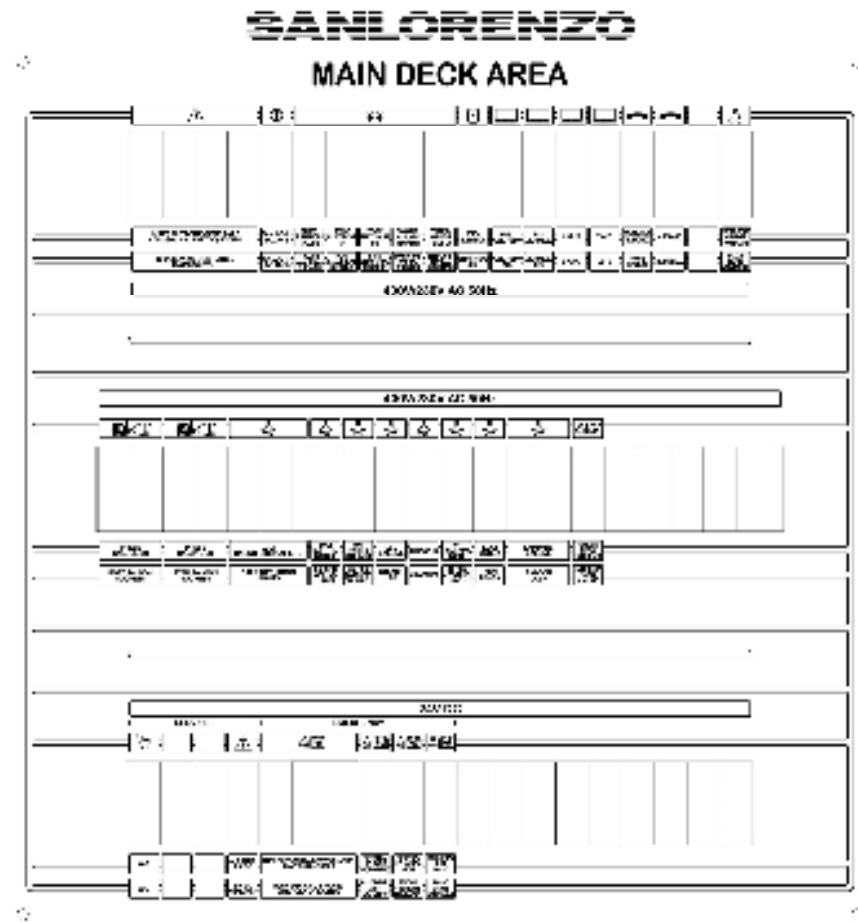
The electrical panel for the management of the power supply from the port quay is located in the engine room and exactly in the technical room on the port side of Yacht.

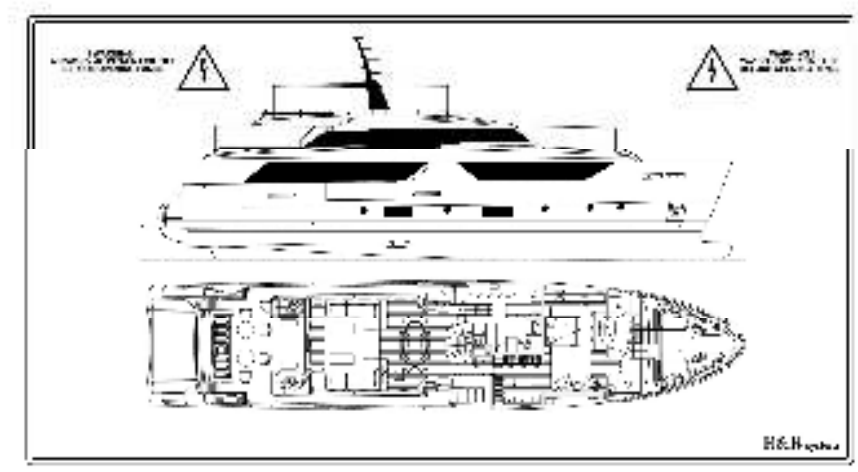


Switchboard – Upper Deck

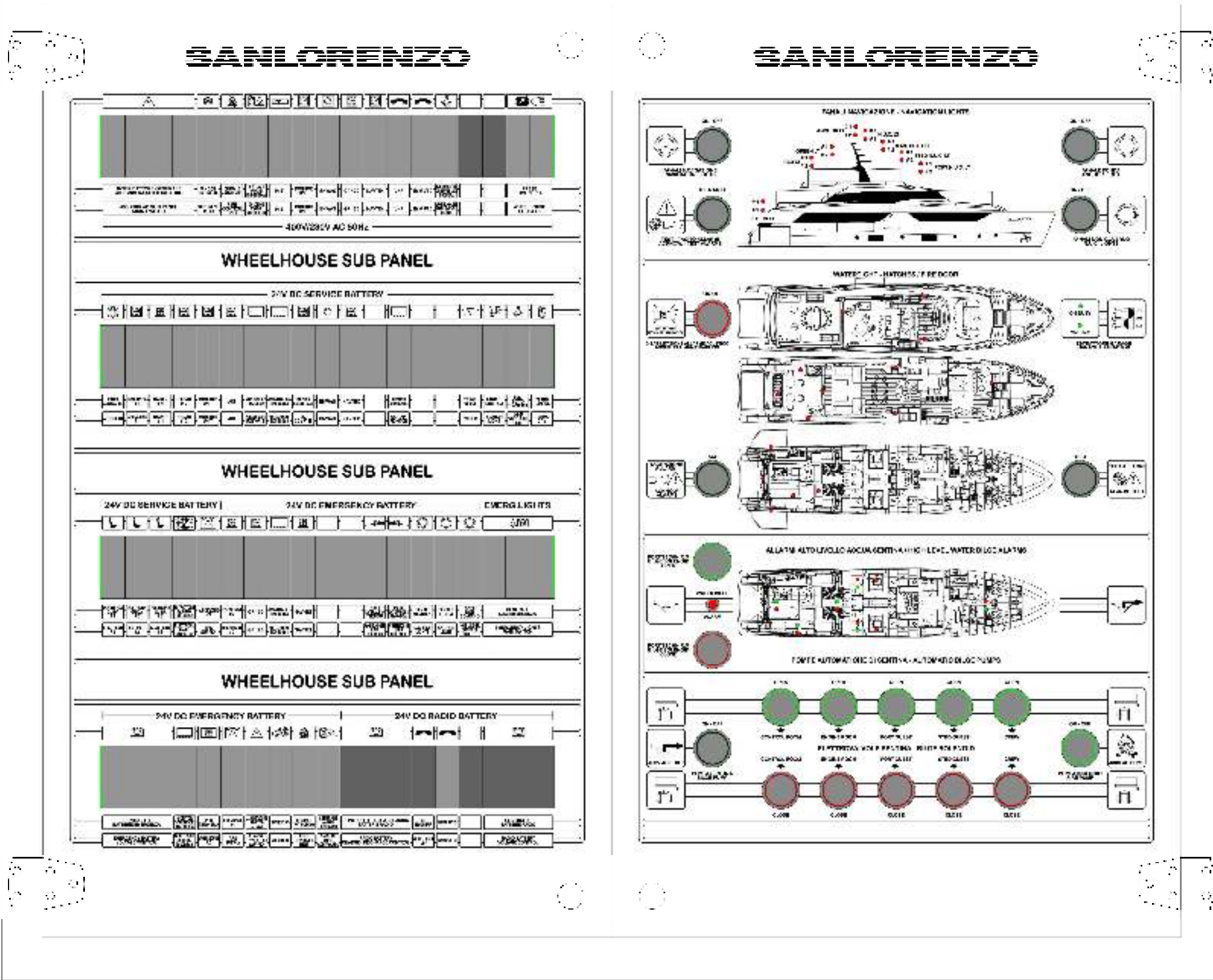


Switchboard – Main Deck

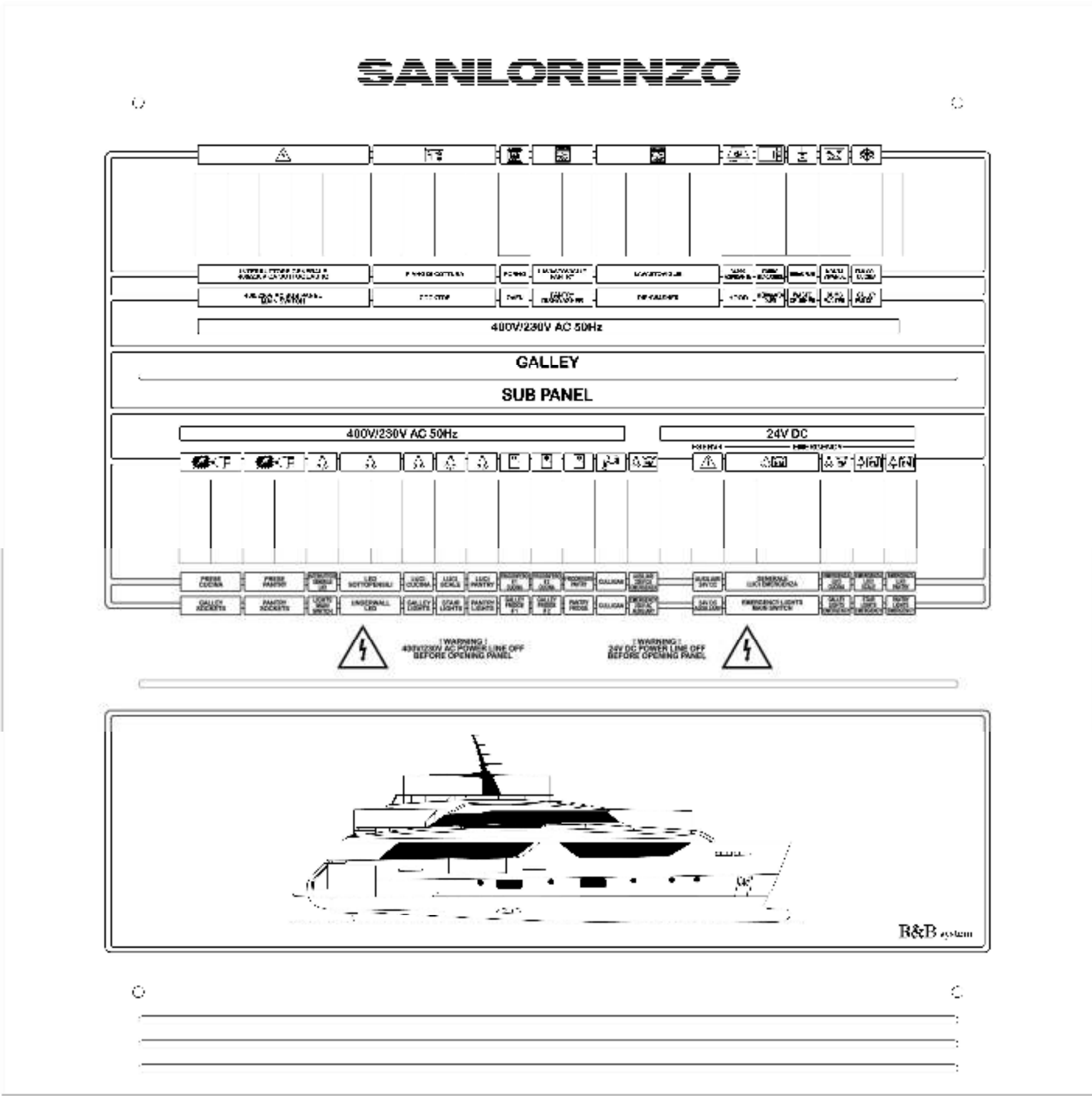




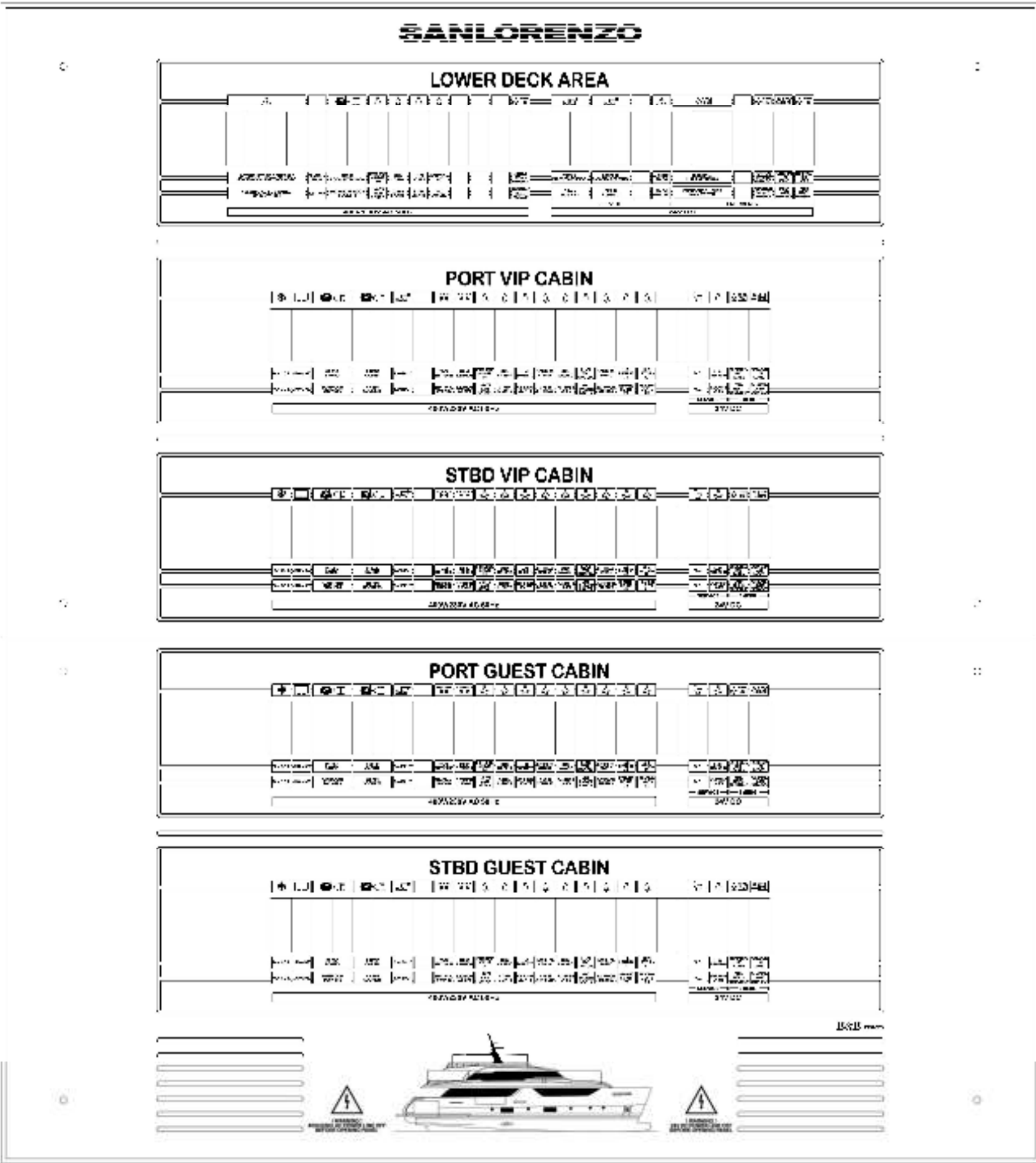
Switchboard – Wheelhouse



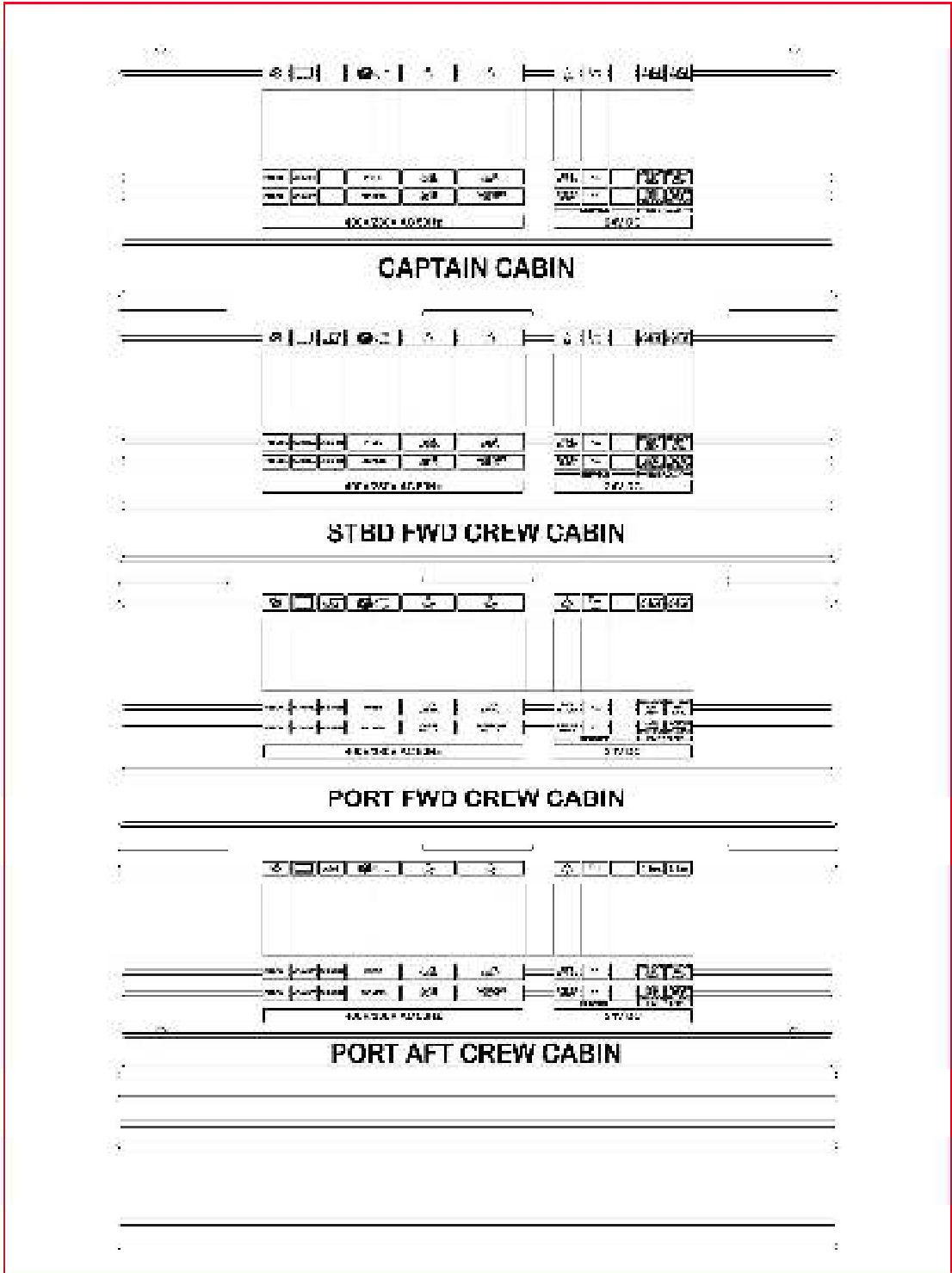
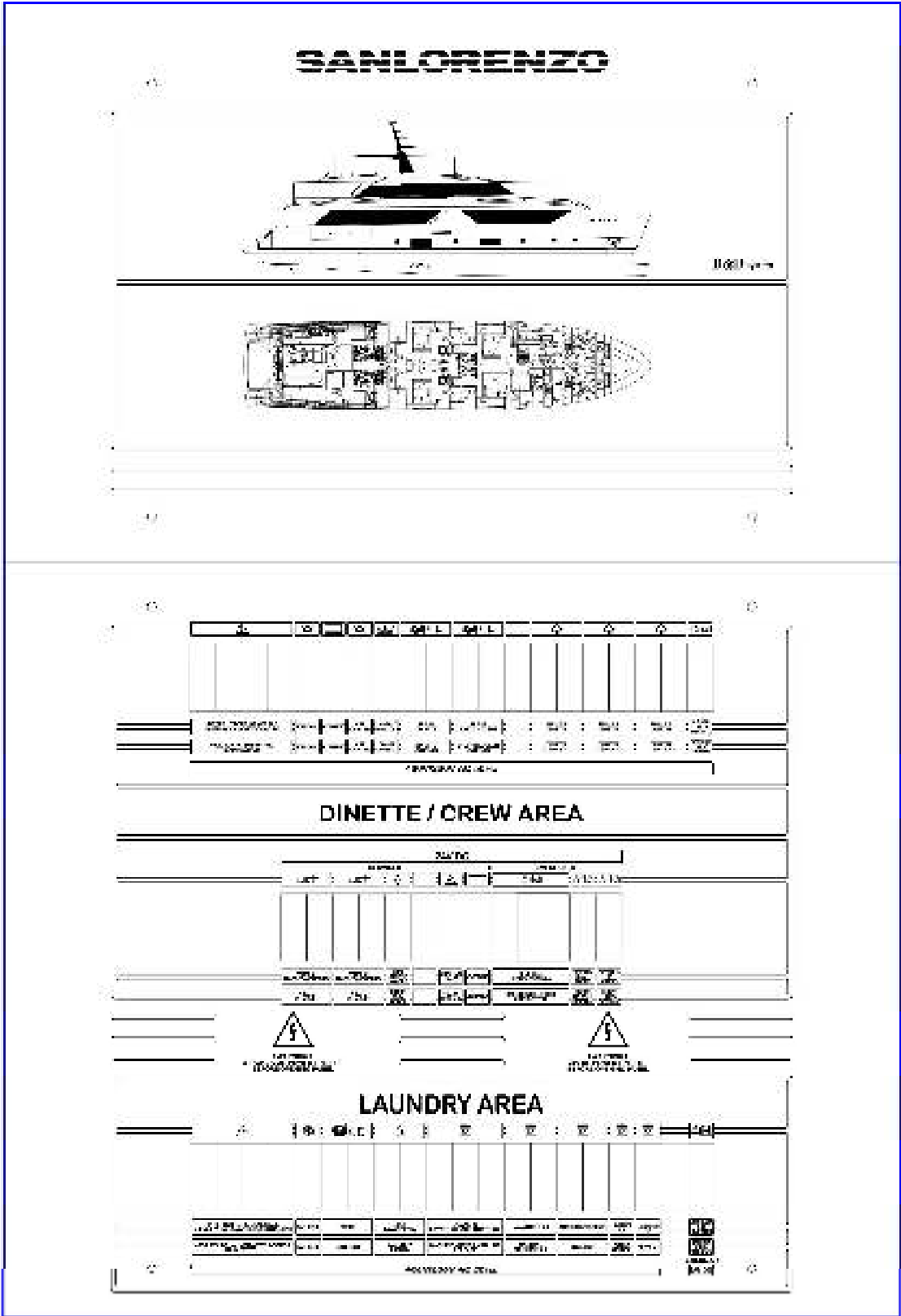
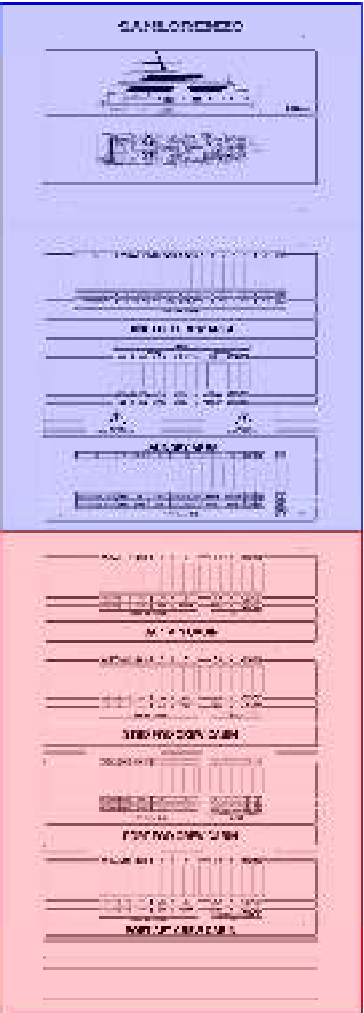
Switchboard – Galley



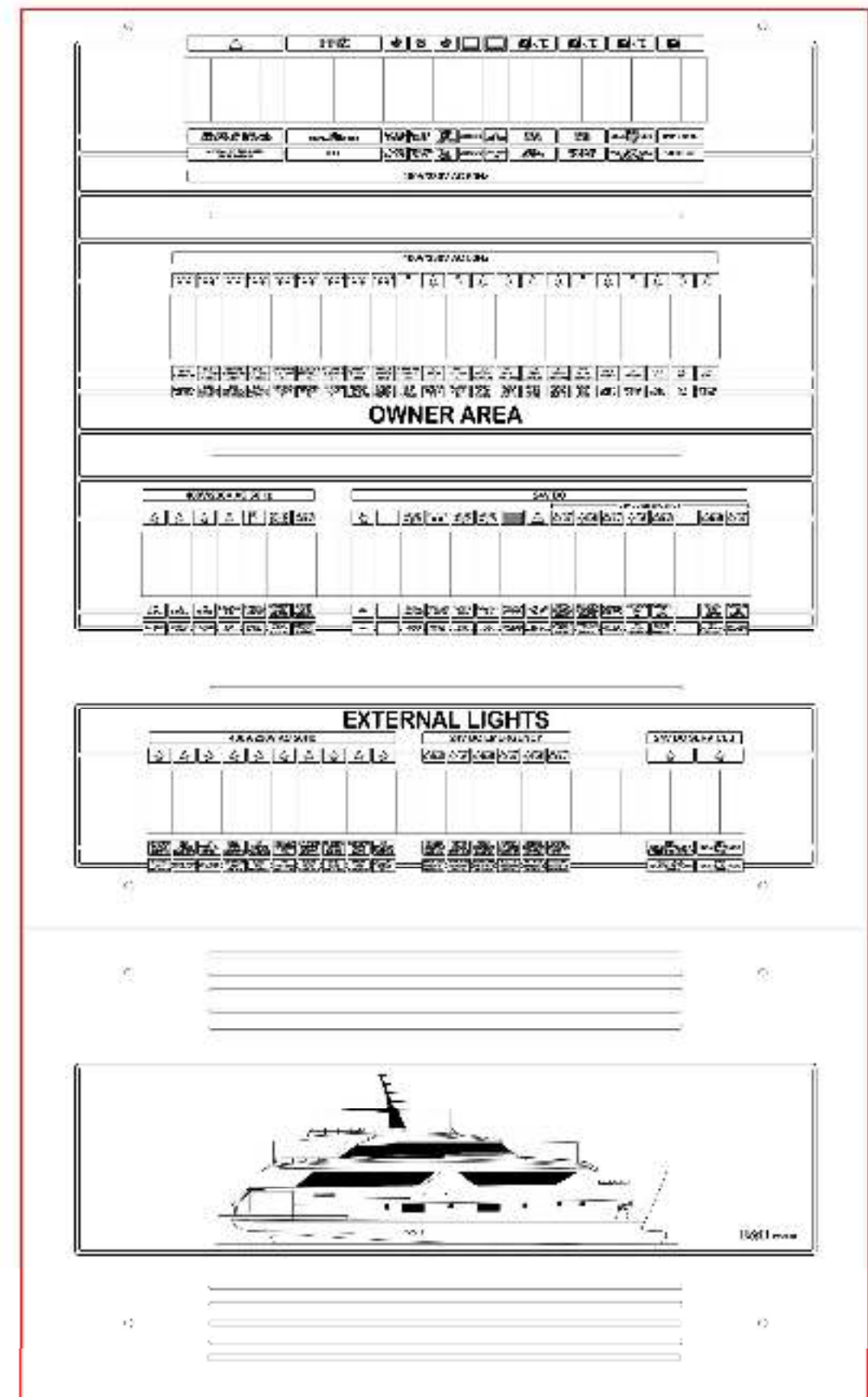
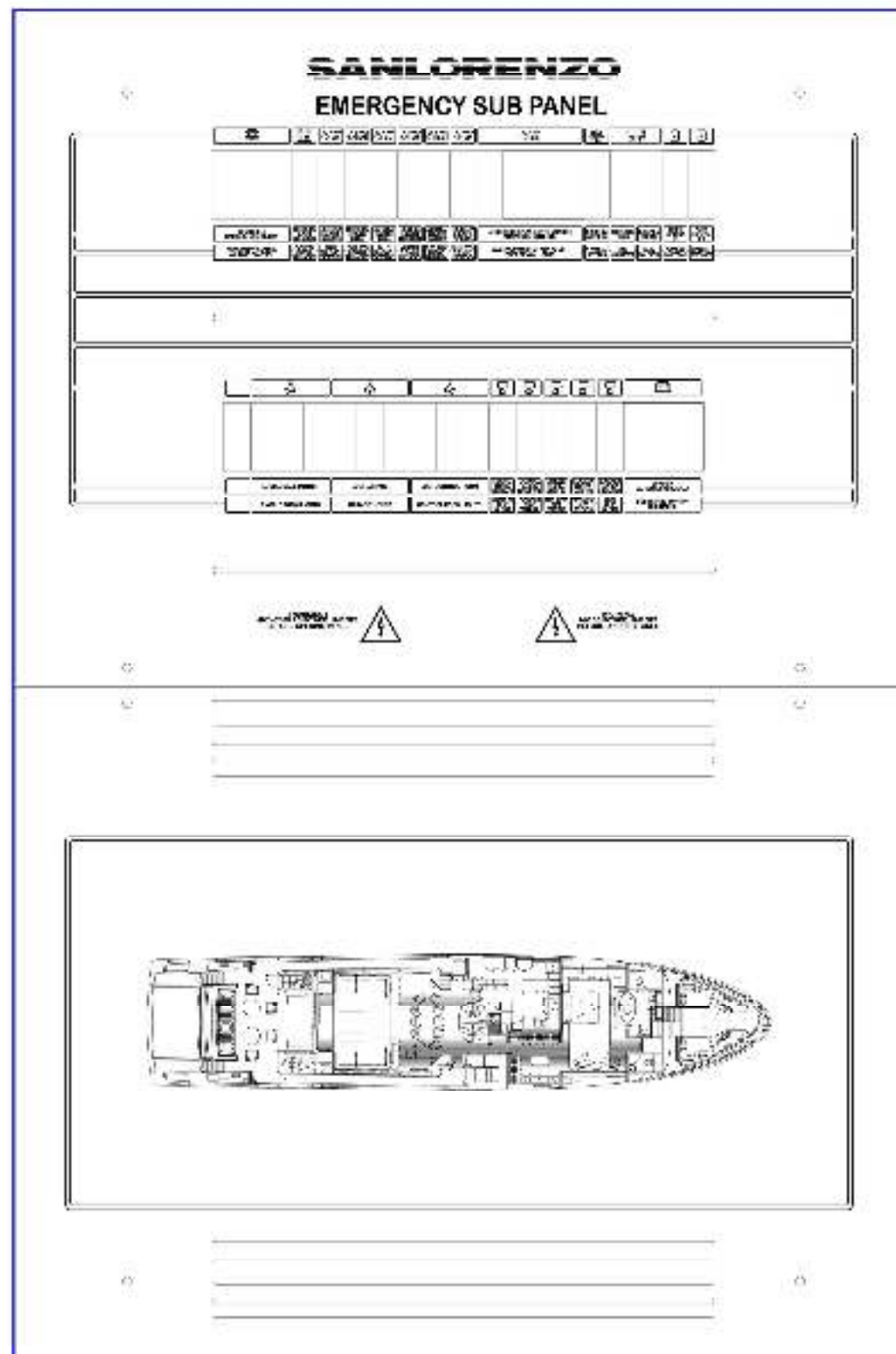
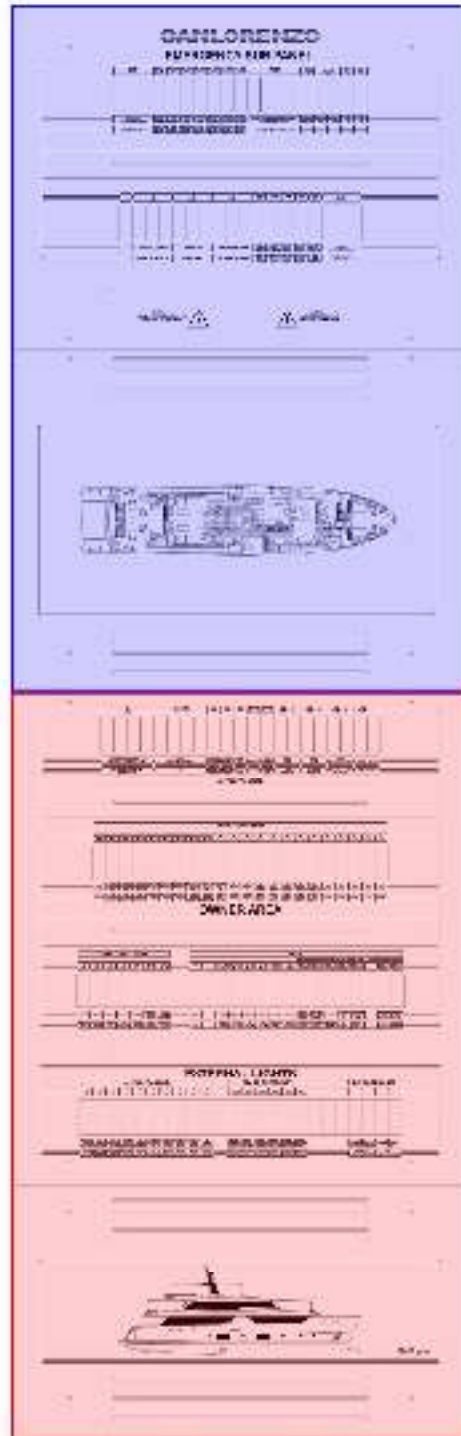
Switchboard – Lower Deck



Switchboard – Crew Cabin



Switchboard – Emergency



Section **D**

SYSTEMS

Propulsion

- Main engines

- Gear boxes

- Shaft lines

- Propellers

- Bow-Thruster

- Stabilising fins

Electric system

- Generators

- Shore power supply

- Batteries

Bilge system

Fire fighting system

Fuel system

Fresh water system

Grey water system

Black water system

Sea water system

Air conditioning system

Scuppers system

PROPULSION

Main engines

The propulsion system of this yacht has been designed to conjugate and exploit the beautiful marine characteristics of the hull, obtaining best performance in terms of speed, consumption and manoeuvrability according to the engine type chosen by you.

The propulsion system of the yacht is provided with two symmetric shaft lines, each one composed of:

- Engine **CATERPILLAR** C32 Acert Tier III, Power 1,193 kW (1,622 mHP) at 2,300 rpm.
- Vulkan Propflex mount
- Gearbox **ZF** mod. 2360 (reduction rate 4:1)
- Coupling flange
- Marinox 17 shaft, diameter 110 mm
- Stainless steel Shaft seal with water lubricated bush
- propeller shaft strut with water lubricated bush
- Five fix blade propellers in NiBrAl

Propeller design and manufacture care ensure high propulsion performance and absence of vibration and cavitation.



CATERPILLAR, world-wide leader in high-performance high-reliability low-weight marine engines manufacture, has carried out propulsion engines and reduction gears commissioning and set up during various sea trials, as well as all necessary calibrations and compensations.

The engines have the following characteristics:

Model		C32 ACERT
Manufacturer		CATERPILLAR
Number of cylinders		12 V (4-Stroke-Cycle Diesel)
Cycle		Electronic fuel injection four stroke Diesel engine
Displacement	l	32.1
Bore	mm	145
Stroke	mm	162
Compression ratio		15:1
Maximum power (at 2,300 rpm)	kW mHP	1,193 1,622
Minimum dry weight	kg	3,152
Rated engine speed	rpm	1,600 ÷ 2,300

CATERPILLAR diesel engines are designed and manufactured with the most modern techniques. Trouble-free operation and high provided power can be achieved by following the schedule maintenance and by using suitable fuels and lubricants.

Each engines will be installed in “Free Standing” configuration on four resilient mountings Vulkan DMR T35, independent one to each other dimensioned and calculated to reduce the vibration.

Injection system fuel distribution will be “Common Rail” type, electronically controlled, with independent cylinder cut out control fuel duplex filter mounted on the engine.

For Engine power Diagram: CATERPILLAR C32 ACERT e Consumption diagram l/h with 2 Engines: CATERPILLAR C32 ACERT, please refer to the engine's builder manual.



The power from the flywheel shall remain within the standard limits, that is up to an air temperature of 50°C (122°F) measured at the entry point of the air purifier, and fuel temperature up to 52°C (125° F) measured at the base of the combustion filter.

The nominal power is equal to the power of the elbows shaft, in accordance with NMMA procedure.

Engine comburent air sucking occur from the engine room air, through filters.

The personnel appointed for operating, checking and supervising the engine should acquire familiarity with such equipment, assisted by SANLORENZO shipyard specialized technicians and by reading carefully the Manufacturer Manuals.



CATERPILLAR C32 ACERT TIER III, installed on board of the yacht, have the following characteristics:

- four stroke cycle;
- direct fuel injection;
- electronically controlled hydraulic injectors;
- turbo-compressor powered engine;
- sea water comburent air post refrigeration system;
- four valves cylinder;
- belt driven 24V alternator;
- engine inherent safety electronic control;
- engine electronic control and monitoring system. ECM system;
- electric starter.

These engines are designed to be electronically controlled by computers. The computers are installed on the engines themselves and control the operating conditions. The control module (ECM), compares the answer from the engine, based on the current operating condition, to the instructions of the operator. This, ensure an accurate control of the fuel injection procedure from the ECM module.

The electronic control system guarantees the following characteristics:

- engine monitoring;
- electronic adjustment of the revolutions;
- automatic air/fuel ratio control;
- determination of the increase of the engine coupling;
- injection stroke control;
- systems diagnosis.



Frontal View



Starboard Side View

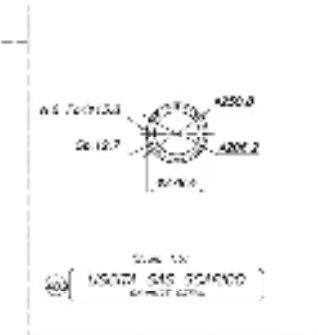
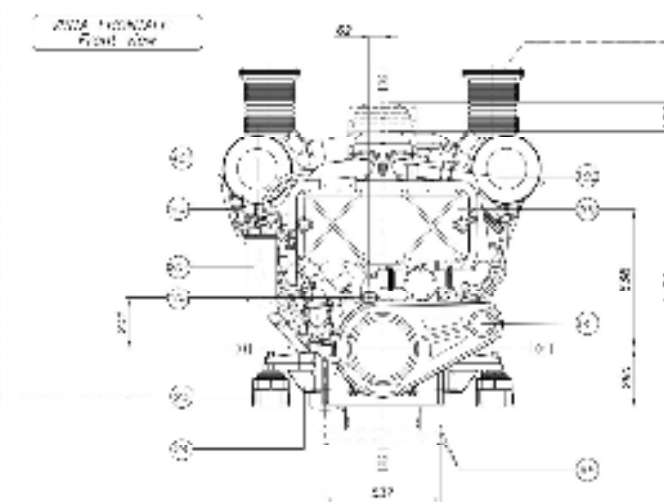
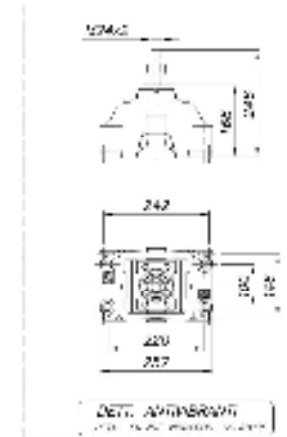
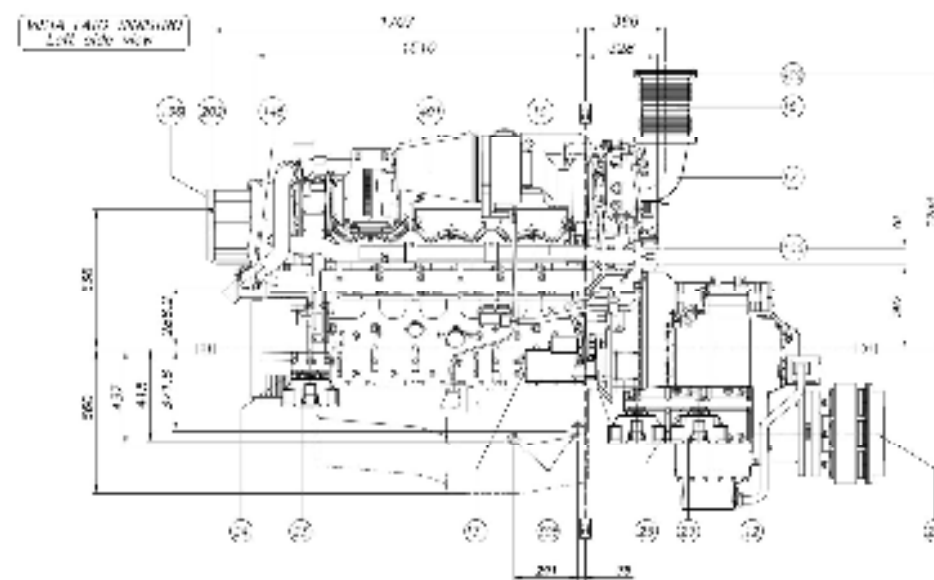


Port Side View

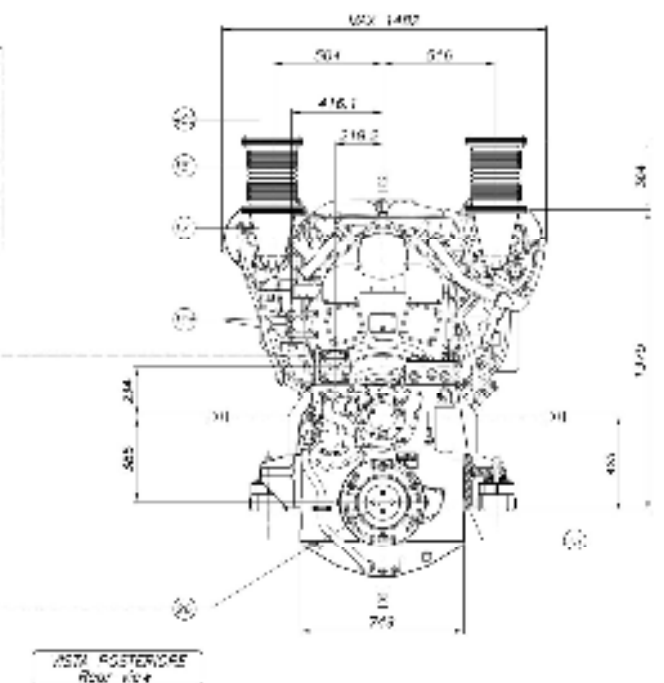
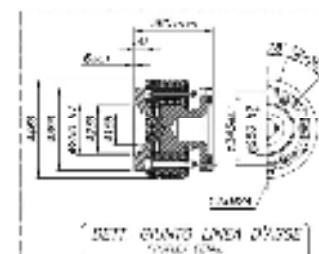
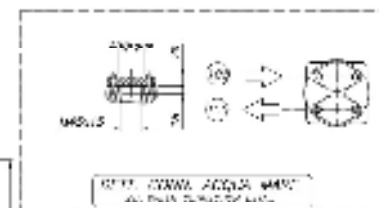
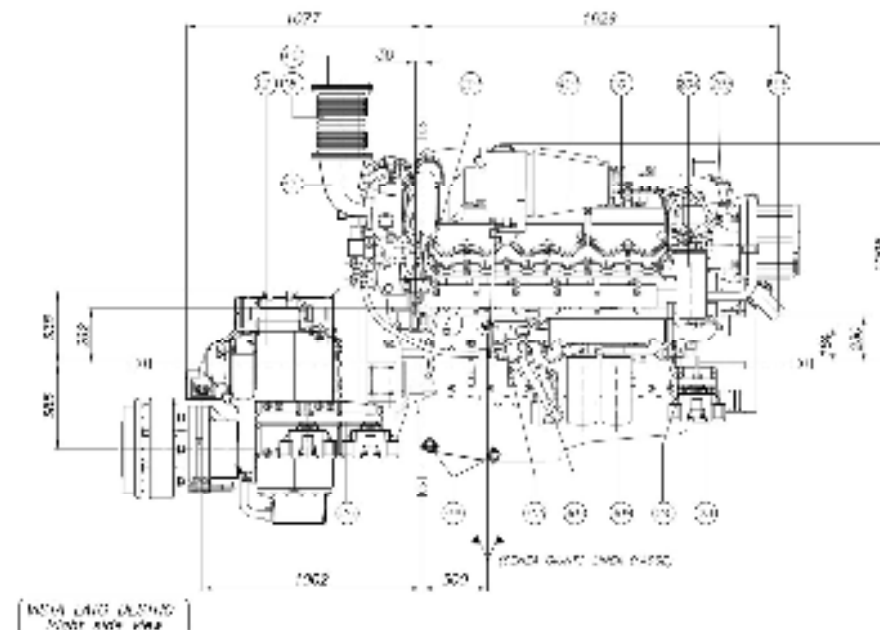
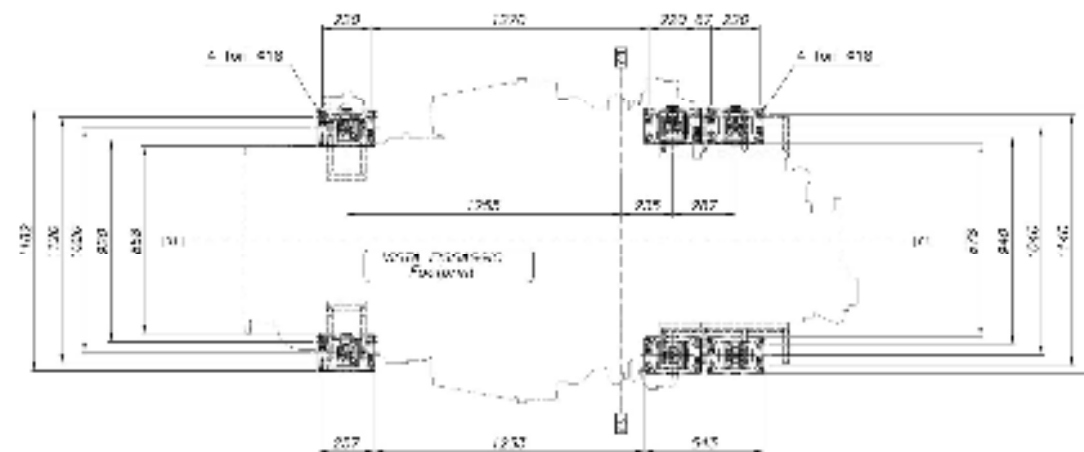
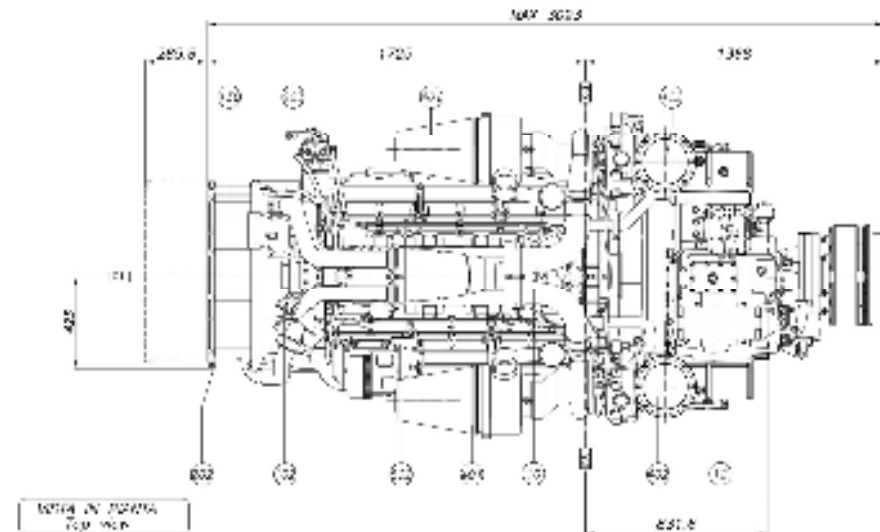


Rear View

CAT C32 Acert - Dimensions



PESO - WEIGHT	
VOLUME A SEXCO (FRESH) 100%	= 1g. 5000
VOLUME DON LIQUID EVALU. WITH FLUIDS	= 1g. 3400
PESQUERA A SEXCO - 50% FRODO (DON SEX CO) 1 SUPERO	= 1g. 1000
PESQUERA DON LIQUID EVAL. SEX CO 1 FLUIDS	= 1g. 1000
CHARRAS FRODO	= 1g. 120
GRAN LINDA PLASER FRODO	= 1g. 250
PESO TOTAL DON LIQUID TOTAL WEIGHT WITH FLUIDS	= 1g. 4950



POS	DESCRIPTION	Q	AM	DOL
20	2000-2001-2002-2003-2004-2005-2006-2007-2008-2009-2010-2011-2012-2013-2014-2015-2016-2017-2018-2019-2020-2021-2022-2023-2024-2025-2026-2027-2028-2029-2030-2031-2032-2033-2034-2035-2036-2037-2038-2039-2040-2041-2042-2043-2044-2045-2046-2047-2048-2049-2050-2051-2052-2053-2054-2055-2056-2057-2058-2059-2060-2061-2062-2063-2064-2065-2066-2067-2068-2069-2070-2071-2072-2073-2074-2075-2076-2077-2078-2079-2080-2081-2082-2083-2084-2085-2086-2087-2088-2089-2090-2091-2092-2093-2094-2095-2096-2097-2098-2099-2100-2101-2102-2103-2104-2105-2106-2107-2108-2109-2110-2111-2112-2113-2114-2115-2116-2117-2118-2119-2120-2121-2122-2123-2124-2125-2126-2127-2128-2129-2130-2131-2132-2133-2134-2135-2136-2137-2138-2139-2140-2141-2142-2143-2144-2145-2146-2147-2148-2149-2150-2151-2152-2153-2154-2155-2156-2157-2158-2159-2160-2161-2162-2163-2164-2165-2166-2167-2168-2169-2170-2171-2172-2173-2174-2175-2176-2177-2178-2179-2180-2181-2182-2183-2184-2185-2186-2187-2188-2189-2190-2191-2192-2193-2194-2195-2196-2197-2198-2199-2200-2201-2202-2203-2204-2205-2206-2207-2208-2209-2210-2211-2212-2213-2214-2215-2216-2217-2218-2219-2220-2221-2222-2223-2224-2225-2226-2227-2228-2229-2230-2231-2232-2233-2234-2235-2236-2237-2238-2239-2240-2241-2242-2243-2244-2245-2246-2247-2248-2249-2250-2251-2252-2253-2254-2255-2256-2257-2258-2259-2260-2261-2262-2263-2264-2265-2266-2267-2268-2269-2270-2271-2272-2273-2274-2275-2276-2277-2278-2279-2280-2281-2282-2283-2284-2285-2286-2287-2288-2289-2290-2291-2292-2293-2294-2295-2296-2297-2298-2299-2300-2301-2302-2303-2304-2305-2306-2307-2308-2309-2310-2311-2312-2313-2314-2315-2316-2317-2318-2319-2320-2321-2322-2323-2324-2325-2326-2327-2328-2329-2330-2331-2332-2333-2334-2335-2336-2337-2338-2339-2340-2341-2342-2343-2344-2345-2346-2347-2348-2349-2350-2351-2352-2353-2354-2355-2356-2357-2358-2359-2360-2361-2362-2363-2364-2365-2366-2367-2368-2369-2370-2371-2372-2373-2374-2375-2376-2377-2378-2379-2380-2381-2382-2383-2384-2385-2386-2387-2388-2389-2390-2391-2392-2393-2394-2395-2396-2397-2398-2399-2400-2401-2402-2403-2404-2405-2406-2407-2408-2409-2410-2411-2412-2413-2414-2415-2416-2417-2418-2419-2420-2421-2422-2423-2424-2425-2426-2427-2428-2429-2430-2431-2432-2433-2434-2435-2436-2437-2438-2439-2440-2441-2442-2443-2444-2445-2446-2447-2448-2449-2450-2451-2452-2453-2454-2455-2456-2457-2458-2459-2460-2461-2462-2463-2464-2465-2466-2467-2468-2469-2470-2471-2472-2473-2474-2475-2476-2477-2478-2479-2480-2481-2482-2483-2484-2485-2486-2487-2488-2489-2490-2491-2492-2493-2494-2495-2496-2497-2498-2499-2500-2501-2502-2503-2504-2505-2506-2507-2508-2509-2510-2511-2512-2513-2514-2515-2516-2517-2518-2519-2520-2521-2522-2523-2524-2525-2526-2527-2528-2529-2530-2531-2532-2533-2534-2535-2536-2537-2538-2539-2540-2541-2542-2543-2544-2545-2546-2547-2548-2549-2550-2551-2552-2553-2554-2555-2556-2557-2558-2559-2560-2561-2562-2563-2564-2565-2566-2567-2568-2569-2570-2571-2572-2573-2574-2575-2576-2577-2578-2579-2580-2581-2582-2583-2584-2585-2586-2587-2588-2589-2590-2591-2592-2593-2594-2595-2596-2597-2598-2599-2600-2601-2602-2603-2604-2605-2606-2607-2608-2609-2610-2611-2612-2613-2614-2615-2616-2617-2618-2619-2620-2621-2622-2623-2624-2625-2626-2627-2628-2629-2630-2631-2632-2633-2634-2635-2636-2637-2638-2639-2640-2641-2642-2643-2644-2645-2646-2647-2648-2649-2650-2651-2652-2653-2654-2655-2656-2657-2658-2659-2660-2661-2662-2663-2664-2665-2666-2667-2668-2669-2670-2671-2672-2673-2674-2675-2676-2677-2678-2679-2680-2681-2682-2683-2684-2685-2686-2687-2688-2689-2690-2691-2692-2693-2694-2695-2696-2697-2698-2699-2700-2701-2702-2703-2704-2705-2706-2707-2708-2709-2710-2711-2712-2713-2714-2715-2716-2717-2718-2719-2720-2721-2722-2723-2724-2725-2726-2727-2728-2729-2730-2731-2732-2733-2734-2735-2736-2737-2738-2739-2740-2741-2742-2743-2744-2745-2746-2747-2748-2749-2750-2751-2752-2753-2754-2755-2756-2757-2758-2759-2760-2761-2762-2763-2764-2765-2766-2767-2768-2769-2770-2771-2772-2773-2774-2775-2776-2777-2778-2779-2780-2781-2782-2783-2784-2785-2786-2787-2788-2789-2790-2791-2792-2793-2794-2795-2796-2797-2798-2799-2800-2801-2802-2803-2804-2805-2806-2807-2808-2809-2810-2811-2812-2813-2814-2815-2816-2817-2818-2819-2820-2821-2822-2823-2824-2825-2826-2827-2828-2829-2830-2831-2832-2833-2834-2835-2836-2837-2838-2839-2840-2841-2842-2843-2844-2845-2846-2847-2848-2849-2850-2851-2852-2853-2854-2855-2856-2857-2858-2859-2860-2861-2862-2863-2864-2865-2866-2867-2868-2869-2870-2871-2872-2873-2874-2875-2876-2877-2878-2879-2880-2881-2882-2883-2884-2885-2886-2887-2888-2889-2890-2891-2892-2893-2894-2895-2896-2897-2898-2899-2900-2901-2902-2903-2904-2905-2906-2907-2908-2909-2910-2911-2912-2913-2914-2915-2916-2917-2918-2919-2920-2921-2922-2923-2924-2925-2926-2927-2928-2929-2930-2931-2932-2933-2934-2935-2936-2937-2938-2939-2940-2941-2942-2943-2944-2945-2946-2947-2948-2949-2950-2951-2952-2953-2954-2955-2956-2957-2958-2959-2960-2961-2962-2963-2964-2965-2966-2967-2968-2969-2970-2971-2972-2973-2974-2975-2976-2977-2978-2979-2980-2981-2982-2983-2984-2985-2986-2987-2988-2989-2990-2991-2992-2993-2994-2995-2996-2997-2998-2999-3000-3001-3002-3003-3004-3005-3006-3007-3008-3009-3010-3011-3012-3013-3014-3015-3016-3017-3018-3019-3020-3021-3022-3023-3024-3025-3026-3027-3028-3029-3030-3031-3032-3033-3034-3035-3036-3037-3038-3039-3040-3041-3042-3043-3044-3045-3046-3047-3048-3049-3050-3051-3052-3053-3054-3055-3056-3057-3058-3059-3060-3061-3062-3063-3064-3065-3066-3067-3068-3069-3070-3071-3072-3073-3074-3075-3076-3077-3078-3079-3080-3081-3082-3083-3084-3085-3086-3087-3088-3089-3090-3091-3092-3093-3094-3095-3096-3097-3098-3099-3100-3101-3102-3103-3104-3105-3106-3107-3108-3109-3110-3111-3112-3113-3114-3115-3116-3117-3118-3119-3120-3121-3122-3123-3124-3125-3126-3127-3128-3129-3130-3131-3132-3133-3134-3135-3136-3137-3138-3139-3140-3141-3142-3143-3144-3145-3146-3147-3148-3149-3150-3151-3152-3153-3154-3155-3156-3157-3158-3159-3160-3161-3162-3163-3164-3165-3166-3167-3168-3169-3170-3171-3172-3173-3174-3175-3176-3177-3178-3179-3180-3181-3182-3183-3184-3185-3186-3187-3188-3189-3190-3191-3192-3193-3194-3195-3196-3197-3198-3199-3200-3201-3202-3203-3204-3205-3206-3207-3208-3209-3210-3211-3212-3213-3214-3215-3216-3217-3218-3219-3220-3221-3222-3223-3224-3225-3226-3227-3228-3229-3230-3231-3232-3233-3234-3235-3236-3237-3238-3239-3240-3241-3242-3243-3244-3245-3246-3247-3248-3249-3250-3251-3252-3253-3254-3255-3256-3257-3258-3259-3260-3261-3262-3263-3264-3265-3266-3267-3268-3269-3270-3271-3272-3273-3274-3275-3276-3277-3278-3279-3280-3281-3282-3283-3284-3285-3286-3287-3288-3289-3290-3291-3292-3293-3294-3295-3296-3297-3298-3299-3300-3301-3302-3303-3304-3305-3306-3307-3308-3309-3310-3311-3312-3313-3314-3315-3316-3317-3318-3319-3320-3321-3322-3323-3324-3325-3326-3327-3328-3329-3330-3331-3332-3333-3334-3335-3336-3337-3338-3339-3340-3341-3342-3343-3344-3345-3346-3347-3348-3349-3350-3351-3352-3353-3354-3355-3356-3357-3358-3359-3360-3361-3362-3363-3364-3365-3366-3367-3368-3369-3370-3371-3372-3373-3374-3375-3376-3377-3378-3379-3380-3381-3382-3383-3384-3385-3386-3387-3388-3389-3390-3391-3392-3393-3394-3395-3396-3397-3398-3399-3400-3401-3402-3403-3404-3405-3406-3407-3408-3409-3410-3411-3412-3413-3414-3415-3416-3417-3418-3419-3420-3421-3422-3423-3424-3425-3426-3427-3428-3429-3430-3431-3432-3433-3434-3435-3436-3437-3438-3439-3440-3441-3442-3443-3444-3445-3446-3447-3448-3449-3450-3451-3452-3453-3454-3455-3456-3457-3458-3459-3460-3461-3462-3463-3464-3465-3466-3467-3468-3469-3470-3471-3472-3473-3474-3475-3476-3477-3478-3479-3480-3481-3482-3483-3484-3485-3486-3487-3488-3489-3490-3491-3492-3493-3494-3495-3496-3497-3498-3499-3500-3501-3502-3503-3504-3505-3506-3507-3508-3509-3510-3511-3512-3513-3514-3515-3516-3517-3518-3519-3520-3521-3522-3523-3524-3525-3526-3527-3528-3529-3530-3531-3532-3533-3534-3535-3536-3537-3538-3539-3540-3541-3542-3543-3544-3545-3546-3547-3548-3549-3550-3551-3552-3553-3554-3555-3556-3557-3558-3559-3560-3561-3562-3563-3564-3565-3566-3567-3568-3569-3570-3571-3572-3573-3574-3575-3576-3577-3578-3579-3580-3581-3582-3583-3584-3585-3586-3587-3588-3589-3590-3591-3592-3593-3594-3595-3596-3597-3598-3599-3600-3601-3602-3603-3604-3605-3606-3607-3608-3609-3610-3611-3612-3613-3614-3615-3616-3617-3618-3619-3620-3621-3622-3623-3624-3625-3626-3627-3628-3629-3630-3631-3632-3633-3634-3635-3636-3637-3638-3639-3640-3641-3642-3643-3644-3645-3646-3647-3648-3649-3650-3651-3652-3653-3654-3655-3656-3657-3658-3659-3660-3661-3662-3663-3664-3665-3666-3667-3668-3669-3670-3671-3672-3673-3674-3675-3676-3677-3678-3679-3680-3681-3682-3683-3684-3685-3686-3687-3688-3689-3690-3691-3692-3693-3694-3695-3696-3697-3698-3699-3700-3701-3702-3703-3704-3705-3706-3707-3708-3709-3710-3711-3712-3713-3714-3715-3716-3717-3718-3719-3720-3721-3722-3723-3724-3725-3726-3727-3728-3729-3730-3731-3732-3733-3734-3735-3736-3737-3738-3739-3740-3741-3742-3743-3744-3745-3746-3747-3748-3749-3750-3751-3752-3753-3754-3755-3756-3757-3758-3759-3760-3761-3762-3763-3764-3765-3766-3767-3768-3769-3770-3771-3772-3773-3774-3775-3776-3777-3778-3779-3780-3781-3782-3783-3784-3785-3786-3787-3788-3789-3790-3791-3792-3793-3794-3795-3796-3797-3798-3799-3800-3801-3802-3803-3804-3805-3806-3807-3808-3809-3810-3811-3812-3813-3814-3815-3816-3817-3818-3819-3820-3821-3822-3823-3824-3825-3826-3827-3828-3829-3830-3831-3832-3833-3834-3835-3836-3837-3838-3839-3840-3841-3842-3843-3844-3845-3846-3847-3848-3849-3850-3851-3852-3853-3854-3855-3856-3857-3858-3859-3860-3861-3862-3863-3864-3865-3866-3867-3868-3869-3870-3871-3872-3873-3874-3875-3876-3877-3878-3879-3880-3881-3882-3883-3884-3885-3886-3887-3888-3889-3890-3891-3892-3893-3894-3895-3896-3897-3898-3899-3900-3901-3902-3903-3904-3905-3906-3907-3908-3909-3910-3911-3912-3913-3914-3915-3916-3917-3918-3919-3920-3921-3922-3923-3924-3925-3926-3927-3928-3929-3930-3931-3932-3933-3934-3935-3936-3937-3938-3939-3940-3941-3942-3943-3944-3945-3946-3947-3948-3949-3950-3951-3952-3953-3954-3955-3956-3957-3958-3959-3960-3961-3962-3963-3964-3965-3966-3967-3968-3969-3970-3971-3972-3973-3974-3975-3976-3977-3978-3979-3980-3981-3982-3983-3984-3985-3986-3987-3988-3989-3990-3991-3992-3993-3994-3995-3996-3997-3998-39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GENERAL NOTE

ALWAYS RECONSTRUCTED A STRONG SUBJECT, WITH STRONG SUB-REASON
ON REASONAL GROUNDS. SEE + GENERAL PRINCIPLES 1 + 2 FOR CHOICE OF GROUNDS
FORUM GROUNDS A STRONG REASON FOR REASONING ABOUT THE DIFFERENT GROUNDS
REASON + GROUNDS IS CHOICE FOR AT LEAST

Control and monitoring panels

Each engine is fitted with a control panel and a monitoring panel. These are fitted over the engine itself.



The electric monitoring panel contains the following monitoring instruments:

- digital revolutions indicator;
- engine oil pressure;
- combustion fuel pressure;
- engine oils temperature;
- transmission oil pressure;
- transmission oil temperature;
- jackets cooling system temperatures.

Functionality details of these indicators are clearly stated on the Constructor Manual. Also on the same manual is indicated the location of the electric/electronic sensors on the engine.

The electronic **CATERPILLAR** system, which controls the propellers from the two main control stations, the main bridge and the flying bridge, requires a series of dedicated equipments, such as:

Engine Control Panel

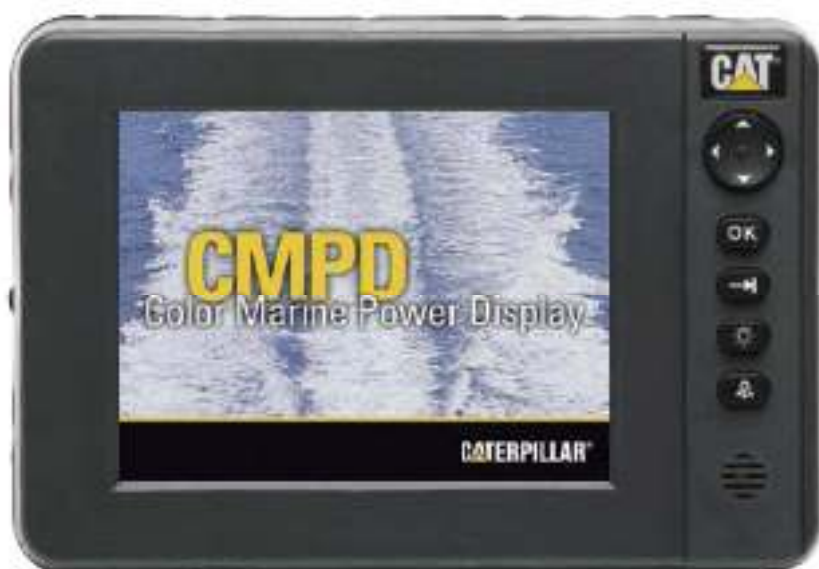
The Cat Marine Power Display (MPD) is a self-contained engine monitoring and display system that is simple to install and provides more information to the operator than previously available single-engine monitoring systems.

- High-resolution, Color VGA 6.5" 640 x 480 dots-per-inch image.
- Transflective screen improves readability by reflecting more or less light as ambient light conditions change.
- Fully graphic LCD display screen – as opposed to a fixed display – allows use of optional screen builder software to customize screen formatting.
- Screen builder software is compatible with Windows PC operating systems.
- Flash programming from user's PC (with Cat Electronic Technician service tool) to MPD provides fast software updates and use of custom screens.

FEATURES

- 4 NTSC/PAL video inputs
- PIP video w/ 30 frames/sec
- 115° viewing
- 4 pushbuttons
- 4-position rocker switch
- WinFlash capability
- Screen Builder software
- 2 external drivers – 300 mA

The Color MPD has five control buttons located on the right front of the display panel. The control buttons (or keys) are listed below along with a description of their associated functions.



DESCRIPTION OF KEYS

SCROLL KEY

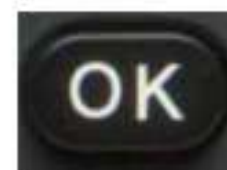
The “**SCROLL KEY**” is used to navigate through screens and menus.

Pressing on an arrow will scroll Up, Down, Left, or Right. These actions are labeled as **UP** arrow, **DOWN** arrow, **LEFT** arrow, and **RIGHT** arrow in the text that follows. Holding down an arrow will allow for fast scrolling in that direction.



OK KEY

The “**OK KEY**” is used to select gauges and menu items. This action will change the look of items displayed on the screen and the data used in these items.



TAB KEY

The “**TAB KEY**” is used to scroll through certain menus, at which time the menu item scrolled to may be selected by pushing the OK KEY.



DISPLAY MODE KEY

The “**DISPLAY MODE KEY**” is used to change the display screen from color (DAY MODE) to monochrome red (NIGHT MODE). Holding this key down will open a “Screen Contrast” menu allowing the user to change the screen contrast.



ALARM ACKNOWLEDGE (OR ALARM) KEY

The “**ALARM KEY**” is used to stop, silence, and clear alarms. Pressing the key once silences the alarm and pressing it again (twice) removes the alarm warning from the screen and display the “MAIN SCREEN.”



Gear boxes

Gears box carry out the following functions:

- engine coupling with propeller shaft and propeller speed reduction;
- rotation direction inversion;
- propeller shaft movement interruption (neutral).

The gears box of your yacht are **ZF** mod. **2360** with reduction rate 4 : 1.



Such documents are extremely important in case of repair works on reversing gears, therefore they should be kept with great care.

General

ZF marine transmissions are designed and manufactured according to the regulations of the various classification societies. The input powers approved by the classification societies depend on the input speed, the class of the craft and the different design regulations currently in force.

Normally, the input shaft rotates in a clockwise direction (looking at the transmission input flange). Transmissions which are specially designed for anti-clockwise input shaft rotation (looking at the transmission input flange) have an arrow on the transmission input end to show the rotation direction.

The total ratio is stamped on the transmission type plate which is located on the back of the transmission housing.

		FRIEDRICHSHAFEN - GERMANY -	
MODEL	SERIAL NO.		
1	2		
PARTS LIST NO.	REFERENCE NO.		
3	4		
RATIO TOTAL	MAIN GEAR	SPUR GEAR	
5			
OIL CAPACITY (dm ³)	MASS DRY (kg)	CLUTCH OIL PRESSURE (bar)	
6	7	9	
PLEASURE	LIGHT	MEDIUM	CONTINUOUS
10	11	12	13
8 OIL CHANGE: AFTER FIRST 50 HOURS OPERATION AND EVERY 100 HOURS OR 12 MONTHS, WHICHEVER OCCURS FIRST. LUBRICANT SEE ZF LIST TE-ML34 SAE 30 CD 140			

1. Model.
2. Serial no.
3. Parts list no.
4. Customer reference no.
5. Total ratio.
6. Oil capacity.
7. Mass (dry).
8. Lubricant, see ZF List of Lubricants.
9. Clutch oil pressure (nominal).
10. Max. power/speed ratio for intermittent operation at very widely differing engine speeds.
11. Max. power/speed ratio for intermittent operation at widely differing engine speeds.
12. Max. power/speed ratio for intermittent operation at differing engine speeds.
13. Max. power/speed ratio for continuous operation at max. engine power.

The ZF 2000 Series transmissions are axially offset reversing and step-up gears of 3-shaft design with a clutch on both the transmission input shaft and the reversing shaft. The general design is the same for all transmissions.

The main components subject to wear can be accessed through an aperture in the housing, even if the transmission is still installed.

The cast transmission housings are extremely torsionally rigid and made from a light metal alloy which is almost totally resistant to salt water. The machined surfaces and tapped holes required for attaching the transmission mountings and the bell housing are included as standard.

In order to ensure long life and quiet operation, all power-transmitting gearing is designed with high safety levels in mind, case-hardened, ground and then specially treated.

The shafts run in anti-friction bearings. A propeller thrust bearing is installed in the transmission to absorb the propeller thrust. The reversing clutches arranged on the input shaft and the reversing shaft are multi-disk clutches with steel/sinter disks. They are operated by means of oil pressure.

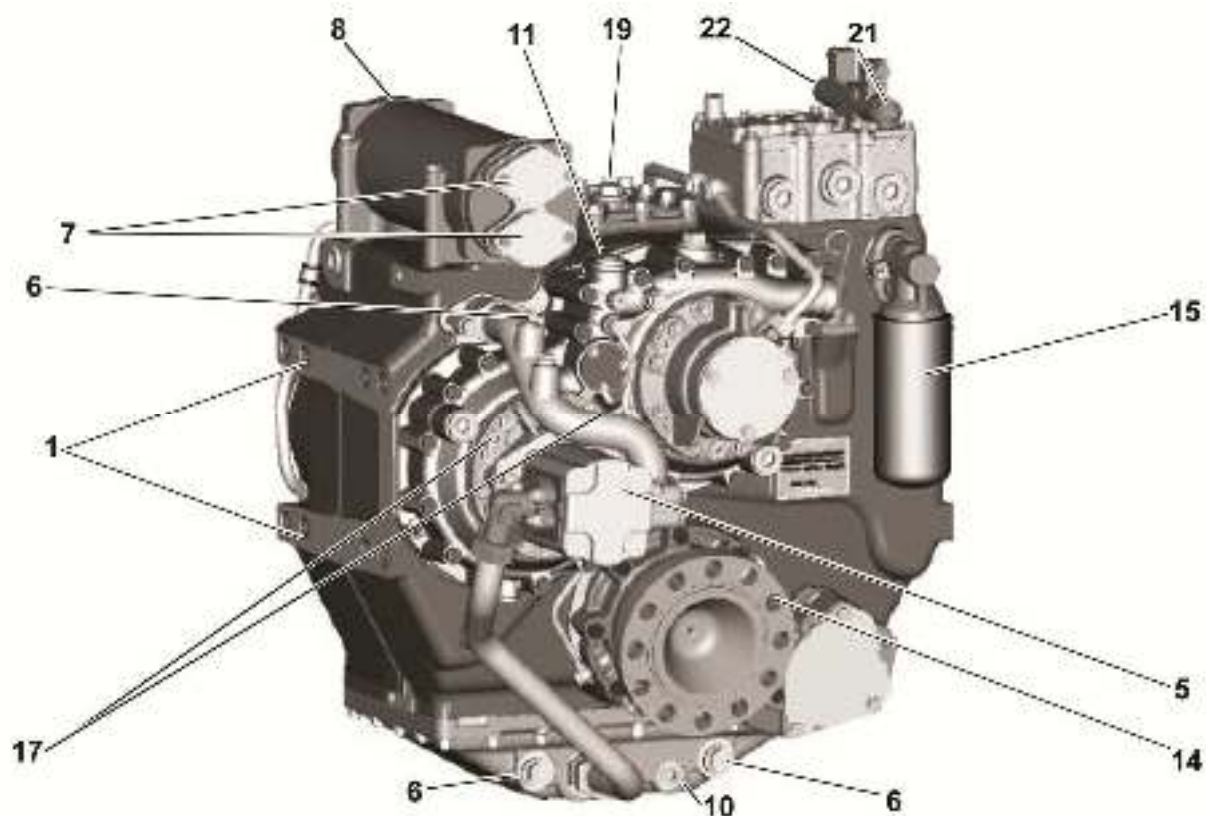
The high delivery rate of the oil pump ensures rapid clutch response. The contact pressure of the clutch is modulated during shifting to ensure smooth operation. At the same time, the flow of oil to the operating clutch is increased, i.e. the heat generated during shifting is quickly dissipated, which means that the clutches can operate efficiently under extreme load. The heat generated in trolling mode is also reliably dissipated, even during continuous operation.

In the event of faults in the operating oil circuit, both multi-disk clutches can be mechanically pressed together by means of three bolts which are accessible from the outside (only when the transmission is at a standstill). This emergency control feature can be used to engage "engine wise" propulsion or "counter-engine wise" propulsion.

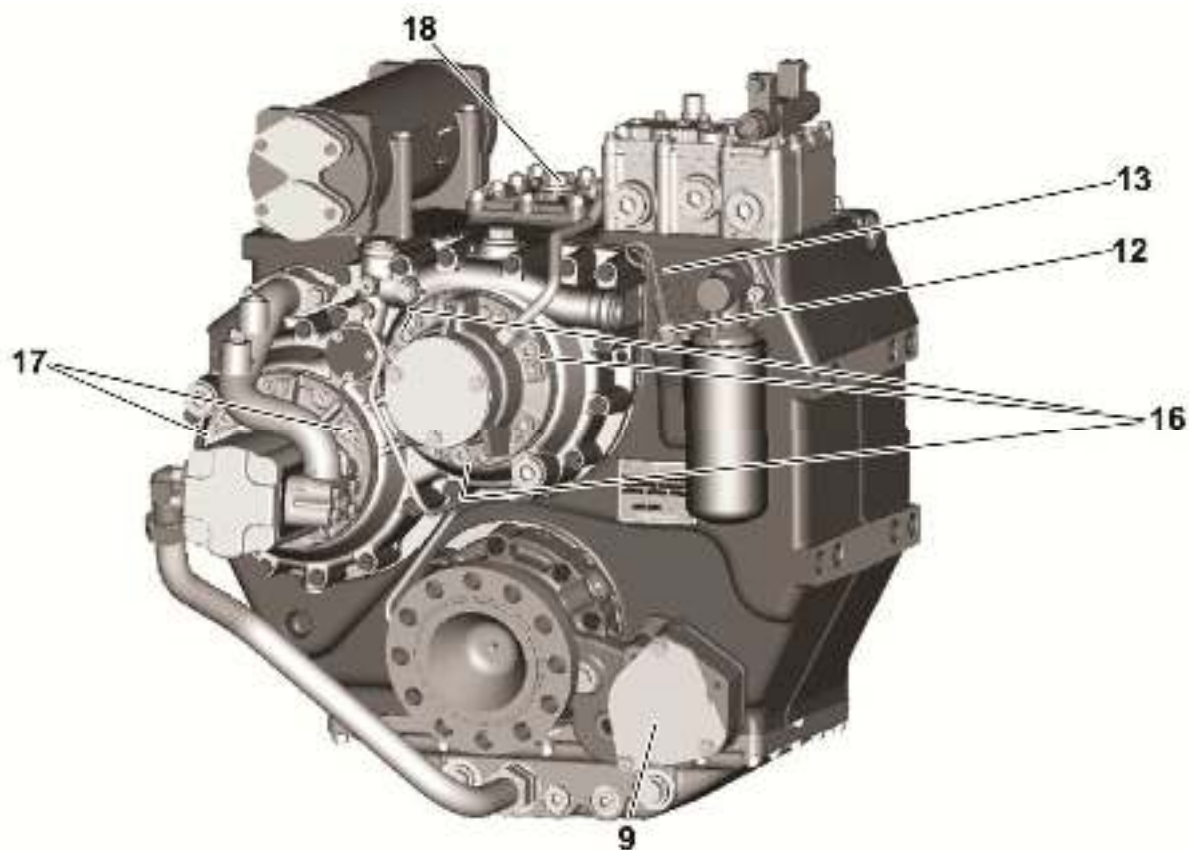
An easily replaceable control unit – fitted with all the valves, slides and control equipment for the operating and lubricating oil circuits – is built onto the transmission housing.

In the basic version of the transmission, the ZF 2000 / ZF 2000 A/V control unit is actuated electrically. If required, can also be supplied with mechanical actuation.

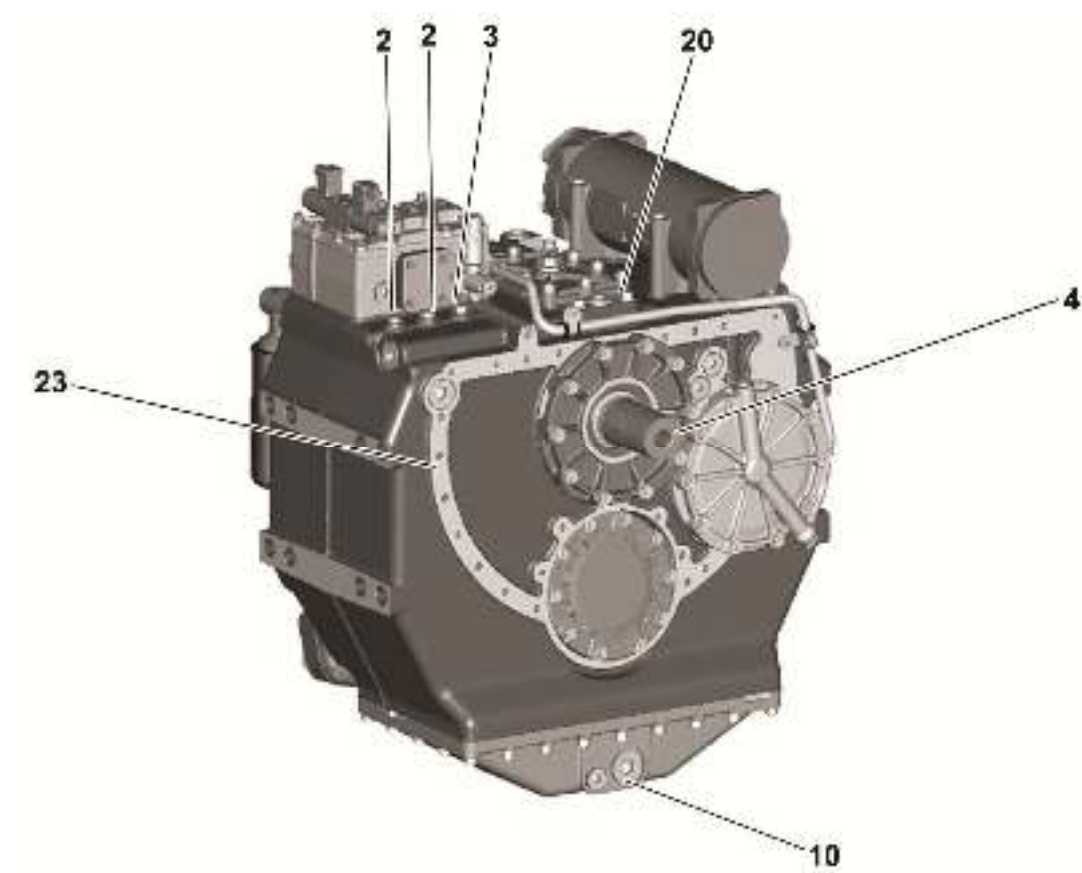
The full amount of engine power can be transmitted in both output rotation directions. The ratio specified on the type plate applies to both output rotation directions, i.e. identical transmission versions can be used in multiengine installations. This set-up requires fewer spare parts and permits the use of standardized reconditioned transmissions.



- 1. Transmission mounting faces.
- 5. Engine-driven oil pump.
- 6. Measuring points for oil temperature.
- 7. Water connection flanges.
- 8. Oil cooler.
- 10. Oil drain plug.
- 11. Measuring point for lubricating oil pressure.
- 14. Output.
- 15. Oil filter.
- 19. Inspection hole cover transmission housing.
- 21. Emergency attuation of the electrical control valve (enginewise rotation).
- 22. Emergency attuation of the electrical control valve (counter-enginewise rotation).



- 9. Position for trailing oil pump.
- 12. Transmission breather.
- 13. Oil dipstick.
- 16. Emergency control (counter-enginewise rotation).
- 17. Emergency control (enginewise rotation).
- 18. Oil filler hole.



- 2. Measuring point for clutch oil pressure.
- 3. Measuring point for clutch oil pressure.
- 4. Input.
- 10. Oil drain plug.
- 20. Measuring point for oil temperature downstream of cooler.
- 23. Flange-mounting face for bell housing.

Shaft line

The shaft line comprises a shaft with sleeve, a stern tube and a propeller bracket.

The shaft is secured to the gear box by a bolted sleeve, which is connected in turn to the gear box.

The shaft is aligned on three points:

- gear box flange;
- stern tube;
- shaft bracket.

The stuffing box head allows shaft seal adjustment, by means of two bolts which press the packing around the shaft.

Adjustment should be carried out when a drop of water filters through the stuffing box every two or three seconds, ensuring shaft-stuffing box contact cooling and lubrication.

Shaft rotation causes wear to the seal (packing), therefore periodical adjustment is necessary. Before the stuffing box head reaches the strut head, add other sealing rings (packing) while the shaft is not rotating.

It is advisable to replace the packing at the beginning of the season, as long inactivity results in dry and unsuitable seal.



WARNING *Do not approach the axles when they are in rotation.*

Two sea water lubricated NBR bushings are located on the shaft bracket and the stern tube. Such bushing should be checked at the beginning and at the end of the season, and whenever abnormal vibrations are detected.

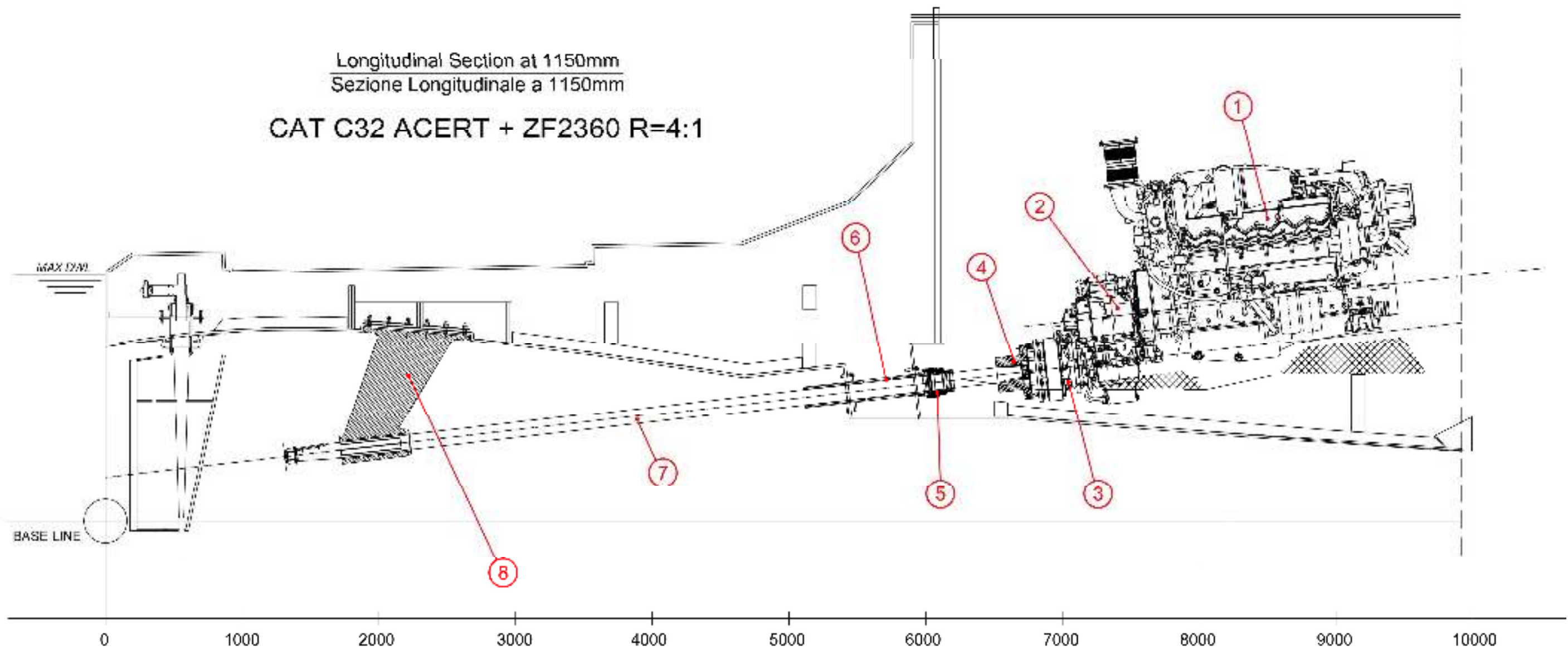
Prolonged navigation in sandy shoals may cause rapid bushings wear.

Stern tube bushing lubrication is carried out by water drawn from the engine sea water cooling system, and then discharged at sea.



WARNING *Stay away from rotating shafts.*

Shaft Line



#	Q.	DE SCRIZIONE	DESCRIPTION	Fornitore & Supplier
①	2	Motore CAT C32 ACERT 1153 Kw@2300 rpm	Main Propulsion CAT C32 ACERT 1153 Kw@2300 rpm	CGT
②	2	Riduttore ZF 2360 R=4:1	Gearbox ZF 2360 R=4:1	CGT
③	2	Giunto Propflex	Joint Propflex	CGI
④	2	Mancione asse D.110 MAN016A.00	Gearbox Flange D.110 MAN016A.00	WORLDYACHTS
⑤	2	Tenuta meccanica MTM 908 D.110	Packing Gland MTM 908 D.110	WORLDYACHTS
⑥	2	Astuccio D.110 AST047.00	Shaft Case D.110 AST047.00	WORLDYACHTS
⑦	2	Asse elica D.110 L.5475 mm AP052.00	Propeller Shaft D.110 L.5475 mm AP052.00	WORLDYACHTS
⑧	2	Supporto asse D. 110	Shaft Bracket D. 110	WORLDYACHTS

Propellers

The five-blade propellers, counter-rotating towards the external direction, have been designed to allow the engines to develop maximum power in normal operation conditions.

Periodically, and after long inactivity, make sure that the propellers are clean; particular dirty propellers reduce considerably yacht's performance and increase the vibration as the fouling cause friction and turbulence in the water surrounding the propeller blade, causing vibrations which propagate along the shaft and to the whole yacht.

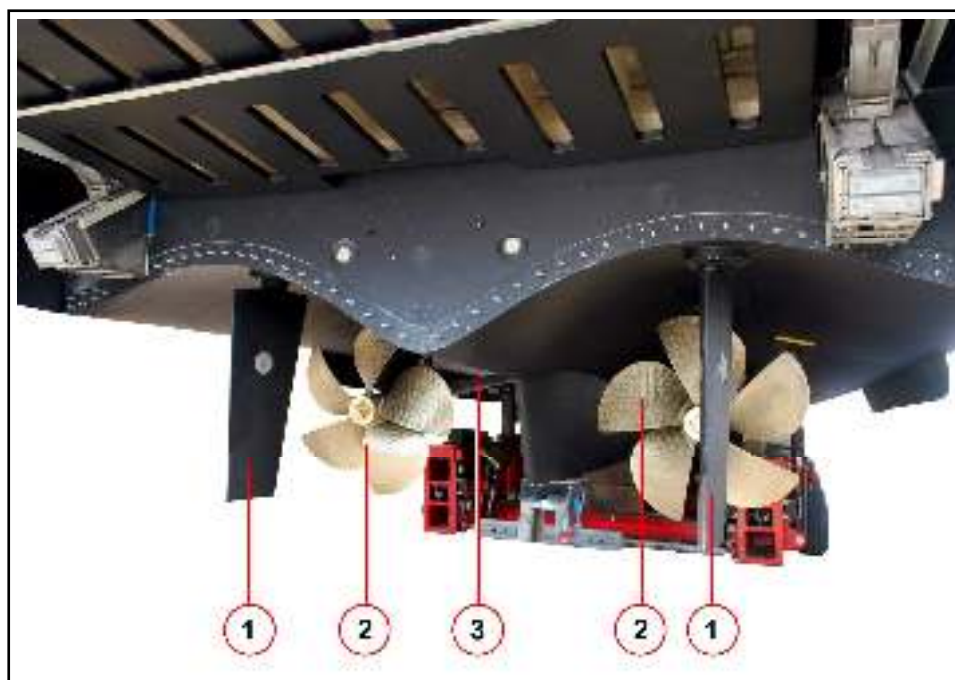
To prevent vibrations, clean propellers and shaft lines more frequently, especially during the summer and in particularly warm waters.



In case of impact with submersed obstacles, check immediately for eventual damages to propellers and shaft lines.

In case of considerable vibrations, reduce the speed and reach the harbour for further inspections, as an increase of vibration may damage both propulsion equipment and yacht's structures.

- 1 – Rudder
- 2 – Propeller
- 3 – Propeller shaft



Bow – Thruster

In order to improve and ease manoeuvres in narrow waters close to the mooring pier and approach manoeuvres in general and to compensate for the drift boat, low pace, due to the wind and current this yacht is designed with Bow-Thruster (type TP60) with two counter-rotating propellers.

The Bow-Thruster is activated by a dedicated electric engine 45kW and powered by 240VAC 3-phases, controlled by means of a dedicated inverter in order to reduce Inrush current.

The system consist of:

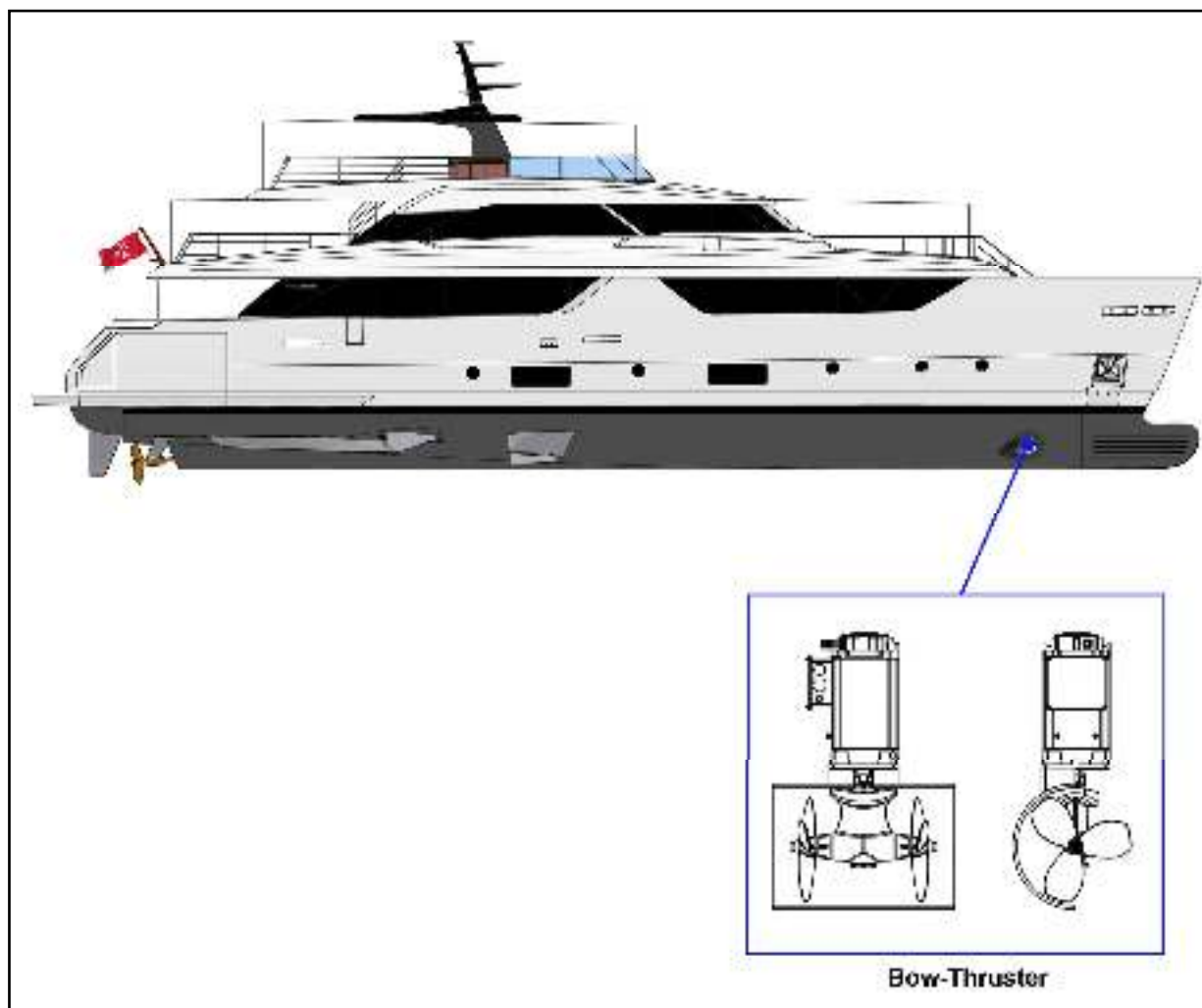
- *Control panel* with buttons and LED's for control of the system;
- *Dualis Electronic Switchboard* which receives the input data and controls the motors;
- *Inverter* modulates the voltage and frequency for perfect speed control;
- *Electrical Motor* supplied in different sizes to drive the propellers;
- *Pod* in aluminum alloy, both lighter and more durable than other systems.

Both the vessel's command stations (Wheelhouse and Flying Bridge) are equipped with two electric control panels with joystick.

Please consider the following practical suggestions for Thruster operation:

- the thruster machinery is simple and robust, but requires a lot of attention from the pilot;
- the thruster is to be used at very low speed of the yacht, or no headway, for higher speeds staggered inverters are recommended, always at low engine speeds, giving stronger reactions;
- to avoid overheating of the system a prolonged continuous use is not recommended (max 10 min.);
- every time the boat is raised to check the cleanliness of the propeller and the status of consumption of protective zincs; in the case that they are consumed by more than 50%, it is necessary to replace them with new ones.

To energise the thruster system, press the magneto-thermal switch on the main switchboard, section uses 230Vac for the Bow-Thruster. Activate the system by means of the ON/OFF button placed on the control panels located in the bridge and in the flying bridge (neither the panel installed on the bridge nor the one installed on the flying bridge takes priority over the other).



Characteristic data	Bow-Thruster
Power at motor axis	45 kW
Propeller diameter	450 mm
Thrust	7.1-8.1 kN
Oil for Pod lubrication	6 liters – ISO VG100



Always stop the system in neutral position for a few seconds before reversing thruster direction.



Remember to de-energise the system after mooring and during normal navigation, pushing stop button on the panel.

Soft Touch Control Panel

The control panel has three soft touch buttons with LED and one low profile proportional joystick.

It is not possible to have more than one control panel in command at any one time.

The command can be transferred to any of the control panels.



Take command button, White = OFF, Blue = ON.

Press the TAKE COMMAND button of a control station to assume command on that control station which will deactivate another panel that was in command.

Press the TAKE COMMAND button a second time on the same panel to turn off the control station if the thruster is no longer needed.

When a Power Management System input is used, the Blue LED will flash until power is available to the thruster.



Dimmer button, to adjust the brightness of the panel.



Acknowledge Alarm button, silences the buzzer.
The LED is red as long as the alarm is active.

OPTIONAL FEATURE: Double click on this button to Lock / Unlock the panel.

PROPORTIONAL JOYSTICK: Move the joystick on the active control station for the desired direction and thrust.

To operate the thrusters, acting on the central location of their joysticks control lever, the system is blocked and stopped; maneuvering to the right, the propellers will push the water towards the boat by moving left to right and vice versa.

The ability to take advantage of these excellent maneuvering qualities depends primarily on the "familiarity" with the yacht. Drilling in open water in a short time you can acquire an enviable ease in operation, being able to carry out very difficult moorings with great confidence even in areas crowded with boats, or in not very favorable conditions.

Canbus Control Panel

The CANBUS panel has two electro-mechanics buttons with LED and one low profile proportional joystick.

It is not possible to have more than one control panel in command at any one time.

The command can be transferred to any of the control panels.



Take command button, White = OFF, Blue = ON.

Press the TAKE COMMAND button of a control station to have the command on that control station. If a different control station was active, it is switched OFF and replaced by the new one. Press again the TAKE COMMAND button on the same panel for turning off the control station at the end of the use of the thruster. When the Power Management System input is used, the Blue LED is blinking up to the consent for the use of the electric power for the thruster.



Acknowledge Alarm button, stops the buzzer. The led is red until the alarm is active.

OPTIONAL FEATURE: Hold press the button to switch between three level of brightness of the button LEDs.

PROPORTIONAL JOYSTICK: Move the joystick on the active control station for the desired direction and thrust.

TROUBLESHOOTING

FAILURE	SYSTEM CONDITION	CAUSE	SOLUTION
CMC control panel OFF	None of the CMC control stations are ON.	The 24 V supply of DUALIS ELECTRA™ switchboard is missing.	Check the 24V power supply to the DUALIS ELECTRA™ switchboard.
			Check the fuses in DUALIS ELECTRA™ switchboard.
	Only a CMC control panel is OFF.	The communication between DUALIS ELECTRA™ switchboard and panel or the panel itself are broken.	Check that the pin in the cable connectors aren't damaged.
			Check the right placement of the rubber seals in the connectors.
			Open the connectors shells and check for the quality of all the electrical connections.
			Check that the cable is not damaged. Measure the electrical resistance of each conductor to verify the electrical continuity.
		The control panel is broken.	Replace the control panel.
The take command button blinks when pressed.	CMC control panel cannot take command of the thruster.	The PMS has not given the consensus to start the thruster system.	Check if the X5 and X6 are ON.
			Check the right working condition of the PMS.
			Check the quality of all the electrical connections.
An alarm occurs when the take command button is pressed.		An alarm is active.	Check the quality of the connections related to the alarm inputs (see previous section).

FAILURE	SYSTEM CONDITION	CAUSE	SOLUTION
ONLY FOR SOFT TOUCH CONTROL PANEL. All buttons on the CMC control panel are blinking	CMC control panel disabled.	The CMC control panel is not in the PLC TCP/IP network.	Check that the panel is connected to the right ethernet switchboard.
			Check that the pin in the cable connectors are not damaged.
			The panel has not been configured.
ONLY FOR CANBUS CONTROL PANEL. Alarm buttons on all but one of the CMC control panel are blinking	The CMC control panel on which the alarm is not blinking is disabled.	The CMC control panel is not in the CANBUS network.	Check the panels connec- tions to the network.
			Check that the pin in the cable connectors are not damaged.
The station is not able to take command of the thruster (ready signal OFF).	The system works from the CMC control panel.	A CMC control station is in command.	Release the command from the CMC control panel.
		The take command re- quest is not properly re- ceived by the PLC.	Check the quality of all the electrical connections.
		The PLC cannot assign the command to the sta- tion (no PMS consen- sus).	Check if the X5 and X6 are ON.
			Check the right working condition of the PMS.
The ON/OFF or the proportional station cannot properly command the thruster.	The system works from the CMC control panel.	The control signal has not properly sent to the PLC.	Check if the control signals are sent with respect of the CMC specifications.
			Check if any alarms/warnings occur in the inverter.
	The system does not work properly also from the CMC control panel.	The frequency converter or the motor do not work properly.	Check the integrity of the motor.
			Check the integrity of the propeller.

Stabilising fins (Electra LR190)

In order to increase the comfort onboard the yacht, while underway in conditions of non-calm sea, the yacht is provided with an optimal stabilising fin system, with lifting surface of 1.90 m² per fin, able to reduce (dampening) the yacht's roll.

Solid state inertial sensors feed the control processor with vessel roll and heel angle data in real time. The stabilising fin's rake is adjusted more than 200 times per second to create constant roll-damping counterthrust. In order to supply optimum performance, the fins are installed so as not to be subject to turbulence generated by the hull's motion; for this purpose each fin must be able to rotate freely for $\pm 35^\circ$ and having a minimum operation rotation of $\pm 25^\circ$.

The speed is the most important factor for the stabilizing system. The rolling phenomenon must be stopped before starting. The processor extremely rapid response time is the key to the system's extraordinary performance.



The fins are installed in such manner so they do not protrude laterally beyond the main cross section of the yacht, in order not to obstacle approach manoeuvres.

From the construction point of view, all openings and hull passages of the system have been carefully studied by SANLORENZO S.p.A. shipyard and by the supplier CMC Marine, and they have been object of specific approval by Classification Society.

The fins are made of epoxy resin, key-mounted on hi-resistance steel shafts. These shafts are actuated by brushless motor.



The system is able to reduce the rolling phenomenon, also while the yacht is anchored, in such condition, pay maximum attention making sure that swimmers or small yachts are not in the system's operation range, due to the extreme danger caused by the fins movement.

Stabilis Electra is a new stabilizer system that reduce the roll motion of the yacht, both during navigation and at anchor, controlling the movement of two special side fins.

The system consists of:



1. **Touch – screen monitor:** allow the system control as well as the monitoring of all the system parameters. It is generally used as main control panel.
2. **Soft touch panel or CanOPEN panel:** for basic control of the system, generally used as secondary control station (optional).
3. **ECO (Electronic Control Unit) panel:** contains the brain (aka «CPU») which reads all data and commands the actuators movement.
4. **EPU (Electronic Power Unit) panels:** Contain the power electronic modules (aka «Servo») to feed the electric motors.
5. **Inertia platform:** Feeds yachts' movement data to the ECO panel.
6. **Actuators:** supplied in different sizes.
7. **Fins:** the fins are the active part of the system. They allow to transform an alternative motion in a righting moment for the yacht.



It is strictly forbidden the use of the Stabilizers system in Anchor mode if the boat is docked or when the yacht is inside an harbour.

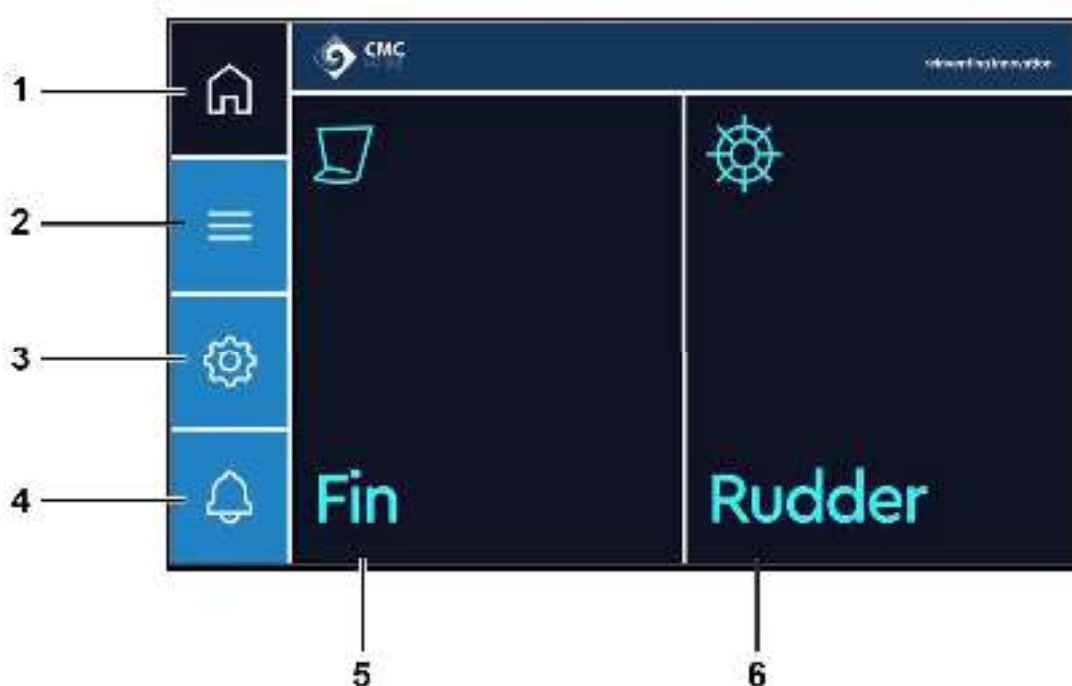
It is strictly forbidden the use of the system in Anchor mode with objects or other boats in the proximity of the hull.

It is strictly forbidden to keep the system ON if someone is in the water near the boat.

TOUCHSCREEN DISPLAY

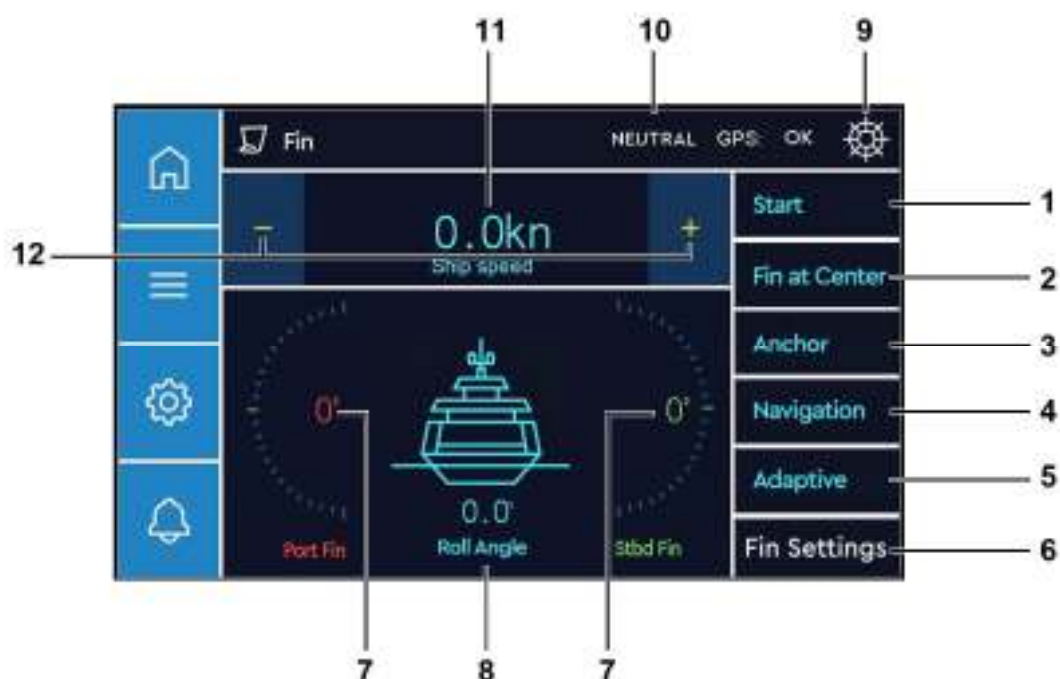
The touch screen monitor is the graphical user interface of the stabilizer system. This chapter describes the Touch-screen monitor interface for systems with Stabilis Electra.

➤ Home Page



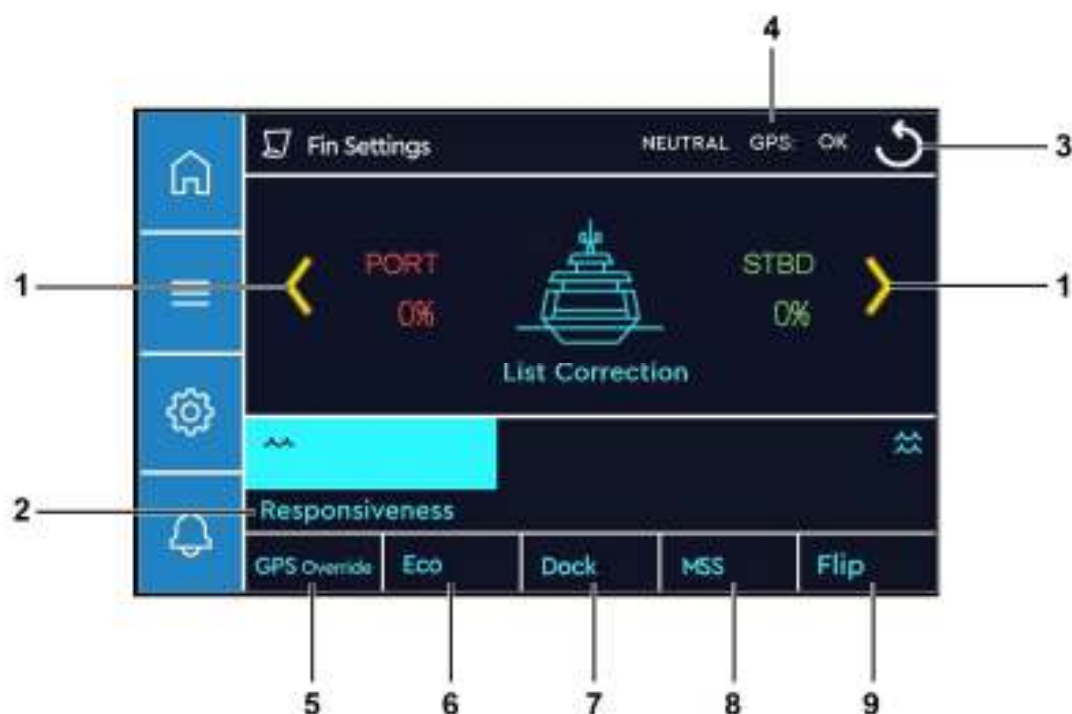
1. HOME PAGE
2. SERVICE PAGE
3. GENERALE SETTING PAGE
4. ALARMS PAGE
5. FIN PAGE
6. RUDDER CAGE: access reservated only if CMC Rudder System is on board

➤ **Fin Page**



1. START / STOP : System status
2. FIN AT CENTER . Move and hold Fins at 0°
3. ANCHOR : Activate/deactivate Anchor Mode
4. NAVIGATION : Activate/deactivate Navigation Mode
5. ADAPTIVE : Activate/deactivate Mode
6. FIN SETTINGS : access to “Fin Setting” page
7. FIN ANGLE : Current port/starboard angle fin read by the motor encoder
8. ROLL ANGLE : Current roll angle
9. RUDDER PAGE : access reserved only if the CMC Rudder System is on board
10. NEUTRAL / AHEAD / ASTERN : Throttle Status
11. SPEED : Speed value supplied by GPS
12. CHANGE SPEED : available during GPS Override

➤ **Fin Setting Page**



1. LIST CORRECTION : Setting of the list percentage (port/starboard)
2. RESPONSIVENESS : adjust responsiveness
3. BACK : go to FIN PAGE
4. GPS : OK / NO FIX
5. GPS OVERRIDE mode
6. ECO Mode (optional)
7. DOCK Mode (optional)
8. MSS Mode (optional)
9. FLIP Mode (optional)

LIST CORRECTION: Use the Starboard/Port arrows to offset the percentage compensation of the yacht's list. Please note that the system response increases with speed, the setting may require adjustment as the yacht's speed varies.

RESPONSIVENESS: Use the scroll to adjust the system responsiveness. Increase responsiveness as the weather conditions become worse.

➤ GPS Override

GPS OVERRIDE MODE allows to manually set the speed, overlooking the information and the errors of the GPS.

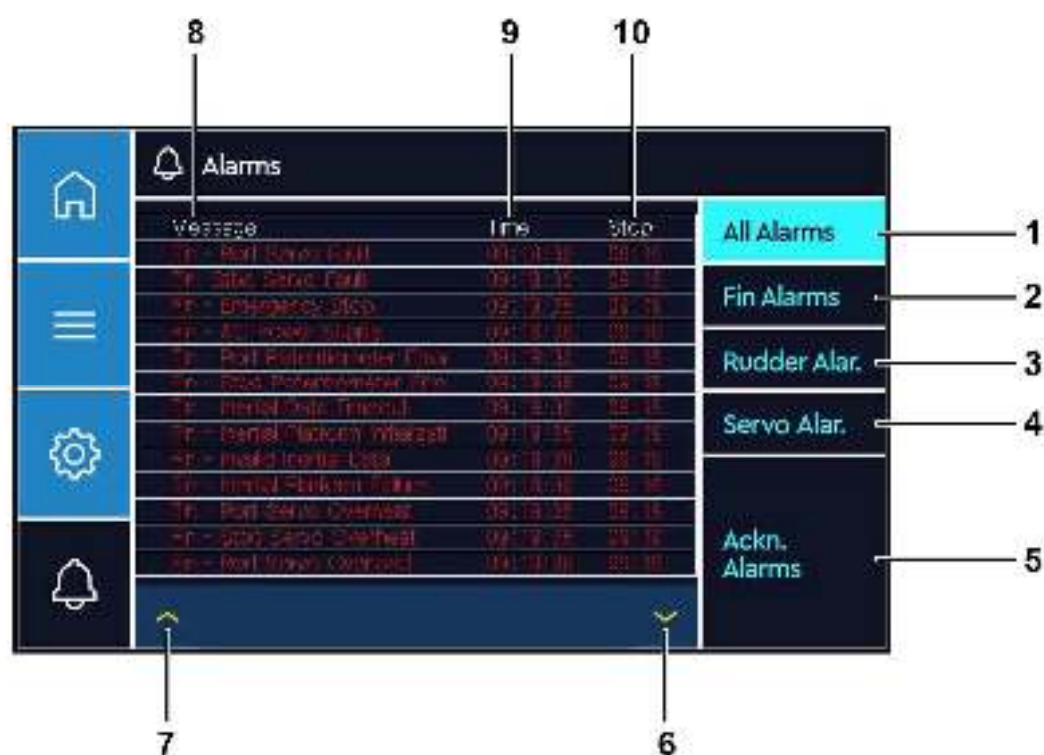
This mode can be activated in the “FIN SETTING PAGE” tapping on “GPS Override”.

To set the speed: go back to the “FIN PAGE” and set it with the +/- icons.

To switch on/off the “GPS override mode”, go to Fin Settings page and toggle on/of the “GPS Override” mode.

Note: “GPS OVERRIDE” and “MSS MODE” cannot be activated together because MSS ON needs the GPS signal to deactivate the “ANCHOR MODE”: when “GPS OVERRIDE” is activated, the MSS MODE is automatically turned off.



➤ **Alarm List Page**

1. ALL ALARMS
2. FIN ALARMS
3. ALARMS
4. SERVO ERROR HISTORY PAGE shows the history and the codes of the servo error. Contact CMC Marine™ for troubleshooting.
5. ACKN. ALARMS : Alarm Acknowledge
6. DOWN : Scroll Down List
7. UP : Scroll Up List
8. MESSAGE : Alarm description
9. TIME : Occurred alarms time
10. STOP : Time alarm reset

➤ **Service page**

Stabilis Electra's software is made up of two components:

- PLC software runs on the PLC,
- HMI software runs on the graphic display.

The Info Page is accessible tapping on "INFO" in the "Service page" and shows:

- the system operating hours,
- the software and firmware PLC and HMI version.



➤ **Advanced Fin Settings page (Password- protected)**

The "Advanced Fin Settings" contains the access to the Advanced Fin Settings page through the user password.

Check with your Builder or CMC Marine™ Service Representative to obtain the password.

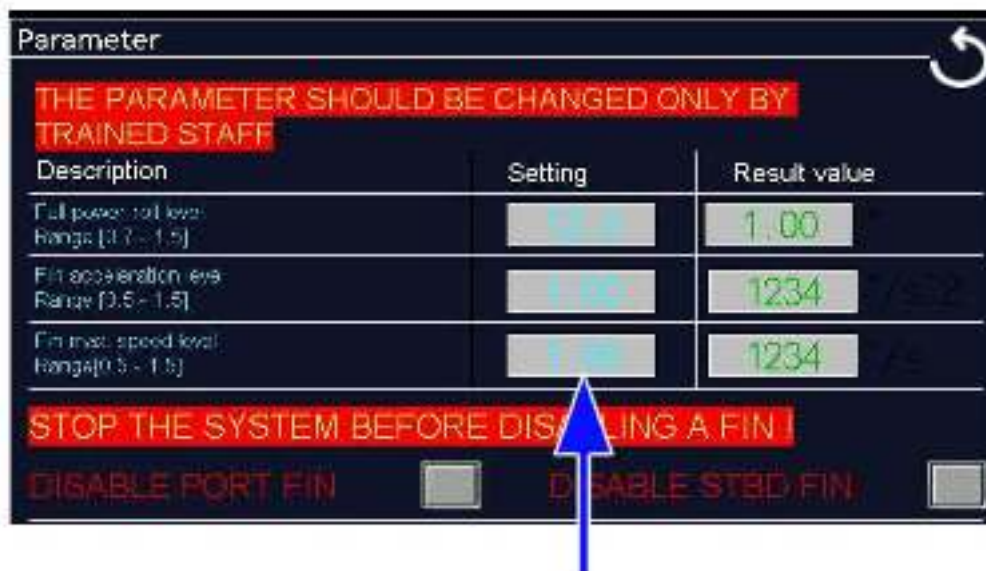




1. SET ENCODER PORT / STDB : realign the encoder angle with the potentiometer angle (hold 2 seconds)
2. RESET ERR POT PORT / STDB : to reset the potentiometer error
3. ZERO POT : potentiometer resetting (hold 2 seconds)
4. POTENTIOMETER ANGLE
5. ENCODER ANGLE
6. PARAMETER page
7. TEST page
8. MANUAL MODE
9. MAINTENANCE page

➤ **PARAMETER PAGE**

In the column 'setting' the user can modify the three values using the ratios in the brackets.



- **Full power roll level** (parameter for ANCHOR MODE). adjusts the roll angle at which the maximum power is provided. A lower value means higher responsiveness.
Only active in AT ANCHOR mode.
- **Fin acceleration level range**: adjusts the maximum acceleration of the fins. A lower value means lower responsiveness.
Only active in AT ANCHOR mode.
- **Fin max. speed level range**: adjusts the maximum speed of the fins. A lower value means lower responsiveness.
Only active in AT ANCHOR mode.

ATTENTION: these parameters should only be altered under the instructions of a CMC Marine's Service representative.

- **Disable P/SB FIN**: From this page can be also disabled one fin, so the system can be used with one fin locked. Stop the system before disabling fin.

➤ TEXT PAGE

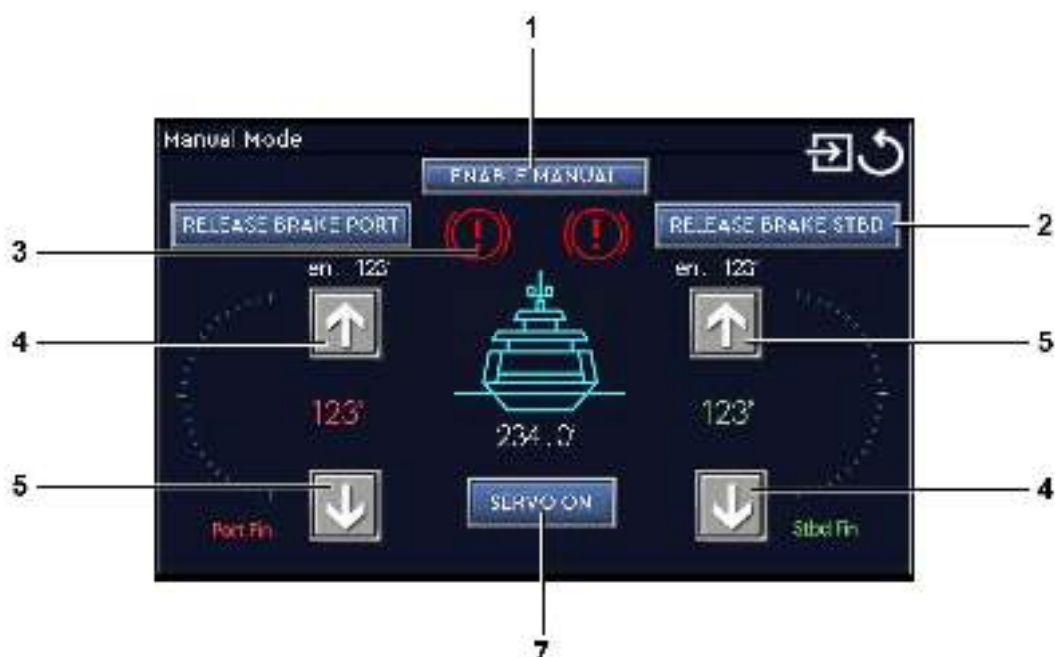
From here tests can be carry out to measure the typical parameter of the system.



ATTENTION: this page should only be accessed under the instructions of a CMC Marine's Service representative.

The 'servo error history' shows the history and the codes of the servo error. Contact CMC Marine™ for troubleshooting.

➤ MANUAL MODE PAGE



1. ENABLE MODE : active/deactive fins in manual mode
2. RELEASE PORT / STBD BRAKE : fins in free flow
3. BRAKE SIGNAL : red – brake active, white – brake released
4. COUNTERCLOCKWISE FIN ROTATION
5. COUNTERCLOCKWISE FIN ROTATION
6. ROLL ANGLE
7. SERVO ON : servo enabled

From the Advanced fin setting page, you can access to "Manual mode page".

To manually move the fins:

1. Mode, first press the ENABLE MANUAL icon to enable the Manual mode.
2. The first step is to press the "SERVO ON" icon to switch on the servo.
3. Then press RELEASE PORT BRAKE or RELEASE STBD BRAKE depending on which side you desire to move, to release the motor brake.
4. You can now use the arrows to control the direction of the fin, press and hold until the required angle has been achieved.

To disable the Manual Mode, reverse the steps shown above, to activate the brake, switch off the Servo, and disable Manual Mode.



Do not release the brakes unless the servo has been already enabled.

If the brake is released while the servo has not been enabled, the fins will be free to move: underway they will align to the flow, at anchor they will move to their "buoyancy position".

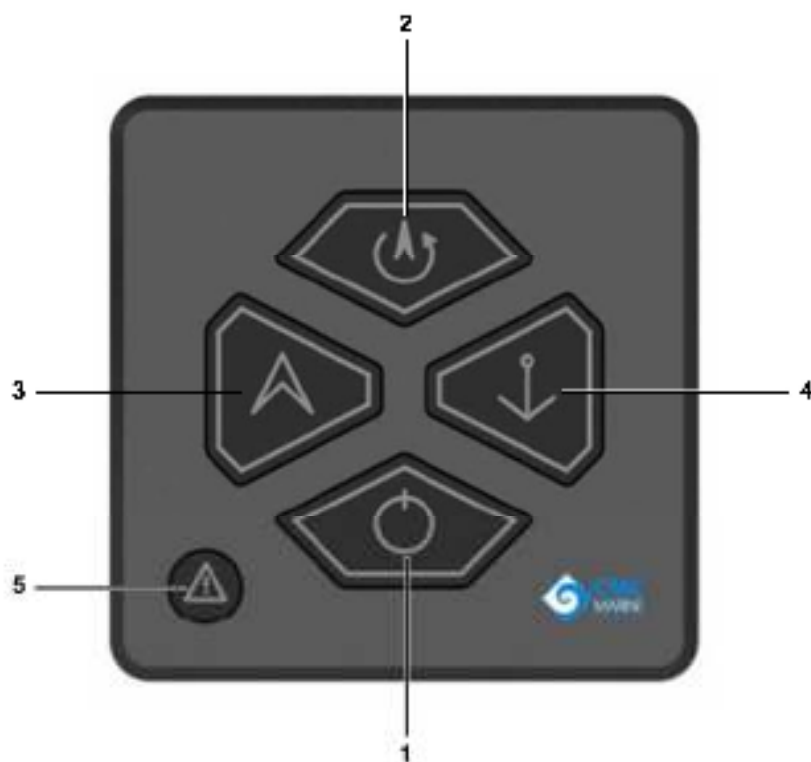
It is strictly forbidden to move the fins with people, objects or other boats in the proximity of the hull.

It is strictly forbidden to move the fins with people, objects or other boats in the proximity of the hull.

Check that both fins are clear to rotate without coming in contact with any kind of obstacle (hull chines, piers, boats, etc.).

➤ **MAINTENANCE USER PAGE**

This page records the 500 hours maintenance dates: follow the instruction above to save the date.

STABILIS ELECTRA OUTDOOR CONTROL PANEL

1. START / STOP
2. FIN AT CENTER
3. NAVIGATION
4. ANCHOR
5. "Acknowledge Alarm", stop the buzzer. The led is red until the alarm is active.
6. Brightness setting of the panel
(the brightness can be adjusted keeping the "Ack button" pressed).

This panel has the same main buttons of the "Home Page" of the touch screen monitor, for basic control of the system.

This panel could be integrated with a thruster control panel: the buttons are the same but just arranged in slightly different way.

Buttons' state:

- White = OFF
- Blue = ON

STABILIS ELECTRA OPERATION

✓ **OPERATION MODES**

The system can work in four different control modes:

- Fin At Center :** Fins are held at center (0 degrees).
- Navigation :** Underway stabilization active, some settings are available for manual adjustment.
- Adaptive :** Underway stabilization active, settings are adjusted automatically by the system
- Anchor :** Stabilization at anchor

✓ **FUNCTIONS & COMMANDS**

START / STOP

From Touch Screen or Control Panel. It turns ON-OFF the control system.

First push: the control system is ready for use and the fins are automatically driven at center.

The switching OFF is not possible if the fins are not at center. Press always FIN AT CENTER before STOP.

FIN AT CENTER

From touch Screen or Control Panel. The system drives and keeps the fins at center.

It must be always used:

- o before maneuvering,
- o before shifting from Navigation/Adaptive to Anchor mode,
- o before switching OFF.

NAVIGATION

From touch Screen or Control Panel. The system stabilizes the yacht during navigation.

Press FIN AT CENTER before NAVIGATION if the system is in Anchor mode.

During Navigation mode, the tunings SEA CONDITIONS and LIST are available.

ADAPTIVE

From touch Screen or Control Panel. Full automatic stabilization mode during navigation.

Press FIN AT CENTER before ADAPTIVE if the system is in Anchor mode.

ANCHOR

From touch Screen or Control Panel. The system stabilizes the yacht at anchor.

Press FIN AT CENTER before ANCHOR if the system is in Navigation or Adaptive mode.

During Anchor mode, the tuning SEA CONDITIONS is available.

SEA CONDITION

Tuning available from the Touch Screen Monitor during "Navigation" and "Anchor" modes. It is **NOT** available during "Adaptive" mode.

It is basically a setting of both speed and angle of the fin rotation. The basic parameters are usually fine for most of the time, but in some special condition (combinations of sea state and boat speed) it can be helpful to modify the parameters to reach the best comfort onboard.

- o Reduce setting if the stabilization system is too "nervous" and a smoother reaction is needed.
- o Increase setting if you need the system more fast and steady.

NOTE: When the stabilization mode is changed the SEA CONDITION comes back to the default value.

LIST

Tuning available from the Touch Screen Monitor only during the "Navigation" mode. It is not available during "Adaptive" and "Anchor" modes.

It allows the compensation of eventual static roll angle of the yacht. Fin angle range $\pm 5^\circ$.

When the stabilization mode is changed the LIST comes back to the default value.

AUTO SPEED

Green=active, Red=alarm. When Auto Speed is active the data from GPS are correctly received by the system, so it can work normally.

When there is a faulty communication data from GPS:

- o the Auto Speed indication becomes red
- o the alarm starts
- o the system works with the same parameters as for maximum speed
- o the range of the fin angle is limited to $\pm 5^\circ$

The user can acknowledge the alarm pressing ALARM RESET.

Auto Speed will change from red to green when the communication from the GPS is stable and the GPS system has a valid fix of the position/speed.

AHEAD / ASTERN INPUTS

The inputs Ahead or Astern from the engines' gearbox or remote control, when active, are visible on the top of the main page of the Touch Screen monitor.

They give additional automations:

- o When Ahead or Astern is active the Anchor mode cannot be activated.
- o When the Anchor mode is active and the input Ahead is received, the system shifts automatically to Fin At Center mode.
- o When the Anchor mode or Navigation/Adaptive mode is active and the input Astern is received, the system shifts automatically to Fin At Center mode.

The automations work only if the system is ON indeed.



In any case the Fin At Center button must be always pressed before maneuvering and before shifting from Navigation/Adaptive to Anchor mode and viceversa.

LOW SPEED

During Navigation mode or Adaptive mode, when the speed of the yacht received by the GPS is lower than 4 kn, the system is automatically driven to Fin At Center mode.



This automatic function cannot work when the Auto Speed is not active (red indication, alarm ON).

In any case the Fin At Center button must be always pressed before maneuvering and before shifting from Navigation/Adaptive to Anchor mode and viceversa.

EMERGENCY MODE

Stabilis Electra™ control system are engineered to be fail-safe. Below, the most important failures and how the system reacts.



Emergency mode must be used with maximum attention, only if necessary and only by trained personnel.

FAILURE	SYSTEM REACTION
Any failure in the control chain	Fins are brought back to the At Center position and locked in place by the brake
Black-out on the AC power supply	The 24V DC power supply is still active: the system releases the brake to allow fin floatation towards the Center position (but not viceversa). After 30 seconds and independently of the fin angle, the brake is activated and remains engaged.
Total Black-out	The brake remains engaged, the fin remains in the position and can be operated through the Emergency manoeuvre .

STABILIS EMERGENCY MANOEUVRE

Tool list

1. Allen key 8mm, length 120mm
2. Allen key 10mm standard length
3. Torque wrench

Overview

Connections to the electrical motor:

1. Power supply cable (1)
2. Electric brake cable (2)
3. Encoder cable (3)



The motor is coupled to the flange with 4 screws M12X45 and 4 M12 washer. The tightening torque for the screws is 90 Nm.



Managing the motor put particularly care with the motor encoder because is the most fragile part of the device.



The system must be powered off (switch off both 220V and 24V breakers).

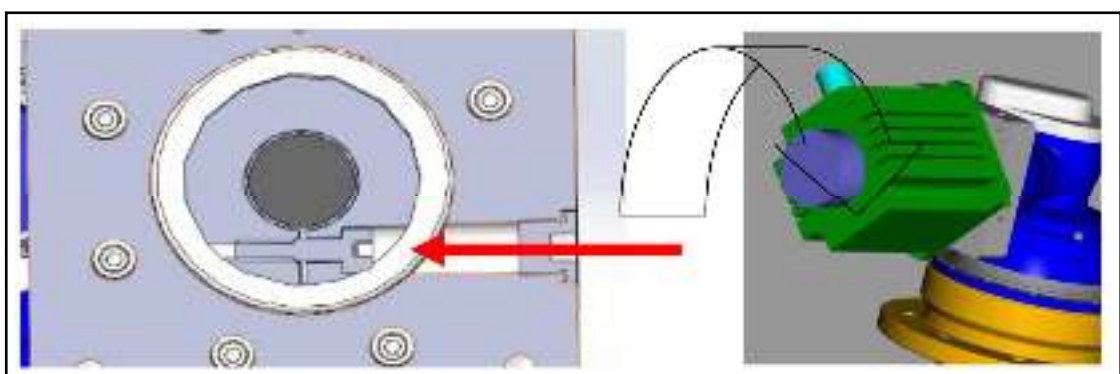
The fins must be locked form outside the hull, otherwise they will rotate for buoyancy forces.

Disassembling of the motor

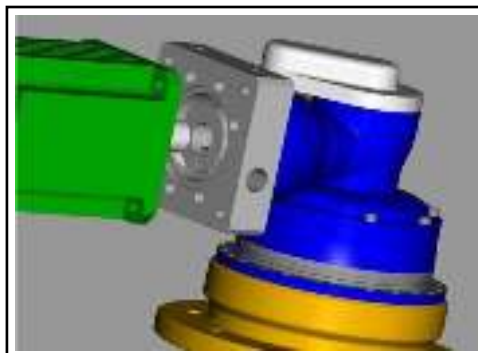
1. Disconnect the 4 electrical cables that are connected to the actuator. Mark and protect every cable.
2. Unscrew the 4 screws (4) M12 on the motor.
3. Unscrew the cap on the flange with an Allen key 10 mm.
4. Check that the screw on the coupling between motor and gear box can be reached by an Allen key.



5. Rotate the motor until the screw on the coupling is aligned with the hole on the motor flange. Unscrew the screw on the coupling between the motor and the gearbox.



6. Remove the motor.



7. Insert the emergency handle in the coupling.
8. Tighten the screw on the coupling at 70Nm.
9. Remove the Allen key and rotate the emergency crank.
10. Put two bolts M12 to keep the handle in position.



TROUBLESHOOTING

This is the list of the alarms that could be displayed on the Touch Screen Monitor and their possible solution.

ALARM	POSSIBLE CAUSE	CONSEQUENCES	RESOLUTION
The Touch Screen Monitor is off	No power supply to the Touch Screen	The system can be driven by the control panel with buttons	Check the fuses FU34 in the ECO switchboard
	No power supply to the stabilization system.	The fins are locked. Stabilization system does not work.	Check the fuses of the main control panel of the yacht and main switch for stabilization system.
	There is no connection between the Touch Screen and the ECO switchboard	The system can be driven by the control panel with buttons.	Check the connection cable between Touch Screen and ECO.
CONNECTIONS TO THE TOUCH SCREEN Check that the Ethernet plug into the rear side of the touch screen is correctly inserted into the socket. The position of the socket is shown on the following figure.  BOTTOM VIEW Ethernet REAR VIEW Power Supply			

ALARM	POSSIBLE CAUSE	CONSEQUENCES	RESOLUTION
The Touch Screen Monitor is ON, but it does not react to inputs and does not show any data	The PLC in the ECO panel is not in RUN mode.	The fans are locked. Stabilization system does not work.	Check the fuse FU31 inside the ECO unit. Switch the PLC in RUN mode.
	CHECK THE STATUS OF THE PLC		
	Open the ECO switchboard while it is still powered and individuate the PLC. Check that the lights are ON as in the following picture. If not, please contact CMC Marina™ Service.		
	  ECO R		
	  ECO ETH ECO FX		

ALARM	POSSIBLE CAUSE	CONSEQUENCES	RESOLUTION
Servo Fault, Servo Overload, Servo Overheat, Servo Excessive	Overload on the fins.	The servo shifts OFF for self protection. The fins are free, with a mechanical friction. The flow can move the fins to Center. When the fins reach the Center or are moving away from Center, the motor brakes keep them locked.	Check the fuses FU31-FU32-FU33 in the ECO switchboard. Reset the servo. Switch off the power supply to EPU, wait 15 seconds minimum and then reactivate power supply. If the error occurs several times, contact OMC Marine™ Service.
	Temporary over current, or sudden voltage drop.		
	RESET OF SERVO DRIVE - Open the EPU switchboard related to the servo with alarm. - Switch OFF power supply to servo from magnetothermal switch. - Wait for 15 seconds minimum. - Switch ON power supply to servo from magnetothermal switch.  15 SEC   - Check that the display on the servowamplifier is ON as in the following figure. (001 or 002) 		
Emergency Stop	The emergency button on the ECO unit has been pushed.	The fins are free, with a mechanical friction. The flow can move the fins to Center. When the fins reach the Center or are moving away from Center, the motor brakes keep them locked.	Move the emergency button to ON position to reactivate power supply.
AC Power Supply	There is no power supply to one or both the EPU units, or the voltage is not sufficient.	The fins are free, with a mechanical friction. The flow can move the fins to Center. When the fins reach the Center or are moving away from Center, the motor brakes keep them locked.	Check the cause of lack of power.

ALARM	POSSIBLE CAUSE	CONSEQUENCES	RESOLUTION
Potentiometer Error	The fin encoder sends an unexpected signal.	The system drives the fin to Center (according the motor encoder reference) and then the brake is activated. The system shifts to Manual mode, the Override Manual alarm is shown.	If also the alarm Fault Potentiometer is active, Check the fuse FU41 in the ECO switchboard. Check the fin encoder cable. Switch off the 24Vdc power supply to ECO, wait 30 sec minimum and then reactivate power supply. Reset the Potentiometer error on the touch screen.
	The fin encoder reference does not match with the motor encoder reference.		Reset the Potentiometer error on the touch screen.

CHECK OF THE FIN ENCODER CABLE

- Find the encoder on the actuator and check that the connector is correctly inserted into the socket. Check also that the security ring is tight.



ALARM	POSSIBLE CAUSE	CONSEQUENCES	RESOLUTION
Potentiometer Error	RESET OF THE POTENTIOMETER ERROR ON TOUCH SCREEN 		
	• Reset the Potentiometer error if it is marked in red (port and/or starb). 		
Inertial Data Timeout	Data reception error from the inertial sensors box.	The system cannot read roll angle of the yacht.	Check the connection cable of the inertial sensors box. Check the fuse FU42 in the ECO switchboard. Switch off the 24Vdc power supply to ECO, wait 30 sec minimum and then reactivate power supply.
GPS No Fix	The GPS has lost the fix.	The system works, the performances in navigation mode are limited. The Auto Speed indicator turns to red.	Wash/flush GPS receiver box. Contact the supplier of the GPS system if necessary.
GPS Timeout	The system does not receive the GPS signal.	The system works, the performances in navigation mode are limited. The Auto Speed indicator turns to red.	Check the GPS system. Contact the supplier of the GPS system if necessary.
Fin not at Zero	The fins are not at zero during astern maneuvering.	If the yacht proceed backward at high speed can be produced damages on the fins.	Verify that system is active (Start) and check the EPU power supply.
Crankshaft Sensor disabled	The crankshaft sensor or the input from engines is not active.	The system keeps working but does not move automatically the fins to zero during astern maneuvering and does not exclude automatically the Anchor mode when the yacht is motoring.	Always press 'Fin at center' before maneuvering. Contact CMC MARINE™ Service.

Specifications

This stabilizer system is composed by two fins actuated by brushless motors and managed by a dedicated control system.

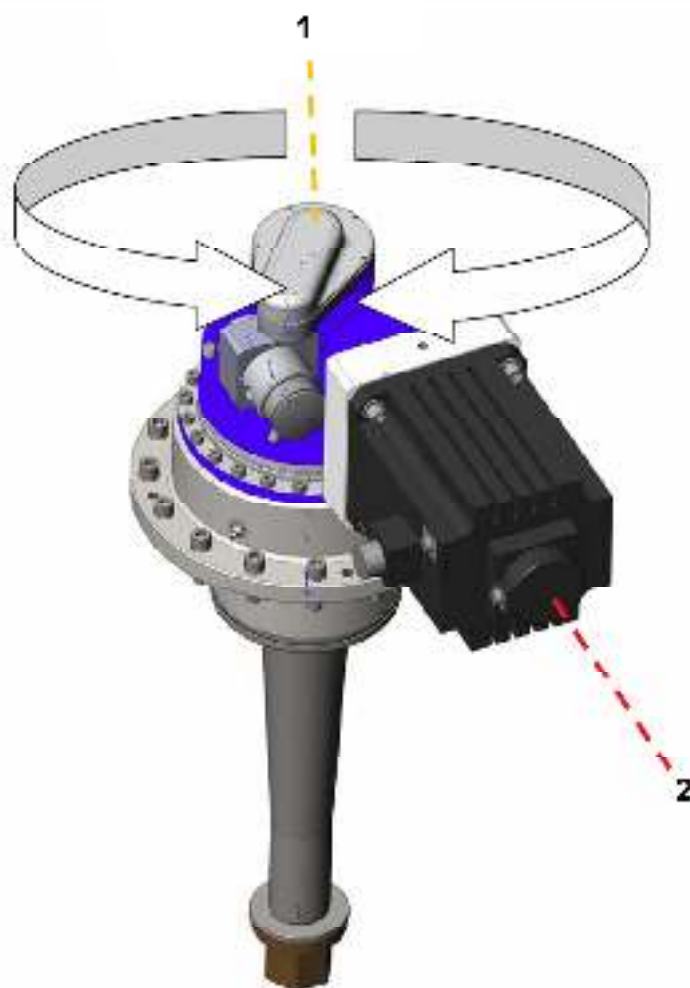
Fin

Fin Model:	190LR
Fin Size:	1.90 m ²
Fin fitting:	CNC machined oversized self-locking taper.
Fin weight:	180 kg.

Actuator

Electrical motor:	5.0 kW brushless with electromagnetic brake.
Fin rotational speed:	From 0°/s to 80°/s.
Power supply:	220Vac or 400Vac 3-phase, 50Hz÷60Hz.
Max power absorption in underway condition:	1.2 kW/actuator at max.
Max power absorption at anchor condition:	3.0 kW/actuator at max working condition with power peaks of 10.0 kW/actuator for t < 0.4 s.
Main Shaft:	100 mm diameter, duplex stainless steel.
Upper and Lower Bearing:	Heavy-duty tapered roller bearings.
Lower Seal:	Two double-lip seals for maximum seawater protection. Easily accessible in a 316 stainless steel retainer.
Upper Seal:	Double-lip, spring-loaded seal for grease retention. The double-lip design also provides protection of the fin actuator against external contaminants.
Lubrication:	Pre-packed with water-resistant grease for lifetime maintenance-free service. Reduction gearbox lubricated with mineral oil.
Fit & Finish Throughout:	Painted white RAL 9016 / polished.
	Actuator weight: 235 kg.

Mechanical Actuator – Onboard positioning

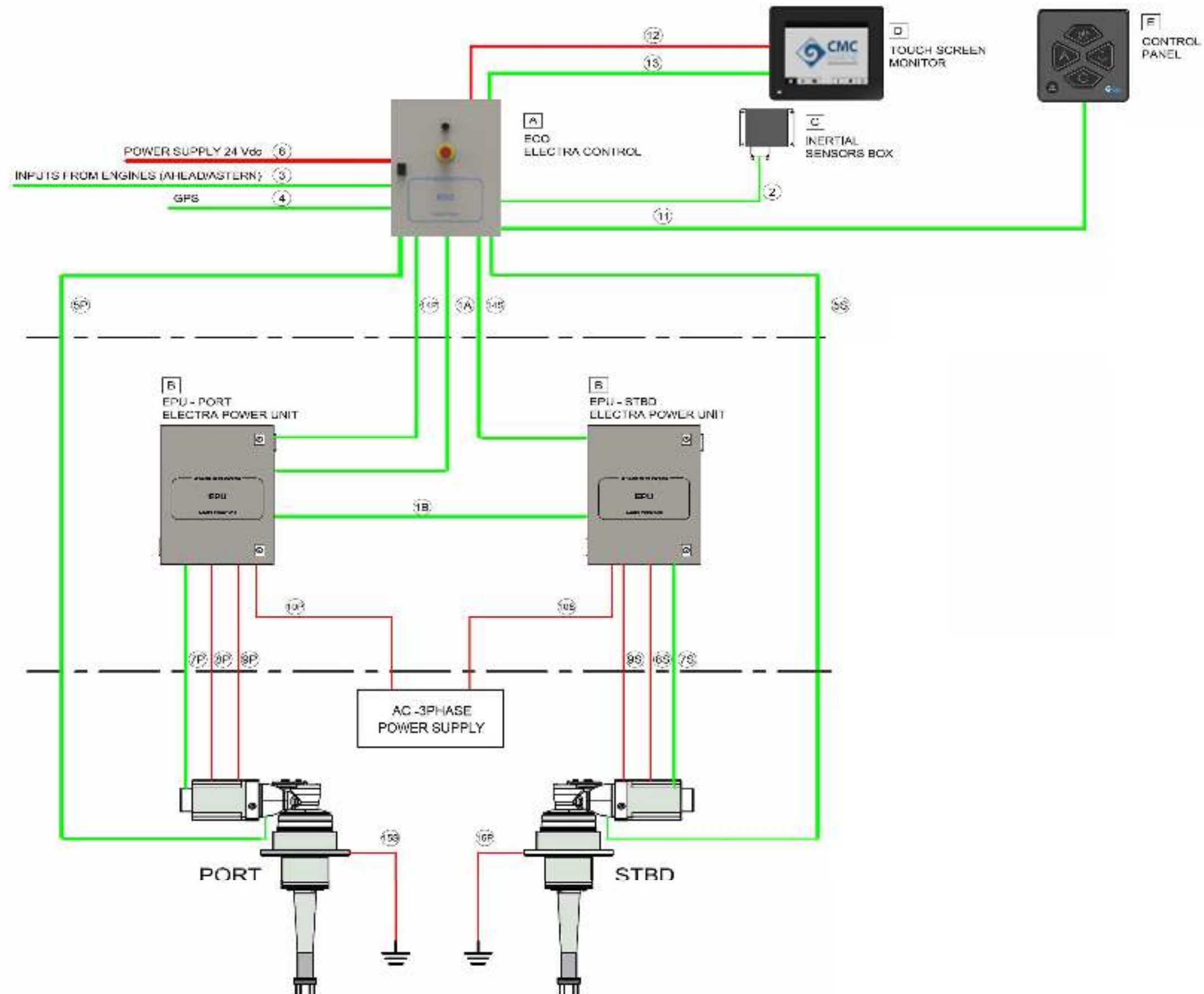


The actuator can be arranged in any position around the shaft axis (1). 16 steps given by the holes in the flange: no restriction for position of the electrical motor axis (2). No matter if the arrangement is not symmetrical between port and starboard.



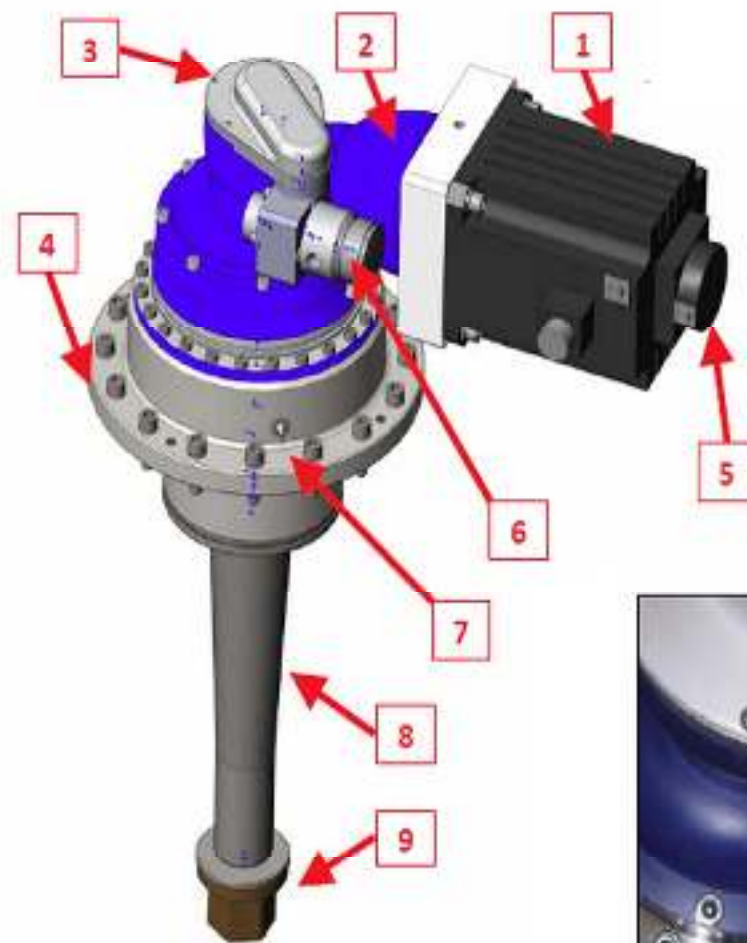
Use the position given by the technical office of the yard for correct housing and easy access to the encoder and to the lubrication port.

Stabyizer Diagram



Stabilising Fins – Details

- FOR BEST READING ITEMS ARE NOT CORRECTLY SCALED
- USE TWO SEPARATE WIREWAYS FOR POWER CABLES **RED** AND SIGNAL CABLES **GREEN**
- THE SIGNAL CABLES **GREEN** MUST NOT BE INSTALLED CLOSE TO POWER CABLES OR OTHER SOURCES OF ELECTROMAGNETIC INTERFERENCE
- CABLES 1A - 1B AVAILABLE 20 m LENGTH AS OPTIONAL
- CABLES 5P - 5S AVAILABLE 20 m AND 30 m LENGTH AS OPTIONAL
- CABLES 7P - 7S - 8P - 8S - 9P - 9S AVAILABLE 10 m, 20m AND 30m LENGTH AS OPTIONAL
- THE INERTIAL SENSORS BOX (C) MUST BE INSTALLED ON THE LONGITUDINAL CENTER PLANE OF THE SHIP, THE INSTALLATION MUST BE EASILY ACCESSIBLE, AWAY FROM ANY SOURCE OF ELECTRICAL INTERFERENCE AND VIBRATION, PROTECTED FROM ACCIDENTAL SHOCKS



1. Brushless electrical motor
2. Planetary gearbox
3. Protection for gears and belt
4. Main flange
5. Motor encoder
6. Fin Encoder
7. Lubrication port
8. Tapered shaft
9. Nut and washer for fixing of the fin
10. Visual indicator of the fin angle (0° in the picture. Looking from the top, the reference mark is aligned with the tangent of the screw head)

GENERAL CABLE LIST					
REF	CABLE	DESCRIPTION	QTY	LENGTH (m)	SUPPLIED BY
1A		Fiber Optic Cable	1	10 m	CMC
1B		Fiber Optic Cable	1	10 m	CMC
2	5x0.5 sh	Inertial sensors control box	1	20 m	CMC
3	3x1	Inputs from engines (Ahead/Astern)	1	-	YARD
4	1X2X0.34 sh	GPS connections	1	-	YARD
5P	4x2x0.25 sh	Fin encoder - Port	1	10 m	CMC
5S	4x2x0.25 sh	Fin encoder - Stbd	1	10 m	CMC
6	3x2,5	Power supply 24V	1	-	-
7P	8x0,5 sh	Motor encoder - Port	1	5 m	CMC
7S	8x0,5 sh	Motor encoder - Stbd	1	5 m	CMC
8P	2x1,0	Motor brake - Port	1	5 m	CMC
8S	2x1,0	Motor brake - Stbd	1	5 m	CMC
9P	4x4 sh	Motor power - Port	1	5 m	CMC
9S	4x4 sh	Motor power - Stbd	1	5 m	CMC
10S	4x10	AC-3PHASE Power Supply	1	MAX 30 m	YARD
10P	4x10	AC-3PHASE Power Supply	1	MAX 30 m	YARD
11	16x1,0	Control Panel	1	10 m	CMC
12	3x1,5	Touch Screen Power Supply	1	-	-
13	CAT 5 FTP crossed ethernet cable	Touch Screen Communication	1	MAX 40 m	YARD
14P	7X1,5	EPU signal cable - Port	1	-	YARD
14S	7X1,5	EPU signal cable - Stbd	1	-	YARD
15P	1x10	Actuator Ground Cable - Port	1	-	YARD
15S	1x10	Actuator Ground Cable - Stbd	1	-	YARD

Electric System

General characteristics

The electric system realised on board **SD 118** has been designed with maximum care for personnel and guests' safety as well as for protecting yacht's equipment. The electric system has been realised by using high quality material, fully responding to specific regulations.

Electric system design and realisation has been carried in compliance with RINA and with "Comitato Elettrotecnico Italiano" regulations (CEI rules).

All electric cables are insulated with non-fire propagating plastic material; they are tagged for maximum maintenance operation convenience and are laid in aluminium raceways or in self-extinguishing PVC sheaths and ducts for further protection.

The most important safety contribution is given by system operation modality and regular maintenance. The electric system, as all other systems installed on board, undergoes hull stress and vibration, is located in a high saline humidity environment and therefore it needs periodical inspection.

The electric system, if not correctly used, incorrectly preserved and not maintained, is statistically one of the most frequent causes of fire on board.

Yacht's equipment is supplied by separate electric systems, with different characteristics:

- **24 VDC**;
- **230 VAC**, 1 phase 50 Hz;
- **400 VAC**, 3 phases 50 Hz.

The DC systems are supplied by battery packs, which supply power separately to the engines (for commands/controls, ignition) (24V), on-board utilities (24V), generators (12V) and emergency utilities (24V) and radio (12V) and are charged by a battery charger or by the alternators driven by the various diesel engines.

AC systems (230V-400V) can be supplied by shore system or by the yacht's generators.

The main switchboard, located in Control Room, provides management and distribution of power supply coming from shore supply and generators, by means of control circuits, interlocking and power supply sources priority settings.



Warning: high risk!

It's recommended not to use electrical or electronic devices whose technical characteristics are not suitable to the equipment on board. Do read carefully the label on the device before using it and check that it fits to the tension provided on board.

All circuits supplied by the main switchboard are protected against short-circuits and overloads (they are equipped with magneto-thermal switches, unipolar for 24V and bipolar for 230V, tripolar+N for 400V, differential switch with sensibility 0.03A which protects 230V-400V systems); they are labelled and most of them are provided with a green operation lamp on the bridge switchboard.

The battery chargers, which are powered from the main switchboard, recharge the on-board batteries in buffer mode. The engine ignition batteries and the on-board generator start-up batteries kick in when the alternators driven by the engines are not in operation.



Warning: high risk of electric shock!

Remember to cut out current, before handling any inner component of electric systems.



WARNING: *any electric system (even low voltage systems), if improperly managed, or subject to excessive loads, might cause short circuits and dangerous overheating, leading to potential fires.*

Only a competent electrician shall carry out all installation, modification or maintenance procedures of the electric system.

The system should be checked at least every two years.



Measures to minimize dangers of electric shocks or fire

Open the ground supply connection switch of the unit, before connecting or disconnecting the ground supply cable.

Connect the ground supply cable to the inlet socket of the unit before connecting it to the ground supply source.

Disconnect the ground supply cable from the ground power source first.

Should the reverse polarity become active, disconnect the cable immediately.

The ground power supply socket cover should be tightly shut.

Do not modify the ground power supply cables. Only use suitable connectors.



Warnings for the 24 Vdc electric system

NEVER work on the system when this is alive.

NEVER modify the electric system of the unit, or its relevant drawings. Only a competent marine electrical engineer should carry out installation, modification and maintenance procedures.

NEVER change or modify the nominal power supply intensity of the over-voltage protection devices.

NEVER install or replace electric appliances or devices with components that exceed the nominal power supply intensity of the system.

NEVER leave the unit unattended while the electric system is powered (except for the bilge water pump, the fire protection system and the alarm system).



Warnings for the 230-400 Vac – 50 Hz electric system

NEVER modify the electric system of the unit, or its drawings. Only a competent marine electrical engineer should carry out installation, modification and maintenance procedures. The system should be inspected at least every two years.

the system is not being used, the ground power supply should be disconnected.

Ground the metal containers or protection boxes of the electrical equipments to the protection conductor device of the unit (green, with or without yellow line, conductor).

Only use electric equipments with double isolation and ground connection (earth).

Should the reverse polarity become active, disconnect the electric system. Correct the problem before reconnecting the power on the unit.

Generators

Your yacht is equipped with two generators: **KOHLER 70EF0ZDJ** 70kW, 400-230VAC, 50Hz, 3-phases (port and starboard side); operated by diesel engines, enclosed in soundproof cases.



KOHLER generators are installed in the engine room. Generators start up occur by means of their own batteries, which are located in protection boxes near the generators.

The generators suck combustion air directly from the engine room; the exhaust gasses, cooled down by cooling water, through a separator, are separated again in gas and water and separately discharged in yacht's lateral outlets.

The diesel generator are supplied with fuel from the daily tank and filtered through dedicated individual **RACOR** type filters complete with water filter alarm sensors that send signals to the wheelhouse; excess fuel is fed back into the main tank.

The engines are cooled down, through heat exchanger, by sea water which is sucked though an independent sea chest, and equipped with check valve and filter.

Sea chest filter should be cleaned periodically, according to system operation and sucked water contamination conditions (algae, mucilage, etc.).

Before cleaning the filter, remember to close the sea chest check valve, re-open the valve after having cleaned the filter.

Decision-Maker 3500

Provides advanced control, system monitoring, and system diagnostics for optimum performance and compatibility. The **Decision-Maker 3500** controller uses a patented hybrid voltage regulator and unique software logic to manage alternator thermal overload protection features normally requiring additional hardware. Additional features include:



- **AC Output Voltage Regulator Adjustment.** The voltage adjustment provides a maximum adjustment of $\pm 10\%$ of the system voltage.
- **Alarm Silence.** The controller can be set up to silence the alarm horn only when in the AUTO mode for NFPA-110 application or Always for user convenience.
- **Alternator Protection.** The controller provides generator set overload and short circuit protection matched to each alternator for the particular voltage/phase configuration.
- **Automatic Restart.** The controller automatic restart feature initiates the start routine and recrank after a failed start attempt.
- **Cyclic Cranking.** The controller has programmable cyclic cranking.
- **ECM Diagnostics.** The controller displays engine.
- **ECM fault code** descriptions to help in engine troubleshooting.
- **Engine Start Aid.** The configurable starting aid feature provides customized control for an optional engine starting aid.
- **Event Logging.** The controller keeps a record (up to 1000 entries) for warning and shutdown faults. This fault information becomes a stored record of system events and can be reset.
- **Historical Data Logging.** Total number of successful starts of the generator is recorded and displayed.
- **Integrated Hybrid Voltage Regulator.** The voltage regulator provides $\pm 0.5\%$ no-load to full-load RMS voltage regulation with three-phase sensing.

- **Lamp Test.** Press the alarm silence/lamp test button to verify functionality of the indicator lights.
- **LCD Display.** Adjustable contrast for improving visibility.
- **Measurement Units.** The controller provides selection of English or metric displays.
- **Power Metering.** Controller graphical display provides voltage, current, power factor, kW, kVA, and kVAR.
- **Programming Access (USB).** Provides software upgrades and diagnostics with PC software tools.
- **Remote Reset.** The remote reset function supports acknowledging and resetting faults and allows restarting of the generator set without going to the master control switch off/reset position.
- **Run Time Hourmeter.** The generator set run time is displayed.
- **Time Delay Engine Cooldown (TDEC).** The TDEC provides a time delay before the generator set shuts down.
- **Time Delay Engine Start (TDES).** The TDES provides a time delay before the generator set starts.
- **Voltage Selection Menu.** This menu provides the capability of quickly switching generator output voltage. Requires initial activation using SiteTecht software.
NOTE: Generator set output leads may require reconnection.
- **Paralleling Functions:**
 - Bus sensing.
 - First on logic.
 - Synchronizing.
 - Communication based isochronous load sharing.
 - Droop load sharing.
 - External controlled load sharing via analogic bias signals.

Shore Power Supply

This yacht is equipped with one shore supply socket with cable and plug for connecting to Voltage three-phase, 50Hz, 125A shore power system and one manual socket with cable and plug for connecting to Voltage three-phase, 50Hz, 125A.

One of the above mentioned sockets, which is connected to the glendenning, and the relevant cable stowed in the aft peak on the winding drum glendenning provided for pourpose, is of the watertight guided type.

Sometimes the pier sockets dimensions or voltages are not compatible with the boat equipment; in this event, refer to the harbour master office for asking information concerning a new plug or an adapter.



Do not allow the end of the ground power supply cable to float on water. This could cause an electric field, which could cause injury or death to bathers close by.



Disconnect all circuits and shore power supply before carrying out interventions of the electric system.

Before connecting to shore power supply, make sure that plug and socket are dry and not damaged.

It is advisable not to leave the shore supply cable connected while the yacht is unattended.

When replacing the plug, remember to connect the yellow-green wire to ground.

Batteries

Batteries for yacht's services, main engines start up, engines control electronics supply and generators start up are located in special protective containers in the engine room.

All the batteries are recharged by battery chargers provided for the purpose and powered via the main switchboard, from on shore or by starting up the on-board generators.

Alternatively, the main engine ignition and generator start-up batteries may be recharged by their respective alternators.



Periodically check the efficiency of each battery.



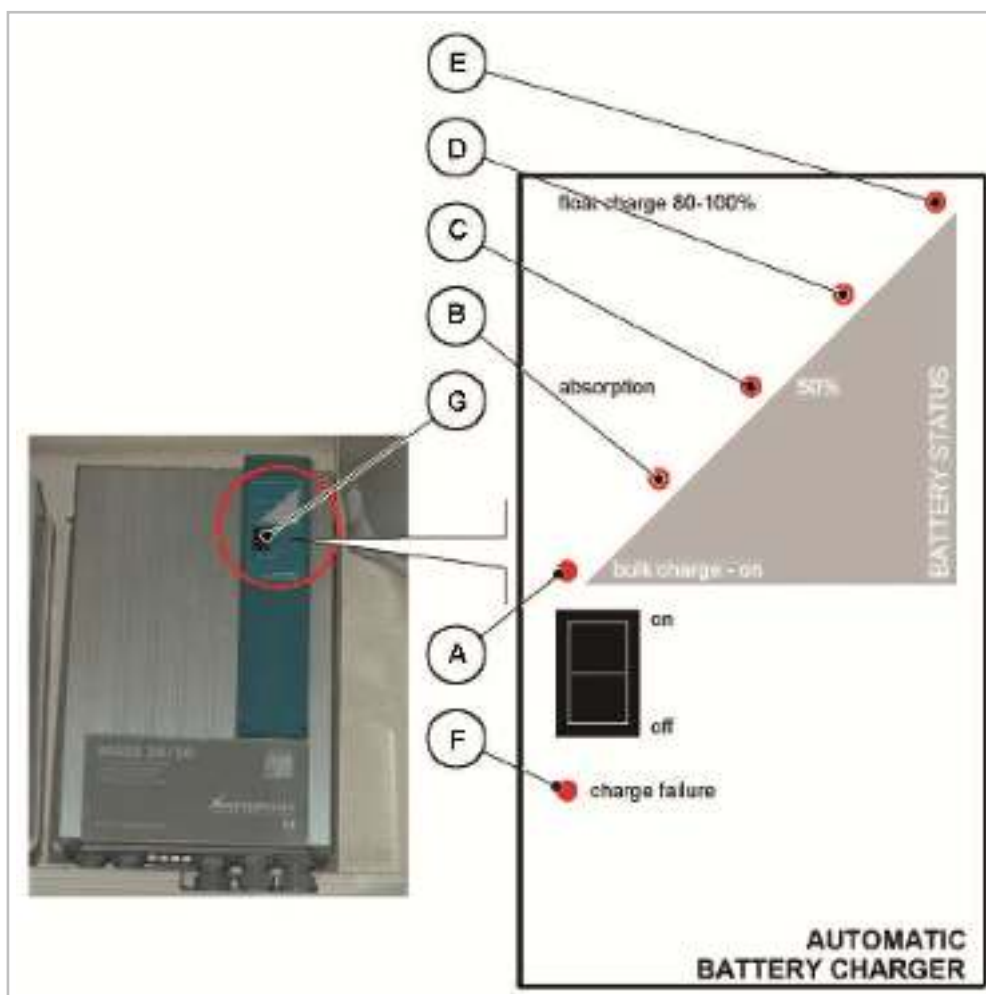
When the yacht is not used, the batteries should always be kept charged by means of periodical re-charging.

If the charge decreases below the minimum, batteries may get irremediably damaged.

- Batteries charger
- N°1 MASS 24/100A battery charger for the services batteries.
 - N°1 MASS 24/50A battery charger for the main engine starting batteries.
 - N°2 MASS 24/50A battery charger for emergency batteries.
 - N°1 CHARGE MASTER 12/25-2 battery charger for D.G. starting batteries.
 - N°1 CHARGE MASTER 12/15 battery charger for radio emergency battery.
 - N°1 MASS 24Vdc/140Ah batteries (n°2 for each engine).
- Batteries
 - N°4 - EXIDE EP2100, 12V-240Ah batteries (N°2 for each engine).
 - N°4 - EXIDE ES1200, 12V-110Ah batteries (N°2 for each D.G.).
 - N°1 - EXIDE ES650, 12V-56Ah battery (radio emergency).
 - N°8 - EXIDE ES1100-6, 6V-200Ah batteries for emergency.
 - N°24 - EXIDE 6EPZV330, 330Ah batteries for services.

The battery charger is equipped with a switch (G) and five lamps (A, B, C, D, E) which show the battery charge level, and a warning lamp (F) which shows charge failure.

Set the switch (G) to ON in order to start up and energise the battery charger. One of the leds on the front part will light up and the charge will start immediately. Set the switch (G) to OFF in order to stop the battery charger.



Leds ON	Meaning
Normal operation	
A	Rapid charge / Discharged battery
A e B	Battery charged at 25%
A, B e C	Battery charged at 50%
A, B e D	Battery charged at 75%
A, B, C, D and E	Maximum charge 5 hours after start up
Fault condition	
F and A	Battery signalling fault
F and B	Battery charger temperature is too high
F and C	Short-circuit indication (25% reduction of charge)
F and D	Short-circuit fault (voltage too low/high)
F and E	Temperature signalling fault

Battery Breaker Panels

Each battery group is provided with a battery breaker disconnecting switch which enables to insulate the group itself from the system it supplies.



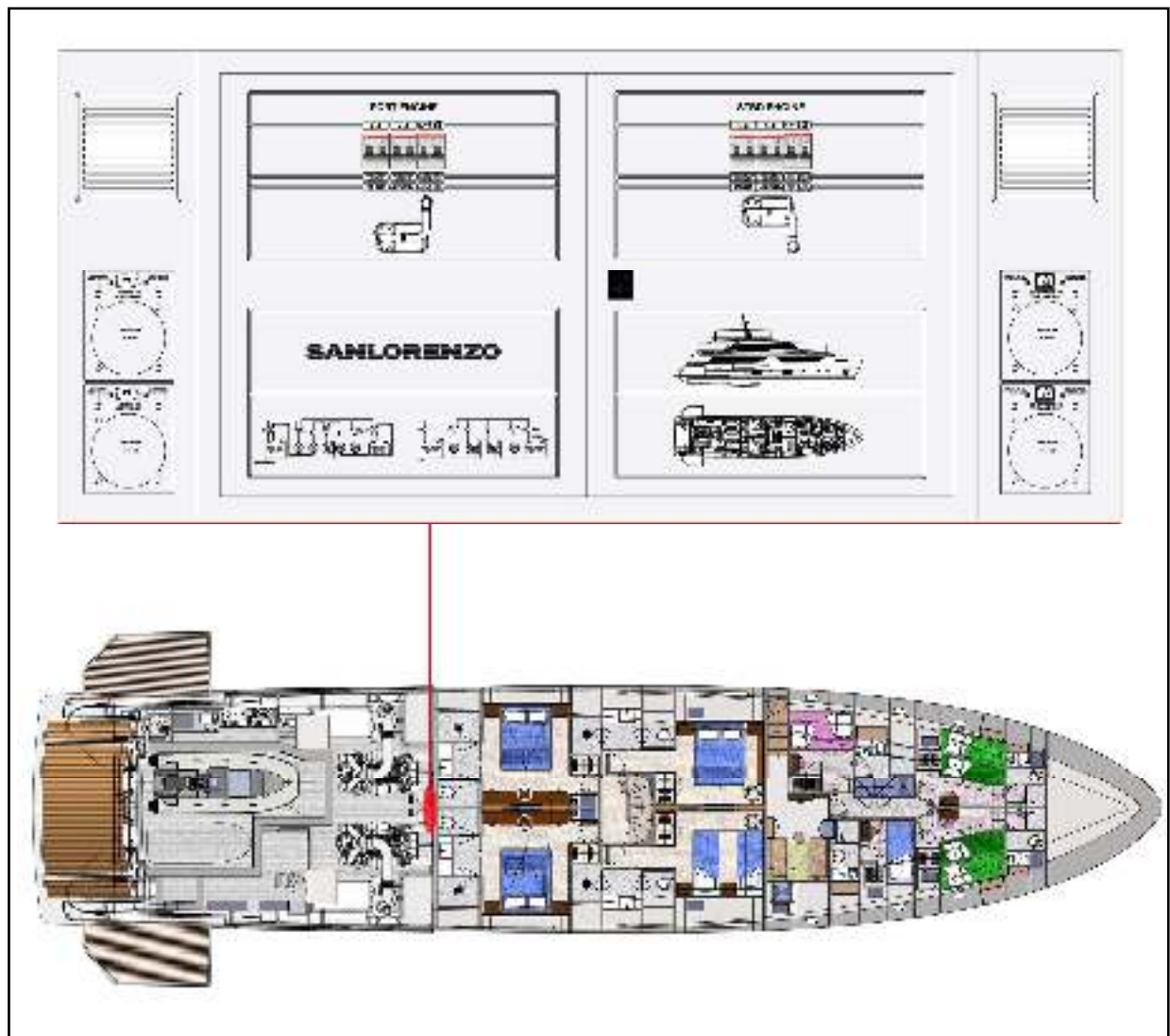
Never disconnect the battery breakers while the engines are running, as this operation may damage the engine alternators.

As far as the main engines starting systems are concerned, in case of one the engines dedicated battery group malfunction, it is possible to start it connecting it to the battery group dedicated to the other engine.

The same possibility is available for the battery groups of the current generators starting system.



The above mentioned operation can be only performed in case of emergency and only for short periods of time.



Battery recharge

To provide energy a battery must be constantly recharged by means of a source of electricity. On board your boat this source is the battery recharger. Always use only the recharging system installed on board; absolutely avoid recharging your batteries by means of other external systems. Your boat is in fact provided with electronic rechargers capable of recharging the battery without requiring external aid.

The most common cause of battery failure, apart from a mechanical break-down, is plate sulphating. When the battery is discharged, lead sulphate crystals are formed on the plate. As the discharging procedure continues the amount of sulphate on the plates increases until it becomes a white layer of 'white lead sulphate'. As this continues, besides discharging the battery completely, the electrochemical activity of the battery itself is interrupted.

In this condition, the battery can no longer be used or will be recharged much less than its stated capacity. This is why a lead battery should never be completely discharged else it will be destroyed.

It is also important to remember that even prolonged and repeated although partial discharging, give rise to the same phenomenon although it will take longer to happen and the results will be less apparent.

Another cause of battery exhaustion is an excessively low electrolyte level. Generally speaking, when this occurs it is due to the evaporation of its water and not of its acid, which evaporates at much higher temperatures. Moreover, the lower part of the plate ends up functioning in a liquid that is much more acid than it should be which could damage the plates themselves. These two phenomena cause conductive material to fall onto the bottom of the battery which could short-circuit the battery element, making both it and the battery itself unusable.

For maintenance operations concerning the electrical system components do refer to the section "Maintenance" of this Manual.

Bilge System

Bilge suction electric-pumps are used to suck water from bilges, discharging it outboard.

The bilge system is composed by two different systems, operating in different ways:

- **Centralised system:**
N°2 electric-pumps (230-400V) located in Control Room, connected in parallel, to be used alternately by opening/closing special three-way valves and N°1 electric-pump (emergency 24V) located in the chain locker / forepeak, which suck water present in various bilge areas, through suction baskets, discharging it outboard.
- **Automatic electric-pumps:**
various submersible automatically controlled electric-pumps, installed in various bilge areas, suck automatically the water, discharging it outboard.

Our yacht is divided into five compartment, each one equipped with both bilge systems equipment:

- **Control room:** N°1 automatic electric-pump and N°1 centralised system suction basket;
- **Engine room:** N°1 automatic electric-pump, N°2 centralised system suction basket (one of them portable), N°1 suction basket connected to the main engine cooling system piping;
- **Port Vip / Guest cabin:** N°1 automatic electric-pump and N°1 centralised system suction basket;
- **Stdb Vip / Guest cabin:** N°1 automatic electric-pump and N°1 centralised system suction basket;
- **Crew cabins:** N°1 automatic electric-pump and N°1 centralised system suction basket.

Each compartment of the yacht is provided with 3 floating switches for:

- automatic electric-pump START/STOP;
- bilge water high level alarm.

The high level alarm is showed by means of a buzzer and a LED on the:

- monitoring system touch panels located in wheelhouse and crew dinette;
- main electrical panel located in Control Room;
- emergency panel located in wheelhouse.



Check periodically bilge system outboard discharges, which are essential for ensuring yacht's floatability.

In case of emergency it is possible to suck water from the engine room bilge by using the sea water pumps of each main engine (See also "Sea water system").

When the bilge is empty, remember to set the valves to sea suction, to ensure engine organs operation.

Bilge system with automatic pumps

The electric pumps are directly installed in the lowest part of the bilge and can suck almost all the water present in the bilge.

By operating the magneto-thermal switches, located on the main switchboard, the bilge pumps are supplied by service batteries and operate automatically.

To activate the automatic electric bilge pumps manually, use the control buttons on the vessel monitoring system touch screens located in wheelhouse, on the control room main electrical panel and on the wheelhouse emergency panel.

A filtering net, located on the pumps suction, prevents sucking foreign matters, avoiding piping clogging.

The electric-pumps discharge directly outboard, through an outlet equipped with check and not return valves.



While the service batteries knife switch is set to "OFF", automatic pumps are disabled.

Before leaving the yacht unattended, make sure that the service batteries knife switch is set to "ON" and that bilge automatic pumps are activated on the main switchboard.

Make sure that the bilges are free from rags and other matters which may clog pump suctions, causing serious damages to the pumps.

Before serving the automatic electric bilge pumps, disconnect the dedicated magneto-thermal switches on the main switchboard.



Eventual hydrocarbon present in the bilge should be collected, kept in cans and discharged in port.

Do not discharge bilge water mixed with oil or fuel into the sea, as it cause serious environmental pollution.

Centralized bilge system

The centralized system is designed to aspirate the water from the bilges of the various compartments.

It chiefly comprises a piping system for aspiration from the individual compartments, a manifold with valves for isolating the bilge to be drawn on, a protection filter and an electric pump powered via a magneto-thermal switch on the main switchboard.

The electric suction pump, which is situated in control room works in manual mode from the main switchboard. Each suction line is fitted with a foot valve with grid.

In the event of electric pump failure, suction from the bilges can be achieved, by operating the valves, with a manual lever pump situated near the electric suction pump.

As a further redundancy, the electric suction pump belonging to the bilge system is arranged in by-pass with the plumbed-in main fire-fighting system pump.

For maintenance operations concerning the bilge system do refer to the section “Maintenance” of this Manual.

Bilge electric pumps operation check

- Check operation of each bilge electric-pump, filling the bilge with clean water until each electric-pump starts up, and checking the correct outboard discharge.
- Check the correct operation of bridge alarm activation float.

Fire Fighting System

SANLORENZO S.p.A. shipyard, for all aspects under its own responsibility, has made all possible design and manufacture efforts to minimise the risk of fire on board.

Particular attention has been paid for choosing materials and during manufacture/assembly.

A regular and correct maintenance of the systems and a prudent behaviour of all personnel are indispensable measures for preventing any risks of fire.

In most cases, fires on board are caused by lack of maintenance and checks, and by low sensibility for trying and remove any possibility of causing direct or indirect fire.

The probability of fighting a fire successfully, depends on the ability to prevent and avoid any condition that may favour a fire to spread, and by intervention rapidity.



Do not stow flammable materials such as gas cylinders, hydrocarbon containers, spray cans, etc. in the engine room.

These objects should be stored in ventilated areas, possibly outside.

Do not store such materials below deck.

Plumbed-In Fire Hydrant System

Five fire-fighting stations (hydrants), complete with intercept valve for hose connection, hose and nozzels, have been installed on board (Upper and Main Deck), more in detail:

- station located near the stairs leading down to the main deck (1),
- station located exactly in the Control Room entrance (2),
- amidship starboard side (3),
- amidship port side (4),
- forepeak starboard side (5).



Given their length, the hoses are enough to cover the vessel's full length making it possible to reach a source of fire at any point on board.

The four hydrants are supplied by a seawater circuit, being drawn up by an electric pump (400V, three-phase, 50Hz) situated on starboard side of control room, by dedicated suction interceptable via valves from the main services seacock pipe.

As a redundancy in the event of the main electric fire-fighting pump fault, a second electric pump (emergency pump), situated in the forepeak and working by dedicated suction interceptable by means of valves from the main seacock pipe, can be used.



Test periodically the system and check pump efficiency; make sure that the water jet power has not decreased compared to the original jet.

If the jet does not comply with this condition, make sure that:

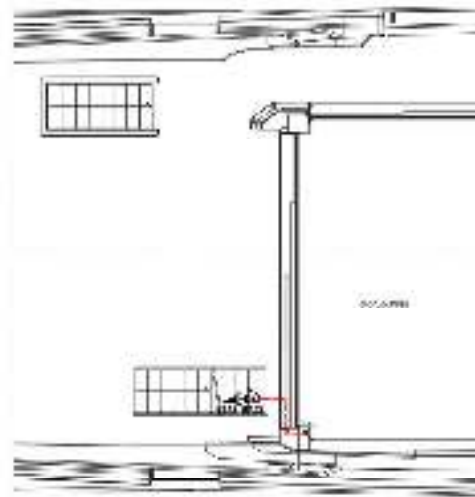
- the sea chest is not clogged;
- the filter is clean;
- the electro-pumps work in the right way;
- the piping does not leak;
- delivery check valves are open.



Do not use water for extinguishing fires near switchboard or energised electric circuits, as it may cause electrocution or short-circuits which can propagate the fire.

As this system is operated very rarely, in order to preserve the main fire fighting electric pump operativeness and efficiency, a branch of the system provides anchor chain washing, for operating the system desultorily.

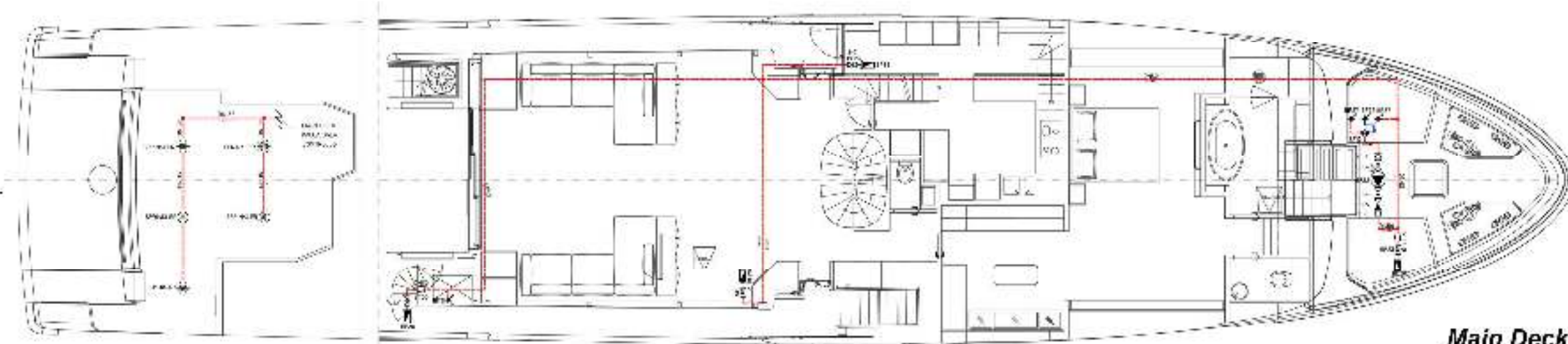
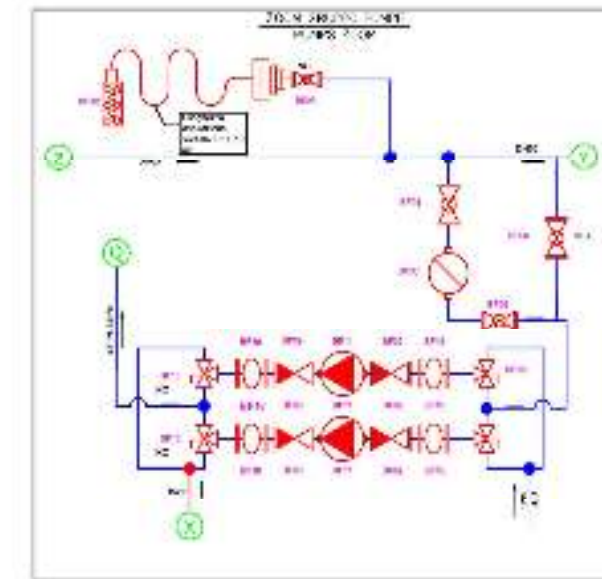
Bilge / Fire-Fighting System



Upper Deck

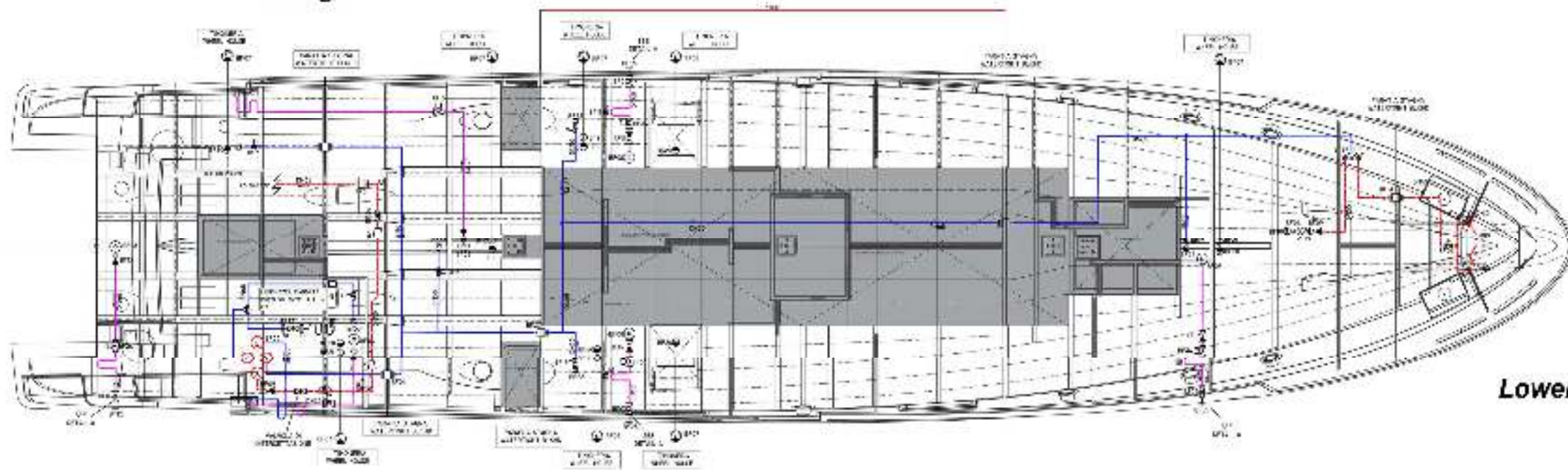
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Main Deck

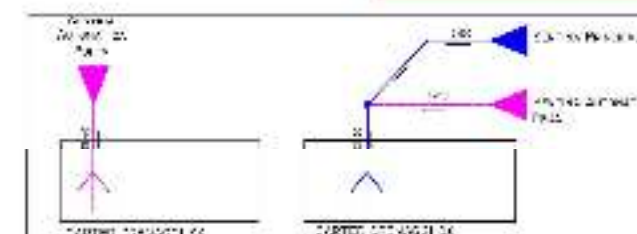
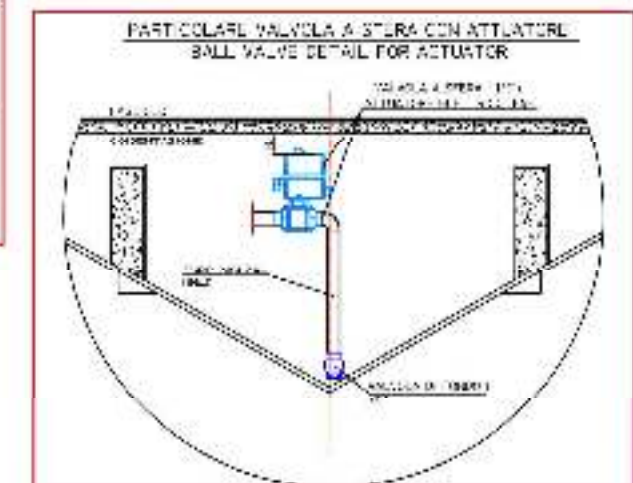
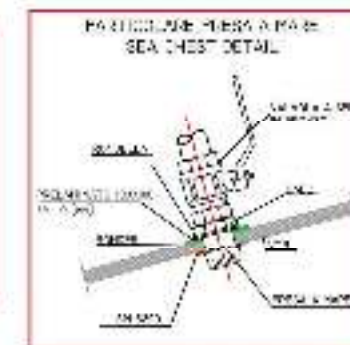
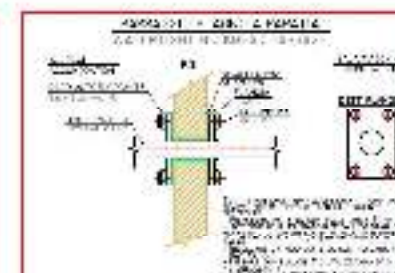
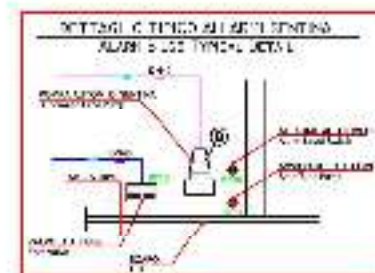
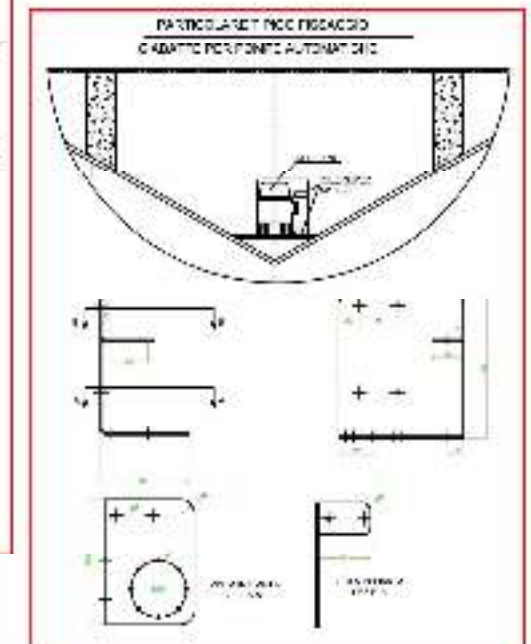
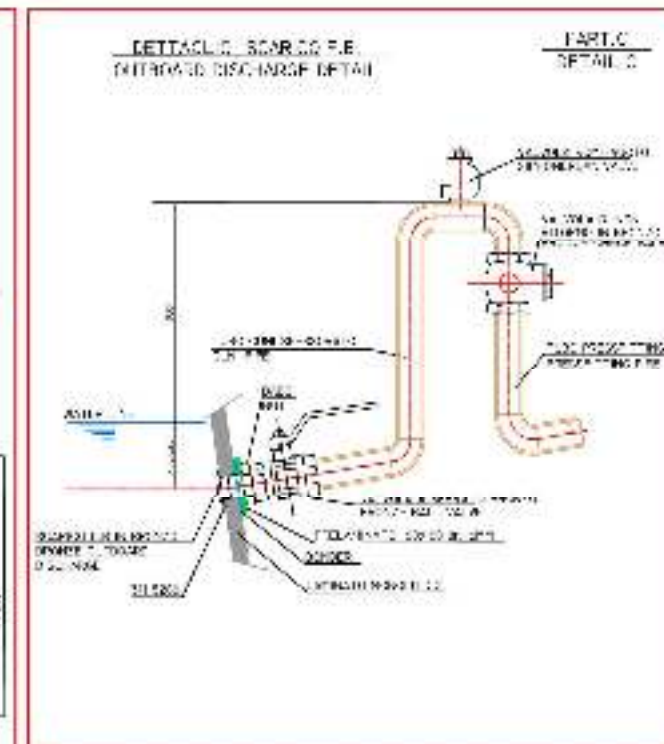
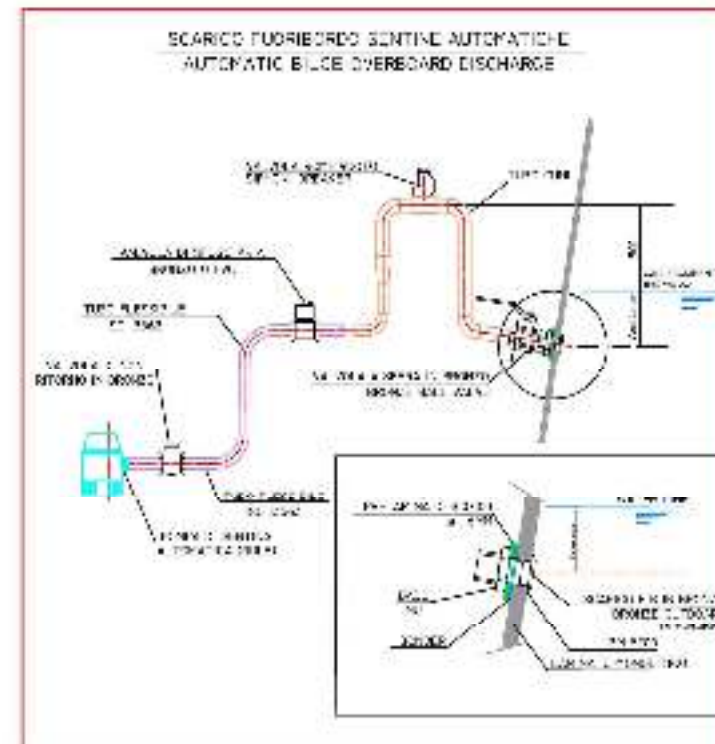
Garage



Lower Deck

Bilge / Fire-Fighting System - Details

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NOVEC – Fixed Fire-Fighting System – Engine Room

The engine room protection is guaranteed by a NOVEC pressurized fire-fighting system.

NOVEC is a colourless, odourless, liquefied and compressed gas; therefore, before activating the system (**by express order of the Captain**), it is necessary that nobody is in the room.

The agent NOVEC is stored as a liquid but is discharged as a gas because it has a quite low boiling point.



The NOVEC fire-fighting system is planned to extinguish fires of classes A and B.

Description *(see attached scheme)*

As soon as **the signal of a serious danger** caused by a fire has been received, the system NOVEC can be activated (**Captain's Responsibility**).

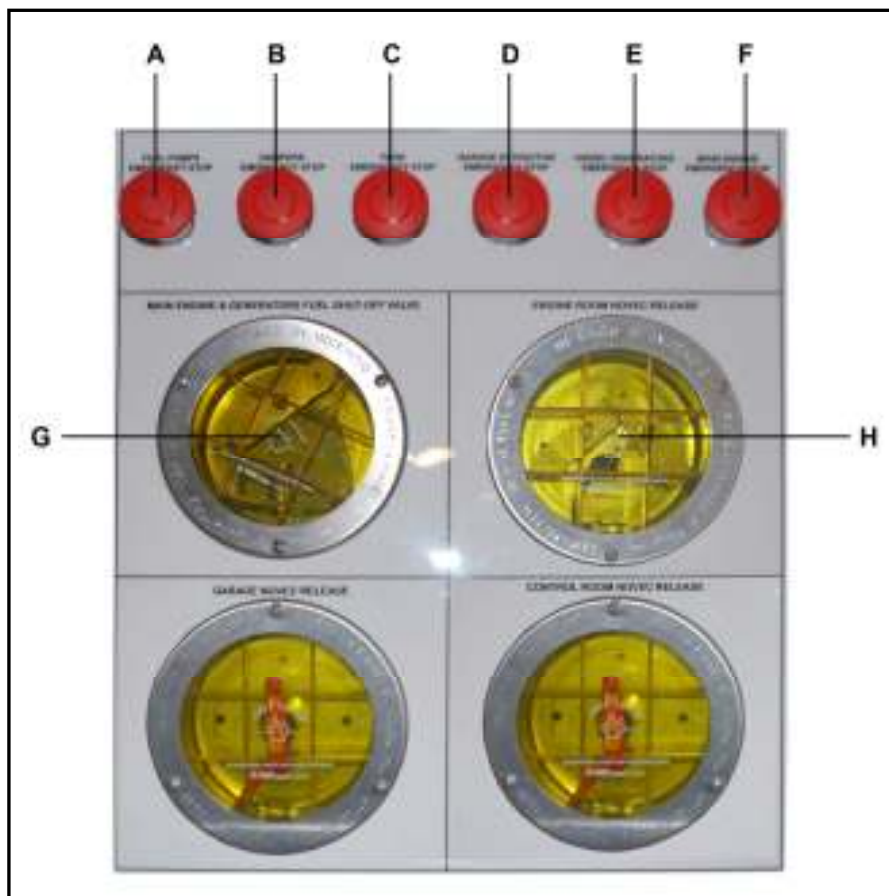
The fire-fighting system is mainly composed of:

- n° 1 NOVEC pressurized cylinder (200 lb);
- n° 1 hand release system (handles) located inside a safety plexiglass box;
- n° 1 Pilot Nitrogen Cylinder;
- Nitrogen Discharge Retarder;
- pressure switch, optical-acoustic alarm and other equipment;
- valve / discharge nozzle.

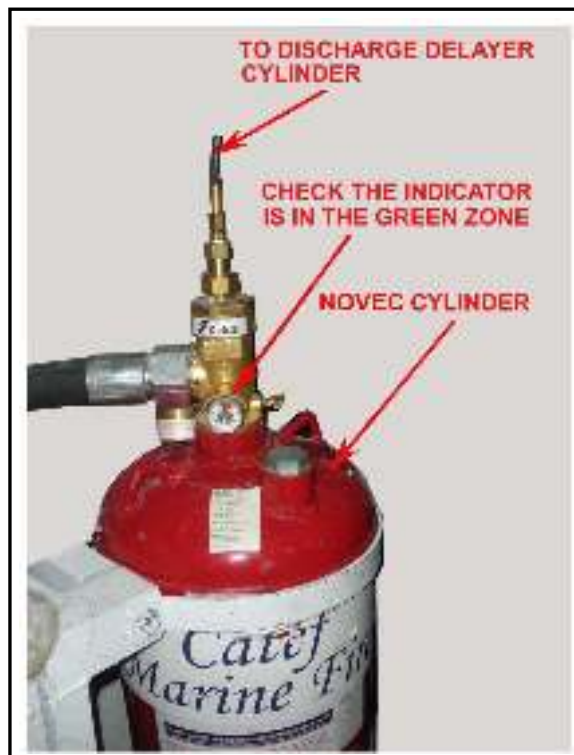
Normal functioning

The fire-fighting system can be activated using the release system (handles) placed in the aft cockpit starboard side, operating as follows:

1. press the follow emergency pushbutton:
 - **A**: stop of fuel pumps;
 - **B**: stop the dampers;
 - **C**: stop the fans;
 - **D**: stop the garage extractor;
 - **E**: stop the diesel generator;
 - **F**: stop the main engines.
2. break the plexiglass safety box (**G**) and pull the handles to open the system interception fuel shut-off valves of MM.PP. and EE.GG..
3. break the plexiglass safety box (**H**) and pull the handles for Engine Room NOVEC release.



Pulling the extinguishing gas release handle (H) the power is delivered, by means of a flexible cable, to the lever in charge of activating a mechanism controlling the gas discharge from the cylinder located in the control room (stbd side of the Yacht).



*If the agent does not discharge it is necessary to operate directly on the lever placed on the delayer cylinder (see detail in § “**Alternative functioning**”).*



*The system has been measured so that the necessary gas is discharged in max. **9 SECONDS**.*



*The system is equipped with a discharge delayer cylinder which enables to delay the agent discharge of **25 SECONDS**.*



Do check the overhauling shown on the labels placed on the various system components and always refer to the Manufacturer's Manual.

Alternative functioning

In particular emergency circumstances the agent **NOVEC** installed on board can be discharged operating directly on the release placed at the top of the pilot nitrogen cylinder, operating as follows:

1. Open the ball valve placed at the top of the pilot cylinder; this will activate the Alarms.
2. Operate the release lever placed on the valve of the pilot cylinder to discharge the extinguishing gas into the protected space.



If the agent does not discharge it is necessary to operate directly on the lever placed on the delayer cylinder (activate the discharge using the control head).

This activation don't give any kind of delay (EMERGENCY MODE).



In case of contact with the extinguishing agent is recommended to wash the body portion with warm water for at least 15 minutes.



The re-charging operation following the cylinder discharge, as well as the replacement and/or overhauling of the fire-fighting system, must be carried out by qualified personnel.



Do avoid unnecessary exposures to vapours, fumes and combustion products.

NOVEC – Fixed Fire-Fighting System – Garage and Control Room

The internal protection of the Garage and Control Room is guaranteed by the dedicated (one for each area) NOVEC pressurized fire-fighting systems.

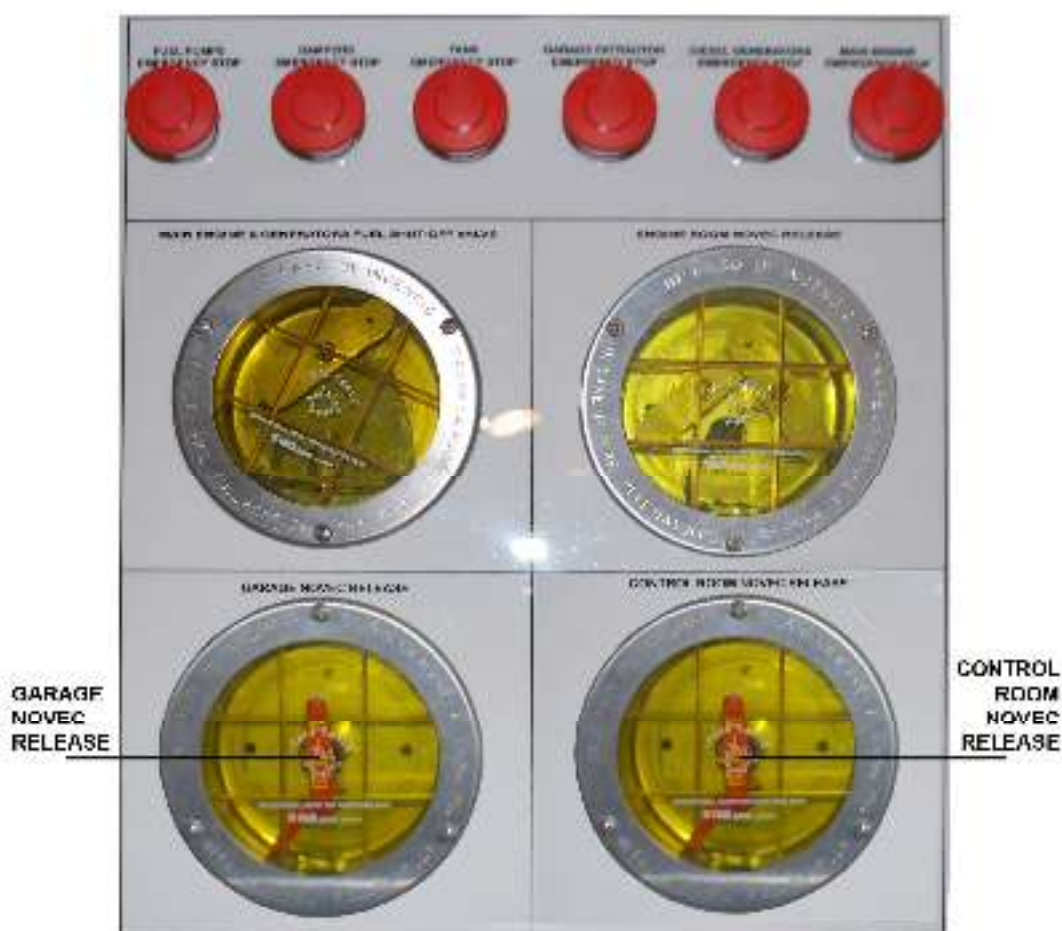
Each system (pre-engineerized) is composed of:

- n° 1 cylinder placed in tender room which can discharge directly in the room;
- n° 1 manual release system (handle) placed in a plexiglass safety box.

Functioning

The fire-fighting system can be activated as follows:

- **automatically**: breaking of the phial placed on the top of the cylinder which starts the alarm signal;
- **manually**: operate the handle located on the aft cockpit starboard side; the breaking of the plexiglass box will start the alarm.



Instructions to follow after the agent's discharge

In order to avoid the agent discharge decomposition products in the adjoining areas it is necessary to protect the room (engine and/or garage) forbidding the entrance until the combustion products have been dispersed.



After having discharged the NOVEC it is necessary to ventilate the room before entering it.



Do refer to the Manufacturer's Manual in order to carry out the maintenance programme.

Restoration of the NOVEC fire-fighting system after discharge

After the activation of the fire-fighting system in the engine and/or garage area and the extinction of the fire it is necessary to restore the essential conditions for a safe navigation, in order to reach the nearest harbour where the appropriate inspections can be carried out.

Navigation can start again only if the deactivated systems have been brought back to their normal functioning conditions.

Therefore it is necessary:

- activate the fuel pumps;
- activate the MM.PP.;
- activate the DD.GG.;
- open the fans and extractors shutters;
- activate the extractors / fans.



*The restoration of the fire-fighting system in order to start the navigation again is recommended **only if** the fire has not damaged the Yacht's structure or other important Yacht's equipment. In this case, or in case of doubt, it is essential to wait for the rescue without trying to start the navigation again.*



Do remember that, after the NOVEC fire-fighting system restoration, the NOVEC cylinders will be empty and not available in case of another fire. Therefore, after having reached the harbour, it is compulsory to have the cylinder re-charged by authorized personnel.

Smoke and gasoline vapours detection system

The smoke and gasoline vapours system can detect fires and gasoline vapours in all areas and decks of Your Yacht.

This system operates the smoke and gasoline vapours detection through:

- n. 1 microprocessor alarm station AUTRONICA (model AUTOPRIME).
- n. 1 control station.

Smoke detection

The fire-fighting system is composed of three independent detection loops, which depend from a microprocessor alarm station AUTRONICA mod. AUTOPRIME.

Each loop consists in optical smoke sensors, in thermal-speed sensors or in mixed technology (optical/thermal) ones depending on the area in which the sensor is installed.

In explosion risky premises (garage) the sensors work in Eex-I and are interfaced by a separation barrier placed outside the explosion premises.

All the sensors are univocal addressed in order to be recognized by the station.

The station continuously controls sensors and loops in order to detect possible failures or malfunctions, signalling them on the control panel located in the wheelhouse.

Each area of the Yacht (technical rooms housing electrical equipment included) is monitored by a sensor.

Emergency call points are positioned near accesses and emergency exits.

In the wheelhouse has been placed a general alarm control panel consisting of two pushbuttons for the activation of the emergency signal (7 short sharp sounds and 1 long sharp sound) and single sharp sounds.

The sirens are placed on two independent loops monitored by the station.

The first loop includes the sirens in the crew area and the sirens / flashing lights in the engines room and in the garage. This loop activates suddenly, in order to enable the prompt intervention of the personnel.

The second loop includes the sirens placed in the owner's area and operates 2 minutes later than the first one, in order to avoid the diffusion of possible false alarms.

The entire system is controlled by a control panel equipped with a keyboard and a LCD backlit screen which signals the alarms and the anomalies as well, indicating exactly, for each event, the its possible cause and giving furthermore the history of past alarms and failures.

By means of the control panel it is possible to perform a complete diagnostic of the entire system and to check the components, if necessary.

The accesses to the various functions are protected by a password, in order to avoid their accidental activation or changes in the operative parameters.

The system is supplied by 230 VAC from board and by an internal emergency buffer battery which guarantees an autonomy of more than 3 hours.

Gasoline vapour detection

The system is based on a microprocessor control station AUTOPRIME and on two explosive mixture sensors working in Eex-d.

The station continuously controls the sensors data and shows the measured percentage on displays.

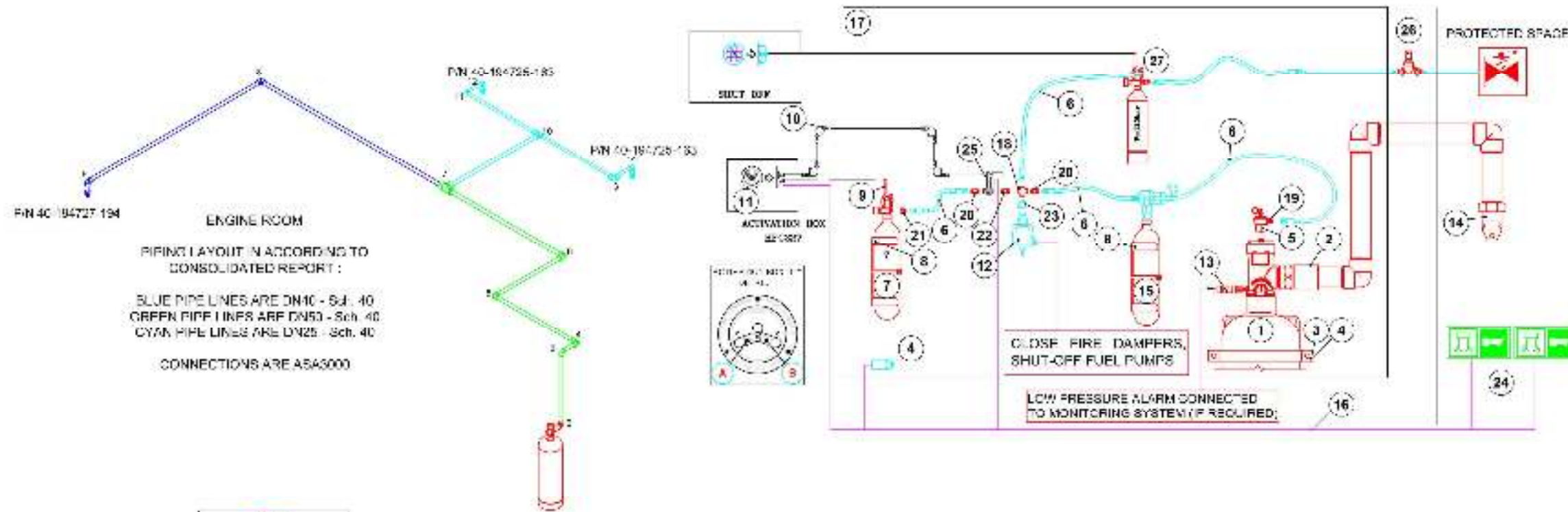
Besides the system performs the sensors diagnostic.

The displays and the pushbuttons are placed on the front of the station.

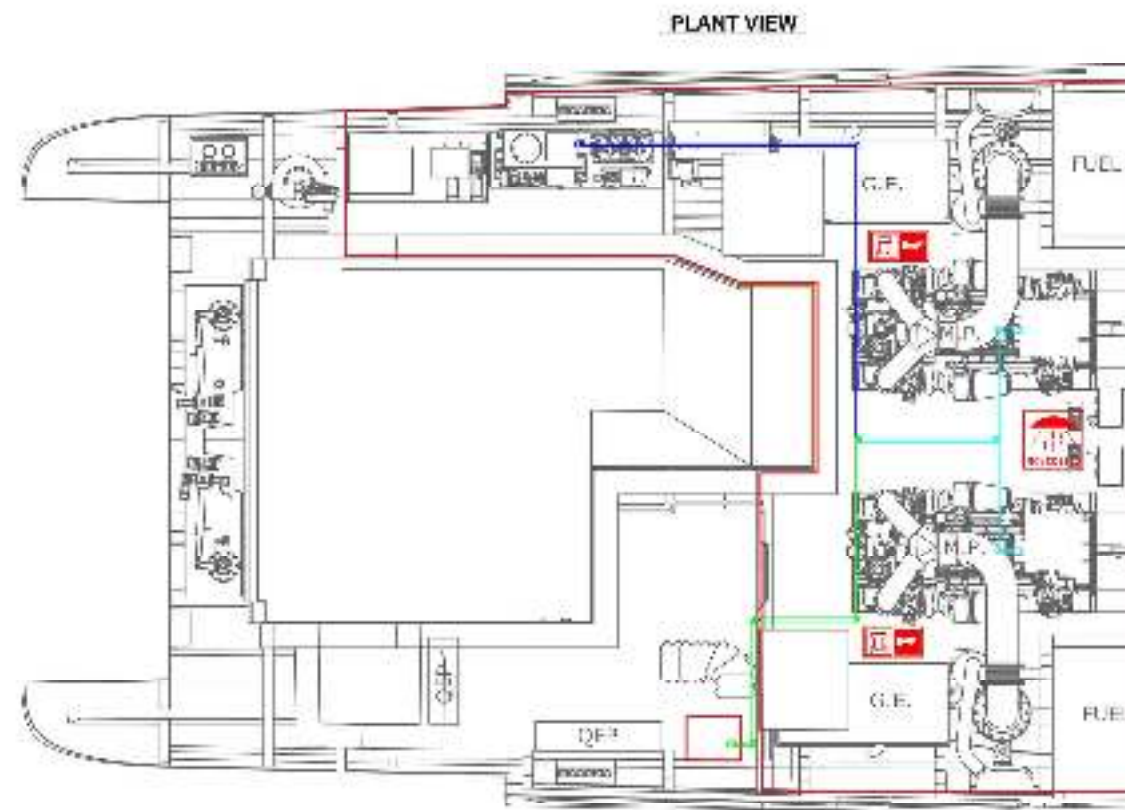
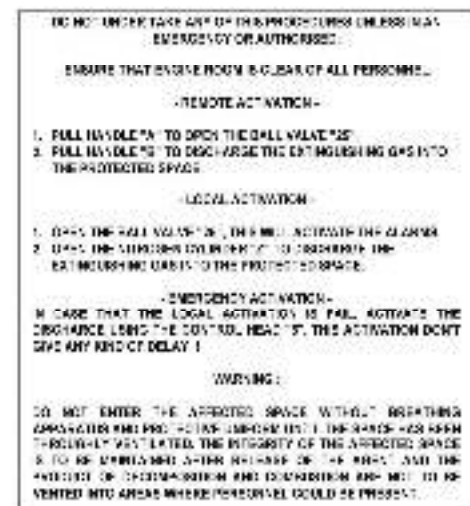
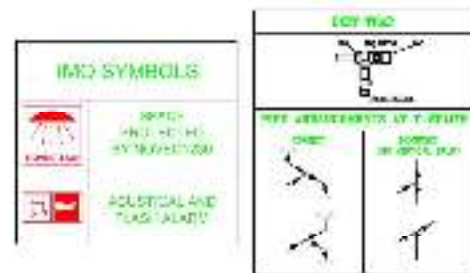
The system is supplied by the emergency battery of the Yacht, in order to guarantee the maximum operation safety.



NOVEC - Fixed Fire-Fighting System



ITEM NO	DESCRIPTION	QTY
1	200 lb Cylinder with Service and Pressure Gauge	1
2	2" Valve, 1/2" NPT, 1/2" O.D.	1
3	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
4	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
5	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
6	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
7	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
8	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
9	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
10	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
11	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
12	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
13	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
14	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
15	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
16	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
17	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
18	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
19	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
20	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
21	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
22	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
23	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
24	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
25	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
26	1/2" Valve, 1/2" NPT, 1/2" O.D.	1
27	1/2" Valve, 1/2" NPT, 1/2" O.D.	1



NOTES TO THE FINANCIAL STATEMENTS OF THE UNIVERSITY OF CALIFORNIA, BERKELEY
 SYSTEMS WILL BE ALLOWED ONLY WHERE PROTECTED AREAS AND IN CHANGING STORAGE AREAS
 NOTES TO DISCLOSED CONSOLIDATED REPORT FOR LENGTH, CLARITY AND LAYOUT OF THE FINANCIAL STATEMENTS.

CONCLUSÃO: O REPORT DE SOBREVIVÊNCIA PARA A LINFOMA DE HODGKIN É DETERMINADO PELA COMBINAÇÃO DA TELA DE DIAGNÓSTICO.

NOTE: THE PRODUCT CAN NOT BE SEPARATED FROM INCARCERATED CONCENTRATED BATTERY. THE INSTALLATION OF POWER MUST BE DONE IN TRANSMISSION WITH THE CLOSER TO THE PRODUCT. A DIFFERENTIAL AND GEAR AND SUPPLEMENTARY SYSTEMS CAN BE.

[illegible]

AN ALARM IN THE EVENT OF POWER FAILURE OF THE ALARM SYSTEM IS TO BE GIVEN.

*IF SUBJECTS ARE ASKED TO DO THIS FIRST-ROUND TYPE

THE ARRANGEMENT OF THE CIRCUITS AND THEIR ELECTRICAL PROTECTION ARE TO BE SUCH THAT THE FAILURE OF ONE OR MORE OF THE CIRCUITS DOES NOT AFFECT THE OPERATION OF THE OTHERS.

FLOODING FACTOR				
LOCALIZATION	NO. OF CONTAINERS (%)	VOLUME OF MATERIAL (MC)	Q. AMOUNT AVAILABLE TO FLOW (MG)	TYPE OF EXTINGUISHER TYPE OF EXTINGUISHER
ENGINE ROOM VOLUME	0.05 @ 0°C	00	02	45-100/25-101

IMPORTANTE: I MANIFESTI DI CODIFICA NON RISPONDEBILITÀ IN CASO DI DETERMINAZIONE DEL RAGGIUNTO RISPETTO AL PRESENTE PROGETTO. ()
 IMPORTANT: I MANIFESTI DO NOT ACCEPT ANY RESPONSIBILITY FOR REVISIONS OF MODIFICATION TO THE PROJECTS ON ()
 I MANIFESTI DI CODIFICA NON RISPONDEBILITÀ IN CASO DI DETERMINAZIONE DEL RAGGIUNTO RISPETTO AL PRESENTE PROGETTO. ()
 I MANIFESTI DO NOT ACCEPT ANY RESPONSIBILITY FOR REVISIONS OF MODIFICATION TO THE PROJECTS ON ()

Fuel system

A system comprising the following has been built into the vessel to take on fuel, store it and deliver it to the various users (MM.EE. and DD.GG.):

- dock fuelling station (2, one for each side);
- built-in storage tanks (2);
- built-in daily tank (1);
- air vent piping (2, one for each side);
- fuel delivery piping (for MM.EE. and DD.GG.);
- fuel transfer piping.

There are two fuelling stations (port and starboard, in the aft area of the deckhouse) for fuel oil on board from the dock. These pipes will allow to stock fuel directly into the main tank.

The built-in storage tanks are located into the vessel bottom beneath the guest cabins.

The tank is equipped with:

- steel inspection hatches (some of them fitted with connections for the fuelling lines, for air vent line, for the sounding sensor and for the transfer pipes).
- sounding sensor with electric indicator of capacity on deck.

The steel inspection hatches are mounted on the tank top; some of them are fitted with:

- air vent pipes;
- pipe for fuel return from the generators;
- fuel transfer pipes;
- pipe for fuel return from the main engines;
- sounding sensor.

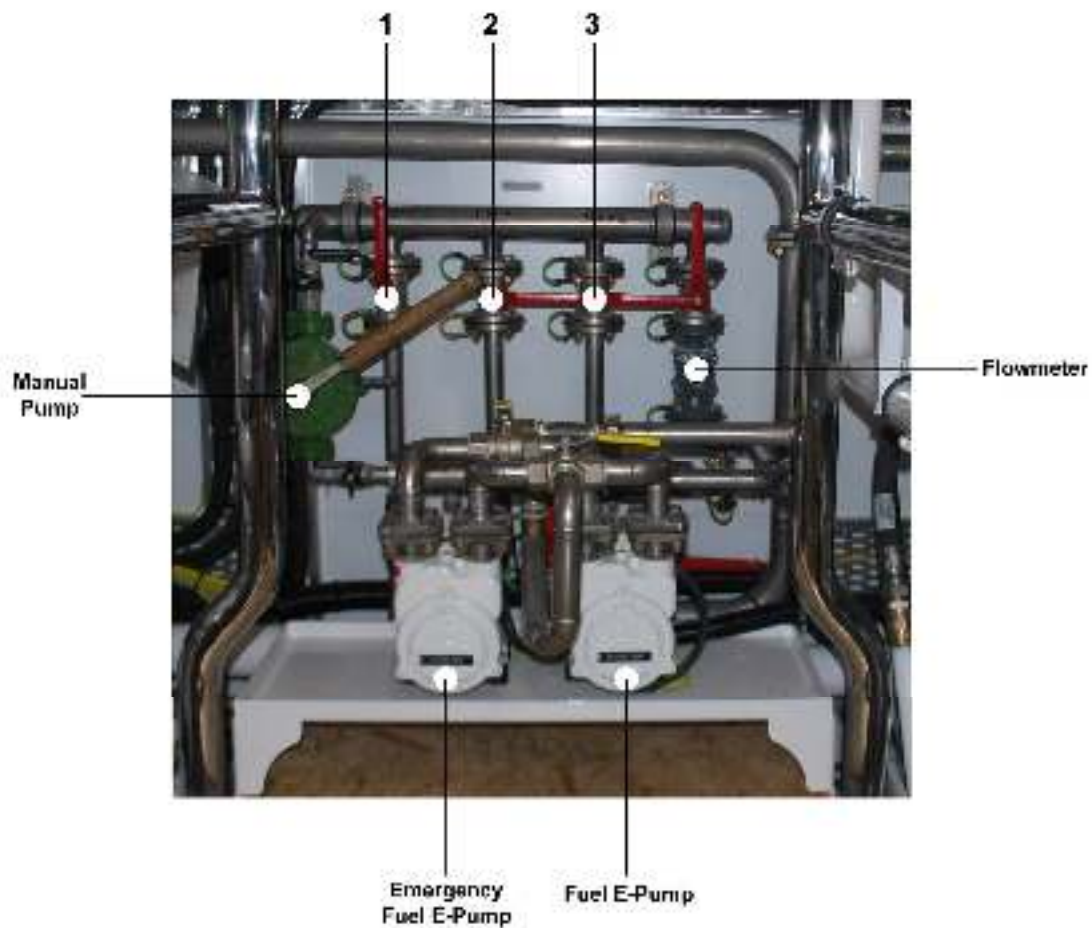
The built-in daily tank has an overall capacity of about 2,500 litres and is equipped with an aluminium manhole for inspections.

On the main electrical switchboard are installed analogic indicators for the fuel quantity collected into the main and daily tanks (one for each) directly connected to the sounding sensor. The same indicator is repeated by digital indicators on the touch screen of the monitoring system. On the monitoring system are shown the high level alarm and the low level alarm.

The system enables both delivery to the individual users and fuel return to the tank from the diesel generators and the main engines.

In order to transfer fuel from the main tank to the daily tank and viceversa the system comprises:

- an electro-pump 400V-50Hz (flow rate 6 m³/h);
- an emergency electro-pump 24VDC (flow rate 6.3 m³/h);
- a manual pump located near the electro-pumps (flow rate 1.4 m³/h).



1. Daily Tank Delivery
2. Structural Tank Delivery
3. Diesel Delivery

The total capacity of tanks is about 37,000 litres. However, due to the yacht's movement in rough sea and to the quantity of fuel which cannot be sucked by the pumps, only 85% of the total capacity should be considered available.

During refuelling, the fuel generates great quantity of foam. To be sure of having refuelled completely the tanks, wait a few minutes and top up, as the upper part of the tank was occupied by the foam instead of fuel.

Refuel the boat several hours before departure; this precaution will give any impurity or water present in the liquid the time to settle down, because heavier than the fuel oil, and eventually settling down in service tanks.



Fuel leakage may cause a fire. Check periodically the system integrity.

Due to the high temperature in the engine room, oil or fuel leakage may evaporate causing serious risk of fire.

Smoking on board is forbidden during refuelling.

Always keep an extinguisher and biodegradable detergents/hydrocarbon absorbers ready to hand in the vicinity of fuel filling station to deal with any spillage onto the deck or into the sea.

Monitor the tank levels constantly during refuelling. Any spillage into the sea must be avoided.

Do not breathe in fuel vapours. Air areas where there is any vapour stagnation.



During refuelling, make sure that fuel does not come out from the refuelling plug or through the vent.

Do not spill fuel into the environment: it is highly polluting.



Stop all engines during refuelling.

The fuel sucked from the daily tank, before reaching the engines flows through separator filters, type **RACOR**, which retain impurities and separate the water eventually present in the fuel.

RACOR separation filters are installed for main engines (two filters for each engine) and **RACOR** separation filters are installed for generators (one filter for each diesel generator). The main engines filtering system is equipped with an alarm sensor that signals the filter malfunction (clogged filter / high water level) on the monitoring system located in the wheelhouse. The lower part of **RACOR** filters include a transparent sump with cup and drain valve, which allow water and impurities drain.



Every marina has toxic waste disposal dedicated areas. It is recommended to dispose waste that can contaminate the environment (such as exhausted oil, fuel oil, oily liquids, batteries, etc.), very carefully.

Prior to performing any job in the engine room, de-energise the automatic bilge pump, in order to prevent that accidental fuel, lubricant or other liquid leakage pollute the surrounding waters.

Fuel purifier system (ALFA LAVAL)

This system is composed of the following components.

- A self-priming volumetric electric pump.
- An ALFA LAVAL fuel purifier, model MIB 303.

The fuel transfer operation, from storage tanks to daily tank, can be carried out other than with the transfer pumps, (in this case, there is no certainty that the fuel transferred to the daily tank is really clean) also using the ALFA LAVAL system. This allow to transfer filtered fuel and, therefore, to extend the life of main engines and generators.



After having correctly set the valves position on the distribution manifold fuel treatment through the ALFA LAVAL filter can be performed drawing fuel from the main storage tanks. The filtered fuel can be sent to the daily tank or to the main storage tanks too.

The specific handbook supplied by the purifier's manufacturer shows start up and operation instructions, as well as the operational modes.

Alfa-Laval separator

The use of separator is restricted to removal of water and solids from fuel oil.



Maximum permissible separation temperature is + 70 °C.

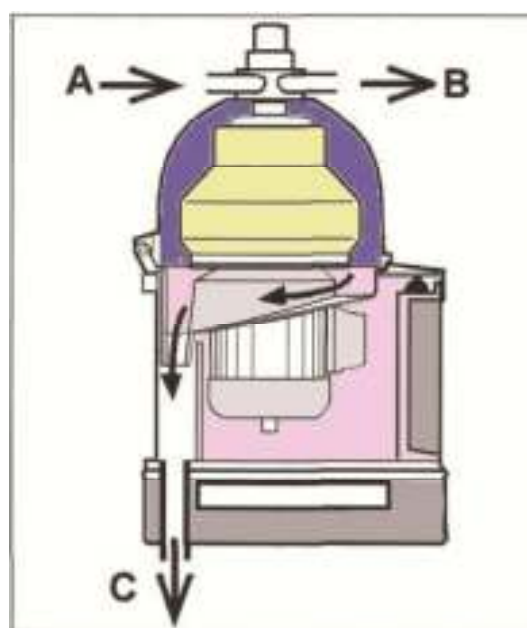
If the liquid to be cleaned contains only small amounts of water and solids, the purifier bowl can be changed to a clarifier, by means of optional conversion kit to replace purifier parts in bowl.

Operating instructions

In order to learn the correct running strategy, firstly it's necessary to read carefully following informations, before starting the separation process.

1. Decide if the separator will be operated as clarifier or purifier.
2. if the amount of water and sludge in the oil is unknown, start the process in clarifier mode.
3. Run the separator for 1-2 hours.

Stop the separator and discharge (about 1 lt) from outlet (C) into a glass bottle or similar to check the water content. In event of amounts of water, operate the separator as a purifier.



- A. Inlet uncleaned fuel oil.
- B. Outlet cleaned fuel oil.
- C. Water outlet.

DESCRIPTION OF MAIN PARTS

A general view of the separator is shown in the figure beside.

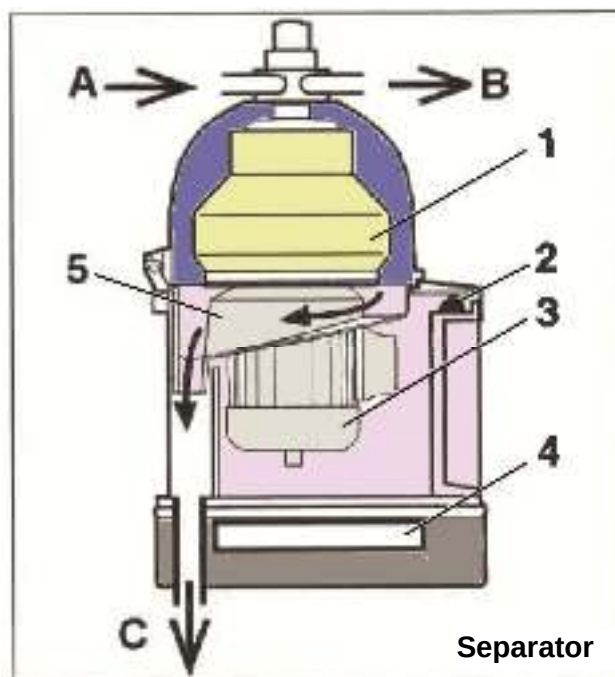
The oil is fed to the oil inlet (A) and down to the rotating bowl (1), where the separator takes place.

The cleaned oil leaves the separator at (B).

If the separator is working as purifier, the separator water runs down the water collection duct (5) to the water outlet (C).

The bowl (1) and motor (3) are suspended on three vibration dampers (2).

The frequency converter (4) for the motor is mounted at the bottom of the separator.



- A. Dirty oil inlet
- B. Clean oil outlet
- C. Outlet water
- 1. Bowl
- 2. Vibration dampers
- 3. Electric motor
- 4. Frequency converter
- 5. Water collection duct



The separator is able to work as clarifier or purifier.

Choose the clarifier when the amount of water is absent in the oil, or only small amounts, at delivery the separator is mounted for working as clarifier.

Choose the purifier mode in event of great amounts of water in the oil.

WORKING PRINCIPLE

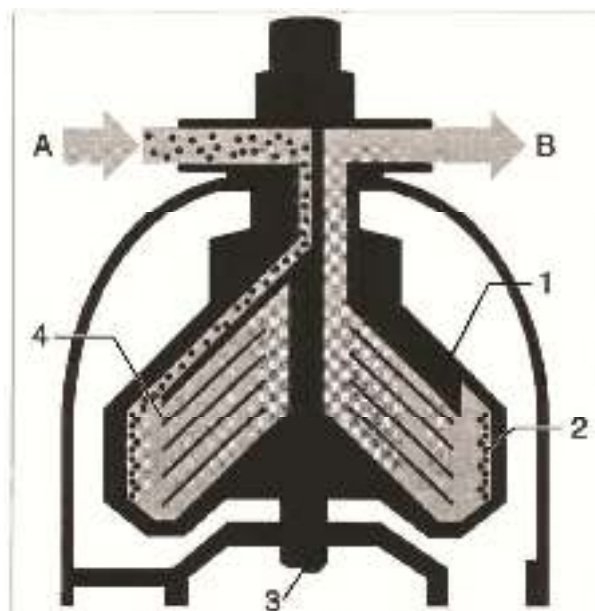
Clarifier mode

The unseparated oil continuously enters at (A) and flows into the bowl (1).

The particles (2) are separated and deposited on the bowl wall.

The cleaned oil is forced inner to the centre of the bowl up to a paring disc (not illustrated). Since the oil is rotating, the stationary paring disc works as a pump which forces the oil out through outlet (B) with a constant pressure.

The particles depositated on the bowl wall are removed by hand periodically.

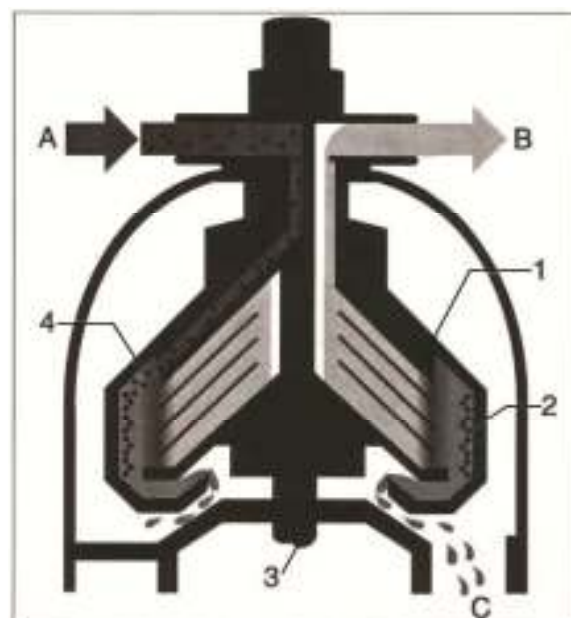


Clarifier bowl

Purifier mode

Operating the separator as purifier, it's compulsory to insert a water seal before starting oil feed, to avoid the oil flows out through the water outlet (C).

Otherwise, the separation principle is similar to clarifier mode, except the separated water, which is more heavy than oil, leaves through the underside of the bowl at (C).



Purifier bowl

Fuel quality

The quality of the fuel is crucial for a good performance of the yacht's engines.

The fuel should be purchased from reliable high sale filling stations, for both the quality and a probable short stay of the fuel inside the shore tank, in order to avoid fuel contamination inside the filling station tank.

The fuel should at least comply with national and international standards for commercial fuels, such as:

- **EN590** (with environmental and low temperature resistance requirements complying to national regulations);
- **ASTM-D975-No 1-D, 2-D;**
- **JIS KK 2204.**

Sulphur content: according to national regulations.

Low density fuels ("urban diesel" in Sweden and "city diesel" in Finland) may reduce the engine power for approximately 5% and increasing oil consumption for approximately 2-3%.

For maintenance operations concerning the fuel system components do refer to the section "Maintenance" of this Manual.



DO NOT smoke during refuelling.

During refuelling, keep a portable fire extinguisher and an absorbing cloth nearby, in order to remove minor fuel spillage.

During refuelling, monitor the tanks levels.

Avoid spilling fuel into the sea.

Fuel leakage may cause a fire. Check periodically the condition and integrity of the supply system.

Due to high temperature that can be reached inside the engine room, any oil or fuel leaks may evaporate, generating a serious risk of fire.



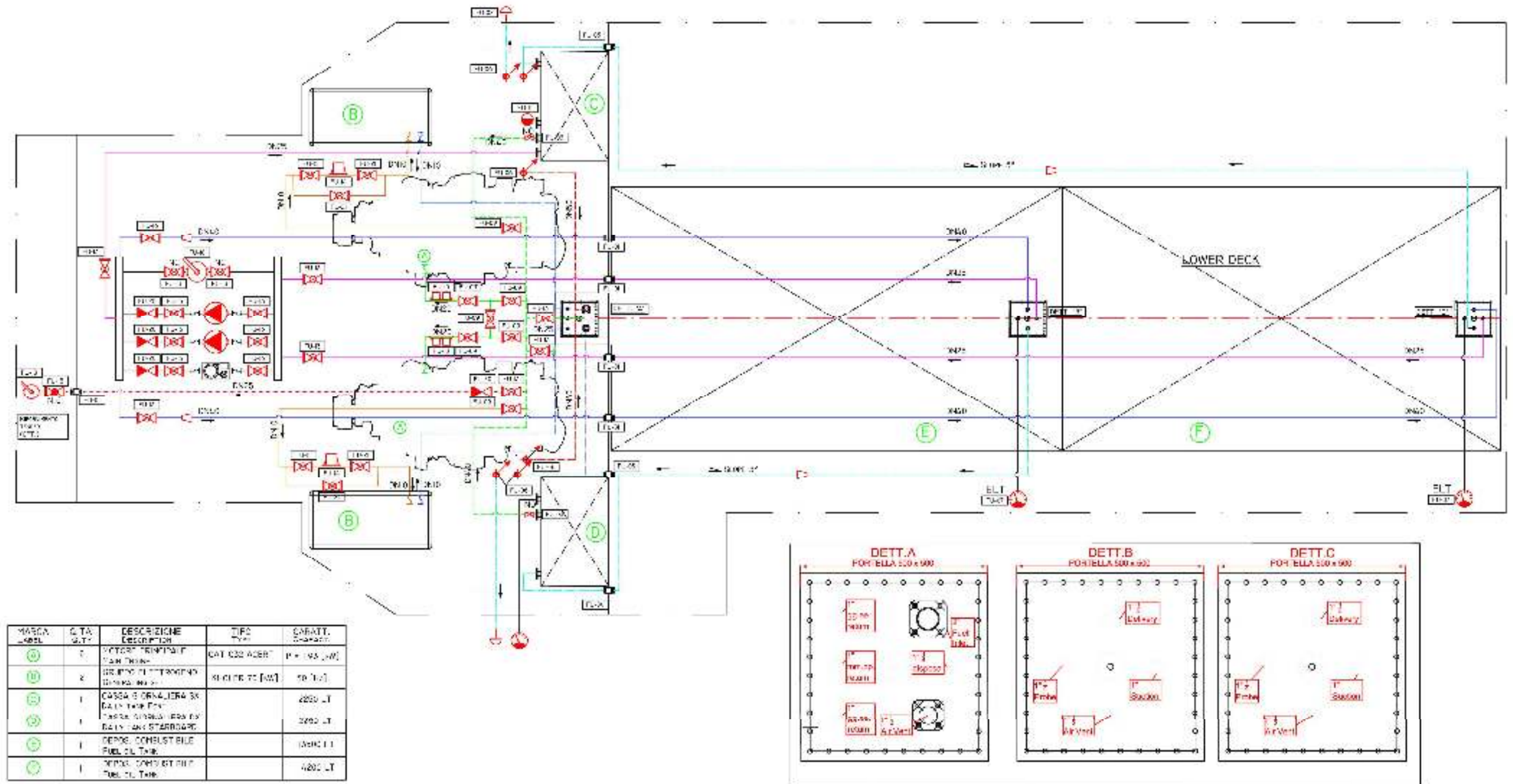
DO NOT inhale fuel vapours.

Ventilate abundantly the yacht's rooms, if saturated with fuel vapour pockets.



During refuelling and fuel transfer, use all the safety precautions required to prevent any leakage, even minimal, of products, which might pollute the surrounding environment.

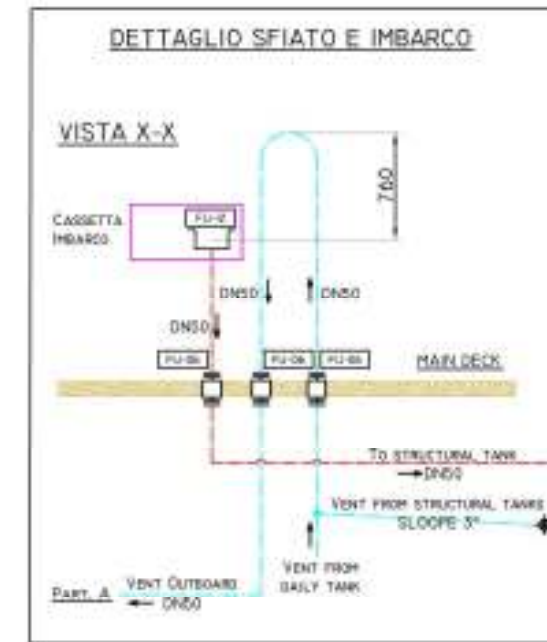
Fuel System



Fuel System - Details

SIMBOLO SYMBOLS	MARCA LABEL	CODICE CODE	DESCRIZIONE DESCRIPTION	ART. ART.	FORNIT. SUPPLIER
	FU01		PASSAGGIO STAGNO PARATIA DN25 W/T SHG PENETRATION NOZZ		
	FU02		SFOGO ARIA CON RETE TAGLIAFUOCO DN40 VENT PIPE WITH FLAME TRAP NOZZ		
	FU03		VALVOLA A RAPIDA CHIUSURA DN40 PN16 QUICK CLOSING VALVE PN16 NOZZ		
	FU04		PASSAGGIO PONTE DN40 DECK PENETRATION NOZZ		
	FU05		PASSAGGIO STAGNO PARATIA DN50 W/T SHG PENETRATION NOZZ		
	FU06		PASSAGGIO PONTE DN50 DECK PENETRATION NOZZ		
	FU07		VALVOLA A SFERA DN50 BALL VALVE NOZZ		
	FU08		PANNELLO DI MONITORAGGIO ALLARMI ALARM MONITORING SYSTEM		
	FU09		VALVOLA A SFERA DN20 BALL VALVE NOZZ		
	FU10		FILTRO GASOLIO MOTORI PRINCIPALI MAIN ENGINES FUEL FILTER		
	FU11		LIVELLO VISIVO CASSA GIORNALIERA VISUAL LEVEL		
	FU12		IMBARCO GASOLIO DN50 FUEL INLET NOZZ		
	FU13		VALVOLA A SFERA DN25 BALL VALVE NOZZ		
	FU14	02050198	FILTRO GASOLIO RACOR 550 MAM DIESEL GENERATOR FUEL FILTER		APG MARINE
	FU15	02040157	GRUPPO POMPE TRAXACO FUEL TRANSFER PUMP	CP300 230V+24 V 50 HZ	GIANNESCHI
	FU17		PASSAGGIO PONTE DN50 DECK PENETRATION NOZZ		
	FU18	02040037	POMPA MANUALE GASOLIO EMERGENCY TRANSFER PUMP		GIANNESCHI
	FU19	02030302	SERBATOIO LEAKAGE E DRENAGGI 16 LT. ENGINES LEAKAGE AND DRAINAGE TANK 16 LT.		
	FJ20		VALVOLA A NON RITORNO DN25 CHECK VALVE NOZZ		
	FJ21		VALVOLA A SFERA DN40 BALL VALVE NOZZ		

LEGENDA LABEL	
	TUBO SU - TUBO GIU PIPE UP - PIPE DOWN
	DRENAGGI TROPPO PIENO OVERFLOW
	IMBARCO NAFTA FUEL INLET
	SFIATI CASSE TANK AIR VENT
	TRASFERITA CASSA STOC. GIORNALIERA TRANSFER FROM TANK TO DAILY TANK
	ALIMENTAZIONE/FEED MMPP ALIMENTAZIONE/FEED GGEE
	RITORNI NAFTA GGEE GGEE FUEL RETURNS
	MANDATA CASSE STRUTTURALI DELIVERY TO THE STRUCTURAL TANKS
	BILANCIAMENTO BALANCING
	OPTIONAL OPTIONAL



NOTE / NOTES

TUBI DI SFIATO IN AISI 304 PRESS FITTING O SALDATI
VENT PIPES AISI 304 PRESS FITTING OR WELDED

TUBI DI TRASFERIMENTO IN AISI 304 PRESS FITTING O SALDATI
TRANSFER PIPES AISI 304 PRESS FITTING OR WELDED

TUBI MMPP e GGEE IN FLESSIBILE OMOLOGATO 15540 - 15541
PIPES APPROVED FLEXIBLE 15540 - 15541

TUTTI GLI O-RING DEVONO ESSERE IN NBR
ALL THE O-RINGS MUST BE NBR

LE GIUNZIONI STRAUB IN SALA MACCHINE DEVONO ESSERE IN METAL GRIPNBR FIRE FENCE
STRAUB CONNECTIONS IN ENGINE ROOM MUST BE METAL GRIP NBR FIRE FENCE

TUBI FLESSIBILI PER BREVI COLLEGAMENTI CON I MOTORI
FLEXIBLE HOSE TYPE FOR SHORT CONNECTION TO ENGINES

TABELLA PRESSIONI PRESSURE TABLE	
PRESSIONE DI ESERCIZIO WORKING PRESSURE	3.5 BAR
PRESSIONE DI PROVA PRESSURE TEST	4.2 BAR
PRESSIONE NOMINALE NOMINAL PRESSURE	6.0 BAR

Fresh Water System

The fresh water system of Your Yacht is supplied by a built-in tank with an overall capacity of about 4,000 litres situated in the forward area of the bilge.

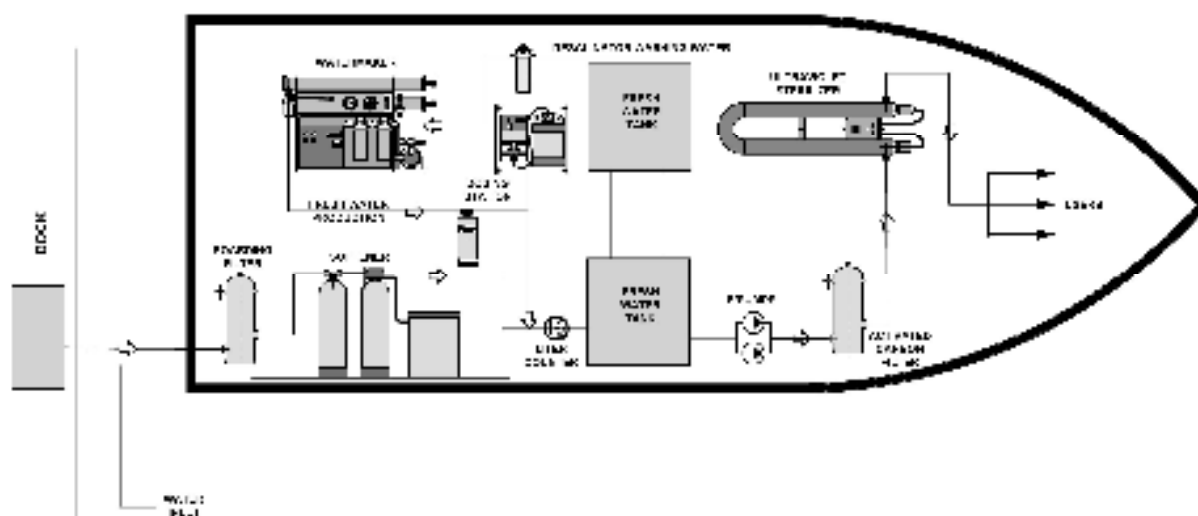
The tank is equipped with:

- inspection hatches for the internal cleaning;
- air vent on the port side of the boat;
- sounding sensor.

To fill the tank the system is equipped with:

- Water Softener (IDROMAR – Idrofast 9100);
- Watermaker (IDROMAR – MC5S);
- Active Carbon Filter;
- UV Sterilizer (SH5000);
- docking filling station on the stern transom area (port side).

The connection diagram of the various devices that make up the Fresh Water System is shown in the following figure:



On the main switch board is installed an analogic level indicator directly connected to the sounding sensor. The same indication is repeated by means of digital indicator on the touch screen of the monitoring system. On the monitoring system is shown the low level alarm.

The system is fitted also with a three-way valve on the filling line that makes possible to supply the on-board circuit via a pressure reducer (to prevent excess pressure in the circuit) directly from the dock.



To avoid damages to tanks and system, refill the system by gravity with proper water pressure and flow rate.

From the tank, before being sucked by the electro-pumps (autoclaves) located on garage, the fresh water is filtered through a pipeline fitted with filter.

The electro-pumps (autoclaves) ensure continuous presence of fresh water at constant pressure in the whole distribution network and are automatically started/stopped by a pressure switch located near the expansion tank, which controls the circuit pressure.

Are installed two electric-pumps (400Vac, one by-pass other), both with variable delivery capacity, electronically adjustable and able to be cut out by means of valves when the system pressure drops below a set value, the second electric-pump starts automatically, to ensure sufficient pressure to all equipment.

In addition to supplying the bathrooms in all the cabins, the galley sinks and appliances, and the flying bridge appliances, the fresh water system also supplies the wheelhouse wipers (via an electric valve), the stern swim platform and fly bridge shower, the washing machine, the air-conditioning chiller, the bows washing bridge workstation.

The fresh water main distribution line is sectioned in local lines by means of distribution manifolds equipped with valves. In this way is possible disconnect every each in case of failure or maintenance.

The outputs of the electro-pumps are connected to a distribution manifold along the boat's side. Secondary manifolds branches out from the main one to supply the single areas. Every system of the area is fitted with an accessible cut out valve placed near the involved area. All valves are labelled.

The fresh water tank must be hygienized periodically, by adding a specific disinfectant solution from the filling plug every two refilling operations.

If the yacht is not used for a long time or during winter storage, it is advisable to pour a concentrated quantity of disinfectant and then discharging the system by opening all equipment.

This procedure allows disinfecting the whole system, avoiding colony of bacteria.



Check periodically the system as a leakage may cause damages and flooding.

The fresh water circuit, and in particular the tanks, must be hygienized periodically, by adding a specific disinfectant solution from the filling plug.

Hot water system

The hot water circuit draws cold fresh water off from the main fresh water distribution line and conveys it to the two electric boilers installed on board (in the crew area) (capacity 100 litres each, power supply 230Vac) to be heated, and distributes it via another piping circuit, completely lagged, to the various conveniences, all of which may be individually shut off. The pressure in the hot fresh water system is thus the same as in the cold fresh water system. The yacht is fitted also with a hot water recirculation electric-pump.

The heaters have been connected up in such a way as to be able to work either both at the same time or in alternation, or else independently.

The boilers are fitted with by-pass valves on pipe connection to allow their maintenance avoiding to stop the entire system.

The boilers are powered by the on-board generators or by the shore supply, and the magneto-thermal switch on the main switchboard, 230Vac distribution, must be tripped to turn them on.



Both boilers' thermostats are set at a temperature of 60°C.

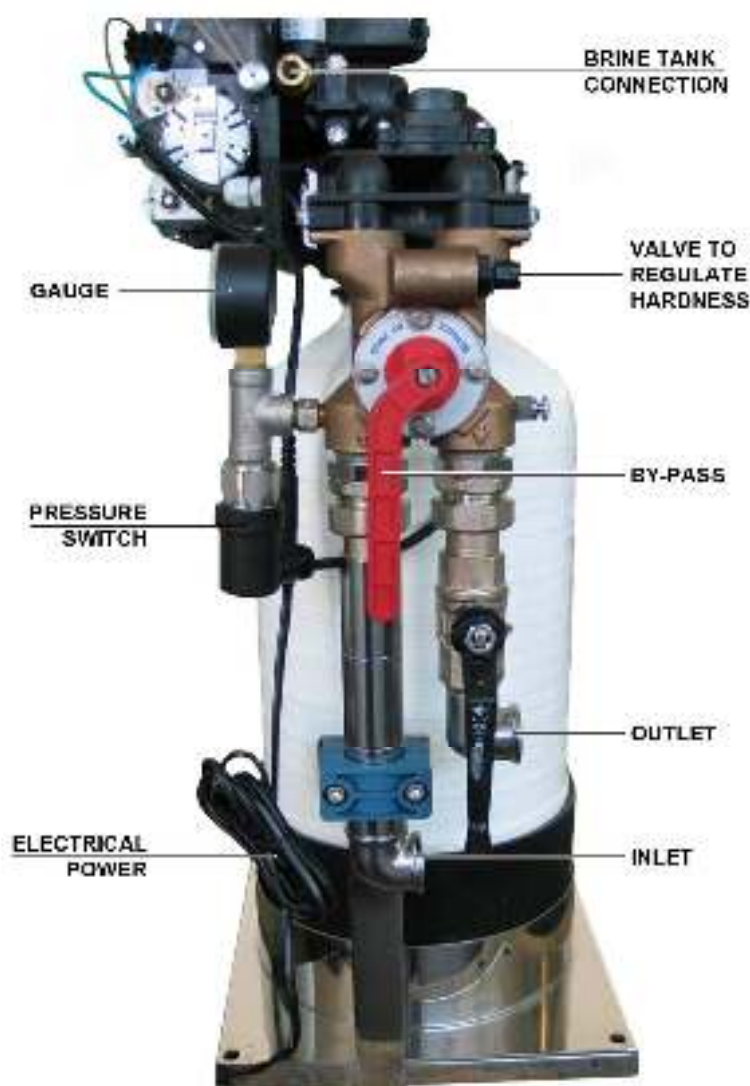
The SANLORENZO S.p.A. shipyard declines all liability for damage caused by setting them at a different temperature.

Softener

The yacht is equipped with Softener (**IDROFAST 9100**) which delivers up to 3.5 m³/h of water, at the minimum operating pressure of 1.5 bar (maximum operating pressure 5 bar).

One of the most common defects of water is its hardness, this system serves to eliminate the hardness of the water and thus to eliminate the limestone contained in it.

The installation of a softener is very straightforward, and the only adjustments to be made to the controller program are setting the water hardness. This softener requires salt for regeneration.



The Softening Process

Hardness in water is caused by the presence of dissolved salts of Calcium and Magnesium. In order to overcome the problems associated with the use of hard water, these salts must be removed. This process is called "Softening". One means of removing the salts is to exchange them for soluble Sodium salts. This technique is known as "Ion Exchange Softening".

IN SERVICE

In order to soften the water, it is passed through a bed of Ion Exchange resin beads which are contained inside a vertical cylindrical vessel. These beads are made of a synthetic material, and are usually amber or dark brown in colour and between 0.5 and 1.0 mm in diameter.

As the water flows down through the resin, the Calcium and Magnesium in the hard water are progressively exchanged for Sodium, with the result that the water which flows out of the unit contains only Sodium salts, which are not scale-forming. The Calcium and Magnesium remain, attached to the resin.

The Ion Exchange resin does not have an unlimited capacity for exchanging Calcium and Magnesium, so to keep the exit water soft it is necessary to periodically "Regenerate" the resin to restore its capacity to soften the water.

REGENERATION

The softening process can be reversed if a strong solution of Sodium Chloride (i.e. Common salt dissolved in water -'brine') is passed through the resin.

The high concentration of Sodium allows it to exchange for the Calcium and Magnesium held on the resin, and these are then carried away to drain. The resin is left full of Sodium to enable it to soften water again.

The regeneration process consists of four stages:

Backwash - Water flows upwards through the resin bed, and out to a drain. As it does so, it loosens the ion exchange beads, removes any resin 'fines' (i.e. small pieces of broken beads etc.) and cleans off any particles of dirt or pipework corrosion products which may have accumulated during the service cycle.

Brine injection/Displacement Rinse - During the first part of this stage, the concentrated salt solution is drawn from the salt storage tank, blended with water to reduce the concentration to the correct level, and passed down through the resin. When the required quantity of brine has been drawn in, the water flows alone to push the remaining brine through the resin at the correct rate, and ensure that all the resin sees the right amount of regenerant.

Fast Rinse - This follows the brine injection and displacement rinse stage, and entails rinsing away the residual brine and Calcium and Magnesium salts from the resin and re-packing the resin bed down. This is done down with water flowing through the resin in the direction of service.

Salt Tank Refill - Following the fast rinse, a quantity of water sufficient to dissolve the correct amount of salt for the next regeneration is returned to the salt tank. When this has finished, the unit automatically returns to service.

METER CONTROL OF REGENERATION INITIATION

A water meter is installed in the outlet from the softener, to measure the volume of water which passes to service. This meter drives a turbine, the movement of which is measured by a magnetic sensor which sends signals to the regeneration controller. The controller microprocessor uses this information to calculate when the unit should be regenerated.

DESCRIPTION OF PLANT COMPONENTS

- **SOFTENER VESSEL AND INTERNALS**

The pressure vessel which contains the ion exchange resin is made from a fibre-glass/epoxy resin outer layer surrounding an inner, seamless shell made from Polyethylene, Polypropylene or ABS. Vessels have a threaded hole at the top, of 2.5 inch nominal diameter, into which the control valve fits.

- **CONTROL VALVE**

The control valve is mounted on top of one vessel, and directs the water flow in and out of the resin beds during the service and regeneration cycles. A manifold assembly is mounted on top of vessel number 2. This is then connected to the main control valve via a pair of plastic interconnect pipes.

- **REGENERATION CONTROLLER AND TRANSFORMER**

The regeneration controller is attached to the valve, and is contained in a plastic housing. The controller is powered by 24VAC electricity, and a separate, wall mounted transformer is connected to the valve to reduce normal mains voltage down from 240V so that all the electrical supply in and around the control valve runs at a safe 24V.

- **BRINE SYSTEM**

The brine system consists of a moulded polyethylene tank into which the brine well and brine pick up are assembled. The tank forms the salt storage container.

The brine pick-up tube within the brine well is connected to the pipe from the control valve which sucks the brine from the tank. At the bottom of the brine pick-up tube is an air check valve. This serves to prevent air entering the valve when all the brine has been drawn in. Air in the system could cause spurting and 'hammering' at the taps or outlets.

- **BRINE MEASURING SYSTEM**

The Brine Measuring System consists of a measuring tank, brine high level float, brine solenoid and brine pick up. These need to be fed from a Salt Storage Tank with fully saturated brine solution.

OPERATION

NORMAL OPERATION

Operation is completely automatic, with regeneration being initiated whenever required by the meter controller.

REFILLING WITH SALT

Since salt is used in the regeneration process, the level of salt in the brine tank will fall after each regeneration. It should not be allowed to fall below 75 mm above the bottom of the cabinet. When the low level is approached (or more often, as a routine) the cabinet should be refilled to the top with pellet salt or 3/4 filled with granular salt.

MANUAL REGENERATION

There can be occasions during the life of equipment such as softeners when the original design basis is not applicable. For example, after a shutdown period, the demand for soft water may far exceed the design flow rate for a short period. Another possibility is that the Water Service Plc finds it necessary to change to a supply which is much harder than that normally received. It will then be necessary to give the unit an additional regeneration to ensure that soft water continues to be available to service.

This additional regeneration is initiated manually.

Remove the timer cover and index the program wheel knob approximately 6mm clockwise until a loud click is heard. Regeneration will proceed automatically immediately. However due to the slow rotation of the program wheel and piston motors, it will be some minutes until regeneration water starts to pass to drain.

BY-PASSING THE SOFTENER

There may be occasions when it is desirable to by-pass the softener, to allow hard water to go to service, or if the softener is for some reason not performing properly.

To do this, open the by-pass isolating valve and close the softener outlet isolating valve. The softener can still be safely regenerated with the valves in this position. If for any reason, it is desired to completely isolate the softener, it will be necessary to close the inlet isolating valve as well.

TEMPORARY SHUT-DOWN

If the softener needs to be taken out of service for some time for any other reason, it is recommended that one or two simple procedures be undertaken to ensure the return to operation is as smooth as possible.

1. The unit should be left with both columns fully regenerated.
2. The electrical supply should be turned off.
3. The inlet and outlet manual isolating valves should be closed.
4. The valve should be drained of water if there is the possibility of the system freezing.

On restart, it is recommended that both columns the unit be regenerated again before being put into service.

CHANGES IN INCOMING WATER

There is a much greater tendency these days for the Water supply Plc to use water from more than one source. It is unlikely that two sources will have similar chemical composition, and so when this happens the raw water being fed to the unit may also change. It is suggested that routine monitoring be undertaken to check whether this is the case. If variation greater than 5% is found, it will be necessary to check whether the settings for regeneration need changing. Resetting is the same as setting.

ROUTINE MONITORING

The following recommendations are made to help the user of the softener confirm that it is performing as required, and to give early warning of possible problems. The operation of the softener is completely automatic, and should not require adjustment (except as above).

Weekly

Check the treated water hardness with a hardness test kit.
Inspect the level of salt in the salt tank and refill if necessary.

Monthly

Check raw water hardness, and record. Compare with original hardness and adjust volume capacity setting if required.

Annually

Inspect and clean/replace as necessary the brine injector, piston and the internal seals. This should be performed by a competent engineer familiar with Fleck valves.

Water Maker

The yacht is equipped with two water makers (**IDROMAR MC5S type**) which delivers up to 500 l/h of drinking water, allowing fresh water availability also during long cruises.

The water makers are located on the port side of engine room connected by means of filter and check valve to the sea water main services manifold.



To energise supply the water maker, operate the magneto-thermal switch located on the main switchboard, 230Vac equipment section. The machinery includes the control panel which allows checking the system conditions and setting various functions.

The fresh water produced by the water maker satisfies all physical, chemical and bacteriological requirements, therefore it can be used for cooking.

The water produced by the water maker supplies directly the fresh water tank through dedicated piping, ensuring continuous availability of fresh water.

If the yacht is not used for a long time, close the sea chest valve and rinse the machinery internal circuit to avoid corrosion caused by the saline. For this purpose the water maker is equipped with direct connection to the fresh water system and/or mobile connection for machinery internal washing after operation.



It is advisable to start the water maker system only while the sea water chest valve is open.

The water maker must not be put into operation in ports or near coasts with high population density, this to avoid clogging the osmotic membrane prematurely and sometimes irreparably. It is advisable to operate the plant offshore or during navigation, to avoid frequent replacement of the pre-treatment filters since, even if the water is clear, being close to the coasts there is a greater concentration of colloidal parts and particles in suspension.

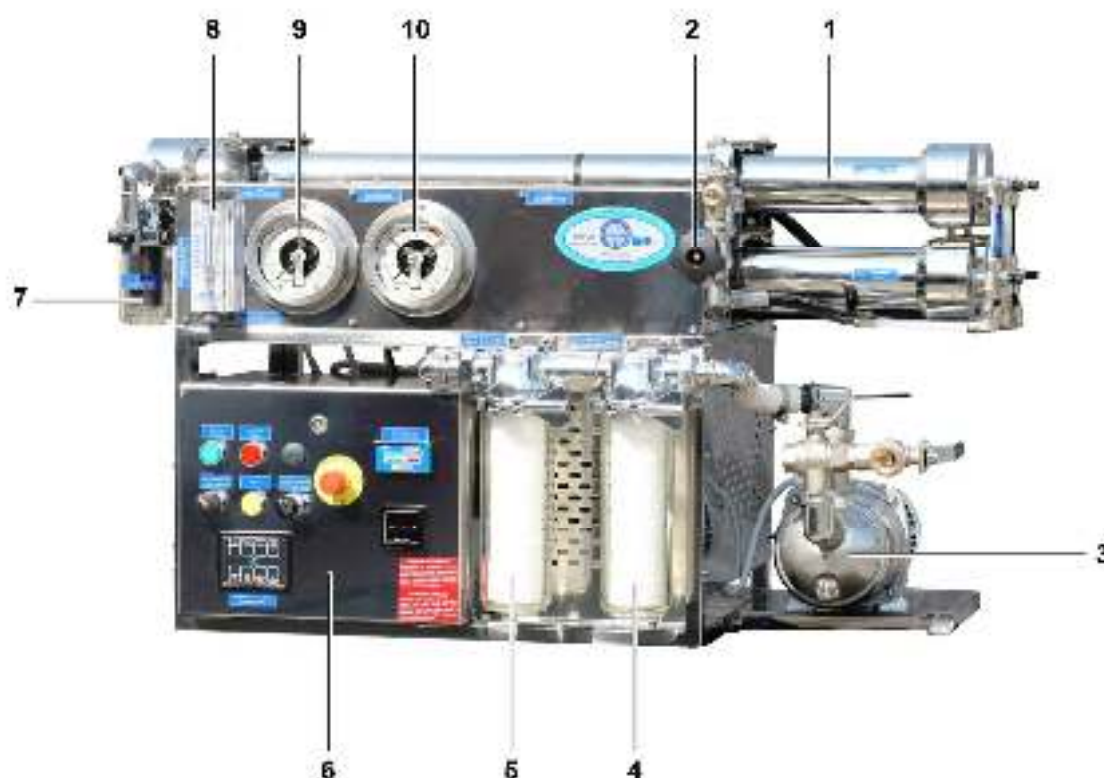
*When manoeuvring, where anchor weighing causes seabed sand movement, **the water maker system must be stopped.***

If the system remains stopped for more than 5 days, it is necessary to wash it with fresh water for 15 minutes.



For correct operation of the water maker and for all procedures concerning trouble check, maintenance, alarms, end-of-season regeneration cycle end washing and winter storage, please refer to the Manual supplied by the Manufacturer.

Main Components



- 1) Membrane Group
- 2) Pressure Regulation valve
- 3) Low Pressure Pump
- 4) Filter 5 micron
- 5) Filter 20 micron
- 6) Electric Box
- 7) Carbon Filter
- 8) Flowmeter
- 9) Low Pressure Manometer
- 10) High Pressure Manometer

ELECTRIC BOX

- START button (green)
- STOP button (red)
- Low Pressure Pump Manual Selector
- High Pressure Pump Key Selector
(Key inside the Panel – ONLY EMERGENCY)
- RESET button (yellow)
- Salinometer
- Alarms
- Hour Counter

1. Position the valve on the seawater intake.
2. Press the green START button.
3. Check that the low pressure gauge is about 3 ATE.
4. After 30 seconds the high pressure pump will start; at this point, increase the pressure until it reaches 60 BAR by turning the pressure adjustment valve clockwise. The 2 red hands are the minimum and maximum levels of the alarms.
5. At this point the system is fully operational; then check the production on the flowmeter and the salinity on the salinometer (green led salinity correct - red led high salinity).

1. Power up the system, the red loop will light up.
2. Place the 3-way valve V1 in the sea intake position. Make sure that the RG1 regeneration valve lever is in the correct position. (in the direction of the arrival of the flow).
3. Start the system using the start button, wait a few seconds for the pressure to stabilize in any case above 10 Bar to prevent the low pressure alarm from intervening, and bring the pressure to 60 ATE closing very slowly, clockwise, the needle valve.
4. From this moment the system is operational for as long as you wish. If the system stops, the yellow reset light will turn on. To restart, press the RESET button. Normally this alarm occurs when there are anomalies in the pressure (either too low below 10 Bar, or too high above 70 Bar).

It may happen that the first water produced has a salinity higher than 800 PPM and therefore it will be automatically discharged by the solenoid valve, but this only for a few minutes. The solenoid valve automatically drains the water produced when the red LEDs of the salinometer light up. When the green ones are turned on, the water produced is automatically sent to the storage boxes. Before stopping the production cycle, slowly lower the pressure to about 24 Bar. After lowering the pressure, the system can be stopped electrically by pressing the production stop button

It is important to carry out the final washing after each production.

If during production you want to taste the water produced, you need to take the water by opening the regeneration valve installed after the activated carbon filter.

ALARMS

- If the low pressure alarm light comes on, restart the unit by pressing the yellow RESET button and increase the pressure until it passes the red needle on the meter.
- If the high pressure alarm light comes on, check the pressure with the pressure adjustment valve, perhaps too closed; turn it anticlockwise and bring the pressure to 60 Bar.
- If during the start-up of the system the red LEDs of the salinometer light up and the production is discharged overboard this is normal in fact after about 30 seconds as soon as the water is of good quality the green LEDs will light up and the water produced to the tanks.

Ordinary Maintenance

1. If the system is not used, it is necessary to perform a weekly wash in fresh water to wash the membranes as inside the membranes the fresh water from the last wash tends to stagnate and therefore a bacteriological proliferation occurs. Performing a new washing, clean water is sent into the membranes and this serves to keep them always efficient.
2. Check the cartridge prefilters periodically. If their appearance is dark in color it is necessary to replace them.
3. Check the hours of operation using the hour counter installed on the electrical panel and every 500 hours of operation replace the oil in the high pressure pump.
At the rear of the high pressure pump there is the drain valve; therefore it is necessary to connect a rubber hose to the valve hose connector and then open it to eliminate the oil. To facilitate emptying, remove the level plug. The content is about 500 grams. After emptying, close the drain valve and fill the pump from the hole where the level plug is inserted.

Special Maintenance

At the end of each season it is necessary to carry out maintenance on the membranes.

This maintenance consists in washing the membranes with 2 different products.

The first step is carried out with the powder product which serves to descale the membranes.

The second step is carried out with the liquid product which is used to sterilize and keep the membranes sterilized.

Dissolve the contents of the bottle containing the CTR powder in a clean bucket by mixing 10 liters with fresh water until completely dissolved.

Transfer the CTR product into a 100 liter container of fresh water.

It is important that this container is at least 100 liters as the regeneration takes place in a closed circuit, so if there was a small volume of water there would be overheating and this would be harmful to the membranes.

1. Connect a hose dia. 25mm to the hose connector of the suction valves from the regeneration tank. Place the 3-way valves in the correct position to allow correct suction of the regeneration product.
2. Connect a hose dia. 20mm to the hose connector of the 3-way sea discharge valve and position the valve lever towards the return pipe to the regeneration tank.
3. Connect a tube to the sampling valve by opening the valve and closing the production valve to the storage tank.

At this point, everything is ready to start washing the closed circuit membranes. To start regeneration, press the green start button and stabilize the pressure on the pressure gauge at about 15 ATE via the VRP regulating valve. The duration of the wash is 30 minutes. After this washing, carry out the exact same procedure using the FLD liquid product for the sterilization and maintenance of the membranes. The duration for this second step is 10 minutes.

Membrane Replacement

- To replace the membranes it is necessary to unscrew the screws that hold the membrane cap mounted.
- Remove the cap by pulling it outwards.
- Extract the membrane to be replaced (during this operation, it is important to respect the o-ring position on the membrane).
- Insert the new membrane into the container and put a little silicone grease on both the o-ring and the membrane cap.
- Reinsert the cap and tighten the screws.
- Restart the system and run at a pressure of 15 Bar for about an hour with fresh water.

Membrane Storage Treatment

If the system has to remain inactive for a period of more than 1 month, carry out the wintering and conservation treatment of the membranes.

It is performed exactly like the regeneration with the valves and pipes in the same position. The only difference is the product to use and the amount of water to use.

Pour 100 liters of fresh water into the container and pour the contents of the IDRO55 bottle (liquid product).

Be careful to handle this product, do not smell it closely or rub your eyes if you have wet your hands with the aforementioned product.

Restart the system without changing the pressure and recirculate the product for about 15 minutes in a closed circuit, always at 15 Bar pressure.

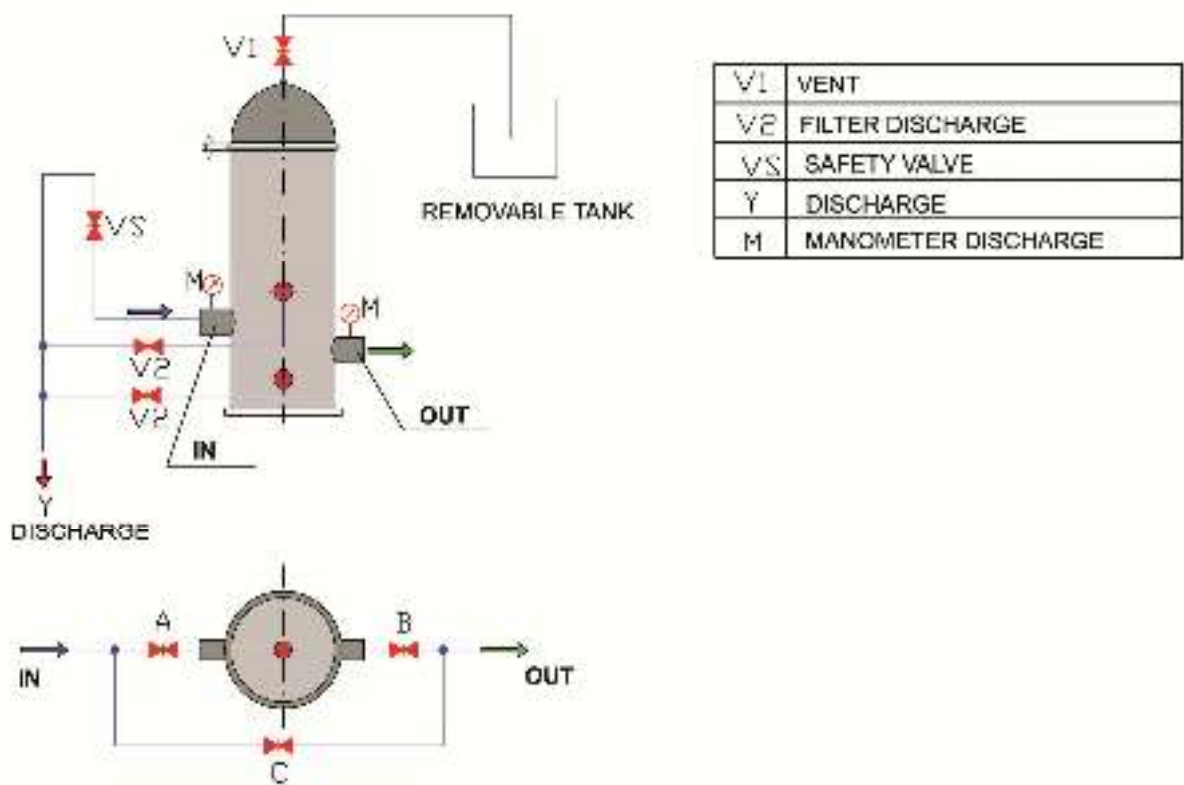
After the time has elapsed, empty the regeneration tank by unloading the product overboard as previously described.

At this point the regeneration and wintering are finished, return the valves to the operating position and remove the hoses used for regeneration. Valve V1 must be held in a horizontal (closed) position.

Activated carbon filter

At the peer, the boat can be hooked up to the dockside water line by using the dedicated manifold connection. The water supplied will go into the structural tank or directly to the various users through an activated carbon filter (4000 l/h production capacity) that will eliminate all impurities, making the water potable and ready to be used.

The filter connection diagram is as follows:



Prior to using the activated carbon filter, read the manufacturer manual that provides all the necessary installation, maintenance and operation information.



The Manufacturer is not liable for any damages to individuals and/or properties due to repair/maintenance performed by non-qualified personnel.

The Manufacturer is not liable for any damages to individuals and/or properties caused by modifications, additional not required accessories or devices of any type installed on the equipment; in particular if they can in any way alter the filter original operation.



The operating pressure shall never exceed the design pressure indicated on the nameplate. Should be expected that the operating pressure may exceed, for any reason, the design pressure, even if for short periods, a safety valve and/or a burst disk shall be installed to protect the equipment.



The operating temperature shall never exceed the design temperature indicated on the nameplate.

The system design temperature and the corresponding gasket chemical with the fluid to be filtered, shall never differ from the characteristics indicated by the gasket Manufacturer.



Prior to initiating any activities on the filter, the operator shall wear adequate protective gloves and clothing, as well as face shields.

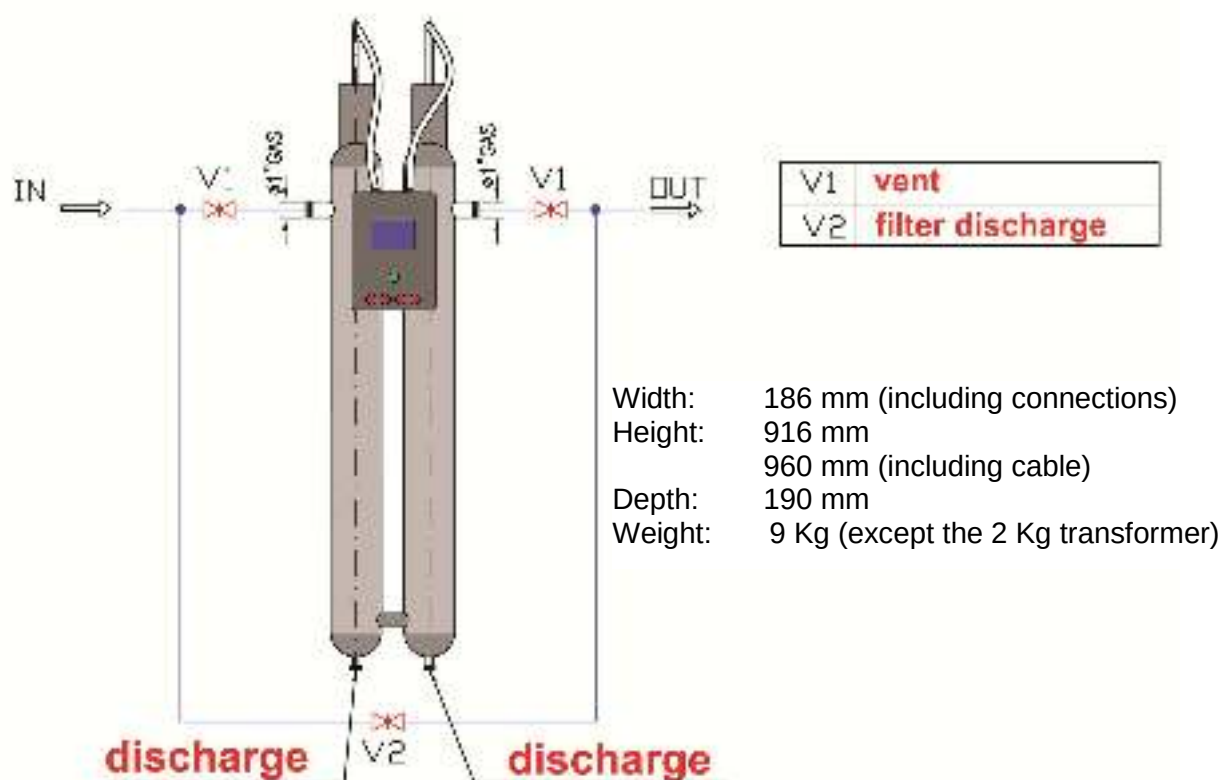
In case of high temperatures, protective gloves and clothing shall be compatible

For maintenance operations concerning the activated carbon filter components do refer to the section "Maintenance" of user's Manual.

UV Sterilizer SH-5000

The freshwater, sucked by the two pumps from the structural tank, prior to be delivered to the boat, goes through a sterilizing filter to remove the bacteria from the pre-filtered water. This device can grant a 99% bacteria content reduction at nominal flow.

The filter connection diagram is as:



The system consists of two horizontal sterilizing chambers. The material of all parts in contact with fluid can be either stainless steel or quartz glass. The chamber contains, longitudinally a quartz protection sleeve that accommodates the 38 W germicide lamp.

The purpose of the protection quartz is to thermally isolate the lamp from the fluid so that the lamp can operate at the optimal temperature conditions.

The germicide lamps are of the mercury vapor, low pressure type, with an emission peak at $2537 \cdot 10^{-10}$ m. wavelength; the lamp casing is designed to absorb the $1800 \cdot 10^{-10}$ m emission peak, preventing the ozone formation in the ambient air.

The sterilizer is powered at 24Vdc by a transformer/power supply to be connected to the 230Vac section of the main switchboard.

The sterilizer system power load is about 80W at nominal voltage, and the output power is about 38W.

The sterilizer system is designed for a maximum flow of 5000 l/h with prefiltered water.

Ambient conditions for the correct operation of the system:

Ambient temperature, minimum :	5°C
Ambient temperature, maximum :	50°C
Water temperature, minimum :	5°C
Water temperature, maximum :	35°C
Relative humidity, maximum :	95%



Prior to using the equipment, read the manufacturer manual that provides all the necessary installation, maintenance and operation information.



During the guarantee period, any repair performed by non-qualified personnel or if non original parts are used will void the guarantee.



The Manufacturer is not liable for any damages to individuals and/or properties due to repair/maintenance performed by non-qualified personnel.

The Manufacturer is not liable for any damages to individuals and/or properties caused by modifications, additional not required accessories or devices of any type installed on the equipment; in particular if they can in any way alter the sterilizer original operation.

Sterilizer signal lights and controls

Switch: Lamp on.

“Line” Led: Green Led, on when the sterilizer is energized.

“Lamp” Led: Yellow Led, on when the lamp is on.

“Alarm” Led: Red Led, on when the lamp is blown or exhausted.



The UV lamp light may cause burns to unprotected skin or eyes. It is recommended not to plug in the equipment prior to installing the lamp inside the housing complete with protection caps.



To clean the sterilizer, do not use corrosive products, acids, scrubbers or steel brushes.

Do not wash down the equipment by using direct water jets or high pressure.



The germicide lamps are fragile and in case of breakage may cause serious injuries; the lamps must be handled with care, and adequate protective wearing gloves. In addition, the lamps contain some mercury and shall be disposed of in compliance with current regulations.



Prior to removing the lamp from its housing, de-energize the equipment at the control panel, to prevent electrical shocks.



In the event of lack of water for several hours, it is recommended to turn the equipment off.

When the sterilizer is off, in the event that the ambient temperature drops below 0°C, it is necessary to de-energize the equipment and drain all water contained inside the chamber to prevent damages from freezing.

In case of a stoppage longer than two days, it is recommended to sanitize the equipment, following a procedure to sterilize the piping (by using a sodium hypochlorite solution, having a quantity and concentration that depends on the size and characteristics of the distribution system).

Assembly sequence

Before installing the equipment it is advisable to remove the quartz pipe (if installed) from the disinfection chamber, in order to avoid accidental breakage.

To disassemble the quartz pipe, proceed as follows:

- unfasten the pressure sleeve, rotating it counter-clockwise until complete extraction;
 - extract the quartz with extreme care and store it in a safe place.
- Caution: sometimes the O-ring tends to adhere to the pipe's external surface.



The quartz pipes are fragile and may cause serious injuries in case of breakage. Therefore they shall be handled with care, protecting the hands with gloves that ensure adequate protection.

Positioning the equipment

- Before installing the equipment make sure that, above it, there is enough space for easily extract the quartz lamp, and in general enough space for maintenance;
- prepare a power socket near the equipment for inserting the transformer/power supply plug;
- make sure that the area is protected against accidental sprays or condense dripping;
- do not place the UV filter in particularly hot, humid and dusty places.



Maintenance easiness and reliability over time of the UV filter depend on a correct and rational positioning.

Water connections

Raw water inlet and treated water outlet connections are composed of two male threaded sections.

A sleeve, closed by a plug and used for complete emptying, is placed below the disinfection chamber.

To ensure a correct water connection, proceed as follows:

- install two check valves on the system, one on the inlet and one on the outlet;
- it is advisable to install a bypass that allows shutting down the system for maintenance while keeping the water flowing;
- it is advisable to place an anti-vibration joint between system's inlet and outlet piping: any vibrations on the system may cause breakage of the quartz pipe and/or compromise the pressure sleeve seal;
- do not place the UV filter in particularly hot, humid and dusty places.



The UV filter shall never be subject to mechanical stress from the piping.

Quartz assembly

Once completed the water connection, reinsert the quartz pipe inside the disinfection chamber.

Pay extreme attention while reinserting the pipe, in order to avoid accidental breakage.

Place the vaseline-lubricated O-ring outside the quartz.

Insert the pressure sleeve and screw it clockwise without forcing.

System pressurization

After having inserted the quartz, pressurize the system, slowly and with great care, making sure that no leakage occurs.

Lamp assembly in the UV filter

Insert the germ-killer lamp inside the quartz protection pipe.

Install the four-pole connector coming from the control switchboard to the lamp's head.



Germicide lamps are frail and, in case of breakage, can cause serious injuries; therefore they shall be handled with care, protecting your hands with gloves that ensure appropriate protection.

They contain appreciable quantities of mercury and shall be disposed of according to applicable regulations.



Avoid touching the lamp's quartz with your fingers, in order to prevent leaving greasy traces; if required, clean with a soft cloth and alcohol.

Screw the closure ring nut on the pressure sleeve.

Screw the cable gland cap on the closure ring nut, making sure not to twist the cable.
Tighten the cable gland cap, still making sure not to twist the cable.

Connection to power network

The UV filter requires 220V power supply at 50/60Hz.

The UV filter shall be connected inserting the transformer/power supply plug into the power supply socket.



It is not allowed to extend or replace the cables between power supply, UV filter and lamp supplied with the system with cables of different types.

While the transformer is inserted in the socket, the UV filter is energized; therefore the green LED shall illuminate.

Setting the switch placed on the ballast to “ON”, the lamp illuminates; while the lamp is illuminated, the yellow LED on the ballast shall illuminate.

The can remain always on with no problems for the UV filter.

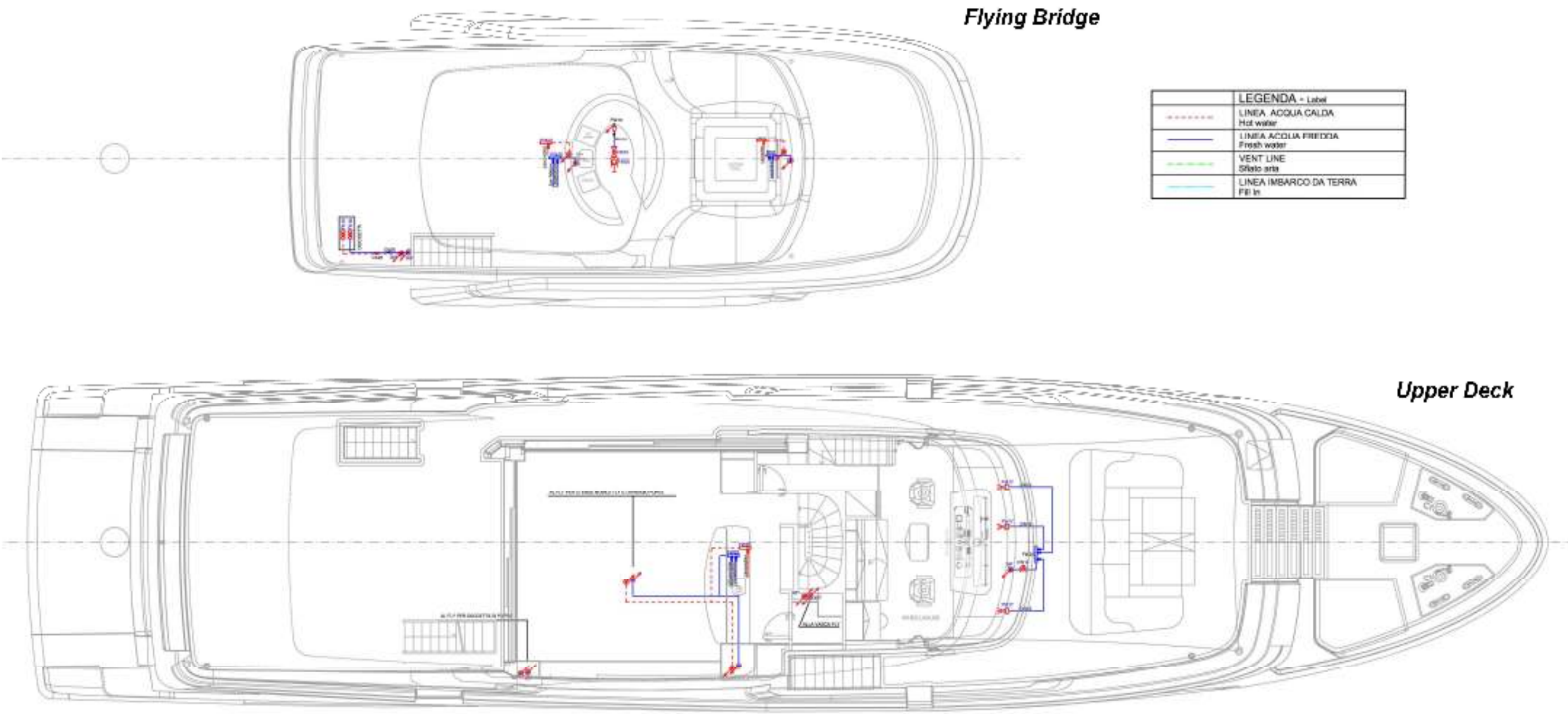


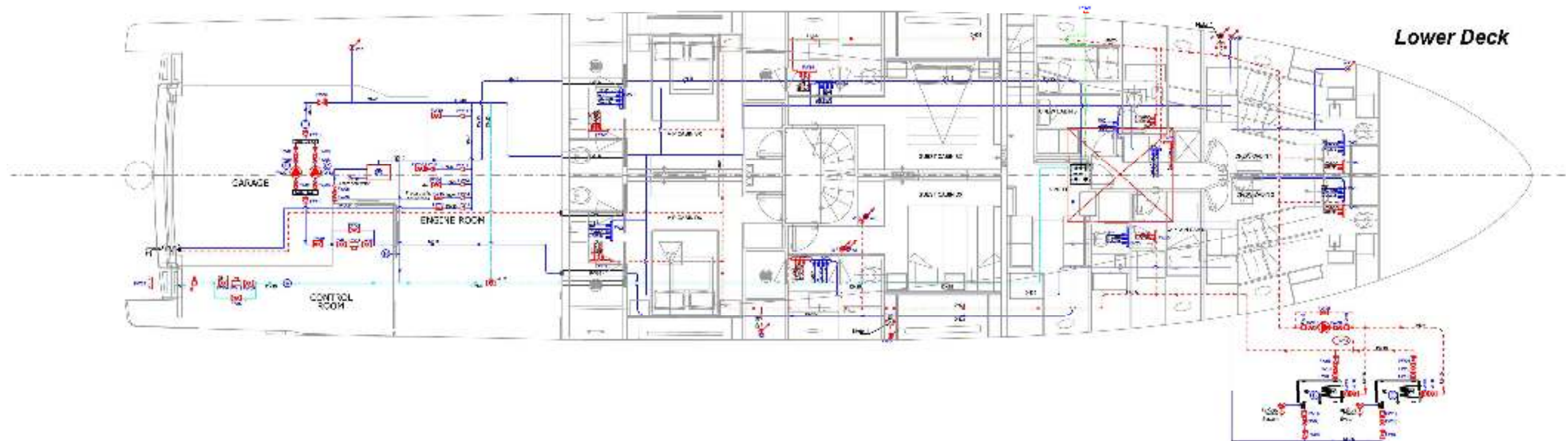
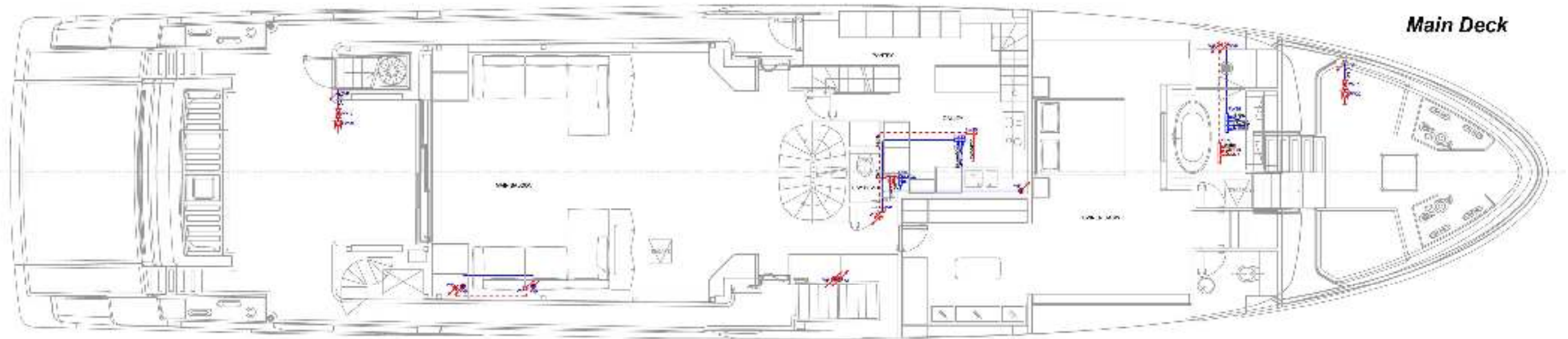
Before extracting the lamps from their seats or inspecting them, always disconnect the power supply from the switchboard.



The UV rays of the lamps' light may cause serious burns to unprotected skin and eyes, therefore avoid looking directly at the operating lamps, and use goggles specifically suitable for protecting against ultraviolet rays.

Fresh Water System - Flying Bridge / Upper Deck



Fresh Water System - Main Deck / Lower Deck

Fresh Water System - Details

SIMBOLO SYMBOLS	MARCA LABEL	Q.TA QTY	D. DESCRIZIONE DESCRIPTION
	FW01 FW02	1	VALVOLA A 3 VIE A "L" 3 Way Valve "L" Type DN 25 VALVOLA A 3 VIE A "L" 3 Way Valve "L" Type DN 32
	FW03	1	IMRARC0 Filter
	FW04	1	PASSAGGIO STAGNO A PARATIA/PONTE Waterlight Passage
	FW04	1	REDUTTORE DI PRESSIONE DN25 Pressure Reducer DN25
	FW05	11	PASSAGGIO A PARATIA Bulkhead passage
	FW06	8	VALVOLA DI NON RITORNO DN 25 Net return Valve DN 25
	FW07	2	VALVOLA DI NON RITORNO DN 40 Net return Valve DN 40
	FW08	22	VALVOLA A SFERA DN 25 Ball valve DN 25
	FW09	5	VALVOLA A SFERA DN 40 Ball valve DN 40
	FW10	3	VALVOLA A SFERA DN 15 Ball valve DN 15
	FW11	1	FILTRO A GEFTELLI DN 25 Strainer DN 25
	FW12	1	FILTRO A GEFTELLI DN 40 Strainer DN 40
	FW13	1	POMPA 230/400V DN25 Fresh Water Pump 230/400V DN25
	FW14	6	REDUTTORE DI DIAMETRO
	FW15	1	VALVOLA A SFERA CON TAPPO DN 25 Ball valve with cap DN 25
	FW16	1	ELETTROVALVOLA PER LAVAGGIO VETRI Solenoid Valve
	FW17	3	UGELLO LAVAVETRI Nozzle
	FW19	12	GIUNTO FLESSIBILE NP16 Flexible Coupling NP16
	FW20	1	ATTACCO RAPIDO PER LAVAGGIO PONTE Quick connection for cock washing
	FW21	2	VALVOLA DI SICUREZZA Safe valve
	FW22	1	POMPA SECONDARIA 24V DN25 Secondary Fresh Water Pump 24V DN25
	FW23	1	VASO DI ESPANSIONE 20 LI Expansion box
	FW24	1	SFIATO CASSE Tank Air Vent
	FW25	21	CLARINETTO DI DISTRIBUZIONE ACQUA Water supply manifold
	FW25	1	DOCCIA ESTERNA Outdoor shower

MARCA Label	Q.TA' Qty	DESCRIZIONE Description	TIPO Type	GARANTIA Warranty
	1	SERBATOIO CENTRALE ACQUA DOLCE Fresh Water Tank		
	2	BOILER ORIZZONTALE Boiler	GIANNESCHI	200 LT
	1	GRUPPO ACQUA CON ESEPP 230V Water pressure system		
	1	GRUPPO ACQUA CON ESEPP 21V Water pressure system		
	1	DISSALATORE Watermaker		
	1	ADDOLCITORE Water pressure system		
	1	TRATTAMENTO ACQUA UV UV Sterilizer		
	1	STAZIONE DOSATRICE IONI ARGENTO Silver ion dosing unit		

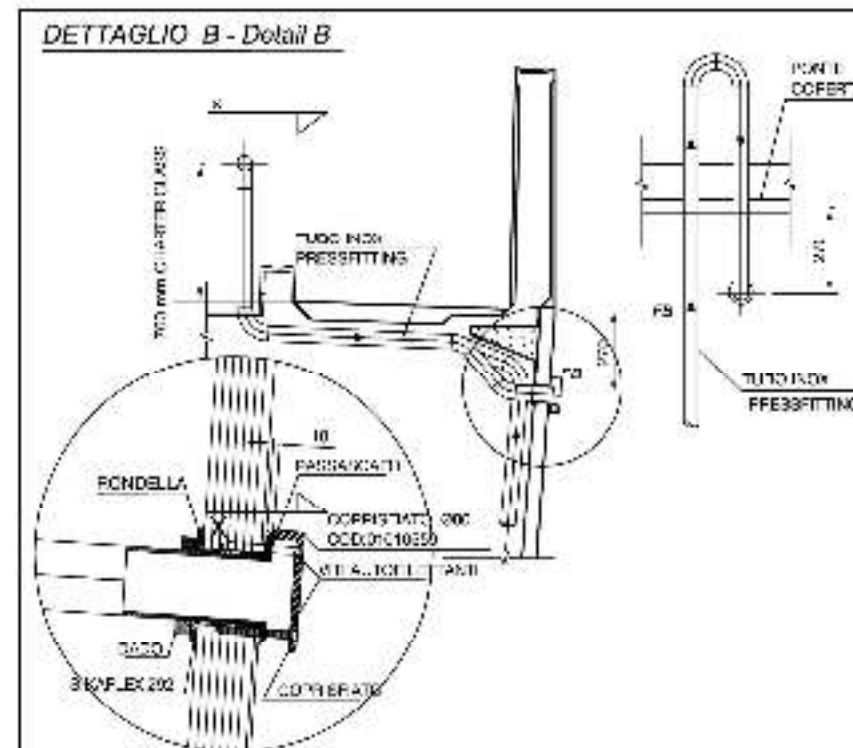
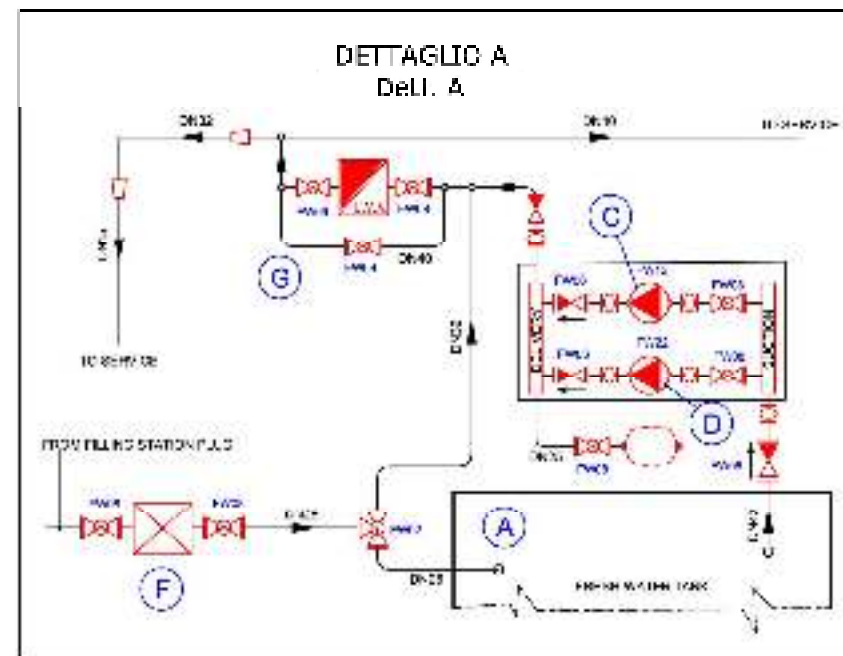
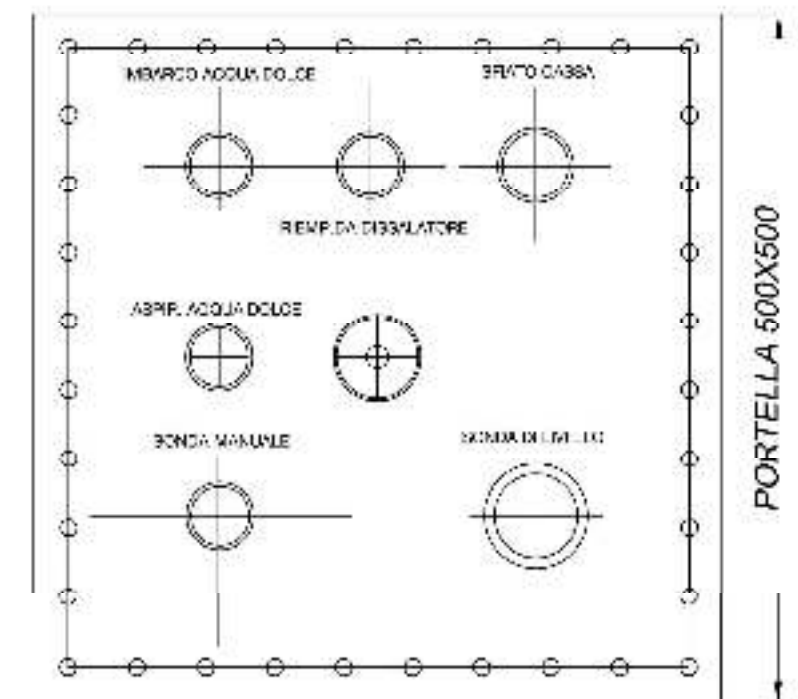


TABELLA PRESSIONI PRESSURE TABLE	
PRESSIONE DI ESERCIZIO Working pressure	3.3 BAR
PRESSIONE DI PROVA Pressure test	4.2 BAR
PRESSIONE NOMINALE Nominal pressure	5.0 BAR

NOTE GENERALI
GENERAL NOTES

- | | |
|-----|---|
| 1) | TUTTA LA SULLUMINER SARA' IN AISI 304, USARE NOMINALE "E" FLANGE 252. ALTERNATIVE IN AISI 316, "FLANGE 242" can be used on approved connections. |
| 2) | I GIUNTI DI ISOLAZIONE DI GIUNTA AVENDO LA FLANGIA DIA. PH. 15 E SARANNO DI TIPO APPROVATO.
FLANGED EXPANSION JOINT IN 1.5 IN. LINE USED (OF APPROVED TYPE) |
| 3) | TUTTE LE MACCHINE ROTANTI SARANNO MONTATE SU ALTERNATIVI E CONNESSE ELASTICAMENTE ALLA RELATIVA TUBAZIONE.
All rotating machinery will be on resilient mounting and connected with elastic joint to the pipe system. |
| 4) | LA CONTINUITA' ELETTRICA DOVRA' ESSERE MANTENUTA LUNGO TUTTA LA TUBAZIONE E RA QUESTA E LA TUBAZIONE A SEPARI.
Electrical drainage should be provided for all metal piping and from this to null connection. |
| 5) | PER L'INSTALLAZIONE DEI COMPONENTI FAR RIFERIMENTO AI MANUALI D'INSTALLAZIONE E ALLE INDICAZIONI DEL FORNITORE DEI MACCHINARI.
During the assembly phase to be referring to the construction instructions. |
| 6) | OPPURE I TUBI FLESSIBILI SARANNO DI TIPO RESISTENTE AL FUOCO, IN ACCORDO ALLA ISO 15549/15541 E CERTIFICATI DAL FORNITORE ADATTI ALL'USO SECONDO GLI STANDARD NATIONAL AND INTERNATIONAL.
LE TUBAZIONI FLESSIBILI DEVONO ESSERE "FIRE" COMPATIBILI.
Any flexible hoses should be fire resistant, in compliance with ISO 15549/15541 and certified suitable for use by the manufacturer in compliance with national or international standards. Flexible hoses are to be as short as possible. |
| 7) | A SCOPERTA DEL COLLO DELLA TRATTORIA DEL CANTIERE LA CASSA ACQUA DOVE' DOVRA' ESSERE POSIZIONATO AD UNA QUOTA SUPERIORE DI 750 MM RISPETTO AL PUNTO DI COPERTA.
The top of the trap, for fresh water tank air vents, should be 750 mm above the deck. |
| 8) | A VALVOLA TRE VIE DOVRA' AVER LA LIA DI IMBACCO DOVE' ESSERE INS. ALLA LA IN ZONA FACILMENTE RAGGIUNGIBILE.
CONDIVIDERE A BORDO IL PENSIZIAMENTO CON L'UFFICIO TECNICO.
The three-way valve on the filling pipe have to be installed in easily accessible position. The valve position must be shared with the technical department. |
| 9) | IL RISSAGGIO DELLE STAFFE DELLE TUBAZIONI DOVRA' AVVENIRE MEDIANTE INCOLLAGGIO.
The bracket of piping must be bonded. |
| 10) | I COLLETTORI DOVRANNO ESSERE INSTALLATI IN ZONE FACILMENTE ACCESSIBILI.
Manholes have to be installed in easily accessible position. |
| 11) | TUTTA LA SULLUMINER SARA' IN AISI-304, USARE NOMINALE "E" FLANGE 252.
All parts are in AISI 316, "FLANGE 242" can be used on approved connections. |
| 12) | DOVE NECESSARIO LE GIUNZIONI SARANNO IN EPDM.
Where necessary rubber gasket should be in EPDM. |
| 13) | LE TUBAZIONI DELL'ACQUA CALDA DOVRANNO ESSERE COIBENTATE.
Hot water piping will be insulated. |



Grey Water System

A 3,000 lt structural tank is located on bottom hull aft for grey and black piping systems. Such tank is equipped with an steel inspection hatches, one of them is equipped with:

- air vent pipe;
- grey water delivery line;
- black water delivery line;
- sensor level;
- washing tank connection.

The grey water system (showers, sinks, bidet) is a gravity hydraulic system.

For galley appliances is allowed the chance to discharge directly outboard or in tank by means of three ways valve (using dedicated SANISPLIT).

However the users' discharges are equipped with hydraulic siphon in order to avoid odours from pipe and SANISPLIT.

The flybridge sink are directly discharged outboard.

A smell filter (MATRIX) is installed, for this reason, on air vent the tank, before of discharging outboard.

N°4 osculated drain boxes are installed throughout the boat, dedicated to collecting the condensate of the various Fan Coils. These then discharge directly overboard through a special filter and dedicated valve.

The collection boxes are located on the bottom and exactly:

- V.I.P. cabin starboard side;
- V.I.P. cabin port side;
- CAPTAIN cabin starboard side;
- LAUDRY.

On board, in the double bottom, 7 SANISPLIT are installed in easily reachable and interlocking areas. Their function is collecting and sending to tank the discharges of following areas:

- V.I.P. cabin starboard side (sink discharge, shower and bidet);
- V.I.P. cabin port side (sink discharge, shower and bidet);
- GUEST cabin starboard side (sink discharge, shower and bidet);
- GUEST cabin port side (sink discharge, shower and bidet);
- CAPTAIN cabin starboard side (sink discharge, shower and bidet);
- LAUDRY (sink discharge, shower);
- CREW cabin starboard side (sink discharge, shower).

SANISPLIT is an automatic soil and waste water pumping unit, which collects the water from the kitchen and bathroom and pumps it into the collection tanks, even without the necessary gradient. SANISPLIT includes a powerful and silent centrifugal pump controlled by a pressure switch, which starts the pump as soon as the water arrives in the unit's collection tank.

The SANISPLIT pump has a delivery of 350 lt./min. and is capable of pumping soil and waste water to a height of up to 11 metres and a distance of 90 metres.

SANISPLIT also features the "rapid disengagement" of the removable part of the unit, containing the pump-pressure switch assembly, from the fixed part, containing the collection tank and to which the piping is connected. This ensures that the maintenance of the system is simple, quick and clean.

SANISPLIT also incorporates a check valve preventing the pumped water from returning into the unit. There are no leaks from the pipes during the disengagement of the removable unit.



Warning: in order to allow a correct operation of siphon of sinks, showers and bidet, during no operation, open taps of all users.

Do not throw nylon or other non-biodegradable products into the toilet.

The grey/black water collection tank should be cleaned and disinfected frequently in order to ensure good performance and avoid formation of sediments and deposits.

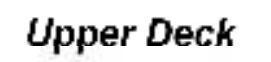
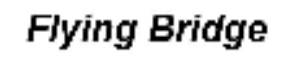


The possibility of discharging galley directly outboard, depends on local regulations.

Avoid, if possible, the discharge of grey/black waters in harbour, inside protect areas or nearly of beach; discharge only after having verify the position of the boat.

Before of coming back at harbour, check the level tank, during back navigation and, if necessary, discharge the tank at open sea, verifying the correct position of the boat.

For maintenance operations concerning the grey/black water system components do refer to the section "Maintenance" of this Manual.



Grey Water System - Main Deck / Lower Deck

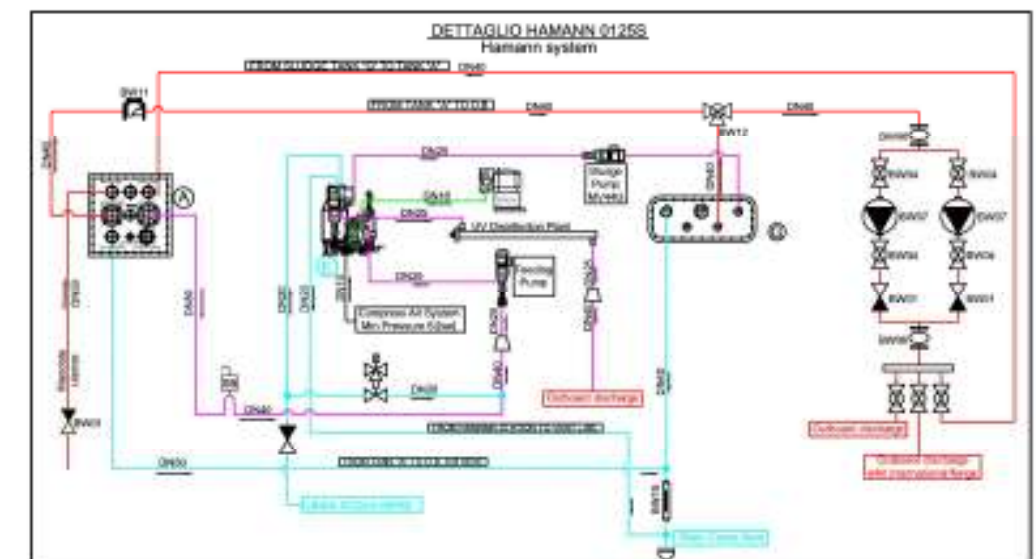
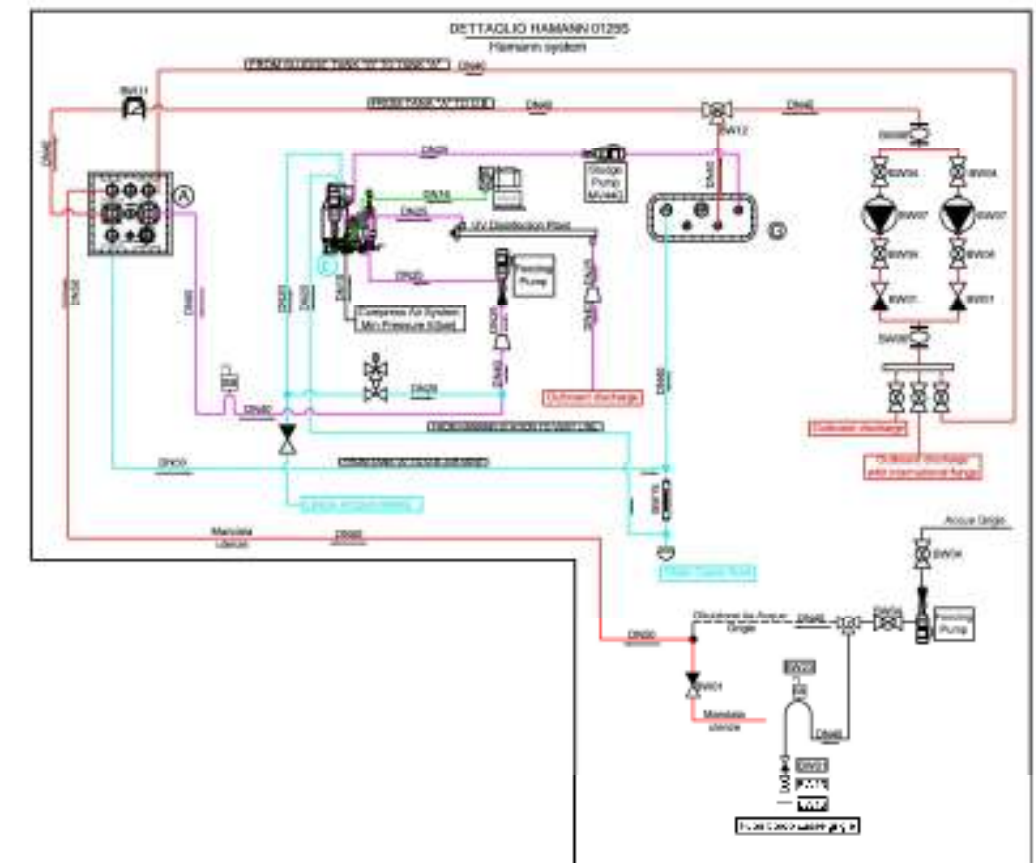
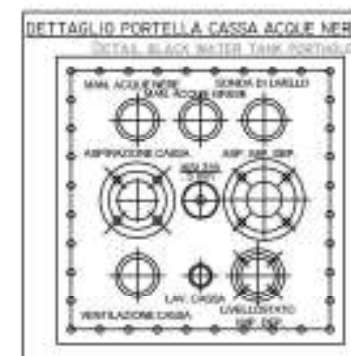
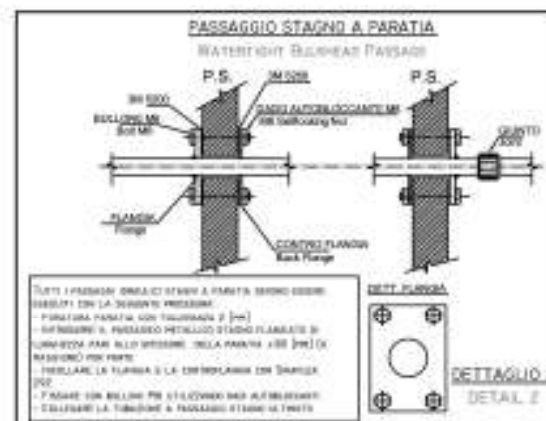
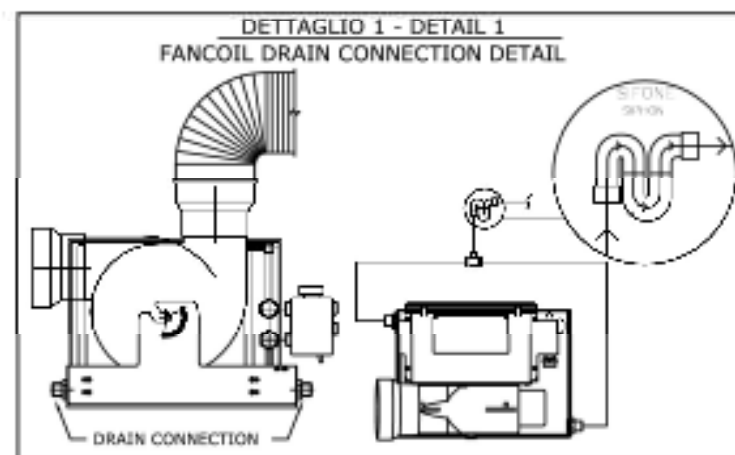
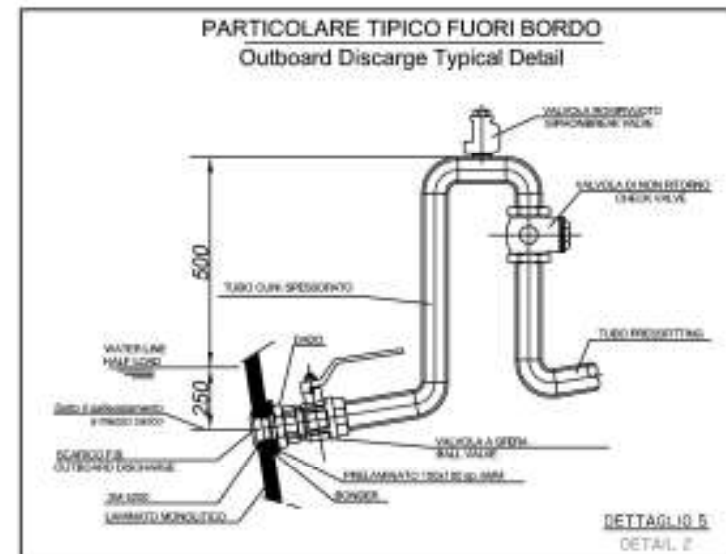
Grey Water System - Details

SIMBOLO SYMBOLS	MARCA LABEL	Q. TA Q. TY	DESCRIZIONE DESCRIPTION
	GW01	2	VALVOLA A 3 VIE A "L" 1/2" Way Valve "L" Type DN 25
	GW02	2	SCARICO FUORI BORDO/Outboard discharge DN 40
	GW03	2	SCARICO FUORI BORDO/Outboard discharge DN 25
		vedi dwg	PASSAGGIO STAGNO A PARATIAPONTE Watertight Passage
	GW04	3	CASSETTA OSCULATI DI RACCOLTA DRENAGGI R.1 L/1 Osculati's Drain Box
	GW05	vedi dwg	PASSAGGIO A. PARATIA Bulkhead passage
	GW06	4	SANISPLIT NK3 230V Sanisplit NK3 230V
	GW07	vedi dwg	RIDOTTORE DI DIAMETRO

LEGENDA LABEL	
	DRENAGGIO FANCOIL Drain fancoil
	LINEA ACQUE GRIGIE Black water line
	SPRATO ARIA Vent Line
	SCARICO CASSETTE WHALE Discharge line

NOTE GENERALI
GENERAL NOTES

- | | |
|----|--|
| 1 | INSTALLAZIONE DEI SISTEMI DI TRATTAMENTO DELL'ACQUA ELETTRICI, PER
L'ACQUA DI RIVELAZIONE, ELETTRICI E PER L'ACQUA DI TRATTAMENTO CONTINUO. |
| 2 | PROVA DI TRATTAMENTO DELL'ACQUA DI RIVELAZIONE E TRATTAMENTO CONTINUO
PER VERIFICARE IL CORRETTO FUNZIONAMENTO DEI SISTEMI DI TRATTAMENTO DELL'ACQUA. |
| 3 | PROVA DI TRATTAMENTO DELL'ACQUA DI RIVELAZIONE E TRATTAMENTO CONTINUO
PER VERIFICARE IL CORRETTO FUNZIONAMENTO DEI SISTEMI DI TRATTAMENTO DELL'ACQUA. |
| 4 | LA CONNESSIONE ELETTRICA DOVRÀ ESSERE MANUTENUTA LUNGO TUTTA LA
LUNGHEZZA E TRA QUANTO LA CONNESSIONE ELETTRICA E IL SCUDO
ELETTRICO DEVRÀ ESSERE PER IL CORRETTO FUNZIONAMENTO DEI SISTEMI DI TRATTAMENTO DELL'ACQUA. |
| 5 | PER L'INSTALLAZIONE DEI MACCHINARI FARSI RIFERIMENTO ALLE ISTRUZIONI E
AI MANUALI DEI COSTRUTTORI.
For installation operations refer to constructor manual instructions. |
| 6 | TUTTI GLI SCACCHI PER "GRANDE" DOVRANNO AVERE UN SISTEMA DI
SOSTEGNO EFFICIENTE PER LE SINGOLE VASCE PER LA COLLEZIONE
PRINCIPALE.
All "gravity" discharges must have an efficient water trap system both of single units and
as a sub-system. |
| 7 | REALIZZAZIONE TASCHE DI TRATTAMENTO LUNGO I COLLETTORI
Problems of inspection must be prevented. |
| 8 | USARE DIVERSI TIPI DI IMBALLA PRODOTTORI DI SCHIUMA PER LE VASCE DI TRATTAMENTO
LUNGO I COLLETTORI PER LA TRATTAMENTO DELL'ACQUA DI RIVELAZIONE E TRATTAMENTO CONTINUO. |
| 9 | PREVEDERE UN SISTEMA DI DISCARICO PER I FANGHI, ELETTRICI, E TRATTAMENTO CONTINUO
ELETTRICI, E TRATTAMENTO CONTINUO E TRATTAMENTO CONTINUO E TRATTAMENTO CONTINUO. |
| 10 | PER GLI SCACCHI ELETTRICI DOVRANNO ESSERE PRESENTI LE TECNICHE
PER GLI SCACCHI ELETTRICI E TRATTAMENTO CONTINUO E TRATTAMENTO CONTINUO. |
| 11 | IL SISTEMA DI TRATTAMENTO DELL'ACQUA DI RIVELAZIONE E TRATTAMENTO CONTINUO
DEVE ESSERE PRODOTTO PER IL CORRETTO FUNZIONAMENTO DEI SISTEMI DI TRATTAMENTO DELL'ACQUA. |



Black Water System

A 3,000 lt structural tank is located on bottom hull aft for grey and black piping systems. Such tank is equipped with an steel inspection hatches, one of them is equipped with:

- air vent pipe;
- grey water delivery line;
- black water delivery line;
- sensor level;
- washing tank connection.

TECMA WC system discharge is designed to discharge in grey and black water tank.

N° 2 Electric-Pumps self-priming 230Vac are installed nearly of the structural tank for emptying of grey and black water tank.

Such pumps powered by the main switchboard, are able to discharge tank, when treatment sewage is switched off, by means of actuation of hand valves, in following modes:

- direct discharge outboard by means of hull valve, under water line;
- shore discharge by means of manifold with international connection, located in tender-compartment.

An automatic stop is installed for protection of such EE/PP, in event of operation mode ON forgotten.

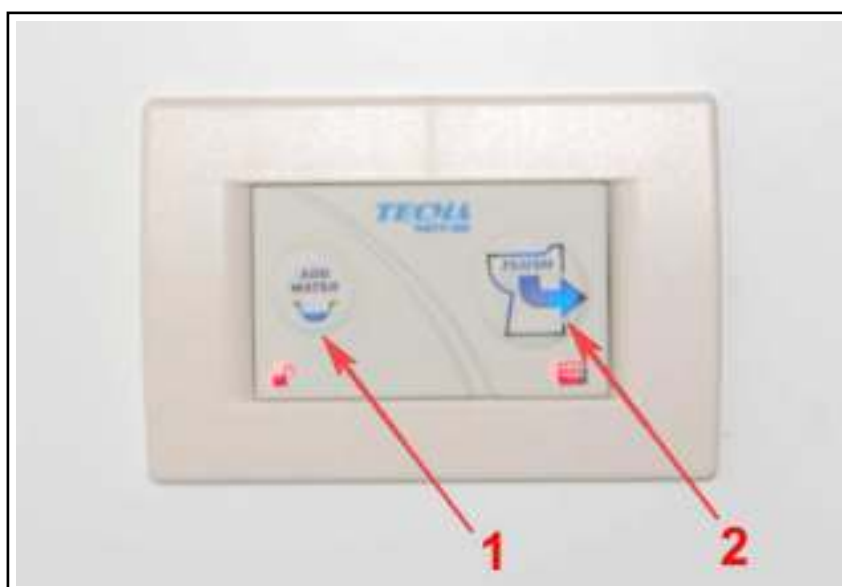
Filling and washing and suction (with fresh water) of W.C. is allowed by means of autoclaves.

WC TECMA discharge is carried out by a macerating pump located on the toilet base, which draws the sewage either into the collection grey/black tank.
Each toilet is supplied with fresh water.

For every each WC the system foreseen interception valve in the way to allow maintenance operation on every each WC without stop the use of the rest of the system.

To control the toilet, operate the switchboard located near the equipment, as follows:

- press the toilet pushbutton (1) “before use”, a certain quantity of water flows into the toilet to wet the ceramic and creating a water bed, avoiding that organic remains dirty the ceramic;
- after using the toilet press the pushbutton (2) “after use”. A greater quantity of water flows into the water ejector supply circuit for emptying the toilet (sewage suction) due to a mincing pump located on the toilet base. A few seconds after suction, new water flows into the toilet in order to fill the siphons and therefore avoiding unpleasant smells;



The advantage given by this W.C. control panel is that the toilet remains completely dry when not used.

If the panel indicates “tank full”, the system (if activated) inhibits toilet operation (and operation of all toilets connected to the same set of collection tanks), by disabling pushbuttons (1) and (2).

The panel rear (which can be reached by disassemble the panel from its compartment) carries an electric trimmer for adjusting the quantity of water necessary for emptying the toilet.



Warning: in order to allow a correct operation of siphon of sinks, showers and bidet, during no operation, open taps of all users.

Do not throw nylon or other non-biodegradable products into the toilet.

The grey/black water collection tank should be cleaned and disinfected frequently in order to ensure good performance and avoid formation of sediments and deposits.



The possibility of discharging galley directly outboard, depends on local regulations.

Avoid, if possible, the discharge of grey/black waters in harbour, inside protect areas or nearly of beach; discharge only after having verify the position of the boat.

Before of coming back at harbour, check the level tank, during back navigation and, if necessary, discharge the tank at open sea, verifying the correct position of the boat.

For maintenance operations concerning the grey/black water system components do refer to the section “Maintenance” of this Manual.

Sewage Treatment System

A system **ECOMAR 45S** (4500 lt/day) sewage treatment system, has been prepared on board to treat sewage prior to discharging it into the sea.

The system is composed by:

- transfer electric-pump;
- seawater suction pump;
- motorised three-way valve;
- treatment unit control switchboard;
- dosing unit comprising:
 - dosing pump;
 - chlorine solution tank;
 - sensor indicating the chlorine solution level in the tank provided for it.



FUNCTION DESCRIPTION

All waste water, such as grey and black water, will flow into a holding tank. When the sewage reaches a designated start-level, the sewage treatment plant will start operating.

Separation and removal

With placement of the suction pipe about 300 mm above the tank bottom, there is space at the bottom of the tank for sludge collection. The sludge can be rinsed and pumped out of the treatment plant when the ship sails in unrestricted waters, which is usually outside the 12 mile zone. This can be done either through flushing out the treatment plant several times or by pumping out the sludge when a sludge pump exists. The sludge can also be pumped to an on-shore facility.

Oxidation

While intaking sewage through the macerator, it will pass an air-vent at the highest point in the piping loop and will be aerated, mixing one third air into the sewage volume.

Enlargement of surface area

Enlargement of the surface area occurs through both the macerator, which cuts all solids into 1mm pieces and a mechanical force created through a strong turbulence, which, when the sewage flowing into the chamber of the mixing and draining pump meets the propellers directing flow in an opposite direction, will further break the organic material into smaller pieces. On the optimally enlarged surface area follows the physical/chemical reactions.

Oxidation and Disinfection

Sewage that is in the reaction chamber of the mixing and draining pump consists of two-thirds sewage water and one-third air plus Sodium Hypochlorite, which was placed by the dosing pump into the water before flowing into the reaction chamber. Through the oxygen (air) and the Chlorine that are then present, organic particles will be oxidized, and bacteria will be destroyed.

Reaction Time

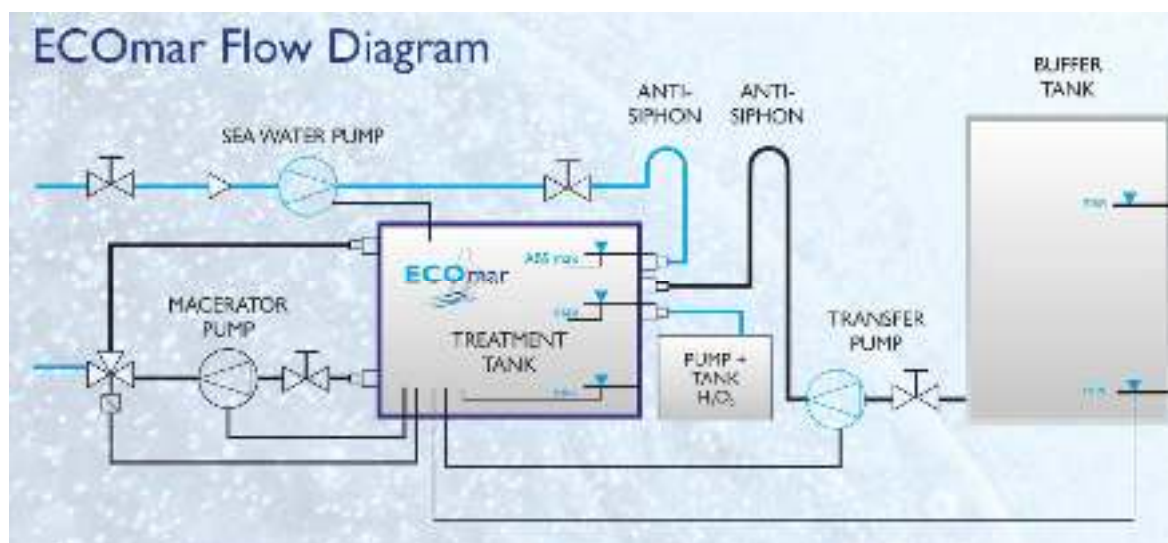
To increase the efficiency of oxidizing organic matter and destroying bacteria beyond the reaction chamber, the reaction time will be prolonged by allowing the sewage to flow into a whirl container. The sewage flowing out of the whirl container will have been treated according to standards and regulations set by IMO, USCG II, Croatian-register, Sweden, Finland and RINA, each of whom have provided Certification. The efficiency of the flow-through-method is dependant upon high quality materials, such as a hypalon stator, tungsten cutting knives, stainless steel cutting rings, stainless steel AISI 316 for all parts in contact with chlorine and sewage, Teflon diaphragms in the dosing pumps, etc.

When a sludge pump exists

The sludge pump serves to insure easier sludge removal as well as to provide additional pumping effort in the case of an emergency or alarm. When the vessel is operating in unrestricted waters, the plant can be switched from TREATMENT MODE (formerly called HARBOUR MODE) over to DISCHARGE MODE (formerly called SEA MODE), so that the tank can be emptied with only the sludge pump.

The term HARBOUR MODE that was used in previous versions of manual and at front door labels of the e-box has changed to TREATMENT MODE. The term SEA MODE changed to DISCHARGE MODE which means that sewage will be discharged untreated.

PRINCIPLE FLOW DIAGRAM



LAID-UP TIME AND CLEANING OF THE TANK



If operating with chlorine it is necessary to protect hands with rubber gloves and eyes with glasses. splashes of chlorine should immediately be washed away with a lot of water.

Laid-up time

The treatment plant will have to be flushed and pumped out. The following flanges should be unscrewed in order to drain away any remaining fluids:

- flange between macerator and tank;
- flange between macerator and pump;
- flange between pump and whirl-container;
- flange between whirl-container and outlet.

The pump's screwed sealing plug should be opened.

Dosing Station:

- Dismount the chlorine/dosing tank together with footvalve, empty all chlorine and flush the tank with clear water.
- Mount chlorine/dosing tank, fill with clear water and empty the tank with the chlorine dosing pump for about 5 to 10 minutes manually while plant is in TREATMENT MODE.

Cleaning of the tank

A monthly cleaning of the tank will be required to prevent sludge from either settling in the tank on the side opposite to the inlet or in the suction pipe of the macerator. Consequently, the tank's cleaning-inlet should be connected with a water hose with sufficient pressure. The pump out can be done by either running the treatment plant or using a sludge pump when it exists. The sludge can also be pumped out to an onshore facility or into unrestricted water when out to sea.

Beware of local regulations for discharging as they can differ from IMO/MARPOL regulation!



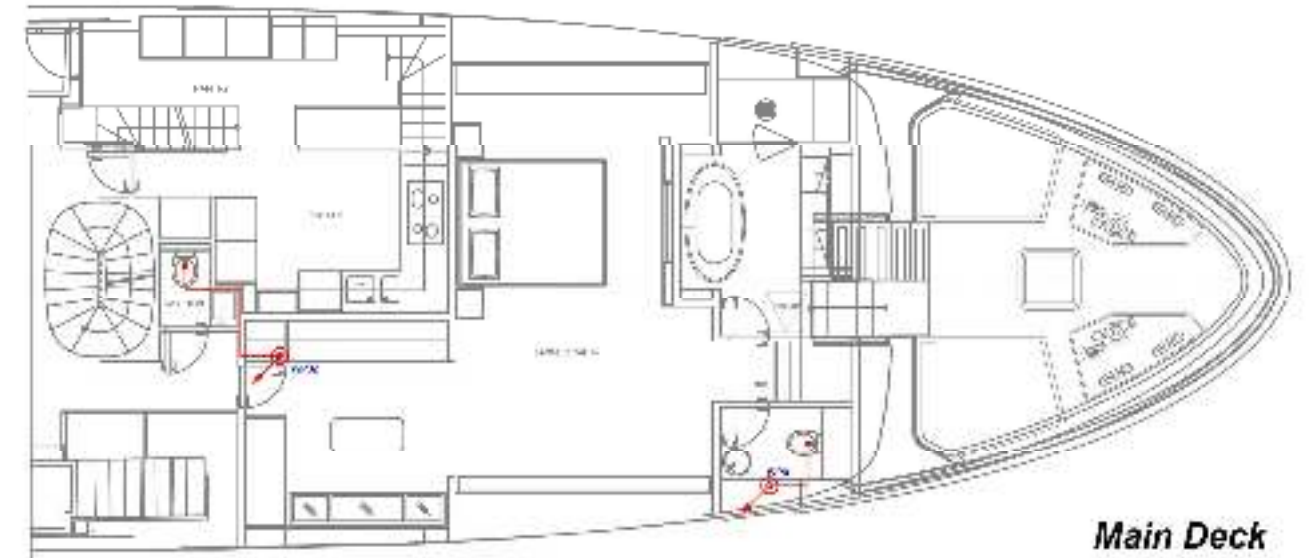
Because of its caustic properties, the chlorine solution can irritate the eye and skin and give rise to breathing difficulties.

Consult the Builder's Manual for the instruction to be followed in the event of accidental contact with said solution.

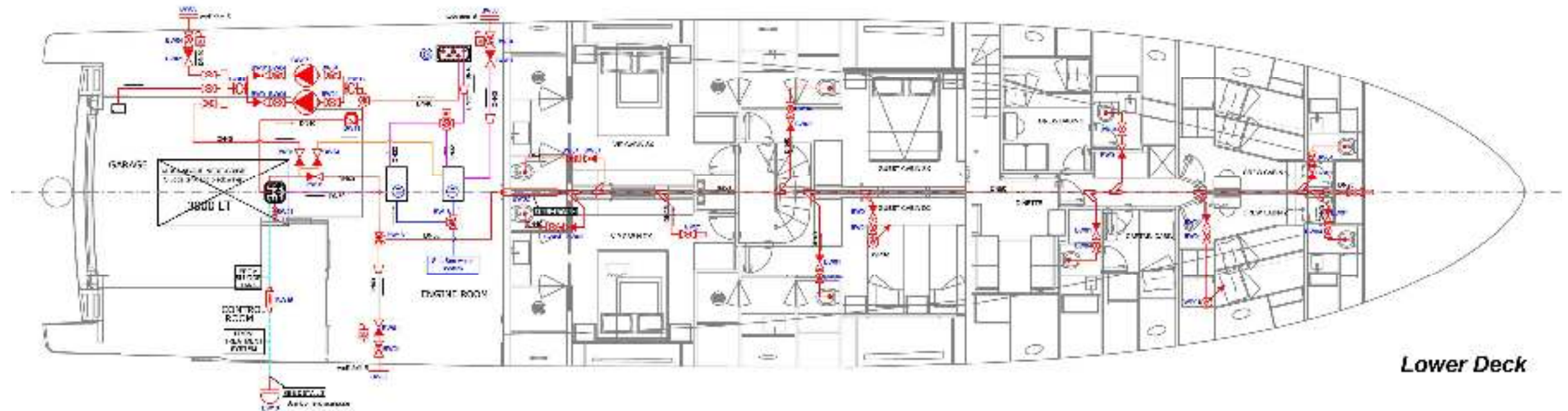
For further details about the system's operation or that of its individual components, consult the Builder's Manual.

Black Water System

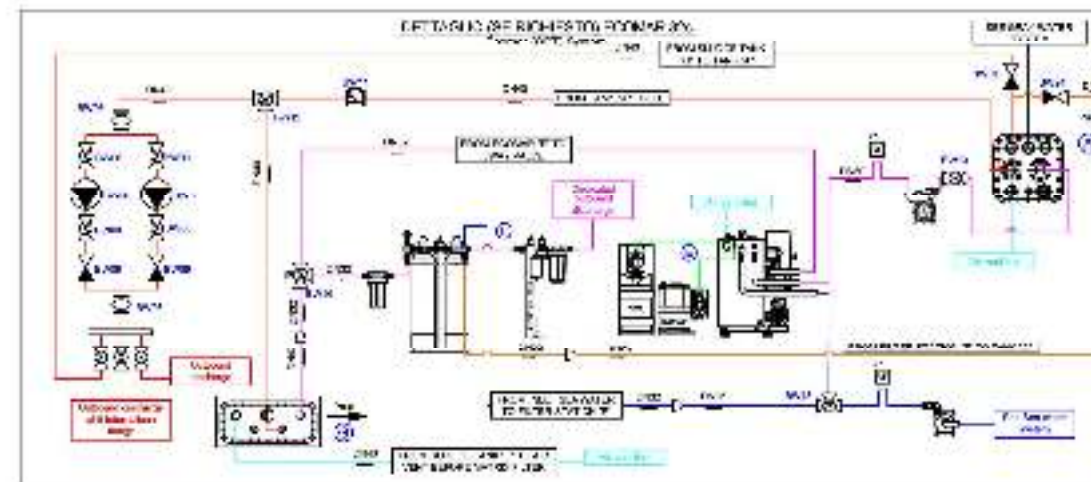
練習問題集 List 1	
	1. $\text{CH}_3\text{COCl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl}$ Acetyl Chloride + Water
	2. $\text{CH}_3\text{Br} \rightarrow \text{CH}_3^+ + \text{Br}^-$ Bromomethane
	3. $\text{CH}_3\text{COCl} + \text{CH}_3\text{OH} \rightarrow \text{CH}_3\text{COOCH}_3 + \text{HCl}$ Acetyl Chloride + Methanol
	4. $\text{CH}_3\text{COCl} \rightarrow \text{CH}_3\text{CO}^+ + \text{Cl}^-$ Acetyl Chloride
	5. $\text{CH}_3\text{COCl} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COOCH}_3 + \text{HCl}$ Acetyl Chloride + Acetic Acid
	6. $\text{CH}_3\text{COCl} \rightarrow \text{CH}_3\text{CO}^+ + \text{Cl}^-$ Acetyl Chloride
	7. $\text{CH}_3\text{COCl} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COOCH}_3 + \text{HCl}$ Acetyl Chloride + Acetic Acid
	8. $\text{CH}_3\text{COCl} \rightarrow \text{CH}_3\text{CO}^+ + \text{Cl}^-$ Acetyl Chloride



Main Deck



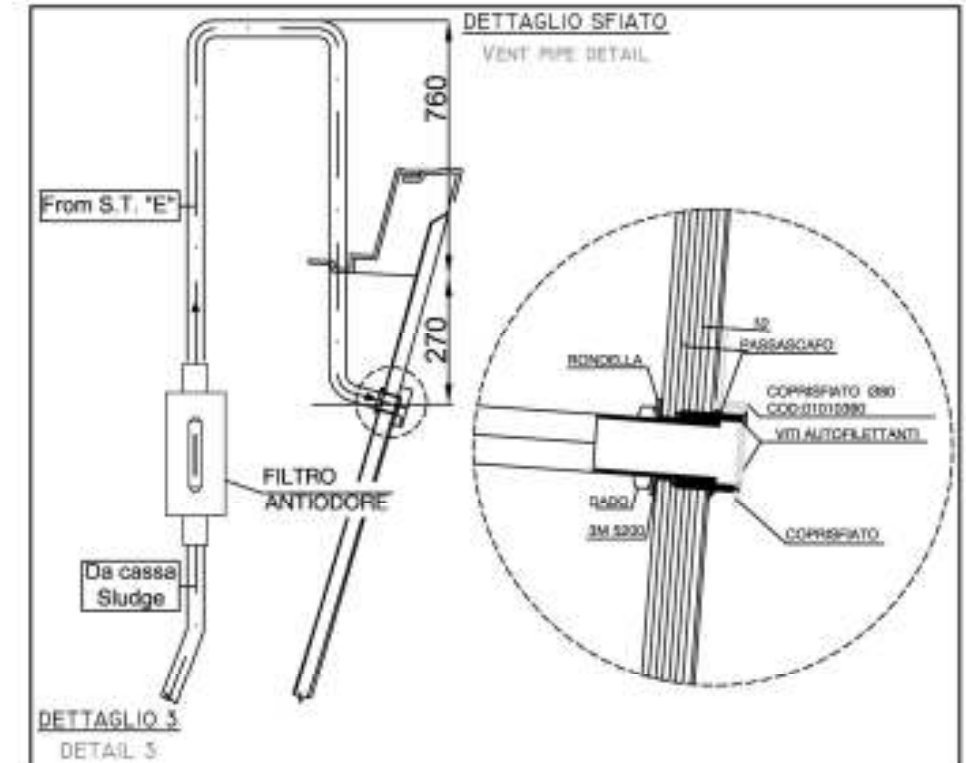
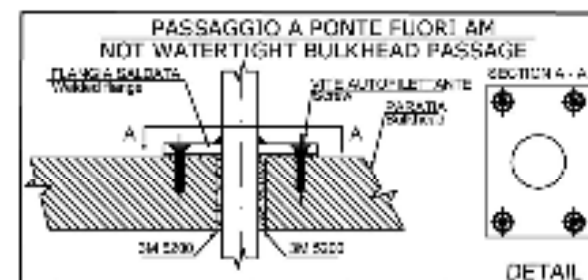
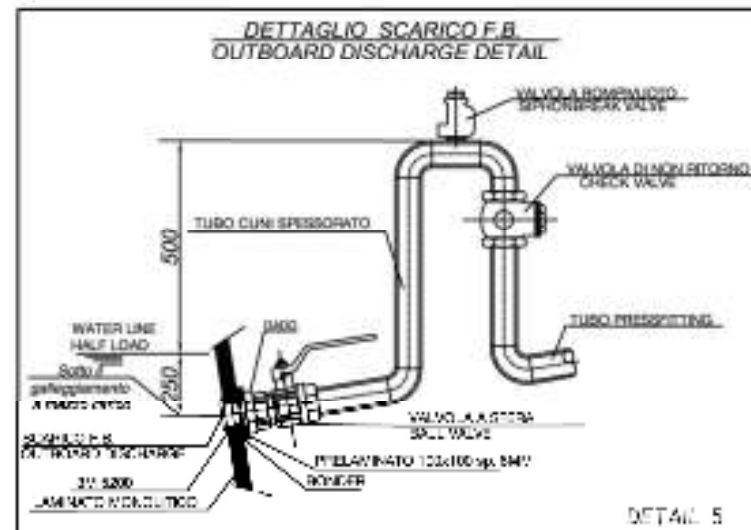
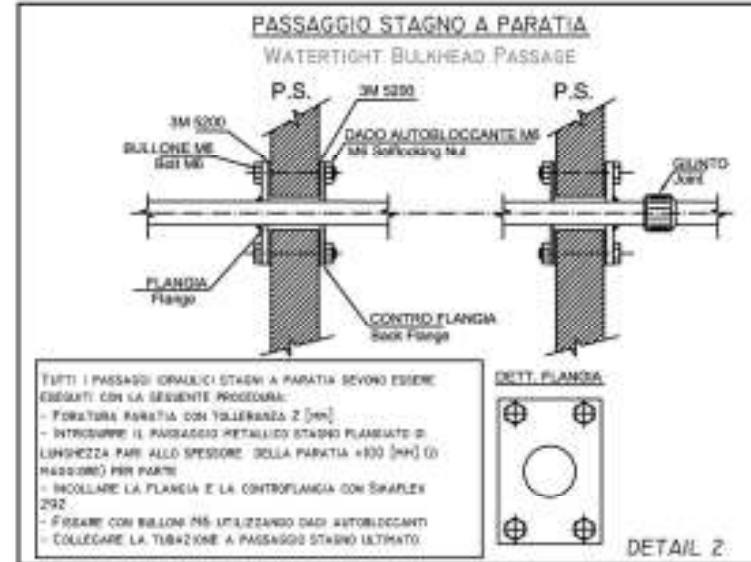
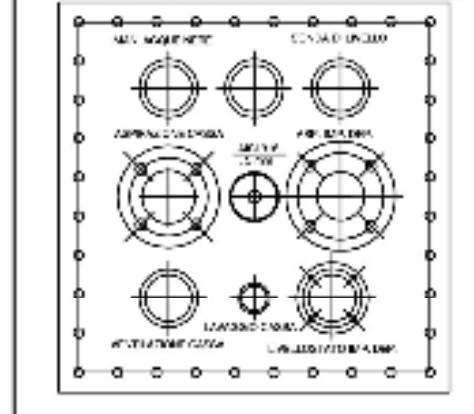
Lower Deck



Black Water System - Details

simbolo symbol	sigla label	codice code	descrizione description	art. art.	materiale material
	BW01	---	VALVOLA DI NON RETORNO DN 40 Check valve NO 40		1X BRONZO 2X OTTONE
	BW02	---	PASSAGGIO PONTE Check Passage		AI81304
	BW04	---	VALVOLA A SFERA DN 40 BALL VALVE NO 40		BRONZO BRONZE
	BW06	---	SCARICO FUORI BORDO DN 40 Outboard discharge NO 40		BRONZO BRONZE
	BW06	---	GIUNTO ELASTICO DN 40 Flexible coupling NO 40		
	BW07	---	POMPA MACERATRICE PARON ATTACCH 1 1/2" 238437 Reducer pump SLUDGE 1 1/2" 238437	VM 60	50/60 Hz
	BW08	---	CONNESSIONE AD Y Y connection		
	BW09	---	LIVELLO ELETTRICO Transducer gray water tank		
	BW10	---	SCARICO SFIATO FUORI BORDO DN 50 Outboard Vent discharge NO 50		OTTONE BRASS
	BW11	---	VALVOLA DI NON RETORNO A PALLA DN 40 Check ball valve NO 40		OTTONE BRASS
	BW12	---	VALVOLA A 3 VIE FLUSSO A 1/2" DN 40 Free way ball valve 1/2" flow NO 40		OTTONE BRASS
	BW13	---	VALVOLA A SFERA DN 50 Ball valve NO 50		PVC PVC
	BW14	---	CONNESSIONE SCARICO ACQUE NERE DN 40 Black water discharge connection 1 1/2" 40		
	BW15	---	FILTRO ANTOIDORE MATRIX SMALL DN 40 Odourlock filter Matrix	MATRIX SMALL	
	BW16	---	SCARICO FUORI BORDO DN 50 Outboard discharge NO 50		BRONZO BRONZE
	BW17	---	VALVOLA A SFERA DN 40 BALL VALVE NO 40		BRONZO BRONZE
	BW18	---	VALVOLA A 3 VIE FLUSSO A 1/2" DN 50 Free way ball valve 1/2" flow NO 50		PVC PVC
	BW19	---	VALVOLA DI NON RETORNO DN 50 Check valve NO 50		BRONZO BRONZE
	BW21	---	VALVOLA A SFERA CON PORTAGOMMA DN 25 Ball Valve no 25 with hose		
	BW23	---	VALVOLA ROMPIVUOTO Siphon Break Valve		
	BW24	---	VALVOLA DI NON RETORNO DN 25 Check valve NO 25		BRONZO BRONZE

DETTAGLIO PORTELLA CASSA ACQUE NERE
DETAIL 1 BLACK WATER TANK PORTICLE



NOTE GENERALI GENERAL NOTES

- 1) TUTTA LA BULLONERIA SARÀ IN AISI-316, USARE FRENAPLETTE "LOCTITE 242".
All bolts are in AISI 316, "LOCTITE 242" is to be used on threaded connections.
- 2) DOVE NECESSARIE LE GUARNIZIONI SARANNO IN EPDM.
EPDM gasket are used where necessary.
- 3) TUTTE LE MACCHINE ROTANTI SARANNO MONTATE SU ANTIVIBRANTI E CONNESSE ELASTICAMENTE ALLA RELATIVA TUBAZIONE.
All rotating machinery will be on resilient mounting and connected with elastic joint to the pipe system.
- 4) LA CONTINUITÀ ELETTRICA DOVRÀ ESSERE MANTENUTA LUNGO TUTTA LA TUBAZIONE E TRA QUESTA E LA CONNESSIONE A SCARICO.
Electrical continuity should be provided for all metal piping and from this to hull connection.
- 5) PER L'INSTALLAZIONE DEI MACCHINARI FARE RIFERIMENTO ALLE ISTRUZIONI DI MANUTENZIONE DEI COSTRUTTORI.
For installation operations referring to construction manual/instructions.
- 6) TUTTE GLI SCARICHI PER L'ACQUA NERA DEVRANNO AVERE UN SISTEMA DI AERAZIONE EFFICIENTE SIA PER IL RINGIO E L'INFIORCHI PER I COLLETTORI PRINCIPALI.
All "grey" discharges must have an efficient aeration system both of single users and main main flow.
- 7) REALIZZARE TASCHE DI ISPEZIONE LUNGO I COLLETTORI.
Facile of inspection must to be provided.
- 8) USARE DETERMINATA QUANTITÀ PRODUZIONE DI SCHUMA NELLE LAVABILI E LAVASTOVIGLIE.
Use foam producer and use used for dish washes & washer machine.
- 9) PREVEDERE UN SISTEMA DI SCARICO PER I RIFIUTI DI TRATTAMENTO ACQUA, I RIFIUTI DI TRATTAMENTO E I RIFIUTI DI TRATTAMENTO DI SCARICO.
A drainage system will be provided for the cold, hot, and frozen, and frozen, and frozen.
- 10) PER SCARICHI BLU, RISPONDERE AL VEDERE SEMPLIFICHE TECNICO.
For apparatus of drainage see technical specifications.
- 11) IL FISSAGGIO DEI TACCI E IL STANTE DELLE TUBAZIONI IMPIANTI DEVRANNO AVERE MECCANISMO INCOLLABILE.
The bracket of piping must be bonded.

Sea Water System

The yacht's sea water system provides sea water distribution to various equipment which uses sea water for their operation or for cooling.

There is one main services seawater suction pipes, fitted with filter, on which the connection points (each with its own intercept valve) for the following systems are foreseen:

- air conditioning cooling system;
- water maker system;
- fire-fighting system.

Independent sea chests, each with its own filters and check valves, are provided for the engine and generator cooling systems.

In case of emergency each engine sea chests are sired and designed to supply both engines.

Main engines and diesel generators cooling system sea chests are located on the hull bottom, amidships, between the two engines.

The cooling seawater aspirated by the engines is subsequently conveyed both to the gear boxes (to cool the lube oil) and to the engine exhaust gas discharge points (to cool the gases).

Sea water delivery to main engines exhaust ducts causes both gas cooling and noise/smoke reduction before passing through the hull exhaust.

For the same reason, also the generators cooling sea water is mixed with exhaust gasses, in a special muffler, and then separated again before being discharged outboard.

Two independent sea chests with their own filter and check valve are fitted to feed the main fire-fighting electro pump and emergency fire-fighting motor pump.

If necessary is possible to suck great quantity of water from the engine room bilge by using the engines cooling system suction. For this operation, close the engine sea water chests, open the check valves for engine room bilge emptying, equipped with suction filtering foot, and empty the bilge. After operation, stop the engine and reset the system.

When using the system in this configuration, the valves should be rotated completely in OPEN or CLOSED position, and should not be left in intermediate position.

The system's valves all bear use identification tags.



When the bilge is empty, open the sea chest valves again, to avoid damages to engine components.

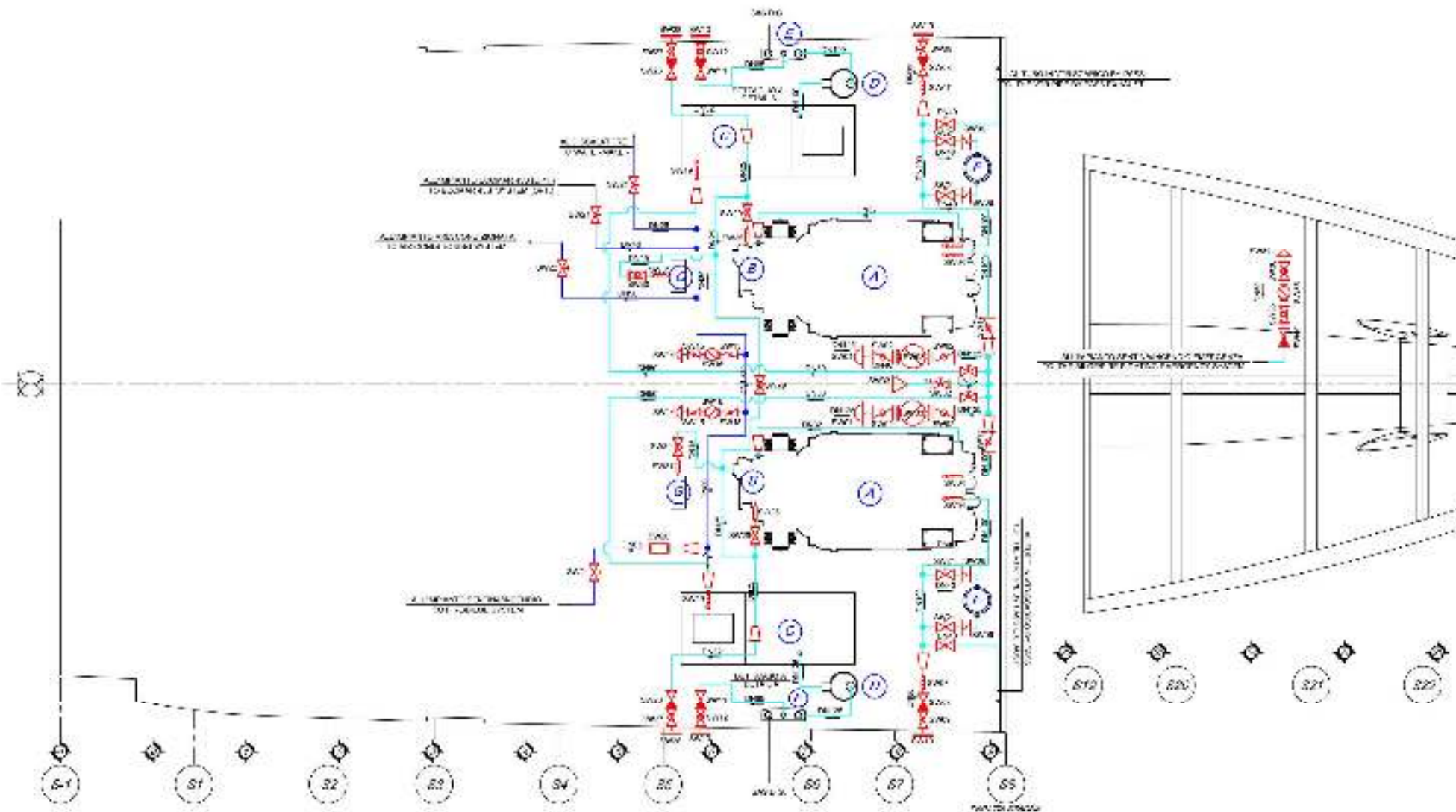
When the yacht is not used, close the sea water chest check valves. Open them again before operating the yacht.



Before opening sea water filter for cleaning operations, close the hull check valve.

Check periodically that the filters on all seawater suction pipes are clean. If necessary, clean or replace them. Remember to operate the appropriate intercept valves prior to conducting maintenance.

Sea Water System

[illegible][illegible]

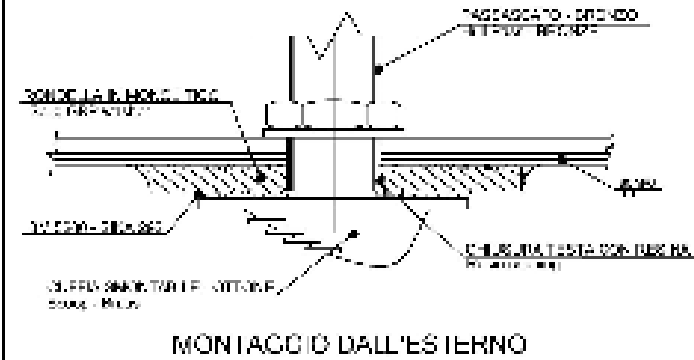
SIMBOLOGIA IMPIANTO	
Type Symbology	
SYMBOL	DESCRIPTION - DESCRIPTION
	INLET AND/OR WASTE Ingresso/uscite
	DIRECTION OF FLOW Direzione di flusso
	REDUCTION TUBE Pipe reduction

MATERIALI / Materials	
APPROVATI PER IL FIRE TEST ROOM - CEMENTO SALVATO - WELDED CEMENTITE - CEMENTO CEMENTITE	APPROVATI PER IL FIRE TEST ROOM - CEMENTO SALVATO - WELDED CEMENTITE - CEMENTO CEMENTITE

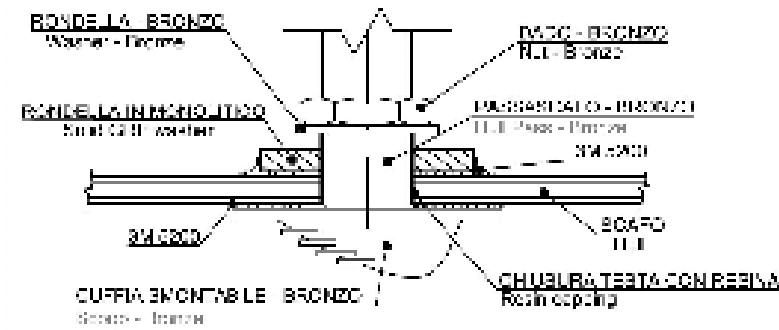
TABELLA PRESSIONI - Pressure Table	
PRESSIONE DI PROVA - PRESSURE TEST	3.0 BAR



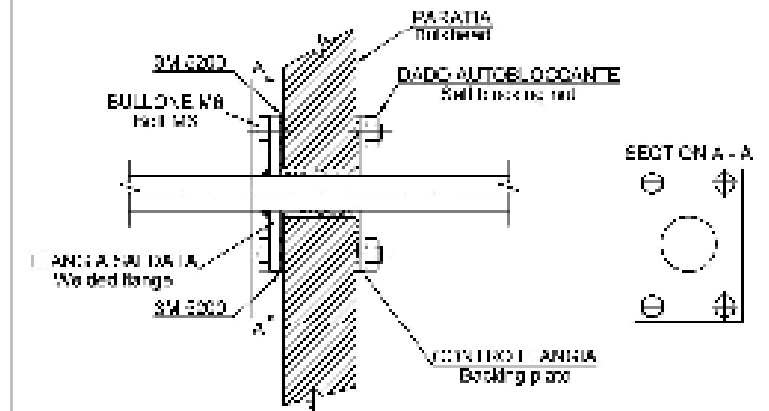
PARTICOLARE TIPICO PRESA A MARE
Sea Water Intake Typical Detail



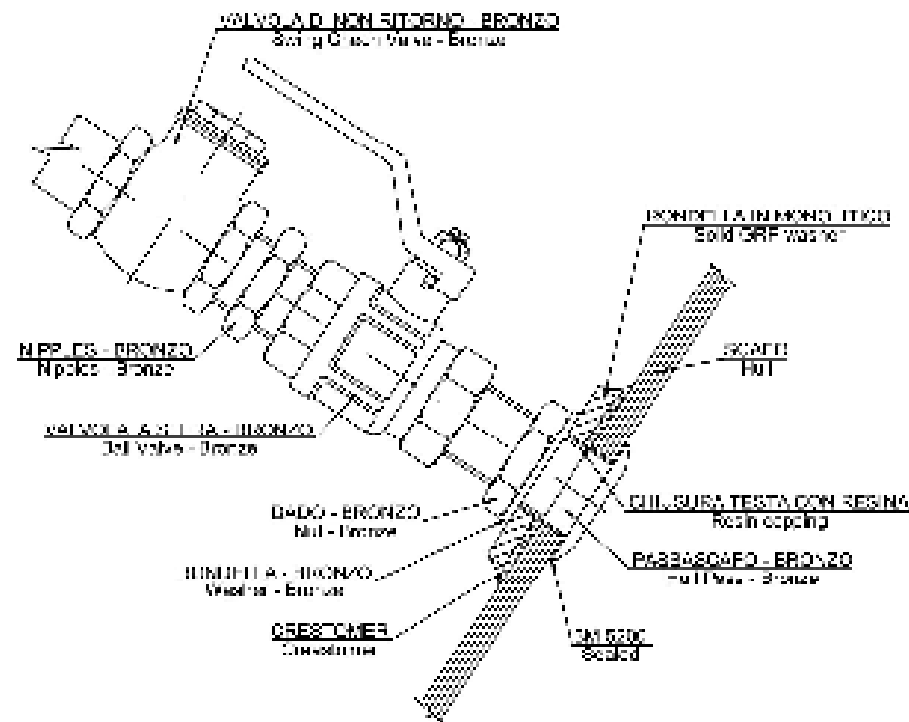
PARTICOLARE TIPICO PRESA A MARE
Sea Water Intake Typical Detail



PASSAGGI STAGNI A PARATIA BLK watertight Penetration



PARTICOLARE TIPICO FUORI BORDO
Outboard Discharge Typical Detail



NOTE GENERALE / General notes

[illegible]

TABELLA CONVERSIONE DA Ø NOMINALE A Ø FORNITORE ESTERNO IN MM.
Conversion Table Diameter of System

[illegible]

Air Conditioning System

The air conditioning system can operate in a “summer cycle” and in a “winter cycle”.

The system is realised with treated water circulation, using a heat pump refrigeration power unit (with sea water condensation).

The air conditioning system is composed of: n°2 chiller unit 144,000 BTU/H located in engine room equipped with n°2 sea water electric/pumps and n°2 recirculating electro/pumps.

The various system operation phases are summarised as follows:

- the refrigerating power unit with electric refrigerating compressors, cools down in the summer and heats up in the winter the circulation fresh water (treated water) by means of a sea water heat exchanger. Such circuit is provided with 2 sea water circulation electric pumps;
- 2 treated water electric pumps draw water from the power unit and sends it into the distribution circuit towards the various fan-coils located in air-conditioned rooms;
- the fan-coils allow thermal exchange between treated water and the surrounding environment, thus modifying the temperature and the hygrometric grade of the involved room;
- the conditioned air can be diffused locally or distributed by a system of thermally insulated piping;
- the system is equipped with monitoring and control electric devices for a sequential start up of the compressors, according to the thermal load;
- the air treatment units, together with various devices that compose the air conditioning system, are mounted on resilient supports, where possible.

The air conditioning system is a combination of distinct functions carried out by the following components:

- refrigerating power unit: it produces treated water according to the season: summer cycle (cold water production) or winter cycle (hot water cycle);
- fan-coils: provide thermal exchange in the involved room and air diffusion;
- sea water hydraulic circuit; provides thermal exchange (condensation) between sea water and the circuit of cooling fluid that circulates in the electric compressors;
- fresh water hydraulic circuit: provides both thermal exchange with the cooling fluid and thermal exchange to/from various rooms;
- electric system: provides power supply to various devices (refrigerating power unit, electric pumps, fan-coils, etc.) as well as for all adjustment, control, automation and safety purposed of the whole system.

Cold/hot water production is ensured by a cooling system operating as heat pump of the water-to-water system with condensation of the sea water type operated by two sea water electric.

The cooling system is composed of the following main parts:

- evaporators;
- condensers;
- fan-coils.

Evaporators can cool the fresh water down, maintaining the Δt (difference of temperature) for the whole operation cycle.

The condensers, of the sea water type completely made in copper with rolled Cu-Ni internal pipes, are designed to operate with sea water minimum temperature of approximately 9°C.

The cold/hot water is delivered to equipment by means of above mentioned electric pumps.

The system is divided in such manner so various rooms can be separately air conditioned and that fan-coils or fan-coil sets, in each room, are controlled by local electric thermostats for temperature adjustment, with scale in degrees Celsius or Fahrenheit.

To supply the system, operate the magneto-thermal switch located on the main switchboard, 230V equipment section.

The control panel is located near the unit, and carries the following equipment:

- summer/winter switch;
- “Treated water” circulation pump alarms warning lamp;
- sea water pump alarm warning lamp;
- pressure switch lock alarm warning lamp;
- hours counter;
- compressor switches;
- machinery control electronic thermostat;
- water temperature control.

Description and information concerning operation and maintenance are contained in the relevant Manual supplied by the Manufacturer.



Before opening the filter for cleaning, close the sea chest valve. At the end of operation remember to open the sea chest valve again.

The air-conditioning unit diagram showing the conditioned air system and the UTA distribution line system is shown below.

The latter uses 2 units located on the port and stbd side of the wheelhouse that sucks air directly from the outside distributed to lower deck and too Owner cabin by means of conduit lines. The aim is to change the air inside the vessel and to offset the depression created by the bathroom extractors.

The aforementioned units are controlled by a dedicated thermostat positioned in its immediate vicinity and having the same purpose as those used for the air-conditioning system fan coils.

The extractors in the individual bathrooms are activated when the:

- the UTA unit is turned on.



Control panel – Air Conditioning

All fan coils are:

- installed in the way to be easily accessible for maintenance and filter cleaning operation;
- fitted with check valves (on both supply and delivery line) in the way to disconnect every each unit from the system;
- fitted with drip pan for collecting the fan coils condensation.

Chilled water pipes and valves are properly insulated to avoid/reduce the production of condensation.

The A/C system is completed by 4 pumping unit (24Vdc powered) that collect the condensation through drainage lines of lower deck cabins and pump it into the black and grey water tank.

The drainages of fancoils of the Main Deck and Upper Deck are directly discharged outboard through the scupper pipes.



Before starting up the system, make sure that the sea chest valve is open and check filter cleaning conditions.

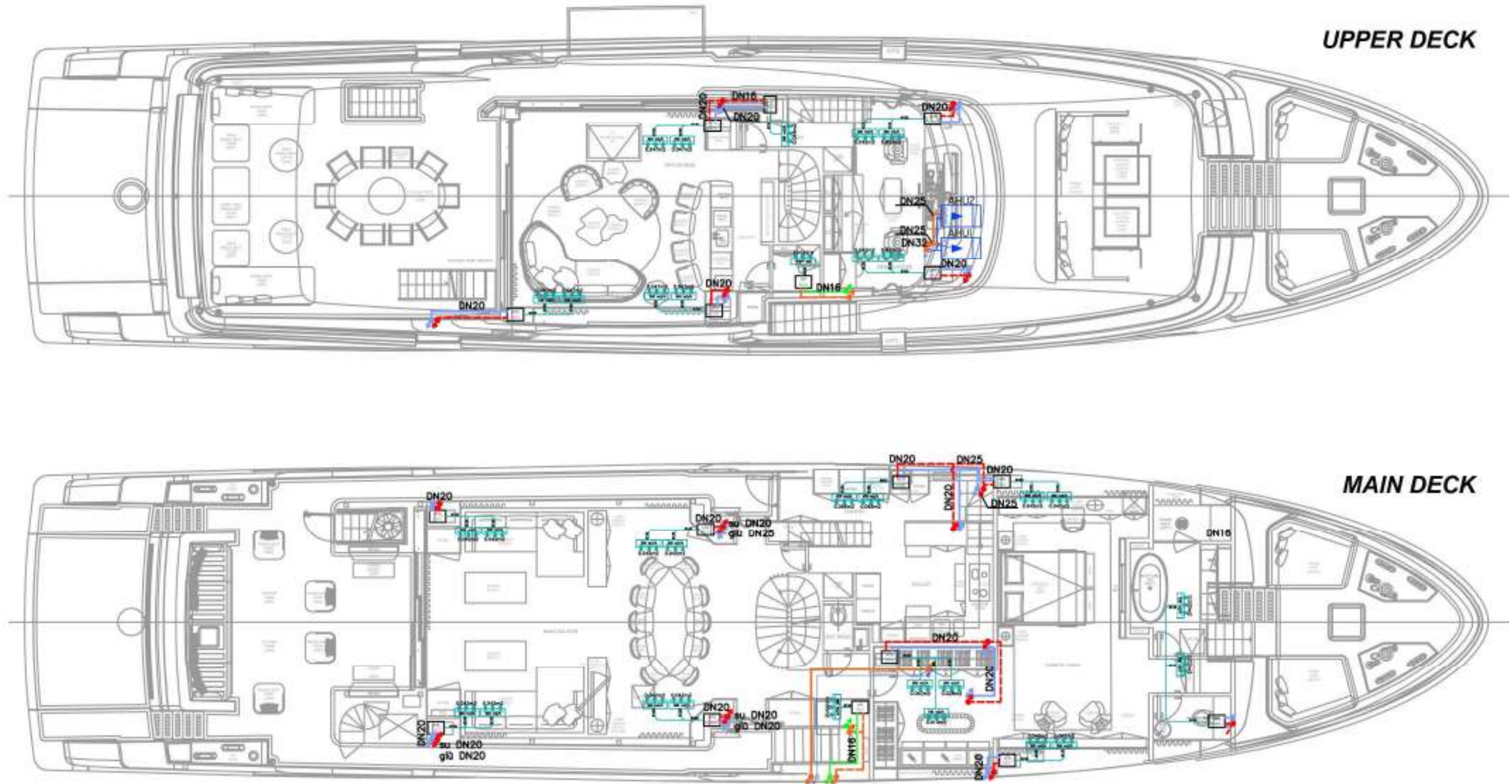
After prolonged inactivity, before starting up, make sure that sea water pumps are not locked.

Operation principle clearness is fundamental for a correct system operation and maintenance, for finding eventual failure and for maintaining the full efficiency of the system.

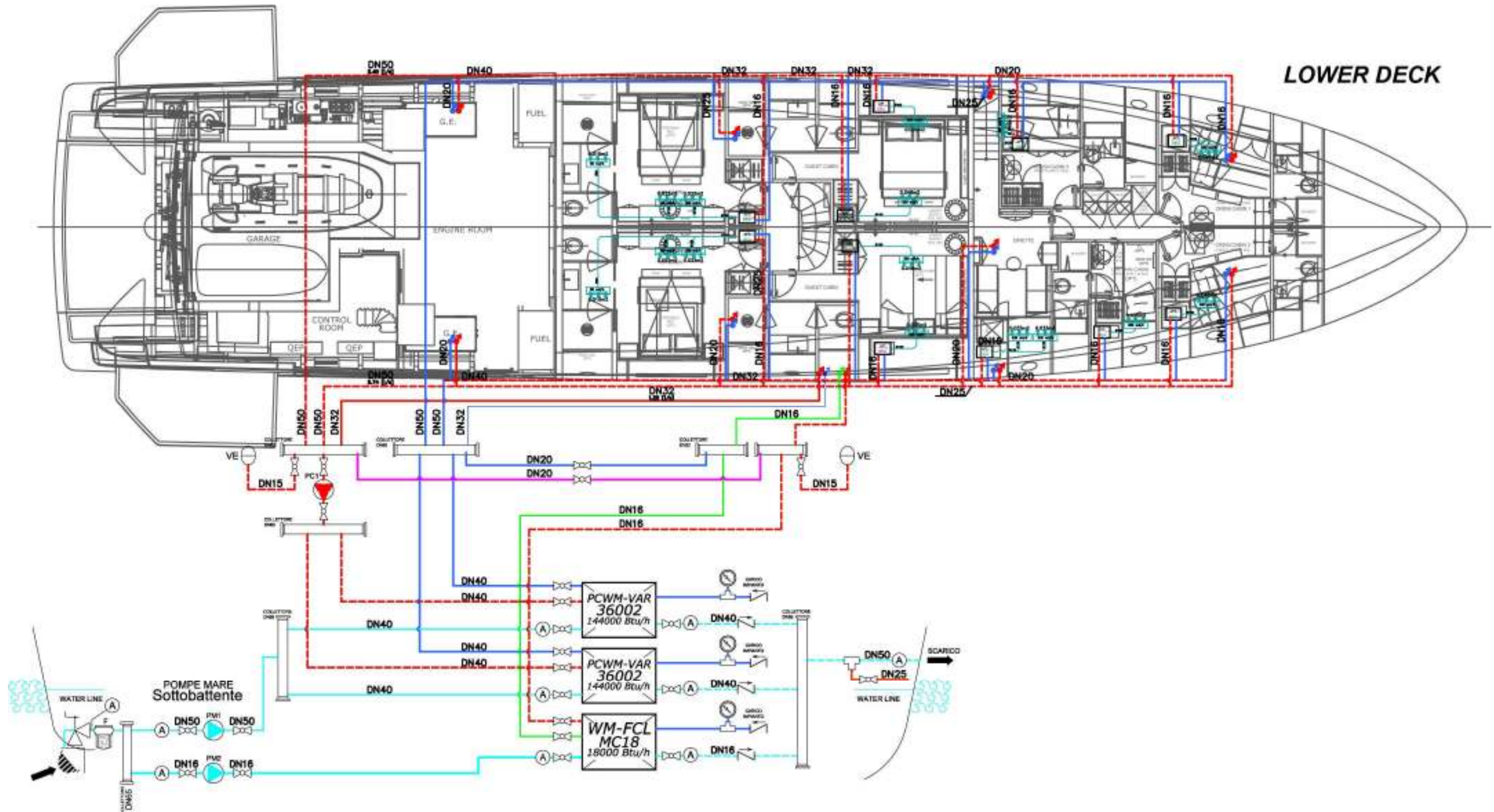
Do not modify the thermostat parameters, which are set by the manufacturer.



When operating the system with shore power supply, make sure that power supply connection can support simultaneous loads of air conditioning system and other yacht's equipment, paying attention that voltage does not descend below 10% during electric compressors start up.

Air Conditioning System – Upper Deck / Main Deck

Air Conditioning System – Lower Deck



Air Conditioning System - Details

LEGENDA									
	FAN COIL UNIT		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER
	WATER PIPE		WATER VALVE		WATER METER		WATER FILTER		WATER STRAINER

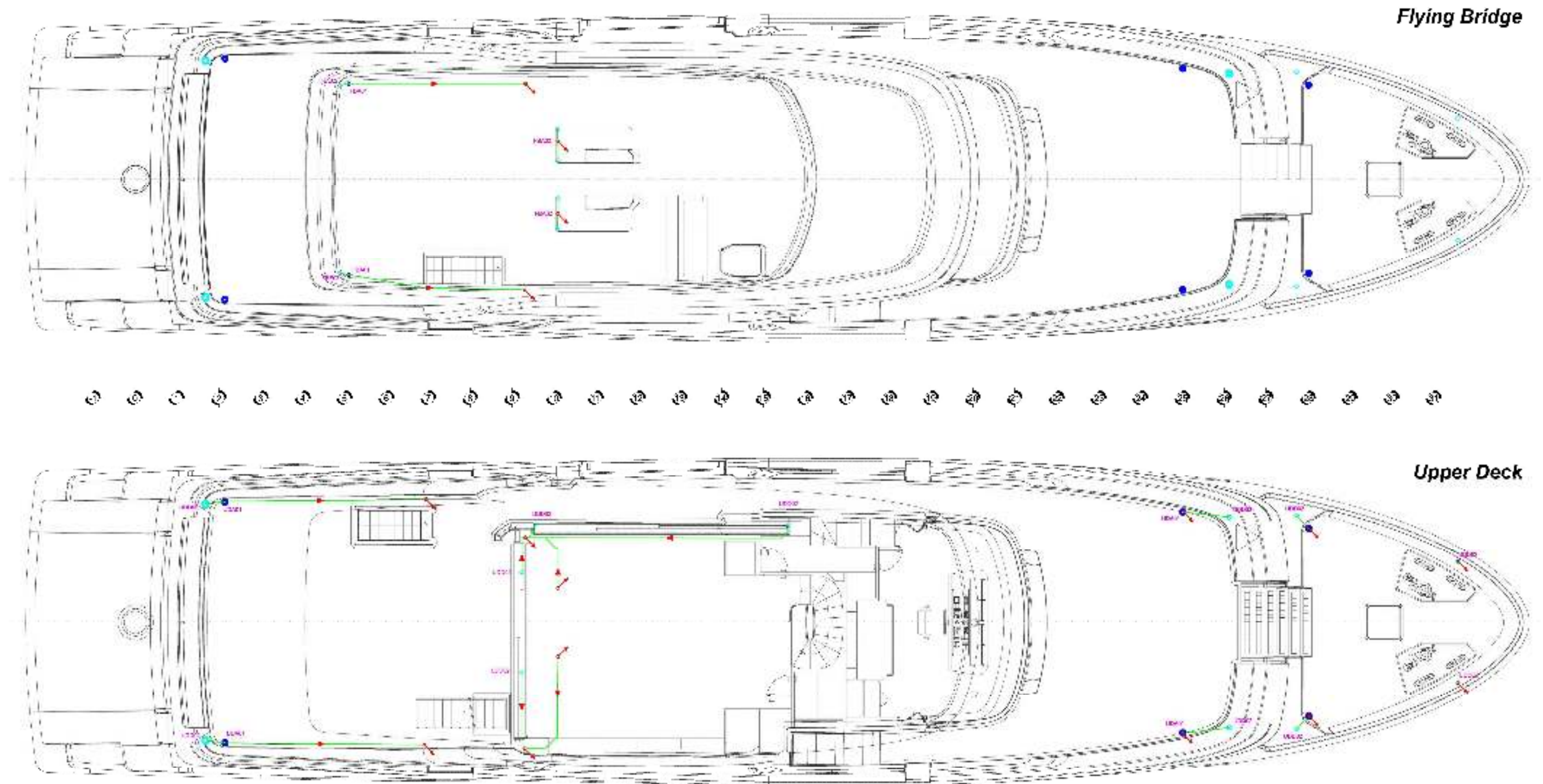
MODEL	WATER FLOW RATE (L/min)	WATER FLOW RATE (GPM)	WATER FLOW RATE (m³/h)	WATER FLOW RATE (USGPM)	WATER FLOW RATE (L/min)	WATER FLOW RATE (GPM)	WATER FLOW RATE (m³/h)	WATER FLOW RATE (USGPM)	WATER FLOW RATE (L/min)	WATER FLOW RATE (GPM)	WATER FLOW RATE (m³/h)	WATER FLOW RATE (USGPM)
AP1	100	1.67	0.01	0.0004	100	1.67	0.01	0.0004	100	1.67	0.01	0.0004
AP2	150	2.50	0.015	0.0006	150	2.50	0.015	0.0006	150	2.50	0.015	0.0006
AP3	200	3.33	0.02	0.0008	200	3.33	0.02	0.0008	200	3.33	0.02	0.0008
AP4	250	4.17	0.025	0.001	250	4.17	0.025	0.001	250	4.17	0.025	0.001
AP5	300	5.00	0.03	0.0012	300	5.00	0.03	0.0012	300	5.00	0.03	0.0012
AP6	350	5.83	0.035	0.0014	350	5.83	0.035	0.0014	350	5.83	0.035	0.0014
AP7	400	6.67	0.04	0.0016	400	6.67	0.04	0.0016	400	6.67	0.04	0.0016
AP8	450	7.50	0.045	0.0018	450	7.50	0.045	0.0018	450	7.50	0.045	0.0018
AP9	500	8.33	0.05	0.002	500	8.33	0.05	0.002	500	8.33	0.05	0.002
AP10	550	9.17	0.055	0.0022	550	9.17	0.055	0.0022	550	9.17	0.055	0.0022
AP11	600	10.00	0.06	0.0024	600	10.00	0.06	0.0024	600	10.00	0.06	0.0024
AP12	650	10.83	0.065	0.0026	650	10.83	0.065	0.0026	650	10.83	0.065	0.0026
AP13	700	11.67	0.07	0.0028	700	11.67	0.07	0.0028	700	11.67	0.07	0.0028
AP14	750	12.50	0.075	0.003	750	12.50	0.075	0.003	750	12.50	0.075	0.003
AP15	800	13.33	0.08	0.0032	800	13.33	0.08	0.0032	800	13.33	0.08	0.0032
AP16	850	14.17	0.085	0.0034	850	14.17	0.085	0.0034	850	14.17	0.085	0.0034
AP17	900	15.00	0.09	0.0036	900	15.00	0.09	0.0036	900	15.00	0.09	0.0036
AP18	950	15.83	0.095	0.0038	950	15.83	0.095	0.0038	950	15.83	0.095	0.0038
AP19	1000	16.67	0.1	0.004	1000	16.67	0.1	0.004	1000	16.67	0.1	0.004
AP20	1050	17.50	0.105	0.0042	1050	17.50	0.105	0.0042	1050	17.50	0.105	0.0042
AP21	1100	18.33	0.11	0.0044	1100	18.33	0.11	0.0044	1100	18.33	0.11	0.0044
AP22	1150	19.17	0.115	0.0046	1150	19.17	0.115	0.0046	1150	19.17	0.115	0.0046
AP23	1200	20.00	0.12	0.0048	1200	20.00	0.12	0.0048	1200	20.00	0.12	0.0048
AP24	1250	20.83	0.125	0.005	1250	20.83	0.125	0.005	1250	20.83	0.125	0.005
AP25	1300	21.67	0.13	0.0052	1300	21.67	0.13	0.0052	1300	21.67	0.13	0.0052
AP26	1350	22.50	0.135	0.0054	1350	22.50	0.135	0.0054	1350	22.50	0.135	0.0054
AP27	1400	23.33	0.14	0.0056	1400	23.33	0.14	0.0056	1400	23.33	0.14	0.0056
AP28	1450	24.17	0.145	0.0058	1450	24.17	0.145	0.0058	1450	24.17	0.145	0.0058
AP29	1500	25.00	0.15	0.006	1500	25.00	0.15	0.006	1500	25.00	0.15	0.006
AP30	1550	25.83	0.155	0.0062	1550	25.83	0.155	0.0062	1550	25.83	0.155	0.0062
AP31	1600	26.67	0.16	0.0064	1600	26.67	0.16	0.0064	1600	26.67	0.16	0.0064
AP32	1650	27.50	0.165	0.0066	1650	27.50	0.165	0.0066	1650	27.50	0.165	0.0066
AP33	1700	28.33	0.17	0.0068	1700	28.33	0.17	0.0068	1700	28.33	0.17	0.0068
AP34	1750	29.17	0.175	0.007	1750	29.17	0.175	0.007	1750	29.17	0.175	0.007
AP35	1800	30.00	0.18	0.0072	1800	30.00	0.18	0.0072	1800	30.00	0.18	0.0072
AP36	1850	30.83	0.185	0.0074	1850	30.83	0.185	0.0074	1850	30.83	0.185	0.0074
AP37	1900	31.67	0.19	0.0076	1900	31.67	0.19	0.0076	1900	31.67	0.19	0.0076
AP38	1950	32.50	0.195	0.0078	1950	32.50	0.195	0.0078	1950	32.50	0.195	0.0078
AP39	2000	33.33	0.2	0.008	2000	33.33	0.2	0.008	2000	33.33	0.2	0.008
AP40	2050	34.17	0.205	0.0082	2050	34.17	0.205	0.0082	2050	34.17	0.205	0.0082
AP41	2100	35.00	0.21	0.0084	2100	35.00	0.21	0.0084	2100	35.00	0.21	0.0084
AP42	2150	35.83	0.215	0.0086	2150	35.83	0.215	0.0086	2150	35.83	0.215	0.0086
AP43	2200	36.67	0.22	0.0088	2200	36.67	0.22	0.0088	2200	36.67	0.22	0.0088
AP44	2250	37.50	0.225	0.009	2250	37.50	0.225	0.009	2250	37.50	0.225	0.009
AP45	2300	38.33	0.23	0.0092	2300	38.33	0.23	0.0092	2300	38.33	0.23	0.0092
AP46	2350	39.17	0.235	0.0094	2350	39.17	0.235	0.0094	2350	39.17	0.235	0.0094
AP47	2400	40.00	0.24	0.0096	2400	40.00	0.24	0.0096	2400	40.00	0.24	0.0096
AP48	2450	40.83	0.245	0.0098	2450	40.83	0.245	0.0098	2450	40.83	0.245	0.0098
AP49	2500	41.67	0.25	0.01	2500	41.67	0.25	0.01	2500	41.67	0.25	0.01
AP50	2550	42.50	0.255	0.0102	2550	42.50	0.255	0.0102	2550	42.50	0.255	0.0102
AP51	2600	43.33	0.26	0.0104	2600	43.33	0.26	0.0104	2600	43.33	0.26	0.0104
AP52	2650	44.17	0.265	0.0106	2650	44.17	0.265	0.0106	2650	44.17	0.265	0.0106
AP53	2700	45.00	0.27	0.0108	2700	45.00	0.27	0.0108	2700	45.00	0.27	0.0108
AP54	2750	45.83	0.275	0.011	2750	45.83	0.275	0.011	2750	45.83	0.275	0.011
AP55	2800	46.67	0.28	0.0112	2800	46.67	0.28	0.0112	2800	46.67	0.28	0.0112
AP56	2850	47.50	0.285	0.0114	2850	47.50	0.285	0.0114	2850	47.50	0.285	0.0114
AP57	2900	48.33	0.29	0.0116	2900	48.33	0.29	0.0116	2900	48.33	0.29	0.0116
AP58	2950	49.17	0.295	0.0118	2950	49.17	0.295	0.0118	2950	49.17	0.295	0.0118
AP59	3000	50.00	0.3	0.012	3000	50.00	0.3	0.012	3000	50.00	0.3	0.012
AP60	3050	50.83	0.305	0.0122	3050	50.83	0.305	0.0122	3050	50.83	0.305	0.0122
AP61	3100	51.67	0.31	0.0124	3100	51.67	0.31	0.0124	3100	51.67	0.31	0.0124
AP62	3150	52.50	0.315	0.0126	3150	52.50	0.315	0.0126	3150	52.50	0.315	0.0126
AP63	3200	53.33	0.32	0.0128	3200	53.33	0.32	0.0128	3200	53.33	0.32	0.0128
AP64	3250	54.17	0.325	0.013	3250	54.17	0.325	0.013	3250	54.17	0.325	0.013
AP65	3300	55.00	0.33	0.0132	3300	55.00	0.33	0.0132	3300	55.00	0.33	0.0132
AP66	3350	55.83	0.335	0.0134	3350	55.83	0.335	0.0134	3350	55.83	0.335	0.0134
AP67	3400	56.67	0.34	0.0136	3400	56.67	0.34	0.0136	3400	56.67	0.34	0.0136
AP68	3450	57.50	0.345	0.0138	3450	57.50	0.345	0.0138	3450	57.50	0.345	0.0138
AP69	3500	58.33	0.35	0.014	3500	58.33	0.35	0.014	3500	58.33	0.35	0.014
AP70	3550	59.17	0.355	0.0142	3550	59.17	0.355	0.0142	3550	59.17	0.355	0.0142
AP71	3600	60.00	0.36	0.0144	3600	60.00	0.36	0.0144	3600	60.00	0.36	0.0144
AP72	3650	60.83	0.365	0.0146	3650	60.83	0.365	0.0146	3650	60.83	0.365	0.0146
AP73	3700	61.67	0.37	0.0148	3700	61.67	0.37	0.0148	3700	61.67	0.37	0.0148
AP74	3750	62.50	0.375	0.015	3750	62.50	0.375	0.015	3750	62.50	0.375	0.015
AP75	3800	63.33	0.38	0.0152	3800	63.33	0.38	0.0152	3800	63.33	0.38	0.0152
AP76	3850	64.17	0.385	0.0154	3850	64.17	0.385	0.0154	3850	64.17	0.385	0.0154
AP77	3900	65.00	0.39	0.0156	3900	65.00	0.39	0.0156	3900	65.00	0.39	0.0156
AP78	3950	65.83	0.395	0.0158	3950	65.83	0.395	0.0158	3950	65.83	0.395	0.0158
AP79	4000	66.67	0.4	0.016	4000	66.67	0.4	0.016	4000	66.67	0.4	0.016
AP80	4050	67.50	0.405	0.0162	4050	67.50	0.405	0.0162	4050	67.50	0.405	0.0162
AP81	4100	68.33	0.41	0.0164	4100	68.33	0.41	0.0164	4100	68.33	0.41	0.0164
AP82	4150	69.17	0.415	0.0166	4150	69.17	0.415	0.0166	4150	69.17	0.415	0.0166
AP83	4200	70.00	0.42	0.0168	4200	70.00	0.42	0.0168	4200	70.00	0.42	0.0168
AP84	4250	70.83	0.425	0.017	4250	70.83	0.425	0.017	4250	70.83	0.425	0.017
AP85	4300	71.67	0.43	0.								

Scuppers System

Special discharge pipes, and scuppers have been installed on open decks, to remove rain or yacht's washing water, in most suitable position to avoid water stagnation.

All scuppers are equipped with a drilled stainless steel protection grid, at deck level, as anti-clogging protection.

Main Deck bulwark is provided with big openings for evacuating great quantities of water (heavy waves).

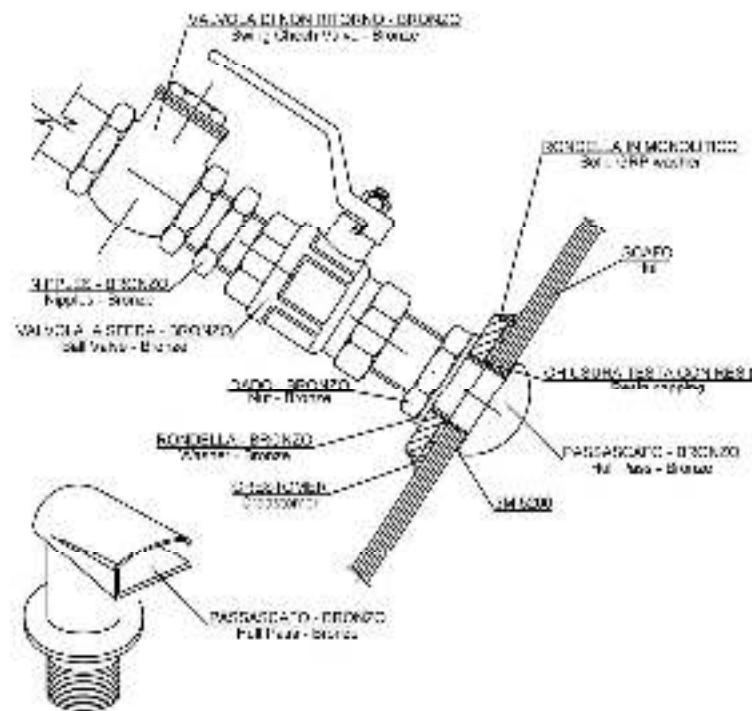
Scuppers System - Flying Bridge / Upper Deck



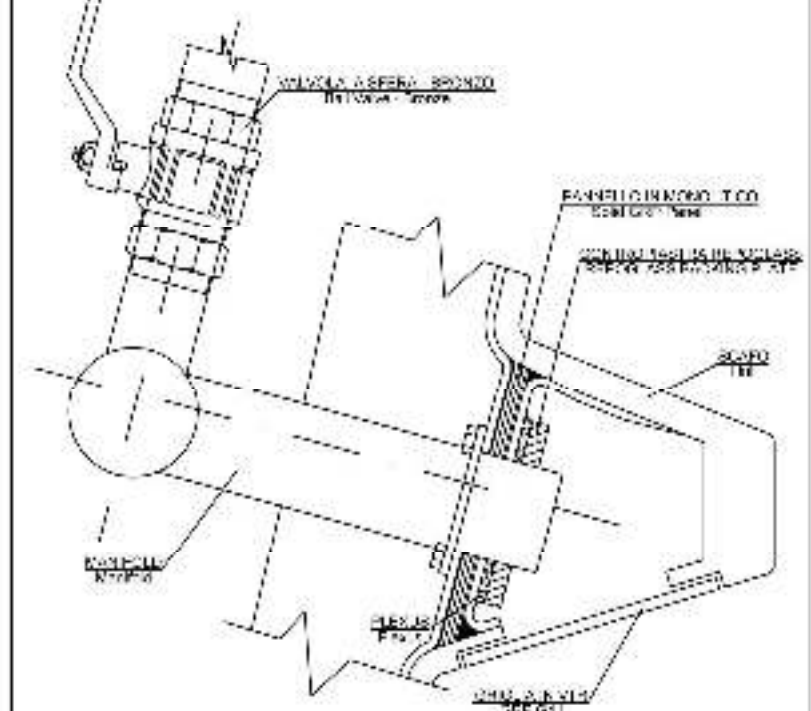
Scuppers System - Details

[illegible]

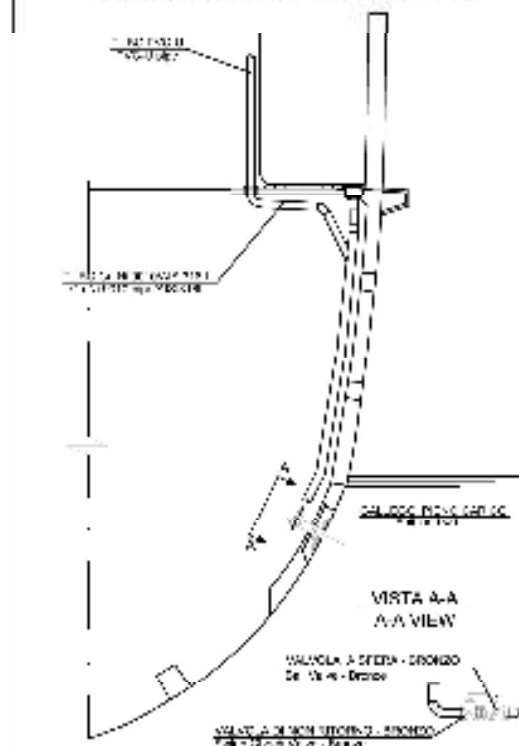
PARTICOLARE TIPICO SCARICO FUORI BORDO
Outboard Discharge Typical Detail



PARTICOLARE TIPICO MANIFOLD OMBRINALI
 n Scupper Manifold Discharge Typical Detail



ENGINE TYPE: 1.8L 16V DOHC I4, 150 HP
Engine model type: connected on below main deck.



Order of angles from typical connection below main deck

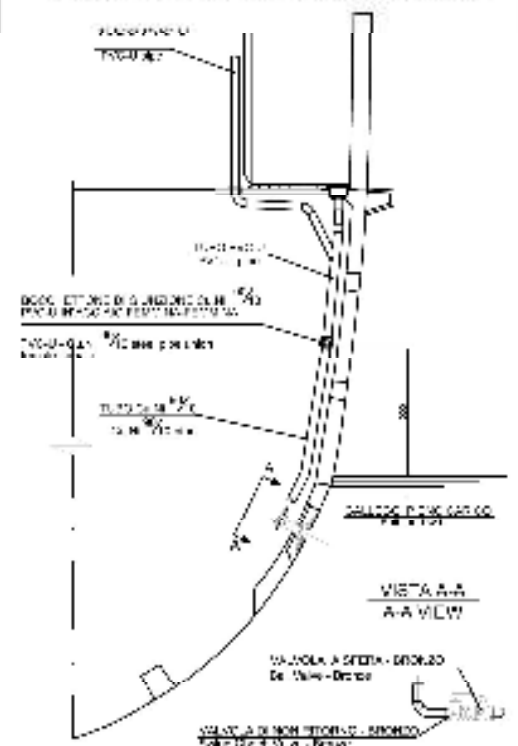


TABELLA CONVERSIONE DA Ø NOMINALE A Ø POLLICI E Ø ESTERNO IN MM.
Conversion Table Diameter of System

[illegible]

NOTE GENERALI / General notes

- 1) TUBERIZATION AND GATE CONNECTIONS: SARIANO IN PAVIA, PN 16 (FOR 1/4" NPT)
 LE TUBAZIONI SONO A MISCELA IL GALLEGGIAMENTO SARIANO IN CUI SI
 PIPING MARKED WITH "001" WILL BE IN PVC-U PN 16 (1/4" NPT) AND
 BEING TO BE USED FOR WATERLINE AND WASTE.
- 2) TUTTA LA BULLONERIA SARIN® N° 304, USARE FRENNI FLETTI
 LOCOTE 242
 ALL BOLTS ARE IN A304. LOCOTE 242 IS TO BE USED ON
 THREADED CONNECTIONS.
- 3) ESTREMI CURA, COME ESSERE PRESA PER LE
 STAFFE PER TUTTO IL PERCORSO DELLA TUBAZIONE.
 PIPE FITTING BRACKETS WILL BE INSTALLED WITH EXTREMELY CARE.
- 4) PREVEDERE DRENAGGI SOTTO TUTTA LA PORTELLA ESTERNA.
 PROVIDE DRAINAGE FOR EXTERNAL HATCH.
- 5) LA CURA N° 14, LE TRONCATURE ESSERE MARCARE A LUNGO
 TUTTA LA TUBAZIONE E LA COLLETTA DI SAGGIO DI 1/4" DEL METALLO
 ELECTRICAL DRAPPAINS FOR ALL PIPING AND FROM THIS TO BE
 IS TO PROVIDED (MATERIAL IN PART).

TABELLA PRESSIONI
Pressure Table

PRESSURE TEST PRESSURED FROM	HOLDUP A BATTENTE
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MATERIALI - Materials

TRANSFORMER APPARATUS MODEL	CONTROL TRANSFORMER MODEL
- TUBULAR TYPE, STRATTON STYLE TUBULE, RESEPTING CO. IN NONO STRUCTURAL PIPE	- CUBIC FRAME STRUCTURE, REPT STYLE TUBULE, RESEPTING TUBES IN NONO STRATTON RESEPTING TUBING
- LUGS IN RESEPTING CO. IN NONO STRUCTURAL PIPE	- RESEPTING TUBING
- AUGER TYPE, STRATTON STYLE TUBULE, RESEPTING CO. IN NONO STRUCTURAL PIPE	

Section **E**

MAINTENANCE

General information

- Safety precautions

Engines Maintenance

- Lubrication system

- Fuel system

- Distribution system

Electric system maintenance

Fuel system Maintenance

Grey / Black water system maintenance

Hull structure and furnishings maintenance

- Characteristics of the teak

- Characteristics of the stainless steel

- Characteristics of internal woodwork

- Characteristics of furniture upholstery

- External upholstery

- Leather care and cleaning tips

Precautions for cold climates

Yacht's long inactivity

Preparing the yacht for winter laying up

Yacht's use after long inactivity

General information

The vessel is fitted with a number of sophisticated equipment and systems, that require a minimum of periodical checks and maintenance in order to operate correctly.

One of the factors that can cause failures is, in general, the intermittent use of the boat and, as a consequence, of the boat equipment.

The experience shows that the regular use of equipment decreases the number of failures; therefore it is advisable to run regularly, for short periods, all equipment.

Daily checks and periodical maintenance are important for maintaining equipment/components in the most efficient conditions; if the periodical maintenance schedule is not correctly followed, the equipment performance can deteriorate, causing a reduction of efficiency and a shorter life. The non-observance of the maintenance schedule can also cause unexpected problems that can decrease the at-sea safety levels.

The maintenance schedule is based on time or running hours intervals.

In case of a long inactivity period (for example during winter) it is advisable to lay up the craft, possibly under cover.



For further instructions on the maintenance program, consult the Manufacturers Technical Manuals, relating to the on board equipments and components. They will often contain important and more detailed information relating to maintenance and preservation.

Safety precautions

Besides following precautions, specific information are given throughout the whole Manual. The purpose of this section is to supply a safety behaviour code for systems' operation and maintenance.

This section includes information for safe maintenance operation on components.

Therefore before operating it is necessary to read and understand any information shown on Manufacture Manuals.

Do not start any maintenance operation before being sure that both operator and other personnel safety is ensured.

In case of doubt concerning maintenance operations, please refer to our Service Department or system Manufacture Service Department.

Operate always with care and in safety conditions.



Prior to performing any maintenance and adjustment tasks, turn all necessary safety devices on and consider informing all personnel operating in proximity.

In particular, put signs in the adjacent areas and secure the accessibility to any item that may cause unexpected damage and injury hazards, if operated.



*Insufficient **information** may cause accidents.*

Before carrying out any operation, inform all people present in the working zone. If working with other people, make sure that they know hand signals which you intend to use. In view of the high level of noise on the site, voice commands should be avoided.

When two or more people are working on the same machine, make sure that everyone knows the operation carried out by other people. Before starting the engine, keep away other people from the dangerous areas (rotating fins and engine belt, tools and cinematic motions, internal and rear part of the machine). Not complying with those precautions may cause serious accidents, even lethal.



Pay maximum **attention and caution** for the whole duration of the intervention. Avoid all possible dangers.



It is extremely dangerous to operate the yacht's controls when drunk or under **influence of medicines**. Do not drink **alcoholic substances** or medicines, before and during operation. Do not take medicines that cause sleepiness.



Cleaning metal components with unsuitable solvents may cause corrosion; use only detergents and thinners of the prescribed type.



Boarding the yacht. Always face the yacht and use the handles and steps. Make sure that steps, handles and rubber soles of the shoes are clean and dry. Do not jump out of the yacht; do not use the controls as support, but use the handles.



Wearing unsuitable **clothing** may cause accidents; do not wear loose clothing as they may easily entangle with moving parts. Wear protective clothing, suitable for the type of operation (helmets, safety boots and glasses, overalls). Button up the cuffs, do not wear ties, scarves and do not leave long hair free.



When restoring metal and non-metal components, **protect your eyes** wearing special safety glasses. Keep away from the zone eventual combustible materials which may ignite in case of sparkles.



Metal splinters projected when machining metal parts, may cause injuries: always wear safety glasses and use a mallet or punch made in soft material.



Faulty **hoisting** equipment operation may cause accidents, therefore their efficiency must be verified. Ensure tackles conformity to local regulations and their suitability to the specific operation.

Suspended loads may fall causing injuries. Do not walk or work under hoisted equipment which are not correctly sustained.



Moving parts of the **engine** are dangerous; do not open the sumps, when operating the yacht.



Operating the **control levers** from outside the bridge may cause serious damages, even lethal: controls must be operated staying in the correct position.



Sparks from the discharge and electric systems can cause explosions and fires. Do not switch on the engines of the boat in closed environments, in the presence of flammable materials, gases or powders.



The fuel is flammable: do not approach flames and do not smoke when refuelling and when working on the engine. Stop the engine, when refuelling. Failure to comply with those precautions may cause fire and injuries.

Do not stock fuel **tanks** on board, in closed premises, as it may cause explosions.



Lubricating oil is toxic: do not swallow Engine oil contains dangerous contaminating agents which can cause skin cancer. Handle the oil as less as possible and protect your skin with creams and gloves. Wash carefully the skin eventually contaminated by oil, with hot water and soap: do not use petrol, fuel oil or oil.



A **battery** with a frozen electrode can explode if used or charged. Do not start the engine if the battery is frozen. To avoid freezing always keep the battery connected.



The **battery** produces explosive gases: avoid flames or sparks and do not smoke in its proximity. Should the battery be used or charged in a closed environment, ensure adequate ventilation. Do not check the charge level of the battery by short-circuiting the terminal blocks with metal tools: only use a densimeter or a voltmeter.



If damaged, the **flexible hydraulic pipes** may cause lethal accidents; carry out periodical checks in order to ensure eventual presence of:

- damaged fittings;
- wear of external covering;
- swelling of external covering;
- bent or flattened pipes;
- armature buried in external covering;
- fittings not correctly positioned.



Gaskets and O-rings incorrectly installed, damaged or worn out may cause leakage and accidents; replace them immediately unless otherwise specified. Do not use trichloroethane or thinner on O-rings and gaskets.



Hydraulic oil spilling, at **operation pressure** may cause injuries: before disconnecting or connecting the flexible pipes, stop the engine and operate the controls for discharging the residual pressure.

Do not start the engine while pipes are disconnected.



High pressure **hydraulic oil** penetrates into the skin: do not verify the presence of oil leakage with your fingers, and do not approach your face. Use a cardboard stripe checking eventual presence of hydraulic oil on it.

If oil penetrates into the skin, call immediately a doctor for specific treatment.



Check periodically the **cylinders** in order to eliminate eventual dirt deposits which may compromise its efficiency. In order to reduce the risk of corrosion, if possible retract the rods whenever leaving or anchoring the yacht.



Hot cooling fluid. When the engine temperature is high, the cooling system is pressurised and the hot liquid may come out, when removing the radiator plug. Therefore, before removing the radiator plug, wait until the system cools down, then rotate the plug to the first mark and discharge the system pressure.



SANLORENZO S.p.A. shipyard declines all responsibility concerning installation and operation of electric, electronic, mechanic equipment carried out by third parties in a manner that is not authorised by the manufacture Yard.

SANLORENZO S.p.A. shipyard declines all responsibility concerning manumissions carried out by third parties on equipment installed in the Yard. Such non-authorised manumissions, besides causing the immediate lose of warranty rights, can cause damages to the craft and to persons on board.

SANLORENZO S.p.A. shipyard declines all responsibility concerning periodical maintenance activities scheduled by the Yard or by Manufacturer Firms, but not carried out, of equipment/components, for which it is necessary to refer to their own Technical Manuals.



Whenever external temperature descends below 0°C fresh or sea water flowing through piping may freeze, breaking the pipes.



Do not disperse any waste avoiding environment contamination and use special disposal areas arranged in marinas.

When working in the engine room, switch magneto-thermal switch of bilge automatic pump off, to prevent that fuel, lubricants and other liquid spilling cause sea pollution.

Engine Maintenance - CAT C32 Acert

Essential requirement for safety operation, low operation costs and conditions for efficient maintenance are:

- fuel quality and quantity;
- clean and dry air for combustion;
- system always kept clean;
- protection of all rubber/synthetic parts, such as seals, sleeves, etc.;
- periodical replacement of all sealing and junction materials, according to the schedule provided by the Manufacturer;
- respect of the schedule provided by the Manufacturer, based upon many years of experience. Specific Manuals supplied by Manufacturers contains the intervention schedule according to hours of operation and yacht use frequency (W1, W2, etc.), as well as precautions if the yacht is not used for prolonged time.

Lubrication system

When using fuel, lubricants and refrigeration fluids, pay maximum attention to maintain perfect cleanliness.



Use only technical fluids approved by the Manufacturer, otherwise the warranty will be invalidated.

Engine oil replacement



*The engine oil is hot, risk of burns!
Do not touch the oil discharge plug with the hands.*



Engine oils are irritant and contaminating liquids, therefore they should be treated and removed carefully.

While the engine is hot, unscrew the oil sump discharge plugs and let discharge completely the exhausted oil.

For this operation use a container which can hold the whole quantity of oil, without overflows.

A reciprocating manual pump is installed on the engine for discharging the oil.

Remove the exhausted oil by pumping it from the sump with the manual pump when the engine has reached the pre-heater temperature. Unscrew the oil discharge plugs on oil filter bowl and let the exhausted oil flow out from the filters.

Screw the discharge plugs after having replaced their sealing rings.



When replacing the oil, replace also the oil filter cartridge.

Oil refilling and check

Do not refill oil above the dipstick MAX mark. Too high oil level may cause engine troubles.

New oil refilling should be carried out through the oil filling plugs. Do not pour too much oil.

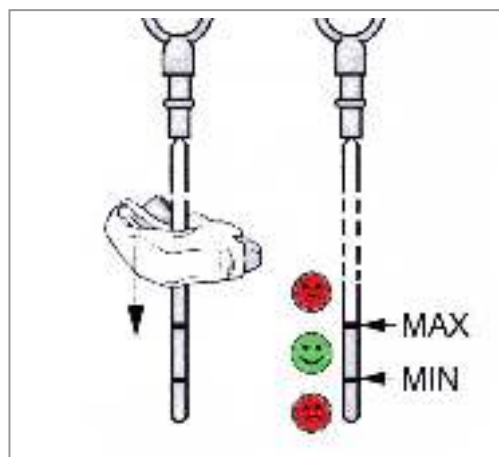
After oil refilling, let the engine run at low speed for a few minutes.

If the oil is not pressurised after 10 minutes, stop immediately the engines.

Check the oil pressure and his stability. Stop the engine. Check the oil level after 20 minutes.

- Extract the dipstick.
- Clean it with a dry, clean cloth.
- Insert the dipstick fully into its compartment.
- Extract the dipstick again.

The level should be between the two marks and must never descend below the lower mark. Top up if necessary.



Do not refill oil above the dipstick MAX mark. Too high oil level may cause engine troubles.

Oil filter replacement

The engine is equipped with a switchable double oil filter, therefore cartridges can be replaced also during operation, however both cartridges should be replaced during the same operation.

For continuous operation, set the switching lever so the two filter halves are operating.

Pay attention to the switching lever position.

Do not leave the control lever in intermediate position, as it may compromise the oil supply.



Unguarded rotating and moving engine components.

Risk of serious injury – danger to life!

- *Take special care when working on a running engine.*

Engine noise above 85 dB (A).

Risk of damage to hearing!

- *Wear ear protectors.*

Hot oil.

Oil can contain combustion residues which are harmful to health.

Risk of injury and poisoning!

- *Wear protective clothing, gloves, and goggles / safety mask.*
- *Avoid contact with skin.*
- *Do not inhale oil vapour.*

Damage to component.

Serious damage to plant!

- *For filter replacement with the engine running, operate the engine at low engine load.*
- *The filter which is to be exchanged must be cut out for a brief period only.*



Do not fill the filters with oil, prior to installation. This oil would not be filtered and would therefore be contaminated. Contaminated oil causes early wear of the engine parts.

Fuel system

The use of fuel which contains water damages the injection system. This problem can be avoided by refuelling immediately after operation when the fuel tank is still hot, preventing condensation.

Discharge periodically the fuel tank condense.



Fuels are combustible.

Risk of fire and explosion!

- *Avoid open flames, electrical sparks and ignition sources.*
- *Do not smoke.*

The system includes also **RACOR** separator filters which separate impurities and condense from the fuel. Do not use fluidity correction agents during winter operation.

Injection pumps

Do not tamper with the injection power unit and with the injection pump. The engine warranty will be void in case of lead seals damages.

It is advisable to contact workshops specialised and authorised by the Manufacturer for carrying out injection pumps maintenance.

Cooling system

- Two separate cooling circuits:
 - Engine coolant.
 - Raw water.
- Coolant cooling by raw water-cooled plate-core heat exchanger.
- Thermostat-controlled coolant system.
- Coolant-cooled / preheated charge air.
- Coolant-cooled fuel return.



Pay maximum attention during cooling liquid discharge while the engine is hot, risk of burns!

Cooling system emptying



Collect the liquid and dispose it according to regulations.

Discharge the cooling liquid only when the engine is cooled down, following the procedure shown in Manufacturers' Manuals.

Cooling system refilling

(only with cold engine)

The engine cooling system must be filled with a mixture of drinking water and ethylene glycol anti-freeze or anti-corrosion product.

Introduce the cooling liquid through the special filling opening. Do not pour cold liquid while the engine is at operation temperature. Make sure that the water/additive proportion is always correct.



For further information, please refer to the Manufacturer's Manual.

Alternator

The alternator is of the maintenance-free type, although it is necessary to protect it against dust and moist.

Follow the following prescriptions to avoid alternator failure.

While the engine is running:

- do not disconnect the battery main switch;
- do not unfasten battery terminals and electric network cable poles;
- if the battery charging warning lamp illuminates during engine operation, stop immediately the engine and eliminate the electric system failure;
- let the engine run only while controlling the charging conditions;
- do not short-circuit the alternator and governor connections one with the other or with ground;
- do not operate the alternator while the battery is disconnected.

Consult the builder's manual for the maintenance operations and inspections.



*The **SANLORENZO S.p.A.** shipyard **declines** any liability for damages caused by the inobservance of what written in the builder's manual.*

Generator Maintenance

Generator features

- Brushless, rotating field design permits power to be obtained from stationary leads.
- Rotor and stator are vacuum impregnated and coated with high-bond epoxy varnish. Varnish helps prevent corrosion in high-humidity areas.
- Rotors are dynamically balanced to minimize vibration.
- Copper windings ensure minimal heat build up. Insulation meets NEMA standards for class H insulation.
- Direct connected to the engine, the generator has sealed precision ball bearings with a precision-machined steel sleeve in the end bracket to prevent shaft misalignment and extend bearing life.
- Mounted on a drip-proof tray.
- Equipped with a four-lead reconnectable stator.
- Voltage adjustment $\pm 2\%$.

Engines features

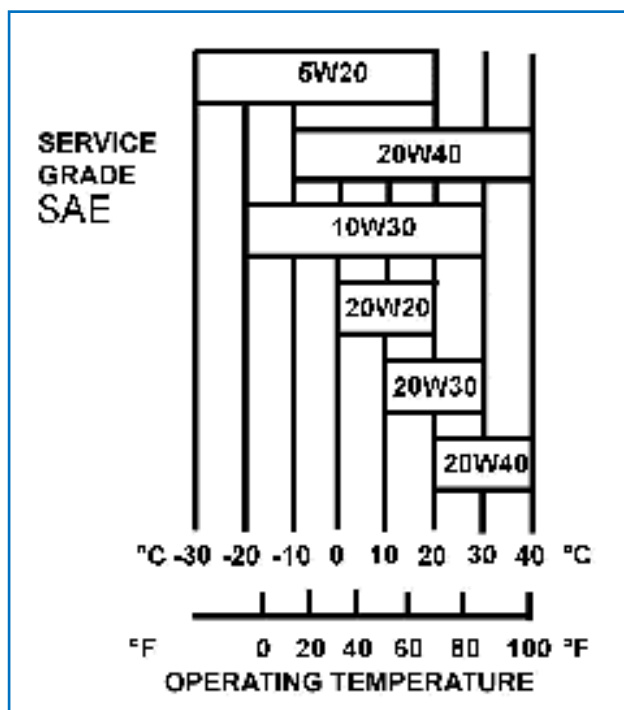
- Diesel fuelled.
- Four cylinder.
- Four cycle.
- Closed cooling system.
- Heat exchanger.
- Frequency regulation of $\pm 5\%$.
- Lifting eyes.
- Electric fuel lift pump.



Lubricating system

Oil Specifications

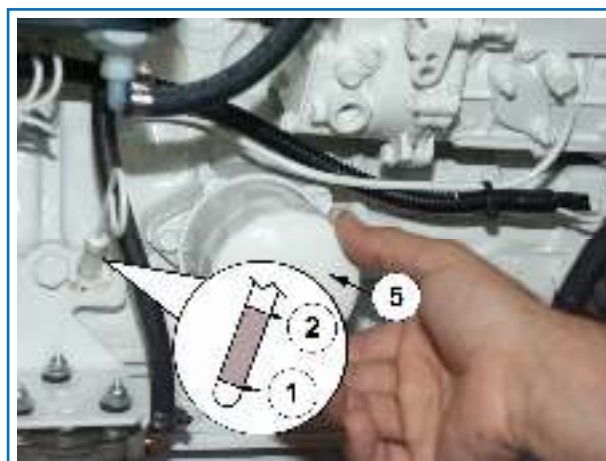
Only use oil in accordance with CD and CC/CD API classification. The illustration shows the recommended SAE viscosity at certain temperature intervals.



The non-respect of the oil specifications can cause unsuitable lubricant oil pressure and difficulties on engine cold start.

Oil check

Check the oil level every day or before each start up, making sure that the level is within safety limits. Remove the dipstick, clean its end, re-insert it completely and extract it again. Keep the oil level between the two marks Min. (1) and Max. (2) on the dipstick, as shown.



Do not start the engines if the oil level is not between the two reference marks.

Oil replacement

If possible, discharge the oil when it is still hot.

– Oil discharge

Set the generator start/stop switch to **STOP**. Disconnect battery charger power supply. Disconnect engine start up battery, starting from the negative terminal (-). Remove the oil discharge flexible pipe (3) from its mounting device. Remove the protection and place it in an exhausted oil container.

– Discharge/refilling procedure with electric pump

Connect the pump to the oil discharge flexible pipe end. Place the pump outlet in a container for the exhausted oil. Remove the oil refilling plugs (4). Open the oil discharge valve on the engine. Discharge completely the engine oil.

– Oil filter replacement

Remove the oil filter (5) by rotating it counter-clock wise with a special spanner. Apply a thin layer of oil to the rubber gasket of the new filter. Assemble the new filter following the supplied instructions.

– Oil top up

Fill with new oil until the level reaches the MAX mark on the dipstick.

– Leakage check

Check again that the generator start/stop switch is set to **STOP**. Connect the generator start up battery, ending with the negative terminal (-). Connect power supply to the battery charger, if present. Start up the generator and check for eventual leakage around the oil filter. Stop the generator and tighten the filter to stop eventual leakage.



Eliminate contaminating and exhausted materials (engine oil, fuel, filter, etc.) respecting the environment.

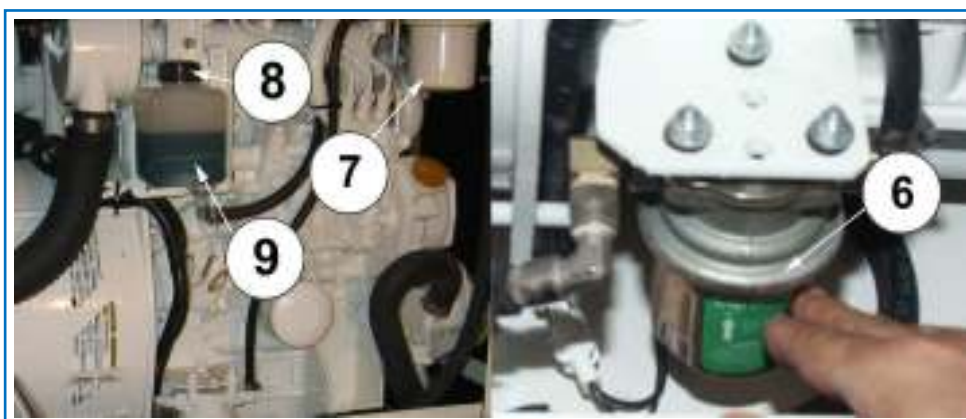
Fuel system

Cleaning / Replacing the fuel prefilter

The diesel fuel, sucked up from the tank, reaches the generator passing through a prefilter. This will hold impurities and separate any water, to prevent them from reaching the engine.

Follow this procedure to clean or replace the prefilter:

- remove the prefilter (6). Clean it or, if necessary, replace it;
- reassemble the element following the disassembly procedure in reverse.



Cleaning / Replacing the fuel filter

- Close the main fuel feeding valve.
- Loosen the fuel filter (7) turning it anticlockwise. Remove the fuel filter and clean any spilled fuel with clean cloths. Dispose of the fuel filter in the appropriate way.
- Clean the contact surface of the fuel filter adaptor.
- Lightly lubricate the seal of the new filter with clean fuel. Screw the filter on the adaptor until the seal makes contact. Tighten the filter by hand by one turn and a half. Always wash your hands after any contact with fuel.
- Open the main fuel feeding valve.
- Purge the system.
- Suck the air out.

Maintaining / Replacing the air filter

The air filter, fitted with sound absorber, reduces the noise caused by air suction and filters the air entering the engine. The sound absorber is connected to the air suction connector.

- Remove the sound screen and unblock the filter retaining system.
- Remove the air suction filter.
- Replace with a new filter.
- Carefully clean both the cover and the base with a clean cloth. Ensure that all the sealing surfaces are correctly coupled and block the retaining system.

Gas Discharge inspection points

Check that all the parts of the discharge system (waste collector, mixing elbow, waste pipe, pipe holder, sound absorber and flapper valve) for cracks, leaks and corrosion.

- Ensure that the flexible joints do not show any signs of softness, cracks, leaks and dents. Replace if necessary.
- Check all metal parts for corrosion or breaks. Replace if necessary.
- Check for loose, corroded or missing clamps. Tighten or replace tube holders and/or brackets if necessary.
- Check the waste outlet for obstructions.
- Visually check the system for any leaks. Check for carbon or soot residuals on the parts of the waste system. This would indicate the presence of leaks, which should be repaired.

Cooling system

Closed Heat exchanger

In a closed cooling system, the seawater circulates through separate chambers within the heat exchanger, or in the engine water-cooling collector system. The seawater mixes with the engine waste and exits through the discharge outlet. The quantity of cooling liquid needed by the engine is 6 litres (the capacity of the liquid collection bowl is 0.24 litres).



Warning: hot cooling liquid or steam can cause bad injuries or death.

Stop the generation and allow it to cool before removing the pressurized cap. Once the system has had time to cool down, carefully loosen the cap to release the pressure.

Let the engine cool down. Release the pressure from the cooling system before removing the cap. To release the pressure, cover the cap (8) with a thick cloth, and slowly turn the cap anticlockwise until the first stop.

Once all the pressure has been released, and the engine is cold, remove the cap.

Ensure that the level of the cooling liquid in the bowl is maintained at about $\frac{1}{4}$ of the bowl capacity. If necessary top up with more cooling solution to maintain the correct level in the bowl. If the cooling system has been emptied, before filling it again, ensure that all bleeding taps are closed and the tube holders are tight. Use a solution of 50% ethylene glycol and 50% of water, softened and clarified, to prevent rust and corrosion and to avoid freezing.

Periodically check the level of the cooling solution in the closed system, by unscrewing the pressure cap. Do not only rely on the level in the collection bowl. Add cooling solution until the level is slightly lower than the opening of the overflow pipe.



Salt water damages.

Salt water rapidly spoils metals.

Deleting any trace from the generator and around it, and eliminate any salt deposits from metal surfaces.



A 50% solution of ethylene glycol can protect from freezing down to -37°C, and from overheating up to +129°C. A lower strength solution might be unable to guarantee appropriate protection against freezing and overheating. A higher strength solution can damage the engine or other components. Do not use alcohol or methanol based cooling solutions, and do not mix them to the cooling liquid. The Manufacturer Operation Manual for the engine contains specifications for the cooling liquid.



Do not top up the cooling liquid while the engine is hot. This could cause the cylinder to jam or the cylinder head to crack. Wait for the engine to cool down.



Pay particular attention to the cooling liquid level. After draining it completely, refill it, until the engine water jacket is completely full.

Washing and cleaning

For optimum protection, drain, wash and refill the cooling system in accordance with the intervals stated in the maintenance program.

Washing and cleaning procedures

- Open the pressure cap and the draining valves of the heat exchanger, the engine block and the cooling system. Allow the system to cool down completely. In some models the drain valves may be hidden behind the belt protection. Remove the pressure cap to facilitate draining.
- Empty, clean and wash the cooling liquid collection bowl.
- Wash the system down with clean water.
- Fill the system with the recommended cooling liquid.

Pressure cap

Closed heat exchanger systems are fitted with a pressure cap to increase the boiling point of the cooling liquid, thus allowing a higher operating temperature. If the cap is leaking, it should be replaced with one with the same pressure strength. Normally the pressure strength is stamped on the cover.

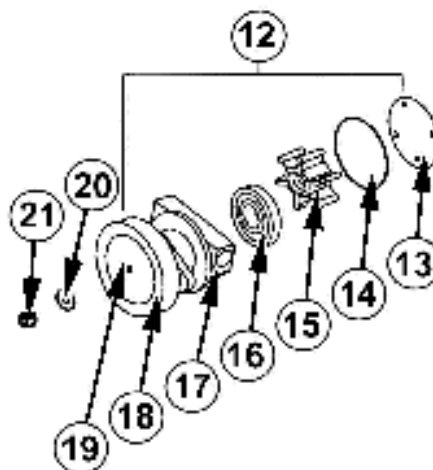
Seawater pump

The seawater pump, operated with the aid of the belt, is located on the service side of the generator. The impeller of the pump should be checked and replaced in accordance with the intervals indicated in the maintenance program. Follow the instructions supplied with the impeller kit. Should they not be available, follow these instructions:

Impeller control and replacement procedure

- Close the sea cock valve.
- Remove the pump cover plate.
- Remove the impeller.
- Check the blades for any damage, cracks, breaks, flattening, or if any of them are missing. They should be straight and flexible.

- 12. Sea water pump.
- 13. Cover plate.
- 14. Gasket.
- 15. Impeller.
- 16. Bearing.
- 17. Encasing.
- 18. Pulley.
- 19. Driving shaft.
- 20. Washer.
- 21. Nut.



- Lubricate the vent with soapy water and install.

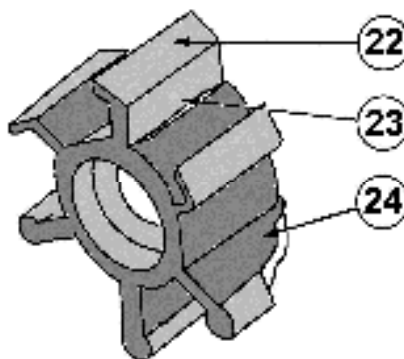


During installation, push and rotate the impeller in the same direction of the engine rotation, until it fits correctly in place.

- Check the cover plate and its gasket for corrosion or damage. Replace if necessary.
- Lubricate the gasket with silicon based grease. Fasten the gasket and the cover plate to the encasing of the seawater pump.
- Open the sea cock valve.

- Start the generator and check for leaks.
- Stop the generator and repair any leaks, or replace any damaged or worn parts.

- 22. Flattened blade.
- 23. Crack.
- 24. Broken blade.



Anticorrosion galvanized anode

The head exchanger is fitted with a galvanized anode (cap) with the aim to prevent electrolytic corrosion caused by seawater. Check and replaced the galvanized anode in accordance with the intervals indicated in the maintenance program. Depending on operating conditions and seawater conditions, the anode might need replacing more frequently.

Replacement

- When the generator is cold, close the sea cock valve, open the drain valve on the engine and drain the cooling liquid in an appropriate container.
- Remove the anode (cap) from the heat exchanger.
- With a metallic brush remove any traces of corrosion from the anode. Replace if necessary.
- Clean the threaded hole of the heat exchanger, and coat the threads of the anticorrosion galvanized anode (cap) with a tube sealant suitable for marine applications.
- Close the engine drain valve and open the sea cock valve. Refill the cooling system.
- Start the generator and check for leaks at the position of the galvanized anode. If seawater comes out from the discharge outlet, the pump is working correctly.

For a correct operation of the generators, it is advisable to read the instructions Manual supplied by the Manufacturer, and however, follow the following precautions:

- before starting the generators, check oil and fresh water level in the internal cooling circuit;
- immediately after start up, let stabilise the engine temperature and apply gradually the electric load to the generator;
- while generators are operating, check periodically the presence of eventual alarms;
- before stopping the generators, it is advisable to let them running for a few minutes without load.

The power generation system of your yacht are equipped with a safety system which disconnects the generators in case of lubrication system or cooling system failure.

Consult the builder's manual for the maintenance operations and inspections.



*The **SANLORENZO S.p.A.** shipyard **declines** any liability for damages caused by the inobservance of what written in the builder's manual.*

Electric System Maintenance

Component	Maintenance	Notes and Cautions
Batteries and battery charger	Periodical checks (at least every month)	When the boat is left unattended for long periods, the service batteries could run down, due to automatic bilge pump frequent operations (in this event, check the shaft stuffing boxes). During long inactivity periods, disconnect the batteries from the system; or, as an alternative, leave the system connected and recharge all the batteries periodically (generator battery included). Check the efficiency of each single battery.
Electric lines and panels	Periodical checks (every six months)	Check electric panel and junction box terminal boards conditions and electric system insulation at least twice a year.
Exterior and interior lights	Periodical cleaning (as required)	Keep the protection screens always clean.



Batteries and terminals check

- Keep the battery 100% charged, recharging it every 2/3 weeks if unused.
- Clean the cover and the body if soiled.
- Check the terminals and grease them if necessary (use vaseline).
- Check to see whether liquid has spilt out of the battery.



Do not disconnect battery cables while generator is running.

Personnel operating on batteries should not wear bracelets, rings and similar accessories.

The following indicative procedures should be carried out by specialised personnel.

- Make sure that battery containers are clean and dry, and that terminals are mounted and externally protected with silicon grease. Clean and grease if necessary.
- Positive and negative cables should be identified before connecting (to avoid sparkles, connect the negative cable after the positive).



Always disconnect the negative terminal (-) before the positive and connect it after the positive (+).

Switchboards check

- Make sure that the internal part of the switchboard is clean.
- Check operation of all switches.



Before operating on the switchboard, inhibit both generators operation and shore power supply.



Warning: risk of electrocution!

Never access the internal parts of switchboards or electric/electronic equipment without having disconnected the power supply.

The SANLORENZO S.p.A. shipyard declines all liability for damage caused by setting them at a different temperature.

Connection and cabling

- Observe the state of the electric system in order to check for corrosion and / or oxidation (cable-ends, fuse holders, etc.).
- Check, where possible, that the bolts fixing the cable ends have been properly tightened.
- Check the condition of the cables, paying particular attention for signs of wear or damage on the cable insulation exterior.
- Check the batteries, in particular, check for signs of oxidation on the terminals and acid on the battery body.
- Maintenance of the terminals is essential since through them the battery provides electric power. In particular, for chemical reasons the positive pole (+) of the battery tends to become covered in sulphur turning white (white sulphate crystals). In time this chemical compound tends to isolate the battery from the system thus greatly decreasing the maximum energy charge that the battery can provide. To avoid this problem, cover the terminals (do the same for the negative pole) with common vaseline: the terminals will thus be permanently protected; check the condition of the vaseline from time to time.
- Carefully check the condition of rubber seals or the waterproof and airtight doors of the electric panels or of the control panels.
- Check the condition of the earth circuit, of the anodes and of the copper plates.



Warning

All checks must be carried out by qualified staff or by the shipyard.

Fuel System Maintenance

Component	Maintenance	Notes and Cautions
Separator filters	Drain separator filters (after every exit at sea, or at least once a month)	The fuel filters are equipped with a transparent bottom tank and a discharge valve for the drainage of water or impurities. <i>Do not discharge the bilge containing drained fuel into the sea: collect it and use toxic waste disposal containers which are available in the harbour.</i>

Fuel contamination

The most common problem related to the fuel is contamination from water, from weed's micro-organisms, bacteria, fungi and various residues. Individually extremely small (100x smaller in diameter than a human hair), they form colonies that grow into long strings and clumps which can clog fuel lines and filters. Algae can survive in the carbon and hydrogen of the fuel and in the oxygen dissolved in the water and on the bottom of the tank. If the tank has water on the bottom, the algae problem can become even more serious. The by-products of algae are acid and sludge which contribute to contamination and corrosion in the tank and engine. Biocides can be very effective in killing off algae, however, the dead organisms accumulate in the bottom of the tank.

After time or in rough water, the contamination in the bottom can become suspended in the fuel causing plugged filters and motor operation problems.

Indicators of contamination are:

- Sludgy fuel filters.
- Frequent filter changes.
- Low rpm at full throttle.
- Dark exhaust smoke.
- Soot on transom.
- Water in fuel/water separator.
- Motor stoppage due to lack of fuel.

Contamination origins

When diesel leaves the refinery, it is clean, with acceptable water content. Every stop the fuel makes on its way from the refinery to your boat is an opportunity to pick up algae and water. By the time it reaches your tank, the fuel has become contaminated to some degree. The problem can become worse in your tank if water is present or condensation is allowed to accumulate. The darkness of a fuel tank, especially in warm climates, is the ideal environment for algae growth.

Problems caused by contamination

Operational problems can occur with only trace amounts of contamination. The most common problem is prematurely plugged fuel filters and plugged water separators. In severe cases, the fuel line can become partially or totally blocked. The micro organisms form deposits on injectors that can lead to inefficient fuel injection and reduced operation economy. These conditions can lead to unanticipated engine stoppage which can be a serious problem especially if you are navigating in tight areas or in bad weather. Corrosion of metal parts is a potentially costly problem that is caused by the acid by-product of algae, hydrogen sulphide. This acid can corrode fuel tanks, injector pumps, injectors and fuel lines and can also attack hoses and tank linings. The problem is particularly severe in fuel tanks where the algae mat traps both moisture and acid against the tank surface. If undetected serious tank damage can result.

Start with clean fuel

The bad things that happen to diesel fuel inside your tank also occur inside the supplier's storage tanks. That isn't a problem for you if the fuel is efficiently filtered on the way to the nozzle. If you have any doubts about the cleanliness of the fuel, pump some into a clean glass jar and let it sit a few minutes. Water and dirt will settle to the bottom. If you see either, filter the fuel before it goes into the tank, or better yet, buy your fuel somewhere else.

Filters

Inspect filters daily. Water in the bowl is visible evidence of water in the tank. If you need to drain the bowl (it has a plug just for this function) more often than every 100 engine hours, the tank needs cleaning. Filters should be changed when they are dirty, but unless your primary filter is fitted with a vacuum gauge (which reveals the extent of restriction) you will need to change the element at some regular engine-hours interval. Your engine manual should specify.

Check tanks for water frequently

If you suspect water contamination in your tank use a water finding paste to check tank bottoms. The water finding paste is a paste that changes colour in a presence of water, spread a thin layer of paste over a dip stick and insert the stick into the lower portion of your tank. Remove any water that is present.

Keep out water

Also minimum quantities of water in diesel fuel can damage the injectors. This is the reason for having a water-separating filter, but too much water in the fuel can overwhelm this filter.

Using fuel contaminated with water is one of the possible ways to bring water into your tanks. It can also enter from the filling cap of the tank or a breather valve or due to condensation forming in the tank. If the tank cap's seal isn't airtight, replace the gasket or the fitting. Due to the daily heating/ cooling cycle, a small amount of moisture will condense out of the air every day. The more air in the tank, the more potential for condensation, so it is a good idea to keep tanks topped off, especially while the boat is not being used.

Prevent algae

The less water there is in the fuel, the less weeds will form. However, keeping all water out of the fuel is improbable, so diesel fuel should always be treated with a biocide to prevent the growth of microbes. Avoid products that contain alcohol, which attacks O-rings and other rubber parts in the fuel system.

Look for source of water

If a significant amount of water is present in your tank. Seals on fuel fill caps and fuel sender units are frequently the problem.

Keep your fuel tanks full

Especially if your boat must remain unused for prolonged periods. A full tank minimizes the formation of water by condensation.

Don't buy fuel soon after a fuel delivery

Contaminants which could be present in your supplier's fuel pump could be pumped into the tank while it is being filled. It takes about twelve hours before the residue suspended in the fuel settle back down to the bottom.

Address any contamination problems

Eliminate any contamination problems as soon as possible.

They won't go away by themselves and can only get worse.

It's better to nip the problem in the bud.



Do not disperse noxious substances into the environment.

Use procedures and equipment suitable for this operation.

Fuel purifier - Maintenance

For all procedures concerning trouble check maintenance, alarms, end of season, regeneration cycle washing and winter storage, please refer to the Manual supplied by the Manufacturer.

Grey / Black water system maintenance

From the operation point of view, this system is very simple and, with just a few precautions, will give good service:

- do not allow the various conveniences' hydraulic syphons to empty (on account of yacht movement or long periods of inactivity); refresh them by pouring a little water into the drainage bowl to prevent a build-up of unpleasant smells;
- use special filtering meshes protecting the various conveniences to avoid, as far as possible, the introduction of solid objects detrimental to the operation of the electric pump and level regulator;
- disinfect the tank and circuits periodically by pouring bactericides down each convenience's drainpipe;
- wash the tank periodically with fresh water and special lime removing detergents, repeating several times so as to prevent the formation of deposits or blockages;
- during periodical tank washing, check that the ballcocks and alarm lights on the bridge switchboard are in good working order by operating the switches that control the electric emptying pump;
- to handle any emergency situations or maintenance requirements, all the system's electric pumps (2 vacuum pumps, 1 sewage emptying pump) are fitted with intercept valves on the suction and delivery lines. Once the emergency has been dealt with or the maintenance operation has been completed, remember to reopen the valves in question to avoid damage to the pumps.

Consult the builder's manual for the maintenance operations and inspections to be conducted on the electric pumps.



*The **SANLORENZO S.p.A.** shipyard **declines** any liability for damages caused by the inobservance of what written in the builder's manual.*

Hull structure and furnishings maintenance

Component	Maintenance	Notes and Cautions
Gel-coat	Periodical cleaning (as required)	In some areas of the boat, the gel-coat may show bubbles, which can break exposing the fibres underneath. The drawback occurs generally in vicinity sharp angles, and depends on air bubbles that, during fabrication, remain entrapped between fibreglass and gel-coat, although Quality Checks are carried out by specialised personnel. Broken gel-coat bubbles are easy to repair by filling the voids and touching up with gel-coat that can be requested to the Service Department. <i>In case of particularly persistent dirt, wash always by using neutral products. Do not use products containing ammonia that turn the surface yellowish.</i> When underway, some structural parts of the boat are subject to bending, and create tension or compression stresses in fibreglass and on gel-coat. The different elasticity of gel-coat and fibreglass, can cause small cracks on the gel-coat surface, in particular in the more stressed areas, e.g. near bollards, stanchions, etc.. Such drawback, however, does not compromise fibreglass mechanical and structural characteristics.

Component	Maintenance	Notes and Cautions
Hull	Periodical cleaning and anti-fouling treatment check (as necessary according to the area where the boat is moored) Check/Restore	The duration of the anti-fouling effects depends mainly on the conditions of the waters where the boat is stationed. To remove the old anti-fouling, do not use sandblasting that may damage gel-coat surface, the first layers of resin applied by the Manufacturer and the anti-osmosis treatment. As suggested by the anti-fouling manufacturers, use paint removers or, as an alternative, wet sanding without damaging the protective treatment and the gel-coat. <i>Small areas of paint may peel off from the propellers even after a short period of operation.</i>

Hull fouling causes considerable speed reduction and may damage the "gel-coat". You should choose an anti-fouling suitable for Your yacht and for the type of water You intend to sail in. Please refer to SANLORENZO Service Department.

Check / Restore

Cleaning and careful check should be carried out while the yacht is in dry-dock. Checks and inspections can be carried out while the yacht is in the water, by means of a diver. Restoration should be carried out in dry-dock.

- Clean carefully hull, rudders and flaps with brushes (in water) or with hydro-cleaner (dry dock) to eliminate seaweed and fouling.
- Inspect the hull coating condition. If necessary, specialised personnel should apply two layers of anti-fouling coating (in dry dock).



When carrying out cleaning and checks while the yacht is in the water, engines and generator start up should be inhibited.

Surface preparation on a yacht already treated

Check carefully the old anti-fouling coating and decide whether a new layer needs to be applied. Make sure that the new product is compatible with the old one. Please refer to SANLORENZO Service Department.

If the old anti-fouling is crusty, thick and scaling, it should be removed completely and the new treatment should be applied as for a new yacht. Use a liquid remover, compatible with fibreglass and remove the old layer with a steel scraper.



Anti-fouling product is toxic. Do not burn it. Use only authorised procedures and refer to specific Authority in case of doubt. Sanding and anti-fouling removal should be carried out wearing suitable protective clothing.

During anti-fouling application make sure that the following hull parts are not coated:

- depth-sounder transducer;
- speed log paddlewheel transducer;
- sacrificial anodes;
- propellers and shaft lines;
- ground plates.



The yacht is delivered already treated with anti-fouling. If you decide to use a different treatment, please refer to SANLORENZO Service department.

Component	Maintenance	Notes and Cautions
Sacrificial anodes	Periodical checks (as necessary according to the area where the boat is moored, but at least every two months)	The underwater metallic items are protected against galvanic corrosion (caused by two dissimilar materials in contact, such as steel and aluminium) by zinc anodes installed on the hull, flaps, and shaft line. In addition to galvanic corrosion, the zinc anode consumption depends on ambient factors such as long chains in the vicinity, metallic hulls or metallic piers, poor isolation of shore electric systems.

Periodical check

This operation should be carried out while the yacht is in dry-dock or by means of a diver if the yacht is in the water.

Carry out anodes (zincs) visual inspection and replace them when consumption is greater than 50%.



Failure to replace worn away anodes will result in corrosion of other metallic fittings.

Before carrying out with this procedure, ensure to take the necessary measures, to prevent the propeller engines and the generator from starting.

Assembly / Disassembly

Sacrificial anodes are fixed to the hull in various positions. It is advisable to clean the anode mounting seat and cover the anodes mounting screws with silicon: this operation will ease worn out anodes replacement. Do not tighten anodes fixing screws with adhesive or other devices which should prevent the removal.

Component	Maintenance	Notes and Cautions
Speed log paddlewheel transducer	Periodical cleaning Replacement (once a year)	Keep always the speed log transducer clean. Fouling of any kind affect its operation and as a consequence the data provided. Water-lubricated paddles duration is approximately one year; they may be damaged and bent due to impact with floating objects or manumission in dry dock. O-rings should not present wear or cuts to ensure sealing. Clean the paddlewheels, by removing the transducer from its hull fitting. If the boat will not be used for long periods, remove the transducer from its hull fitting and replace it with the dummy plug.
Underwater exhausts	Periodical checks (as necessary according to the area where the boat is moored, but at least every month)	Check the underwater exhaust terminal cleanliness conditions periodically. Clean, if necessary. <i>Carbon deposits, marine growths and fouling may affect the engine regular operation, causing performance degradation and serious damages. Inspect periodically the exhausts, making sure that they are free from obstructions. Their obstruction, even if partial, may compromise the corrent engine operation.</i>

Periodical check

This check consists of inspection and eventual fouling removal carried out by a diver. Make sure that propulsion engines are stopped during this check.

Component	Maintenance	Notes and Cautions
Shafts and propellers	Periodical checks (as necessary according to the area where the craft is moored, but at least every three months)	Keep propellers and shafts always clean; calcareous formation, parasites, or the presence of foreign matters such as ropes, barnacles, seaweed, rags, plastic bags cause a propulsion efficiency reduction, propeller damaging cavitation and vibrations, with consequent damages to the shaft seals and stuffing boxes. Inspect and clean propellers and shafts when the boat is out of the water, or by using a diver. During the summer, and when the yacht is in particularly warm sea, checks should be carried out more frequently. During long inactivity periods it is advisable to tighten stuffing box nuts to avoid automatic bilge pump periodical operation. <i>If the yacht is not used for long time, rotate periodically the shafts for a fraction of turn to prevent fouling formation between shaft and bushings.</i>
Shaft bushings	Periodical checks (at least every six months)	The shaft neoprene bushing, may worn out rapidly, mainly if operating in sandy water. When worn out, the bushing causes strong vibrations. When the boat is out of the water, a good technician can easily evaluate the bushing replacement need, by moving the shaft.
Shaft seals	Periodical checks	Check the shaft seals periodically, and in any case, before leaving port and during the navigation (only a slight water dripping is allowed).

Component	Maintenance	Notes and Cautions
Teak deck	Periodical cleaning (as required) Sanding (as required)	Teak deck covering does not require particular maintenance, except for periodical wetting with fresh water. Scratches and stains may occur on the teak, therefore slight sanding should be carried out to eliminate deteriorated areas. Avoid deep and localised sanding which would cause depressions and areas with different colours.
Light alloys and stainless steel	Periodical cleaning (as required)	It is a good rule, to wash down the entire boat after each navigation, in particular all metal parts, that are damaged by salty humidity. Hose handrail stanchion bottom plates, windows, skylights, anchors, cleats, ladder and any other area where sea water may stagnate with plenty of fresh water. <i>Do not use brushes or abrasive rags on metallic fittings, not even on rusty spots, scratches on the surface result in a less shiny appearance.</i> Protect all metal parts with vaseline oil periodically.

Characteristics of the teak

Teak has natural properties, which make it ideal for the marine environment. In the condition seen when your boat is new, the teak shall be a "light golden colour", and you may want to maintain that colour.

However, teak is very light sensitive and will become bleached by prolonged exposure to sunlight.

The best way of maintaining your teak deck in this original colour is to scrub it regularly with a soft bristle deck brush and plenty of FRESH water. This will have the effect of cleaning the pores of the wood and killing any algae growth which occurs when fresh water is present for long period of time. It has been found that if stubborn dirt cannot be removed with water alone, then a scrub with soapy water will usually remove any stains.

Do not be afraid to use plenty of water on the teak deck as this will help to maintain the equilibrium of the wood, especially in hotter climates.

Should you prefer the "light grey colour" of seasoned teak decking, the natural process will stabilise after a few months leaving a light, almost grey appearance to the teak.

The best way of maintaining your teak deck in this condition is to scrub it regularly with a soft bristle deck brush and plenty of SALT WATER.

We do not recommend the use of any proprietary teak finishes i.e. oils, stains or varnishes.

Characteristics of the stainless steel

Stainless steel is not totally resistant to corrosion.

Where stainless steel may be exposed to a corrosive environment, such as marine environments, polluted surroundings or other situation, it is essential to follow these maintenance tips to keep hardware in top shape for many years to come.

- Clean stainless steel hardware frequently with soap and water.
- Any cleaner safe for glass is usually safe for stainless steel.
- Never use coarse abrasive like steel wool or acids or bleach on stainless steel. These may actually cause rusting.
- Remove any rust spots as soon as possible with a brass, silver or chrome cleaner. Pitting can develop under rust that remains on stainless steel for any period.
- Use a good quality car wax for added beauty and protection. Alternatively use one of the special stainless steel polishes available in good chandlery shops.

Never leave stainless steel in contact with iron, steel or other metals, which cause contamination leading to rust or corrosion.

Component	Maintenance	Notes and Cautions
Windshield	Periodical cleaning (as required)	Wash the windshield with fresh water after each navigation. <i>Do not use rags or anything other than water, because salt sediments or soil can damage the surface.</i>
Windshield wipers and windshield washer	Periodical cleaning (as required)	Wash them carefully with fresh water and coat with vaseline oil; grease the spring with silicone grease. Check the rubber blades conditions periodically, and replace the blades if worn; and prevent bad visibility problems.
Flying bridge instrumentation and navigation lights	Periodical cleaning (as required)	Use clean wet rags for cleaning. <i>Do not use chemical or abrasive products.</i> At the end of navigation, after washing and drying, cover instrumentation and equipment.

Characteristics of internal woodwork

Every piece of SANLORENZO wood veneer is unique. As every tree is individual and unique this means that, only rarely, will there be an exact match between individual panels.

- Wood is natural, organic material. Variation in colour and grain structure will occur between every piece of wood veneer.
- Grain direction will give a different appearance of colour when viewed from different angles, which is why, when individual panels are placed next to each other, they often appear to be of a different colour if the grain is running in different directions.
- Exposure to daylight over a period of time will gradually change the colour of veneer. Generally woods such as Cherry will darken.

Caring for veneered surfaces

All material used in our manufacturing process are chosen for their robust performance and will withstand normal wear in a boating environment. SANLORENZO wood veneer finishes should be dusted regularly with a clean dry cloth. More difficult stains can be removed by using a soft cloth, dampened with warm water and gentle detergent; dry and polish immediately with a soft dry cloth.

Wood being an organic material is prone to movement and colour variation caused by environmental conditions. Exposure to ultra violet rays over a period of time will alter the colour of the veneer and the use of window blinds and sunscreens are strongly advised. The use of storage heaters and de-humidifiers will also protect the wood from extreme humidity and temperature levels, which can cause damage to the lacquered finishes.

Characteristics of furniture upholstery

It is important to remember that the upholstery for the sofa and chair slipcovers lightens with the sunlight and therefore, it is important to keep the curtains closed to prevent this from happening, especially in areas adjacent to the windows in the saloon, etc. Follow the instructions for treating the various fabrics as given in the following chart.

Stain or dirty marks

Clean immediately using a damp cloth or a proprietary brand foam cleaner. Be careful not to over-wet upholstery. In the case of stubborn stains, use a proprietary brand stain remover. Make sure the stain remover is adequate for the interested fabric, try first on a small spot before extending the treatment.

General cleaning

Use a special detergent or remove the part from the boat and bring it to a specialised company. It is not advisable to take the covers off the foam and dry-clean these separately. The upholstery has been purpose built to fit into very specific areas and in many cases the foam and timber parts can not be removed from the covers.

Winter maintenance

Keep dry and aired.

Summer maintenance

Vacuum clean regularly.

You can find more specific instruction on permissible and suggested method of cleaning upholstery in the attached table.

Use this table and the SANLORENZO Decor book to identify the affected type of material and then check the suggested care instruction prior to take any cleaning action. In case of troubles in identifying the textile code please contact your SANLORENZO dealer.

If you take on a specialist cleaner, show them the following table, identifying the affected textile to help finding the most appropriate treatment.

External upholstery

The vinyl used for external upholstery is a top quality hardware-ring fabric. The fabric is coated with lacquer finish designed to protect the fabric from soiling. It is therefore very important not to use an abrasive cleaner on the external upholstery. Wipe off any marks with a soapy cloth, any proprietary brand liquid cleaner (the type suitable for cleaning plastic baths) or a spray cleaner. For general day-to-day cleaning use net "flash" cleaning liquid.

Avoid staining the upholstery with sun lotion by protecting the cushions with a towel.

Winter maintenance

Clean the cushions and dry them before winter storage. Spraying the cushions with a wax waterproofing agent can help, but it will need to be cleaned off again in the spring.

Summer maintenance

None required as vinyl has good UV protection.

Prevention of dampness and mildew

Keep cushions dry and clean. A protective coating of waterproof wax will stop dampness and mildew. The vinyl itself is mildew resistant.

Cushions will let water in through the seams if they are not protected against wet conditions. External cushions should therefore be stowed away or protected by the cockpit hood in such conditions. If they have to be left exposed they should be stacked vertically.

Leather care and cleaning tips

There are four stages in leather care that include cleaning, conditioning, polishing, and protection.

There are a good variety of product on the market that are formulated to be compatible with today's water-based leather finishes and will not damage the leather surface. You may use them on all of your finished leather seating. They are available as a sprayable products or pre-moistened wipes. Finally, if you are in doubt, we recommend seeking the advice of a professional leather cleaner.

Keep the leather pores free from dust particles. Wipe completely with a soft rag, preferably once a week. Never use saddle soap, solvents, abrasives or caustic household cleaners such as soap or dish detergent. Common cleaning materials and chemicals can cause severe damage and costly repairs.

If circumstances require to clean up the leather without using a proper leather cleaning product please use a gentle cleaner. Direct sources of heat and extended exposure to direct sunlight may also be harmful to the leather and should be avoided.

Nubuck and suede should be routinely brushed and vacuumed to remove topical soil. Spills should be immediately blotted with an absorbent white cloth whenever they occur. Stains may be buffed with an ultra fine abrasive pad (e.g. 3M White) using just enough pressure to remove the stain. A soft bristle brush will help restore the soft suede or nubuck nap.

Cleaning

The cleaner of your choice should not leave any greasy residue behind. Residue makes leather susceptible to bacteria and can break down the stitching of your item. Before applying anything to your leather item, be certain to test it out for effect and possible colour distortion on an area that isn't visible to the eye. Once you've ascertained whether the leather care product is acceptable to use, apply it to your item. With a slightly dampened cloth, remove the cleaning product. For areas with stitches, there are brushes available on the market. Another cleaning product to consider having in your leather care collection is a nubuck cleaning cloth. They have an astonishing ability to clean and restore leather to its original look.

Conditioning

Leather conditioners are meant for occasional use. They contain fats and/or oils that help lubricate leather and replenish the suppleness. Look for a product that will penetrate the strong fibres in leather, but beware of any that include petroleum or mineral oils. While petroleum by-products won't damage your leather immediately, they do over a period of time. Again, just as with cleaning, keep on the look out for thick, greasy conditioning treatments for the best care of your leather.

Polishing

Polishing is done for special occasions when you want a more glossy finish on your leather. There are a couple things to be wary of when purchasing a polishing agent. Some products contain colouring factors that will brush off on things you come in contact with. Some products also have a tendency to clog the pores in leather or dry leather out. Just as with cleaning, be sure to test out the product on a small area and when ready, buff to a shine.

Protection

Moisture barriers are extremely crucial in preventing rain or other liquid hazards from damaging leather. Stiffness and spouting will happen if leather isn't protected beforehand. There is a drawback in protecting leather with a moisture barrier product. They tend to fill in the pores with a greasiness that makes cleaning, conditioning, and polishing difficult, but it's a necessary process to ensure leather isn't destroyed. Periodically apply a moisture barrier and allow it time to penetrate and dry before using your leather item.

Removing mildew

To remove mildew from leather, create a mixture of one-cup rubbing alcohol per one-cup of water. Wipe the mildew area with a cloth dipped in the diluted alcohol, then allow it to dry. If the mildew persists, use mild soap and water that contains a germicide, then remove with a clean dampened cloth and allow drying.

Wet leather

An important key to keeping leather in top-notch condition is to treat wet leather before it has a chance to dry. Remove any dirt, mud, or other stains with a cleaning agent, then condition while the pores are still fully responsive. It is critical to remember that leather should be dried away from heat.

Storing leather

Remember that leather is a natural material and should never be stored in plastic because it encourages the growth of mildew and bacteria and will ruin the leather. Always store leather in a cool, dry place away from heat.

Removing stains

Fresh stains from things such as blood and food can be cleaned up quickly with a damp cloth. Stains from oil or grease can be lifted by grinding ordinary blackboard chalk, sprinkling the area, and leaving the powder on for a twenty-four hour period. Resist the urge to rub the powder in. After a sufficient time has past, simply use a leather care brush to remove the powder. While fresh stains can be treated and cleaned on board, ground-in stains should be attended to by a professional cleaner who deals in leather.

Ink marks

Treat early for best results.

Finished Leather: Use Ink & Stain Remover product followed eventually by Leather Cleaner product, and then Conditioner and Protector products.

Aniline Leather: Rub a white eraser quickly and firmly over the mark. Test in an inconspicuous area in advance.

Grease

Gently lift excess grease with dry, soft cloth. (see also above).

Finished Leather: Clean with Leather Cleaner product, then treat with Conditioner and Protector product.

Aniline Leather: Sprinkle cornstarch over affected area and cover with dampened white cloth for 4-6 hours; remove cloth and cornstarch. Repeat as needed.

Wax

Place ice cubes in plastic bag over wax until it becomes firm enough to gently lift from surface. Treat any residual marks as grease.

Gum

Gently heat with hair dryer. Use clean white cloth to rub gum from surface onto cloth. Treat any residual marks as grease.

Cleaning products

On the market you could find a good variety of products (3M, Lexol, Zymol, Mothers, Townsed, Sonax, etc.) specifically made for leather care; carefully read and follow the directions on the product label.

Always test these products on an inconspicuous area before using it extensively. Finally, if you are in doubt, we recommend seeking the advice of a professional leather cleaner.

Never use

Saddle soap, abrasive cleanser, soap, furniture polish oil, varnish, or ammonia water. The leather has already been permanently preserved in the tanning process and needs no maintenance other than the simple cleaning recommended.

Most liquids will be initially repelled by the leather, but if left to stand over an extended period of time, they will be absorbed. This is why it's a good idea to wipe up spills immediately.

If the spill is absorbed, it will dissipate over time, just as human skin will absorb and eventually diffuse stains.



The preceding remedies are in accordance with general cleaning industry procedures, cleaning shall be adapted case by case and according to your experience on the subject.

SANLORENZO S.p.A. shipyard cannot therefore guarantee favourable results, and we assume no liability in connection with their use.

MATERIALS FOR INTERIOR FITTINGS



INSTRUCTIONS FOR CLEANING, MAINTENANCE AND USE.

SANLORENZO S.p.A. shipyard cannot therefore guarantee favourable results, and we assume no liability in connection with their use.

CORIAN:

Component	Notes and Warnings
<u>Corian furniture and surfaces 1/2</u>	Corian is a high-durability material and easy to clean. It is by its very nature restorable, since it is uniform throughout its thickness. You should however avoid subjecting it to impact, heat or chemical products. On installation a Corian surface has a shiny satin look. With use, it could take on a softer and lighter look. We ask you to consult our guidelines so that this transformation takes place uniformly.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. Always follow the simplest methods for daily cleaning. Even if liquids cannot penetrate the material, it is best to remove them immediately. Use a damp cloth with a light abrasive cream or a common de-greasing detergent (e.g. Cif) always using a circular motion and finally dry with a soft cloth. 2. For Corian sinks: remove all the oil or grease residues from normal food preparation once or twice a week using a detergent or a solution for solid surfaces. Spray a 3/4 bleach and 1/4 water solution on the sink and leave it to act for a few hours or overnight. The exposure time should be limited to 16 hours. Rinse or clean with a damp cloth in the morning. We recommend you dry sinks carefully after each use in order to prevent the build up of lime scale. 3. Avoid pouring boiling liquids directly into the sink without first opening the cold water tap. 4. Do not place boiling pans (particularly cast iron ones) on Corian surfaces, but always use a pan stand with rubber feet or some form of protection. 5. For surfaces next to the hob: always use a pan with a size corresponding to the ring/plate, positioning it centrally. Too big a pan can damage the surrounding surface. 6. Like any material, Corian could show slight signs of abrasion over time after daily use: avoid cutting or chopping on the surface.

CORIAN:

Component	Notes and Warnings
<u>Corian furniture and surfaces 2/2</u>	1. The darker coloured Corian boards tend to show the signs of everyday wear, scratches and dust more quickly than the lighter colours. 2. Accidental spills of chemical substances such as paint strippers, aggressive cleaning products for metals or ovens, products containing methylene chloride, acids, nail varnish solvents, acetone based products, etc. must be promptly removed by rinsing copiously with water and soap to prevent the work surface from being damaged. In particular, use an acetone-free product and then wash with copious amounts of water to remove nail varnish remover solvents. Uncontrolled or prolonged exposure to chemical products damages the surface. 3. Corian boards used to make fittings on a ship often have reduced thickness: it is therefore risky to place heavy objects above them or for anyone to climb on top of them (e.g. when cleaning). 4. For Corian furniture or surfaces that are directly exposed to sunlight for long periods and could therefore reach high temperatures, it is recommended to wash them only when the temperature has dropped to ambient temperature. When washing use fresh water and be very careful with the direction of the rinsing water jet: not straight on and under pressure, but rather let it slide delicately over the surface.

GRANITE:

Component	Notes and Warnings
<u>Granite surfaces</u>	Granite is one of the strongest and easily maintained natural stones, because it is less sensitive to chemical attacks. Granite surfaces also have a structure that is particularly resistant to abrasion and stress. It also reacts very well with weak acids such as alcohol and vinegar and also some of the stronger ones that may be used in household cleaning, like for example hydrochloric acid, but it can still suffer corrosion process. In fact, the use of these substances is only advisable in extreme cases, after trying fruitlessly with other less powerful and invasive detergents.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. Absolutely avoid using hydrofluoric acid (present in some lime scale remover products) for cleaning. 2. Avoid contact with oil or grease which could cause stains that are difficult to remove. 3. Avoid leaving the surface in contact with citric acid (present in fruit, tomatoes and vegetables for example) for a long time. 4. For shiny or polished surfaces, it is a good idea to only use low concentration products (very diluted) with an acidity level just above neutral, whereas for compacted stone materials which are not very sensitive you can use concentrated solutions to perform more effective cleaning. 5. Following the washing, whether with acidic or base solutions, rinse with copious amounts of water to inhibit the action of the detergent so that the corrosive power of the substances used is neutralized. 6. It is advisable to periodically carry out water repellent treatment using specific products: effective surface protection in fact helps to keep the anti-stain and oil repellent actions active on the material. <u>Important:</u> like the majority of cladding used on a boat, boards in this material can be the "light" type, i.e. reduced thickness. This makes them more sensitive to any changes in temperature with the risk of breaking or cracking, especially in direct contact with a source of excessive heat.

MARBLE:

Component	Notes and Warnings
<u>Marble surfaces</u>	Marble requires a lot of care both in its use and for everything that concerns its treatment and cleaning. In the later case, if the dirt is light and there are no particular hygiene requirements, it must be cleaned with water without the addition of other substances: remember that marble is a stone and thus can absorb substances that may compromise its sheen. If the surfaces are particularly dirty neutral soap can be added to the usual water (better if distilled) Marseilles soap is the one most recommended both because of the product's delicacy (it does not attack the stone) and its effectiveness on stains; it is always recommended to rinse with distilled water and wipe with a wool rag so as to give it its original brilliance.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. When cleaning marble, products containing acidic substances (e.g. denatured alcohol or hydrochloric acid and hydrofluoric acid) must absolutely be avoided. They would have a corrosive effect and make the surface coarse and dull. 2. Remember that the following are also acidic substances to be avoided: lemon juice, vinegar, fruit juice, tomato etc. which can corrode the shiny part of the marble making it dull and coarse. As soon as any of these substances touches the marble surface, remove it with a cloth and immediately wash the area in question with neutral soap. 3. Ammonia, which on the other hand is basic, can be used on marbles and calcareous rocks in general if there are heavy stains for which neutral detergent is not enough. 4. Marble can become dull over time. We recommend you carry out careful polishing treatment cyclically using specific products so that the sheen is preserved over time. <u>Important:</u> like the majority of cladding used on a boat, boards in this material can be the "light" type, i.e. reduced thickness. This makes them more sensitive to any changes in temperature with the risk of breaking or cracking, especially in direct contact with a source of excessive heat.

STONE:

Component	Notes and Warnings
<u>Stone surfaces</u>	Natural stones on the whole derive from rocks that are mainly calcareous, hence with poor resistance to acids, and generally soft and barely compacted. These characteristics make these surfaces very delicate and spongy and therefore prone to easily absorbing aggressive substances which could damage them irreversibly.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. For routine maintenance: delicately remove the dirt that is deposited daily on the surface with a soft damp cloth. Only use neutral detergents diluting them in water according to the quantity of dirt. 2. Acidic detergents must not be used for cleaning as they could ruin and badly stain the material. 3. When cleaning, use cloths or delicate soft sponges: stone surfaces can be scratched extremely easily. 4. Regularly apply (at least every 60 days) specific stain-proofing and anti-decay products, which are fundamental for restoring the protective layer that is normally worn with washing and everyday use. Protecting this material is indispensable for its original features to be preserved over time, for its beauty to be enhanced and for making its surfaces more cleanable. 5. In the case of important damage on the surface or faulty routine maintenance, an extraordinary operation is recommended: this consists of completely removing the treatments previously carried out. This operation should only be carried out if it is necessary and consists of fully restoring the initial treatments applied to the material. Specialists in the sector should be used in this case. <u>Important:</u> like the majority of claddings used on a boat, boards in this material can be the "light" type, i.e. reduced thickness. This makes them more sensitive to any changes in temperature with the risk of breaking or cracking, especially in direct contact with a source of excessive heat.

RIMEX, STAINLESS STEEL AND MIRROR SHINE ALUMINIUM:

Component	Notes and Warnings
<u>Rimex, stainless steel and mirror shine aluminium surfaces</u>	Maintenance of these materials is extremely easy: marks and stains can be generally removed with a non-abrasive soft cloth and a low concentration product (the same as used for cleaning glass and mirrors). In extreme cases, light acidic products such as alcohol and ammonia heavily diluted in water can be used.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. Use a neutral detergent and plenty of hot water to clean very dirty surfaces. Then rinse with clean cold water. The final result can be further improved by drying the surface with a soft cloth; this way you prevent water streaks and marks from forming. 2. For finger prints and marks: use a soft dry cloth and any glass detergent. Solvents can also be used in extreme cases. 3. For tape residues and glues in general: use a soft cloth and solvent first, then rinse with water and/or glass cleaning products. 4. For silicone residues or paint marks: apply a solvent solution onto a soft cloth and act directly on the surface. 5. Avoid contact with products containing bleach. If this happens, immediately remove the product and in any case never let it act for more than thirty minutes. 6. Do not adopt abrasive materials, wire wool or metal brushes for cleaning; these could scratch the surface. 7. Make sure the cloths used during cleaning are not contaminated by grains or fragments of hard material. <u>Important:</u> constant maintenance coupled with proper cleaning of these materials prevents serious surface defects such as marks, scratches and permanent stains from forming. Often more radical operations like mechanical polishing are necessary to remedy this type of defect; but sometimes this remedy can no longer be carried out on finished and installed products.

LEATHER COVERINGS:

Component	Notes and Warnings
<u>Leather covered surfaces and furniture</u>	Here are some useful instructions for preserving the look of leather furniture and surfaces for as long as possible. Leather for furnishing requires careful cleaning and maintenance. The piece of furniture is exposed in daily use to different types of stresses caused by external factors such as body oils, transpiration, absorption of humidity and general wear and tear. Moreover, leather has a limited elasticity, so when it is stretched or deformed it does not completely return to its original shape and because of more or less intense use it could become loose over time.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. Like any other natural material, it can lose its colour. Therefore avoid exposing leather accessories or furniture to direct sunlight for long periods (e.g. when wide windows, doors or balconies are opened). 2. For careful maintenance, we recommend delicate cleaning one a week with a dry cloth or a vacuum cleaner with a soft brush. 3. Absolutely avoid using steam equipment when cleaning. 4. In order to look after its surface look and texture, periodically apply a water based protective cream (indicated for all type of leather, except for chamois leather) which will help you to keep it soft so as to prevent pigmentation and dryness and at the same time protect it from any marks and the dirt that gets deposited every day. 5. When you use household chemical substances, e.g. detergents as glass cleaners, bleach, nail varnish, acetone for removing it, glue, paint and corrosive materials in general, we recommend you protect the leather so as to prevent obstinate marks, which are then difficult to remove, from forming. 6. Certain cosmetic and body care products contain solvents and oils. It would thus be better not to sit on leather seats, sofas or armchairs with wet hair or immediately after applying body care products.

PARQUET:

Component	Notes and Warnings
<u>Parquet flooring</u>	The wood from which parquet is made is a soft natural material that can be subjected to changes or damage once laid on the floor. The care and maintenance of this form of flooring starts with the correct habit of washing it and following the right instructions.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. For the coarsest dirt, just go over it with a vacuum cleaner taking care to change the brush and use a soft bristle one; finally wash it with a well wrung soft cloth following the grain of the wood. 2. For washing parquet, only use water or at the most specific products (diluted in water) that do not contain alcohol or ammonia. 3. There are reviving products (these too to be used diluted in water) to protect the surface. They should be applied with a well wrung soft cloth (always following the grain of the wood). Once dry, go over it with a cotton cloth. <i><u>These products are to be used at your own discretion. In any case, it is recommended to read the instructions and instructions for use carefully.</u></i> 4. In the areas with the heaviest traffic, it is possible that scratches, splintering, dents and grooves left on its surface may become highly visible, especially in those parquets where surface treatment with transparent resin coatings has not been provided for. 5. In the case of points of the flooring being exposed to sunlight (e.g. where wide windows, doors or balconies are opened), avoid long term direct contact: the sun can considerably alter the wood's colour. 6. The temperature and humidity of the room where the parquet is can in turn affect the integrity of the blocks which could squeeze together or expand producing gaps or cracks. 7. After washing your parquet floor, we recommend you carefully dry the blocks so as to prevent them from being impregnated with unexpected infiltration of stagnant water with the risk of swelling.

MOQUETTE:

Component	Notes and Warnings
<u>Moquette flooring</u>	Moquette requires constant careful maintenance because it is subject to different damage and degeneration even in the best quality versions. Because it is a particularly delicate covering material, it requires constant careful maintenance.
	<u>RECOMMENDATIONS FOR MAINTENANCE</u>
	1. Its worst enemies are: damp and dust. The first can cause the development of moulds that make their home in the surface with the consequent production of bad odour. Always ventilate the premises and let the sun into the cabins as soon as possible. Whilst as regards dust, contrary to popular opinion moquette does not produce it, but rather some types hold it back preventing it from dispersing into the air. So it needs to be cleaned all the more. 2. Routine cleaning: regularly clean the moquette every day with a suitable device capable of removing the dust deposited on the surface, such as a carpet beater, vacuum cleaner or similar. 3. Occasional cleaning: this type of maintenance can be carried out easily without resorting to specialist firms and allows you to obtain deep cleaning and a reinvigorated moquette. It should be carried out occasionally, when you consider it appropriate according to the type of material laid, the intensity, the trampling, the laying area and the degree of routine maintenance you have managed to carry out. This operation can be carried out using specific products for washing moquettes: solid detergents (in powder form) which must be applied directly onto the surface or liquid products (diluted in water, usually in the tank of the device you are using). To get deep penetration of the product, always use a brush (by hand or with a polishing type tools), then vacuum clean the surface to remove the detergent to which the dirt has been attached. 4. Extraordinary cleaning: if necessary, when the moquette's conditions require it or when you want to carry out more exacting maintenance, washing the moquette with so-called extraction devices allows even more effective and deep cleaning. This type of planned maintenance operation must be carried out with suitable equipment or by a specialist firm.

WOODS FOR INTERIOR FITTINGS:

Component	Notes and Warnings
<u>Interior fittings: woods 1/2</u>	Suitable resistant materials, which mainly have excellent durability features in environments with a high degree of saline damp, are used for the craft's interior fittings. As regards the multiple quantity of coverings on the craft, one should know that majority of them are made of Okoumé marine plywood (with a lacquered or veneered finish), a material that guarantees the specific conditions of its use in sailing precisely in consideration of the operational limit conditions where it is used, but like all other natural materials requires controlled environmental conditions in terms of temperature range and humidity to be maintained. It is thus useful to take into consideration the small precautions that will help you to preserve the beauty and durability of the interior fittings inside your boat.
	<u>RECOMMENDATIONS FOR MAINTENANCE AND ENVIRONMENTAL CONDITIONS</u>
	1. It is advisable to use products that are non-aggressive, very diluted and have an acidity level only just above neutral for cleaning both lacquered and veneered coverings. Do not apply the detergent directly onto the surface but always through the use of a soft cloth. 2. In the case of furniture that could remain exposed to direct sunlight (e.g. when wide windows, doors or balconies are opened), we recommend you keep them as sheltered as possible to prevent surface discolouring (this is especially valid during the craft's period of use). 3. Avoid the interior rooms being closed for long periods (with the possible formation of excessive damp) and ventilate them regularly. 4. OPTIMAL ENVIRONMENTAL CONDITIONS FOR INTERIOR FITTINGS: ➤ Unfavourable environmental conditions in terms of humidity and temperature must be prevented from forming on board. ➤ Despite the use of high quality materials and insulating treatment, wood components are subject to swelling and shrinking according to the humidity level and the environmental temperature. ➤ In particular: with very low humidity (e.g. cold dry weather outside and heating on) the wood tends to shrink with a variation in the order of 2-3 mm/m (for strips or panels of a limited size). This can lead to cracks opening in the panels' joint area.

WOODS FOR INTERIOR FITTINGS:

Component	Notes and Warnings
<u>Interior fittings:</u> <u>woods 2/2</u>	➤ With very high humidity (e.g. with very humid and cold outside weather, with the heating off) the wood tends to swell, leading to cracking of the surface finishes as well as forcing in the contact points with the adjacent materials. ➤ With high interior humidity and temperatures, the surfaces with transparent shiny varnish can become irreversibly opaque. ➤ Unfavourable environmental conditions also lead to a loss of stability of the panels, which twist and go out of shape with the consequent problems of: the fronts of the cupboards and various furniture jamming, the wall panels, floors, ladders, etc. creaking to a point where the deckhead linings possibly come off their support. In order to best preserve the fittings' aesthetic and functional features over time, the interior environments must always be kept in acceptable climatic conditions, i.e. with temperatures between 18÷28°C and humidity levels of 55÷65%. It is also important to prevent repeated hot-cold changes of temperature through auxiliary heaters; the recommended heating rate for the environments is 1-2°C/hour.

Component	Maintenance	Notes and Cautions
Wood and tapestry	Periodical cleaning (as required)	Light and humidity are the worst enemies of such materials; do not to expose them to direct light as far as feasible, and ventilate the indoor spaces whenever allowed by the weather. It is extremely important the use of window shades, any kind of wood, both natural and coated, will experience discolouring when exposed to sunlight.
Sunbathing cushions	Periodical cleaning (as required)	Remove the cushions from the seats on a regular basis and let their bottom side and the seat surface dry. When washing down and when it is raining, remove the cushions and stow them in a covered and ventilated place. However, when cushions are wet, remove them from their seat to prevent that water or humidity entrapped between cushions and underneath surface affect the gel-coat and deteriorate the cushion cover.

Precautions for cold climates

The use of anti-freeze and preservative liquid is suggested for all climates; its use extends the operation temperature range, reducing the freezing point and increasing the boiling point.

When the external temperature approaches 0°C make sure that cooling circuits are supplied with anti-freeze mixture, to avoid damages.

Otherwise replace the cooling liquid with such mixture, making sure that it reaches the whole circuit (by operating the machinery).

Before filling the system with anti-freeze mixture, wash the cooling circuit.

It is advisable to use anti-freeze liquid. The following table shows the mixture percentage according to the external temperature:

External temperature (up to ...)	Anti-freeze	Water
-8° C	20%	80%
-15° C	30%	70%
-25° C	40%	60%
-35° C	50%	50%

Circuits of systems containing fresh water or sea water (fresh water system, air conditioning, etc) or sea water should be almost completely emptied.

For main machinery, please refer to prescriptions contained in Manufacturers' Manuals.

Yacht's long inactivity

The following list is only a guideline for the customer concerning ordinary maintenance, which should be carried out when the yacht remains inactive for a reasonably long period.

Read carefully the instruction Manuals of single equipment, as they contain particular and important instructions for maintenance and preservation.

- **Propeller Engines**

Before the winter, let fresh water flow through the sea water circuit, check the anti-freeze fluid in the fresh water circuit, remove eventual salt deposits and use special protective sprays.

Carry out periodical maintenance, on engines as shown in specific operation and maintenance Manual, on air and oil filters and on cooling pump impeller. Lubricate and seal to avoid condense and humid air, which may damage valves and cylinders.

- **Generators**

Adopt same precautions as for the engines.

- **Reduction gears / Reversing gears**

Carry out reversing gear scheduled maintenance.

- **Batteries**

Periodically charge the batteries and protect the terminals with vaseline grease.

- **Sun bathing cushions**

Remove all sun bathing cushions, make sure they are dry and store them in a dry and ventilated area.

- **Aluminium and steel**

Wash all metal parts with fresh water and protect them by greasing with a rag moistened with vaseline oil.

- **Wood and internal tapestry cloth**

Cover sofa cushions with perspiring tarpaulins and apply the special covers on all windows, to prevent sunlight from entering the room, as sun radiation damage wood and tapestry colours.

- **Teak deck**

Clean and protect with "Teak Wonder"; in spring, sand and protect with "Teak Wonder" if necessary.

- **Zincs**

Check zincs wear and replace zincs on hull, propeller shafts and flaps, if necessary.

- **Speed log transducer**

Extract the log and insert the special plug.

- **Windshield wipers**
Wash with fresh water and lubricate with vaseline oil. Cover the windows with the protective closures.
- **Anchor windlasses**
Check the reduction gear oil level, protect the electric parts with special protective spray and lubricate the wildcat with silicon grease.
- **Grey/Black waters tank**
Pour sterilising products in basing, shower, bidet discharges. Empty the tanks, clean with fresh water and check floats operation. Empty nearly completely by using alternatively the two electric-pumps.

Pour a sanitary product containing paraformaldehyde (i.e. T5 that is sold in nautical accessories shops) in the toilets and rinse the tank a couple of times with this mixture. Empty the tank completely by using the service electric-pump (washing the whole circuit).
- **Tender outboard engine**
Wash the engine cooling circuit with fresh water. Carry out maintenance as required by the Manufacturer.
- **Bow-Thruster**
Check the oil level and replace the zincs.
- **Hydraulic power units**
Protect with special spray and check the oil level.
- **Hydraulic steering gear**
Protect with special spray and check the oil level.
- **Refrigerators**
Careful cleaning; pour a packet of specific anti-odour products in the compartment.
- **Fire extinguishers**
Check the correct position and preservation conditions.
- **Safety equipment**
Store all external mobile equipment inside the yacht.
- **Fresh water tank**
Wash with disinfectant and empty the fresh water circuit, especially if the yacht is not used during the winter.
- **Fuel tanks**
Clean especially for what concerns eventual water in the fuel. Close the valves of each tank.

Other precautions

- For what concerns the engine room it is advisable to carry out general cleaning, removing all salt deposits from the systems and protecting all electric, mechanical, and hydraulic equipment, by using special protective spray.
- Check all main deck and flying bridge lights and set all protective covers on the equipment.
- Set all cabins, clean and inspect all peaks and stores.
- Check all manhole gaskets and lubricate them with special silicon lubricant.
- Clean the fan coils with an air jet, sucking the dust from inlet and outlet grids.
- Wash the yacht with fresh water.
- Check all equipment: damages, wear, fissures may be caused by incorrect operation or wear due to operation. Repair damaged equipment. Replace equipment if necessary.
- If the yacht is stored outside, it is advisable to cover the yacht with a waterproof tarpaulin; allow ventilation to prevent harmful humidity.
- Whenever a system has been checked and prepared for preservation, or has been set for long inactivity, it is advisable to insulate the system as much as possible from other systems (de-energising, closing check valves, etc.) or closing /protecting it from external environment.

Preparing the yacht for winter laying up

To assist you during boat winterization, we've prepared the following checklist to make it easier for you to prepare for the winter storage. For those whose lat/lon coordinates mean warm weather year-round, this list will hopefully provide a reminder for your annual maintenance. The following checks are by no means exhaustive and are recommended as a minimum level of preparation for a long resting or winter stop.

Storing a boat for the winter or for any long period of time requires preparation, fuel needs to be stabilized, moving parts must be greased, valves need to be fogged, the boat should be cleaned thoroughly, inside and out, and a storage cover placed on it to protect the boat from the elements of outside storage or the hazards of dry storage facilities. There are also problems with mildew to contend with and, in some places the formation of ice can cause serious damage. Your first step in winterizing should be to make a checklist of all items that need to be accomplished.

Remember that the following is only a generic outline of areas which should be of concern to you, for any specific details check the manuals provided with the single equipments and fittings or contact your marine dealer.

Out of water storage

- Pressure wash hull, clean barnacles off props and shafts, rudders, struts and trim tabs.
- Clean all thru-hulls and strainers.
- Open sea water intakes to allow any water to drain.
- Check the hull for blisters and if you find any that should be attended to you might want to open them to drain over the winter. Give the hull a good wax.
- The batteries should be recharged every 30-60 days, to prevent discharging from ruining its efficiency and duration.

In water storage

- Close all sea water intakes and check rudder shafts and PSS seal for leaks, if any get them serviced by qualified technician.
- Check your battery to make sure it is fully charged, clean terminals, add water if necessary and make sure your charging system is working.
- Check bilge pumps to ensure they are working and that float switches properly activate the pumps and that they are not hindered by debris.
- Make sure either to check your boat periodically or have the marina check it and report to you.
- If in an area where the water you are docked or moored in actually freezes, you should have a de-icing device or bubbling system around your boat.

Do not, however, neglect to consult the manuals provided with each major component for manufacture's recommendations on winterization. If you have not done a winterization job before seek out a professional to do the job for you.



For SAFETY reasons (fire risk) we strongly recommend not to leave the boat connected to the VAC shore power, if unattended for long period of time.

Preparing the fuel

Diesel that is left in a storage container over the winter or even a few weeks will naturally break down causing contaminant sedimentation. When the engine is started again in the Spring, these substances can clog up the entire fuel system.

It is suggested that the fuel tank be filled to capacity since all tanks are vented to the atmosphere and moisture in the form of humidity enters through these vents. If there is no air space for the water to condense on the sides of the tank, water is not a problem.

A good diesel additive should be added to the fuel to inhibit the formation of impurity and to stabilize it, preferably each time the tank is filled.

There are also various products that absorb water that could also be poured into the tank. If diesel fuel does go bad, changing the filter, draining and cleaning the tank, and adding new fuel should remedy the problem. Filtering services can rejuvenate old diesel fuel.

Engines

Oil that has been used to run the engine is not as effective in the guarding against problems as fresh oil that has never been used. You should run the engines to warm them up and change the oil while it is warm. This tends to allow impurities to be drained away with the oil. You should also change the oil filters.

For fuller, more effective protection, spray a protective, anti-corrosion liquid onto the electrical contacts and more exposed metallic parts (there are various suitable products available on the market). It is also crucial to replace the oil filters. Wash the engines down with plenty of fresh water. Circulate anti-freeze through the collector using a flexible aspiration hose running from the water pump to the anti-freeze tank.

Wipe down the engine with a shop towel sprayed with a little fogging oil.
Change the fluid in your transmission.

Freeze protection

Antifreeze, for water systems and marine engines, is a necessary winterizing product. You should circulate antifreeze through the engines manifold if your cooling system does not contain all-time permanent fluid. One of the most important aspects of antifreeze products these days is their impact on the environment. Pressure by government and environmental organizations and even from marinas themselves have prompted some manufactures of antifreeze products to switch from a toxic ethylene glycol base to a non-toxic propylene glycol base. Propylene glycol will not harm the environment, plus manufactures who use it say its better for the engine as well. Camco 30% propylene glycol is a non-toxic antifreeze for all engine and water supply systems. It is ideal for winterizing boats, it contains rust and corrosion inhibitors. Burst protection to -50 degrees when used at full strength.

Cleaning the boat

Before the boat is put into storage it should be thoroughly cleaned, including the gel coat, teak, vinyl and carpet.

Barnacles, mussels or scum left on the boat over an extended period of time can become extremely tough and crusty - and very difficult to remove come Spring when you want to be in the water. The same goes for dirt and grease in the interior of the boat.

A good cleaning of vinyl, carpet and other fabrics can prevent mildew from growing and teak will last much longer if it's properly prepared for winter storage. Polishing the boat before storage is also wise because dirt will not stick to a polished fiberglass surface. If a boat doesn't have a protective barrier, the finish deteriorates from there, making a Spring cleaning very difficult.

Preventing mildew growth

Mildew has a habit of showing up on a stored boat. If a boat is stored outside under an unventilated cover, the interior of the boat is a perfect place for mildew to grow. It's warm, dark and moist from humidity. Mildew grows freely, leaving stains and its musty, stale smell behind.

When a boat is shut up underneath a cover, either exposed to the sun outside or inside a garage or storage facility, humidity, with its high amount of water vapor, is what gives rise to mildew and its odor. The water vapor will penetrate the electronics and fabrics of the boat. Any wood in the cabin will absorb water as well. Provide to ensure proper ventilation to avoid as much as possible an excessive degree of humidity inside your boat. Use of any electrical device to control humidity inside your boat is strongly discouraged because of the very high risk of fire. Alternatively you might want to use a air dryer crystals, there is a variety of product available on the market; some of this crystals is boosted with deodorant. Depending on the amount of humidity the efficiency of such crystals may last from 1-to 3 months.

Covering the boat

If a boat is stored outside in the elements, a storage cover is a must to protect the interior of the boat from the weather. Boats that are placed in dry storage also require covers to keep dust or greasy films off the boat's interior. A good 8 to 10 ounce cotton canvas is generally agreed to be sufficient if it's properly supported so that water will not bag. For dry storage, just about any light canvas or polypropylene tarp or fitted cover will suffice. Its purpose is to keep dust and other light matter from accumulating so water proofing and tensile strength are not considerations.

Windshield wiper

When the boat is put into winter storage or in case of long halts, insert a block of wood or plastic beneath the wiper arm to lift the wiper blade off the windshield, thereby preventing damage to the rubber edge.

Steering system and hydraulic system

You should thoroughly inspect the rudders and trim tabs and remove any plant life or barnacles. Grease all fittings and check fluid levels in hydraulic steering and gangway pumps. Check with attached component manual for additional recommendations by the Manufacturers.

Bilges

Make sure the bilges are clean and dry. Use soap, hot water and a stiff brush to clean up any oil spills.

Once the bilges are clean, spray with a moisture displacing lubricant and add a little antifreeze to prevent any water from freezing.

Fresh water system

- Completely drain the fresh water tank and hot water heater.
- Pump a non-toxic antifreeze into the system and turn on all the facets including the shower and any wash-down areas until you see the antifreeze coming out. Also put non-toxic antifreeze in the water heater.
- Alternatively, the system can be left completely empty; in this case take care to completely fill the system and to pressurize it before turning the water heater on again.

Air conditioning

- Close the seacock and remove the inlet water hose from the air conditioner.
- Allow all water to drain from the system.
- Loosen the screws on the pump head to allow the water to drain from the pump.
- Drain and clean the seawater strainer.

We advise you have these operations carried out by a specialized technician.

Ice maker / Fridge

- Disconnect power supply first, then shut off the water supply to the unit.
- Disconnect the water supply connector from the unit (see constructor manual for details about this operation).
- Turn the power on and allow the unit to run for an hour. Remove any cubes that may have been ejected during this period. Turn off the power and ensure the door remain open to allow inside to defrost.
- After it has defrosted, wipe it dry. Ensure door will remain open during storage to avoid formation of bad odours.

Take care of the fridges just as you would do for your house unit when disconnecting them for long time. Turn off the power and ensure the door remain open to allow inside to defrost. After it has defrosted, wipe it dry. Ensure door will remain open during storage to avoid formation of bad odours.

Head

- Pump out the holding tank at an approved facility. While pumping, add fresh water to the bowl and flush several times. Fill the bowl with fresh water and add Holding Tank Deodorant and Cleaner (or similar products that will not harm your system) and let sit for a few minutes.
- Add more water and pump out. Add anti-freeze solvent and pump through the hoses, in the holding tank, valves, maceration pump and in the drain pipe. Check the shower sump, and dry the container thoroughly to prevent any bad odour in the boat.

Interior

Once you have taken care of the system you should remove any valuables, electronics, lines, PFD, fire extinguishers, flares, fenders, etc. Over the winter these items can be cleaned, checked and replaced as necessary.

Open all drawers and lockers and clean thoroughly. Turn cushions up on edge so that air is able to circulate around them or, better yet, bring them home to a climate controlled area. To keep your boat dry and mildew-free you might want to use some of the commercially available odour and moisture absorber products.

Yacht's use after long inactivity

The following list is only a guideline for the Customer concerning ordinary maintenance, which should be carried out when the yacht is used after a long inactivity.

We would remind you that the following list should not be regarded as complete, as additional work may need to be carried out depending on the general state of your vessel and the conditions in which it has been laid up.

Read carefully the instruction Manuals of single equipment, as they contain particular and important instructions for maintenance and commissioning.

Operations:

- check the hull;
- inspect and clean carefully hull, rudders and flaps with brushes (in the water) or by using the water cleaner (in dry dock) to remove algae and fouling;
- check the hull coating condition. If necessary apply 2 coats of suitable anti-fouling; this operation should be carried out in dry dock by specialised personnel;
- check the state of the propellers and inspect for leakage from the shaft line seals;
- check sacrificial anodes and replace if necessary;
- check batteries charge, and make sure that terminals and containers are clean and dry. Recharge batteries by using shore power supply;
- check engine and reversing gear oil, check oil and diesel fuel filters;
- Adjust the tension on the propeller engines belts and generators alternator belts;
- operate all sea water intake valves and open sea water circuit valves, necessary to start each system;
- if the systems have been treated with anti-freeze, run clean water through until it has been completely rinsed out;
- open the fuel circuit valves and refuel. Vent the fuel circuit if necessary;
- inspect the cables and connections, and contact a specialist for the requisite repairs if there are obvious signs of deterioration. If the connections appear to be in poor condition, clean and protect them with a suitable CRC-type liquid. Ensure that the terminal board connections are tight;
- energise all auxiliary equipment of the yacht;
- start the main engines;
- start generators with same care as for the engines.



After yacht's long inactivity, besides above mentioned operation, carry out the following checks:

- a) check all flexible pipes and steering system, flaps and gangway connections;*
- b) heat the engines for some minutes;*
- c) check each bilge electric pump and their operation;*
- d) check operation of black waters electric pumps, grey waters electric pumps and fresh water autoclave, by filling relevant tanks and disinfecting them and letting the fluids circulating into the whole circuit;*
- e) check operation of all navigation instruments.*

GLOSSARY

A

ACCELEROMETER

instrument that shows whether the vessel is increasing or decreasing its speed.

AGROUND

resting on or touching the ground or bottom (usually involuntarily).

ALIGNMENT

straight line passing on two points; an alignment at sea is carried out when sailing on a route passing to two known points (lighthouses, beacons, etc.); typical operation for landing in a shore full of dangerous situation.

ALPHA

term that represents the letter A in the ICAO international phonetic alphabet.

ALTOCUMULUS

cloud at approximately 5,000-6,000 metres, often occurs in layers or patches with wavy, rounded masses or rolls, with distinct borders and shades; if it tends to dissolve, the weather may remain steady, while if the cloud assumes a lens-shape, the wind speed may probably increase.

ALTOSTRATUS

cloud at approximately 6,000–7,500 metres, characterised by a generally uniform grey sheet or layer, which dims significantly the day light.

ANCHOR

a metal device attached to the ship by a line or chain and designed to prevent or slow the drift. There are various types of anchors, depending on the sea bed and vessel characteristics: stocked, Fortress, CQR, Danforth, Hall, Bruce and grapnel.

(to) ANCHOR

to lower the anchor in order to stop the vessel tied to it.

ANCHOR RING

ring located at the anchor stem end, used to connect the chain or cable.

ANCHOR WINDLASS

particular type of winch, usually provided with wildcat, used to ease anchoring operations.

ANCHORAGE

a place suitable for anchoring; a good anchorage must have a sea bed which ensures riding, as well as being repaired from strong wind, surf and waves.

ANCHORING POINT

point where the anchor is lowered, in order to stop the vessel.

ANEMOMETER

instrument that measures the wind speed (if the vessel is not moving, it shows the true wind, while if the vessel is moving, it shows the apparent wind, that is to say the sum of true wind and vessel's speed generated wind).

ANSWERING PENNANT

international code of signals flag. When hoisted alone, it means "I have understood", when hoisted at half mast it means "I received the message, but I have not yet acknowledged". When hoisted by the race jury, it has various means.

ANTICYCLONE

atmospheric high pressure area.

ANTI-FOULING

underbody paint suitable for avoiding attack and fouling of seaweed and particular types of marine fauna (such as barnacles).

ANTI-SKID

treatment or coating applied to the main deck planking in order to avoid skidding.

APPAREL

assembly of operations for preparing the vessel for going at sea (setting the equipment in operation conditions, food storage, etc).

ATMOSPHERIC PRESSURE

the weight of the air above the considered point.

B**(to) BAIL**

to remove water from the bilge, peaks or generally from the lower part of the vessel.

BALLAST

weight at the bottom or the lower portion of the boat to give her stability and/or to provide satisfactory fore and aft trim. Ballast can be placed inside the hull of the boat or externally in a keel. Heavy substances can be loaded by a vessel to improve stability, trimming, sea-keeping and to increase the immersion at the propeller.

BANK

sea bed with reduced depth compared to surrounding areas; when its depth becomes dangerous for the navigation, is called "shoal".

BARNACLES

a shellfish often found on a vessel's bottom which alter characteristics and performance of the vessel itself. In order to prevent barnacles formation, the submerged part of the vessel is coated with anti-fouling paint which contains poisonous salts or pigments.

BAROMETER

instrument measuring the atmospheric pressure.

BATHYMETRIC

line joining points with equal depth.

BAY

wide and protected cove.

BEACON

a lighted or unlighted fixed aid to navigation attached directly to the earth's surface.

BEAM

the width of a vessel at the widest point.

(to be on) BEAM SEA

a situation in which waves strike a boat from the side, causing it to roll unpleasantly.

BEARING

a compass direction, in compass points or degrees, from one point to another. Relative bearing is the direction relative to the heading of the boat with the bow 0 degrees and the stern 180 degrees. True bearing is the direction from the ship relating to true north with north being 0 degrees and south 180 degrees.

BEAUFORT (FRANCIS)

English admiral lived in 1800 who invented the Beaufort scale for wind measurement.

BEAUFORT SCALE

an international number system used to describe wind forces and sea conditions from 0 for a flat calm to 12 for a hurricane (i.e. Force 7 wind means wind with average speed of 28 to 33 knots).

BECALMED

unable to move due to lack of wind; said of a sailing vessel.

BECCACCINO

Italian term which identifies a boat of Snipe class.

BELAYING PIN

removable equipment, similar to a cleat, on which light ropes are tied; tool used for unfastening tied knots.

BELL

acoustic signalling instrument to be used in case of fog, it is compulsory for vessels longer than 12 metres; it means also the external rotating part of a winch, on which the rope is coiled.

BELOW

any room situated beneath the main deck.

BILGE

the compartment at the bottom of the hull of a ship or boat where water collects and must be pumped out of the vessel.

BILGE PUMP

a mechanical, electrical, or manually operated pump used to remove water from the bilge.

BINNACLE

column where the compass used to be installed; nowadays the compass of large vessels is installed in the steering wheel support.

BLOCK

nautical term which identifies a pulley used to gain mechanical advantage; according to the number of pulley it is called single, double, triple block; an openable block, in which it is possible to insert a tensioned cable, is called snatch block.

BOAT

common name for small vessels; according to Italian laws concerning pleasure vessels, the term "boat" identifies a vessel longer than 6 metres and heavy displacement between 3 and 50 tons (above this limit, the vessel is referred as "ship").

BOAT HOOK

a long sturdy pole fitted with a blunt hook at one end designed to catch a line when coming alongside a pier or mooring, or in pushing or fending off.

BORA

strong north-easterly wind, typical of northern Adriatic Sea, it often follows the passage of cold fronts.

BOSUN OR BOATSWAIN

crew member responsible for the crew and for the vessel's equipment; in military Navy, the bosun is the Petty Officer responsible for the seamanly services.

BOTTOM

the underside of the hull that sits in the water.

BOW

the forwardmost or front part of the vessel.

BOWMAN

crew member which carries out and controls operations made at the bow.

BOWPLATE

fitting installed at the bow, generally composed of a single block with connection for stay and anchor chain passage.

BOWS

front part (port or starboard) of the vessel, just aft of the bow.

BOWS SEA

keeping a course in order that the vessel hits the waves with an angle of approximately 45 degrees.

BRAVO

term that represents the letter B in the ICAO international phonetic alphabet.

(to) BREAK GROUND OUT

to lift the anchor from the bottom along its vertical.

BREAKER

upper end of a wave, which breaks in a cascade of spray and foam, pushed by the wind and by the wave itself.

BREAST LINE

a docking line going at approximately a right angle from the boat to the dock, preventing movement away from the dock. Also known as a Waist Line.

BREEZE

thermal wind, perpendicular to the shore line, caused by the difference of temperature between ground and sea, both during the day and at night (land breeze and sea breeze); this term is also used to identify a force 2 wind (light breeze, 4-6 knots) or force 3 (gentle breeze, 7-10 knots) according to the Beaufort scale.

BRICCOLA

venetian term which identifies a wooden pylon used both as channel marker and as mooring post.

BRIEFING

skipper and crew meeting.

(to) BROACH

the turning of a boat broadside to the wind or waves, subjecting it to possible capsizing; a turning or swinging of the boat that puts the beam of the boat against the waves, creating a danger of swamping or capsizing; loss of steering. A knockdown.

BROAD PENNANT

a trapezoidal tapering flag, which can be used for identification or communication.

BROAD REACH

see "pace".

BRUCE

anchor with a single semi-circular fluke and two ears which opens as butterfly wings; invented by Peter Bruce for North Sea platforms, it ensures great effectiveness especially on sandy bottoms.

BULB

weighed fin attached to the underbody in order to give stable balance to the boat (weight stability); the weight of modern pleasure yacht bulbs vary between 40 and 45% of the total weight of the yacht.

BULKHEAD

a name given to any vertical partition or wall which separates different compartments or spaces from one another, also adding strength. Sometimes bulkheads are also watertight, adding to the vessel's safety in case of collision.

BULWARKS

the extension of the ship's side above the level of the weather deck.

BUNK

bed, generally reduced in size, that can be made of a wooden plane mounted to the hull or a sheet supported by an anti-roll system composed of metal tubular structure.

BUOY

a floating object of defined shape and colour, which is anchored at a given position and serves as an aid to navigation.

BUOY ROPE

rope that links a floating object to a submersed one, i.e. connecting a moored vessel to a mooring post, or the line connected to a small buoy that signals the position of the anchor.

BUOYANCY

as the upwards force applied to any object immersed in a liquid, equal to the weight of the displaced water; a vessel floats because its weight is less than the displaced water.

**CABIN**

an enclosed room.

CABOOSE

a small ship's kitchen, or galley on.

CALME

complete absence of wind: force 0 of Beaufort scale.

CAPSTAN

a large winch with a vertical axis.

CAREENING

cleaning and coating the underbody with anti-fouling paint.

CARLINE

part of the longitudinal framing of the hull.

(to) CAULK

to fill wooden vessel seams with oakum and cotton using caulking irons and hammer.

CENTRE OF BALANCE

centre of gravity of the submersed part; called also centre of thrust as it is the point where the hydrostatic thrust is theoretically applied, the centre of balance varies according to the vessel heeling.

CENTRE OF GRAVITY (C.O.G.)

centre of gravity of a body, it represents the point within something at which gravity can be considered to act.

CENTRE OF LATERAL RESISTANCE

geometric centre of lateral section of the underbody of a vessel.

CENTREBOARD

a fibreglass or metal blade projecting through the bottom of the hull in centre which prevents the boat from sliding sideways. In small dinghies it pivots up and back into the centreboard trunk.

CHAIN

metal rings (links) linked one to the other by means of welding, used to connecting the anchor.

CHARLIE

term that represents the letter C in the ICAO international phonetic alphabet.

CHART TABLE

table used for plotting.

CHARTER

hiring a vessel, with or without captain or crew.

CHINESE GYBE

contrary of broach; it can be more dangerous as the sudden bearing away may cause an uncontrollable gybe.

CIRRO CUMULUS

small, well defined, roundish masses increasing from below, they indicate variable weather.

CIRRO STRATUS

horizontal or slightly inclined masses, attenuated towards a part or the whole of their circumference, bent downward or undulated, separate, or in groups, or consisting of small clouds having these characters.

CIRRUS

very high clouds (also above 10000 metres); puffy, like cotton balls with flat bottoms, formed by ice crystals; if they tend to thicken, they may indicate arrival of bad weather (hot front), if they move with southerly wind, they may join lower clouds.

CIRRIPEDS

see barnacles.

CLEAT

a stationary device used to secure mooring lines.

CLINKER

particular construction of a wooden hull, with partial superimposition of planking plates, as in roof tiles.

CLOSE-HAULED INSTRUMENT

instrument which amplifies the wind angle indication compared to the vessels' bow, easing the wind direction reading when sailing to windward.

CLOVE HITCH

two half hitches around a spar or post. Easy way to make a line temporarily fast to a piling or post. The clove hitch can jam under heavy tension, making it difficult to untie.

COACHROOF

the cabin roof, raised above the deck to provide headroom in the cabin. Also trunk.

(to) COCKBILL

lower two anchors, one after the other, on the same chain, in order to avoid drag of the first one.

COCKPIT

space at the middle or at the stern of the vessel's main deck where the crew can sit being protected by bad weather; it includes the tiller or steering wheel, the compass and main lines. The cockpit edges are generally provided with winches for the sheets.

COD LINE

light line formerly used in hammock clews, now used mostly for decorative purposes.

(to) COIL

to lay a line down in circular turns in order to create a spiral; operation must always be carried out following the torsion line direction, generally clockwise.

COLLISION

impact between two vessels, at sea.

COMPASS

instrument essential for navigation, composed of a magnetic needle which is always oriented towards the magnetic North: the magnet is mounted to a graduated 360 degree rose, on which it is possible to read the vessel's course, indicated by a fixed line called "lubber line" it deviates from the North: the whole device is immersed in special liquid which deaden the force of counterblows, inside a spherical container provided with transparent section and eventual light for night operation.

COMPASS BOWL

see "Compass".

COMPASS ROSE

a sixteen-point star and relevant named directions such as north, south, east and west, as well as different names of winds, corresponding to their direction.

COMPASS SWINGING

operation carried out to calculate the compass deviation according to the course, in order to provide necessary correction during plotting.

(to) COMPENSATE

to correct the magnetic deviation, in order that the compass needle indicated the magnetic North.

COMPLETING

arranging internal rooms, equipment, instrumentation, steering devices, etc. after having built the hull.

CONSPICUOUS POINT

any object on the coast, easily visible and identifiable from the sea, such as light-houses, church towers, chimneys, mountains, etc.; main conspicuous points are shown on pilot books.

COURSE

the prescribed compass direction in which a vessel is being steered; the course is expressed in degrees and can be referred to the geographic North (true course) to the terrestrial magnetic North (magnetic course) or to the North shown by the compass (compass course).

COVER

sheet used for protecting the main deck instrumentation.

CQR

type of anchor, whose name is the abbreviation of "secure"; it identifies a small anchor, very effective especially in sandy and muddy bottoms; it is widely used on pleasure boats as main anchor.

CRADLE

frame for supporting a vessel when aground.

CREW

people whose duty is to run the vessel; the skipper, before beginning a cruise, assigns the duties to each crew member.

CRUISING BOAT

decked vessel equipped with one or more cabins.

CUMULO-NIMBUS

large cloud often consisting of great towers or pillars formed in the bank. It can exceed 8,000 metres in height. Its upper part is bright white which opens in a characteristic anvil-shape, while its lower part is similar to nimbostratus: they are usually associated to the passage of a cold front and are characterised by heavy showers and strong winds.

CUMULUS

thick, whitish cloud that exhibits vertical development, If the cloud tends to become dark, it may forecasts a thunderstorm; small and isolated white cumulus that disappear in the hottest hours of the day are a sign of good weather.

CYCLONE

area of noticeable atmospheric pressure decrease, with subsequent inflow of strong winds and unstable weather.

D**DACRON**

name used in the USA for identifying a trademark of tightly woven polyester fibres (in Italy they are called Terital); most of the polyester textures used for manufacturing sails are made in USA, therefore Dacron has become a yachting common term.

DANFORTH

type of flat and articulated-fluke anchor: the large surface of the flukes ensure great effectiveness in sandy and muddy bottoms.

DEAD RUNNING

pace with the wind coming exactly from stern.

DECK

a permanent covering over a compartment, hull or any part of a ship serving as a floor.

DECK STRINGER

outboard strake or strakes of plating on the main deck.

DELTA

term that represents the letter D in the ICAO international phonetic alphabet.

DEPRESSION

area whose atmospheric pressure is lower than normal values.

DEPTH

distance between the sea surface and the seabed, this value is shown in metres, in the charts.

DINETTE

French term which identifies the below deck space occupied by table and sofas, often transformable into a double bed.

DISPLACEMENT

weight of the water displaced by the vessel (Archimede's principle), that is equal to the total weight of the vessel.

(to) DOCK

generally used as synonym of “mooring”, it identifies a mooring to a quay, in order to be able of leaving the yacht directly, without using the tender (on the other hand, it is possible to moor to a buoy and reach the shore with the tender).

DOCKING

mooring to a quay.

DORADE

a horn type of vent designed to let air into a cabin and keep water out.

DOWN BY THE BOW

sink the bow more than in normal trim.

DOWN BY THE STERN

sink the stern more than in normal trim.

DOWN WIND

a boat sailing downwind, away from the wind source with the sails let out all the way, is running with the wind.

(to) DRAG

anchor movement when subject to traction, when it does not grab the bottom.

DRAUGHT

the depth of the boat below the waterline; the amount of vertical distance from a boat's water line to the bottom of its keel.

DUNNAGE

the walking surface of cockpit or cabins, composed of sets of panels, called dunnage.

E**(to) EASE**

to release completely the tension on a rope or a cable.

EAST

one of the four cardinal points of the compass, identified with the letter E, corresponding to 90 degrees on the compass rose.

ECHO

term that represents the letter E in the ICAO international phonetic alphabet.

ECHO-SOUNDER

electronic instrument that measures the sea depth; it is installed on all modern vessels, replacing the sounding line; it operates according to the principle according to which the propagation speed of the sound, in the water, is approximately constant: by measuring the delay of the echo of an ultrasound signal transmitted by a transducer installed in the bilge, the instrument measures the seabed depth.

EIGHT KNOT

a knot in the form of a figure eight, placed in the end of a line to prevent the line from passing through a grommet or a block.

EVOLUTIVE EFFECT OF THE PROPELLER

tendency of a vessel of displacing the bow towards port side while sailing with the engines, with a right-hand propeller, and vice-versa; this effect is particularly noticeable when sailing astern.

EYE

knot used for forming a loop at the end of a rope; there are various types of eyes, the most common is the bowline, used to attach the sheet to the sail.

EYEBOLT

metal ring-shaped support fixed to the main deck for connecting cables, blocks, tackles, etc..

F**FAIRLEAD**

see "Sheet lead".

FALL

end of the rope used for a knot, which is to be tensioned.

FATHOM

British measure indicating the depth, equivalent to 6 feet.

FAULT

breakdown of the hull or vessel's equipment.

FENDER

an air or foam filled bumper hung from the sides of a boat to protect it from rubbing or chafing against a dock or another boat.

FETCH

the distance across water with a wind or waves have travelled.

FIBERGLASS

a composite material made of fine glass fibres woven into a cloth then bonded together with a synthetic plastic or resin.

FIN KEEL

fin extending below the boat, for reducing the drift.

FINS

small flattened projecting surfaces applied on the lower part of the bulb keel, in order to reduce vortex and to increase the lift, without increasing the draft.

FITTINGS

assembly of all metal components of the equipment.

(to) FIX THE POSITION

to calculate the geographic point given by latitude and longitude, corresponding to the vessel's position on the terrestrial surface, at the considered time.

FLOATING ANCHOR

semi-floating object towed in the water in order to retard a vessel's leeway; it is generally composed of a textile cone whose mouth is kept open by a stiff frame, as to act as a mini parachute; the rear end is provided with a smaller opening than the mouth, which allows non-swirling deflux of the water.

FLOATING SAFETY DEVICES

a device used to keep a person afloat. It can be an individual device (life belt, life jacket, etc.) or collective (life buoy and life raft).

FLOOR

Wooden framing perpendicular to the keel, when linking two ribs, it forms a "frame".

FLOORBOARD

the assembly of dunnage installed onboard.

FLUKE

the portion of an anchor that digs securely into the bottom, holding the boat in place.

FLUKE TIP

first part of the anchor that sinks in the seabed.

FLUSH DECK

completely flat main deck, without any other superstructure.

FOG

condensation of atmospheric humidity in small water drops which reduce the visibility down to less than 1 km (if visibility does not reach such limit, it is referred as "mist").

FOGHORN

instrument which issues acoustic signals, to be used in case of fog or low visibility.

FOLL

rope end, opposite the tensioned end, used for making a tie.

FOXTROT

term that represents the letter F in the ICAO international phonetic alphabet.

FOREGUY KICKER

rope used for pulling a rod downwards.

FORTRESS

type of anchor with articulated flukes: it is similar to the Danforth anchor, but it can be completely disassembled.

(to) FOUL

to be tangled, i.e. a line on itself or on other cables or other objects.

FOULING

assembly of vegetal and animal incrustation formed on the underbody and generally speaking in any object immersed in the water for long time.

FRAME

cross section of a vessel, and the wooden ribs that form the shape of the hull.

FREEBOARD

the height of a ship's hull (excluding superstructures) above the waterline.

FRESH WAY

movement of a vessel through the water, when the propulsion thrust has ceased.

FRONT

contact between two air masses with different temperature, humidity and pressure characteristics, it assumes great importance for navigation, as it causes atmospheric variability, rapid changes of wind and sea state. The hot front is composed of hot and humid air that moves above cooler and dryer air; it is associated to wide cloud systems, persistent rains that go on for a few days. The cold front is composed of cold air that moves displacing hotter air; its development is typically vertical, with cumulo-nimbus formation associated with heavy thunderstorms and strong wind gusts. If both air masses have low or null speed, the contact zone is defined as "stationary", while if the whole hot air is pushed upwards and only cold air remains at sea level, the zone is defined as "occluded front", characterised by temperature decrease, heavy rain and weak wind.

G**GALE**

corresponding to force 8 of Beaufort scale (wind at 34-40 knots); the waves can reach a height of 7 metres, wave crests begin to break into spindrift; strong gale, force 9, with wind up to 47 knots.

GATE SEA COCK

a particular system used in special check valves, in order to close rapidly by hand a sea water intake or outlet.

GEL-COAT

polyester resin used for providing fiberglass hull with protection against water and atmospheric and chemical agents.

GEOGRAPHIC COORDINATES

see "latitude" and "longitude".

GIVE WAY

an order to men in a boat to pull with fore force, or to begin pulling, both during crossing (port-tack boat must give way to a star-board-tack boat) and approaching the buoy.

GMT

Greenwich Mean Time.

GOLF

term that represents the letter G in the ICAO international phonetic alphabet.

GONIOMETER

instrument which measures angles.

GPS

Global Positioning System; a radio navigation system used to determine position, in latitude, longitude and height: it is based on the detection carried out by a receiver which detects and decodes radio signals transmitted by at least four satellites on the 24 GPS satellites.

GRAPNEL

small anchor with four flukes, which can be closed as an umbrella, for easing the stowage; it is mainly used of rubber dinghies, tenders and small boats.

GRATING

wooden grid which allows water deflux; it is installed in bathrooms and on cockpit floor, in order to keep the feet dry.

GREAT CIRCLE ROUTE

the largest circle which can be inscribed on a sphere by a plane that cuts through the centre of the sphere. On the earth, the equator is a great circle, as are all the meridians of longitude which pass through both poles. The shortest distance between two points on the earth's surface lies along the great circle which connects the two points; at out latitudes, a great circle route appears as a curve on nautical charts, therefore it cannot be easily followed, therefore it is often approximated with many true course sections.

GREENWICH

location of an important observatory, in London; as an international convention, the meridian passing to Greenwich is the reference meridian 0; the term GMT (Greenwich Mean Time) indicates that the hour is referred to the mean time of Greenwich, used in all calculations, both astronomical and nautical, recently replaced with UTC, with equal value.

GRIPING

A boat that tends to turn towards the wind.

GUDGEON

pin of a hinge; a metal bracket attached to the transom into which a rudder pintle was hung.

GUST

sudden wind blow, often associated with clouds formed by dark nimbo-stratus charged with rain (gust line).

(to) GYBE (JIBE)

turning the boat so that the stern crosses the wind, changing direction. To change direction before the wind onto another tack with the boom coming over by the force of the wind. Caution is needed in this manoeuvre, especially in heavy wind.

H

HALF HITCH

a single turn of line around an object with the end being led back through the bight. It's the basis upon which many nautical knots are constructed.

HALL

type of heavy anchor with articulated flukes: it has high effectiveness on all types of bottoms and is generally used for large vessels.

HAND BEARING COMPASS

portable compass equipped with collimators in order to calculate the angle between the North and the direction of an object seen from the vessel.

HANDRAIL

assembly of lifelines and stanchions which form a rail all around the vessel. Shaped profiles, mounted to the coachroof, on sliding hatch sides or indoor, as a support for the crew.

HATCH

a sliding or hinged opening in the deck, providing people with access to the cabin or space below.

HATCHWAY

opening located on the main deck, provided with closing hatch, which allows the crew to enter the yacht (manhole) or for rooms ventilation; if the hatch is transparent, it is called skylight (especially when small).

(to) HAUL

to pull a line in order to lift a heavy object tied to the opposite end of the line; as a nautical term, it means withdrawing a vessel from the water. To control the direction of a vessel, in order to approach another vessel or the quay.

(to) HAUL IN

to increase the tension of a rope.

(to) HAUL OUT

to take the vessel out of the water.

HAULAGE

to withdraw a vessel from the water, usually for winter storage.

HAULED

a particularly tight tie.

HAWSER

circular opening on the bulwarks, generally at the bow, through which the anchor chain passes; usually the pleasure boats have a single hawser on the main deck plane, located above the chain locker.

HAZE

reduced visibility, but still greater than 1 km: below this value, it is referred to as "fog".

HECTO-PASCAL

modern measuring unit, which replaces the millibar; 1 hPa = 1 mbar.

HEELING

the amount that a boat is tipped over side-to-side, relative to its normal horizontal position.

HOLD

the space for cargo below the deck of the ship.

(to) HOOK

link a rope to a fix point of the vessel.

HOTEL

term that represents the letter H in the ICAO international phonetic alphabet.

HURRICANE

a strong tropical revolving storm of force 12 of Beaufort scale, with winds above 64 knots. In the northern hemisphere hurricanes revolve in a clockwise direction. In the southern hemisphere these storms revolve counter-clockwise and are known as typhoons.

HULL

the main structural body or shell of the boat, not including the deck, keel, mast, or cabin.

I**ICAO**

International Aviation Organisation, international phonetic alphabet used in maritime and air navigation in order to allow comprehension of letters and numbers in all languages.

INDIA

term that represents the letter I in the ICAO international phonetic alphabet.

INTERNATIONAL CODE

set of four letters that identify a vessel.

ISOBAR

line shown on weather charts which connects all points with same pressure.

ISOTHERM

line which connects all points with same temperature.

J

(to) JOIN

to connect two cable one to the other, in order to form a longer cable.

JULIET

term that represents the letter J in the ICAO international phonetic alphabet.

K

K

in Morse code it corresponds to "- . -" (dash-dot-dash): according to the international code of signals the "K" flag hoisted alone it means "I want to communicate with you".

KEDGE

small light anchor used on light vessels, also used for manoeuvring with two anchors.

KEEL

the central structural basis of the hull.

KEELSON

a beam attached to the top of the floors to add strength to the keel on a wooden boat.

KEVLAR

polymeric aromatic aramid fibre, with typical yellowish colour; it has been recently introduced in pleasure yachting both for hull manufacture and in sails and ropes manufacture, due to its extraordinary resistance to traction; in fact, kevlar is five times more resistant than steel, for the same weight, with approximately no elongment under stress.

KILO

term that represents the letter K in the ICAO international phonetic alphabet.

KNOT

the intertwining the parts of one or more ropes; a speed of one nautical mile (1852 metres) per hour.

L

LAMINATED

technique of manufacturing wooden hulls, superimposing thin layers, diagonally one to the other, using special glues in order that the structure will not be deformed due to the water absorption.

(to) LAND

reach the shore for leaving the vessel.

LANDING

place where it is possible to land (mooring to shore); for instance, a tourist landing is a harbour specially equipped for pleasure boats (also known as "marina").

LATITUDE

angular distance measured in degrees north or south of the equator, measured along the meridian of the considered point: latitude and longitude compose the geographic coordinates of a point.

(to) LAY UP

to remove all equipment necessary for navigation from a vessel; this operation precedes an accurate inspection of all components of the yacht.

LEADLINE

a line with a weight on the end used to measure depth. The lead is dropped into the water and marks on the line are read to determine the current water depth.

LEEWAY

sideways displacement of a vessel due to wind effect.

LEEWAY

what a vessel loses by drifting to leeward due to streams.

LENGHT OVERALL

maximum length of a vessel, including overhanging ends that extend beyond the main bow and the main stern.

LIFE LINE

cable situated on the main deck, from bow to stern, to which the life belt clevis must be attached, when sailing in bad weather.

LIFE RAFT

safety device, compulsory for all vessels authorised to sail at more than 6 miles from the shore; it is a collective rescue mean and must be able to carry all people present onboard the vessel; it is generally a self-inflatable pneumatic dinghy, containing particular equipment which allow survival in case of vessel's sinking.

LIFELINE

ropes carried between stanchions, for men to hold on by; the assembly of lifelines (generally two per side, at different height) and stanchions form the handrail.

LIFT

perpendicular component of the force generated on a solid surface (rudder, fin keel, etc.) due to its movement in a fluid; i.e. the fin keel lift is the force that opposes to the drift.

LIFT

a textile rope used for supporting a rod.

LIGHT AIR

very weak wind (1-3 knots) corresponding to force 1 in Beaufort scale.

LIGHT ALLOY

aluminium alloy based material used in masts and boom manufacture: it has been recently introduced in the manufacture of small-medium displacement hulls.

LIGHTHOUSE

light source situated on the shore in a raised position in order to allow night sighting from great distance; each lighthouse emits a particular type of light, white, green or red, with different frequency; such characteristics are shown on charts and on the list of lighthouses and fog signals, in order to ease recognition.

LIMA

term that represents the letter L in the ICAO international phonetic alphabet.

LINE

strong rope, with reduced diameter.

LINK

chain ring.

LOCKER

a chest or box, to stow anything away in.

LOG

a device used to measure the distance travelled and speed through the water.

LOGBOOK

registry on which the skipper records every hour the information and variations concerning the navigation (i.e. route modification, sea conditions, wind speed and variations, etc.).

LONGITUDE

angular distance measured in degrees between the considered point to the Greenwich meridian (international reference), calculated along the parallel passing to the point.

LOOP

passing a rope around a cleat, a ring or any other object, holding both ends; mooring with loop lines is widely used, as it is possible to leave slowly the quay, releasing one of the two ends of the loop and recovering the other end, without the need of unboarding or using the tender.

(to) LOOSE

reducing the tension on a rope; the operation is different from "letting go" a rope, which means to withdraw the whole tension on the rope.

LORAN

long-range navigation system that uses radio signals transmitted at specific times. An onboard receiver computes position by measuring the difference in time of signal reception. This system has now been re-

placed with Loran C, an improved version of the system.

(to) LOWER

to lower sails, flags, etc. loosening the lines to which they are tied.

LUBBER'S LINE

a line on the compass aligned with the centreline of the vessel that indicates the vessel's compass heading. Since it points to the vessel's bow, it enables a course to be steered by bringing the lubber's line to the point on the compass card which indicates the desired course.

(to) LUFF UP

to steer the boat more into the wind, thereby causing the sails to flap or luff.



MAGNETIC VARIATION

angle between geographic North (or true North) and magnetic North.

MAGNETIC DEVIATION

angle between the magnetic North and the direction shown by the compass needle; the deviation is caused by pieces of iron such as engines, anchors, etc. and varies with the course.

MAIN DECK

surface that covers and encloses the hull at the top: in modern fibreglass sailing boats all walking surfaces are made anti-skid by means of either wrinkling realised directly on the mould or with special paints: on more luxurious yachts the main deck is covered with planking made with marine environment resistant wood such as Siamese teak.

MAIN DECK EQUIPMENT

set of equipment, winches, blocks, etc. necessary for main deck operations.

(to) MAKE FAST

to tie a rope securely to a sustain (cleat, ring, lifeline).

(to) MAN

to prepare part of equipment in order to carry out its specific duty.

MARINA

a place where boats can find fuel, water and other services.

MARITIME SIGNAL FITTING

fix or floating structure arranged for supporting a signal.

MARLINE TWINE

small line used for whipping, seizing, and lashing.

MATT

pressed glass fibre used for manufacturing fibreglass vessels.

MAXIMUM BEAM

maximum width of the vessel.

MAY DAY

rescue distress call, generally carried out on channel 16 VHF; from French "m'aider" that means "help me".

MELTEMI

northerly wind that blows in the Egean Sea, mainly during the summer.

MERCATOR (GERHARD KREMER)

Belgian geographer who invented the Mercator projection used for nautical charts.

MERIDIAN

a semi-great circle joining the north and south poles. Known as lines of longitude, they cross the equator and all parallels of latitude at right angles (see also "Greenwich").

MIKE

term that represents the letter M in the ICAO international phonetic alphabet.

MILLIBAR

a unit used to measure the barometric pressure of the atmosphere. 1 millibar equals 1000 dine per square centimetre, or 0.03 inches of mercury.

MISTRAL

strong north-westerly wind, called also with the French term "Mistral", it often generate sea conditions which can be tough for navigation.

(to) MOOR

the act of confining and securing a ship in a particular station, by chains or cables, which are either fastened to the adjacent shore, to a buoy, to anchors etc.

MOORING CABLE

thick cable used for mooring large vessels.

MOORING POST

submersed object, mounted to the bottom, with suitable weight, shape or dimensions for allowing mooring a yacht; in this case the mooring line is called "trip line".

MORSE

a language of "dots" and "dashes" invented by Samuel Morse, used to send messages, either sound using radio waves, or light using a searchlight.

MYLAR

polyester film used for manufacturing sails to be used with light wind: a sail made with a film, instead of using a traditional textile material, has constant elasticity coefficient on all directions.

**NADIR**

the NADIR is the intersection of the perpendicular to the horizon passing through the observer with the invisible celestial hemisphere.

NAUTICAL SCIENCE

art and technique of sailing.

NAUTICAL MILE

distance at sea is measured in nautical miles, which are about 6067 feet, 1.15 statute miles or 1852 meters. Nautical miles have the unique property that a minute of latitude is equal to one nautical mile.

(to) NAVIGATE

to sail on the water with a vessel.

NAVIGATION

the art and science of determining the position of a vessel and the course needed to safely and efficiently move the boat from place to place; 1. astronomical navigation: determining the position of a vessel with the aid of satellite and ephemeris; 2. Coastal navigation: sailing considering conspicuous points and bearing points generally situated along the coast; 3. Dead reckoning navigation: navigation technique estimating position based on last known position, heading, speed, and time, and considering the effect of drift.

NAVIGATION LIGHTS

lights that are turned on at night, located on both sides of the bow, green on starboard side and red on port side, each one covering an angle of 112,5 degrees, from the bow.

NAVIGATIONAL CHART

particular geographic chart that reproduces the terrestrial surface according to Mercator's projection, in which all parallels are perpendicular to meridians; in this type of chart angles equality is maintained, but distances are deformed and increased going from the equator towards the poles.

NIMBOSTRATUS

dark, grey low altitude cloud that produces continuous precipitation in the form of rain or snow. Found in an altitude range from the surface to 1500-2200 meters.

NOOSE

running knot carried out with two half turns opposed and coupled.

NORTH

one of the 4 cardinal compass points, identified with the letter N. North is the direction toward the North Pole and is at 0 degrees on the compass rose.

NOTICES TO MARINERS

bulletins issued by "Istituto Idrografico della Marina" (Navy Hydrographic Institute) showing temporary or permanent variations of information shown on nautical charts, pilot books and on the list of lighthouses, beacons and fog signals.

NOVEMBER

term that represents the letter N in the ICAO international phonetic alphabet.

**OAR**

a pole with a blade at the end used to row a boat.

OFF-SHORE

see "Open sea".

OILSKIN

water repellent clothing used in bad weather conditions.

ONE-OFF

prototype of race boat.

OPEN HAWSE

when a vessel rides by two anchors, which are approximately 90° apart one from the other; This operation is carried out in order to avoid that the vessel rotates due to a change of wind.

OPEN SEA

high sea, out of shore sight.

OSCAR

term that represents the letter O in the ICAO international phonetic alphabet.

OSMOSIS

is the net movement of solvent molecules through a partially permeable membrane into a region of higher solute concentration, in order to equalize the solute concentrations on the two sides.

OVERALL

maximum length of the vessel.

OWNER

the owner of the vessel.

P

PACE

direction of the vessel course compared to the wind; it assumes various nouns while the wind rotates from bow to stern (closed haul windward: 40-45 degrees to apparent wind; windward: 45-55 degrees; close reach windward: 60-80 degrees; reach: 90 degrees; run: 100-130 degrees; broad reach: 140-170 degrees; run free: 180 degrees).

PAINTER

long line used for mooring, fixed to the bow; towing rope.

PAPA

term that represents the letter P in the ICAO international phonetic alphabet.

PARALLEL

also called latitude lines, it is a circumference on the terrestrial surface, parallel to the equator, therefore all point of the same parallel have the same latitude. Parallels are always perpendicular to meridians.

PARALLEL RULE

plotting tool formed by two rulers always kept parallel.

PASCAL

measuring unit of pressure, equal to 1 Newton on square metre.

PEAK

small compartment or storeroom.

PILOT BOAT

small motor boat used for piloting ships in the harbour or for transporting the pilot onboard the ship.

PILOT BOOK

publications issued by the Navy Hydrographic Institute, covering detailed information about coastal navigation, including navigational aids, courses, distances, winds, streams, anchorage and harbours.

PITCHING

rhythmic longitudinal rotation of the boat (the bow raises at the stern sinks, and vice-versa).

(to) PLANE

to gain hydrodynamic lift, causing the boat to lift, rising slightly out of the water, reaching speeds in excess of those normally associated with its waterline length.

PLANKING PLATES

assembly of wooden planks or aluminium, iron or steel sheets, which compose the external part of the hull, covering the framing; there are different types of planking plates, such as superimposed, double, diagonal or clinker.

PLEASURE

navigation carried out for leisure or sport, without profit purposes.

(to) PLOT THE COURSE

set of operations used to follow all navigation phases on the chart (located on the chart table).

PLYWOOD

wooden plate composed of various layers glued and pressed together, with fibres perpendicular one layer from the other in order to reduce structure expansion or deformation: marine plywood is generally realised superimposing mahogany sheets and using marine-environment resistant glue; usually the visible layer is made with teak or other high quality wood, eventually veined with thin black laths (for external parts) or white laths (for internal parts).

POLAR

polar circle: terrestrial parallel which delimits a polar region; the North polar circle is called Arctic polar circle, the south one is called Antarctic polar circle; polar bearing: angle between the vessel's longitudinal axis and a conspicuous point.

POLAR DIAGRAM

graphic that shows the apparent wind speed in all possible paces, considering the true wind speed and the vessel's speed constant.

PORT

left side.

PORTHOLE

round opening in a ship's hull for ventilation, light and other purposes, generally provided with a sturdy glass hatch equipped with watertight gasket.

POST

stempost is the bow line which links the bulwarks. The sternpost is the stern end, generally composed of the transom, which increase the living space onboard and the hull shape stability.

PRAM

service tender.

PROPELLER

propulsion device for engine driven vessels.

PROTRACTOR

graduated circle used for measuring angles, during plotting operations.

(to) PULL AWAY

setting the vessel's course towards the wind propagation direction, therefore increasing the angle between apparent wind and the vessel's course (maximum 180 degrees).

PULLEY

wheel that rotates around a shaft; along its circumference, it carries a race which a cable slides on: the pulley is the central part of blocks.

PULPIT

an elevated guard-rail set up at the bow of a vessel, usually made of steel. When erected at the stern, it is called a pushpit.

PUSHPIT

an elevated guard-rail set up at the stern of a vessel, usually made of steel. When erected at the bow, it is called a pulpit.

Q

QUADRANTS

the four sectors in which the compass rose is divided into: first quadrant, between 0 and 90 degrees, second quadrant, between 90 and 180 degrees, third quadrant, between 180 and 270 degrees and fourth quadrant, between 270 and 360 degrees: i.e. a "wind from the first quadrant" is blowing from a direction between North and East.

QUALIFICATION

official recognition of suitability for steering a vessel (license); authorisation for sailing beyond certain limits (i.e. for a yacht, more than 6 miles from shore line).

QUARTER SEA

waves on a boat's quarter, to a bearing 45° abaft the beam.

QUAY

fix or floating structure which allows vessels' docking.

QUEBEC

term that represents the letter Q in the ICAO international phonetic alphabet.

R

RACE

a competition of skill and seamanship between yachts.

RADAR

Radio Detection and Ranging. An electronic instrument that uses radio waves to find the distance and bearing of other objects. Used to avoid collisions, particularly in times of poor visibility.

RADAR REFLECTOR

an object designed to increase the radio reflectivity of a boat so that it is more visible on radar, therefore it must be installed in the highest position as possible.

(to) RAISE

being able to see the shore line, another vessel or a distant object.

(to) RANGE

to lay a line down in wide coils, in order to avoid entanglement.

RATING

a method of measuring a yacht's expected performance relative to another yacht while racing. Over the years many formulas have been experimented with, in order to enable diverse yachts to race competitively against each using a handicap based upon the rating.

REACH

see pace.

REEF KNOT

also known as the square knot, it is formed by two half hitches in which the ends always fall in line with the outer parts. This knot is used to link two lines.

RELATIVE HUMIDITY

value measured by the hygrometer; it indicates the percentage of steam actually present compared to the total steam that can be contained in the air at room temperature (at 100% there is condensation).

RESERVE

equipment stored on board used to replace worn out or broken parts.

R.I.Na.

Registro Italiano Navale; Italian office which checks, tests and approves nautical equipment and vessels.

RING

fix part of a cable, tied to a bracket (generally an eyebolt, sometimes referred as "ring") or to a ring fixed to a block; the rope, passing to a second block, and again to the pulley of the first block, compose a tackle.

ROADSTEAD

a partly sheltered area of water near a shore in which vessels may ride at.

ROARING FORTIES

a region between 40° South and 50° South where westerly strong winds circle the earth unobstructed by land.

ROLL

the alternating motion of a boat, leaning alternately to port and starboard; the motion of a boat about its fore-and-aft axis. It is one of the most common causes of sea-sick.

ROMEO

term that represents the letter R in the ICAO international phonetic alphabet.

ROPE

generically referred to a textile cable.

(to) ROUND

overtaking an obstacle such a cape or promontory, following a route at safety distance.

ROWLOCK

places cut in the gunwale of a boat for the oar to rest in while pulling.

RUBBING STRAKE

wooden, textile rope or plastic reinforce that runs along the hull, generally at maximum beam height, for protection.

RUDDER

a board-shaped swinging vane, controlled by a tiller or wheel, and attached to the rudderpost or stern for steering and manoeuvring a vessel. The rudder is called "compensated" if the rotation axis is a bit aft of the leading edge; this type of rudder reduces the forces on the tiller or steering wheel.

RUDDER BLADE

submersed part of the rudder which allows to modify the course of the vessel.

RUDDER HEEL

aft end of the hull on which the lower rudder gudgeon is inserted.

RUDDER TRUNK

the trunk fitted in the counter to receive the rudder post into which the tiller is fitted.

(to) RUN AGROUND

a vessel whose keel rests on the seabed.

RUNNING

sailing in run free pace at high speed.



SAFETY HARNESS

the life belt or life jacket is a particular floating device which is worn for ensuring floatability in case of accidental fall into the sea; the life belt, usually equipped with shoulders and legs straps, is used to remain tied to the boat, in case of bad weather or during dangerous operations; it is provided with rope and one or two clevises.

SAILING TO WINDWARD

sailing towards the wind.

SAILOR

man or boy employed in sailing deep-water craft. Word is sometimes loosely used to include men who go to sea. Used officially to denote a seaman whose duty is to plot the course of a vessel.

SANDWICH

particular manufacture technique for pleasure boats, made in various layers (i.e. two layers of fibreglass enclosing a layer of balsa or polyurethane, in order to create a stiff and light structure).

SCOOP

plastic container used for removing water from the boat by hand.

SCOPE

submersed part of the anchor chain, when the anchor is lowered.

(to) SCULL

to impel a boat by one oar at the stern, the boat movement is obtained by moving the oar in an "eight-shape" movement.

SCUPPER

an opening in a deck or cockpit to allow water to run off the deck and drain back into the sea.

SELF-EMPTYING

a cockpit that allows rapid and complete water deflux through the scuppers: it is essentially a safety factor, if a wave (or heavy rain) has filled the cockpit itself.

SELF-INFLATABLE

pneumatic dinghy or raft (contained into a special stiff or soft container) which can be inflated automatically by pulling a special rope; the self-inflatable raft is compulsory for vessels authorised for open sea cruises.

SELF-TAILING

see winch.

SEXTANT

a navigational instrument used to measure the altitude of celestial bodies on the horizon; by means of special tables (ephemerids) that describe the position of stars and planets at each moment of the year, it is possible to calculate the point on the terrestrial surface where the observation is carried out.

SHACKLE

metal (generally stainless steel) rod bent in U-shape with flattened or drilled ends in order to allow insertion of a pin, which can be threaded, blocked by a split pin or other device. U-shaped fitting closed with a pin across the open ends, used to secure two rings.

SHACKLE UNDOER

tool used for unscrewing (or screwing) the shackle pins.

SHAFT

shaft of propeller or rudder: supports (generally made in stainless steel) which allow rotation of various devices.

SHALLOW

see "Bank".

SHANK

main section of the anchor, where other parts are attached to.

SHEEPSHANK

a knot used to temporarily shorten a line.

SHEER

main deck longitudinal bend.

SHEET BEND

joining knot suitable for both large and thin lines; it is also used to tie the flag rope to a halyard.

SHEET LEAD

also called Fairlead; fitting (eyebolt, transmission, fix or mobile block) used to guide a line in a particular direction without chafing, i.e. forwarding a line to the cockpit, setting the sheet in order that arrives to the winch at the right angle.

SHELF

longitudinal part of the vessel framing.

SHELTER

obstacle to wind or sea, which reduces its intensity.

SHIP

generic term which identifies a large vessel; the Italian law defines a ship as a vessel with heavy displacement above 50 tons.

SHOAL

an underwater sand bar or hill that has its top near the surface, therefore dangerous for navigation.

SIERRA

term that represents the letter S in the ICAO international phonetic alphabet.

SINKING

accident that cause the loss of the yacht, or its transformation into a relic.

SIROCCO

a warm wind of the Mediterranean area, either a foehn or a hot southerly wind in advance of a low pressure area moving from the Sahara or Arabian deserts. Called LEVECHE in Spain.

SKEG

an extension of the keel for protection of propeller and rudder.

SKIN FITTING

closable pipe used to suck or discharge water, through an opening situated on the underbody.

SKIPPER

Yacht's captain.

SKYLIGHT

a framing of metal fitted over an opening in a deck, with glass or Plexiglas inserted for the admission of light into a cabin, engine room, etc.

(to) SLACK

release a rope, decreasing its tension.

SLIDING HATCH

opening and relevant door or sliding door to reach belowdeck rooms.

SNATCH BLOCK

a block with a single sheave which is hinged and opens on the side so that the bight of a line can be led into the block and closed without running the whole length through.

SOS

a distress call made by a vessel requiring assistance. These three letters were chosen because they were easy to make and read using Morse Code (... _ _ _ ...).

SOUTH

one of the 4 cardinal compass points, identified with the letter S. South is the direction toward the South Pole and is at 180 degrees on the compass rose.

SOUTH-WEST WIND

south-westerly wind, mostly violent with speed above 40 knots; it generates high waves, especially on the Italian west coast; it can be very dangerous, as it is often followed by perturbations.

SPLICE

to join two lines together by interweaving and tucking their strands over and under each other in various manners. A strong way of joining lines without the use of knots.

SPLIT PIN

metal rod inserted into the hole of a screw or in the eyebolt of a pin, and then bent, in order to fix its position or avoiding detachment.

SPOILER

stiff appendix applied to the transom, which increase the dynamic length of the vessel, or only for easing boarding the vessel from the sea.

SPRINGS

a dock line leading forward or aft, to prevent a vessel from moving ahead or astern. The after bow spring line is attached near the bow and runs aft, where it is attached to the dock. The forward quarter spring line is attached to the quarter of the boat, and runs forward, being attached to the dock near the bow of the boat.

STABILITY

the tendency in a boat to keep an upright position or to return to it when careened over.

STALL

usually air travels smoothly along both sides of a sail, but if the sail is not properly trimmed, the air can leave one of the sides of the sail and begin to stall. Stalled sails are not operating efficiently. Analogously also the rudder can stall, making impossible to steer the vessel.

STANCHION

rod which supports the lifelines, that pass into the special holes; the handrail is composed of stanchions and lifelines.

STARBOARD

right side.

(to) STEER

to drive a vessel.

STEERING WHEEL

wheel connected to the rudder, used for steering the vessel.

STEERING WIRE

steel cables which transmit the controls from the steering wheel to the rudder.

(to) STEM

a term indicating that a vessel is holding her own against a contrary wind or current.

STERN

the back (aftermost) part of a vessel.

STERN GLAND

component in the box in which the propeller shaft rotates.

STERN LIGHT

white light that must be turned on when sailing at night: according to international rules, showing an unbroken light over an arc of the horizon of 135 degrees, alighted to the vessel's longitudinal axis.

STOCK

reinforce situated on the upper part of the anchor, generally extractable and perpendicular to the stem.

STOCKED ANCHOR

typical anchor, suitable for any type of seabed, although heavy and difficult to handle; on pleasure boats only the removable-stock model it is used.

STOPPER

hardware used to fasten a rope, by means of a self-locking geared eccentric element, connected to an external lever to be lifted in order to release the rope (as long as the lever is lowered, the system allows rope movement only in the tensioning direction); by using the stoppers, it is possible to keep tensioned more ropes, using only one winch.

STORM

corresponding to Force 10 of the Beaufort scale (wind at 48-55 knots; during “violent storms” (force 11) the wind can exceed 60 knots.

STRAND

a number of fibres twisted together ready to be laid up into a rope with other strands.

**STRATOCUMULUS
(CUMULO-STRATUS)**

a principal low-level cloud type (cloud genus), predominantly stratiform, at about 3,500-4,000 metres of altitude, in the form of relatively low gray and/or whitish layer, sheet or patch. Its elements are often arranged in bands or rolls that lie across the wind. Light rain, snow, or sleet may fall from stratocumulus.

STREAM

water displacement due to natural causes such as tides or winds.

STUFFING BOX

textile or plastic package which covers the mooring line in order for protecting it from wear; greased cotton oakum which covers the engine propeller shaft inside the strut, in order to avoid that water enters the vessel.

SURF

the breaking of the sea upon the shore; strong surf in the harbour may cause damages to yacht, which may collide one with the other or with the quay.

(to) SWAMP

vessel rolling as much as to be partially submersed, due to a wave, wind gust or wrong manoeuvre.

T**TACHOMETER**

instrument used to measure the speed; the term is not often used, and replaced with speed indicator or log, although the latter is the proper mile counter, but the two instruments are always installed together.

TACK

route section sailed following the same heading.

(to) TACK

to change a boat's direction, bringing the bow through the eye of the wind.

TACKLE

a purchase where two or more blocks are used to increase mechanical advantage, or the power exerted on a line.

TAFFRAIL

upper edge of the stern, connecting the hull and the main deck.

(to) TAKE A BEARING

to “point” an object between two sights of a bearing instrument.

TANGO

term that represents the letter T in the ICAO international phonetic alphabet.

TEAK

tropical hardwood tree, family Verbenaceae; its wood is light and highly resistant, used for manufacturing the main deck planking and indoor furnishing. Particularly valuable is the Siamese teak.

TELL-TALE

see windex.

TENDER

service dinghy or small rubber dinghy used to land when it is not possible to dock.

THIMBLE

a pear-shaped, grooved metal fitting around which an eye splice is made.

TIDE

the predictable, periodic regular rising and lowering of water in some areas due to the pull of the sun and the moon.

TIER

a vessel is tier moored when moored parallel to another vessel already moored to the quay.

(to) TIGHTEN

see "to haul in".

TILLER

rod connected to the rudder in order to rotating it staying in the cockpit: being directly connected and being on the opposite side of the rotation pin, the tiller should be moved in the opposite direction compared to the rudder.

TILLER EXTENSION

small rod or articulated extension, sometimes telescopic, applied to the tiller end; it is used to steer the boat when away from the cockpit, i.e. in small boats, when the helmsman is leaning out on the trapeze.

TIMBERS

on wooden vessels, the frames or ribs of a ship, connected to the keel, which give shape and strength to the ship's hull.

TOE-RAIL

on rowing boats, is the place where oars resting supports are installed: in sailing boats, is a metal or wooden angular structure placed around the main deck external perimeter; it is used by the crew to brace themselves when sitting or leaning out; it is generally provided with holes for water drainage.

TONNAGE

a measure of a vessel's interior volume; therefore it identifies a volume and not a weight, although measured in tons (a tonnage of 1 ton is equal to 2,83 sqm.).

TOPSIDES

the sides of a vessel between the waterline and the deck, but referring also to on or above the deck, including hull, main deck, superstructures etc.

TOW LINE

long mooring (or towing) line, fixed to the stern.

TRANSDUCER

component of an instrument circuit that transforms the received signal into electric current, which is measured in order to supply an indication of the intensity of the considered value (i.e. the log transducer transforms the propeller speed in a low intensity current, which is sent to the power unit and then measured by the power unit itself).

TRANSOM

the athwartship portion of a hull at the stern. The flat, vertical aft end of a ship.

TRAVEL LIFT

bridge-type crane.

TRIM

correct position of the hull compared to the sea surface.

TRIMMER

second rudder positioned along the trailing edge of the fin keel, used in race boats for correcting the trim.

TRIPPING LINE

buoy linked to a buoy rope.

TROUGH

an elongated area of relatively low atmospheric pressure.

TRUE COURSE

a course steered by the compass that has been corrected for variation and deviation, maintaining a constant angle on meridians and parallels, appearing on charts as a straight line; in general all routes carried out by pleasure vessels are "true courses".

TUCK

the form of the hollow in the quarter near the transom or stern-post.

**TURN**

complete turn of a rope around an object.

UCINA

Unione dei Cantieri Nautici ed Affini; Italian association that groups most important nautical operators in Italy.

UNDERBODY

submersed part of the hull.

UNDERBODY APPENDIX

any surface projecting from the hull (bulb keel, bulb, rudder, skeg, trimmer, ecc.).

UNIFORM

term that represents the letter U in the ICAO international phonetic alphabet.

UTC

Universal Time Coordinated; universal reference time, it has recently replaced the GMT, in order to underlining its absolute time reference.

V**VHF**

radio frequency included between 30 and 300 MHz (Very High Frequency); it has become the synonym of radio sets carried on boats, as they operate in this band.

VICTOR

term that represents the letter V in the ICAO international phonetic alphabet.

W**WAKE**

moving waves, track or path that a boat leaves behind it, when moving through the water; the track of disturbed water a boat leaves as it moves.

WARP

the longitudinal threads in canvas and other textiles.

(to) WARP

to move a vessel from one place to another by means of a rope made fast to some fixed object, or to a buoy.

WATER-LINE

the line where the water comes to on the hull of a boat. Generally it is painted some centimetres above the real waterline, in order to easily observe the longitudinal trim of the vessel.

WATERCRAFT

boat or other object manufactured for navigating; the Italian law defines a watercraft as a small pleasure boats, not longer than 6 metres, with engine not exceeding 20 HP in power and heavy displacement not exceeding 3 tons.

WATERLINE LENGTH

the length of the boat at the waterline.

WAVE

oscillations of the sea caused by streams, tides, surf or wind blowing along the surface.

WAVE HEIGHT

elevation of crest above preceding trough, calculated perpendicularly from the horizon.

WAVE LENGTH

the wave length is defined as the horizontal distance between two successive crests or troughs, calculated parallel to the sea surface.

WAYPOINT

a charted feature or chosen position on a chart.

WEATHER HELM

the natural tendency of a sailboat to come up into the wind.

WEATHER REPORT

summarised description of the weather situation of a zone with short-mid term forecast.

WEFT

the transversal threads interwoven with warp, in canvas and other textiles.

(to) WEIGH ANCHOR

to raise anchor in preparation for departure.

WEST

one of the 4 cardinal compass points, identified with the letter W. West is the direction at 270 degrees on the compass rose.

WHARF

a solid structure extending from the shore, used for mooring small/medium displacement vessels.

WHIPPING

to bind the strands of a line with a small cord. Winding twine or heavy thread around the end of a line to keep it from unravelling.

WHISKEY

term that represents the letter W in the ICAO international phonetic alphabet.

WHITECAPS

first white crests formed on the sea, which identify a wind stronger than 15 knots.

WILDCAT

metal pulley with a set of cogged wheels attached to the windlass drum. The cogs are so arranged as to engage individual links of chain and pull them aboard by using the anchor windlass.

WINCH

a metal drum shaped device used to increase hauling power when raising or trimming sails, loading and discharging cargo, or for hauling in lines. A machine that has a drum on which to coil (always clockwise) a rope, cable or chain for hauling, pulling or hoisting. Self-tailing winches are provided with a system located on the upper part of the drum, which throttles automatically the sheet when winding; coffee grinder are large winches, operated by special separate systems.

WIND

movement of air; it can be generated by variation in pressure between various zones, even at great distance (synoptic wind); when it is caused by thermal gradients, it is called "breeze"; true wind is the wind that flows freely, not altered by the presence of the vessel; apparent wind is the sum of true wind and vessel's advancement wind; advancement wind is the wind felt in complete absence of true wind, by any moving object, due to its speed. Catabatic wind is an air current moving downward or down a slope because of cooling especially at night; summed to the land breeze recalled from the sea, it often generate strong gusts.

WIND VANE

see "Windex".

WINDEX

object used for showing the wind direction, composed of an arrow or flag rotating on a pin mounted on the mast head.

WINDWARD

pace carried out maintaining a course as close as possible to the wind direction (approximately 40-45 degrees on left or right of the apparent wind): windward pace is divided into closed haul windward and close reach windward, according to the wind angle (see “pace”).

WINTER STORAGE

taking the vessel out of the water (generally during the winter); in order to carry out all maintenance operations to the vessel and its equipment.

X

X-RAY

term that represents the letter X in the ICAO international phonetic alphabet.

Y

YACHT

luxury sailing or engine-powered vessel, with small/medium displacement.

YACHTING

non-commercial boating activity.

YANKEE

term that represents the letter Y in the ICAO international phonetic alphabet.

(to) YAW

to swing or steer off course, due to the wind or stream.

Z

ZENITH

the point of the celestial sphere which is directly overhead. Its opposite is “nadir”.

ZULU

term that represents the letter Z in the ICAO international phonetic alphabet.

NOTES:

This image shows a full page of blank white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for writing or drawing. There are no margins, text, or other markings present.

